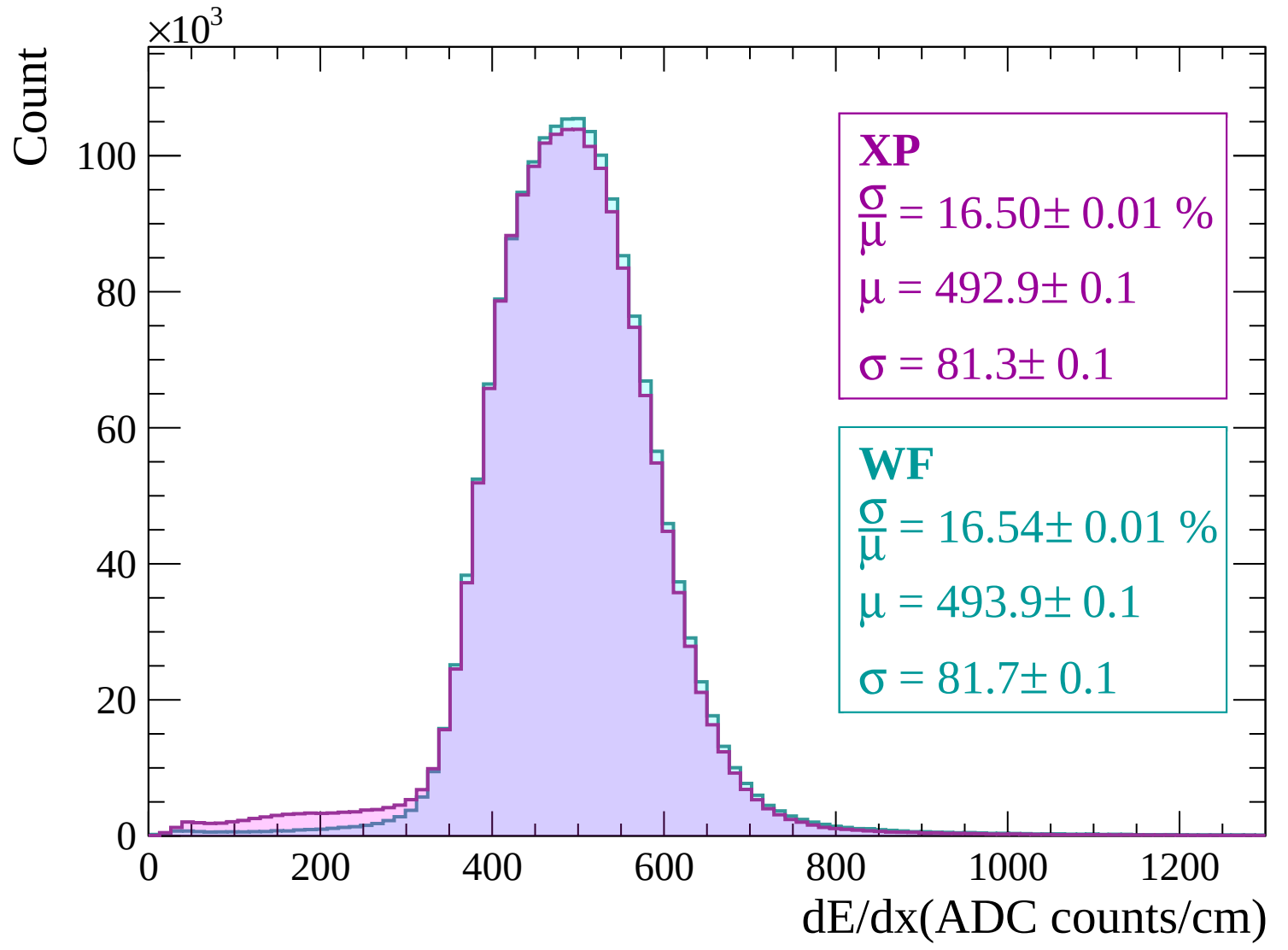
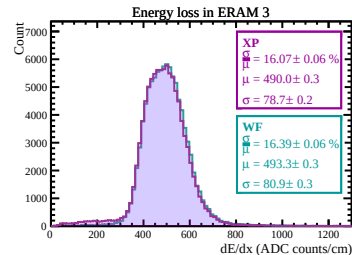
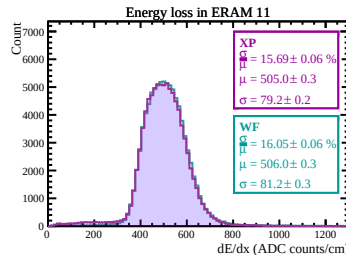
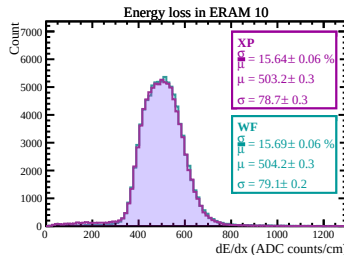
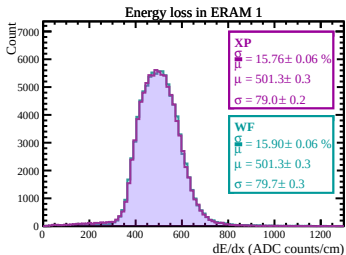
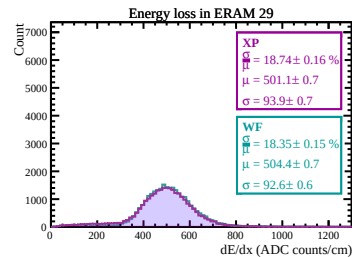
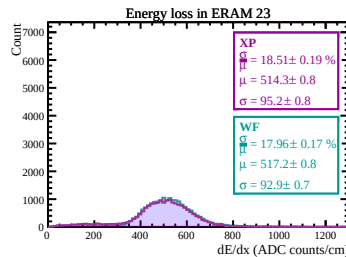
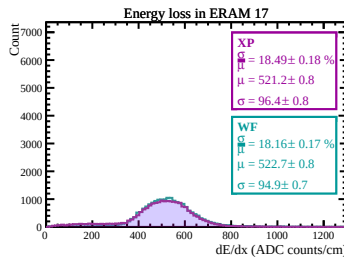
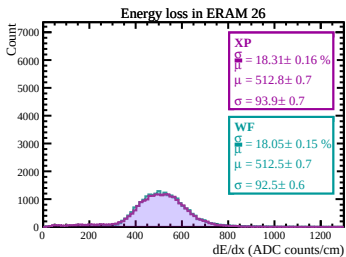
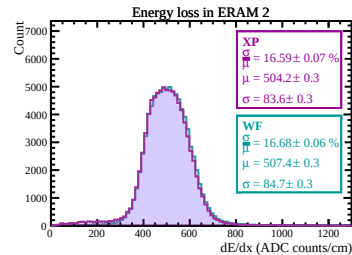
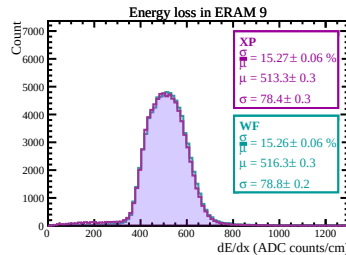
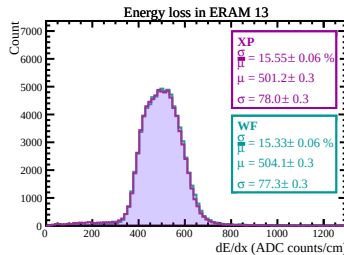
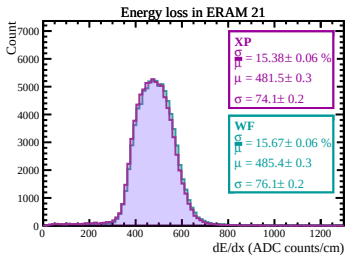
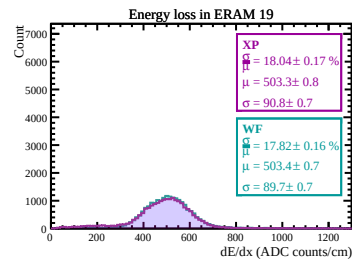
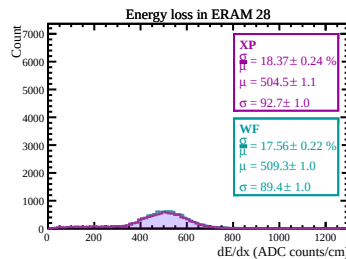
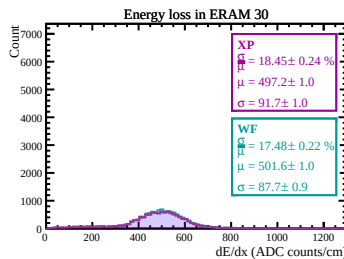
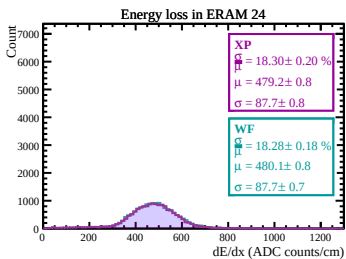
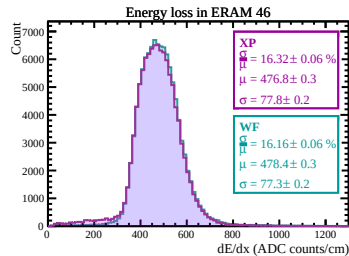
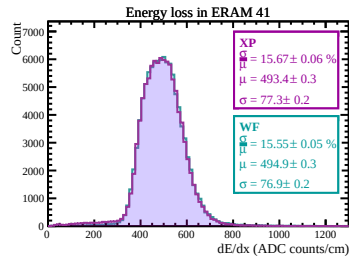
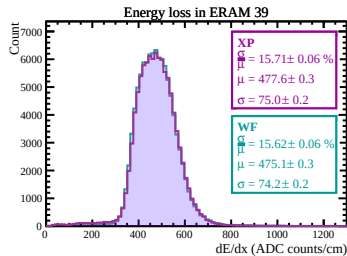
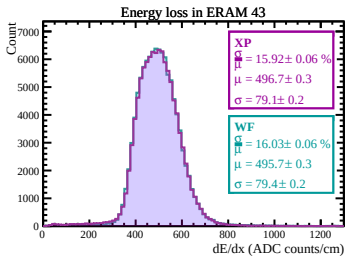
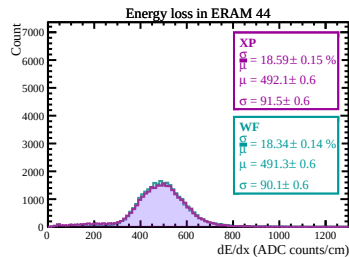
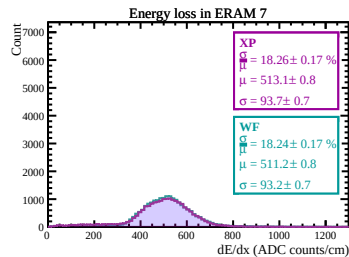
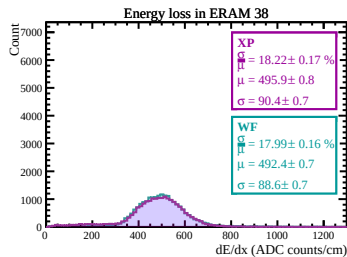
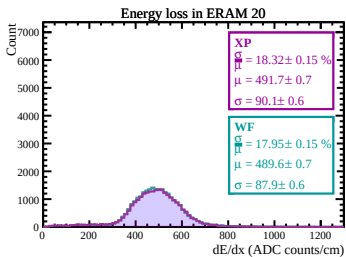
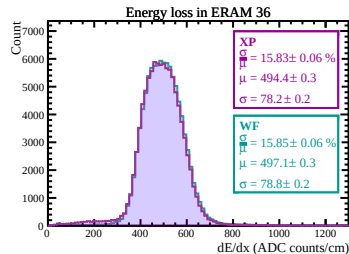
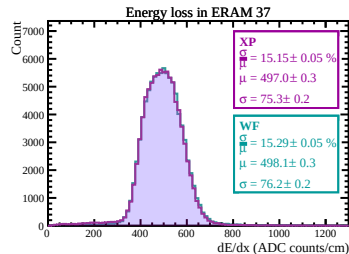
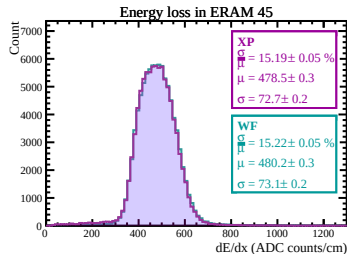
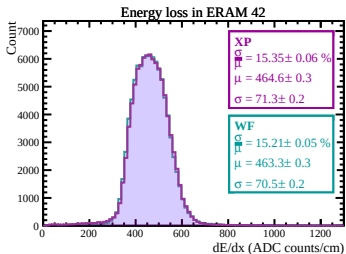
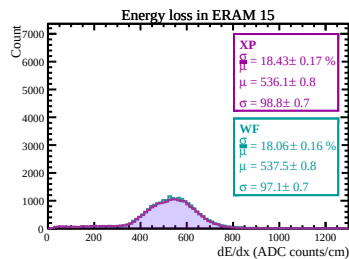
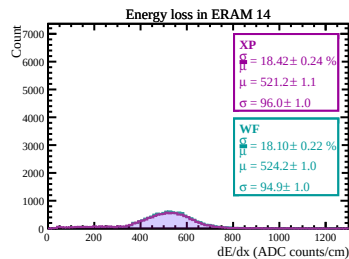
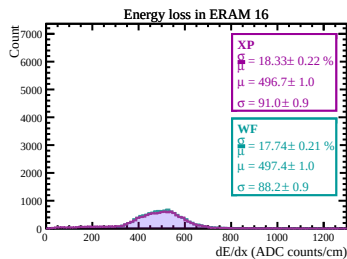
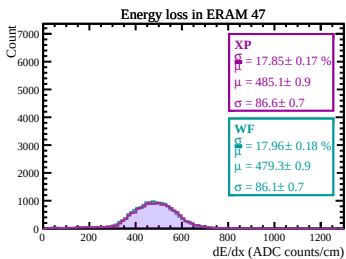
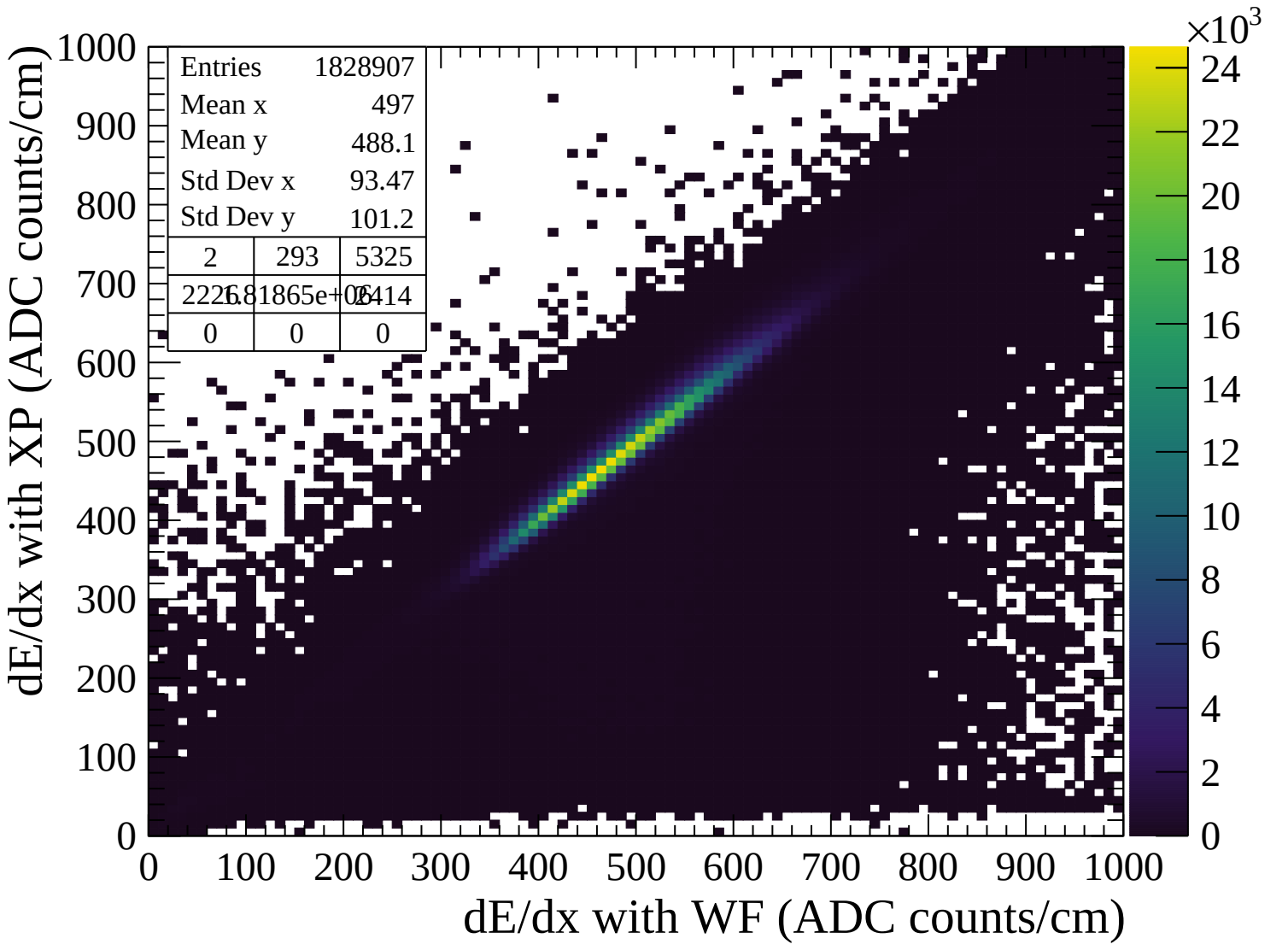
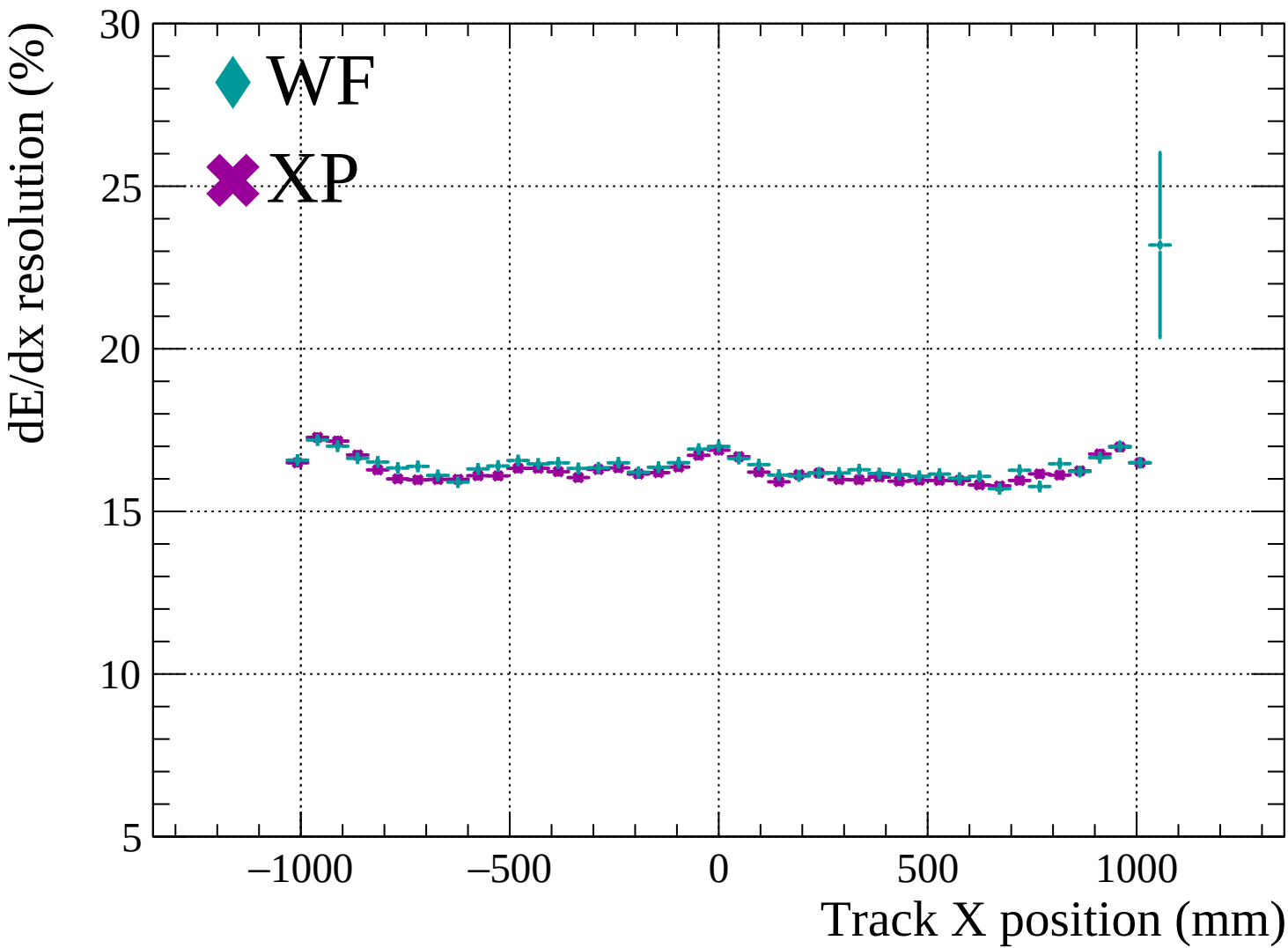


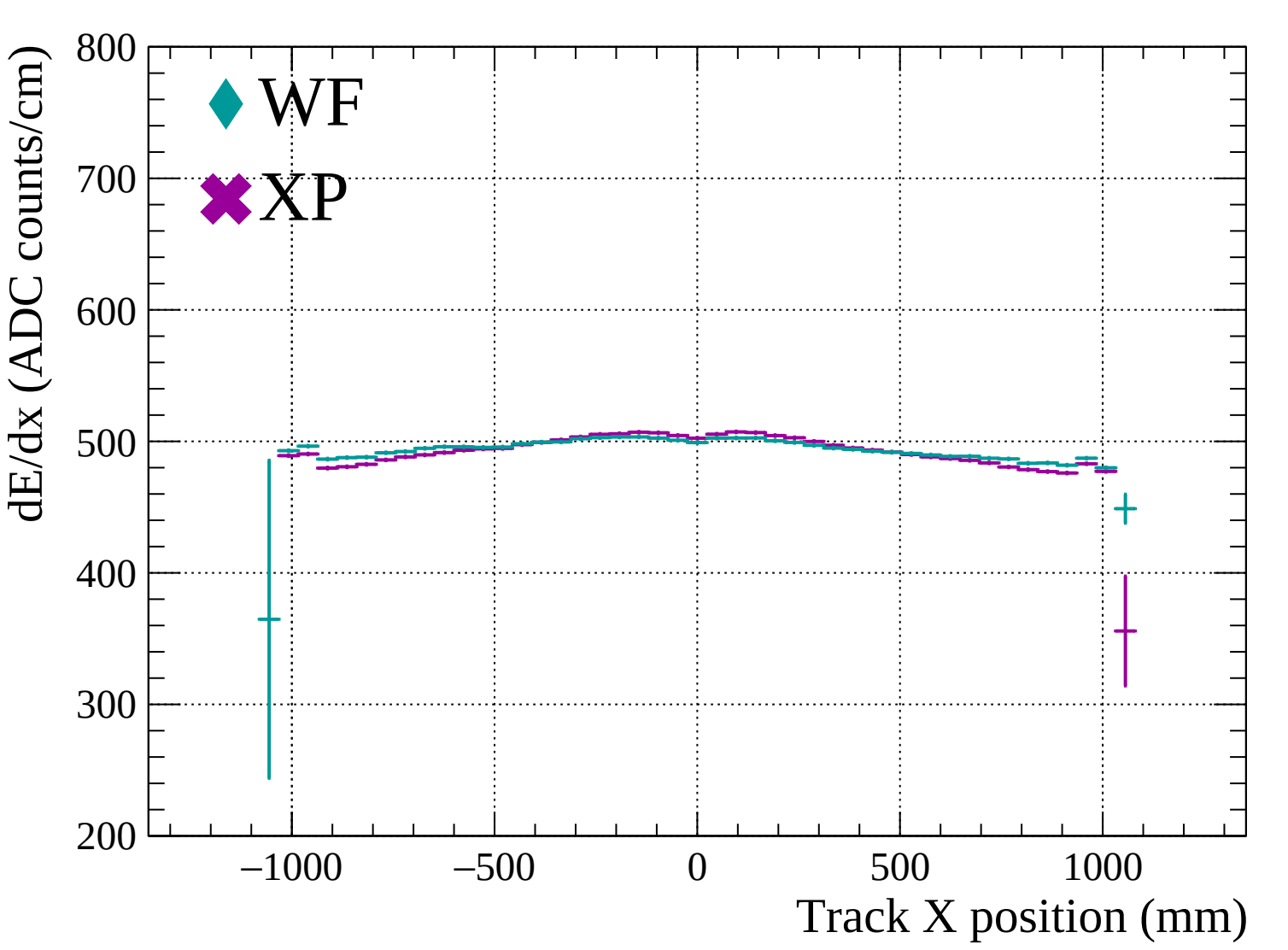


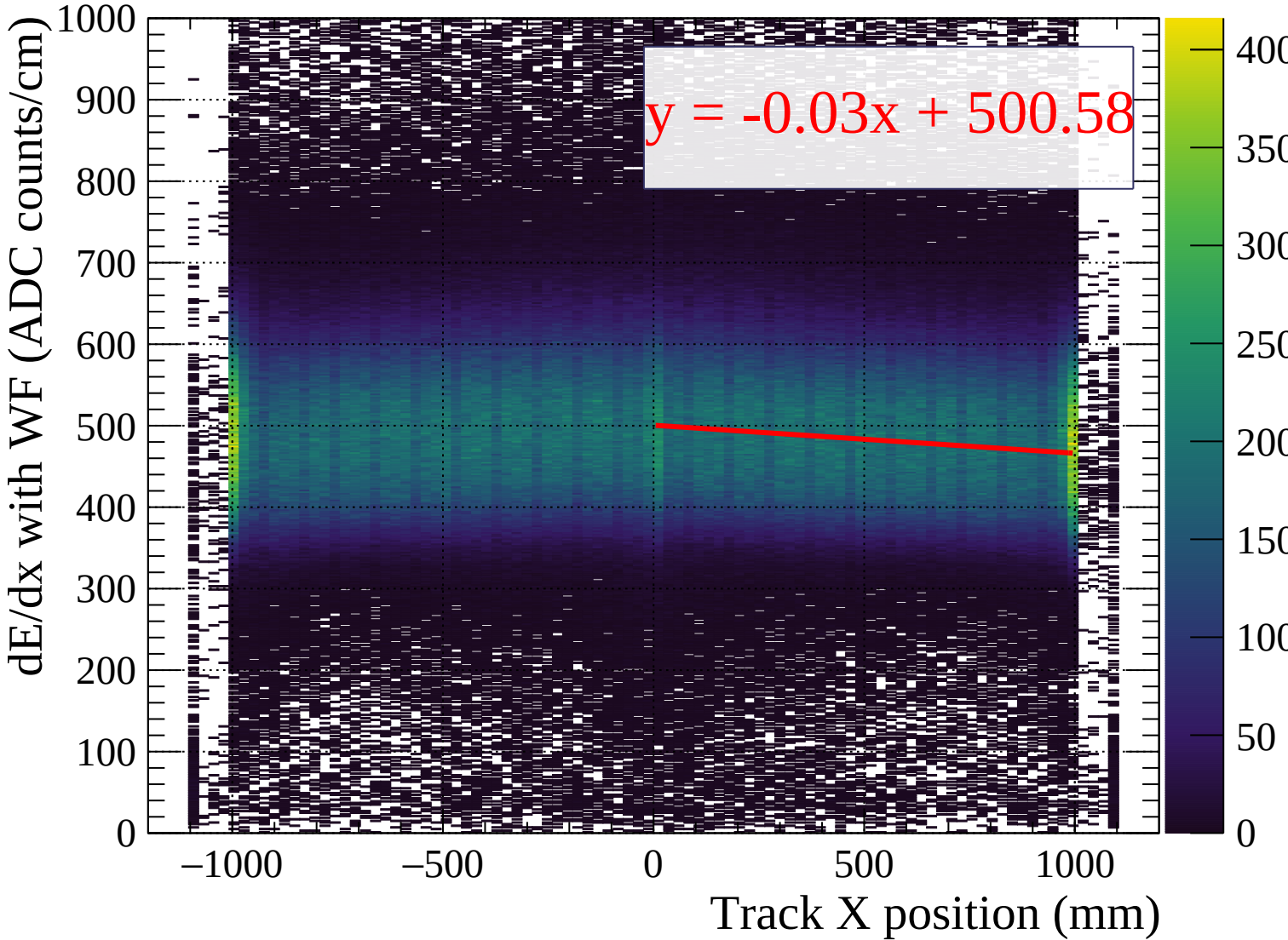


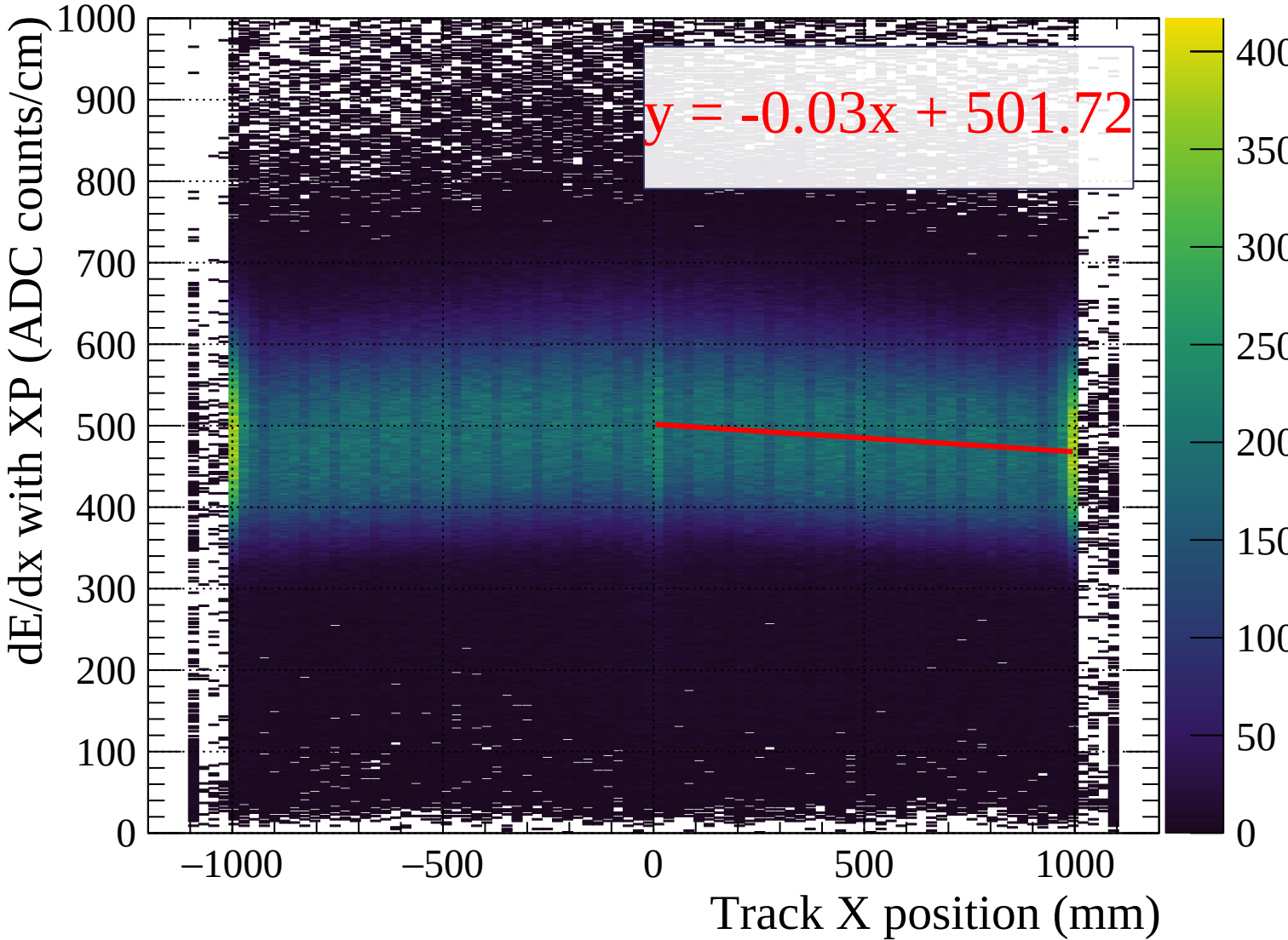


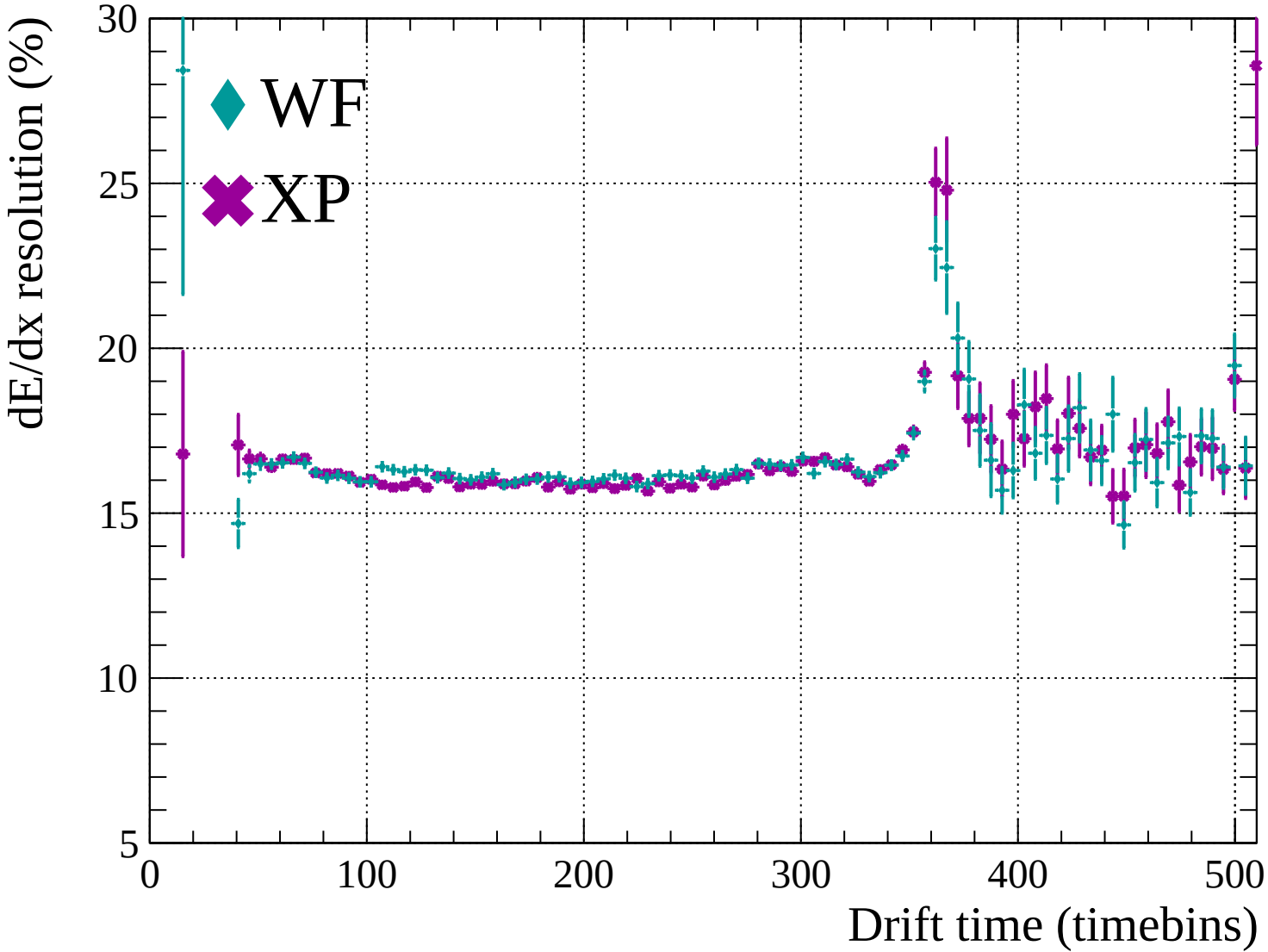


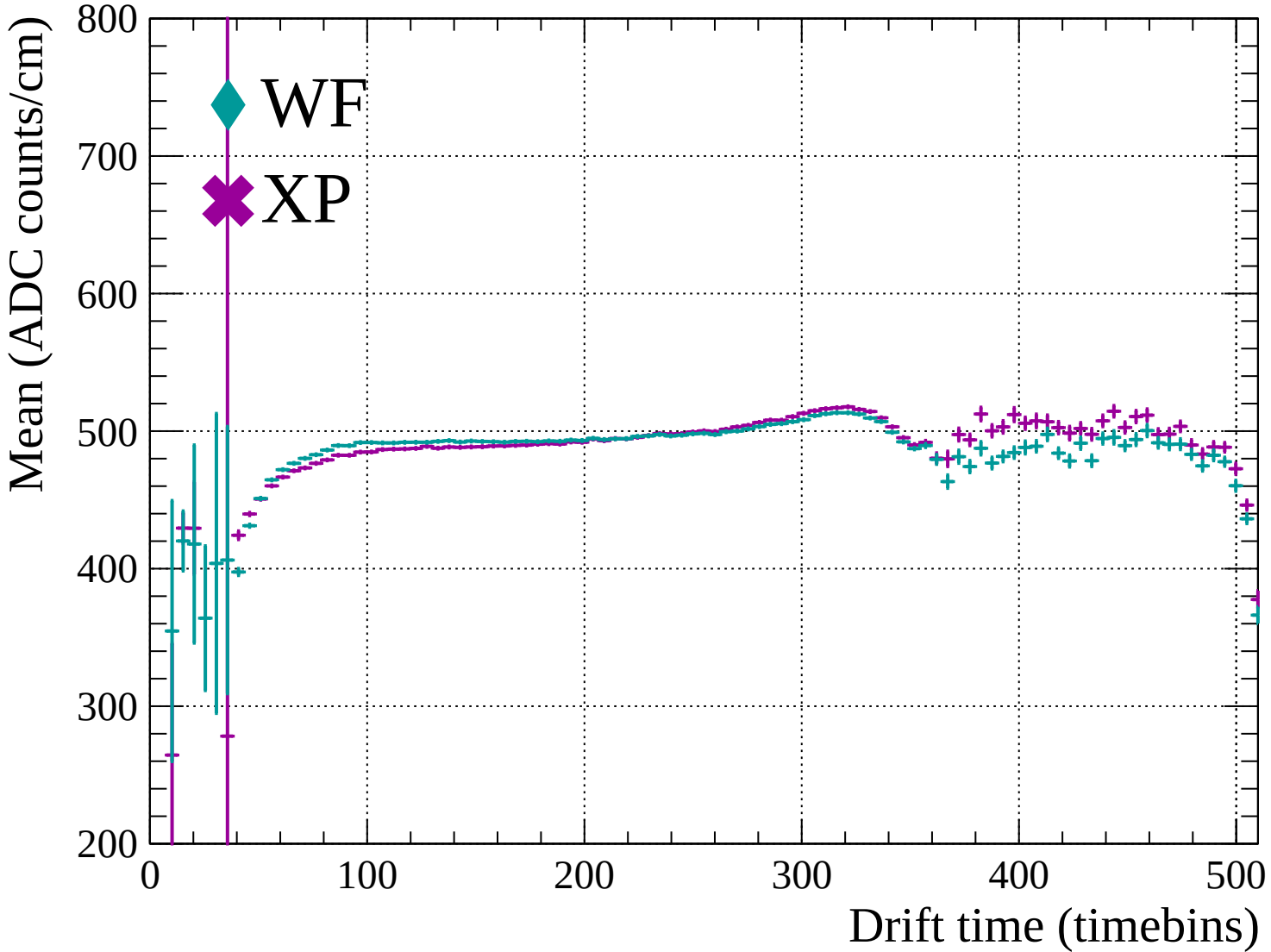


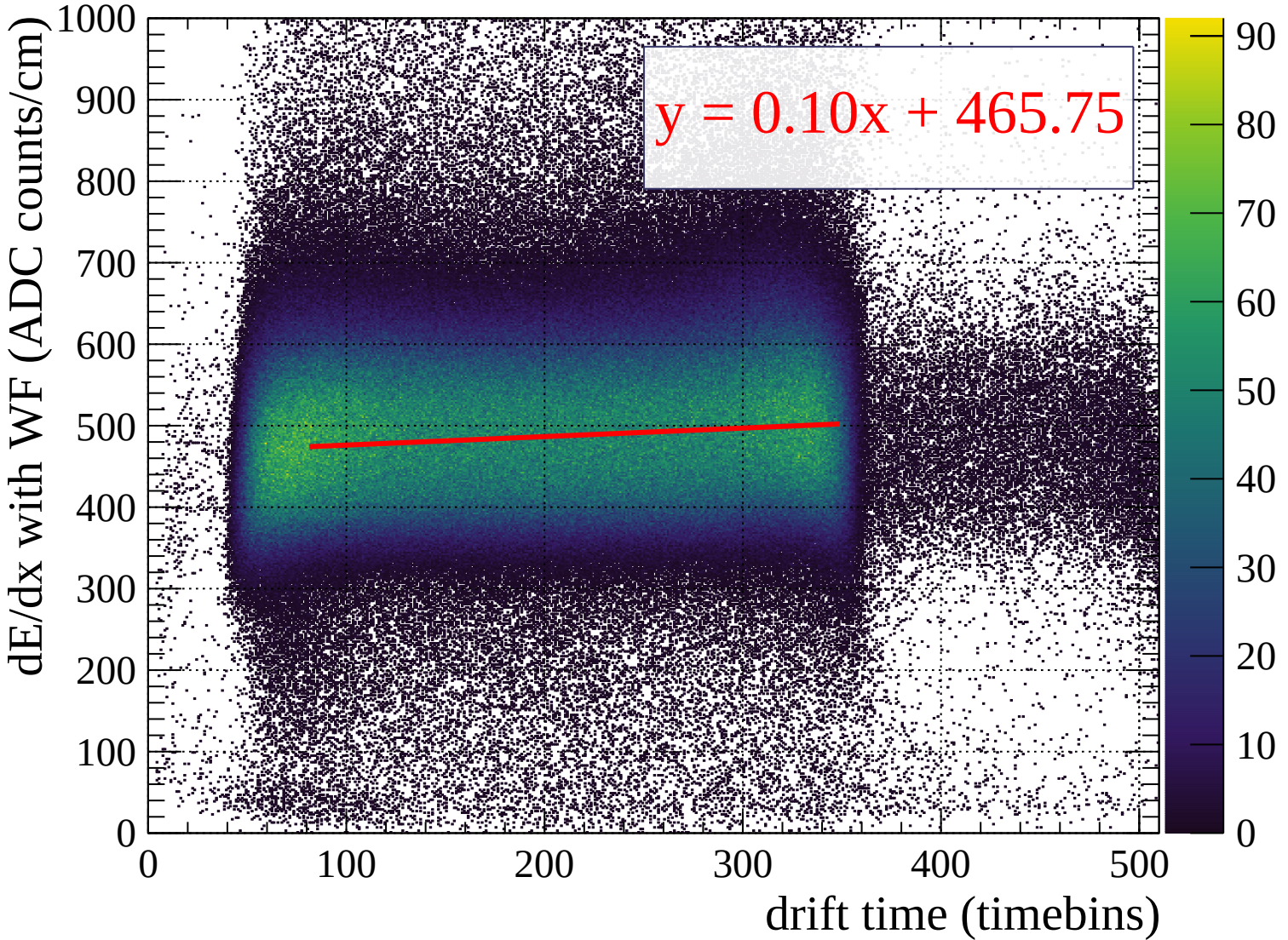


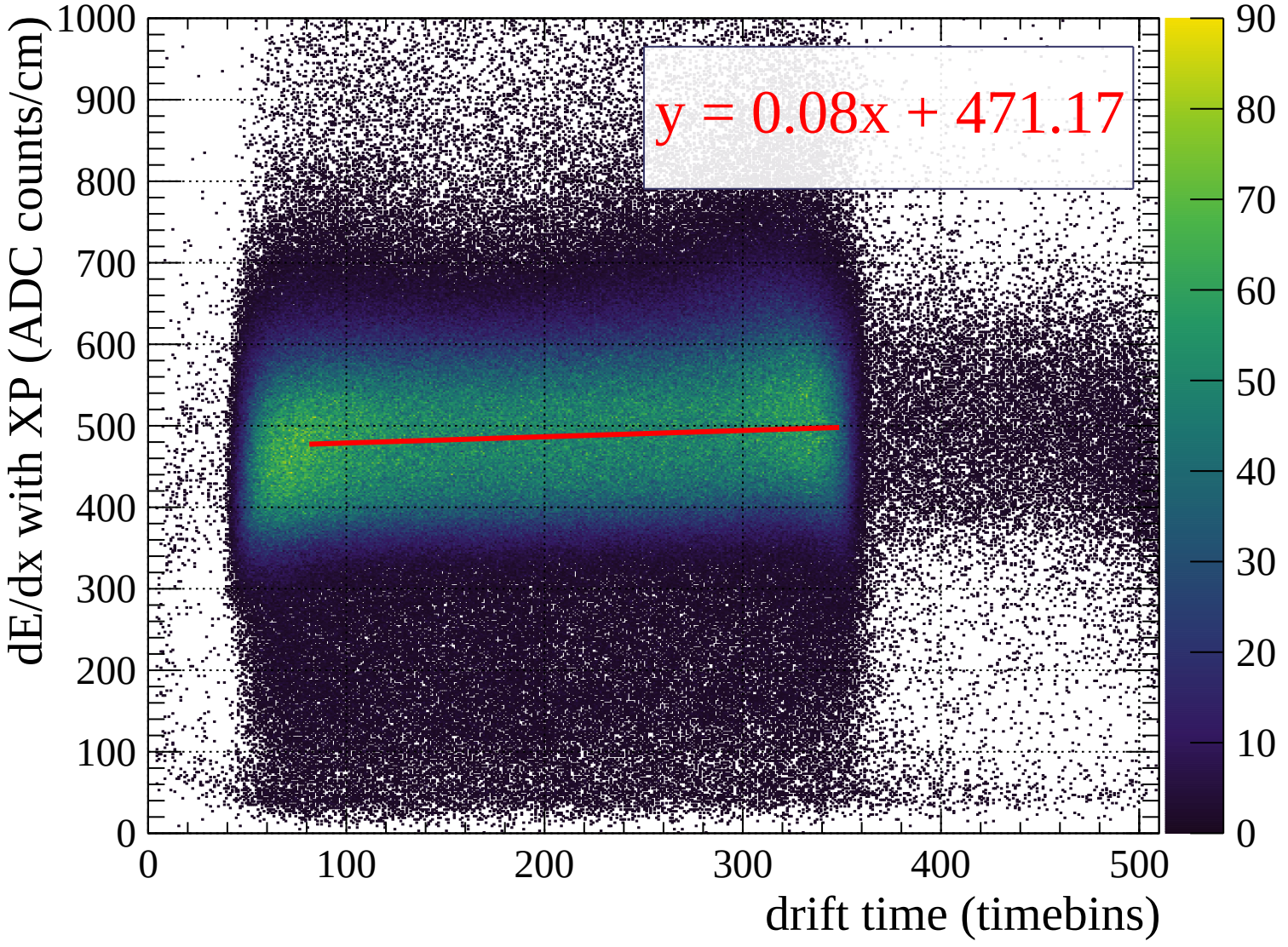


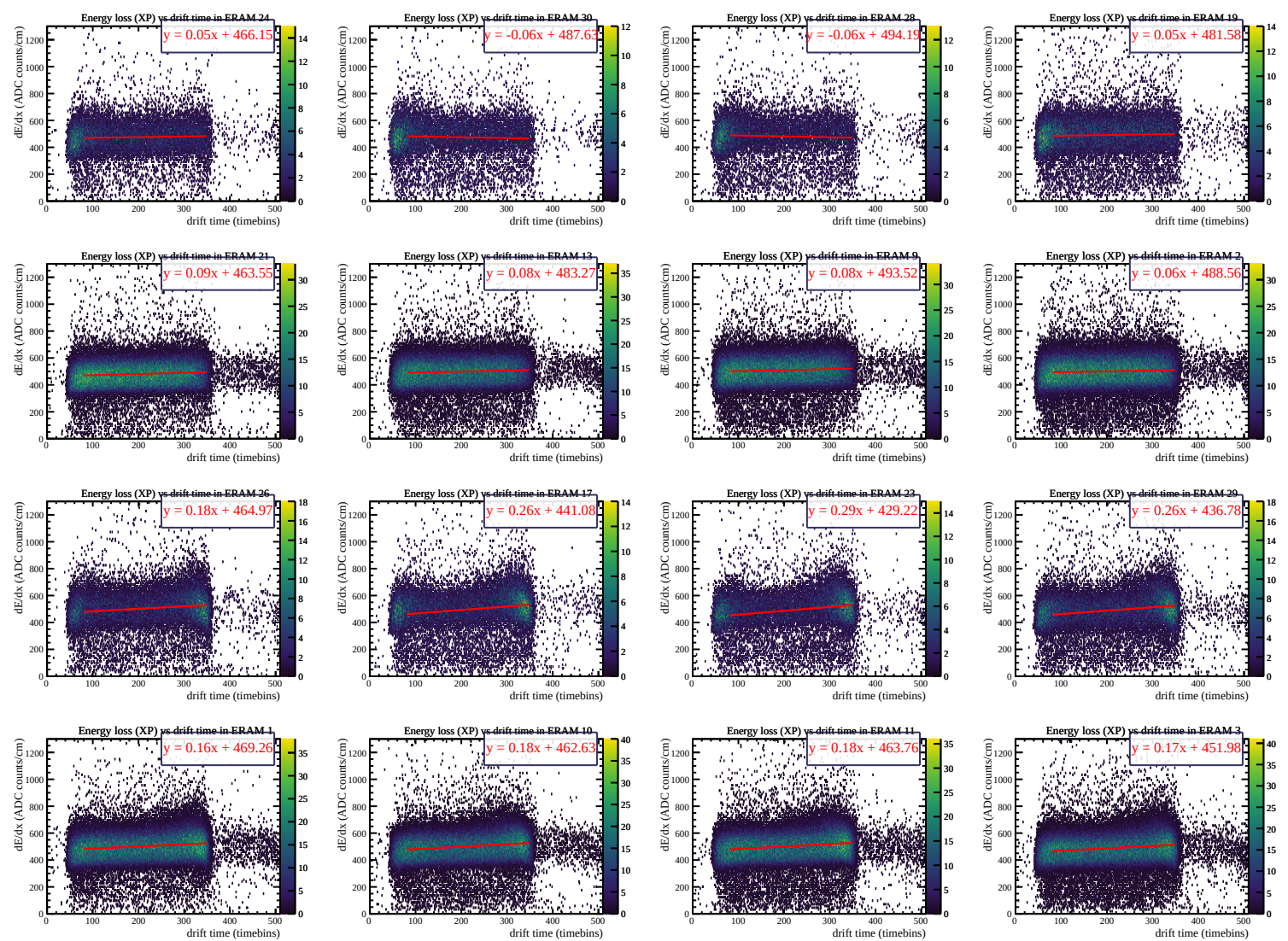


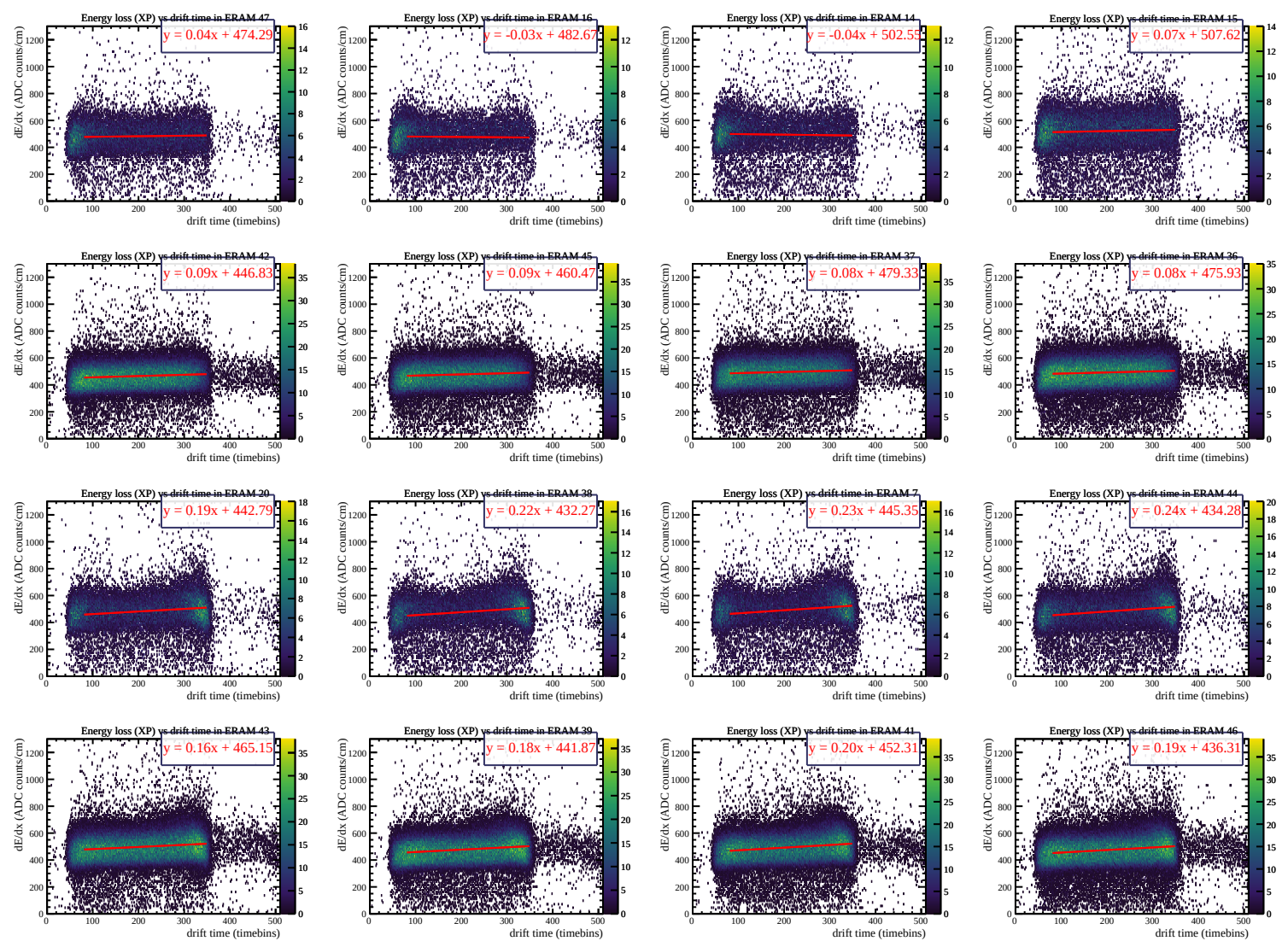


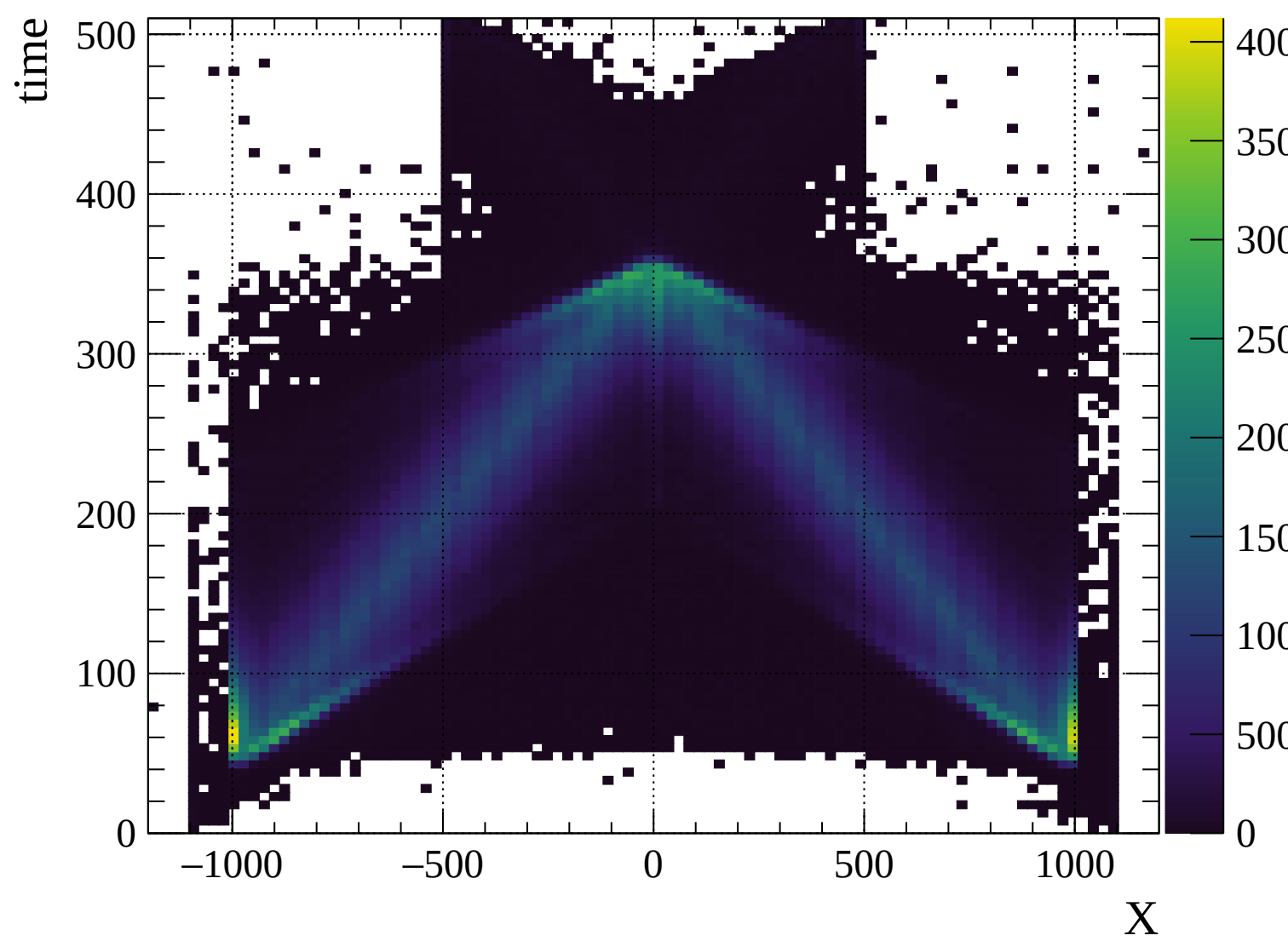


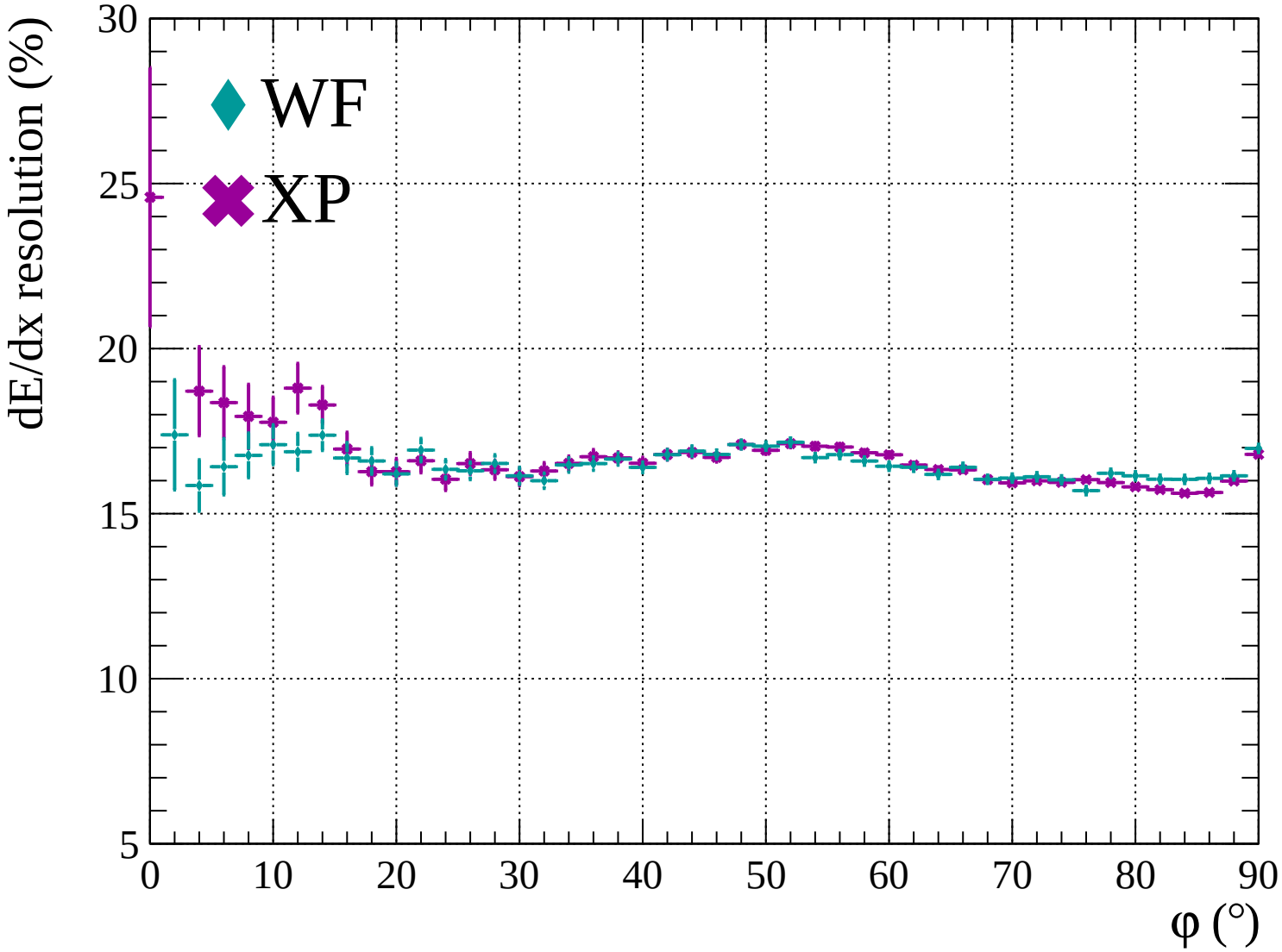


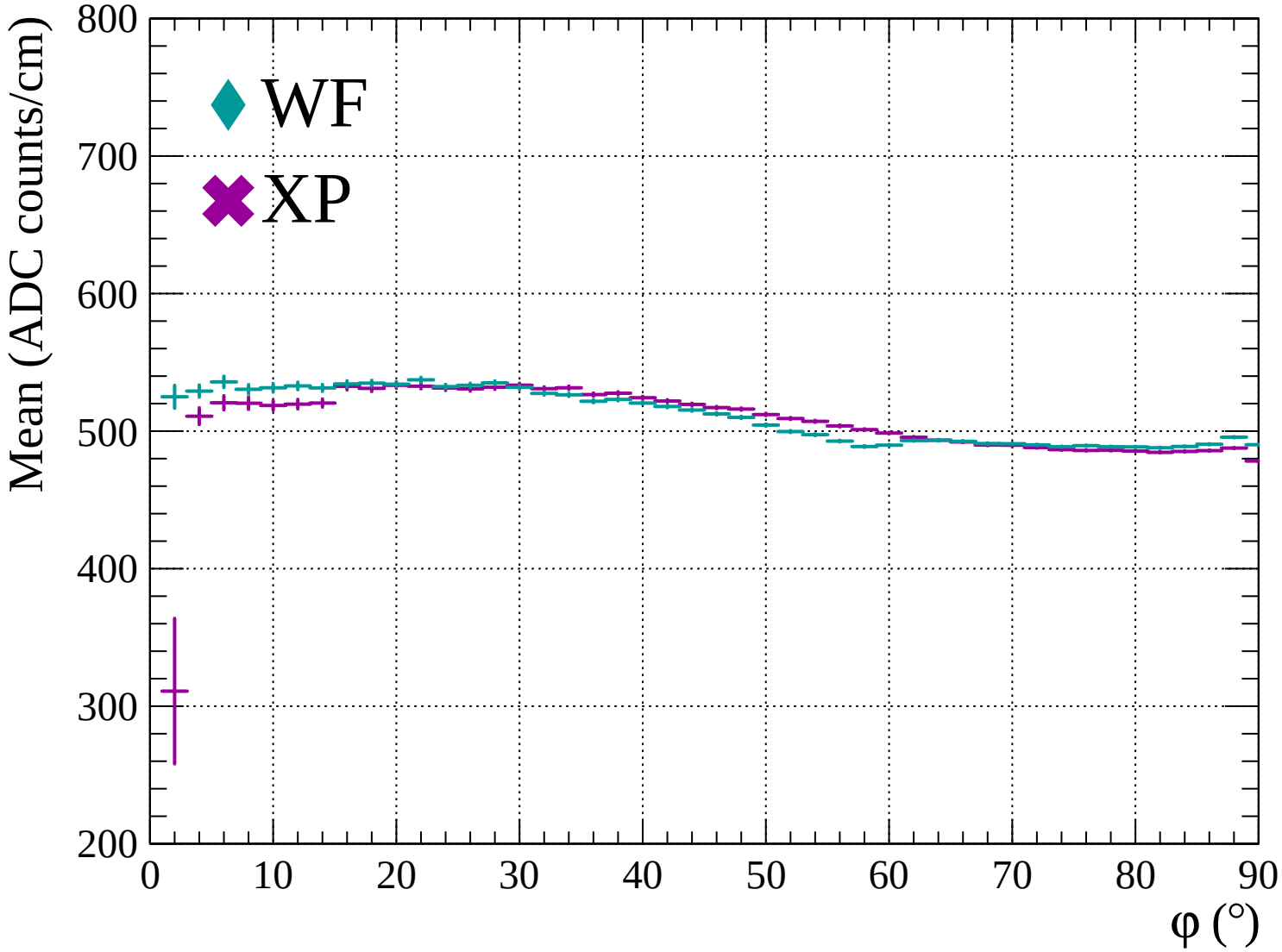


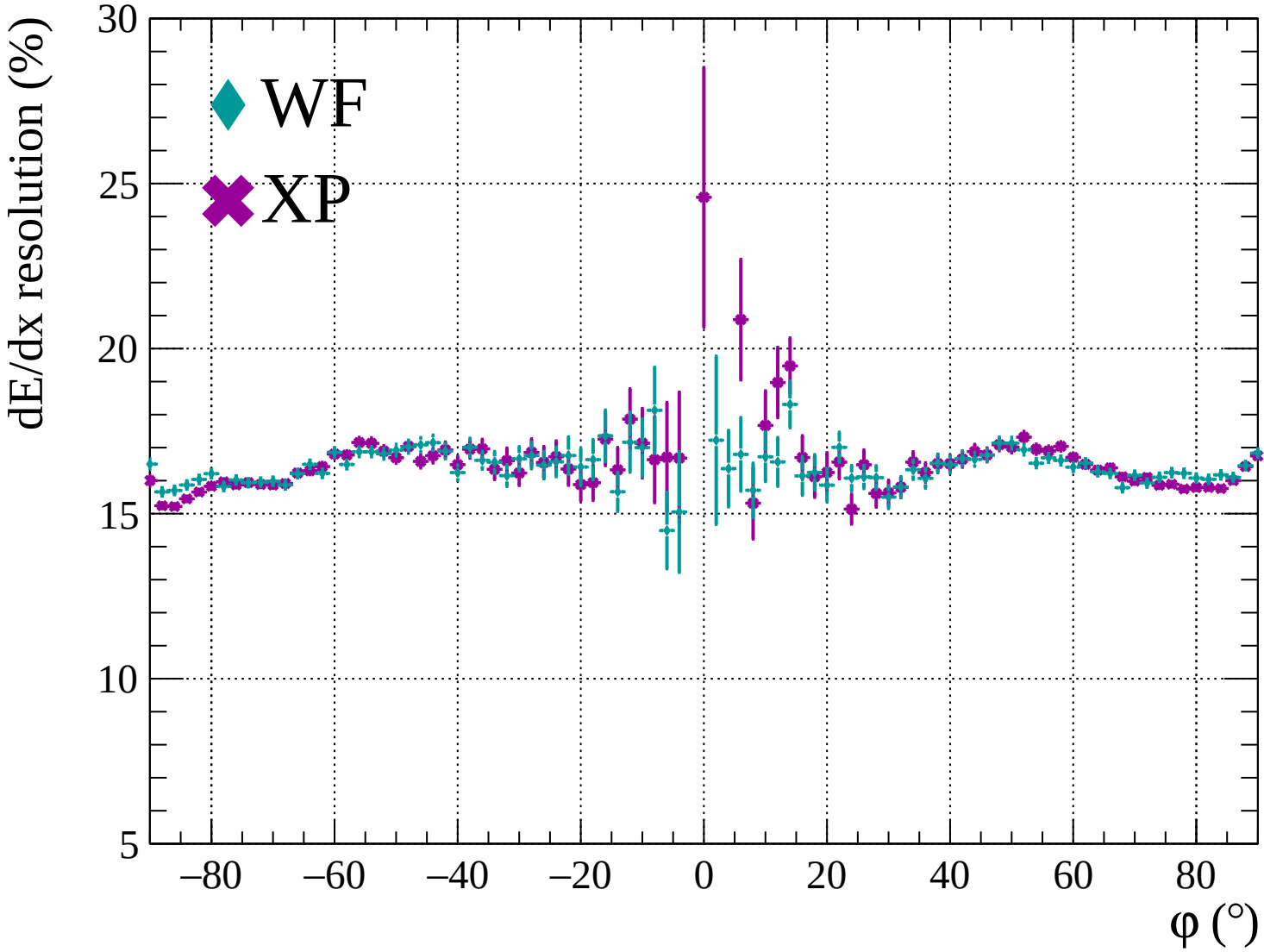


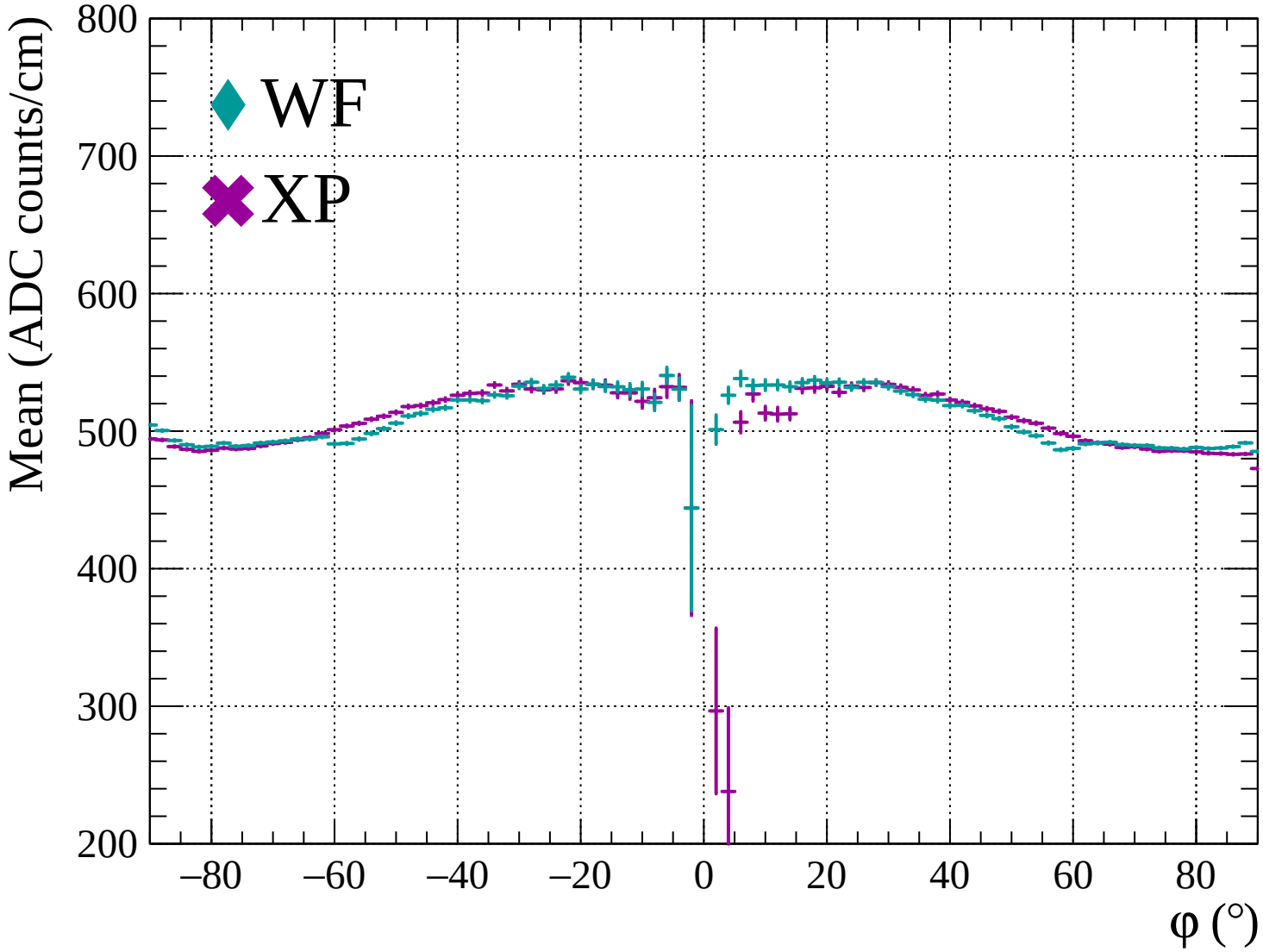


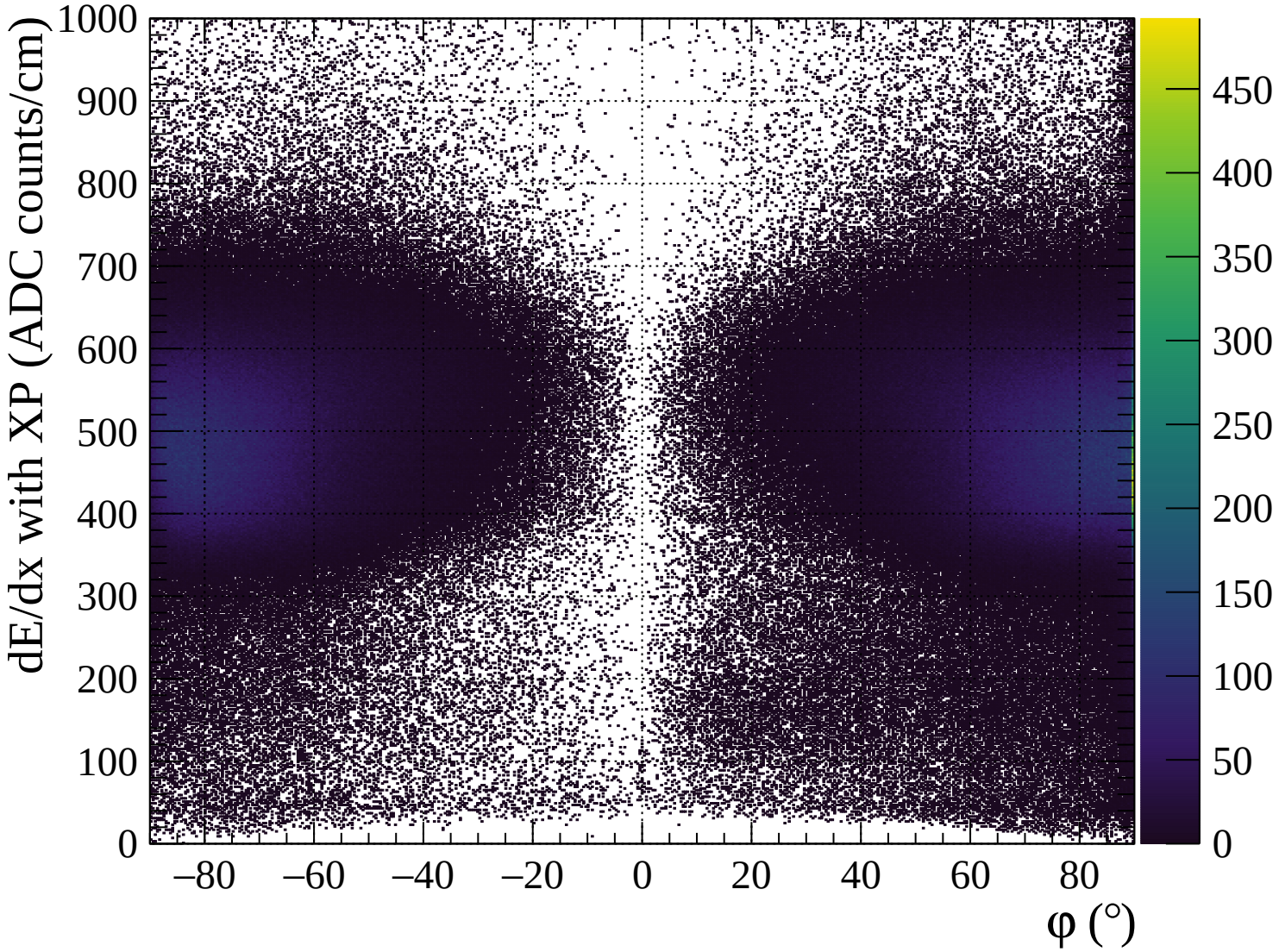


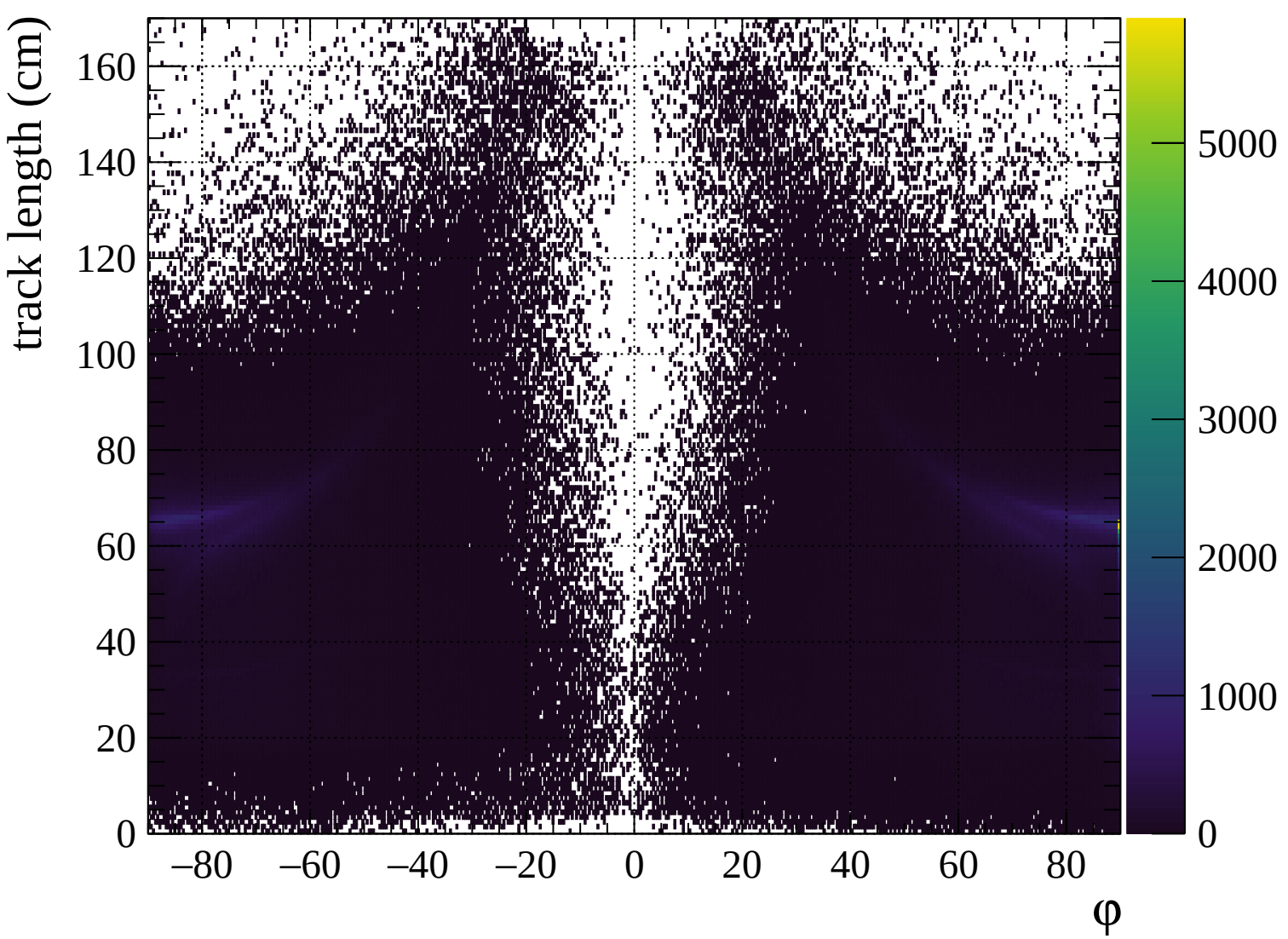


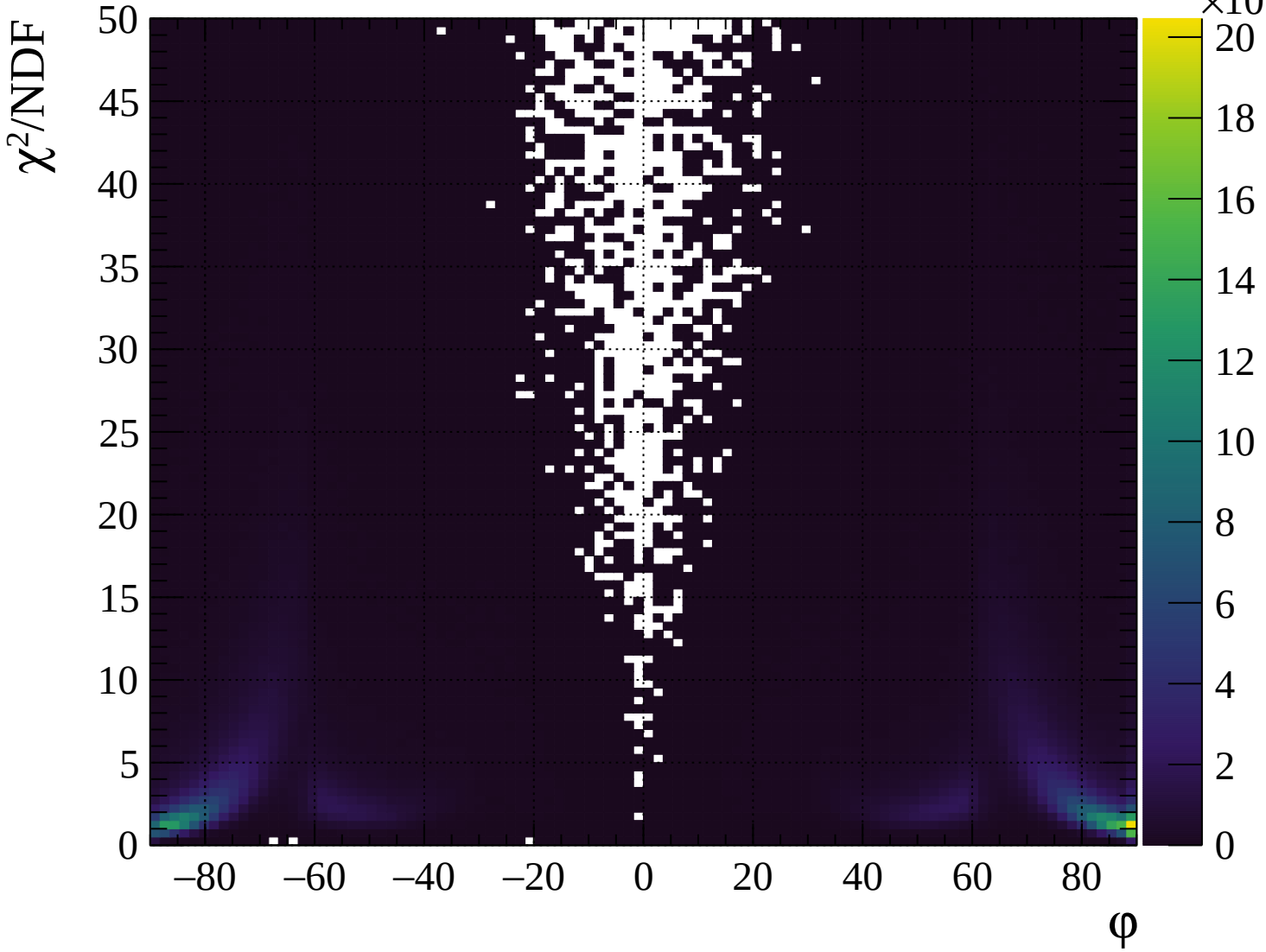


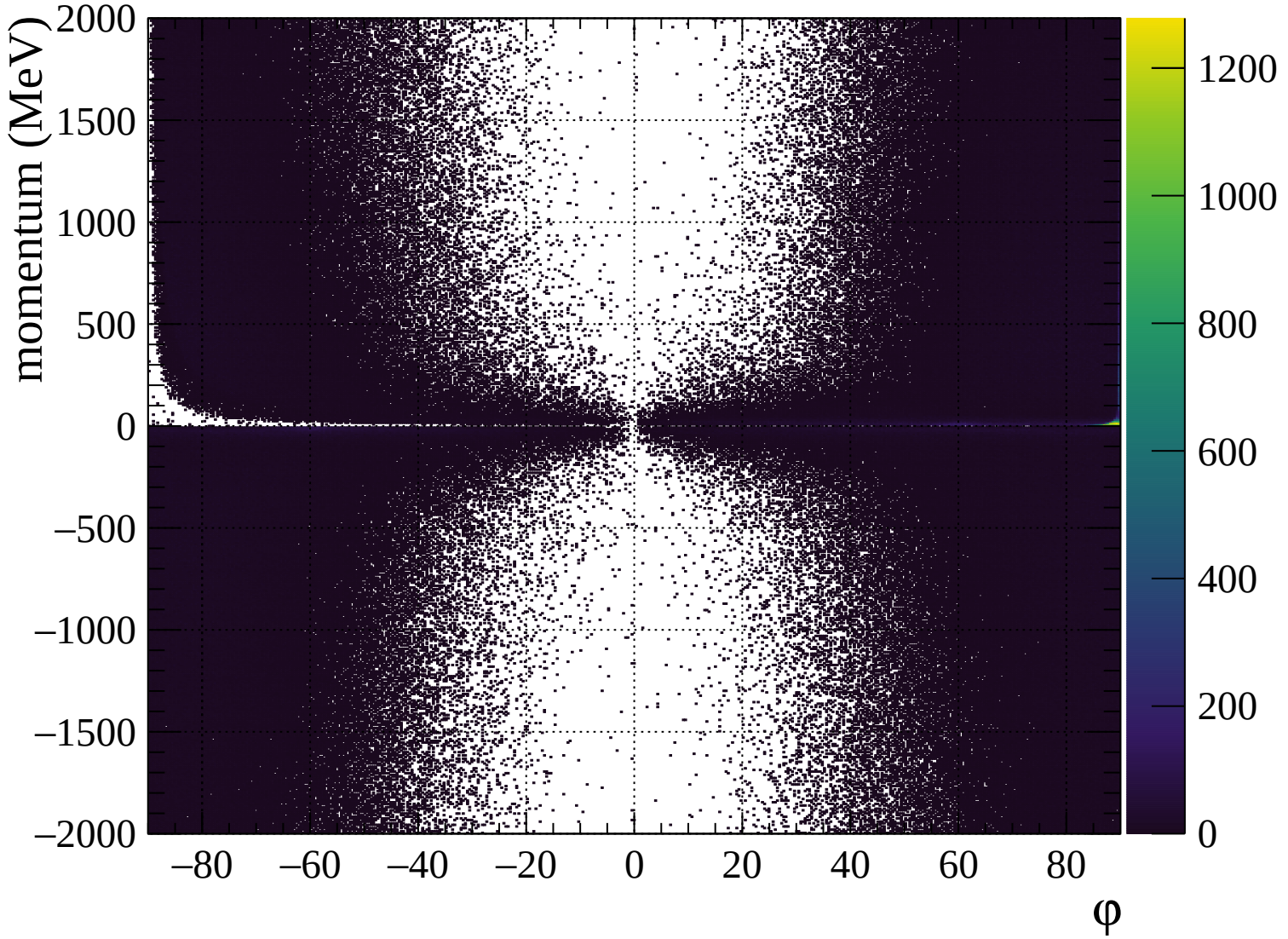


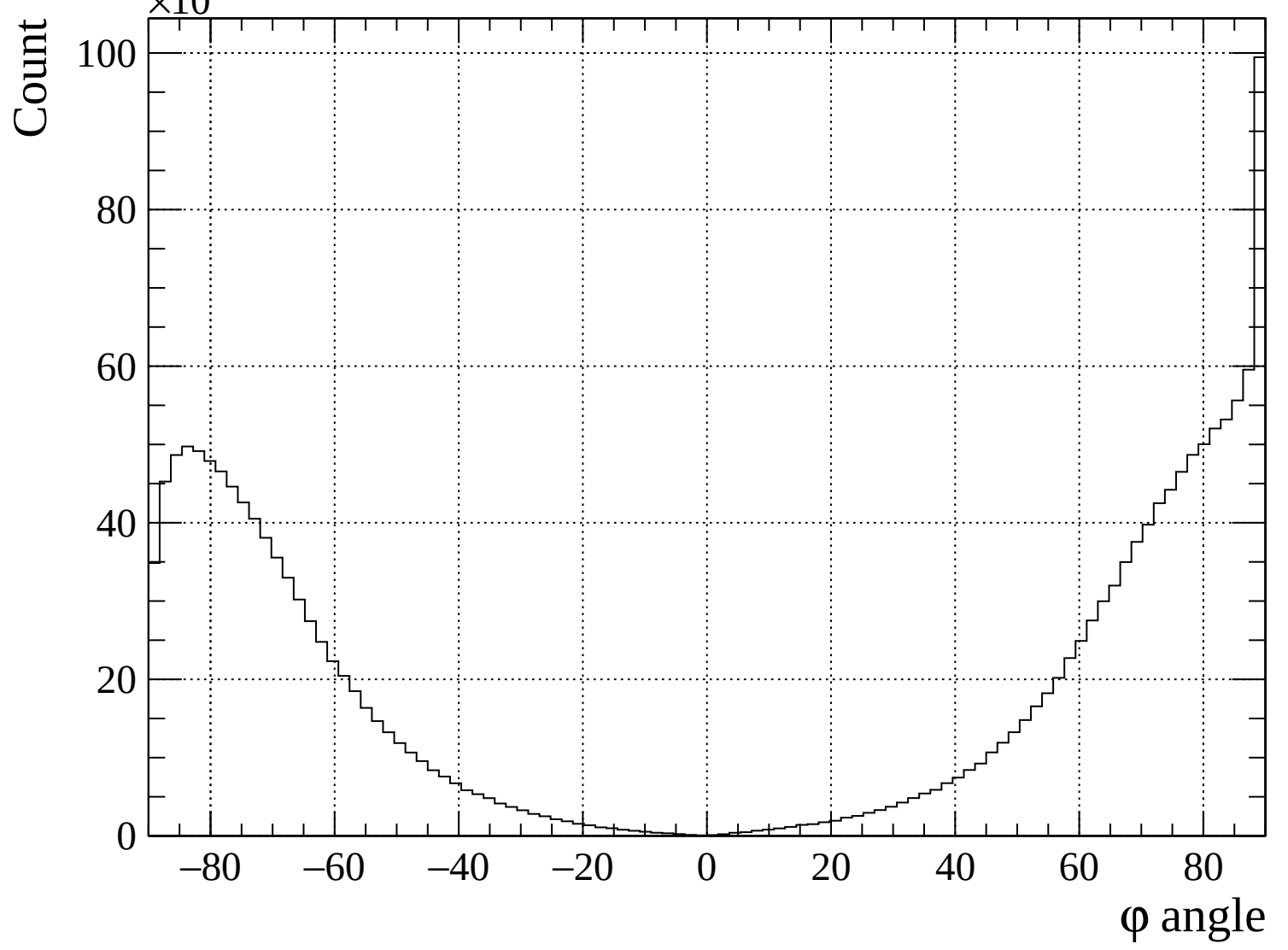


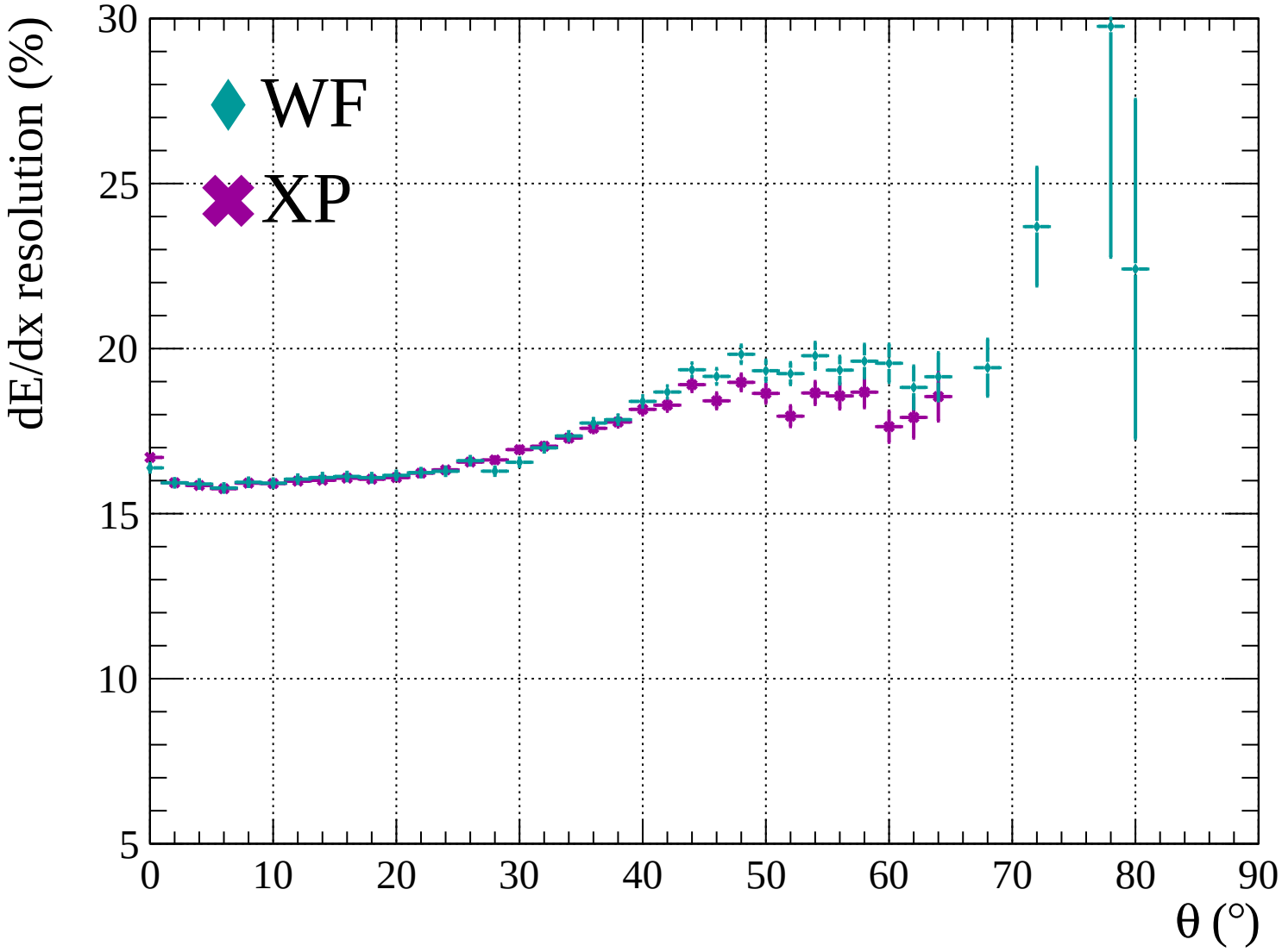


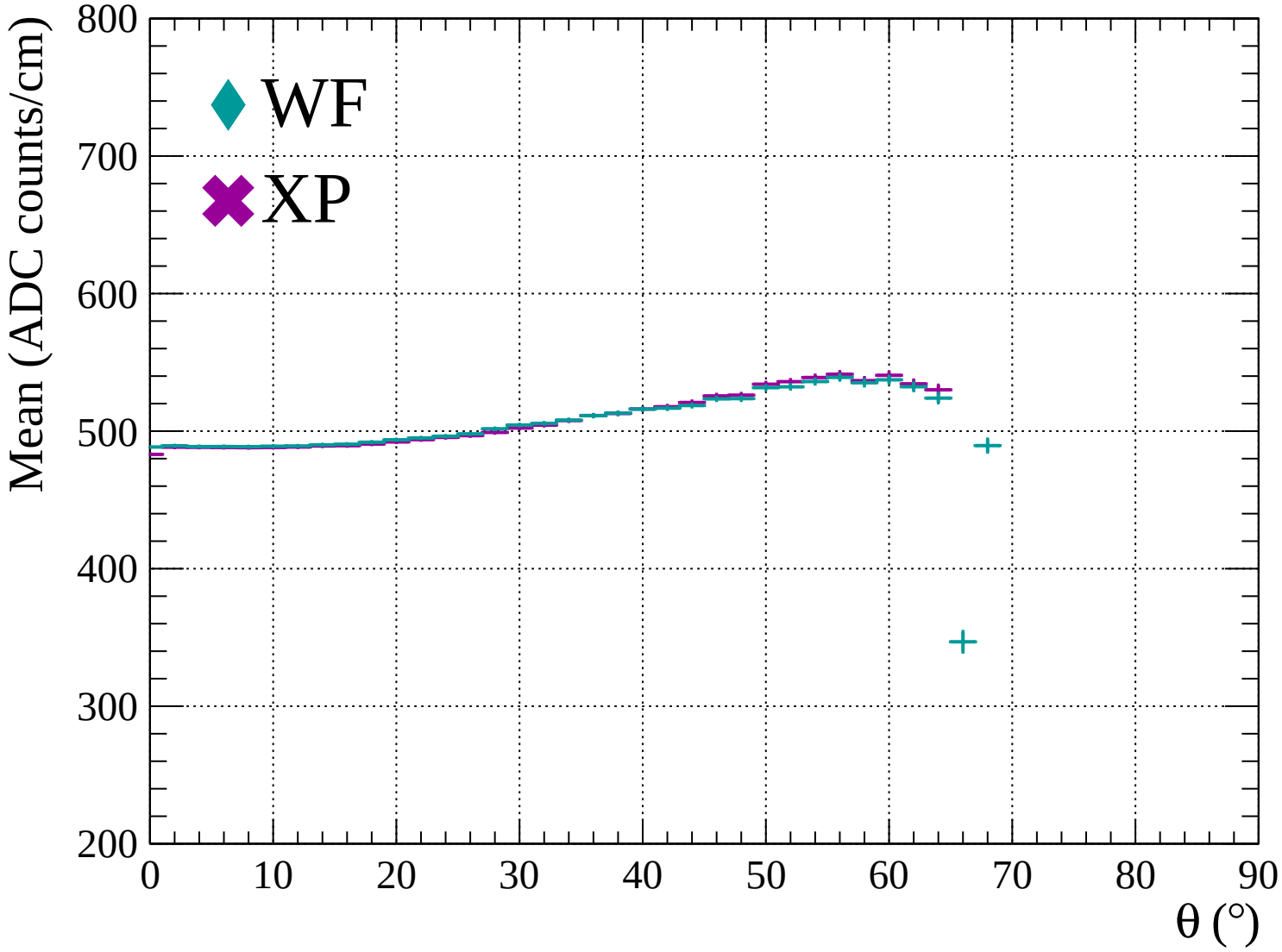


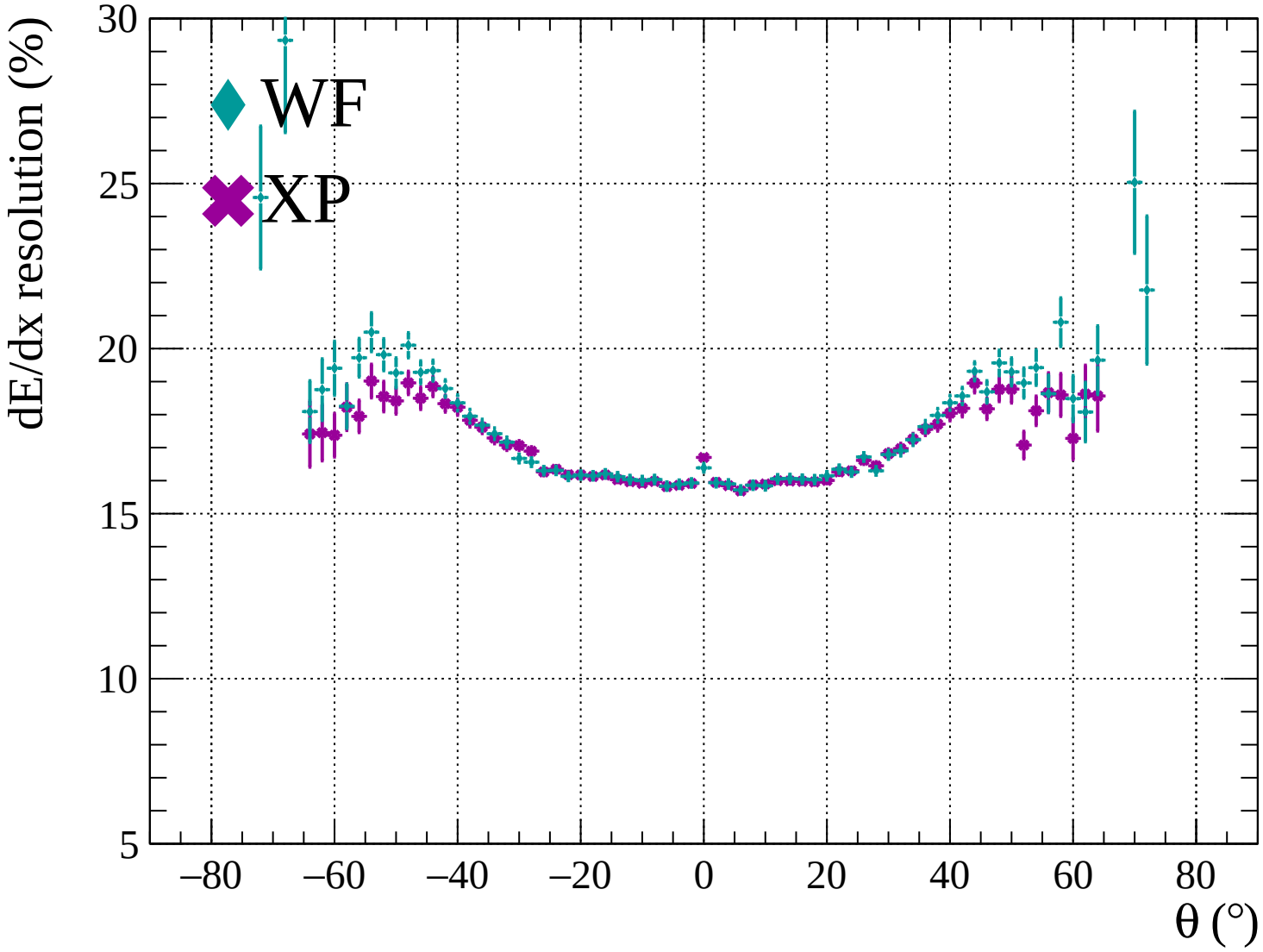


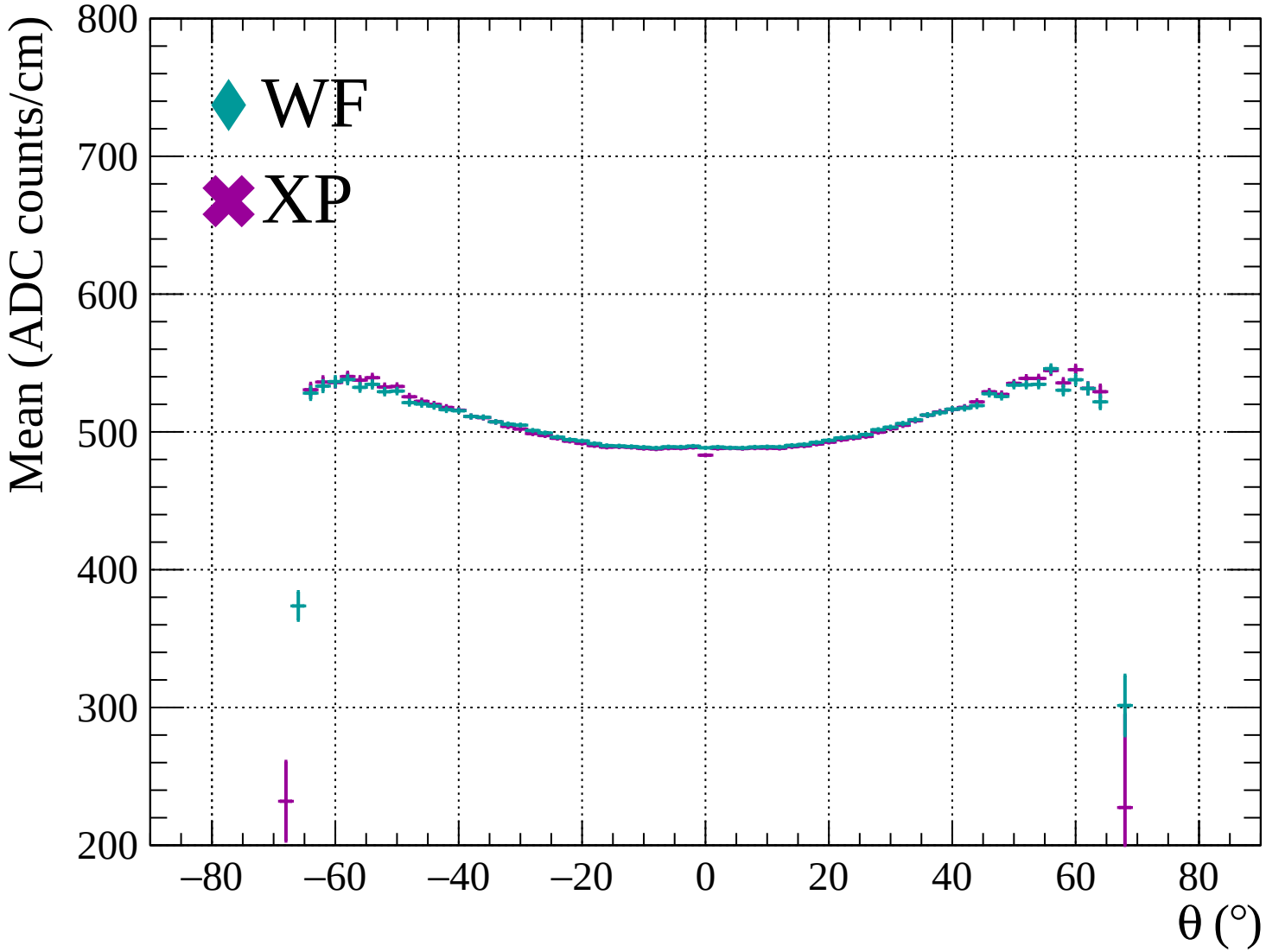


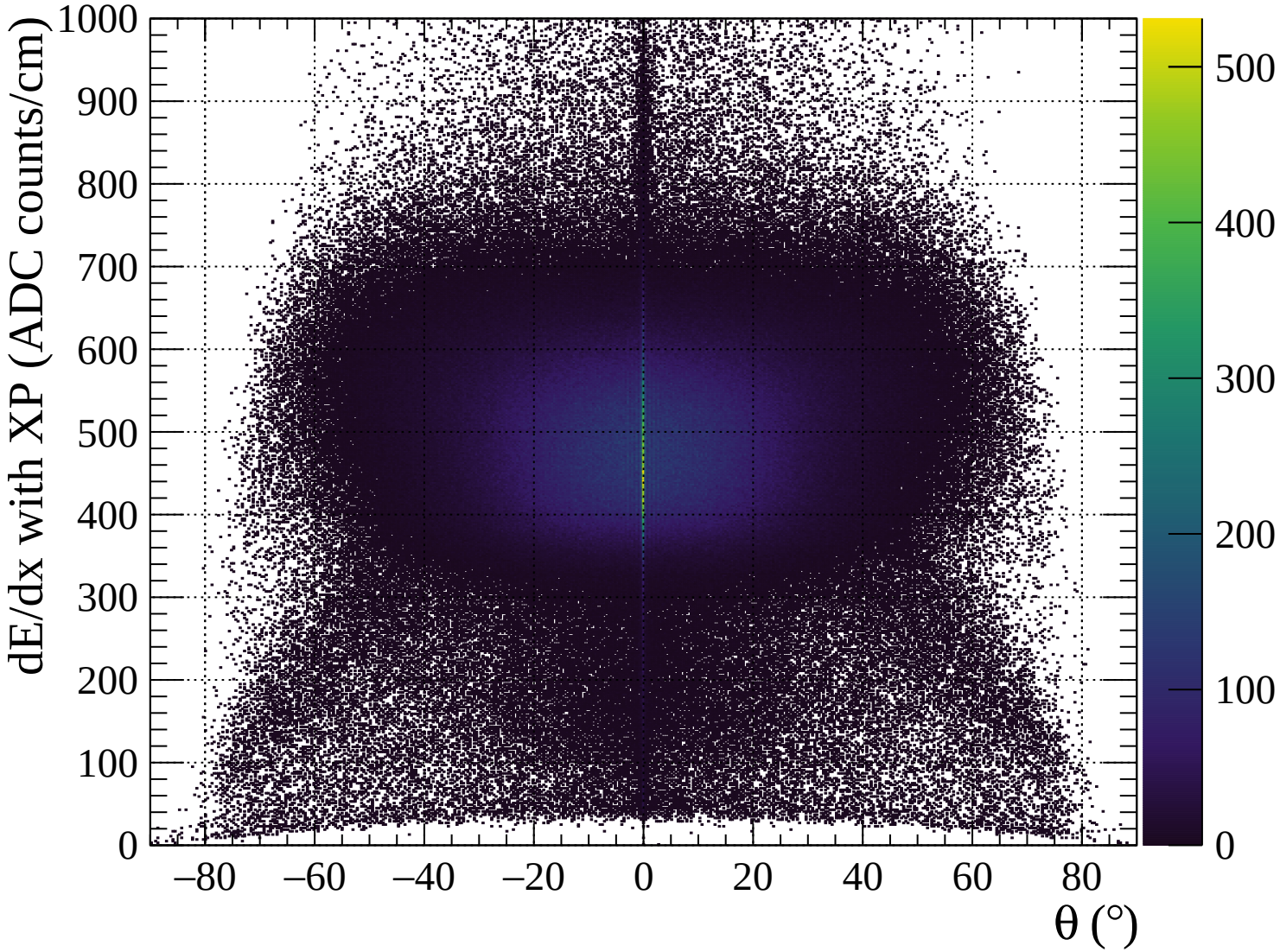


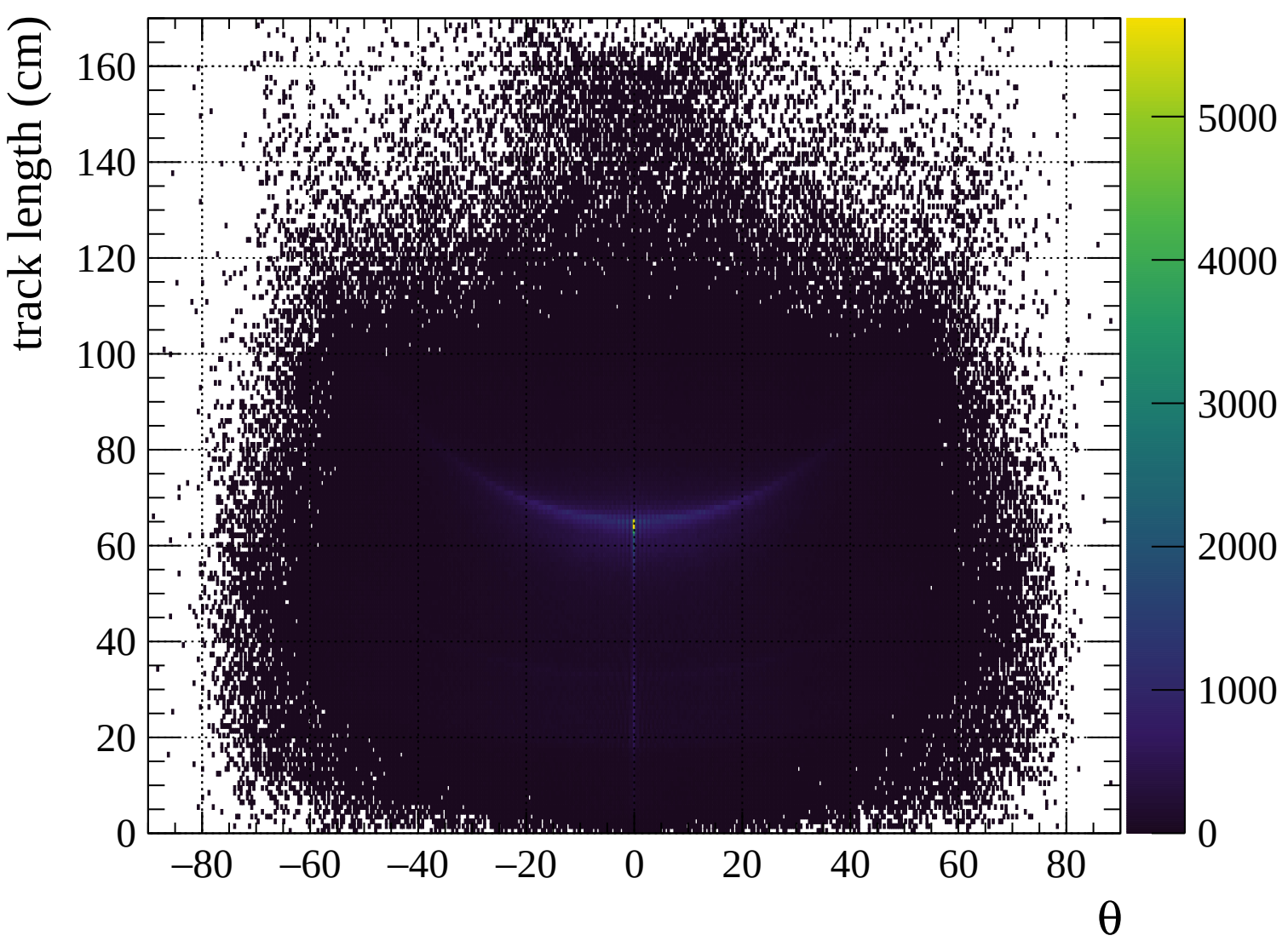


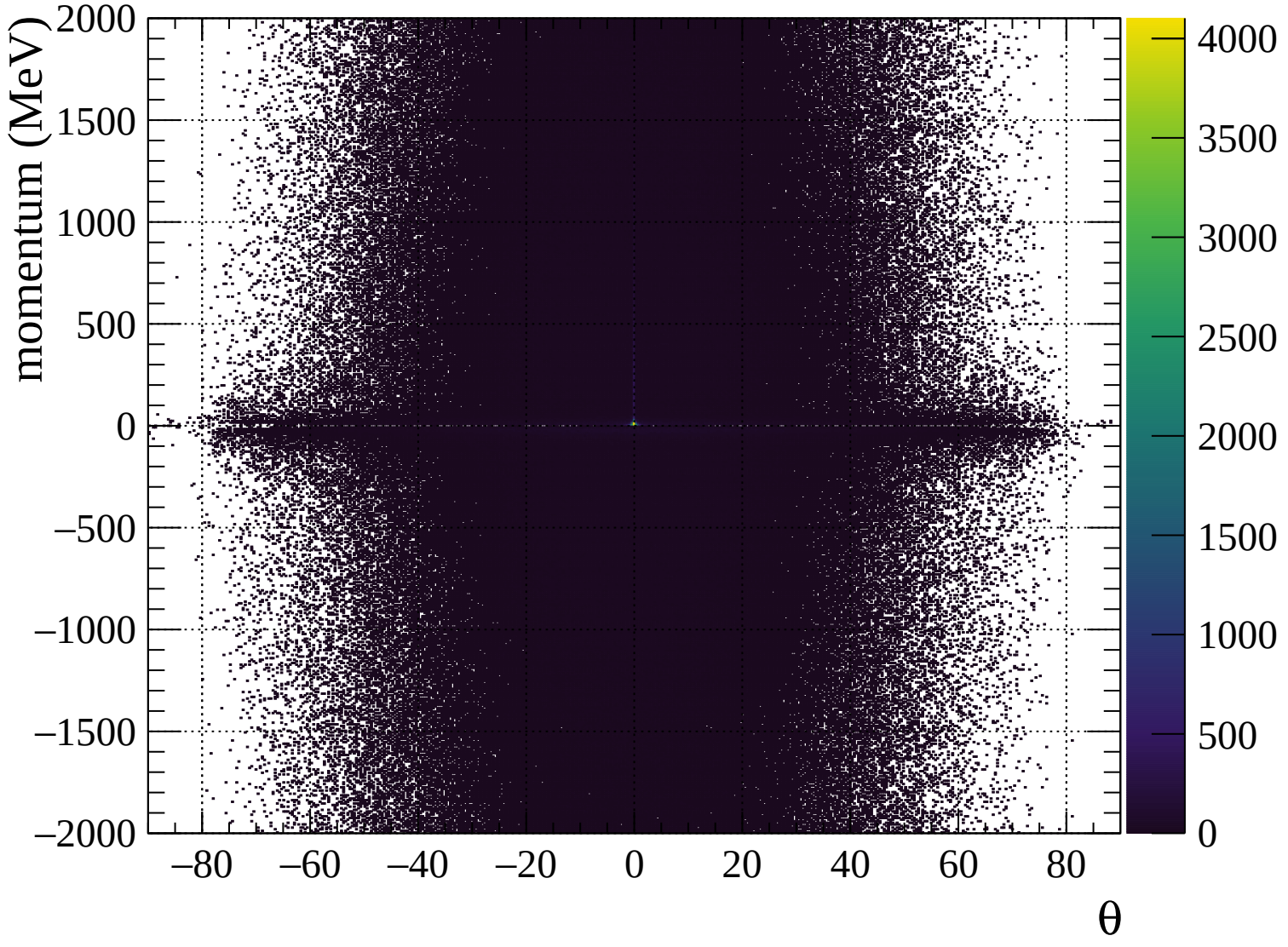


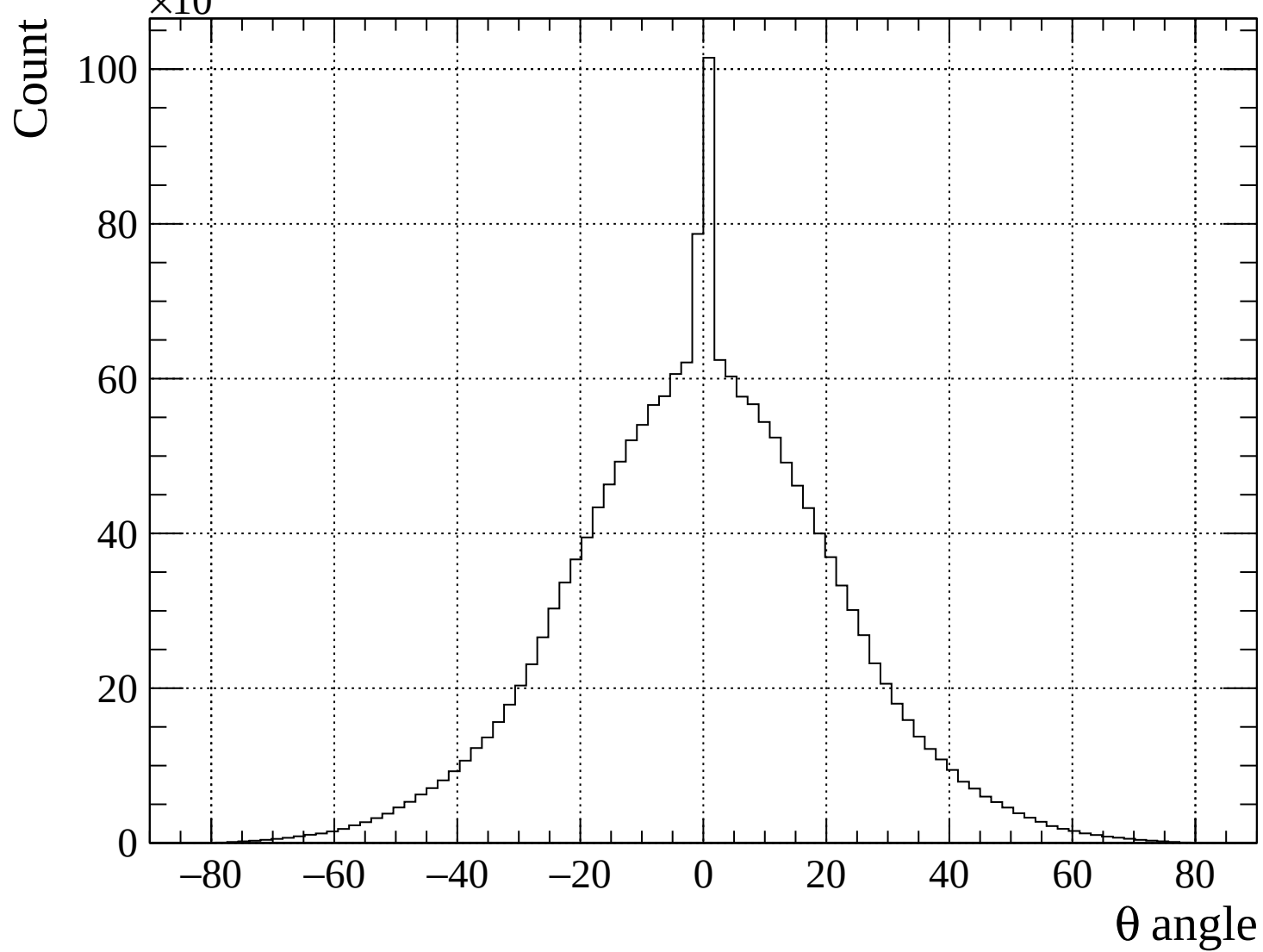


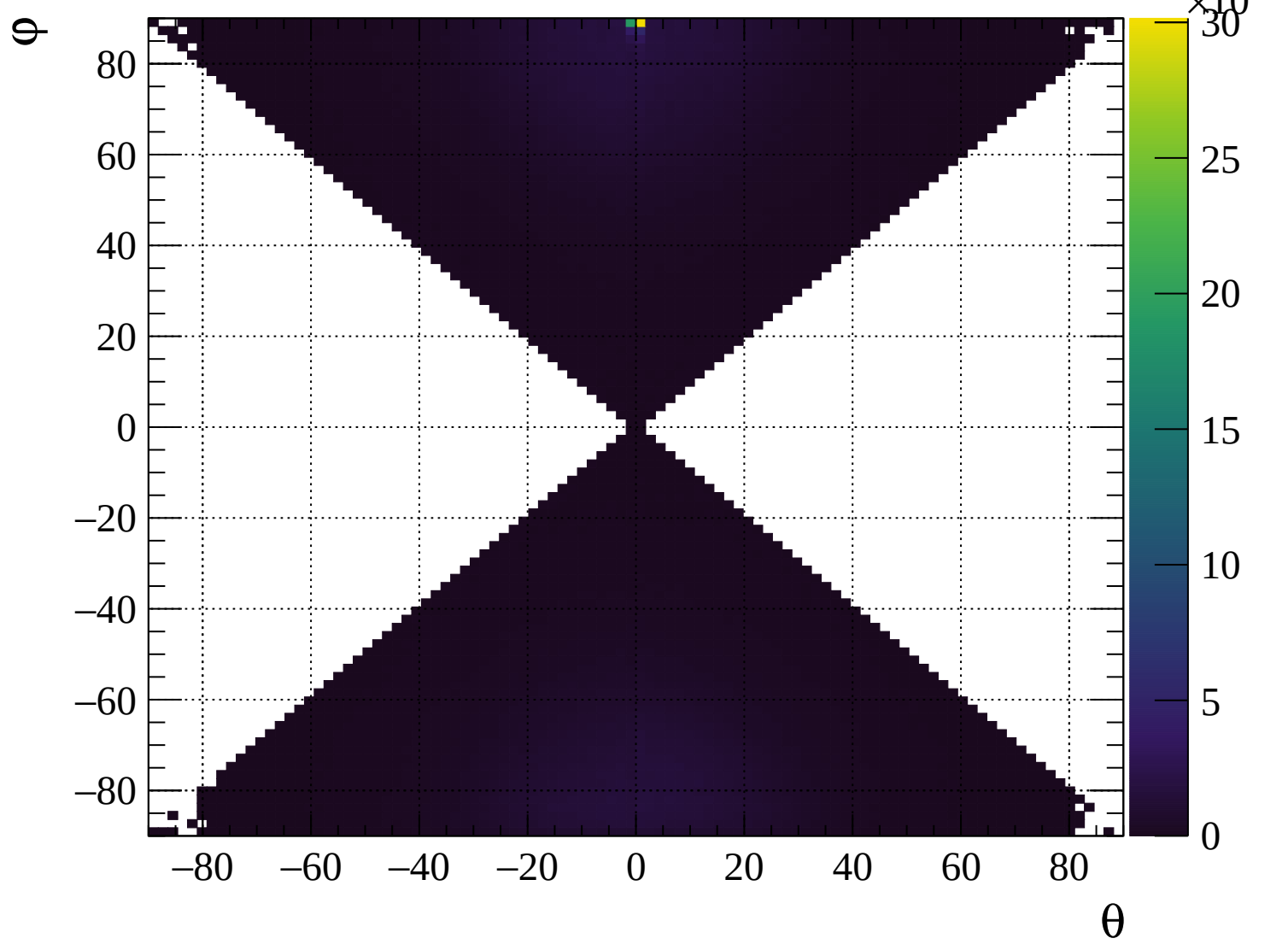


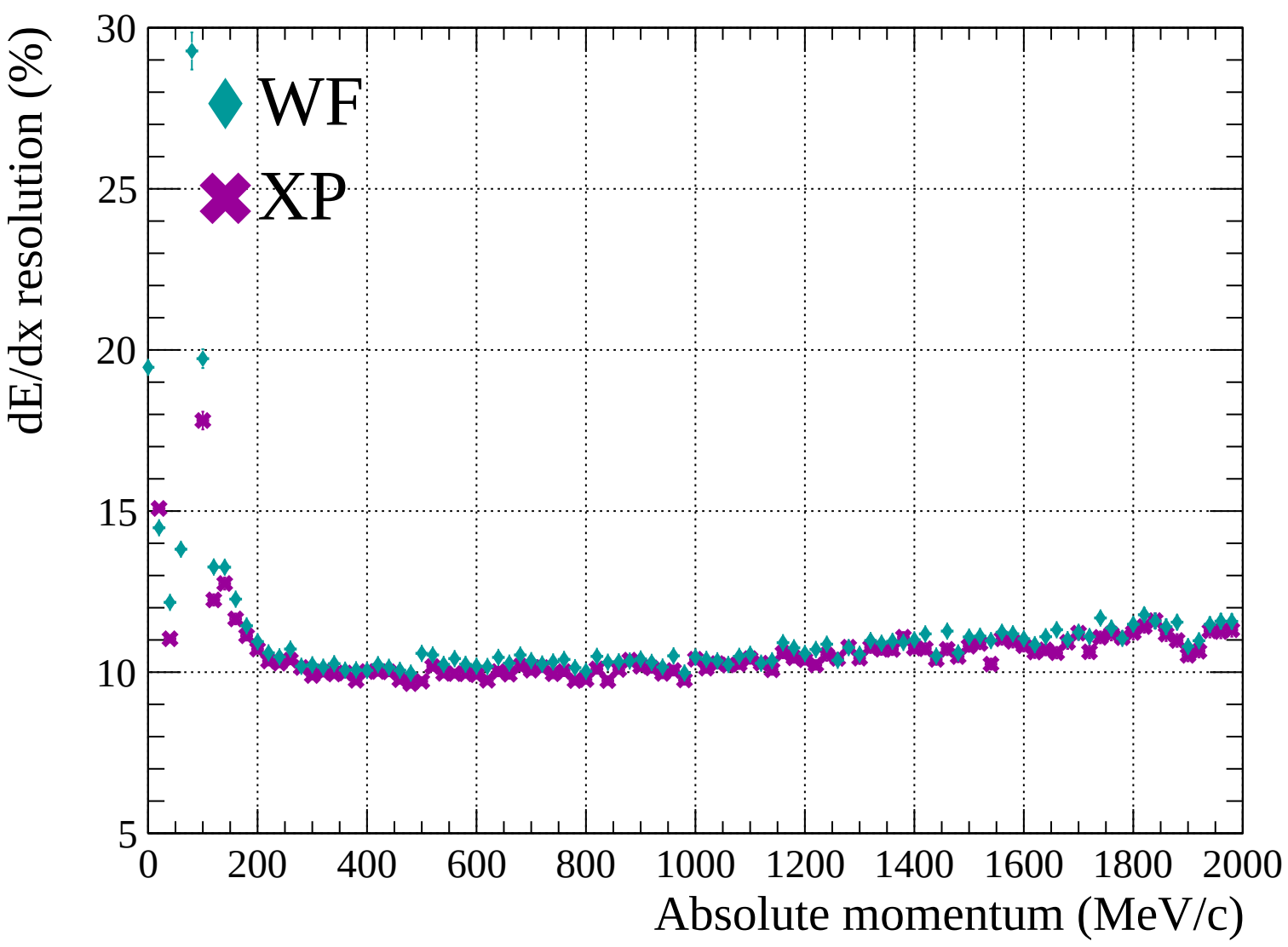


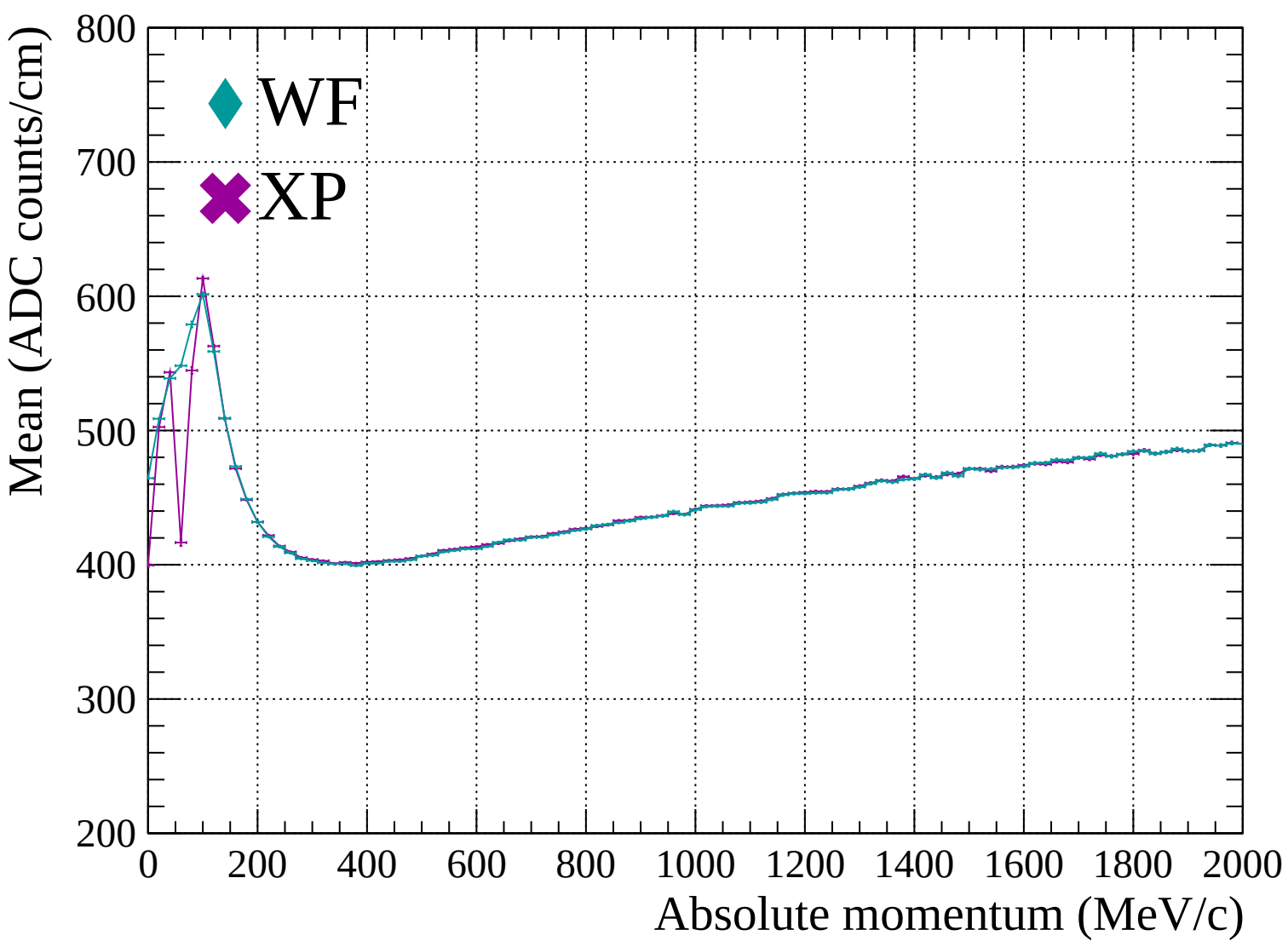


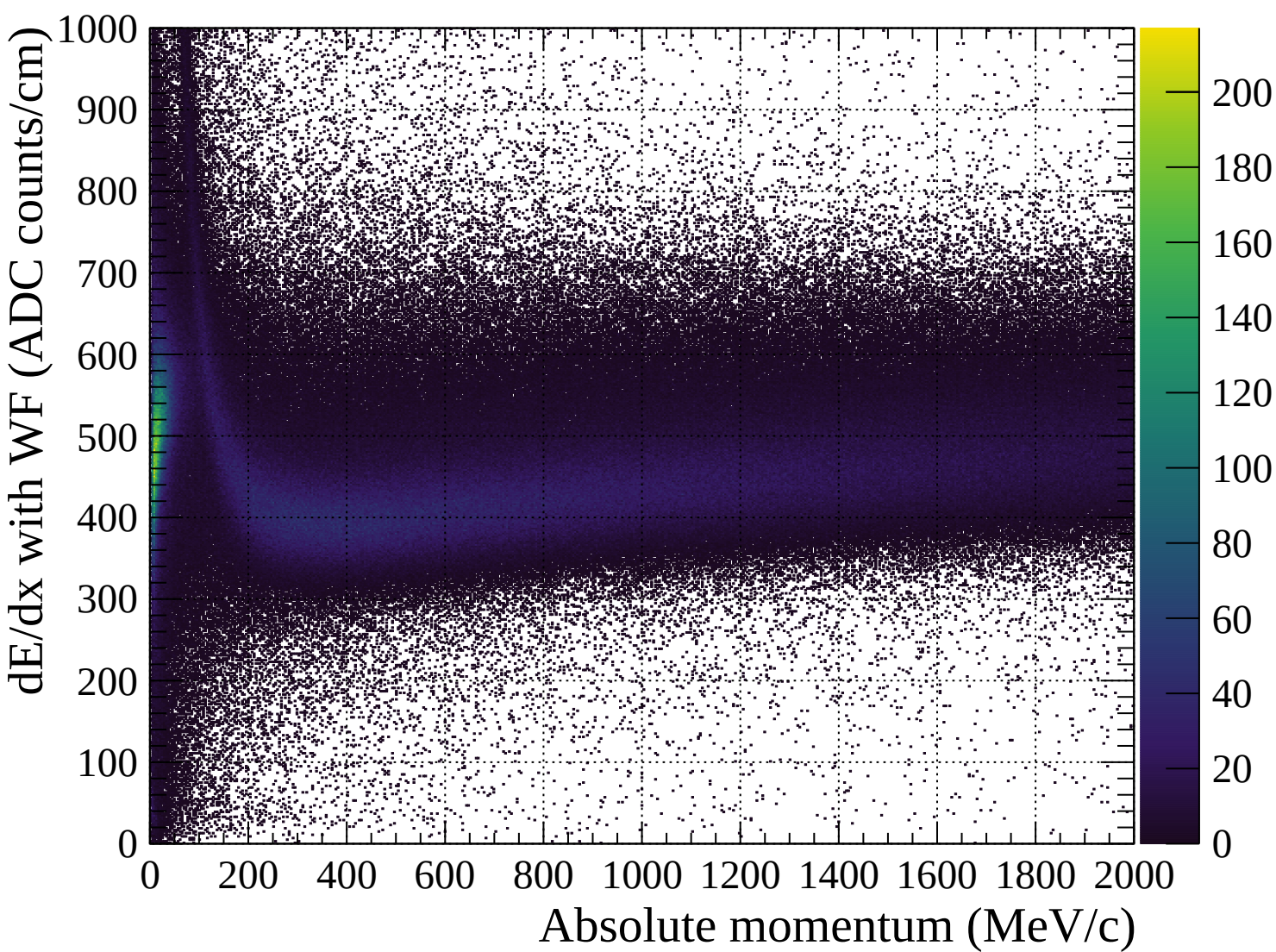


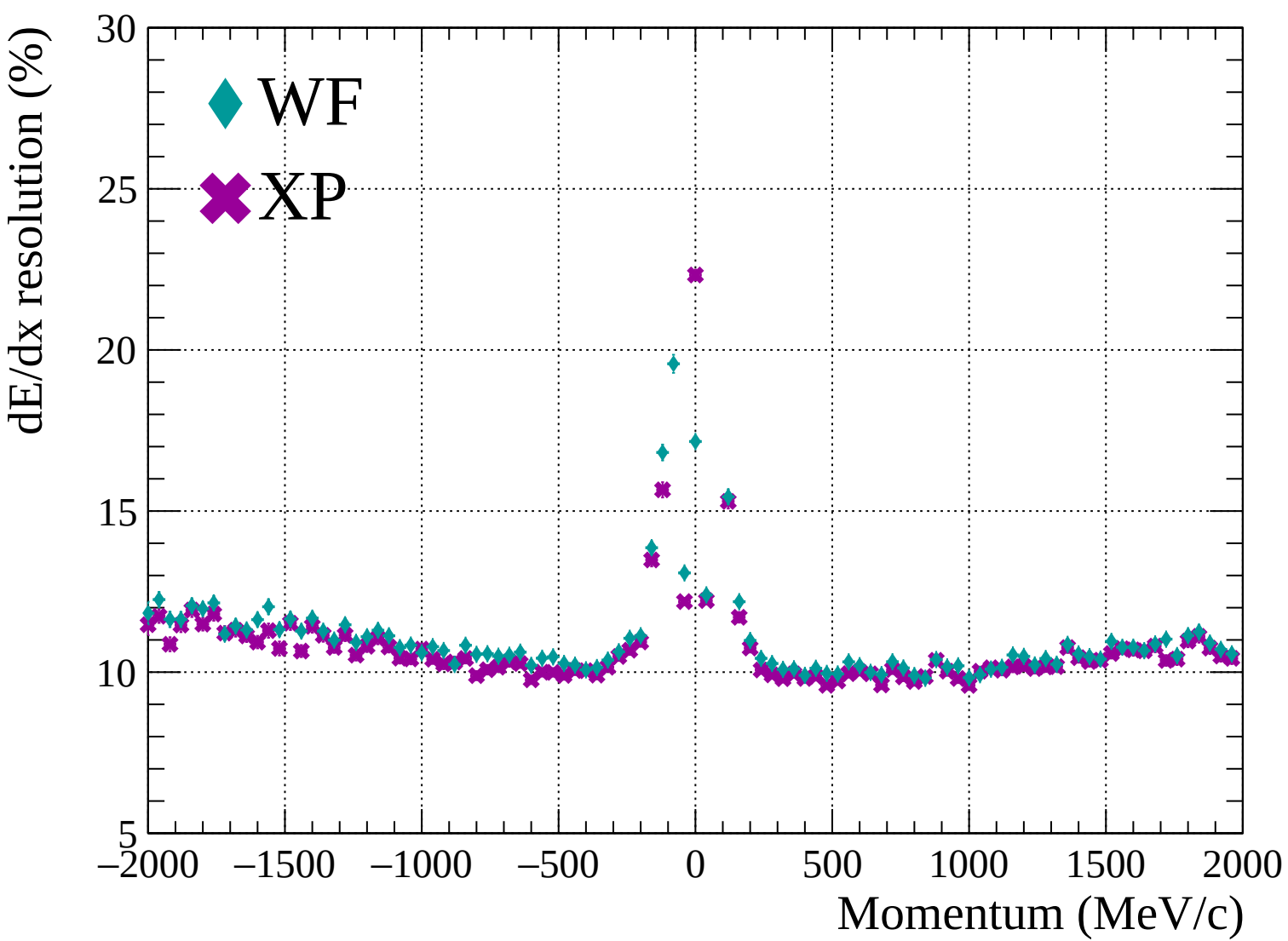


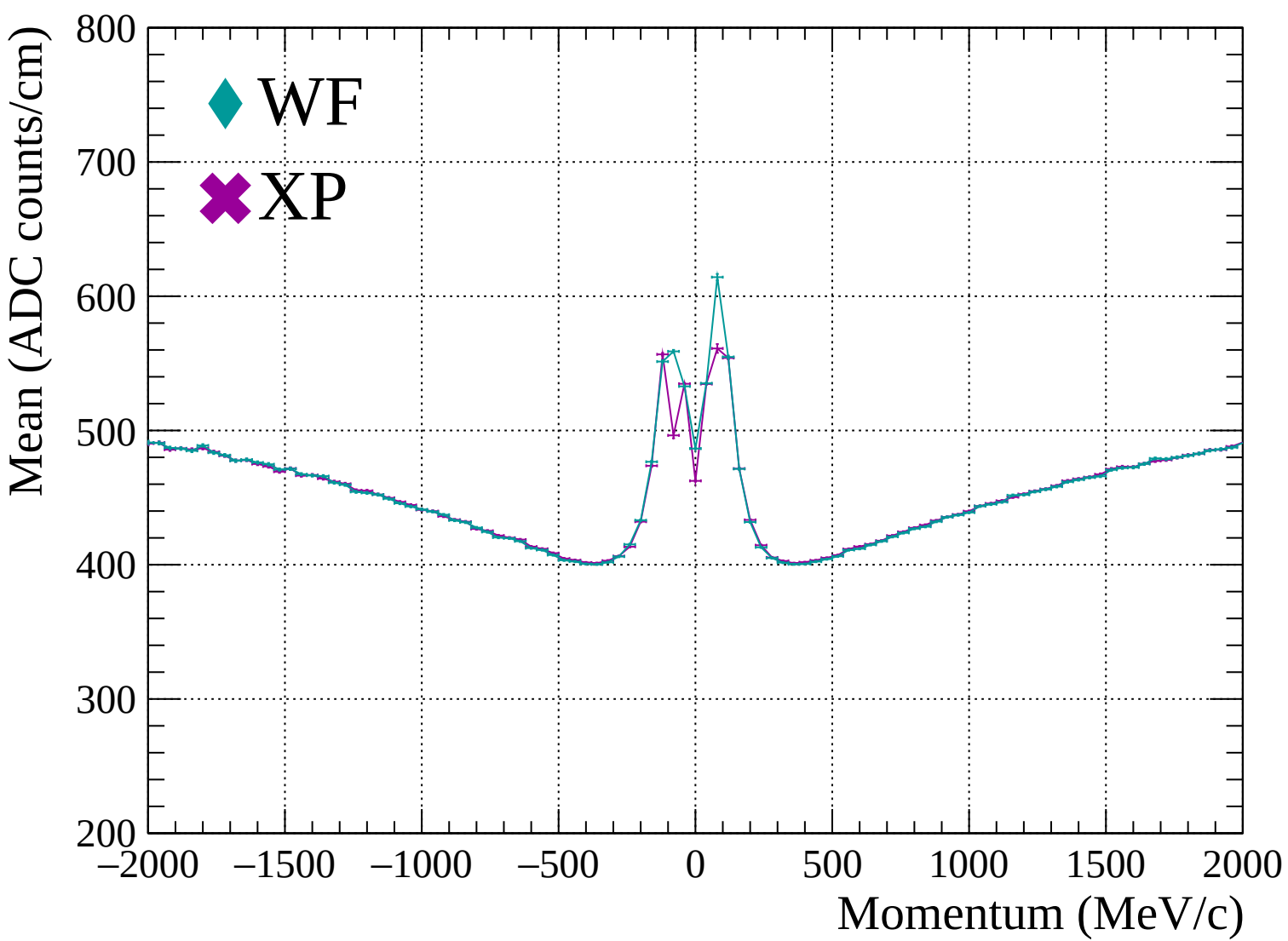


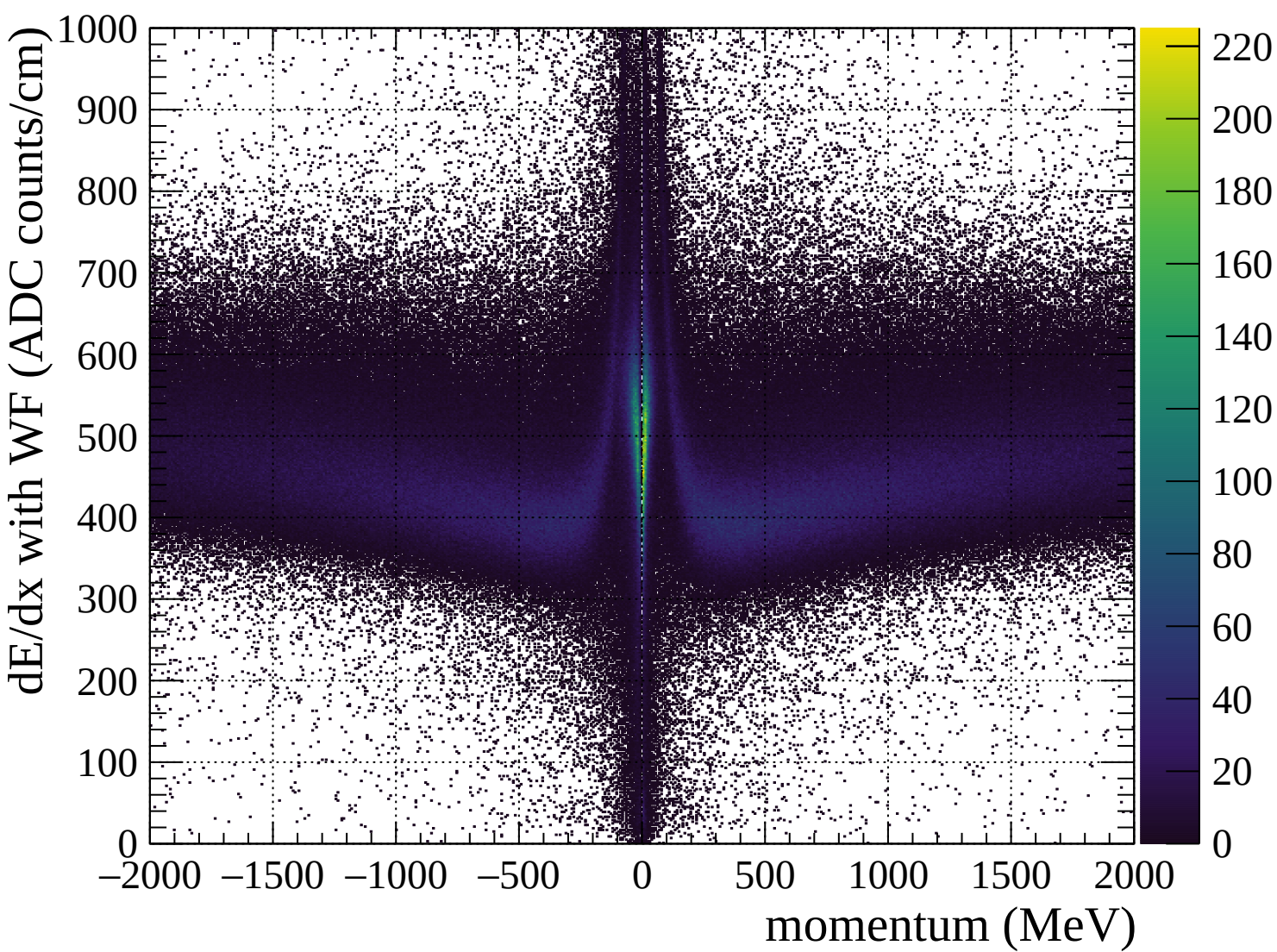


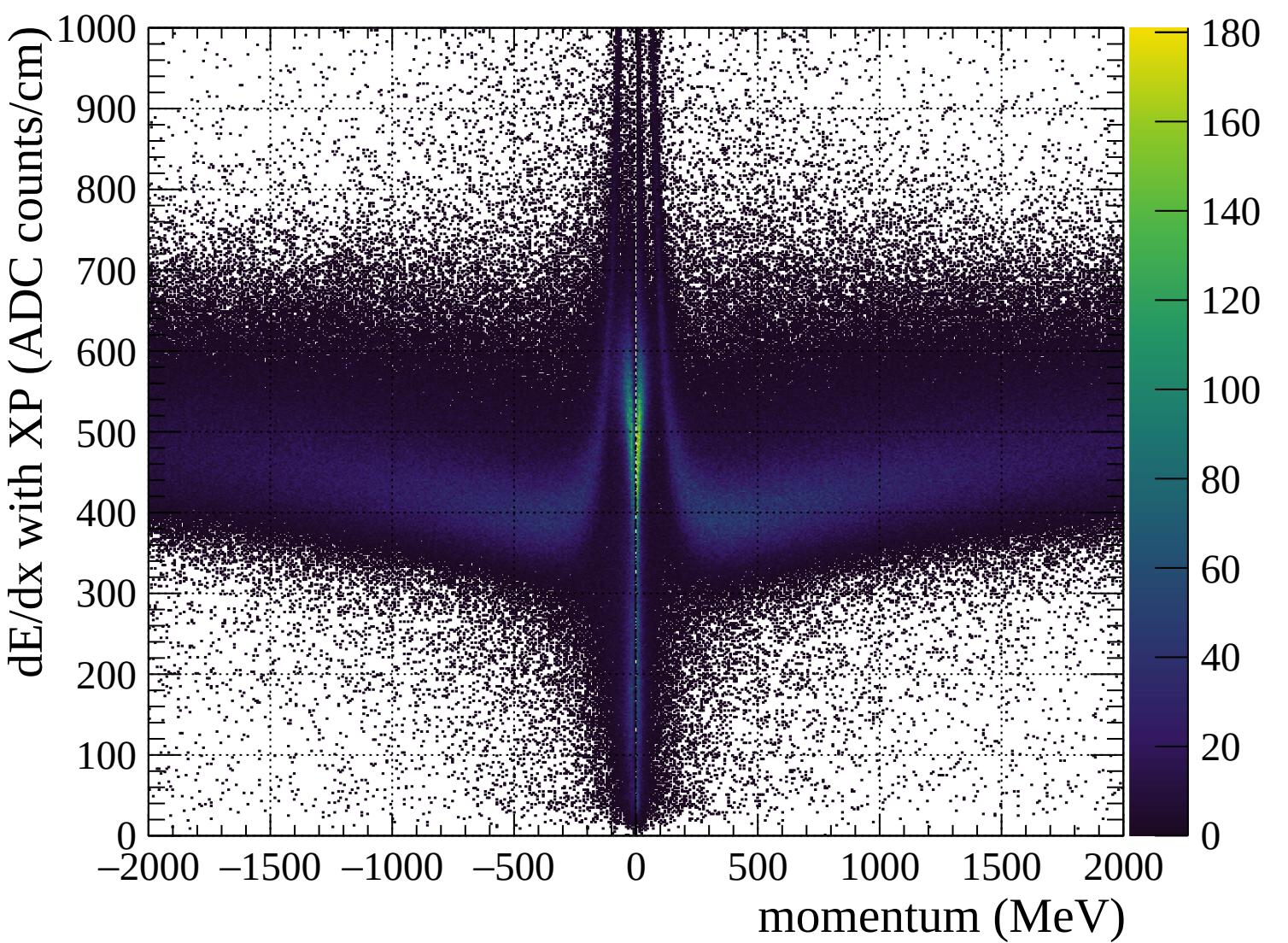


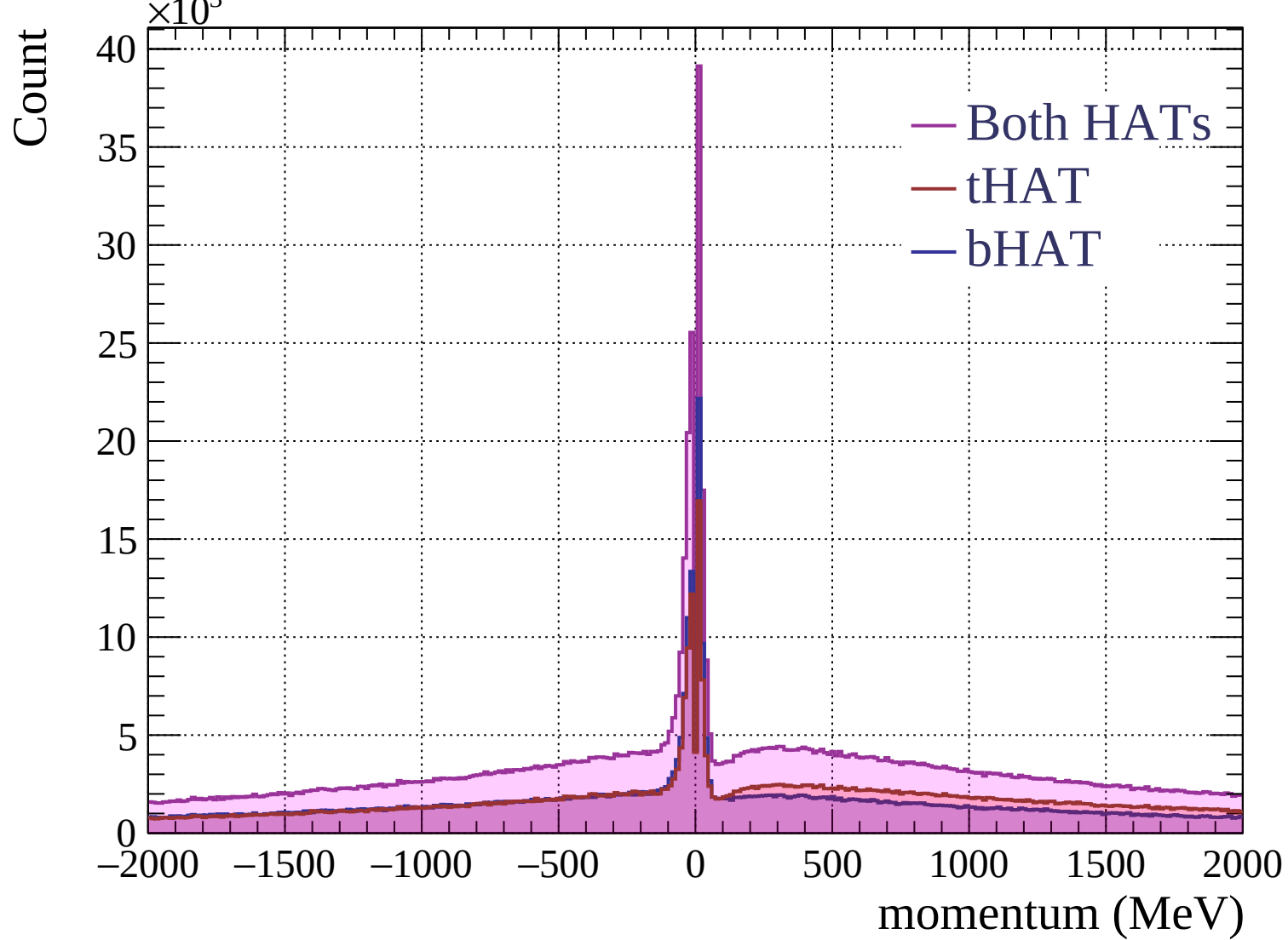


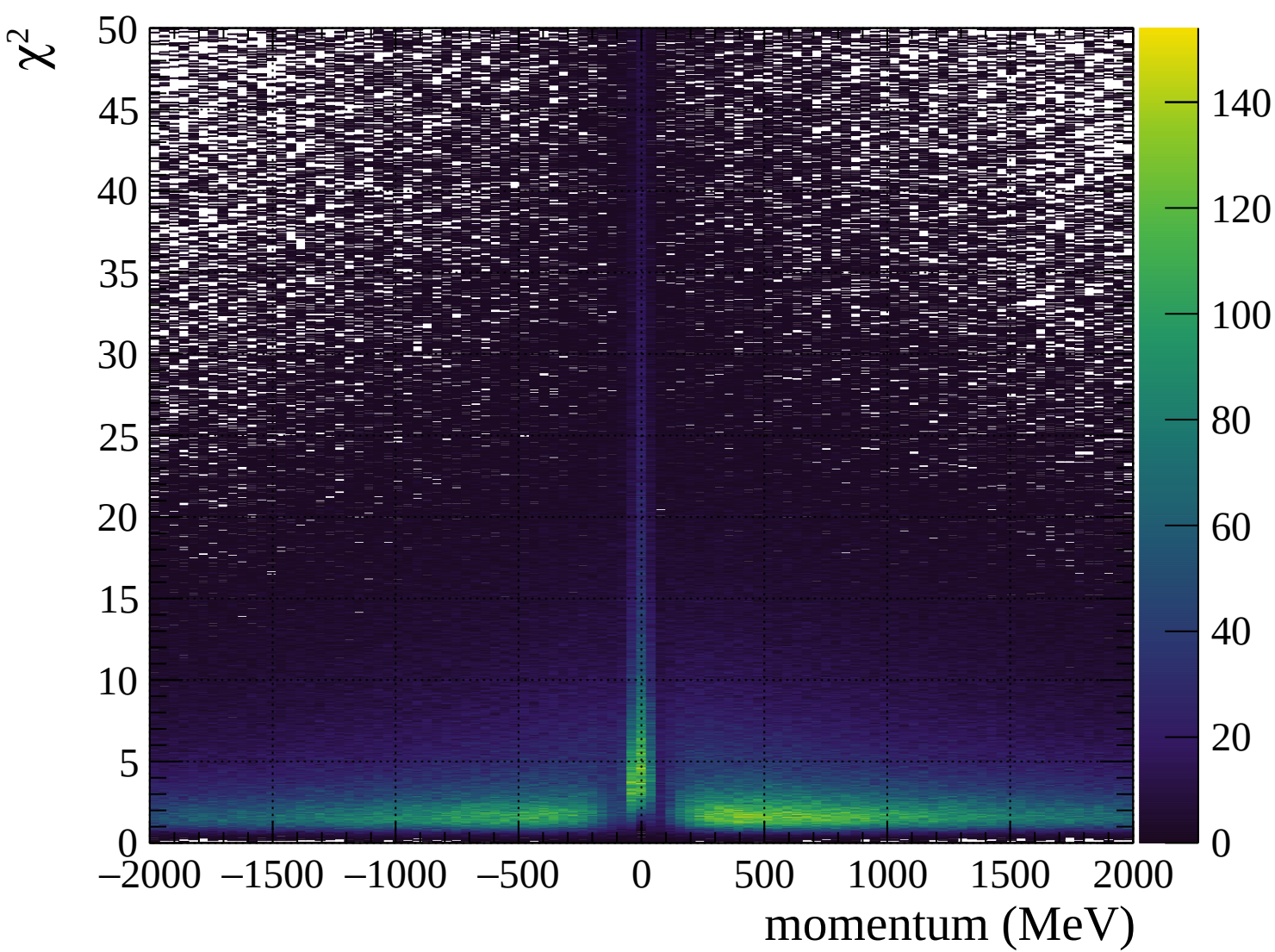




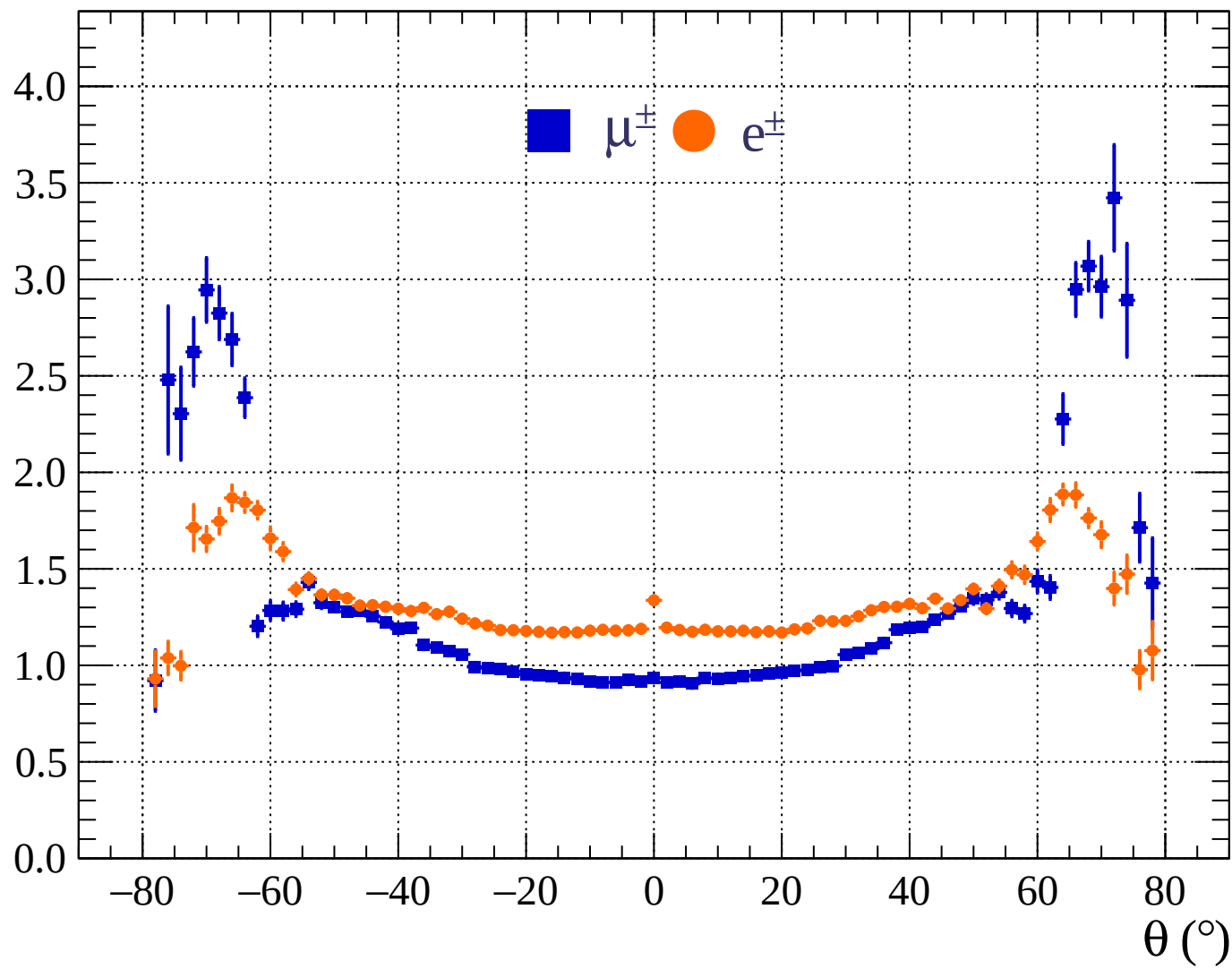




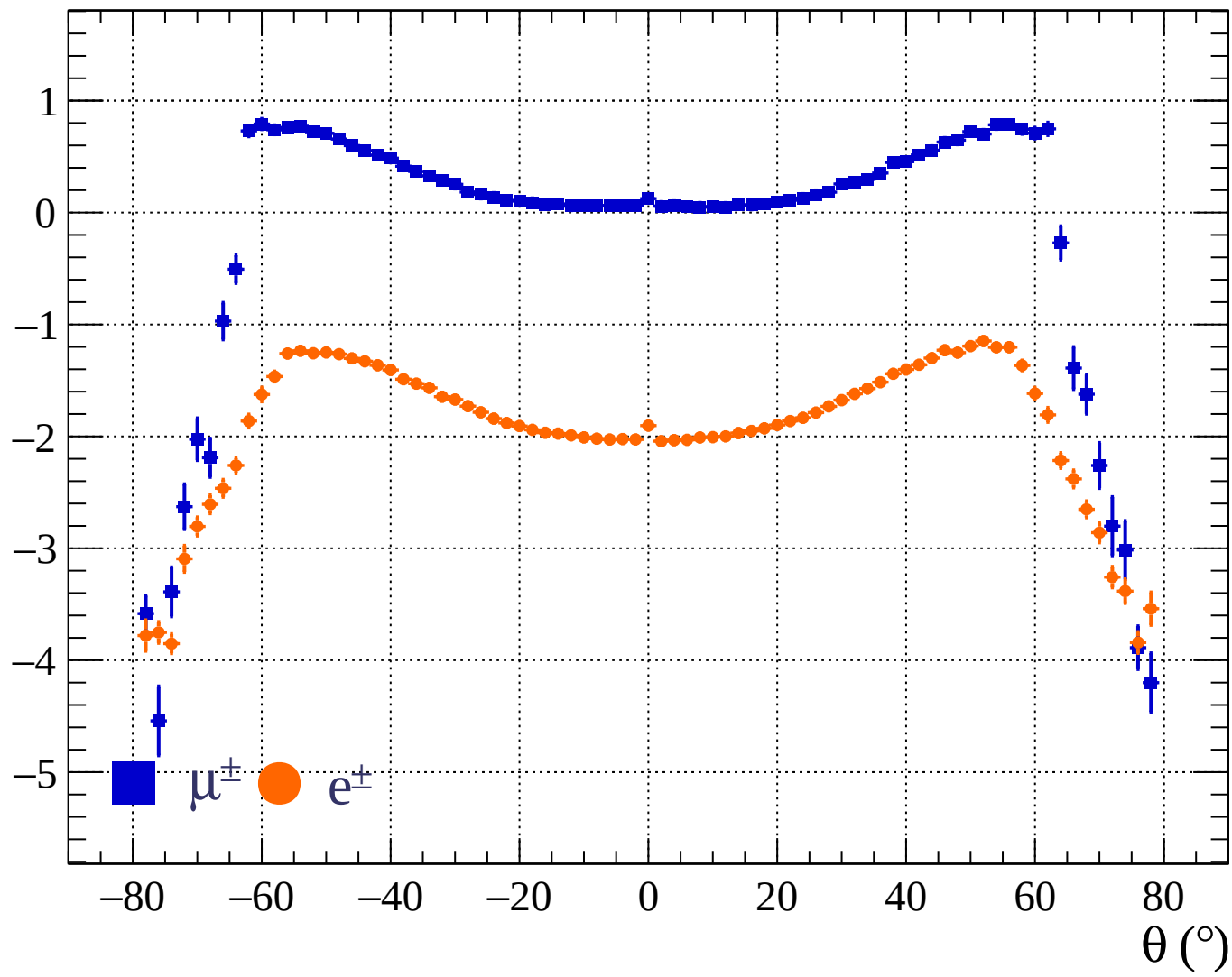




Pulls standard deviation

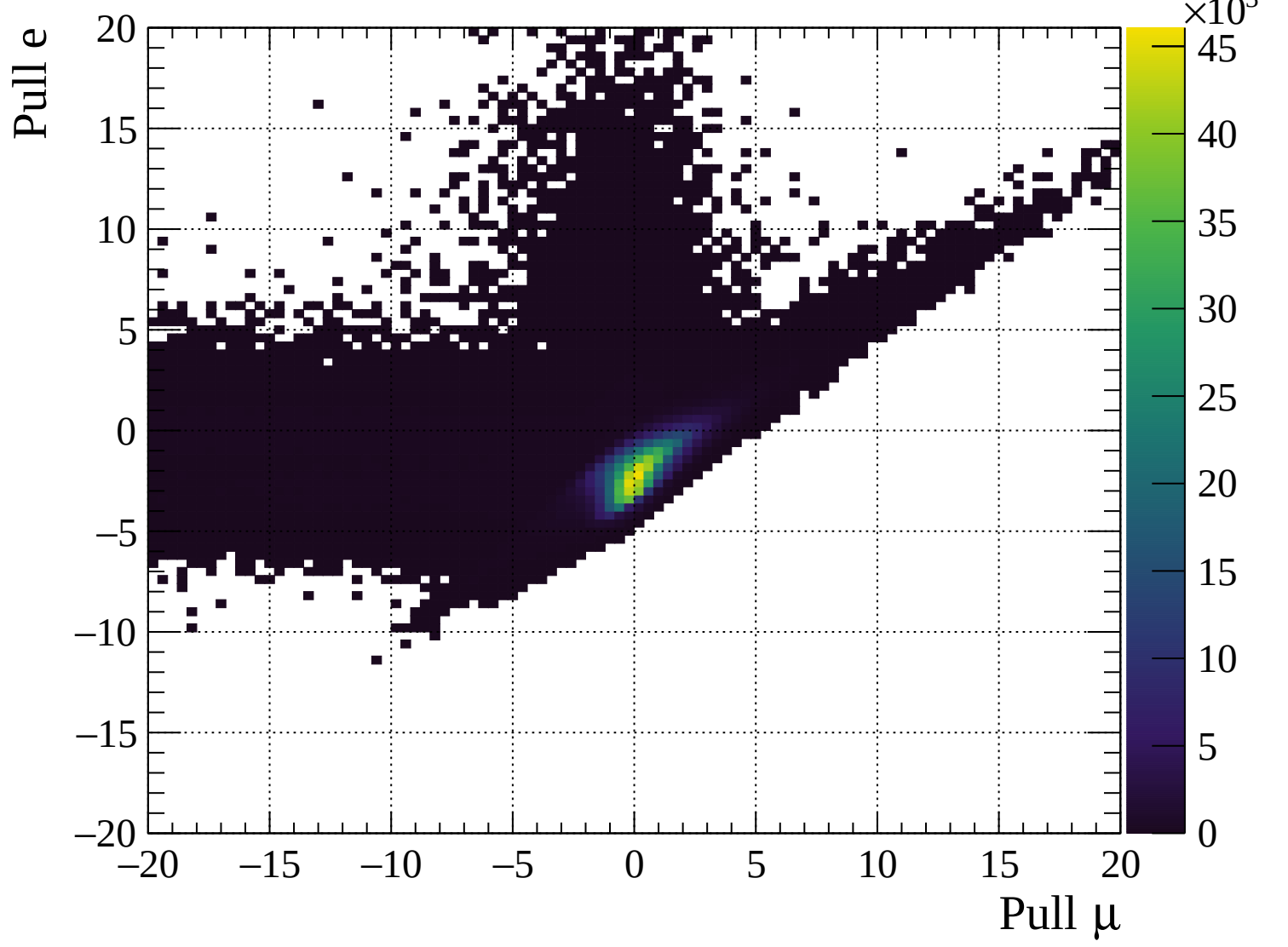


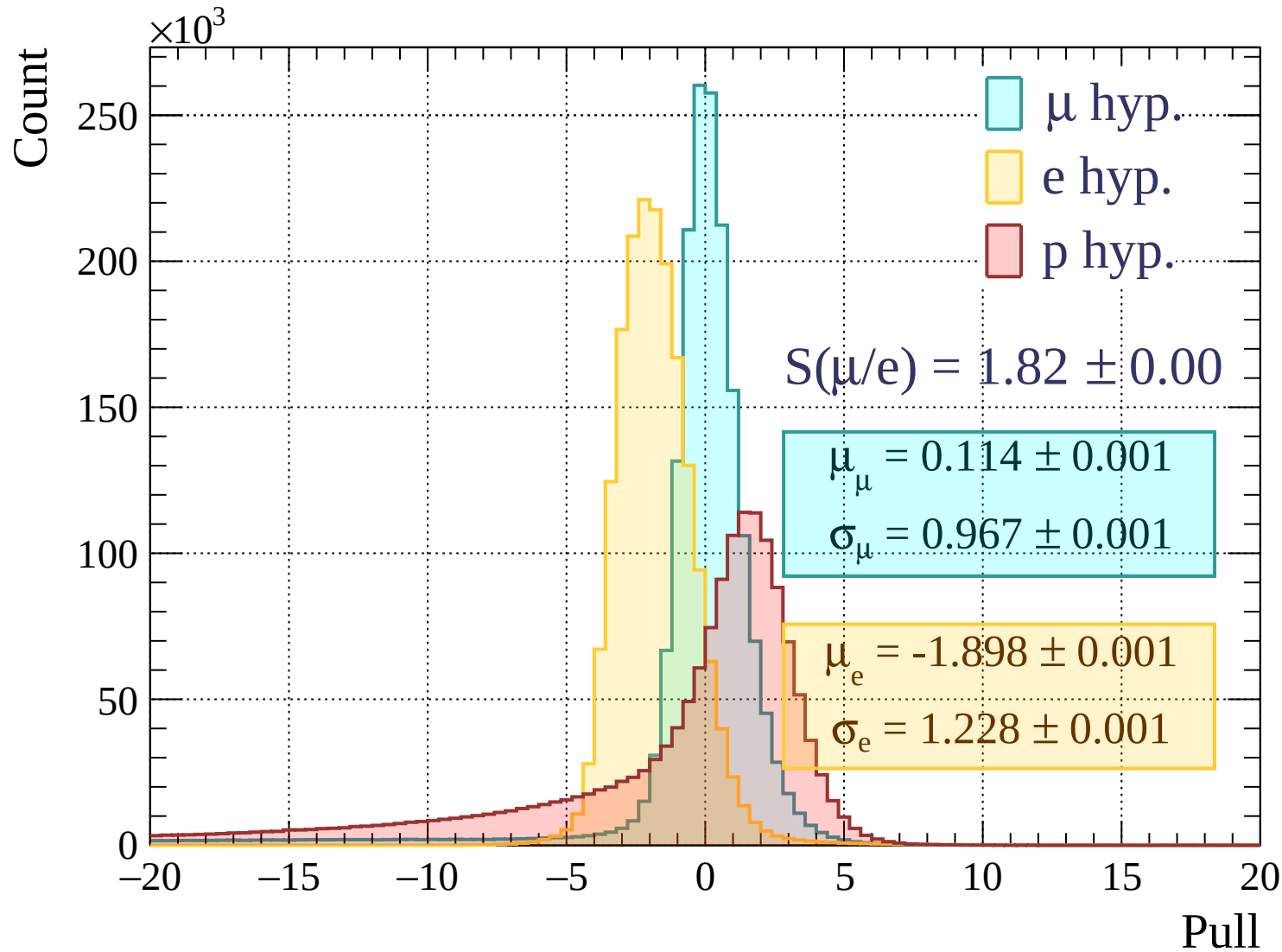
Pulls mean

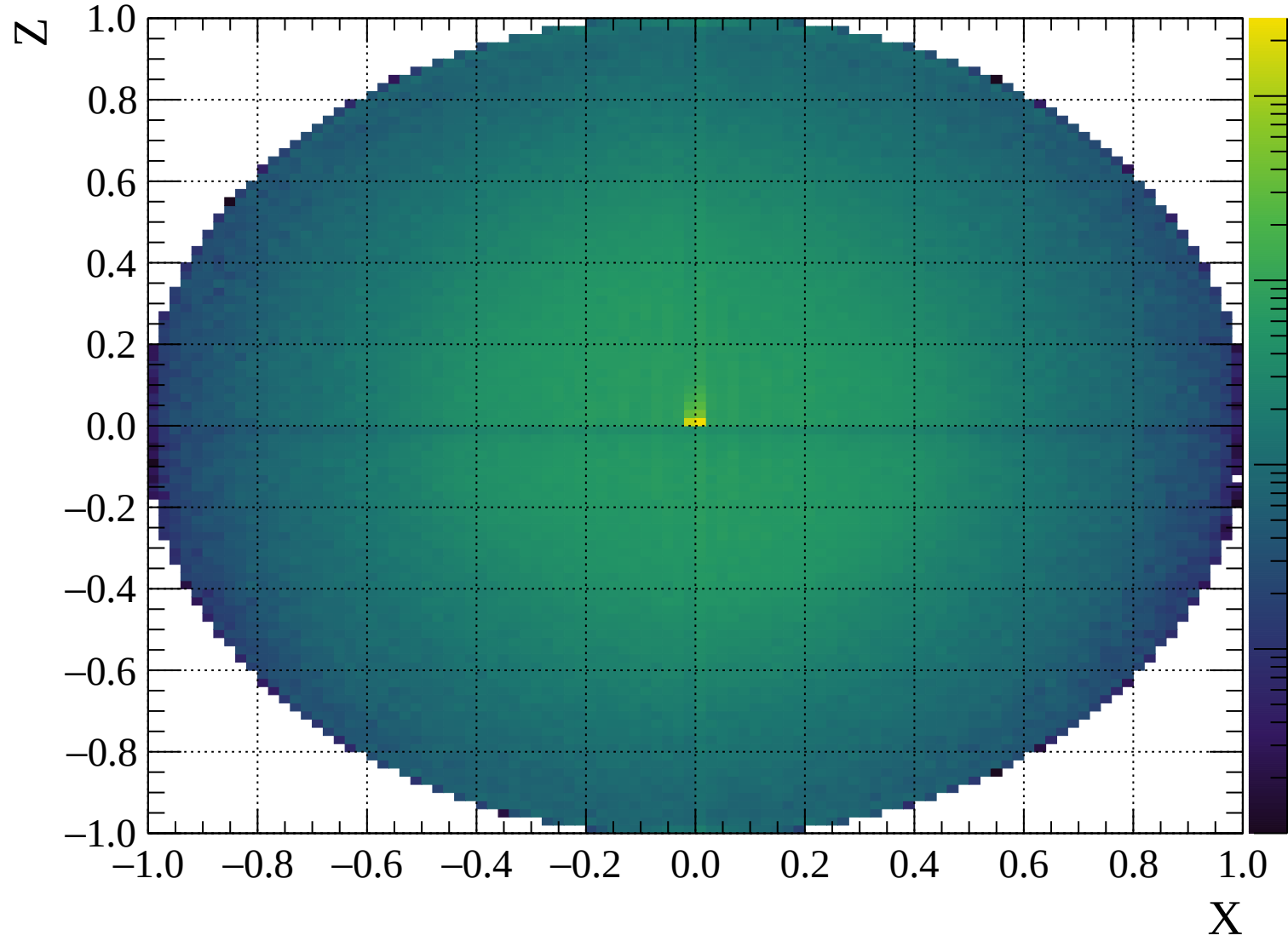


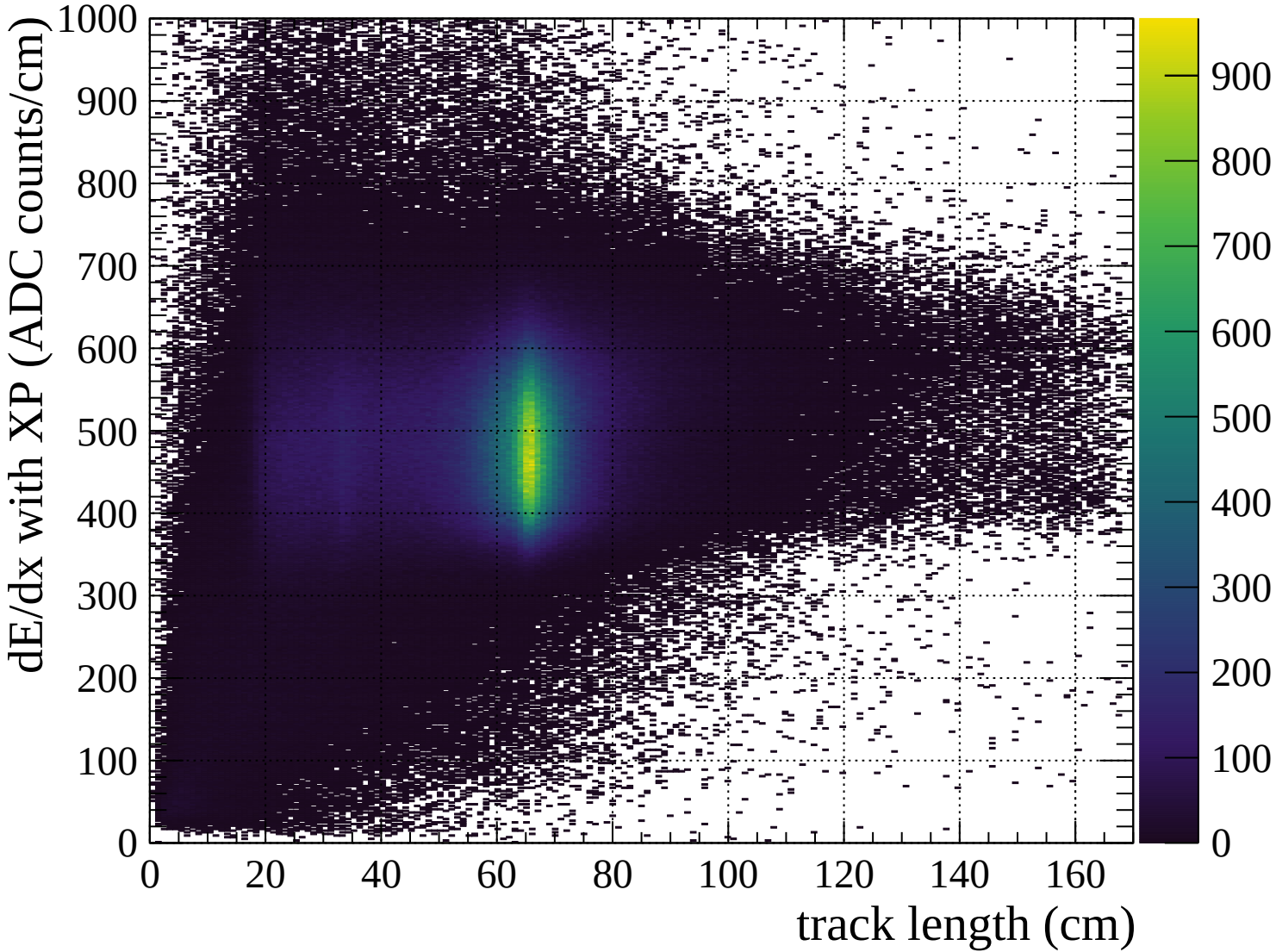
■ $\mu^\pm$ ● $e^\pm$

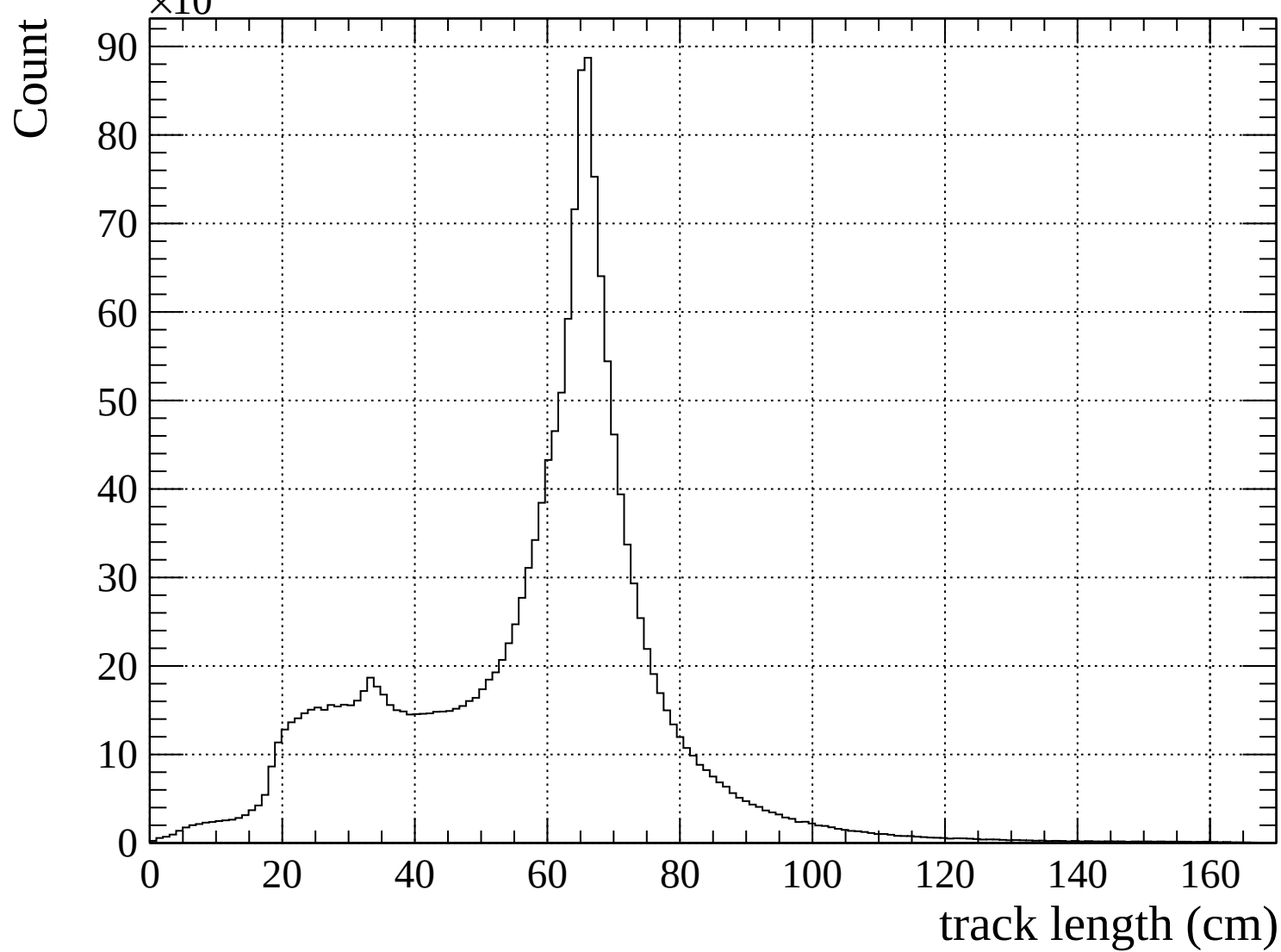
■ $\mu^\pm$ ● $e^\pm$

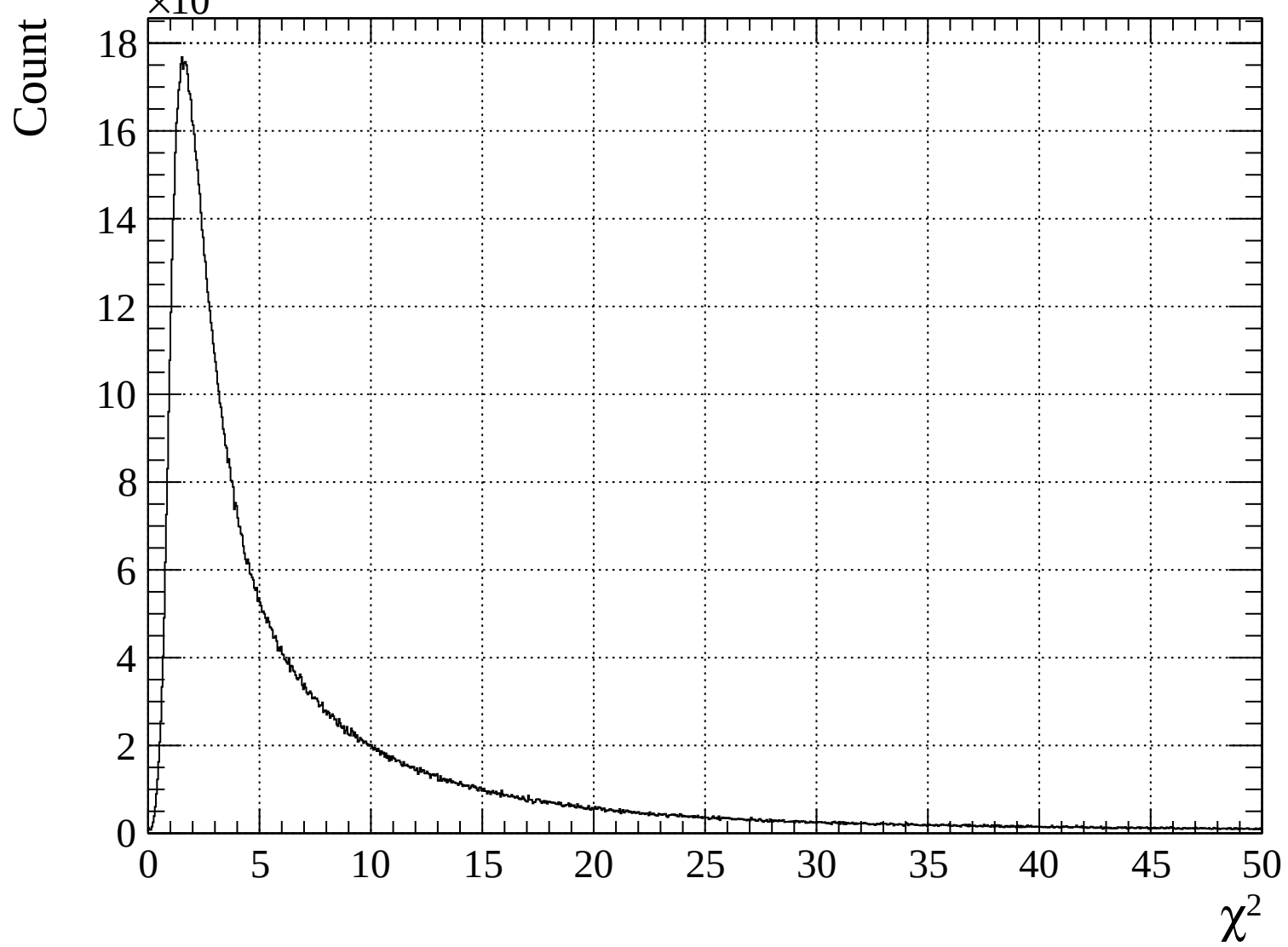


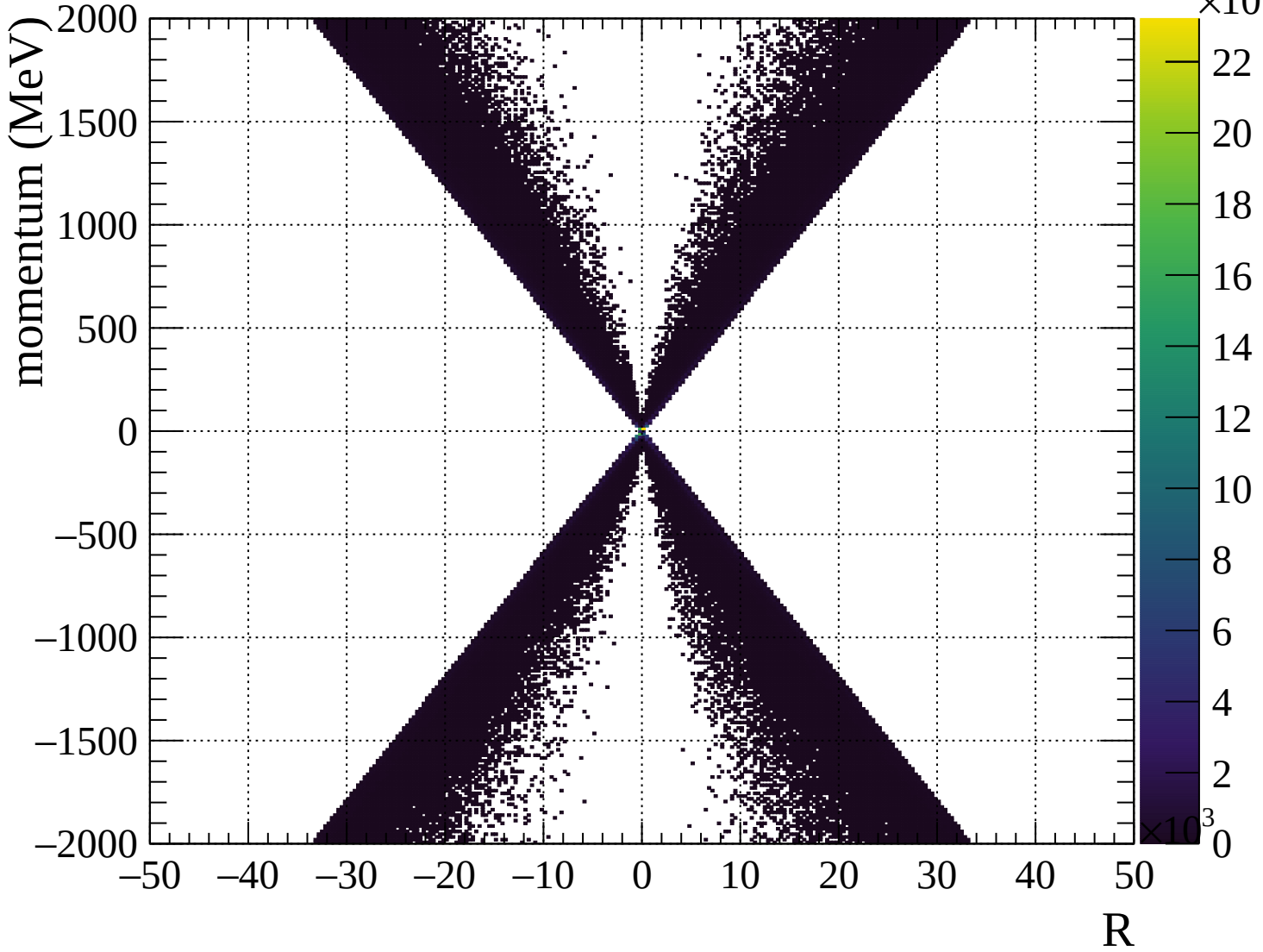


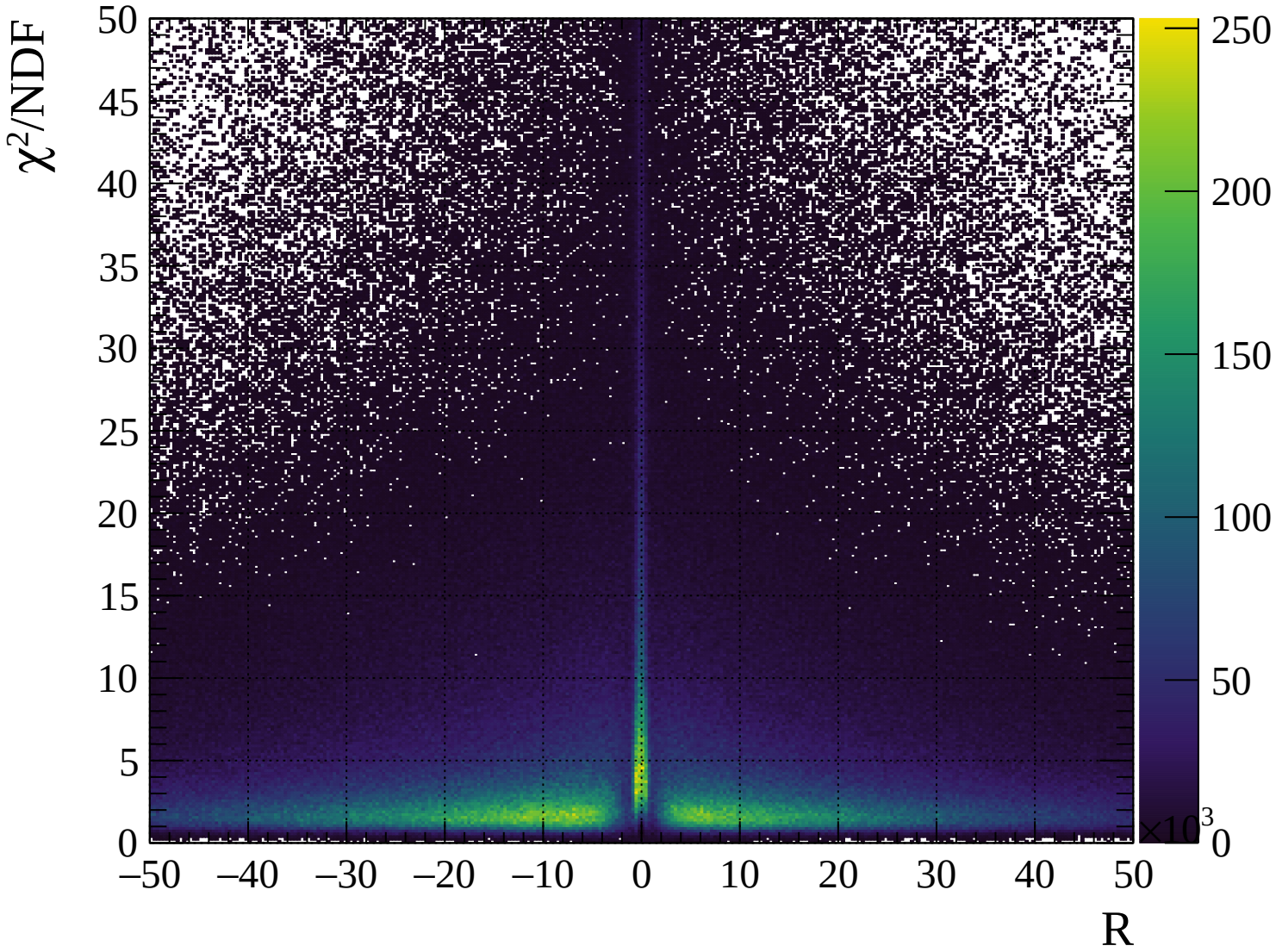








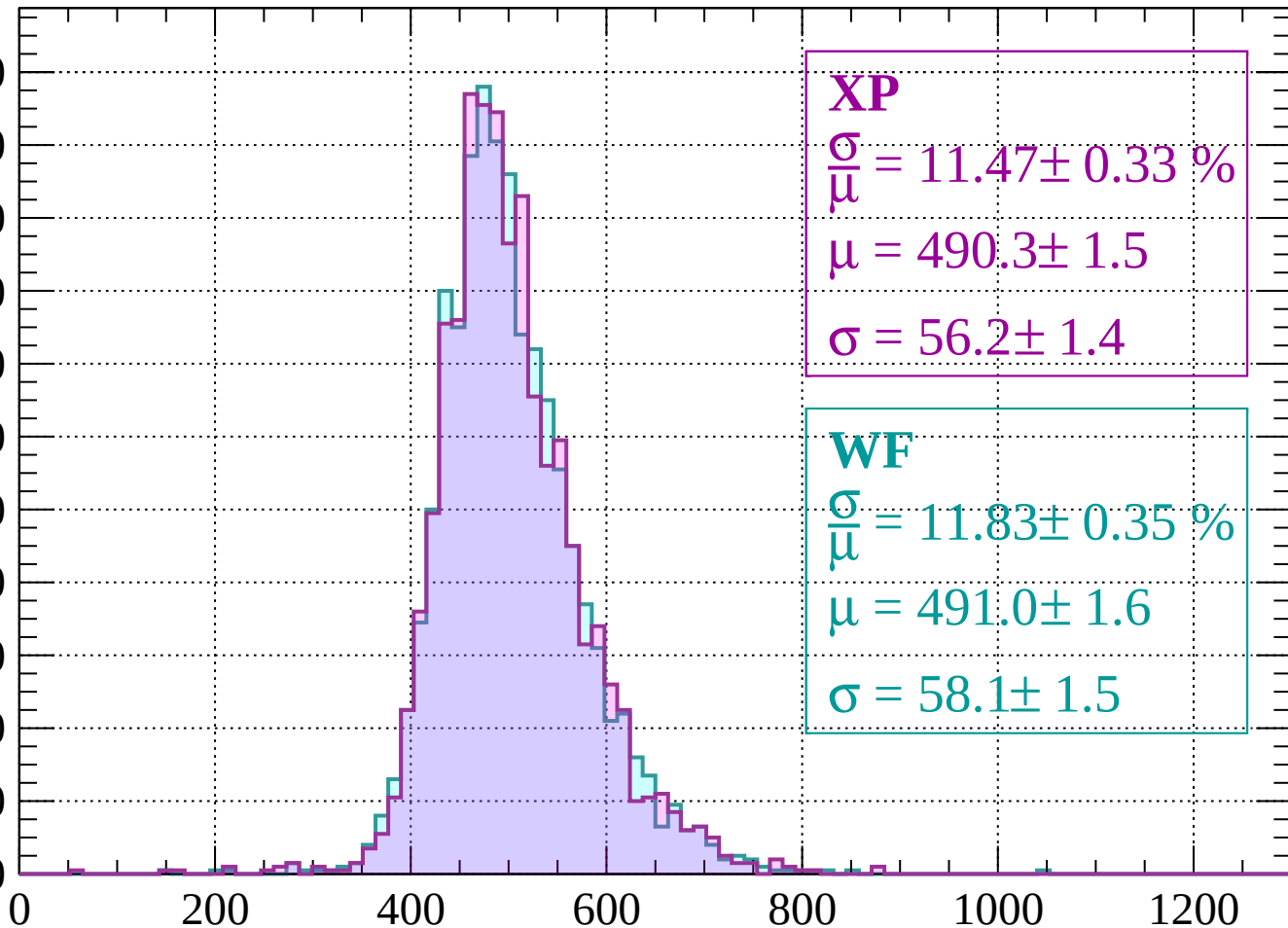




Energy loss | -2000 < p < -1960

Count

220
200
180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 11.47 \pm 0.33 \%$$

$$\mu = 490.3 \pm 1.5$$

$$\sigma = 56.2 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.83 \pm 0.35 \%$$

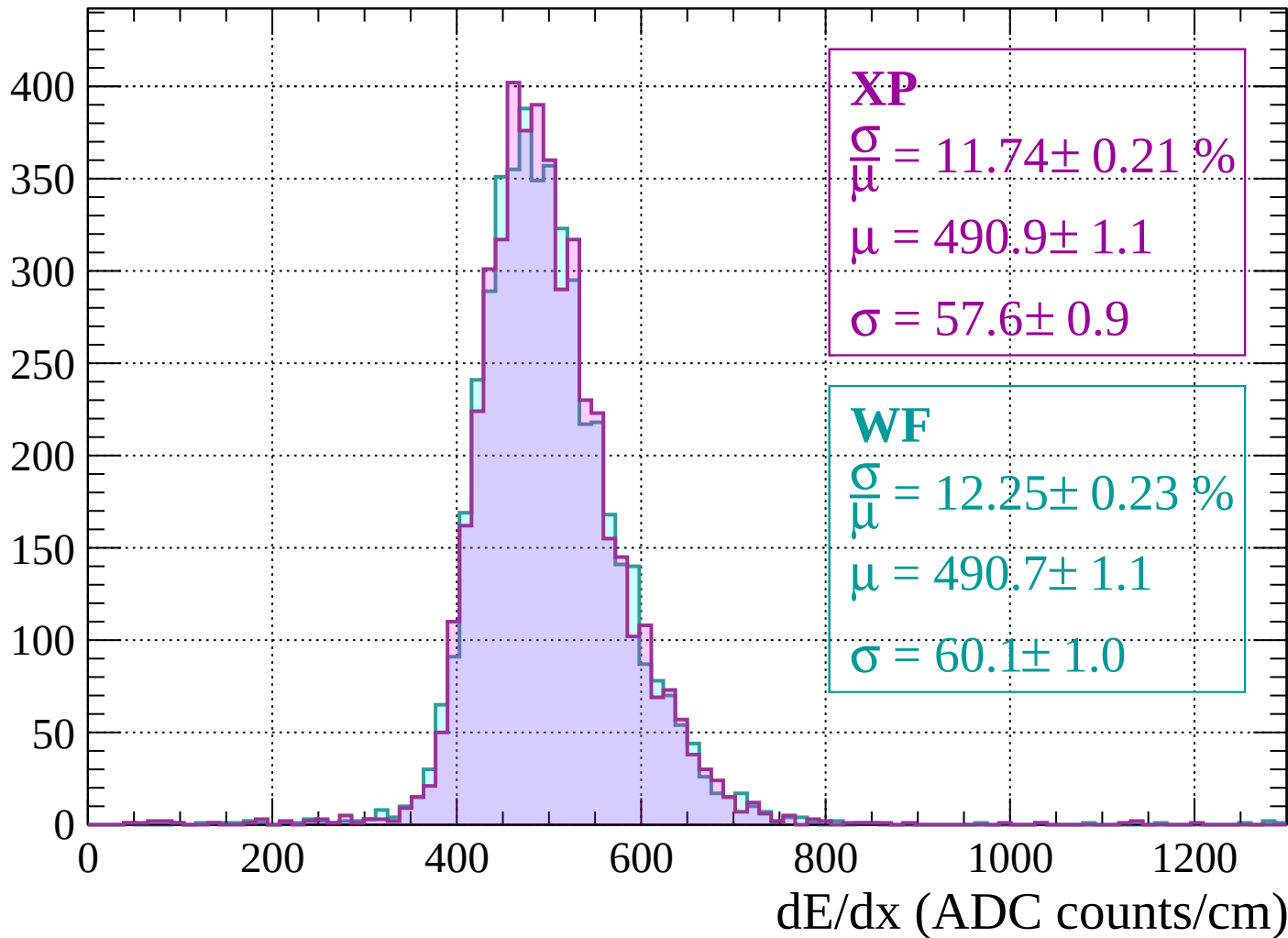
$$\mu = 491.0 \pm 1.6$$

$$\sigma = 58.1 \pm 1.5$$

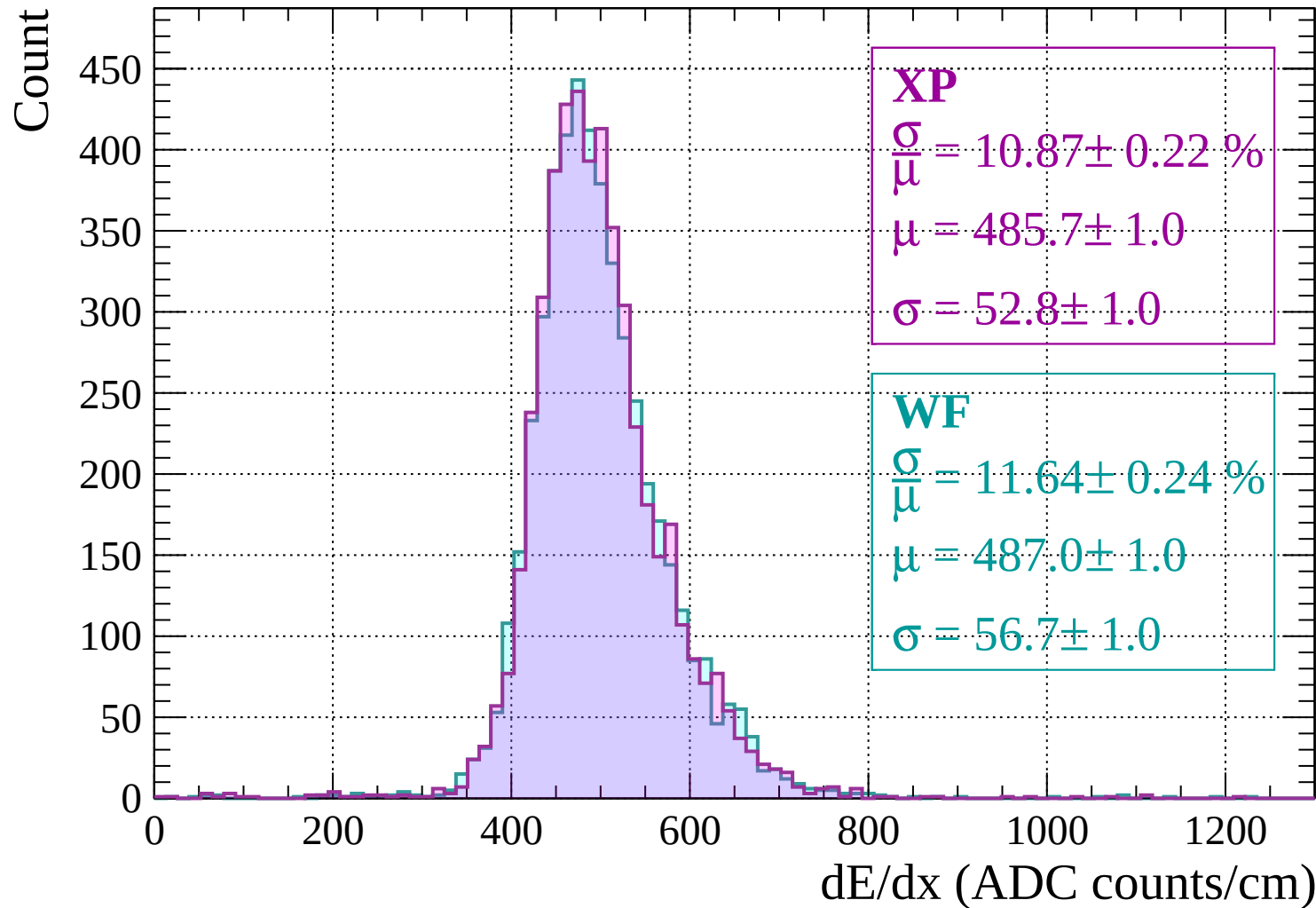
dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

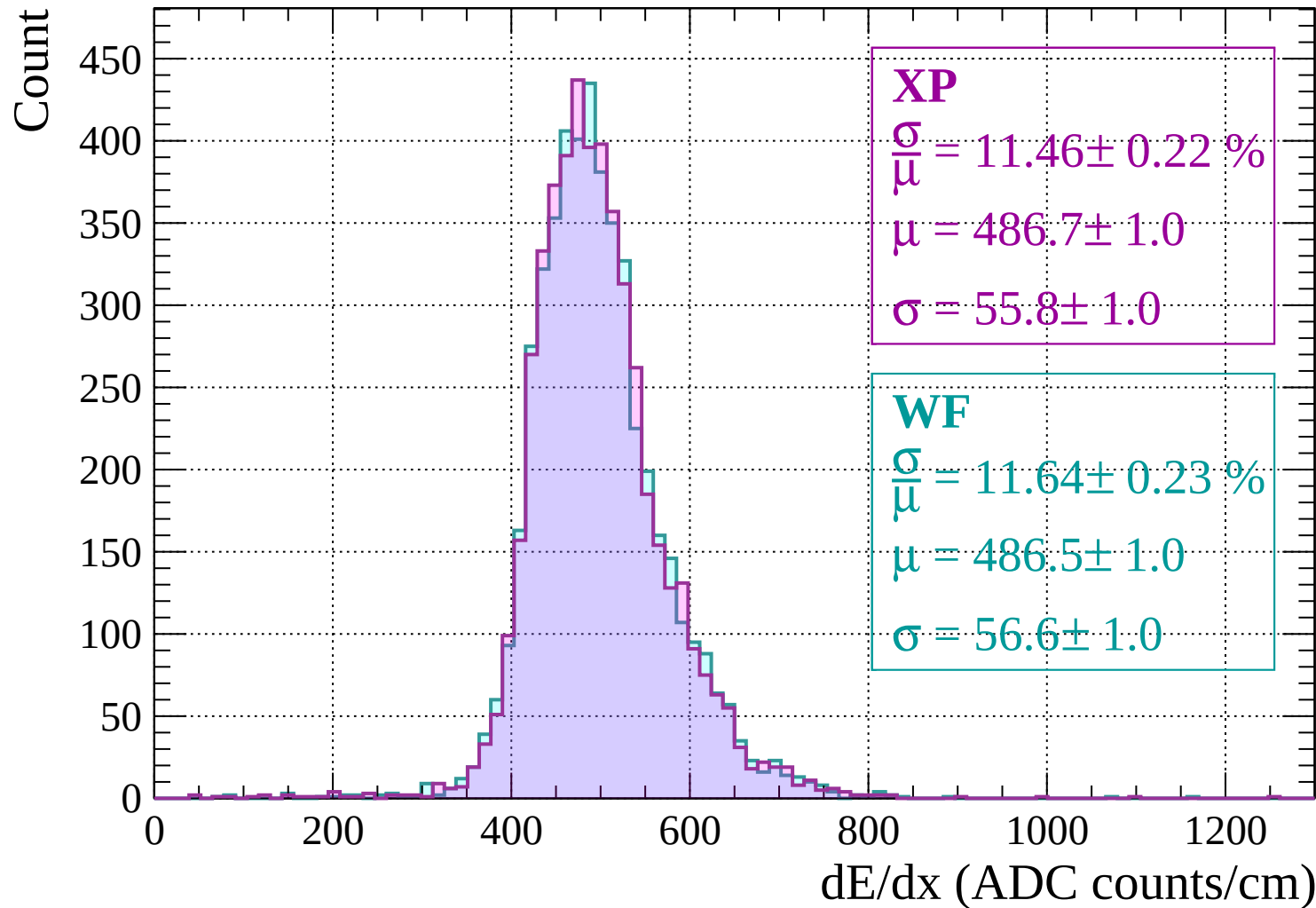
Count



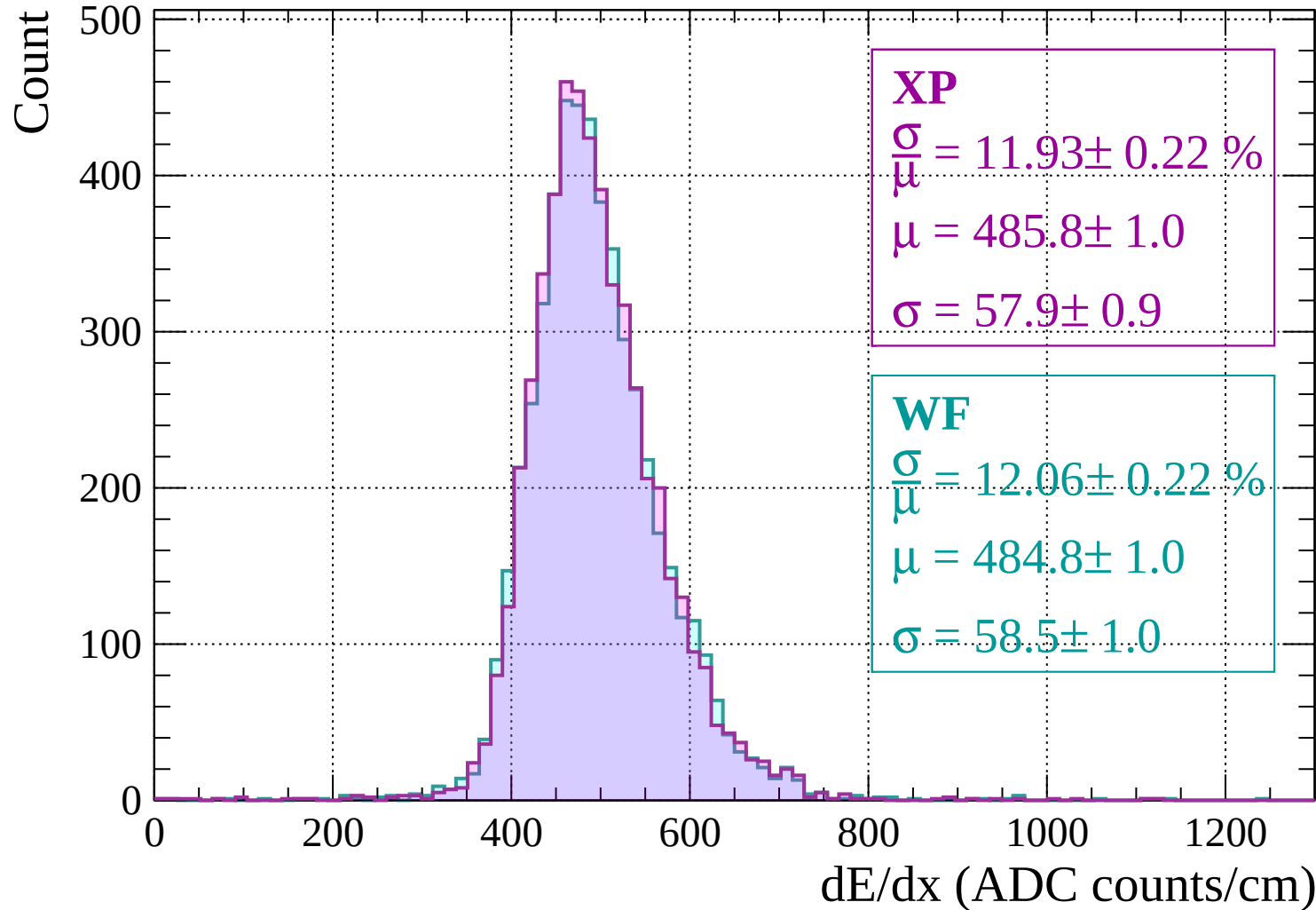
Energy loss | -1920 < p < -1880



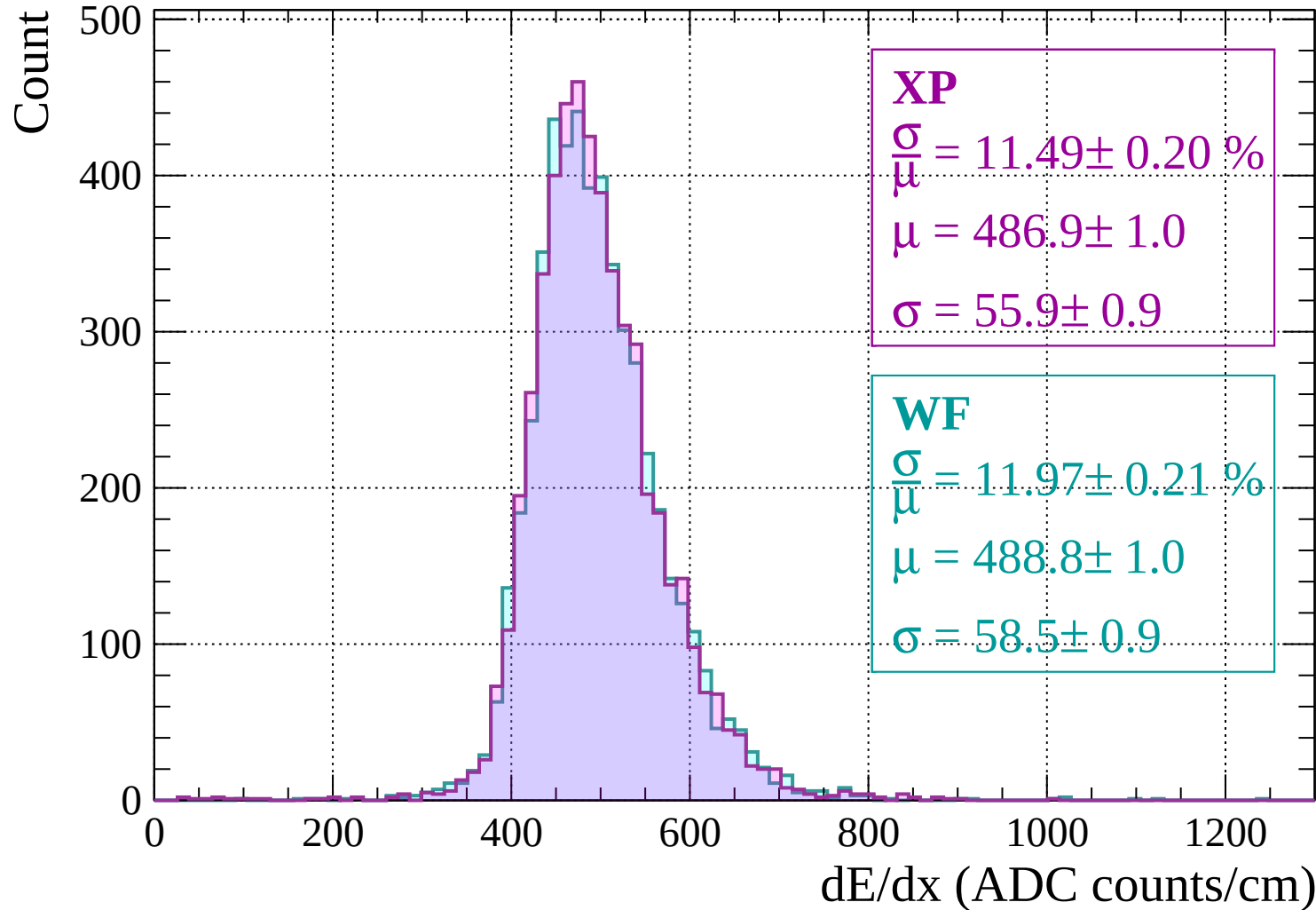
Energy loss | -1880 < p < -1840



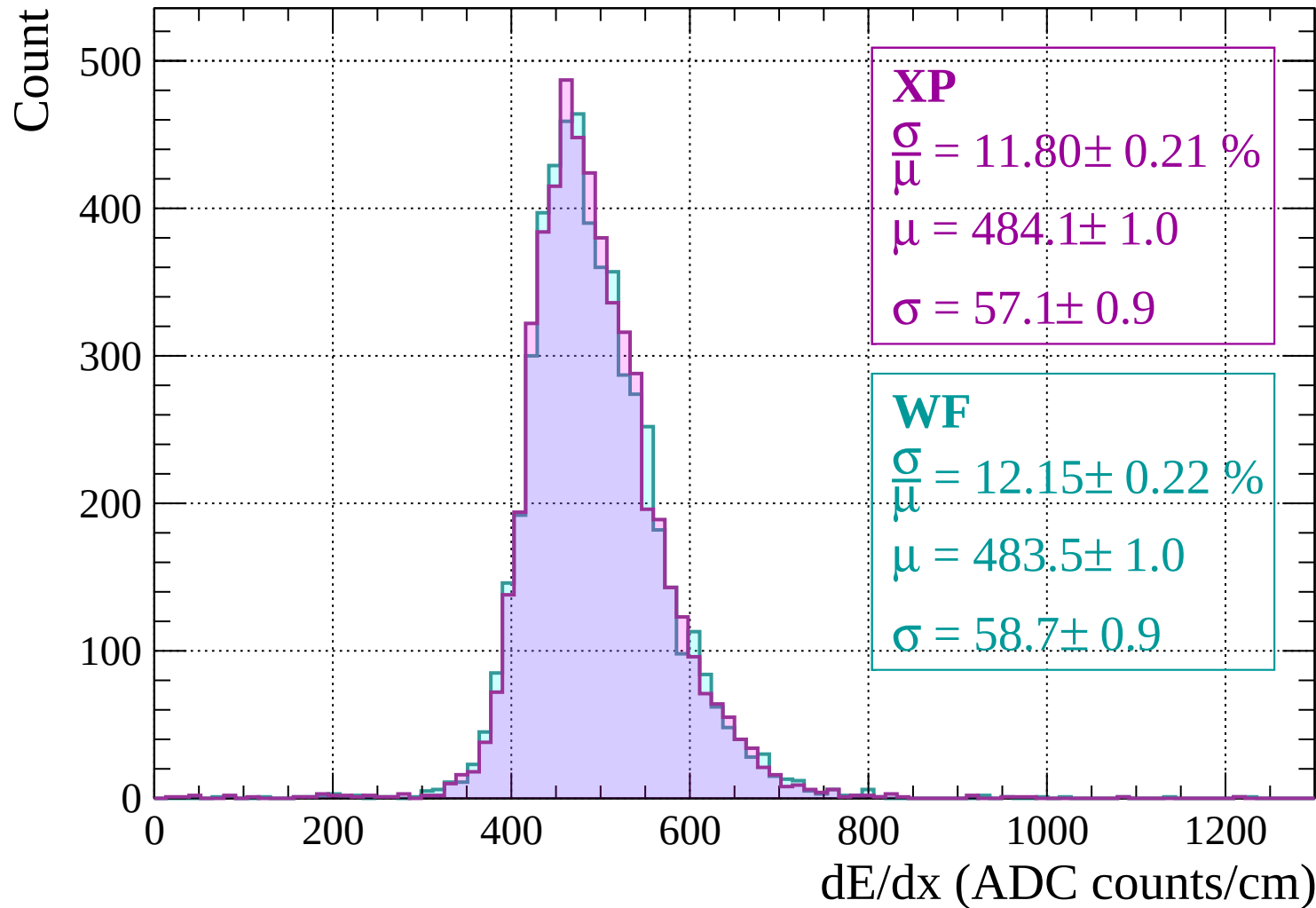
Energy loss | -1840 < p < -1800



Energy loss | -1800 < p < -1760

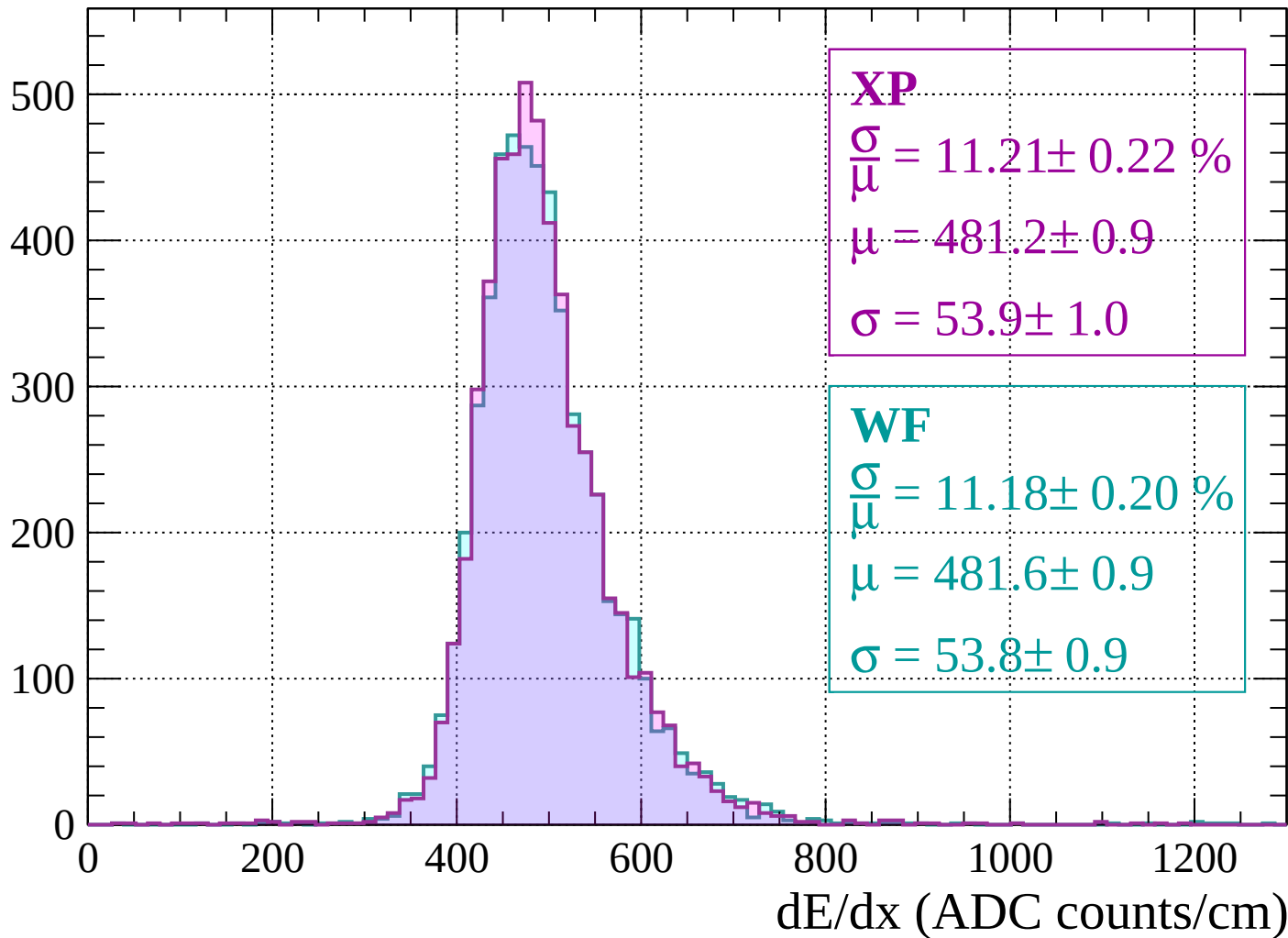


Energy loss | -1760 < p < -1720



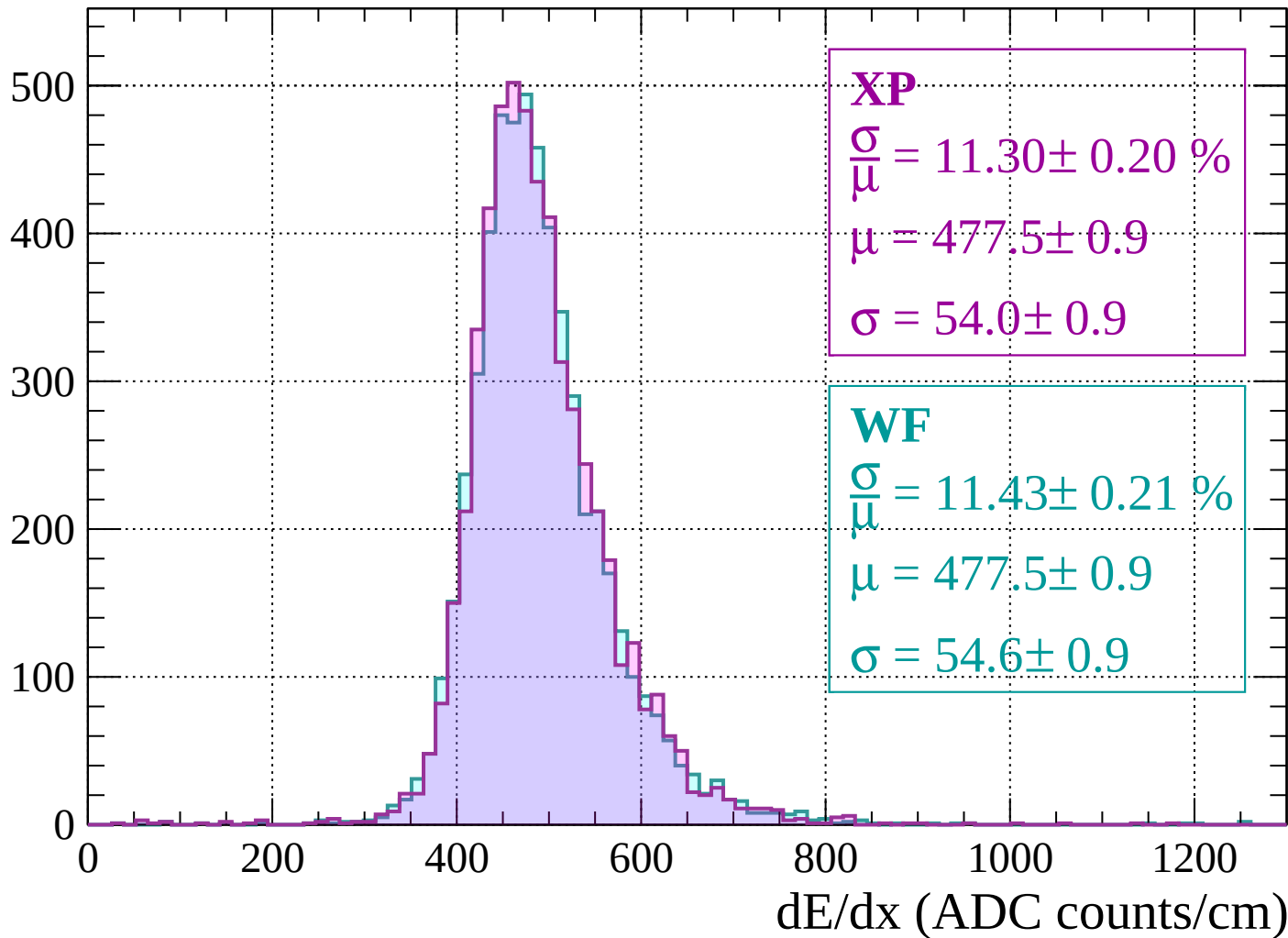
Energy loss | $-1720 < p < -1680$

Count



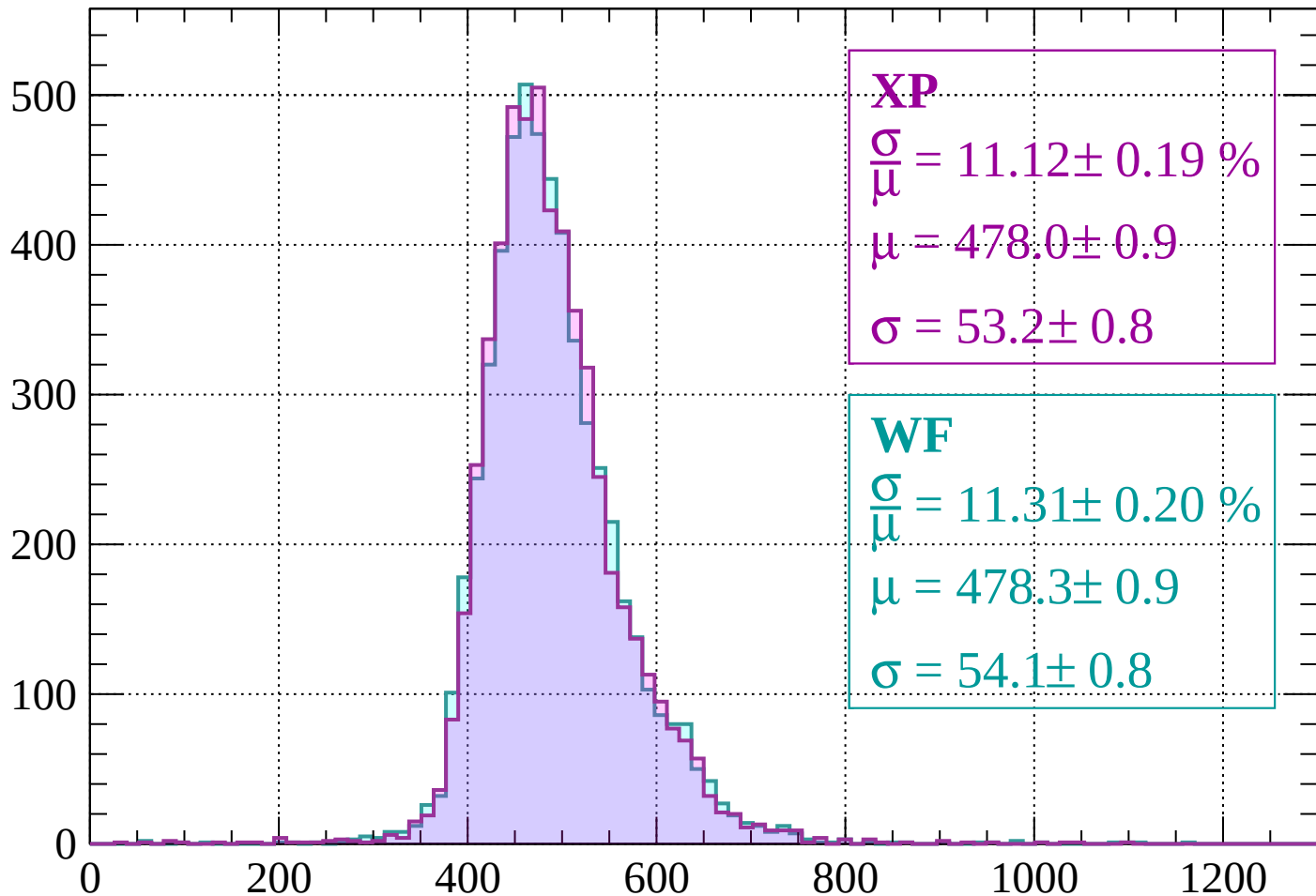
Energy loss | $-1680 < p < -1640$

Count



Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 11.12 \pm 0.19 \%$$

$$\mu = 478.0 \pm 0.9$$

$$\sigma = 53.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.31 \pm 0.20 \%$$

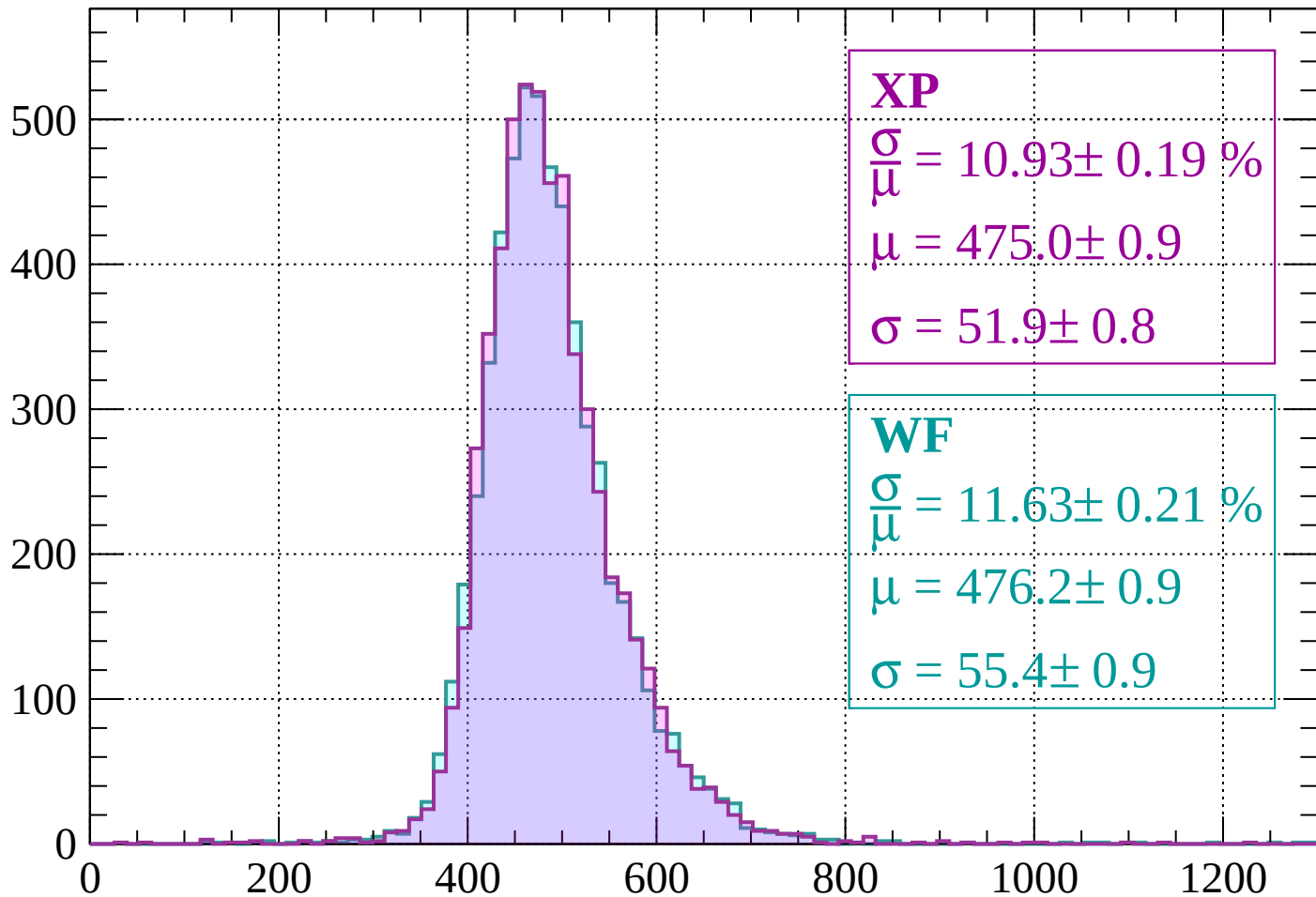
$$\mu = 478.3 \pm 0.9$$

$$\sigma = 54.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

Count



XP

$$\frac{\sigma}{\mu} = 10.93 \pm 0.19 \%$$

$$\mu = 475.0 \pm 0.9$$

$$\sigma = 51.9 \pm 0.8$$

WF

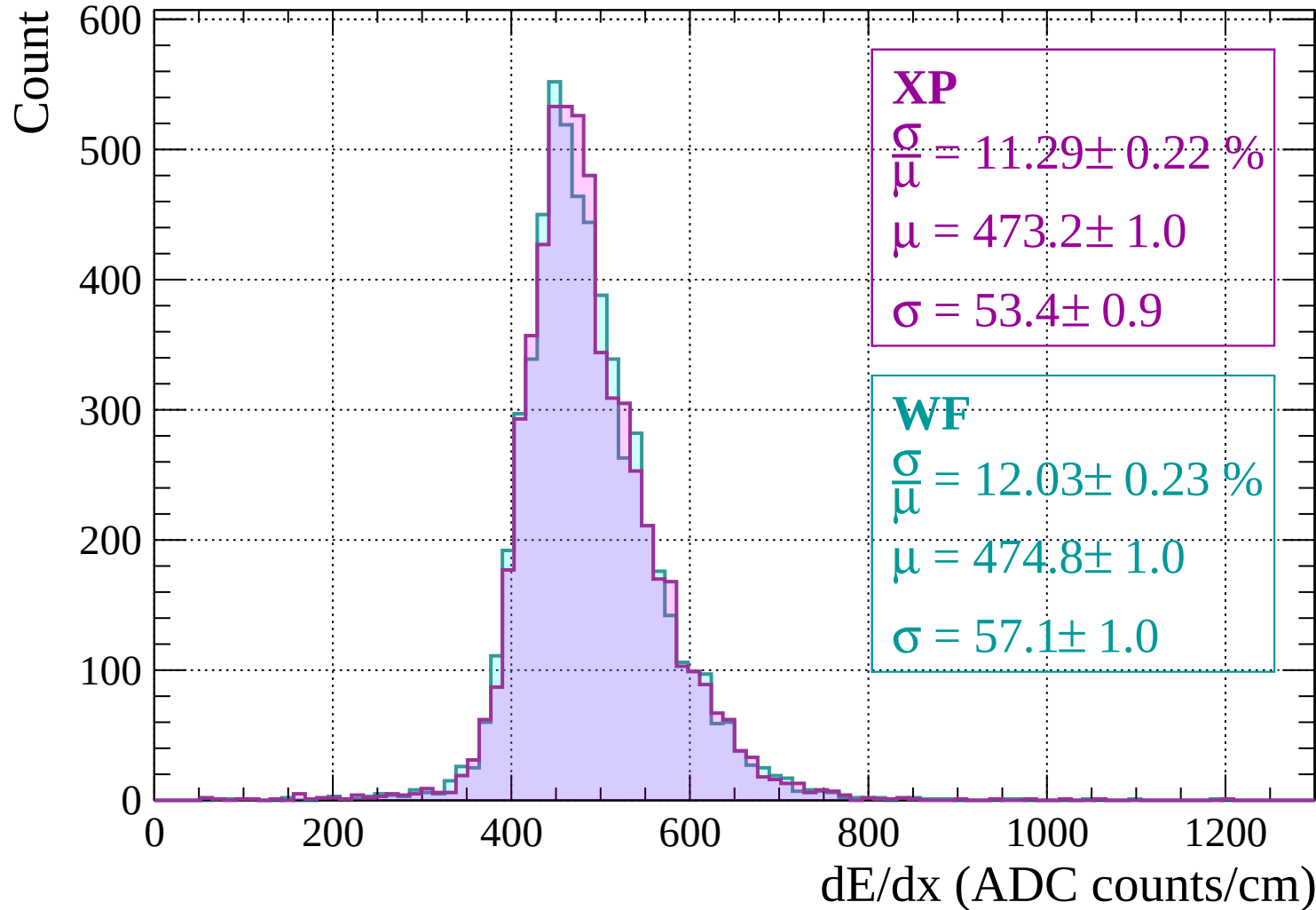
$$\frac{\sigma}{\mu} = 11.63 \pm 0.21 \%$$

$$\mu = 476.2 \pm 0.9$$

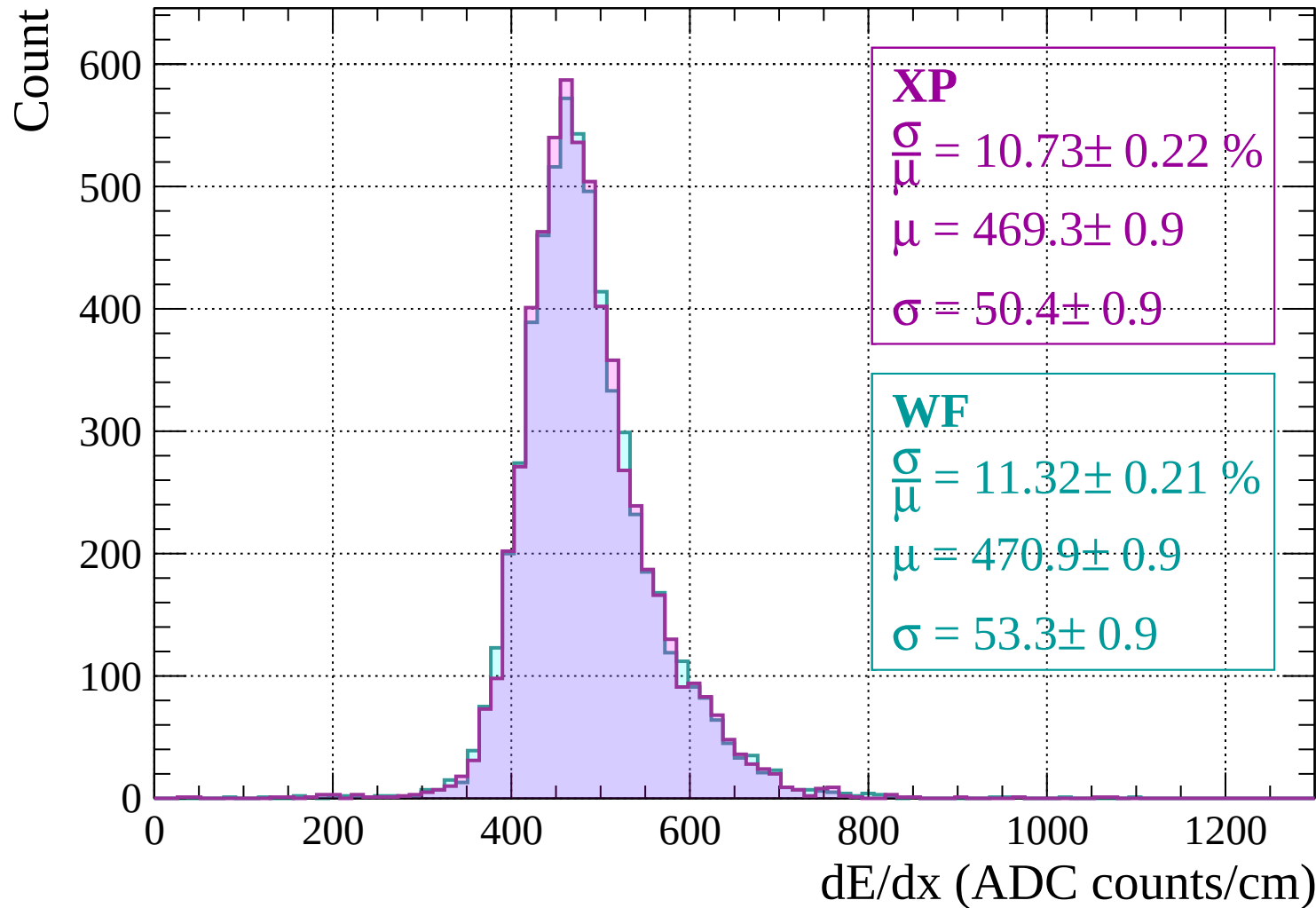
$$\sigma = 55.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1520

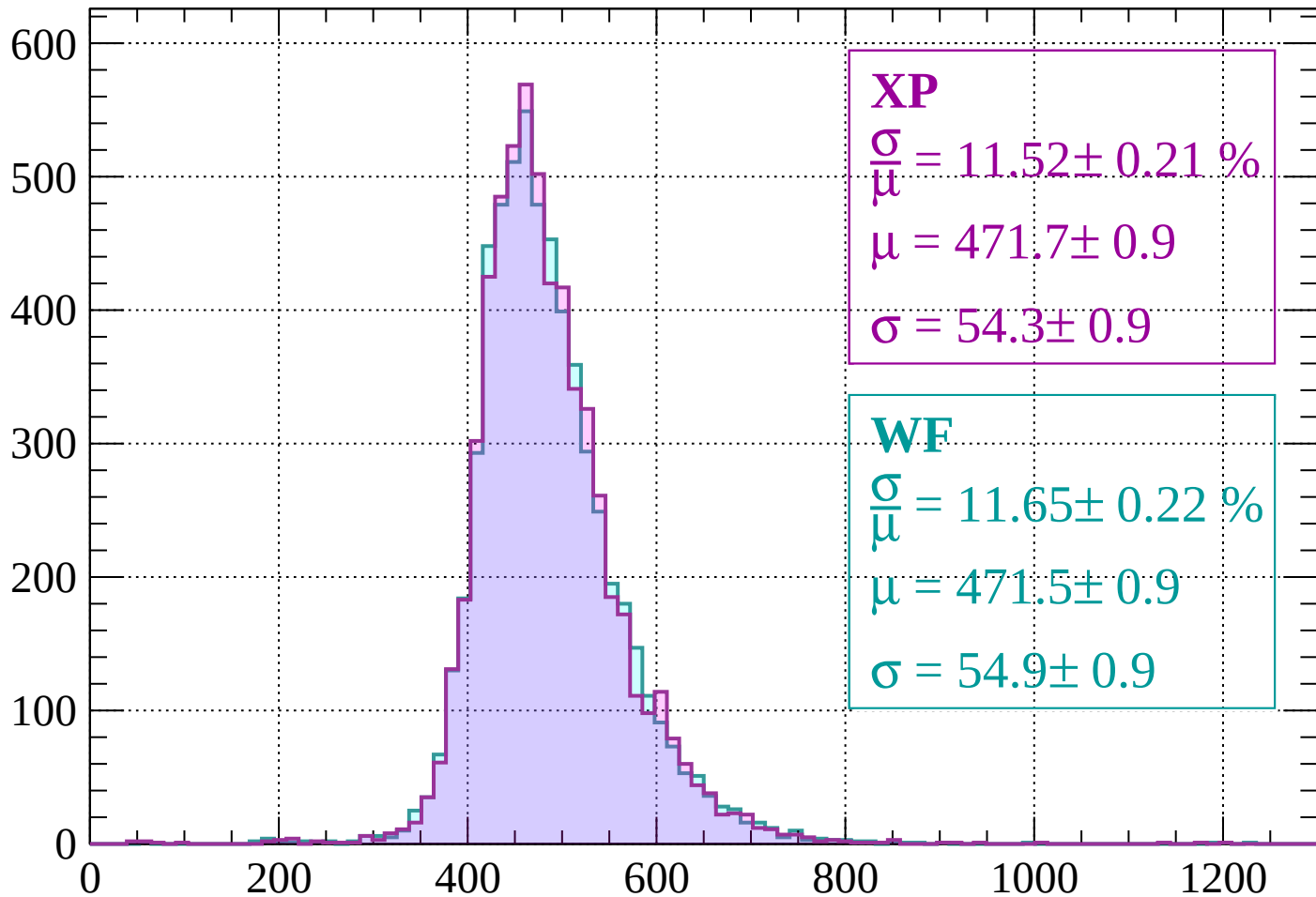


Energy loss | $-1520 < p < -1480$



Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 11.52 \pm 0.21 \%$$

$$\mu = 471.7 \pm 0.9$$

$$\sigma = 54.3 \pm 0.9$$

WF

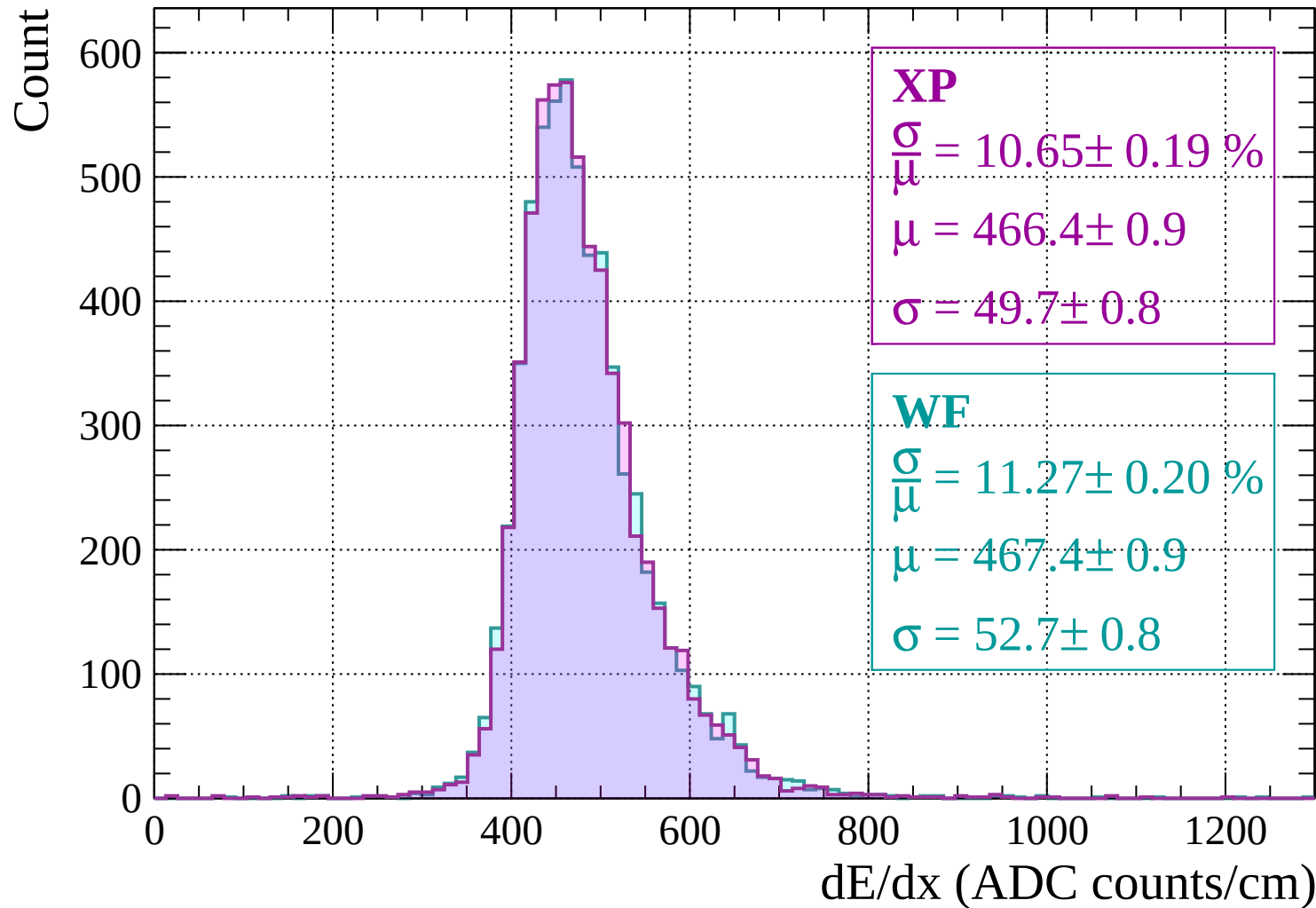
$$\frac{\sigma}{\mu} = 11.65 \pm 0.22 \%$$

$$\mu = 471.5 \pm 0.9$$

$$\sigma = 54.9 \pm 0.9$$

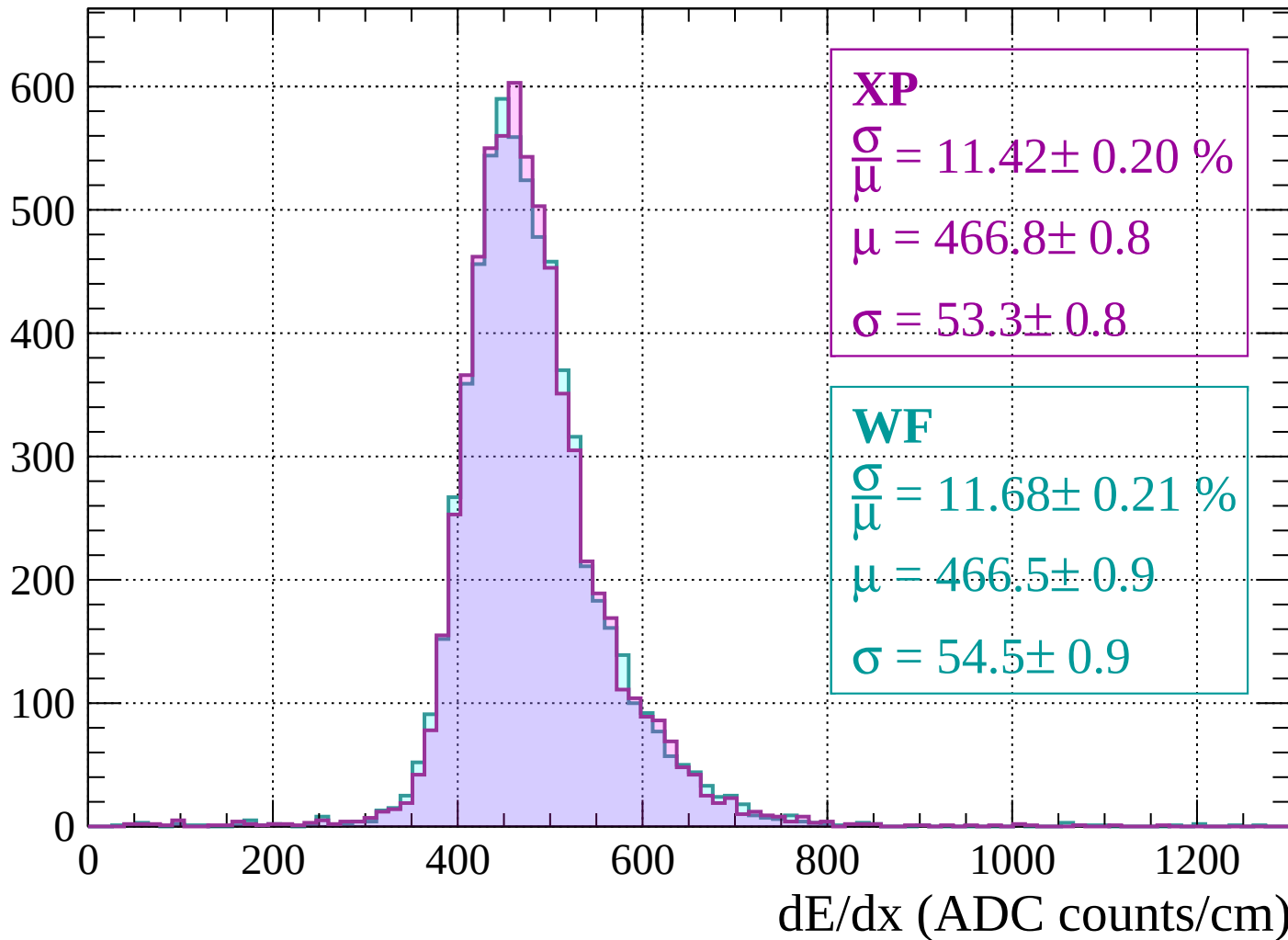
dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$



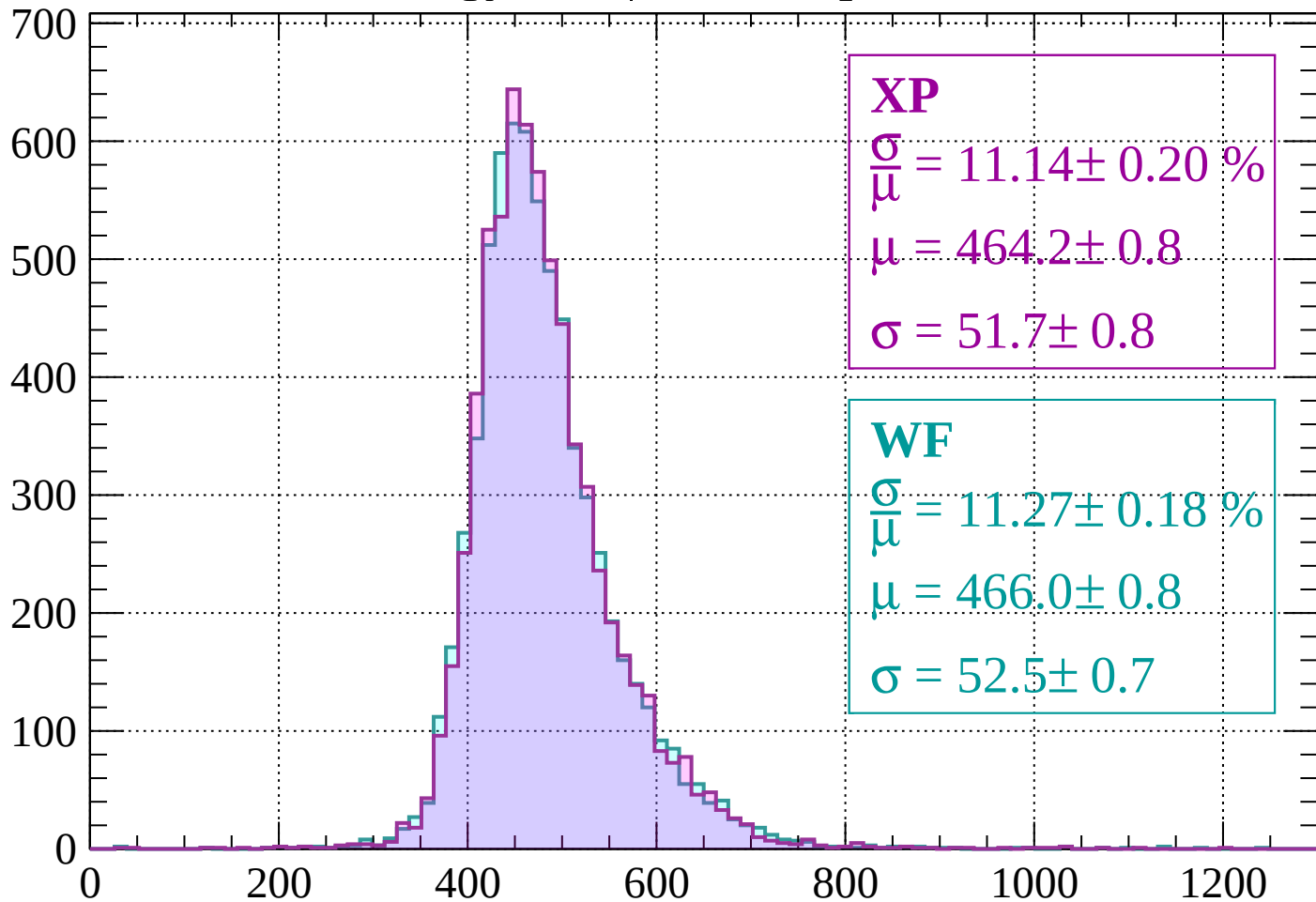
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 11.14 \pm 0.20 \%$$

$$\mu = 464.2 \pm 0.8$$

$$\sigma = 51.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.27 \pm 0.18 \%$$

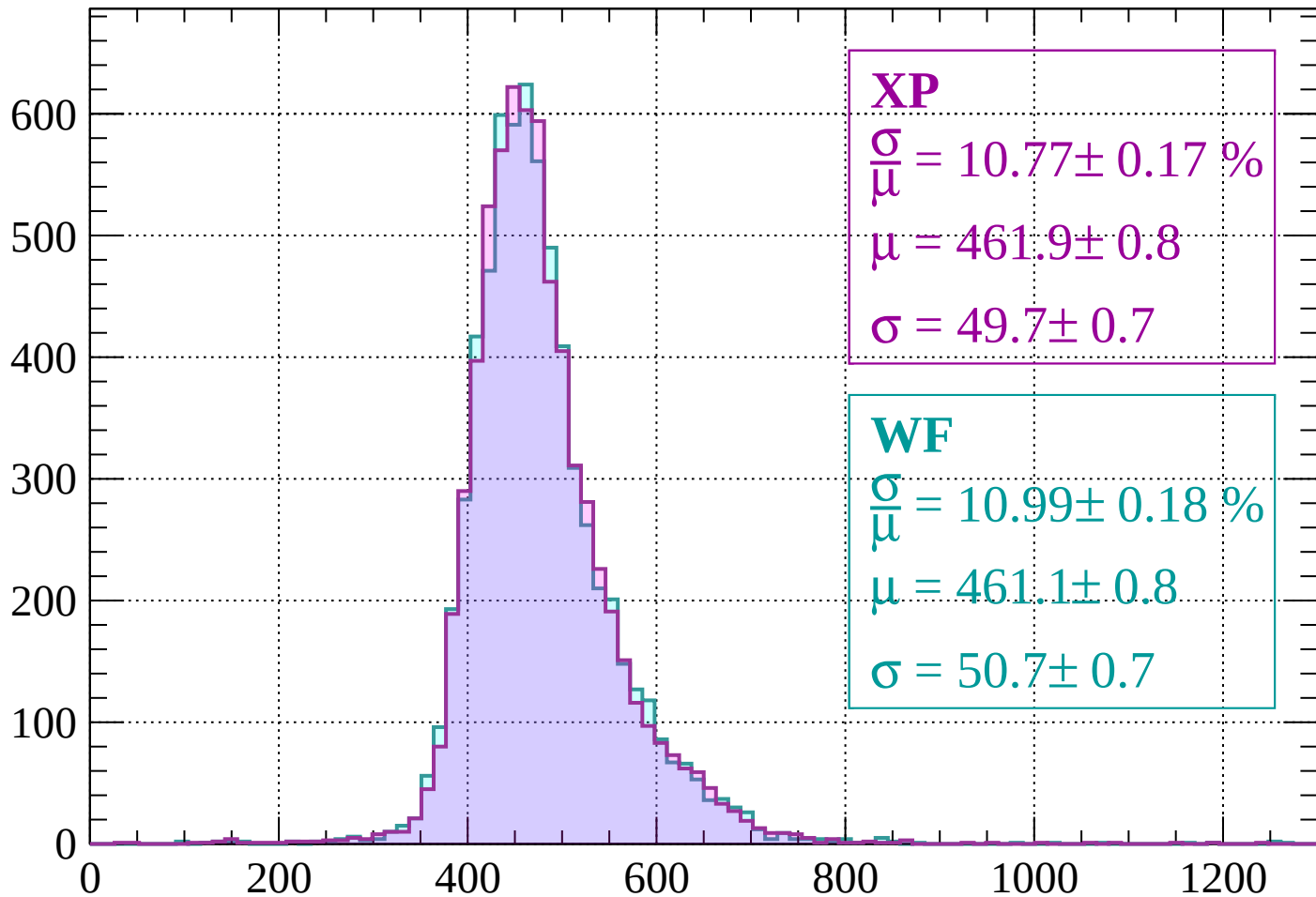
$$\mu = 466.0 \pm 0.8$$

$$\sigma = 52.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 10.77 \pm 0.17 \%$$

$$\mu = 461.9 \pm 0.8$$

$$\sigma = 49.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.99 \pm 0.18 \%$$

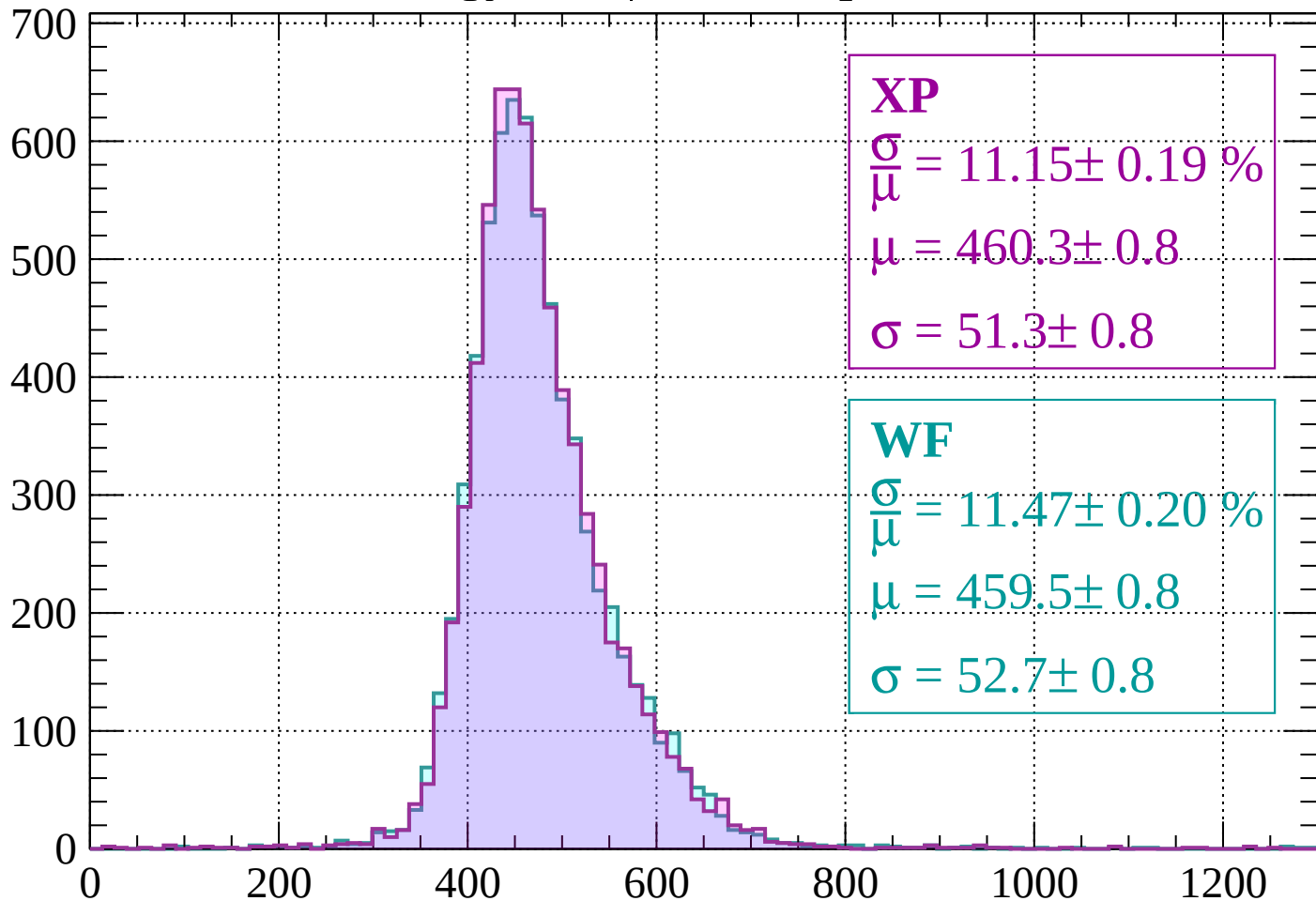
$$\mu = 461.1 \pm 0.8$$

$$\sigma = 50.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1280 < p < -1240

Count



XP

$$\frac{\sigma}{\mu} = 11.15 \pm 0.19 \%$$

$$\mu = 460.3 \pm 0.8$$

$$\sigma = 51.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.47 \pm 0.20 \%$$

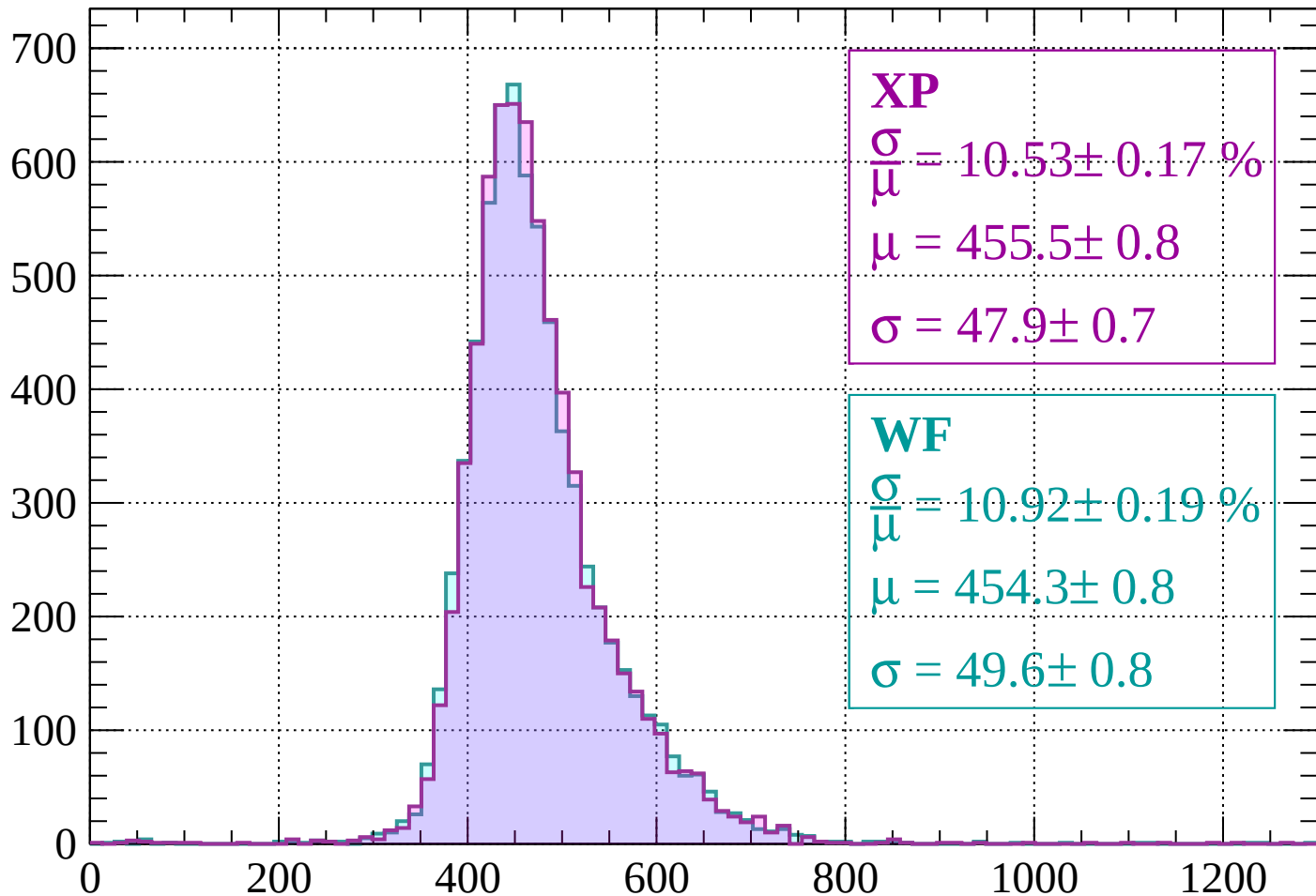
$$\mu = 459.5 \pm 0.8$$

$$\sigma = 52.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 10.53 \pm 0.17 \%$$

$$\mu = 455.5 \pm 0.8$$

$$\sigma = 47.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.92 \pm 0.19 \%$$

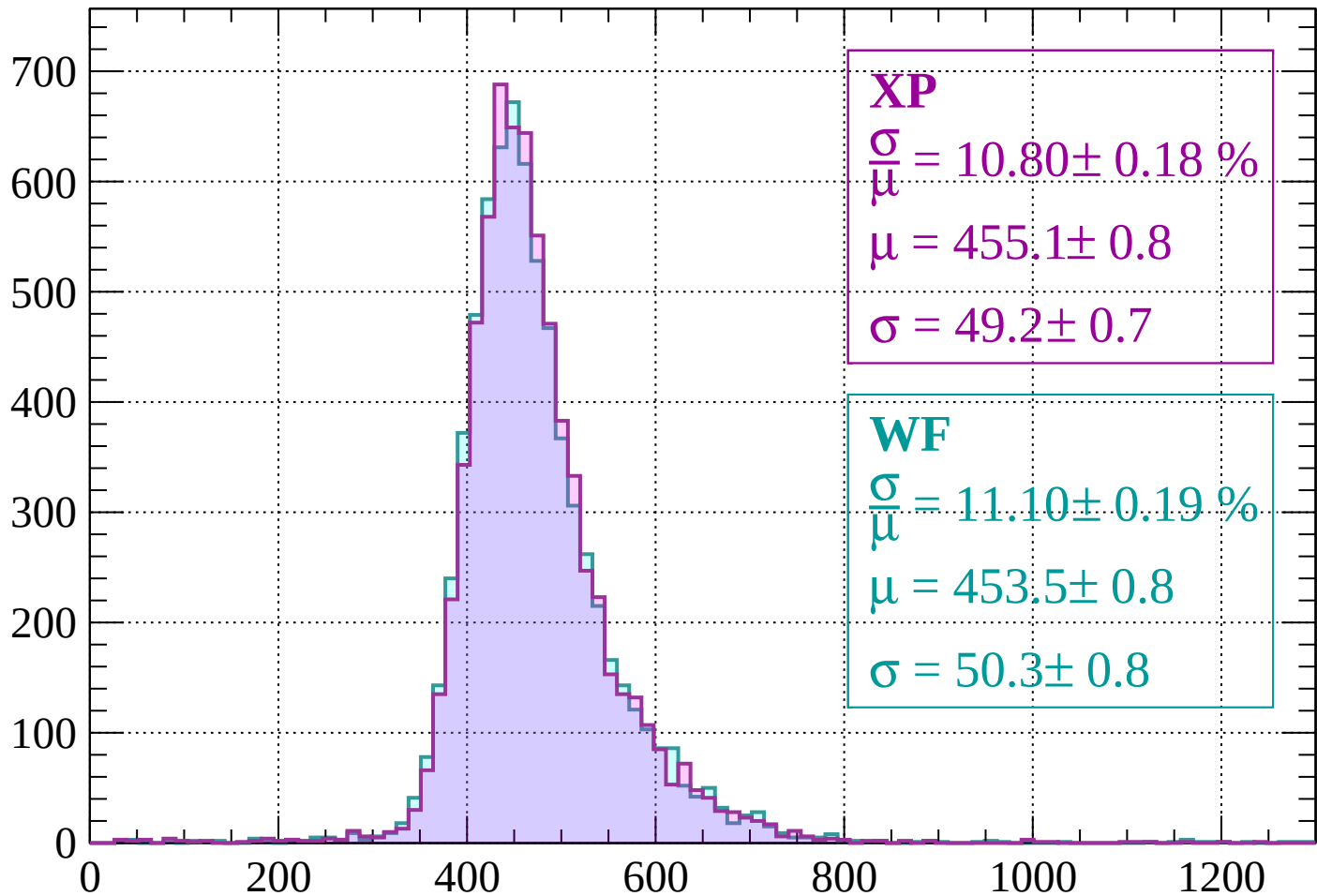
$$\mu = 454.3 \pm 0.8$$

$$\sigma = 49.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 10.80 \pm 0.18 \%$$

$$\mu = 455.1 \pm 0.8$$

$$\sigma = 49.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.10 \pm 0.19 \%$$

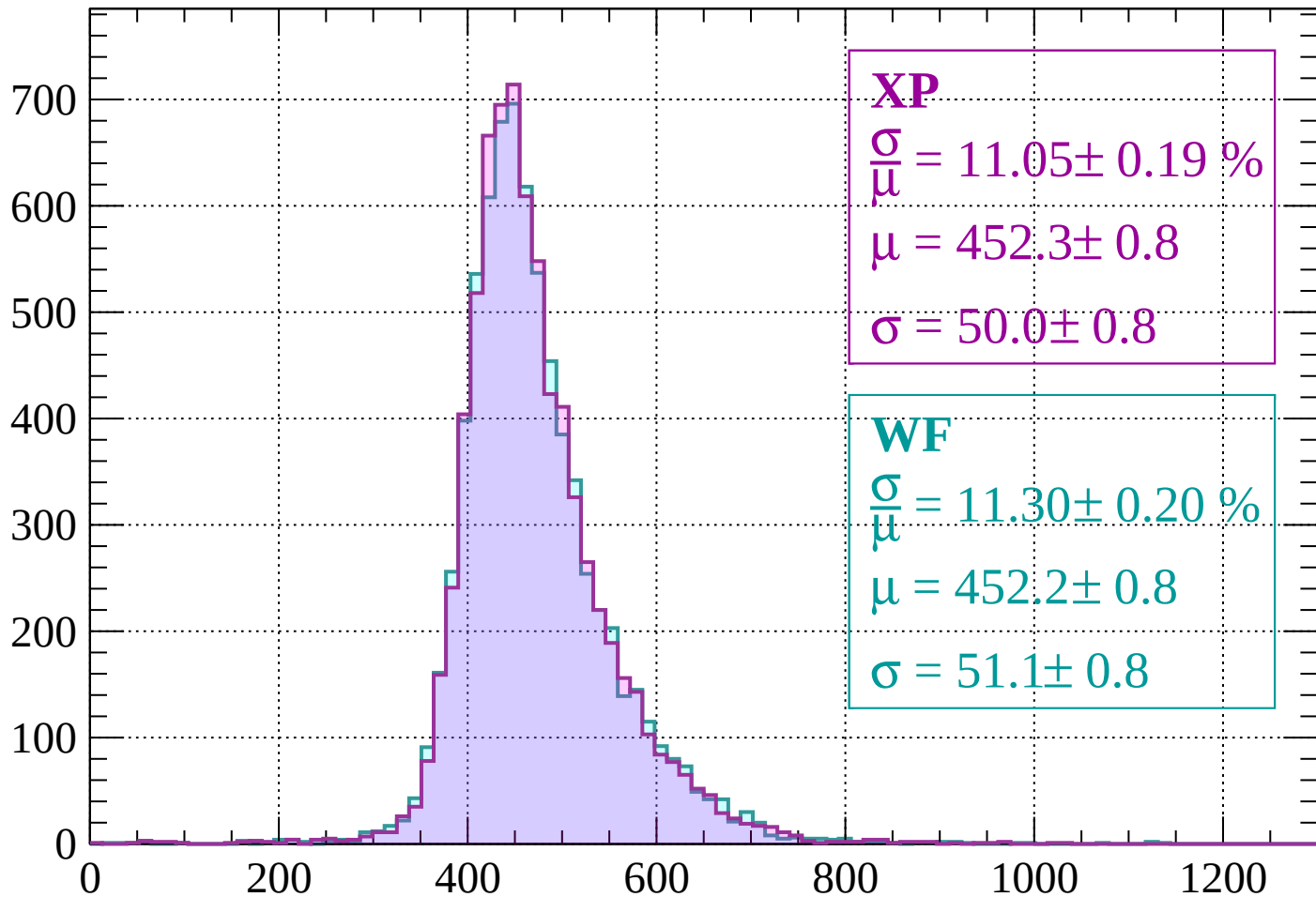
$$\mu = 453.5 \pm 0.8$$

$$\sigma = 50.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 11.05 \pm 0.19 \%$$

$$\mu = 452.3 \pm 0.8$$

$$\sigma = 50.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.30 \pm 0.20 \%$$

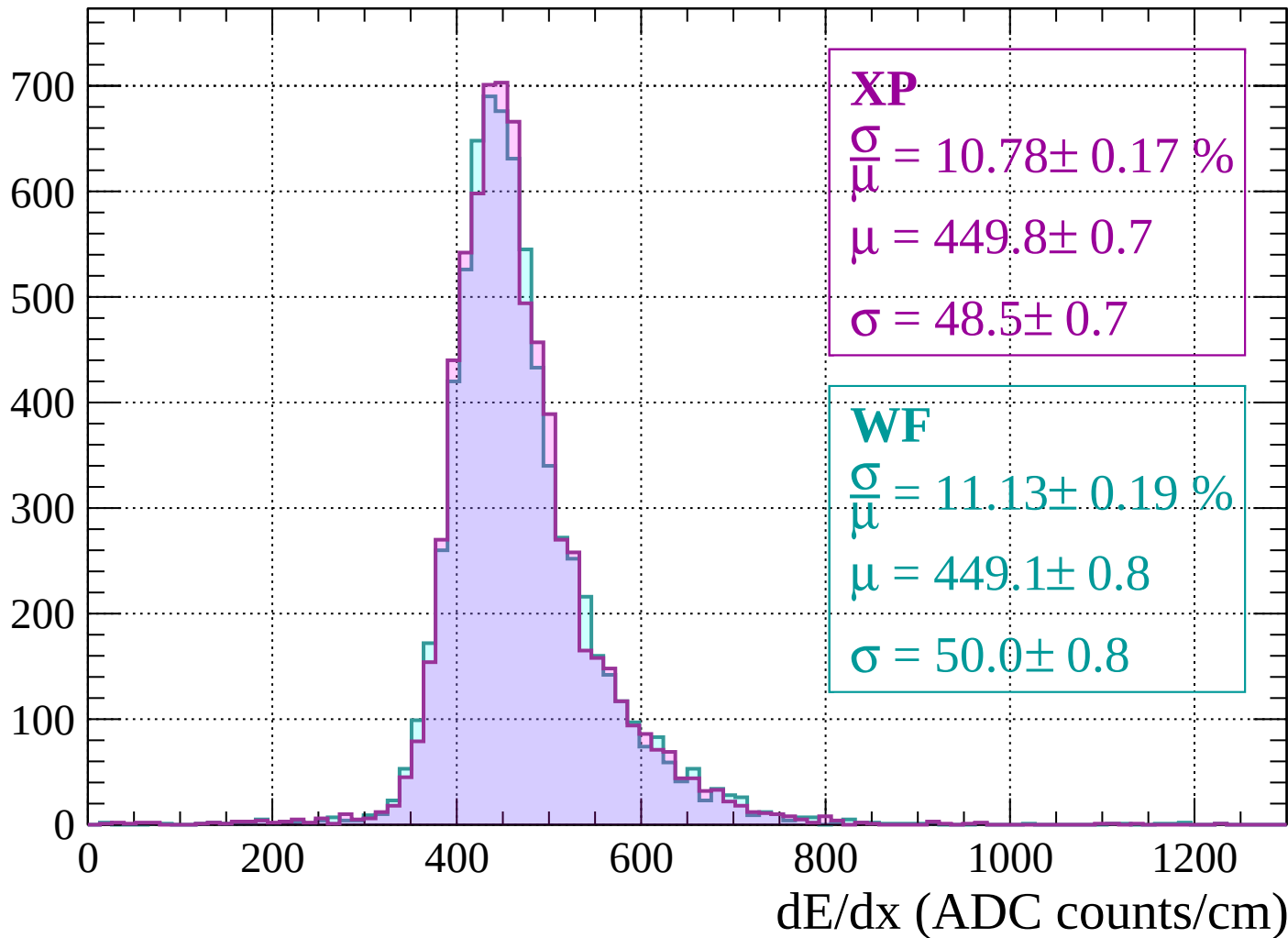
$$\mu = 452.2 \pm 0.8$$

$$\sigma = 51.1 \pm 0.8$$

dE/dx (ADC counts/cm)

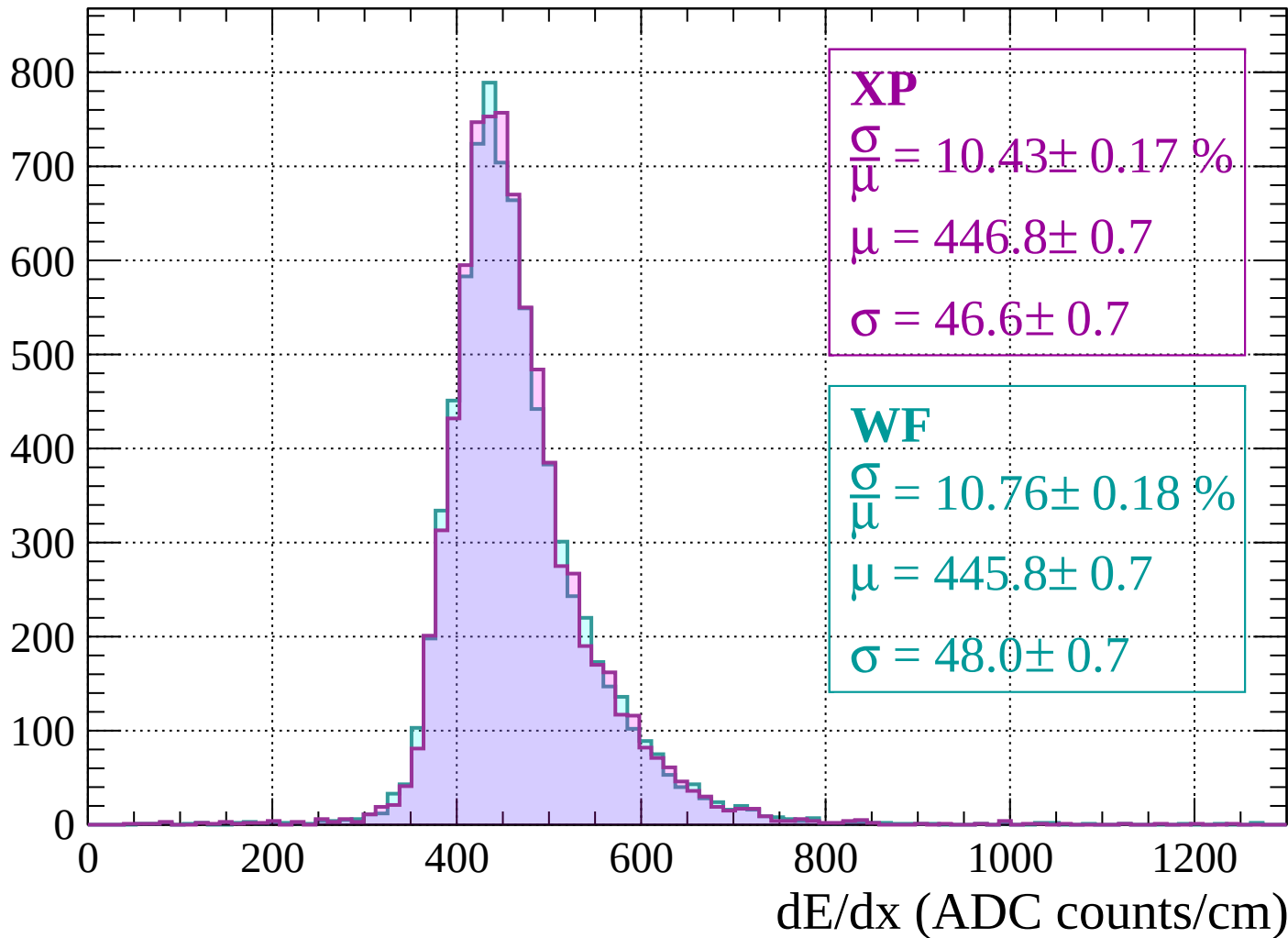
Energy loss | $-1120 < p < -1080$

Count



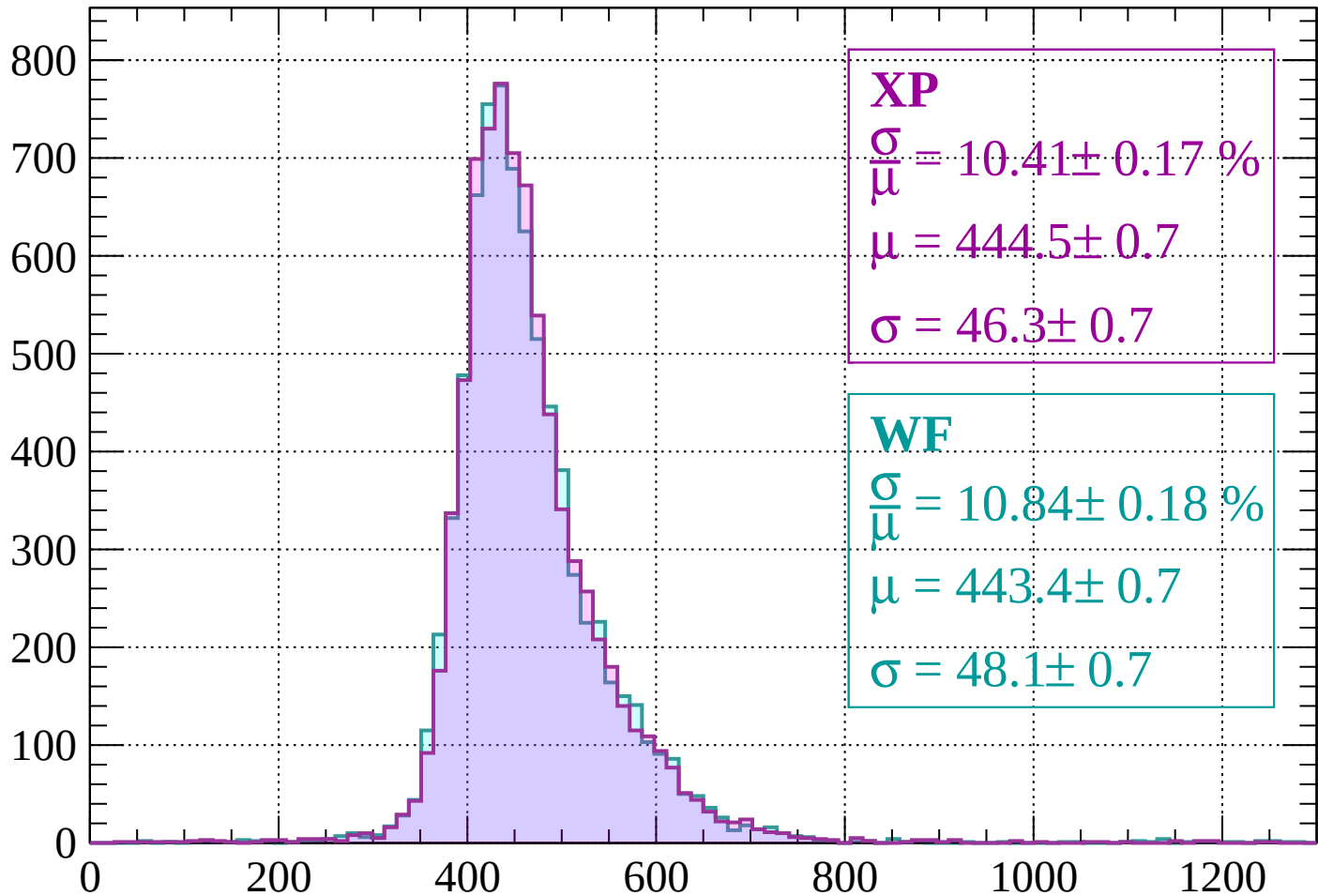
Energy loss | -1080 < p < -1040

Count



Energy loss | -1040 < p < -1000

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 10.41 \pm 0.17 \%$$

$$\mu = 444.5 \pm 0.7$$

$$\sigma = 46.3 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.84 \pm 0.18 \%$$

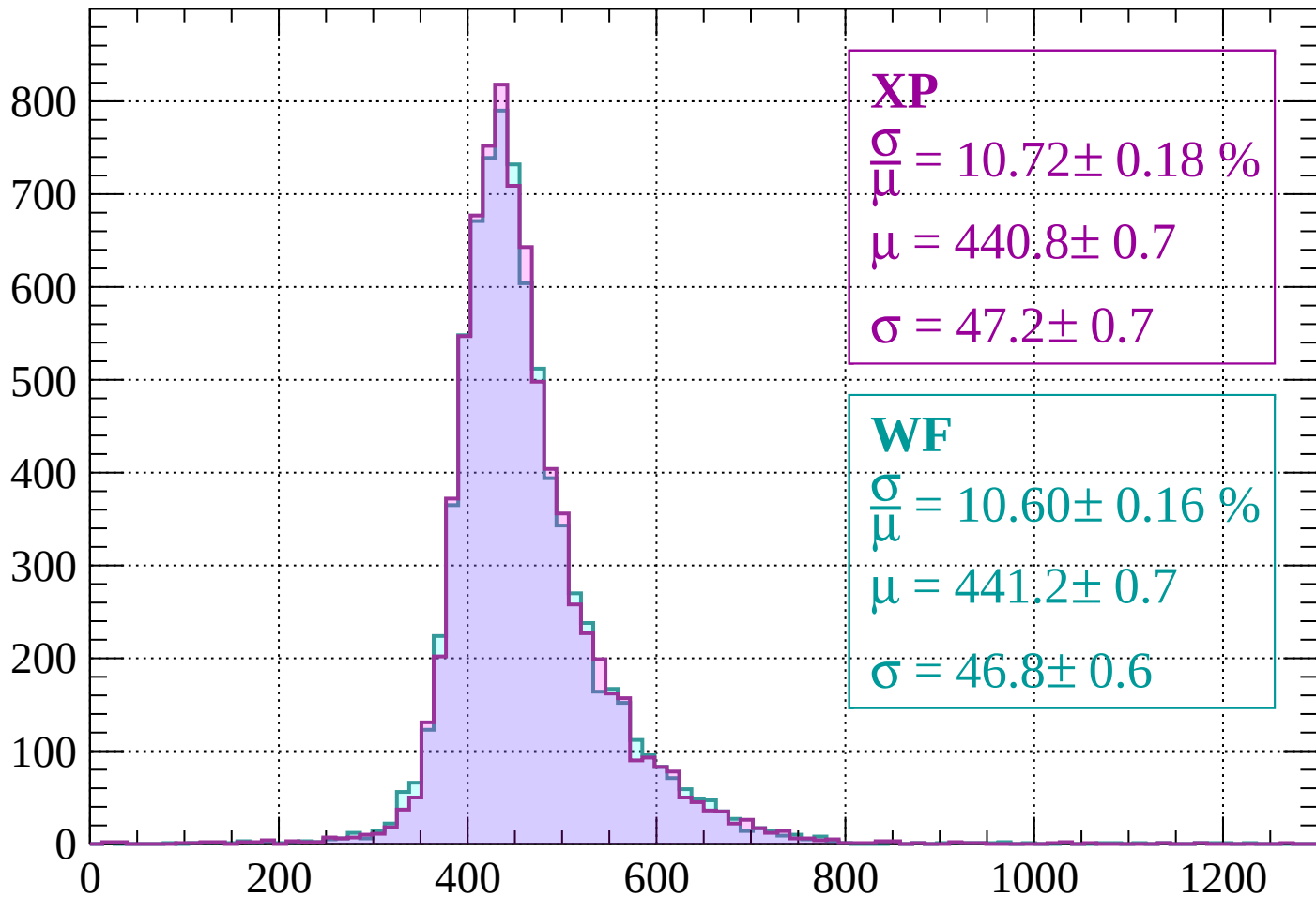
$$\mu = 443.4 \pm 0.7$$

$$\sigma = 48.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 10.72 \pm 0.18 \%$$

$$\mu = 440.8 \pm 0.7$$

$$\sigma = 47.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.60 \pm 0.16 \%$$

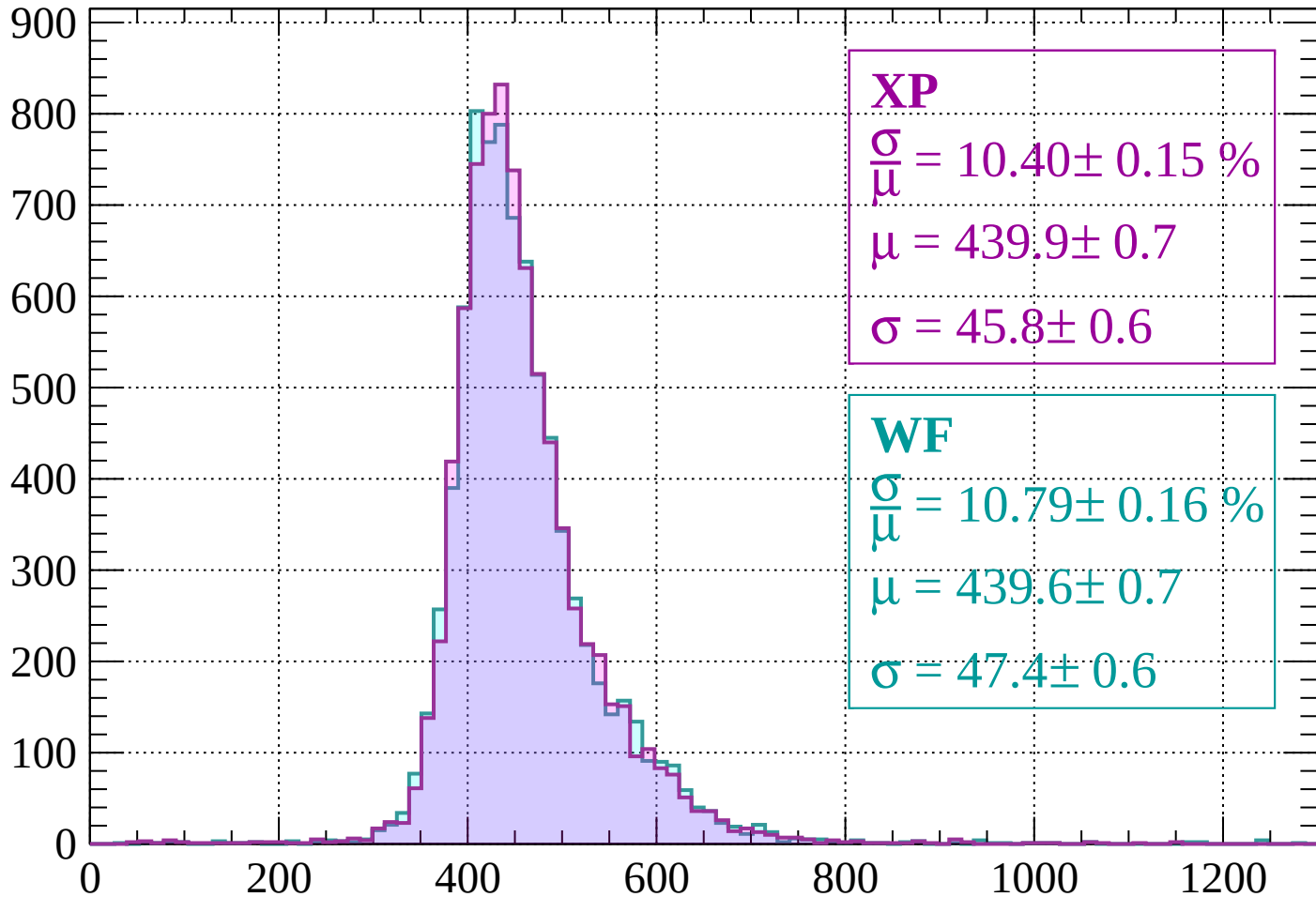
$$\mu = 441.2 \pm 0.7$$

$$\sigma = 46.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 10.40 \pm 0.15 \%$$

$$\mu = 439.9 \pm 0.7$$

$$\sigma = 45.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.79 \pm 0.16 \%$$

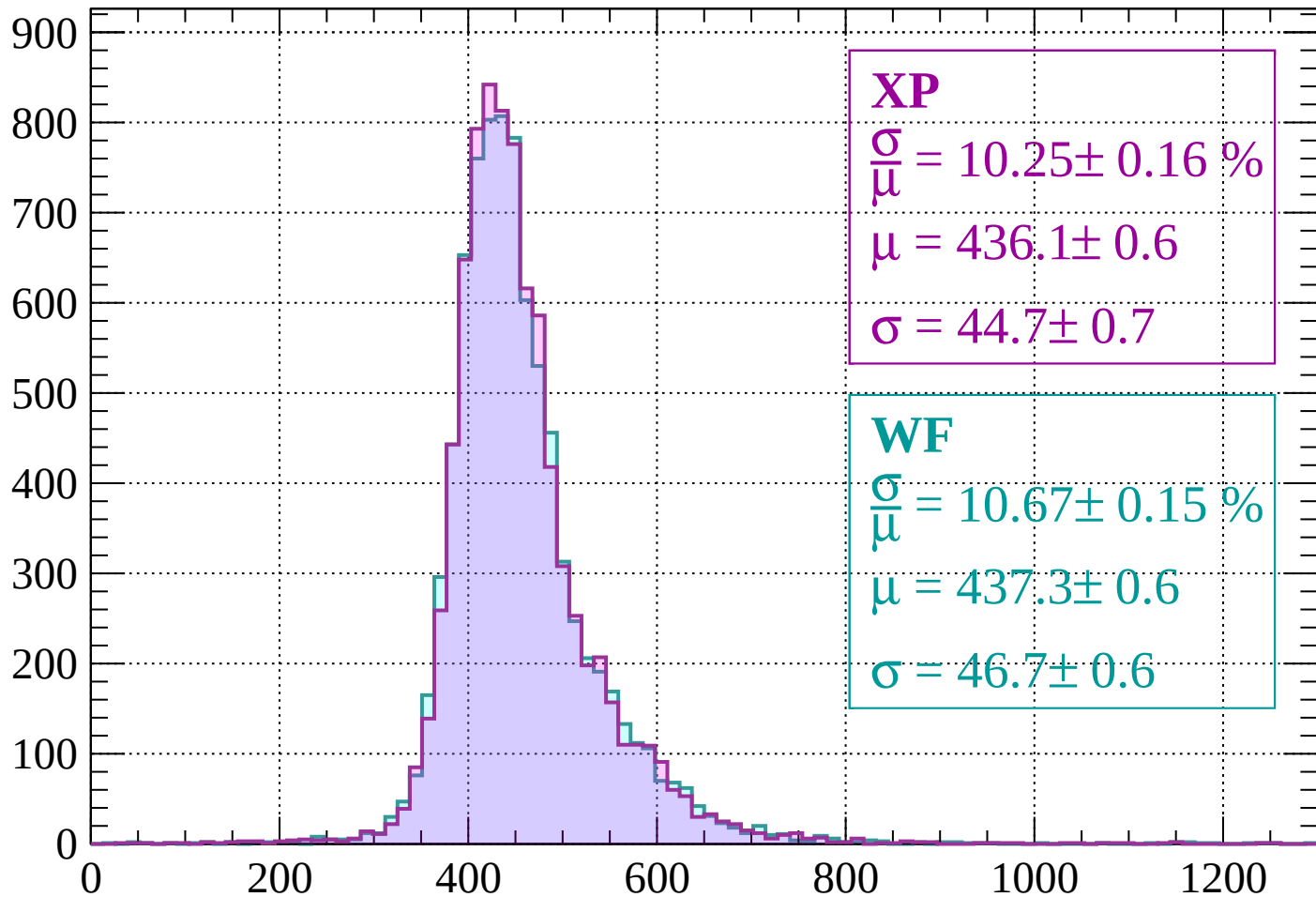
$$\mu = 439.6 \pm 0.7$$

$$\sigma = 47.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 10.25 \pm 0.16 \%$$

$$\mu = 436.1 \pm 0.6$$

$$\sigma = 44.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.15 \%$$

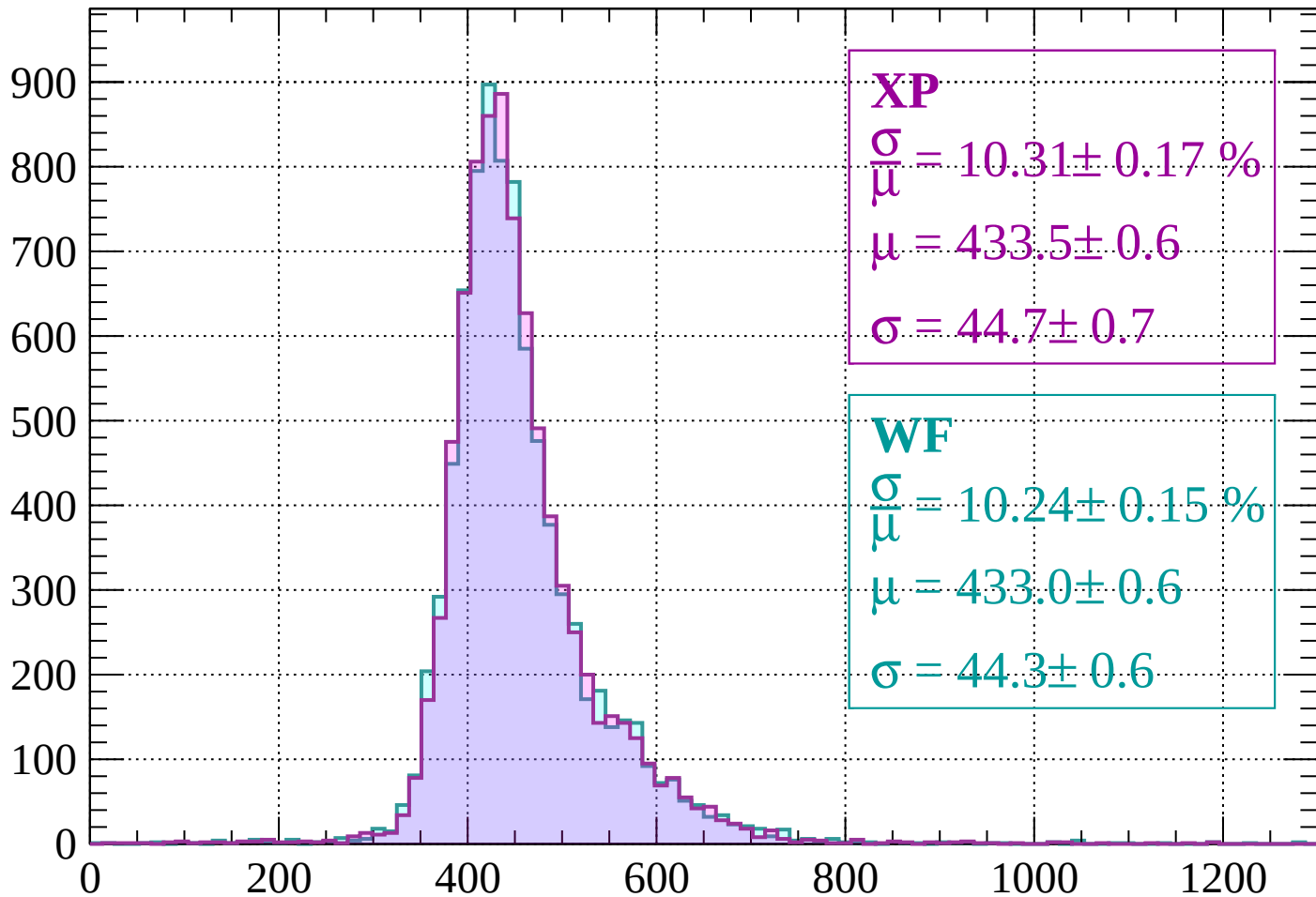
$$\mu = 437.3 \pm 0.6$$

$$\sigma = 46.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 10.31 \pm 0.17 \%$$

$$\mu = 433.5 \pm 0.6$$

$$\sigma = 44.7 \pm 0.7$$

WF

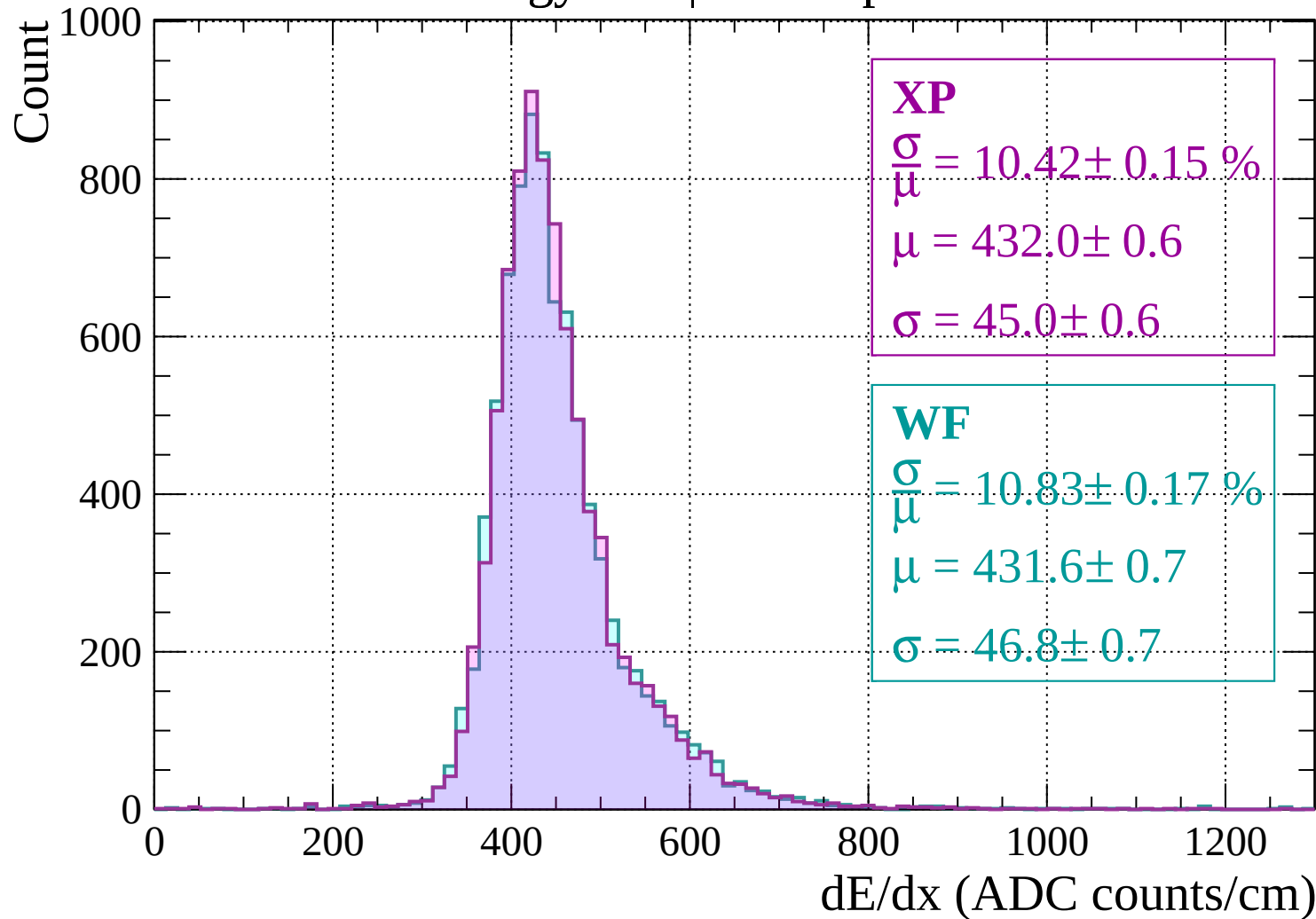
$$\frac{\sigma}{\mu} = 10.24 \pm 0.15 \%$$

$$\mu = 433.0 \pm 0.6$$

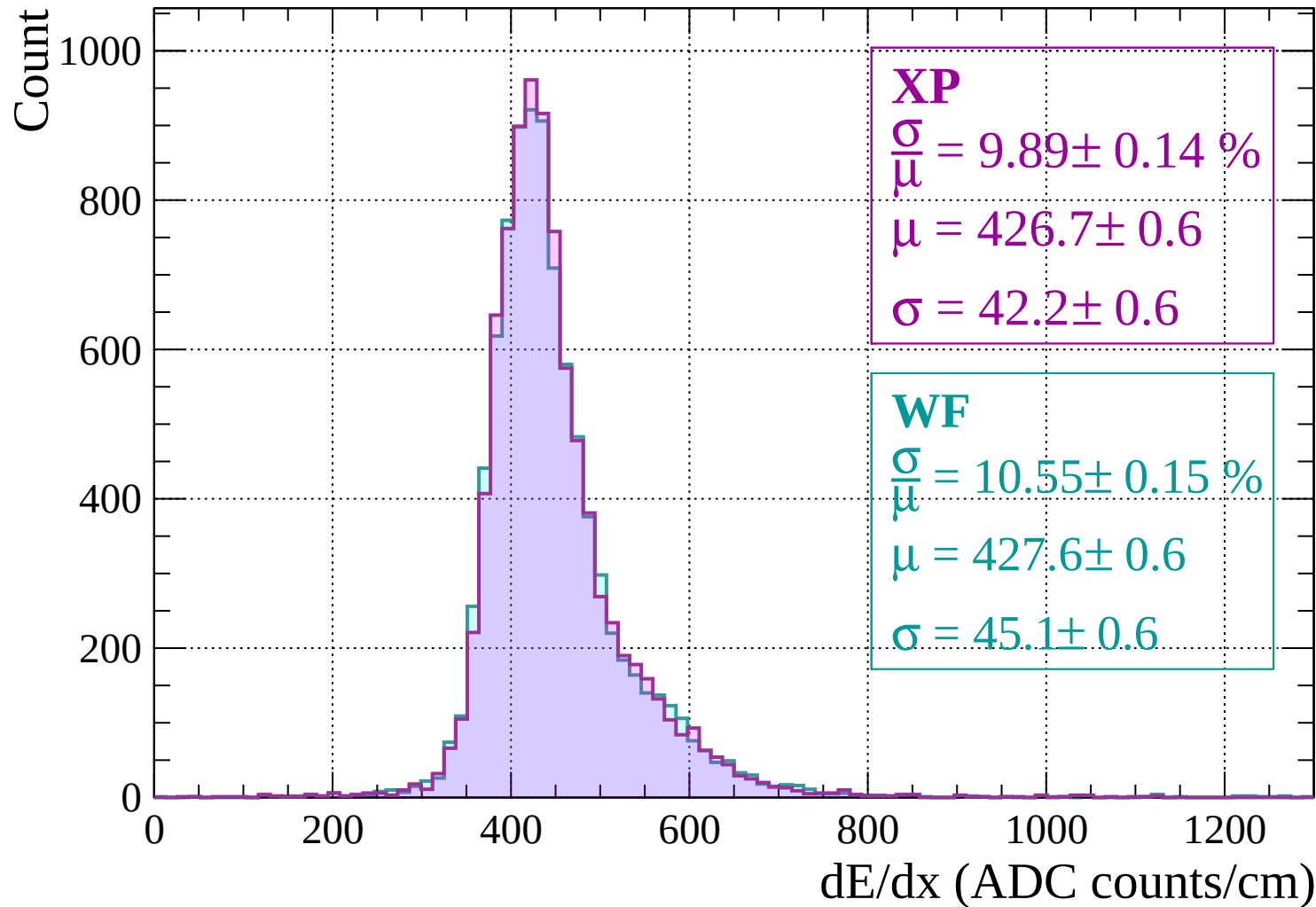
$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)

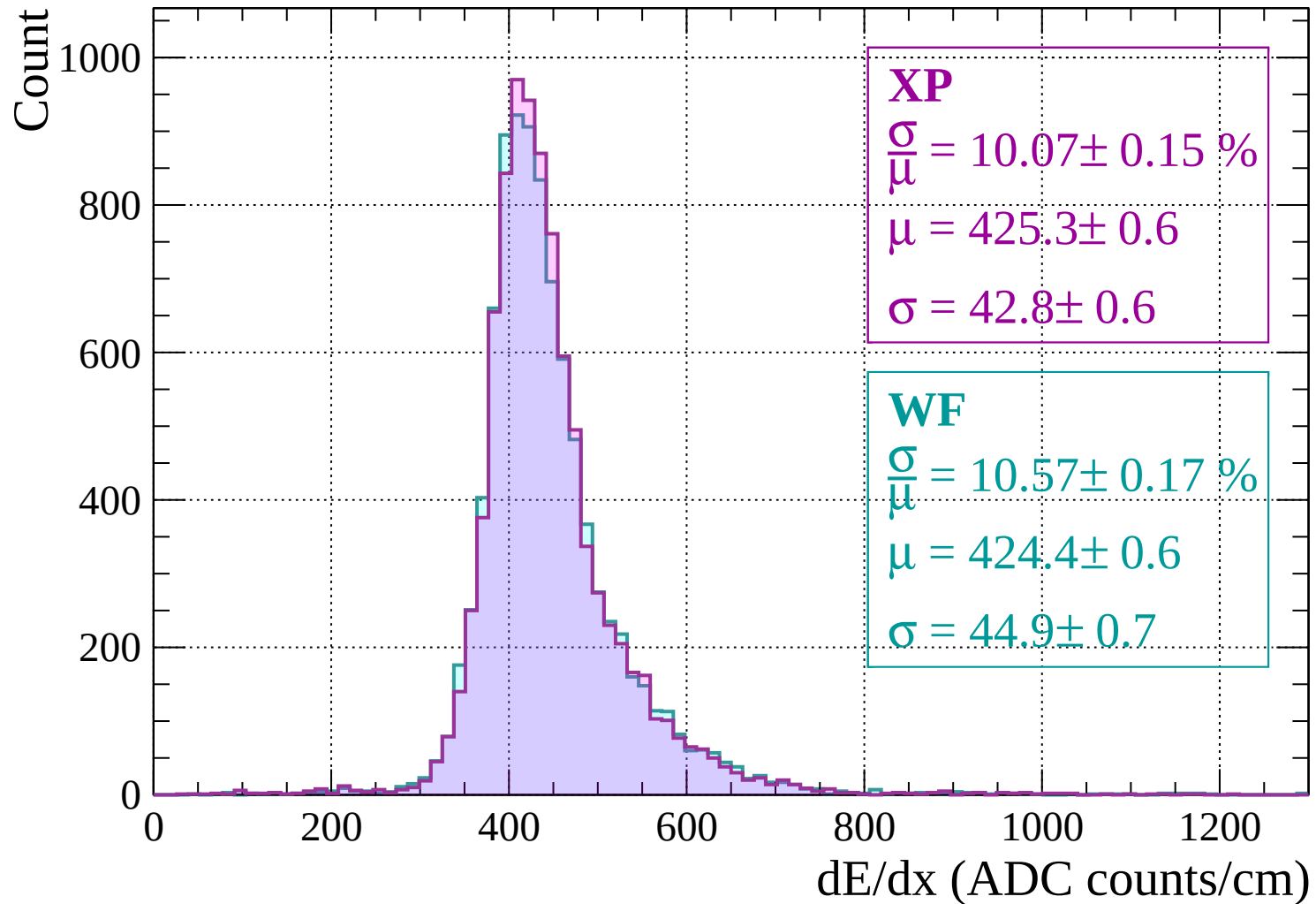
Energy loss | $-840 < p < -800$



Energy loss | $-800 < p < -760$



Energy loss | $-760 < p < -720$



Energy loss | $-720 < p < -680$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.15 \pm 0.15 \%$$

$$\mu = 421.7 \pm 0.6$$

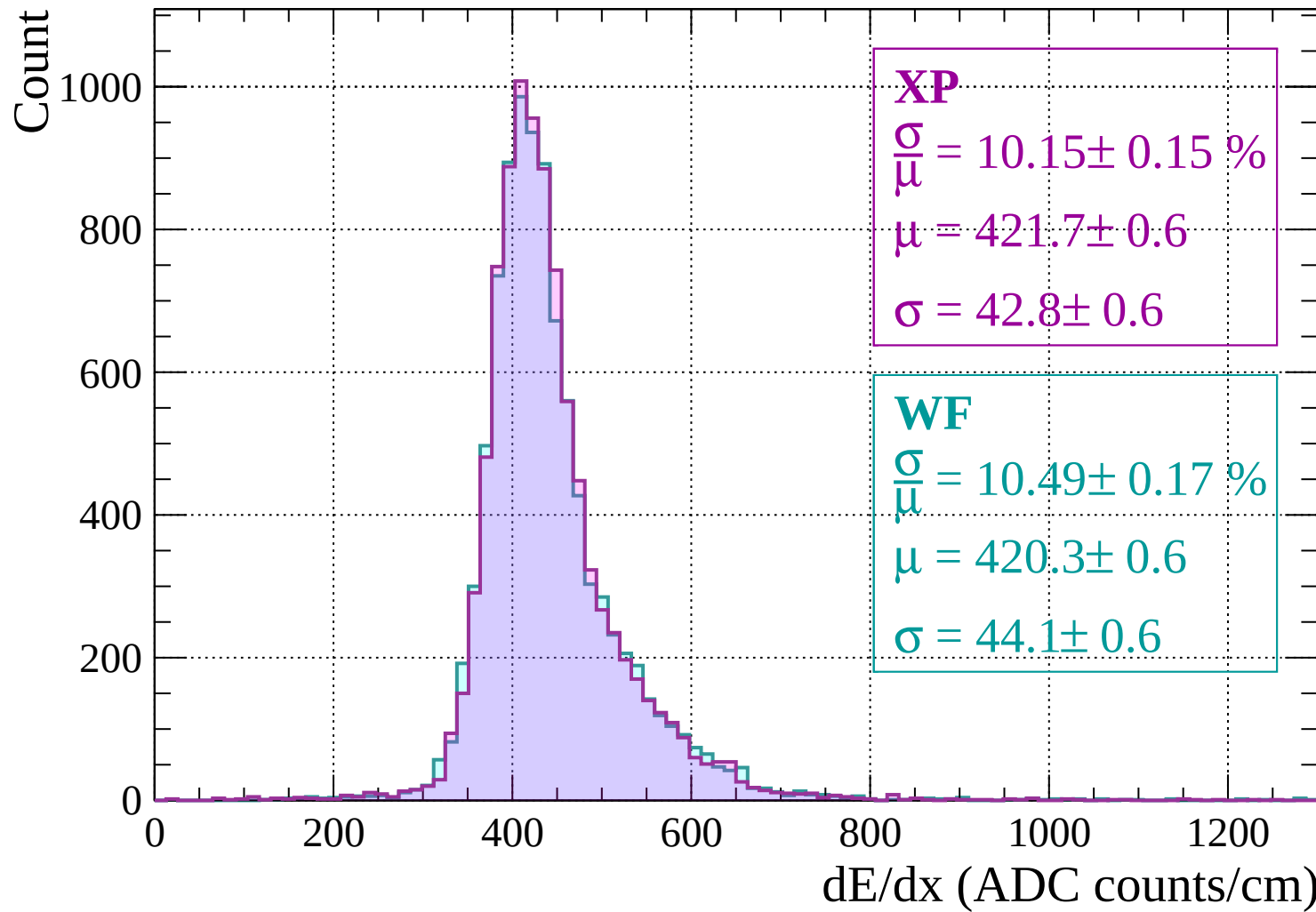
$$\sigma = 42.8 \pm 0.6$$

WF

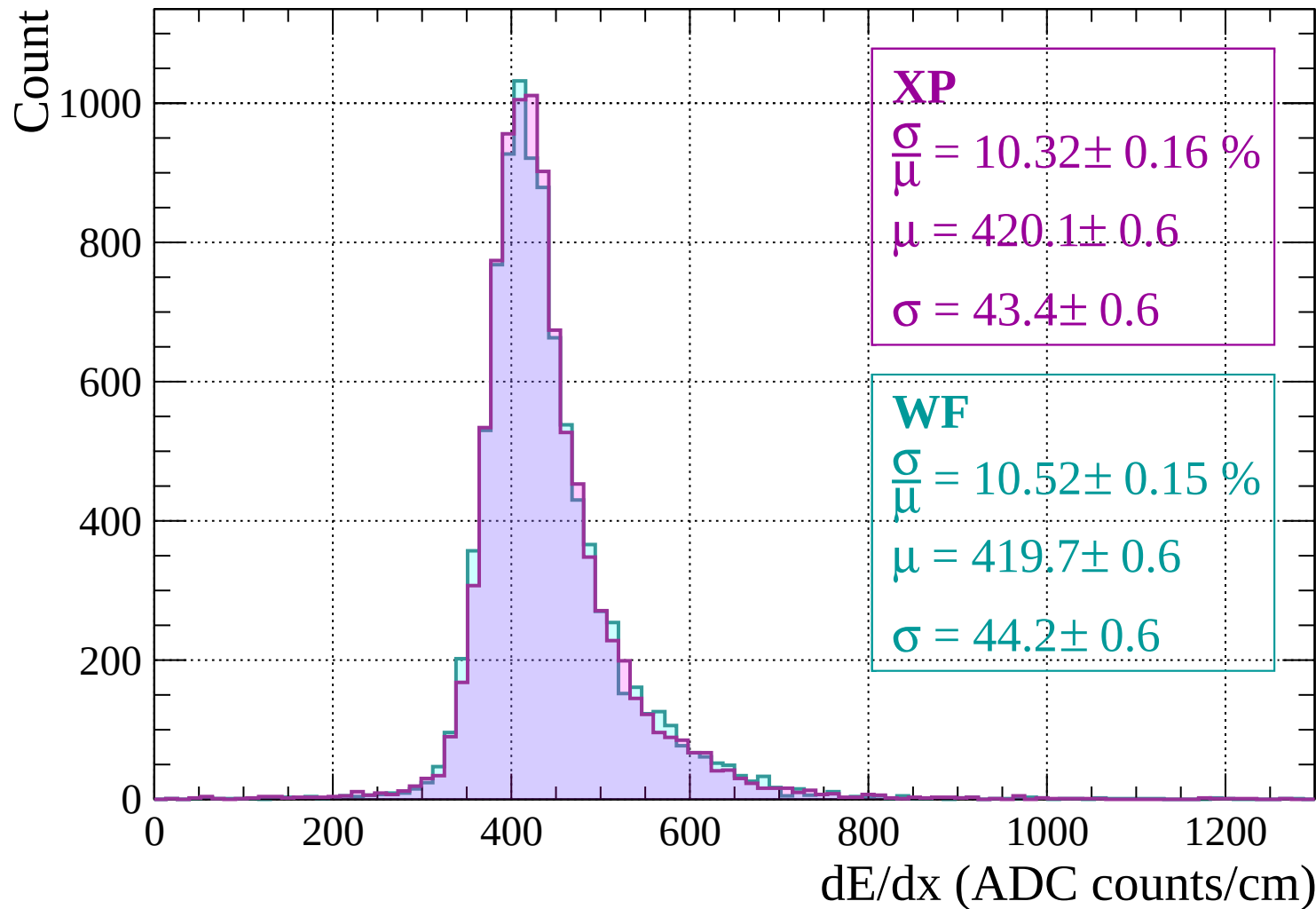
$$\frac{\sigma}{\mu} = 10.49 \pm 0.17 \%$$

$$\mu = 420.3 \pm 0.6$$

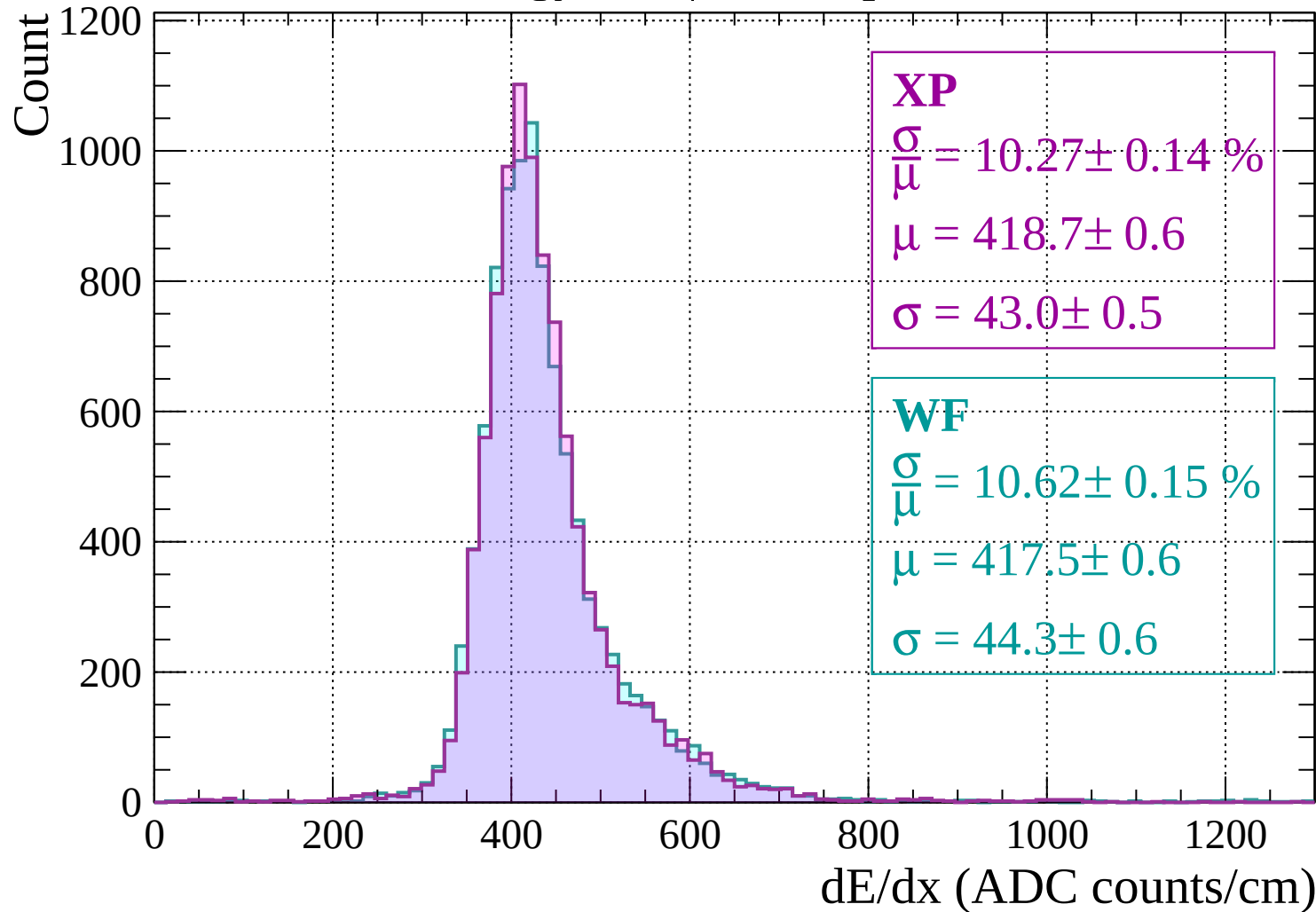
$$\sigma = 44.1 \pm 0.6$$



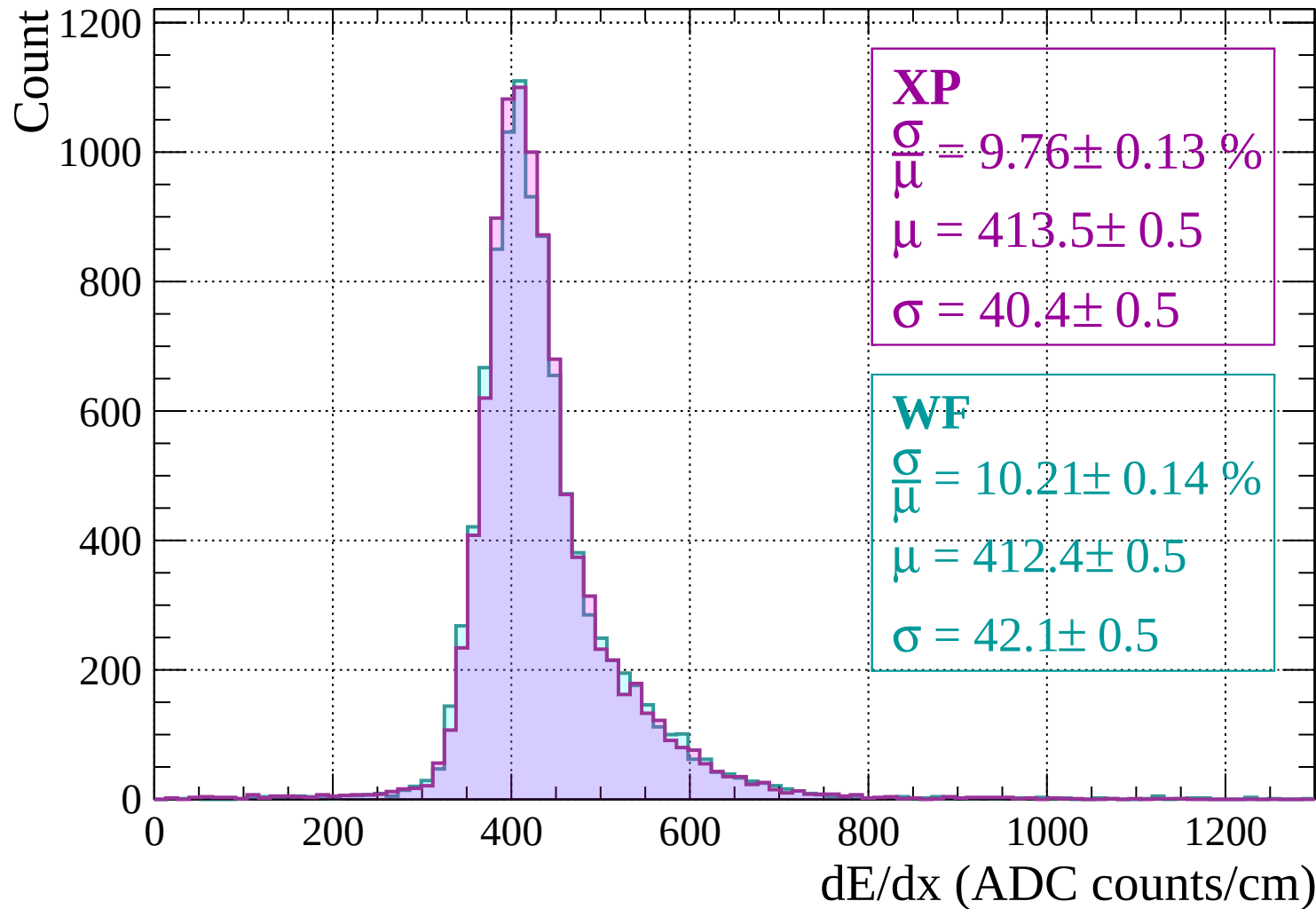
Energy loss | $-680 < p < -640$



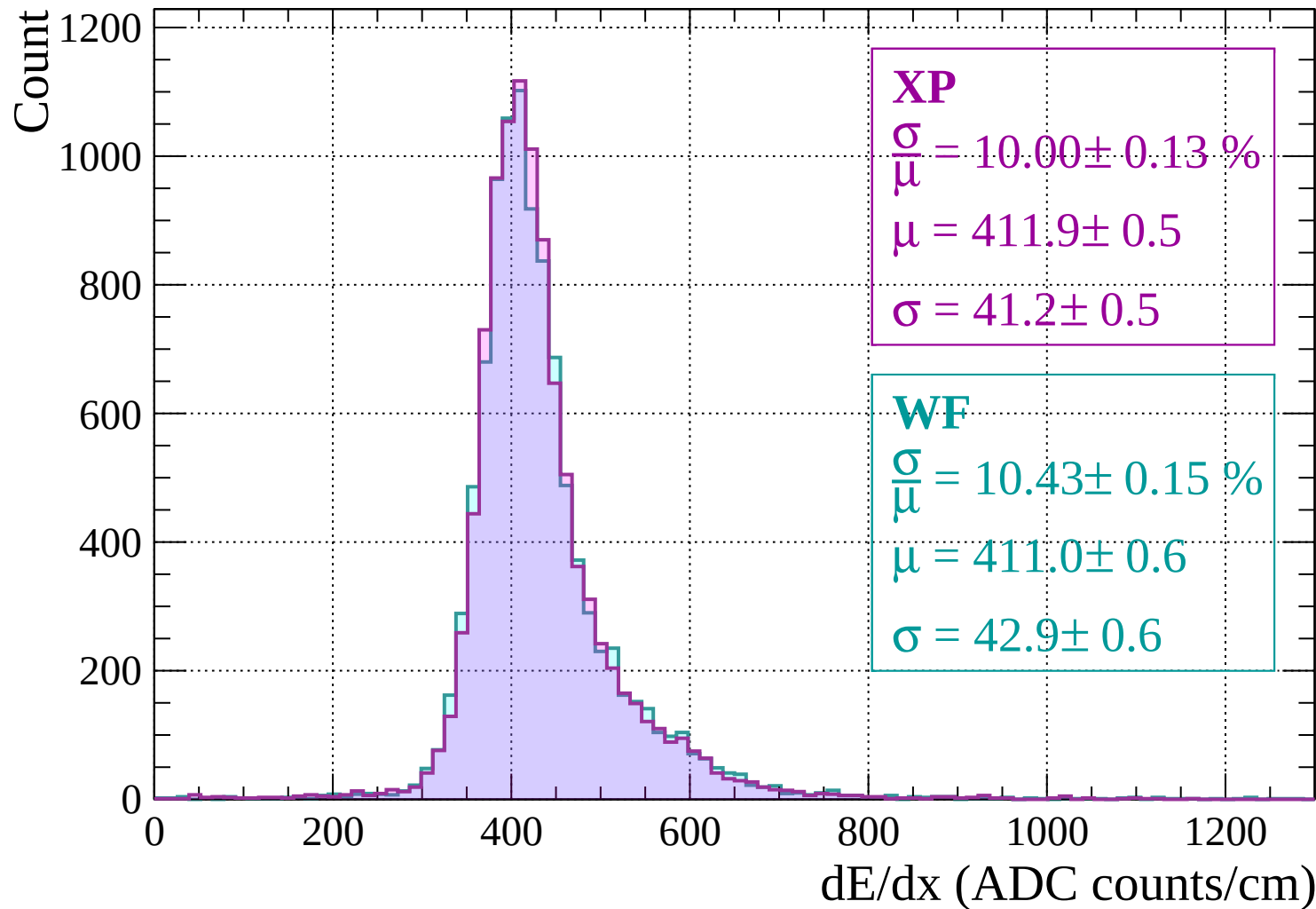
Energy loss | $-640 < p < -600$



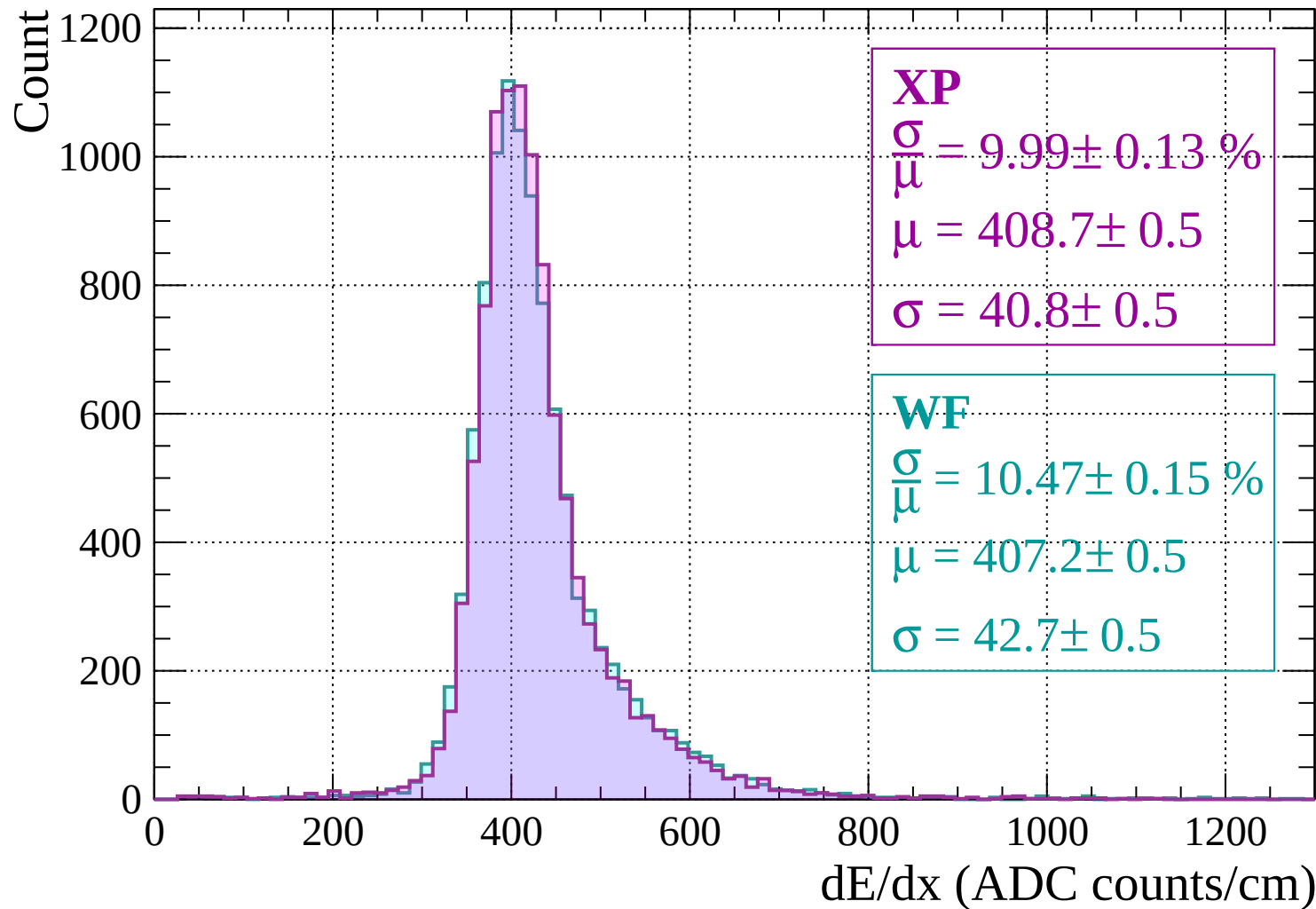
Energy loss | $-600 < p < -560$



Energy loss | $-560 < p < -520$

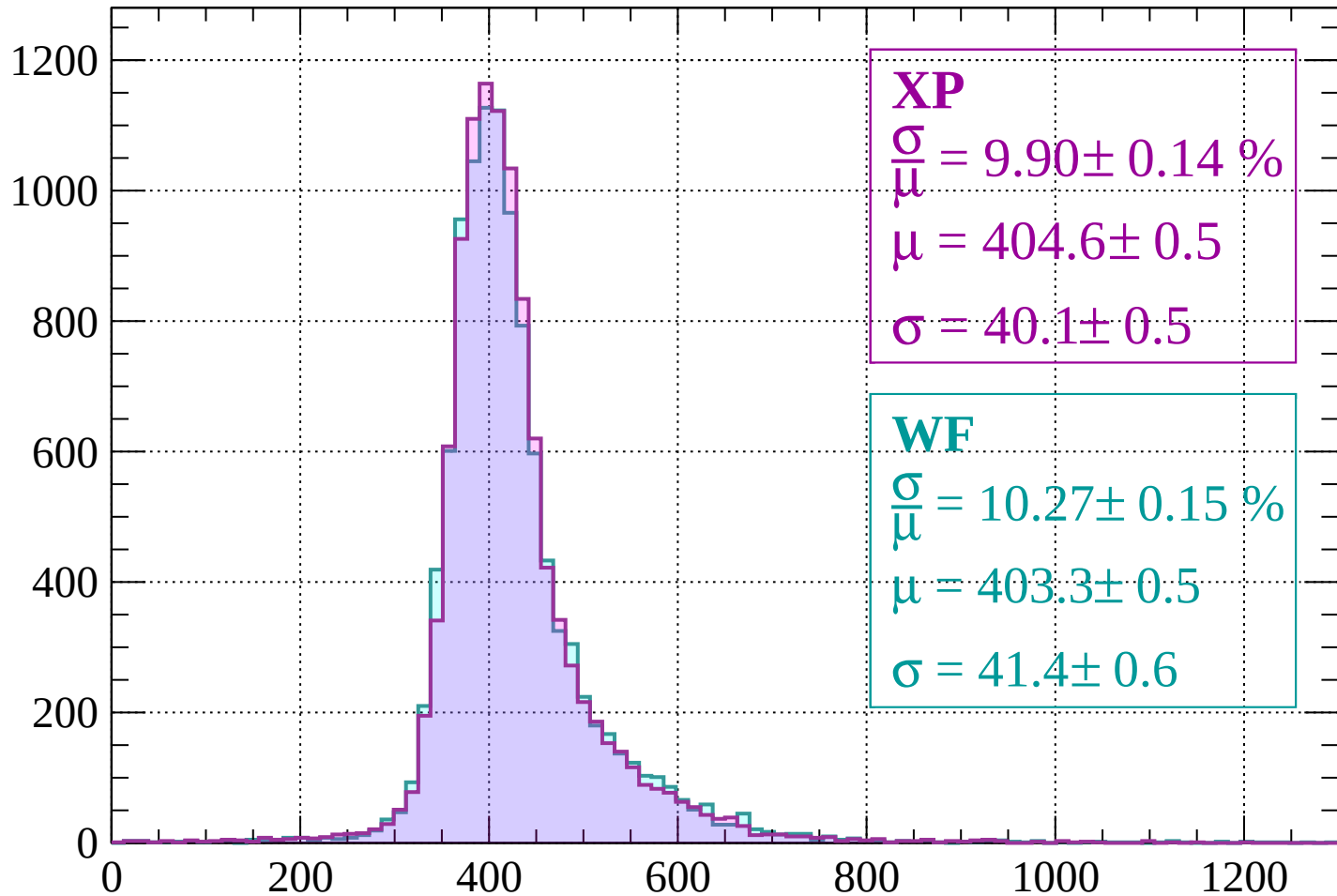


Energy loss | $-520 < p < -480$



Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 9.90 \pm 0.14 \%$$

$$\mu = 404.6 \pm 0.5$$

$$\sigma = 40.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.27 \pm 0.15 \%$$

$$\mu = 403.3 \pm 0.5$$

$$\sigma = 41.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count

1200

1000

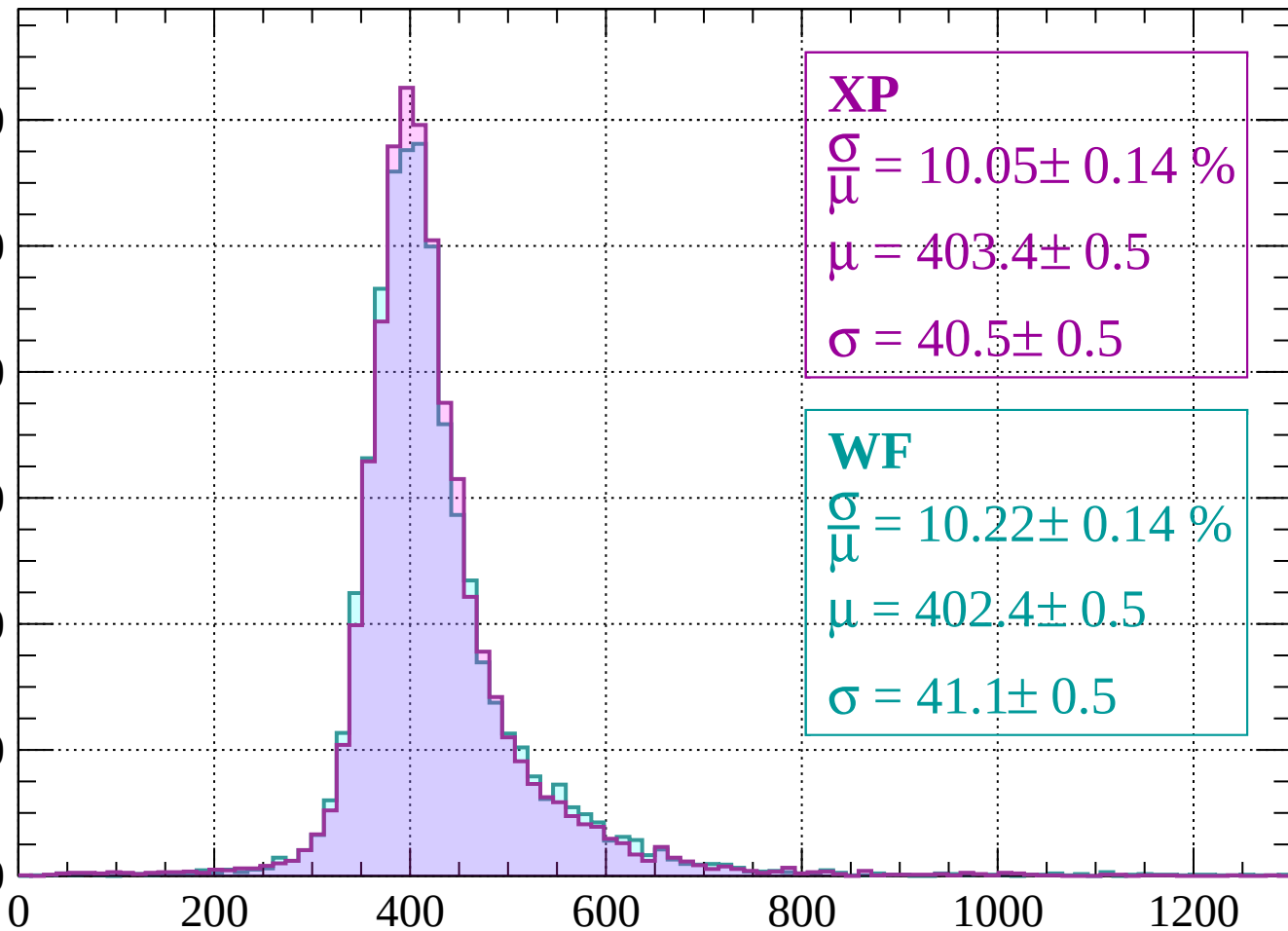
800

600

400

200

0



XP

$\frac{\sigma}{\mu} = 10.05 \pm 0.14 \%$

$\mu = 403.4 \pm 0.5$

$\sigma = 40.5 \pm 0.5$

WF

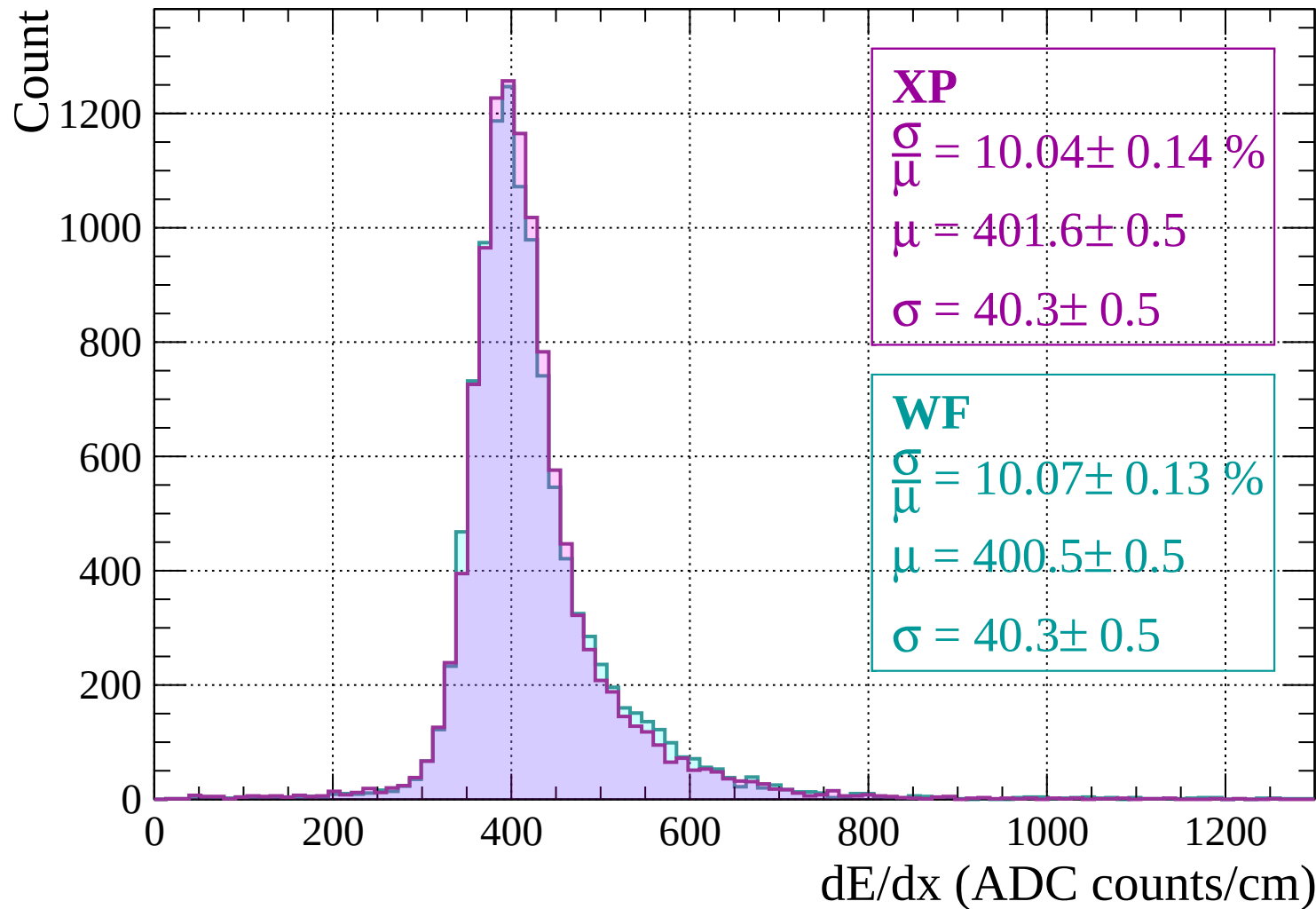
$\frac{\sigma}{\mu} = 10.22 \pm 0.14 \%$

$\mu = 402.4 \pm 0.5$

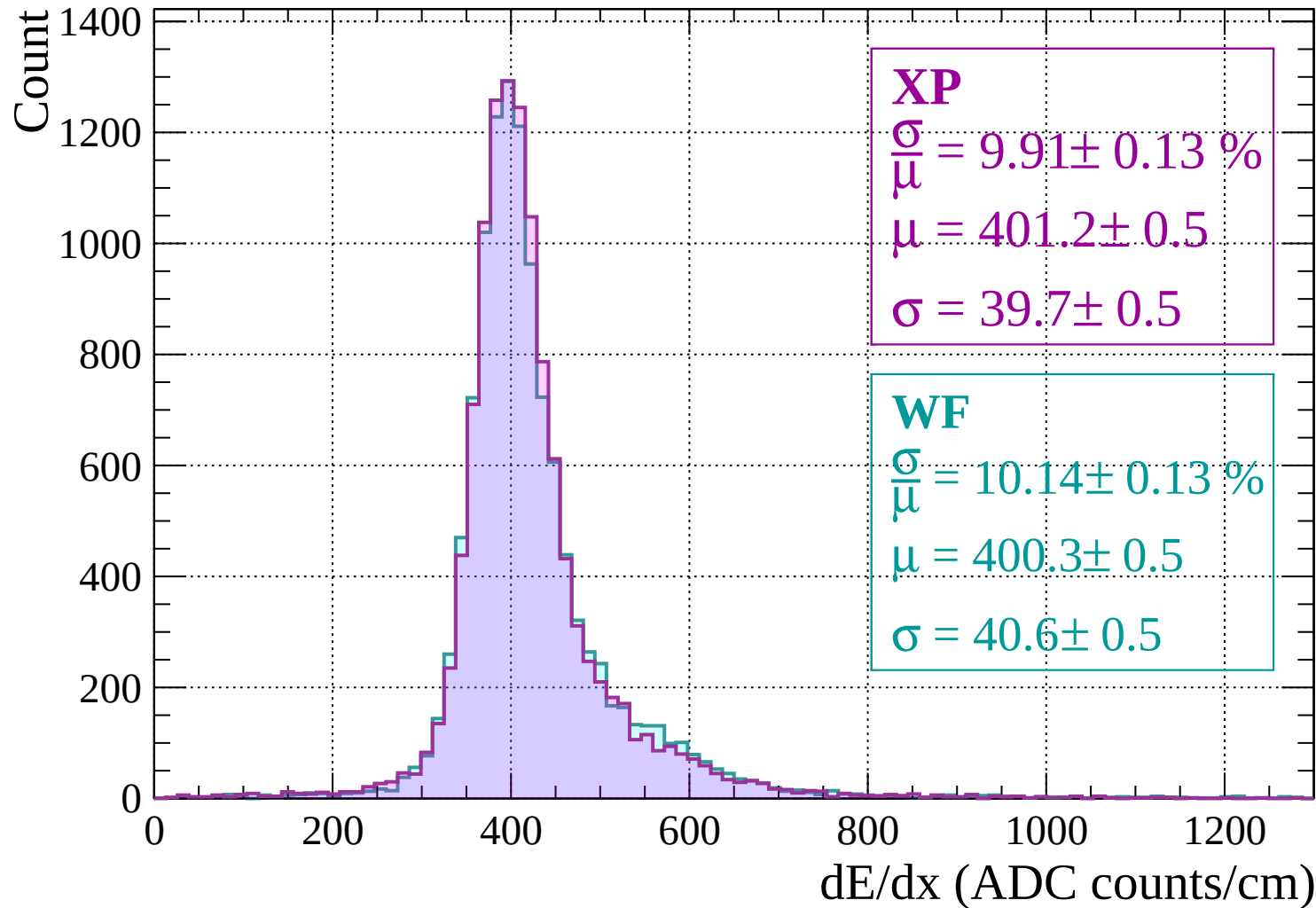
$\sigma = 41.1 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

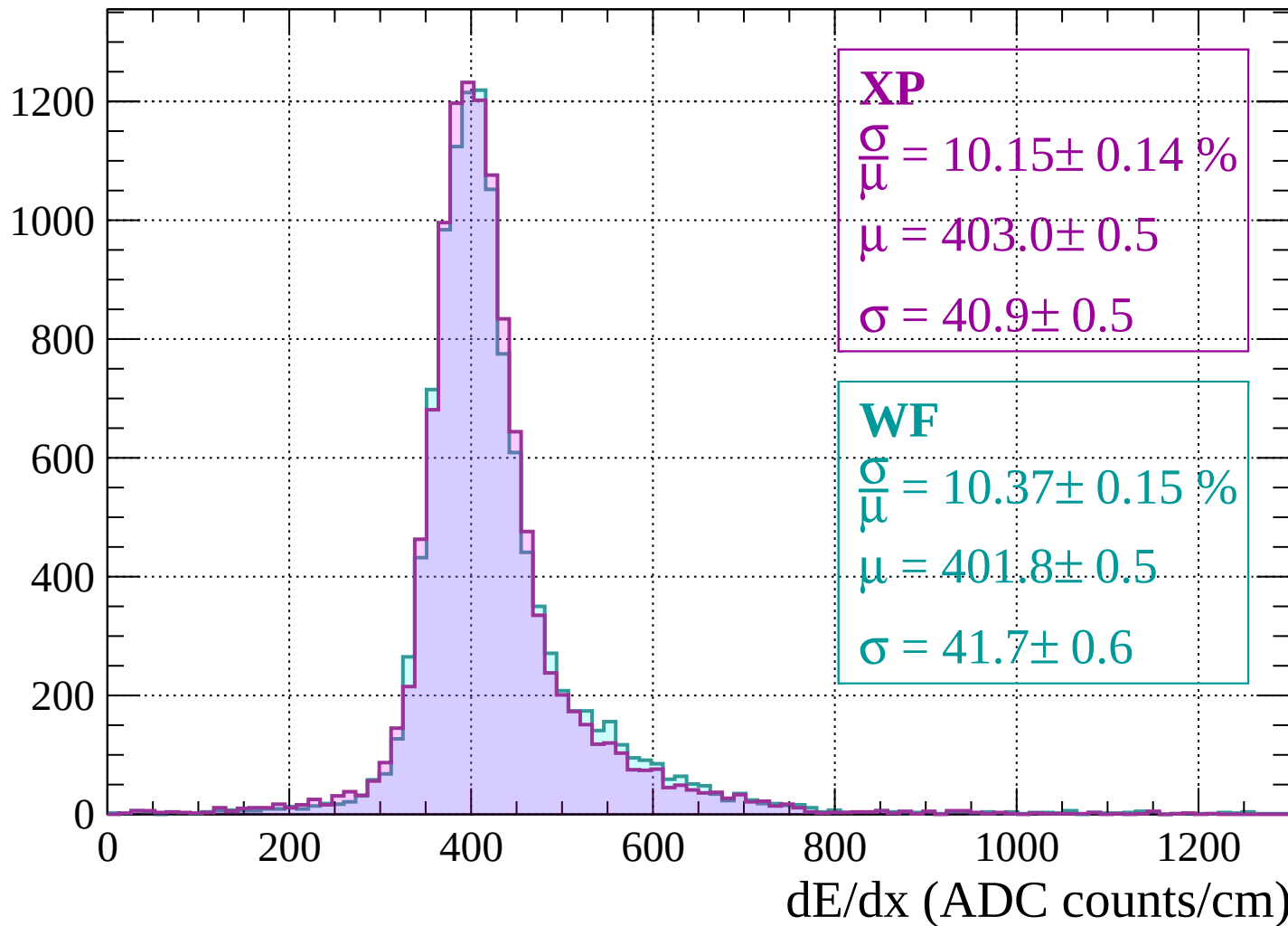


Energy loss | $-360 < p < -320$

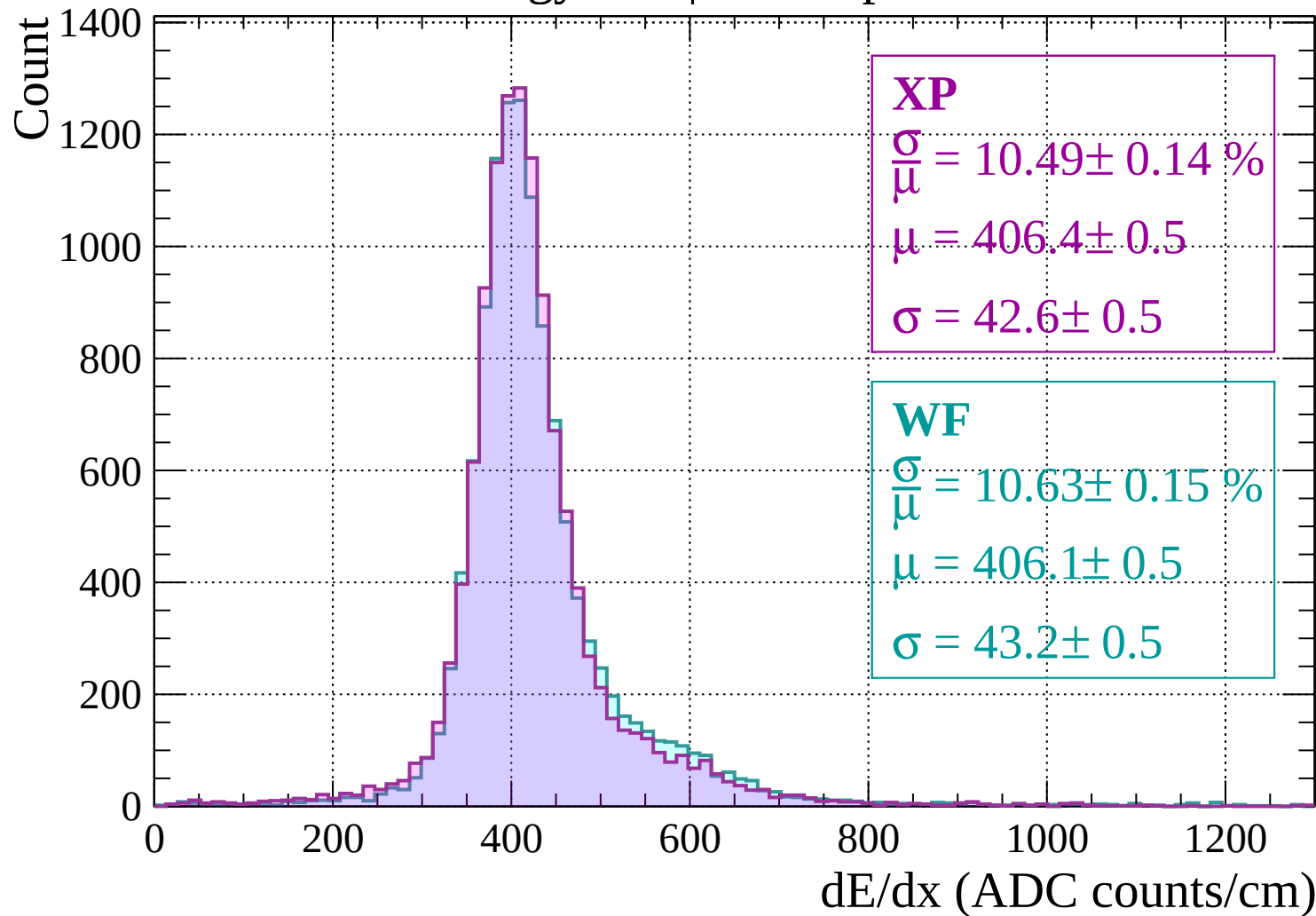


Energy loss | $-320 < p < -280$

Count

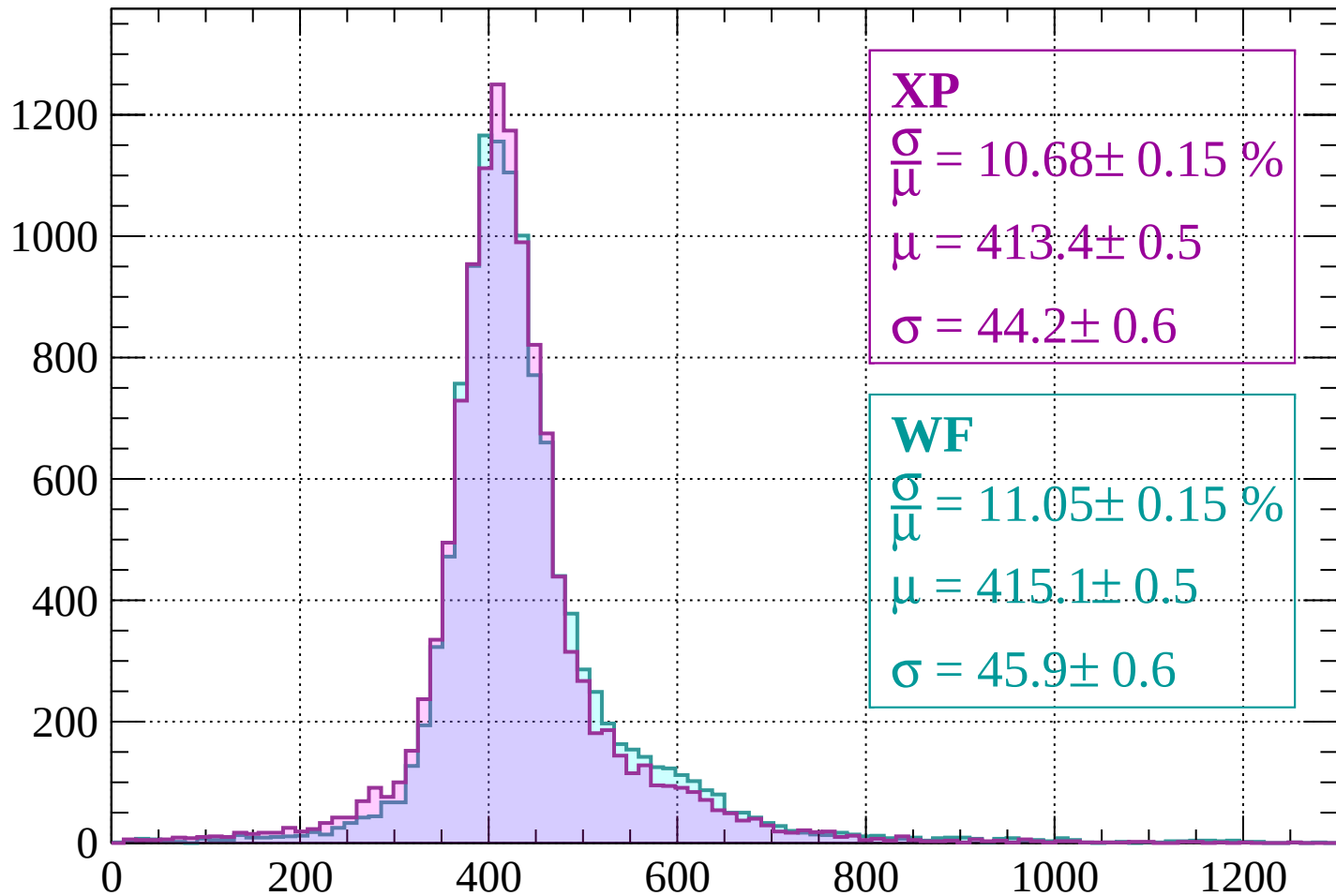


Energy loss | $-280 < p < -240$



Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 10.68 \pm 0.15 \%$$

$$\mu = 413.4 \pm 0.5$$

$$\sigma = 44.2 \pm 0.6$$

WF

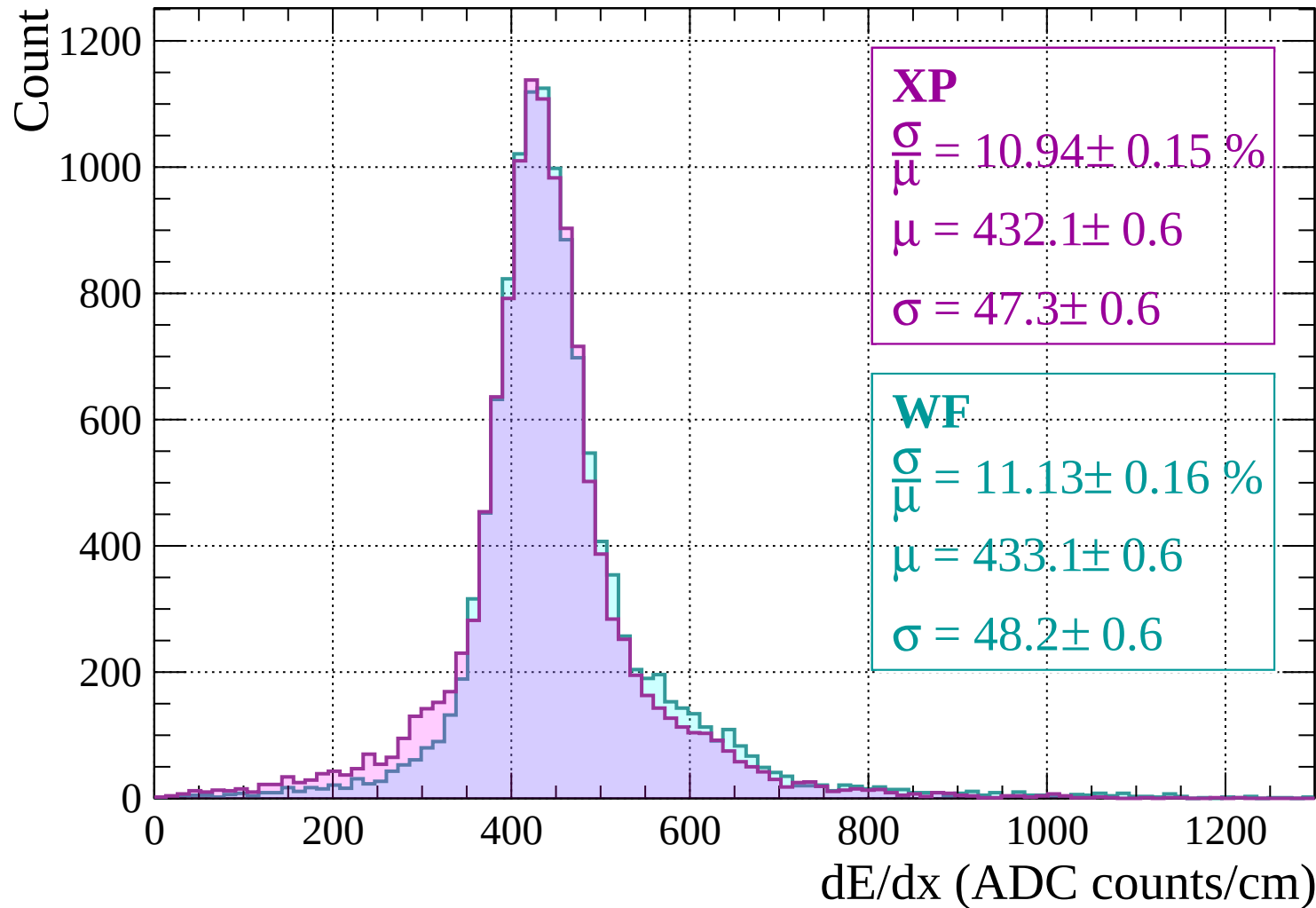
$$\frac{\sigma}{\mu} = 11.05 \pm 0.15 \%$$

$$\mu = 415.1 \pm 0.5$$

$$\sigma = 45.9 \pm 0.6$$

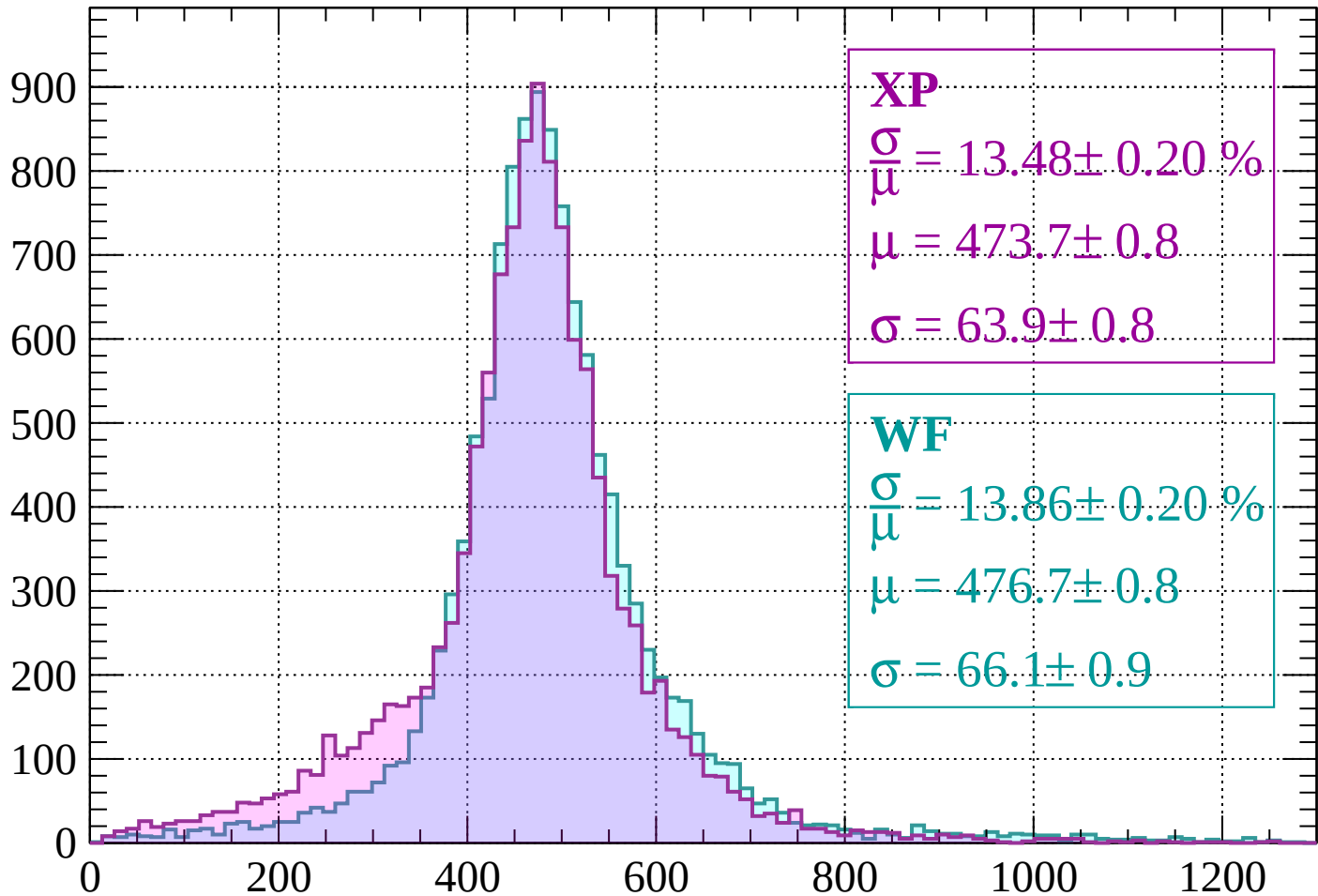
dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$



Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 13.48 \pm 0.20 \%$$

$$\mu = 473.7 \pm 0.8$$

$$\sigma = 63.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 13.86 \pm 0.20 \%$$

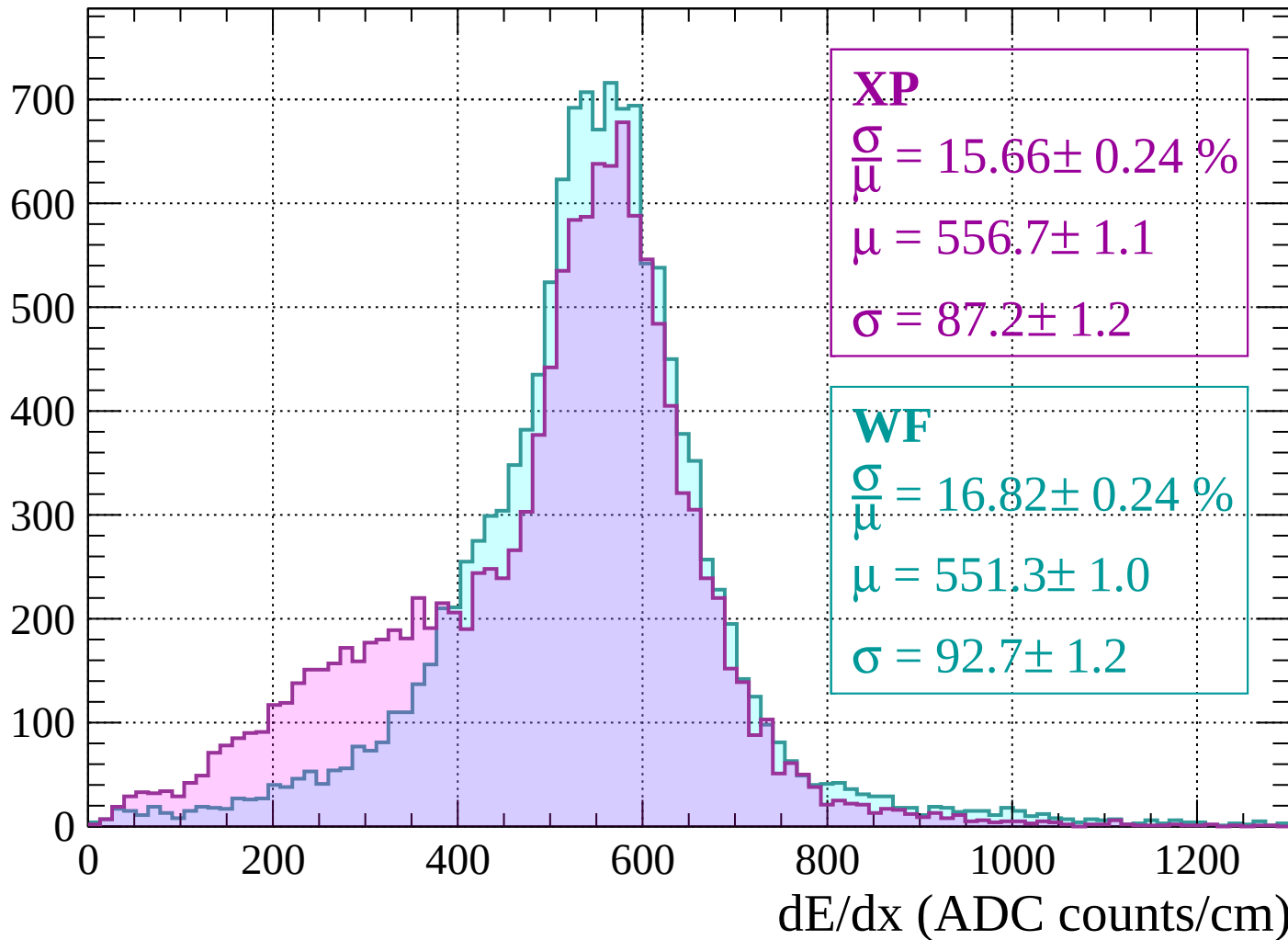
$$\mu = 476.7 \pm 0.8$$

$$\sigma = 66.1 \pm 0.9$$

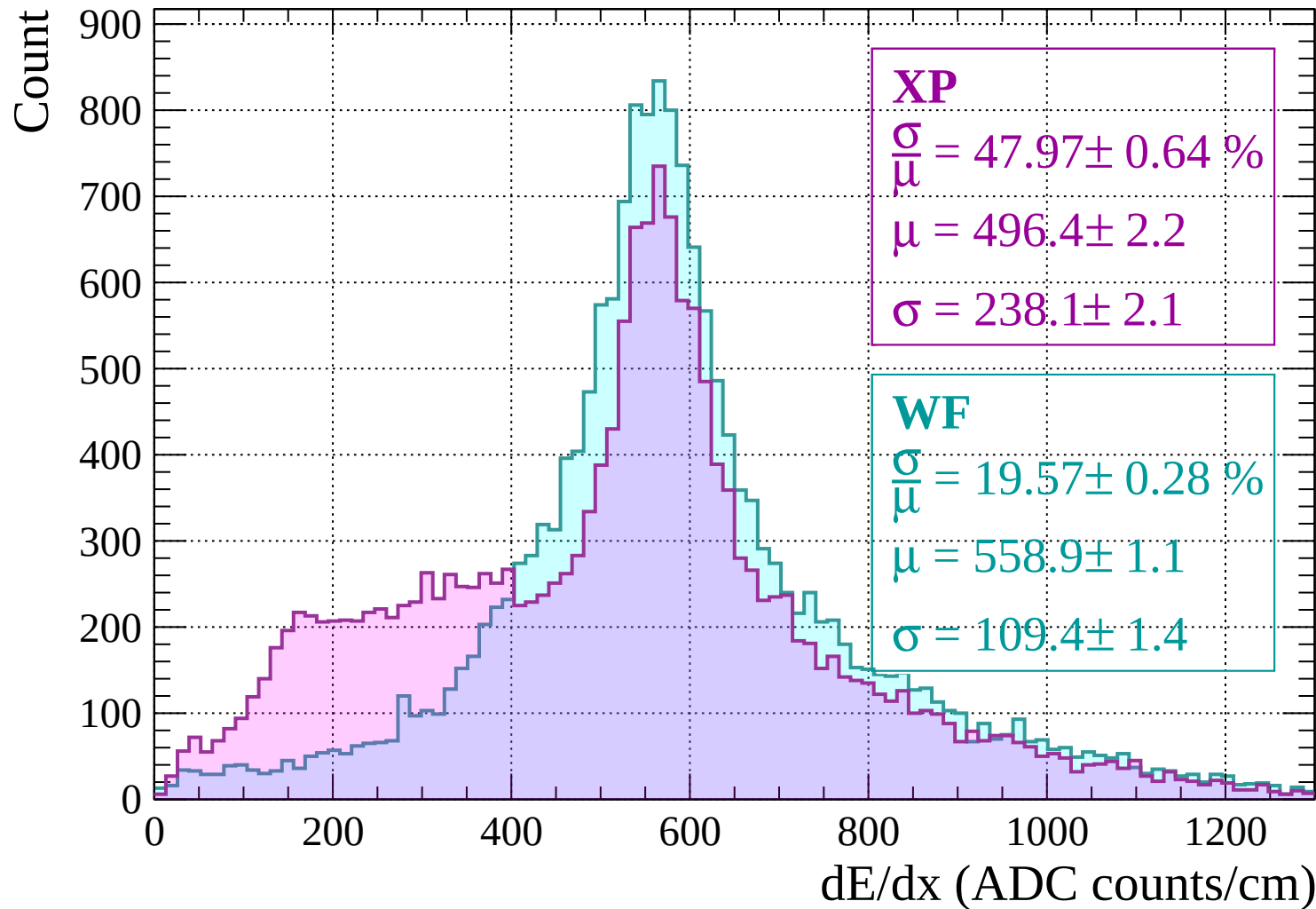
dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

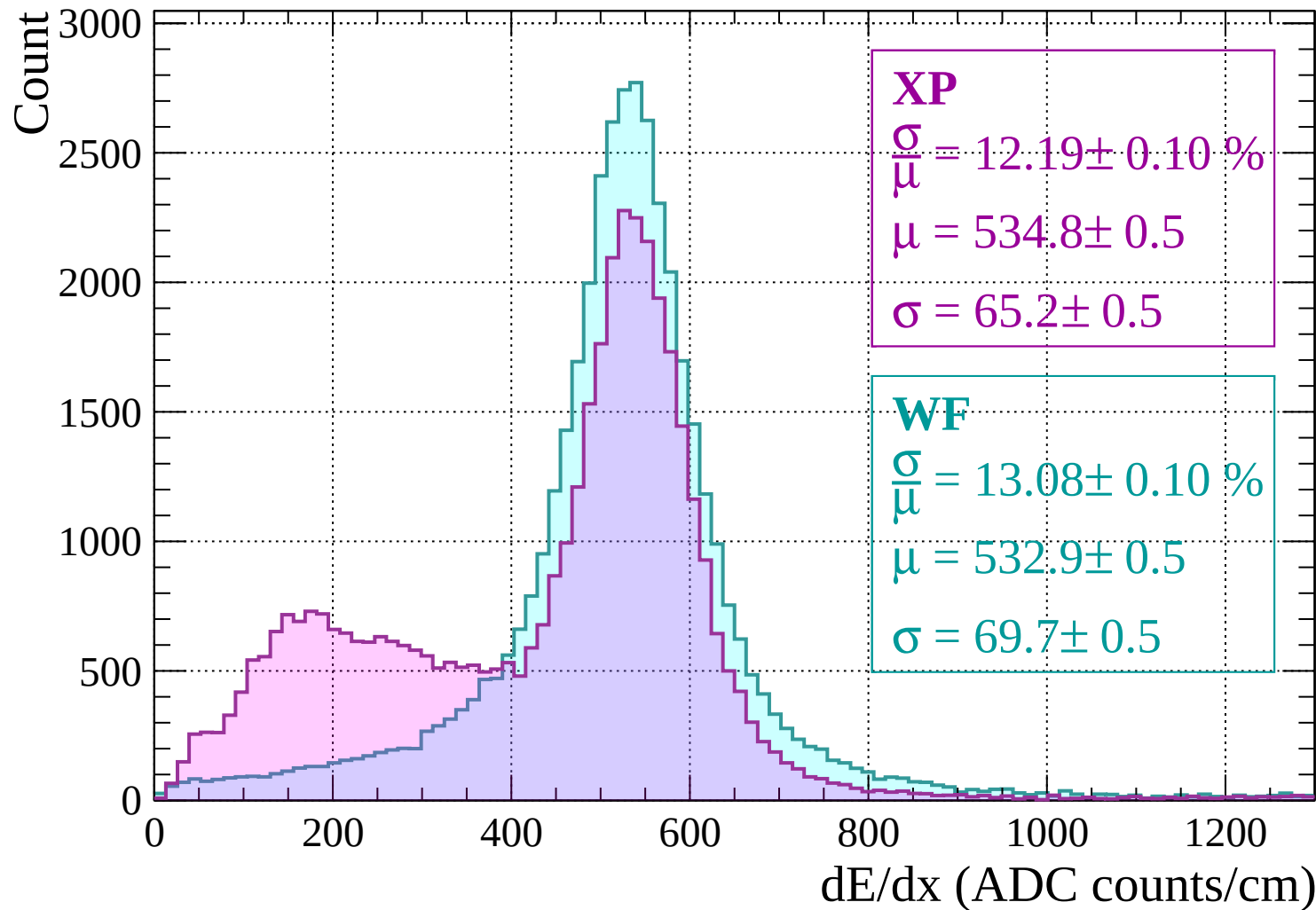
Count



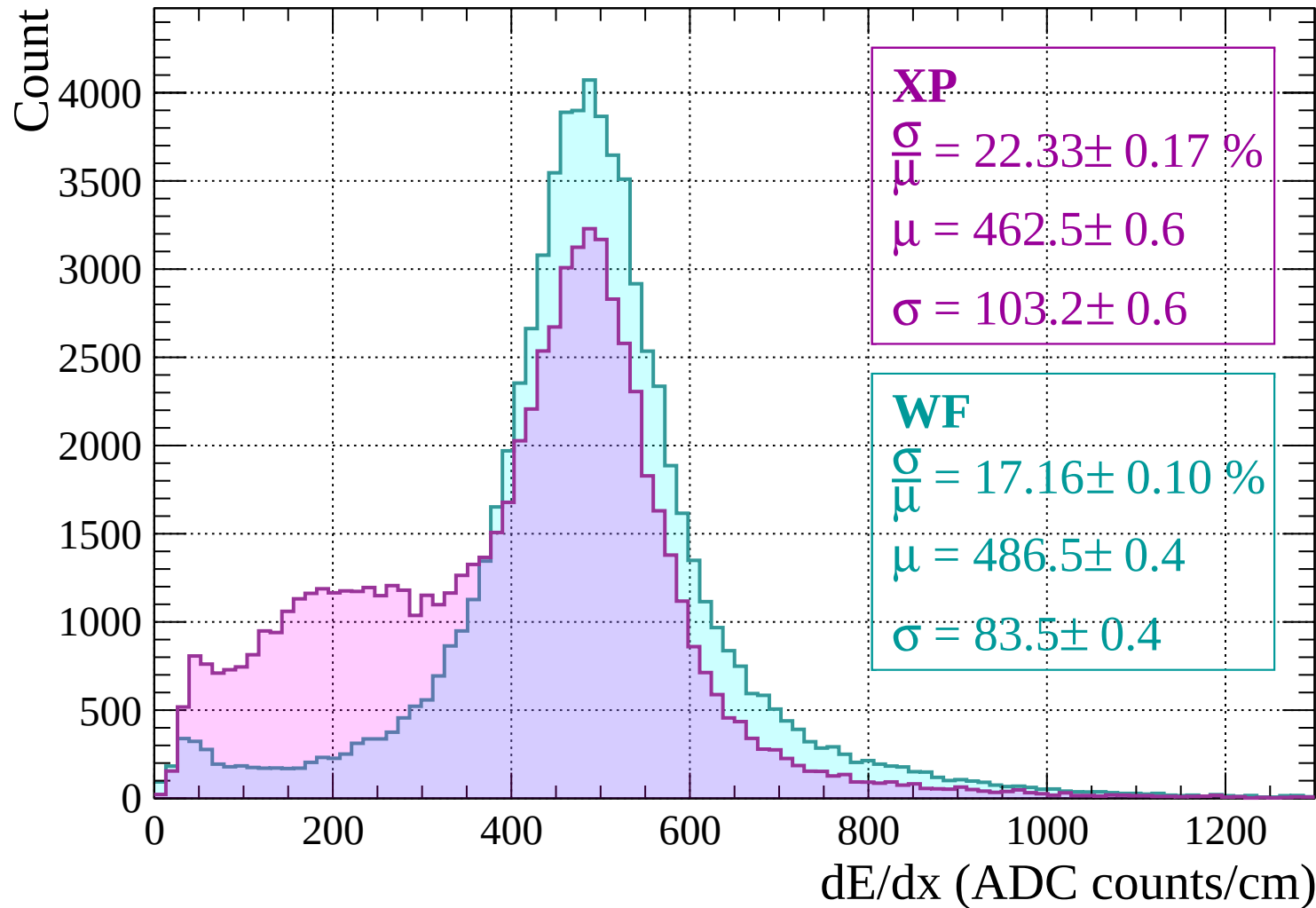
Energy loss | $-80 < p < -40$



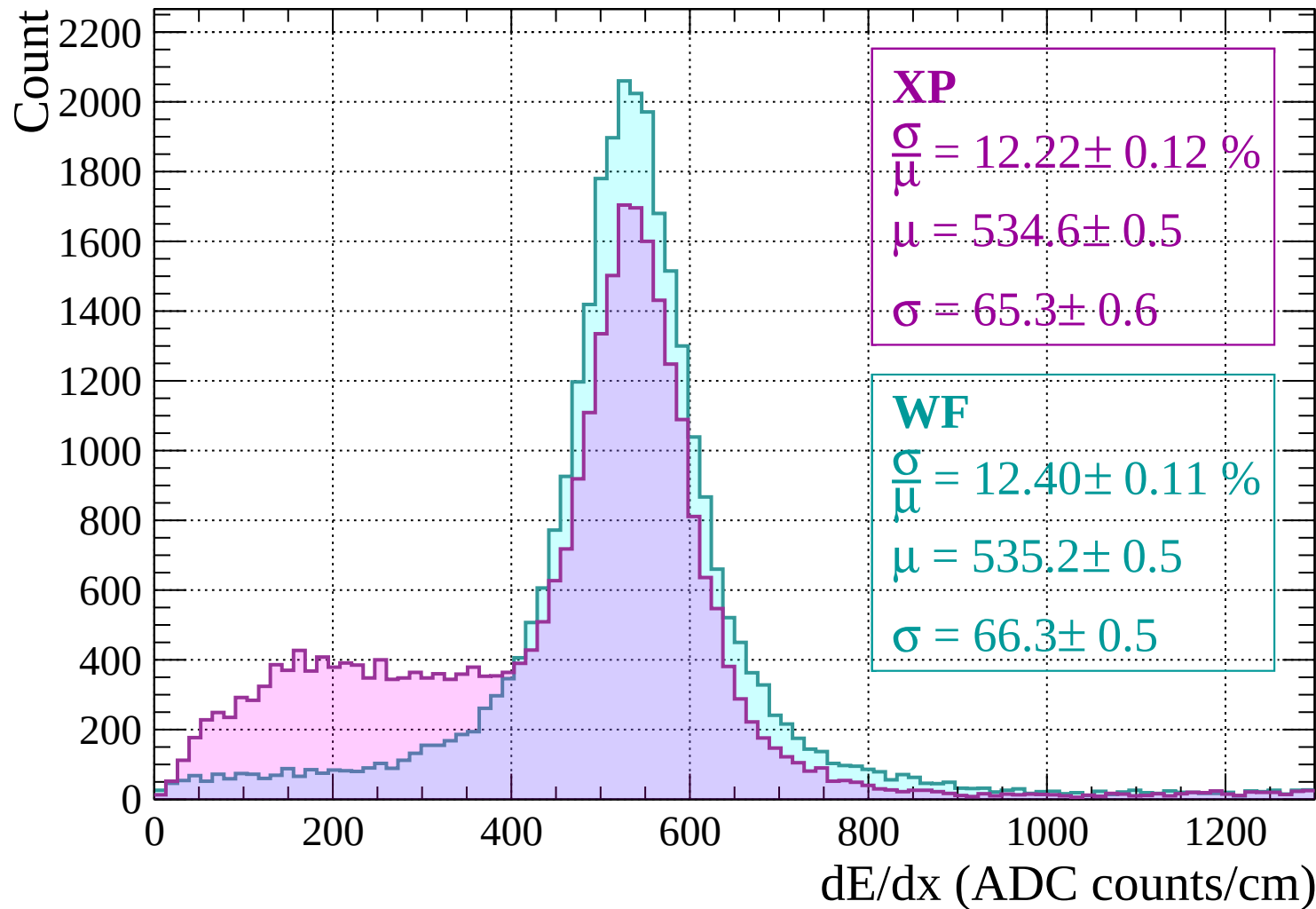
Energy loss | $-40 < p < 0$



Energy loss | $0 < p < 40$

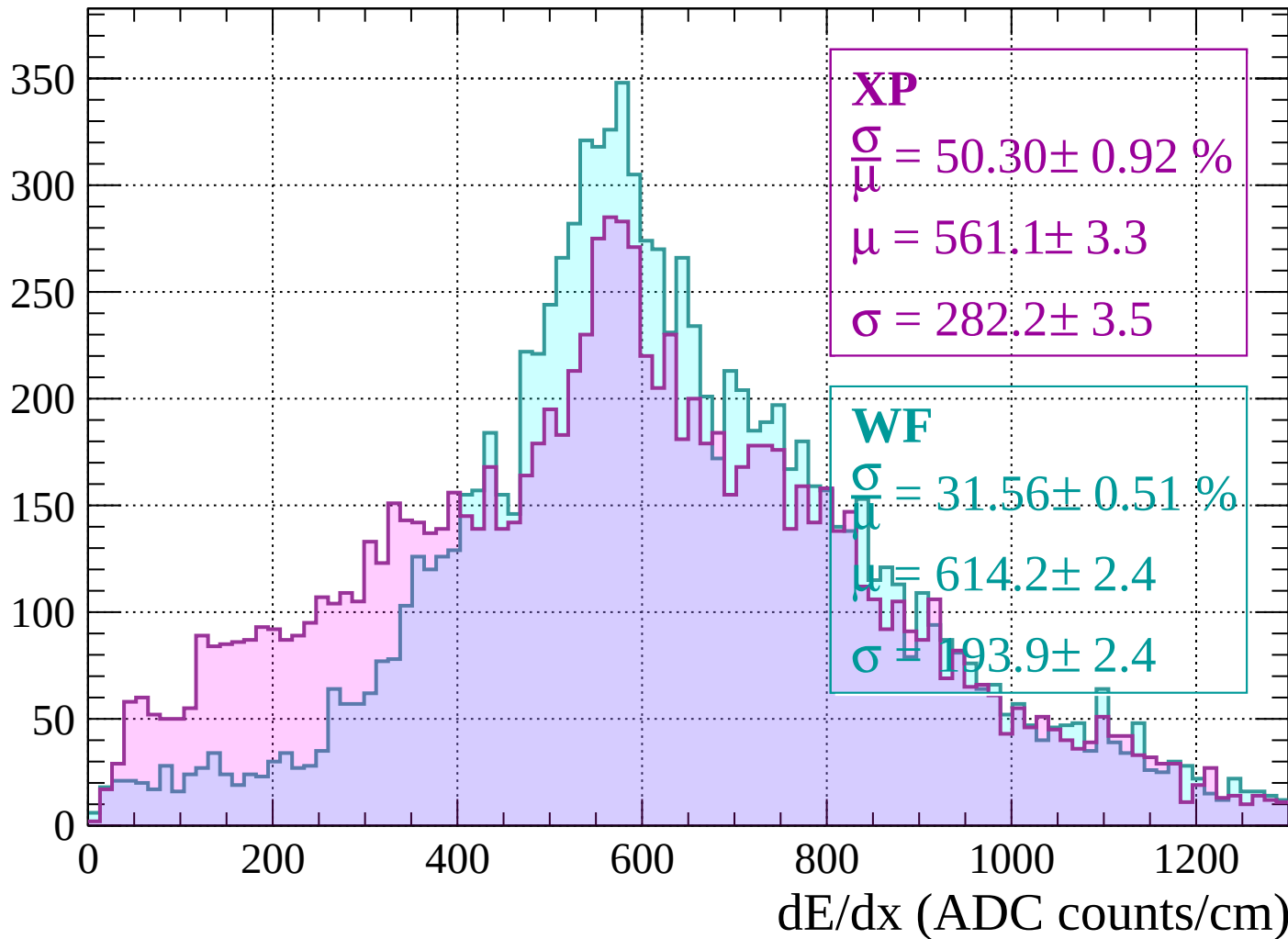


Energy loss | $40 < p < 80$



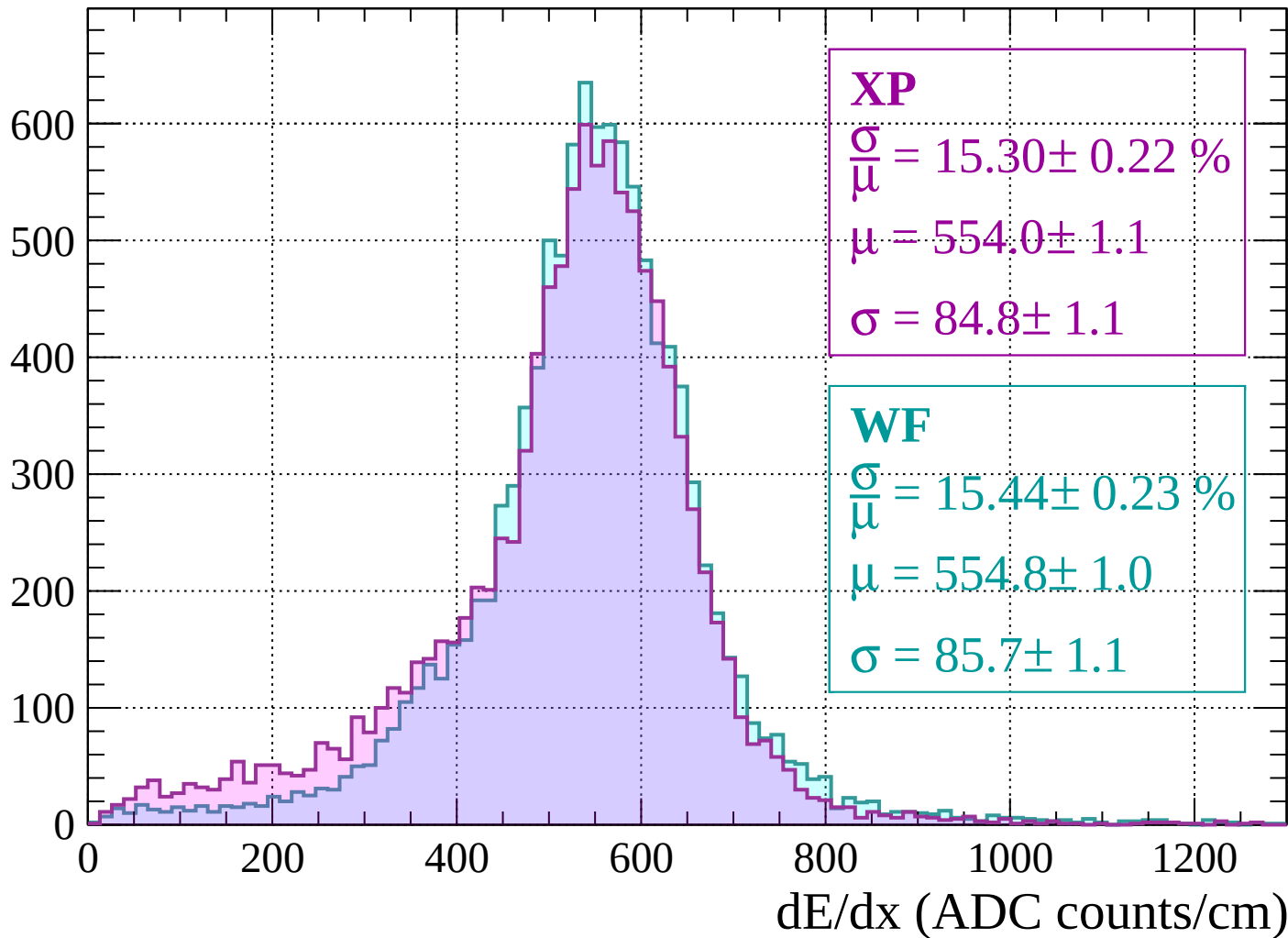
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 11.70 \pm 0.17 \%$$

$$\mu = 471.6 \pm 0.7$$

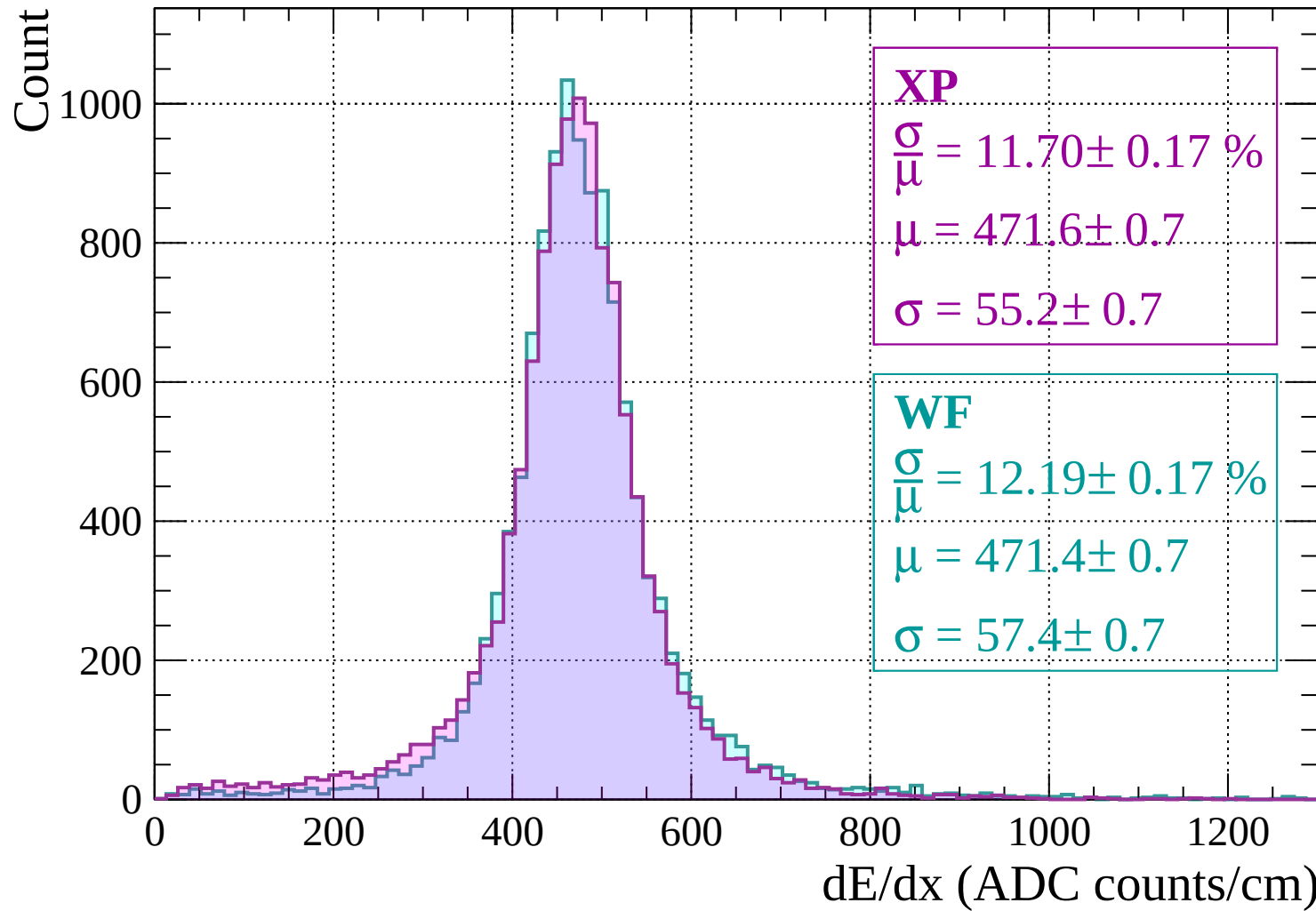
$$\sigma = 55.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 12.19 \pm 0.17 \%$$

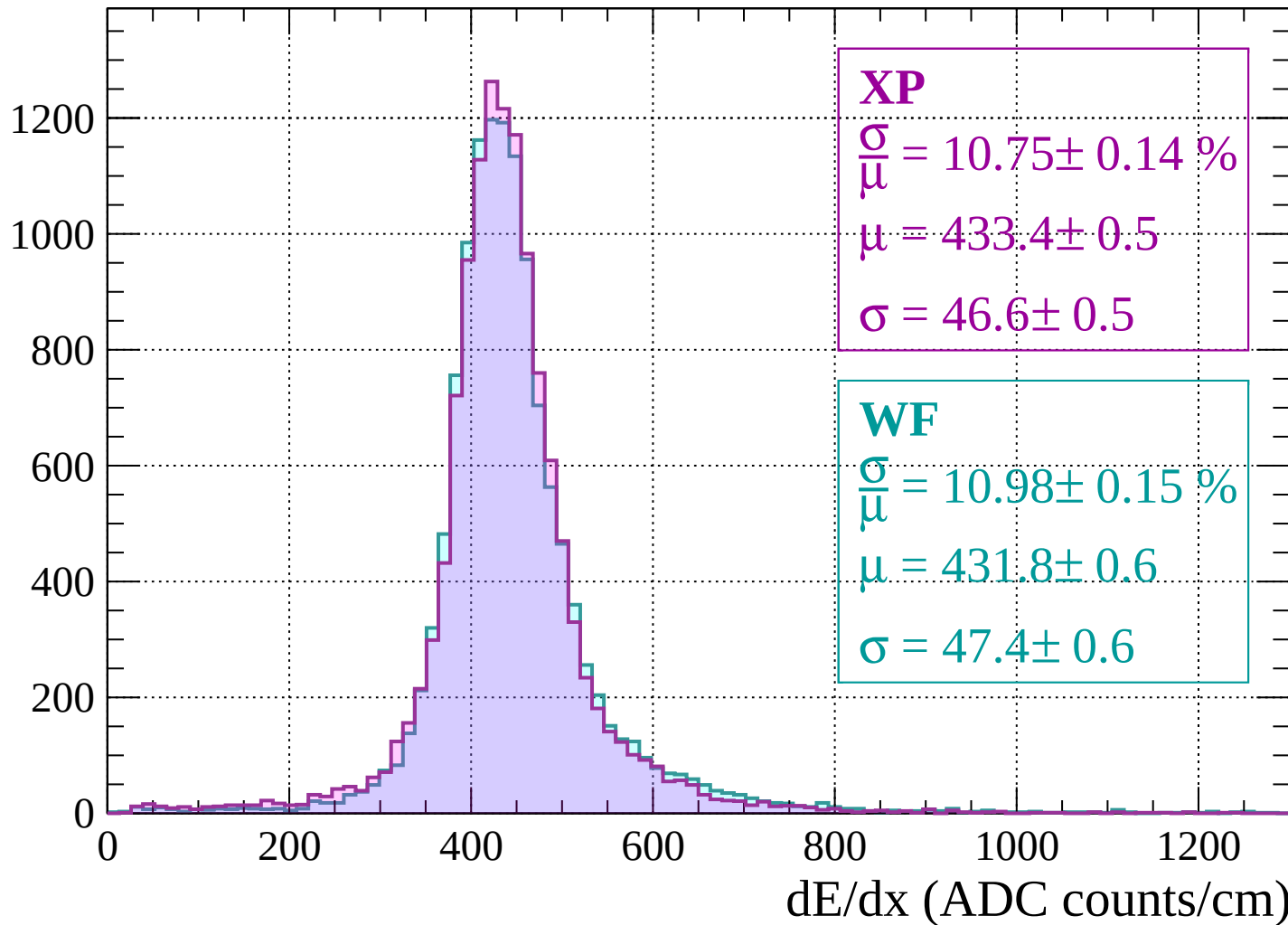
$$\mu = 471.4 \pm 0.7$$

$$\sigma = 57.4 \pm 0.7$$

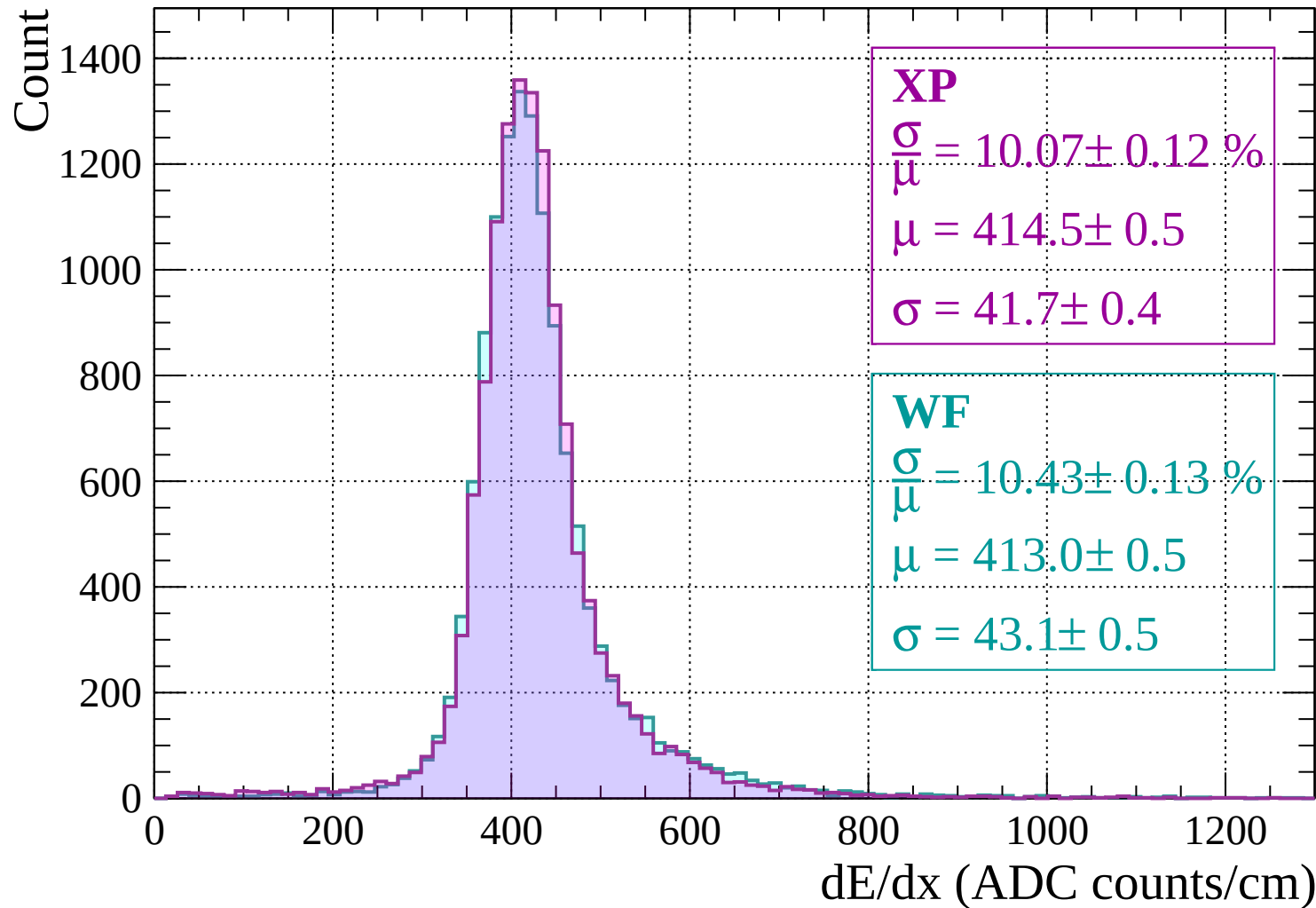


Energy loss | $200 < p < 240$

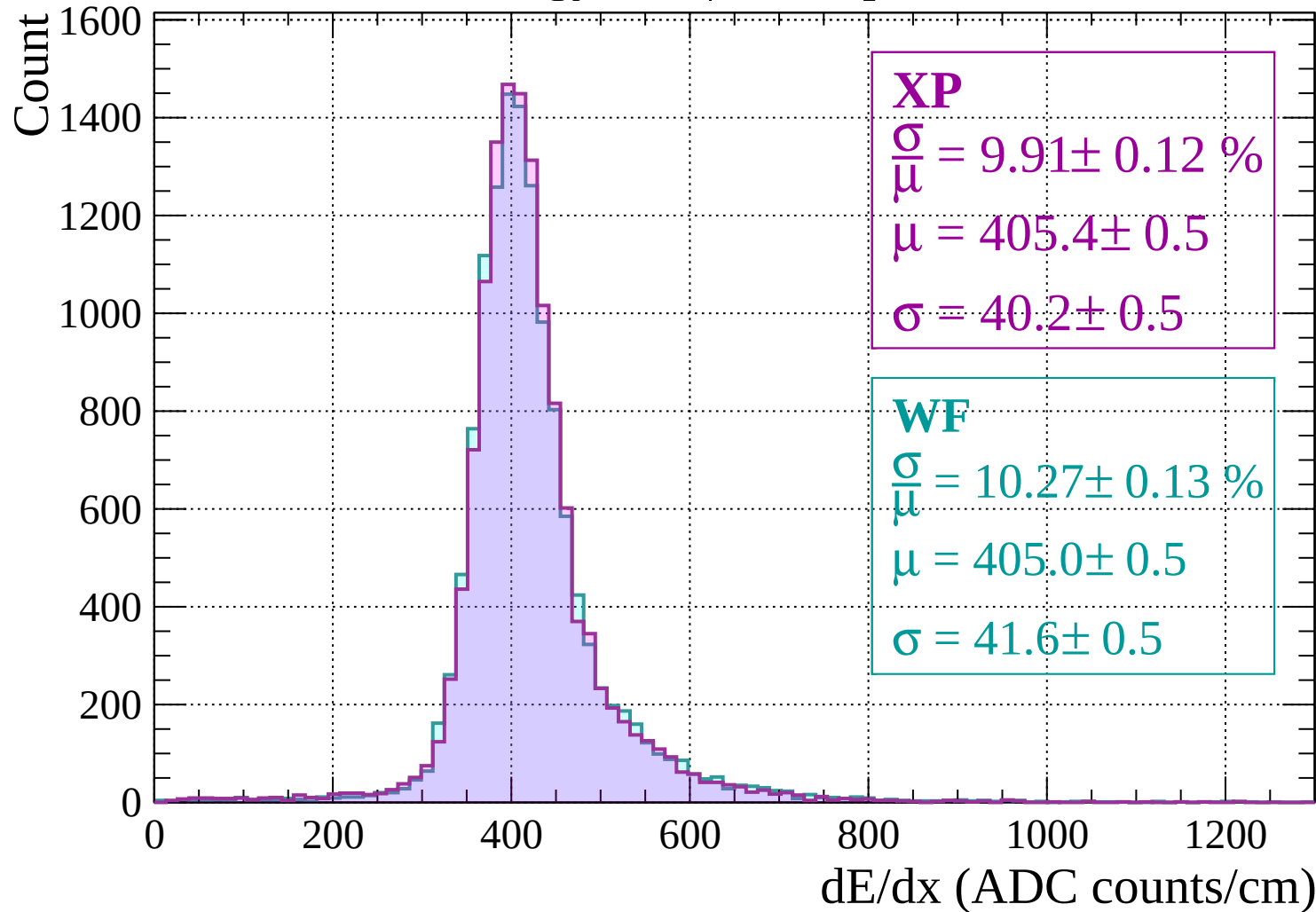
Count



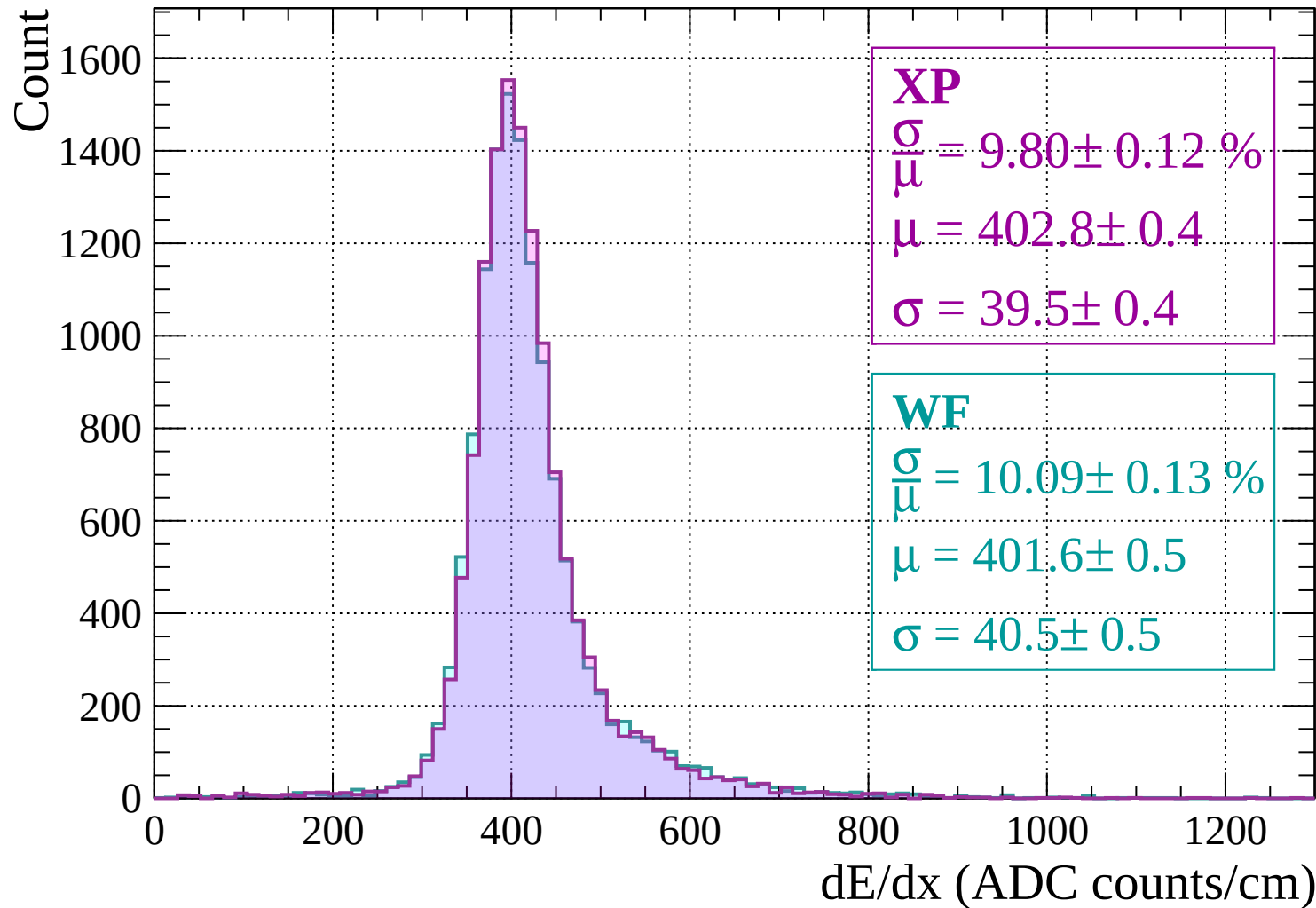
Energy loss | $240 < p < 280$



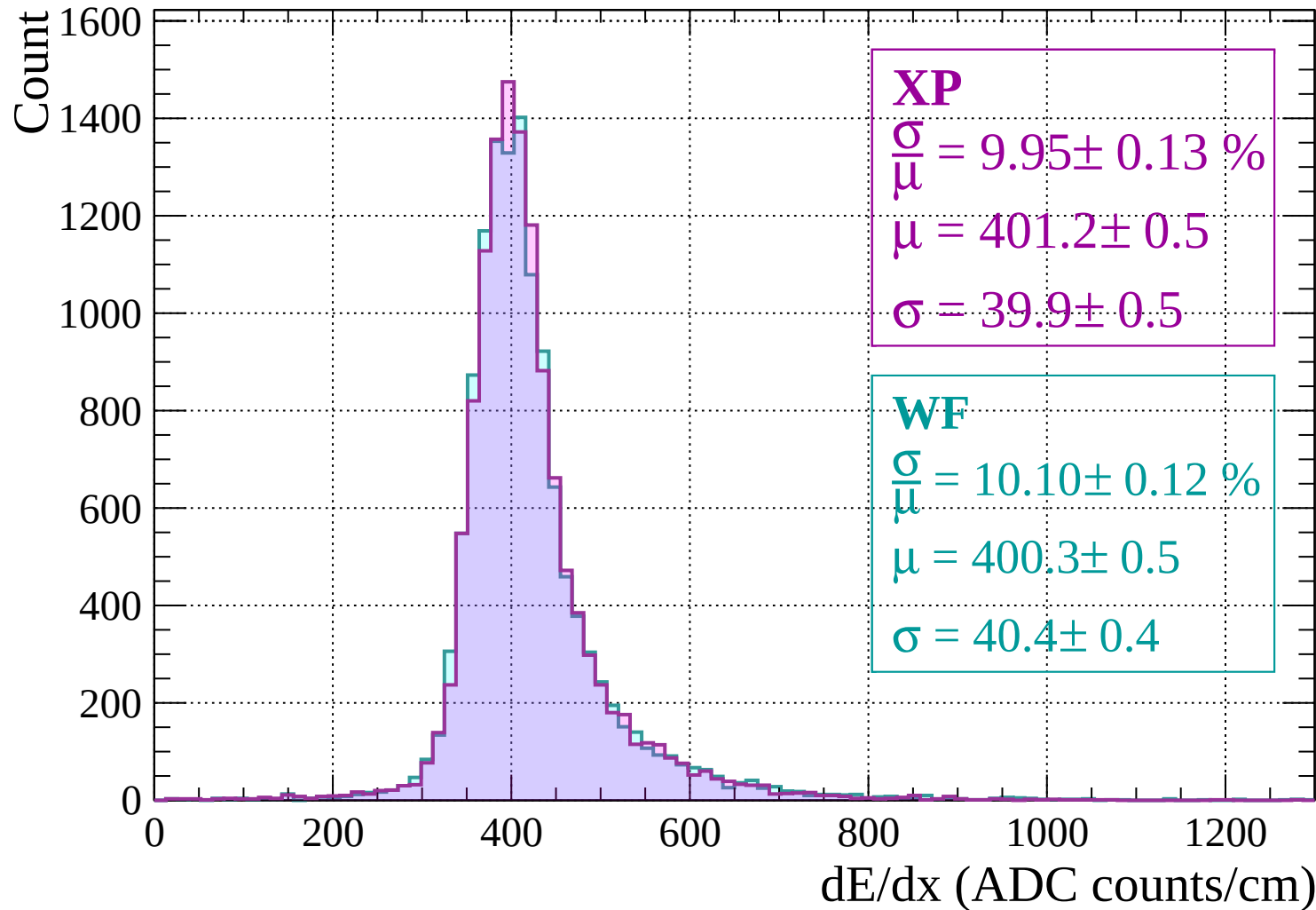
Energy loss | $280 < p < 320$



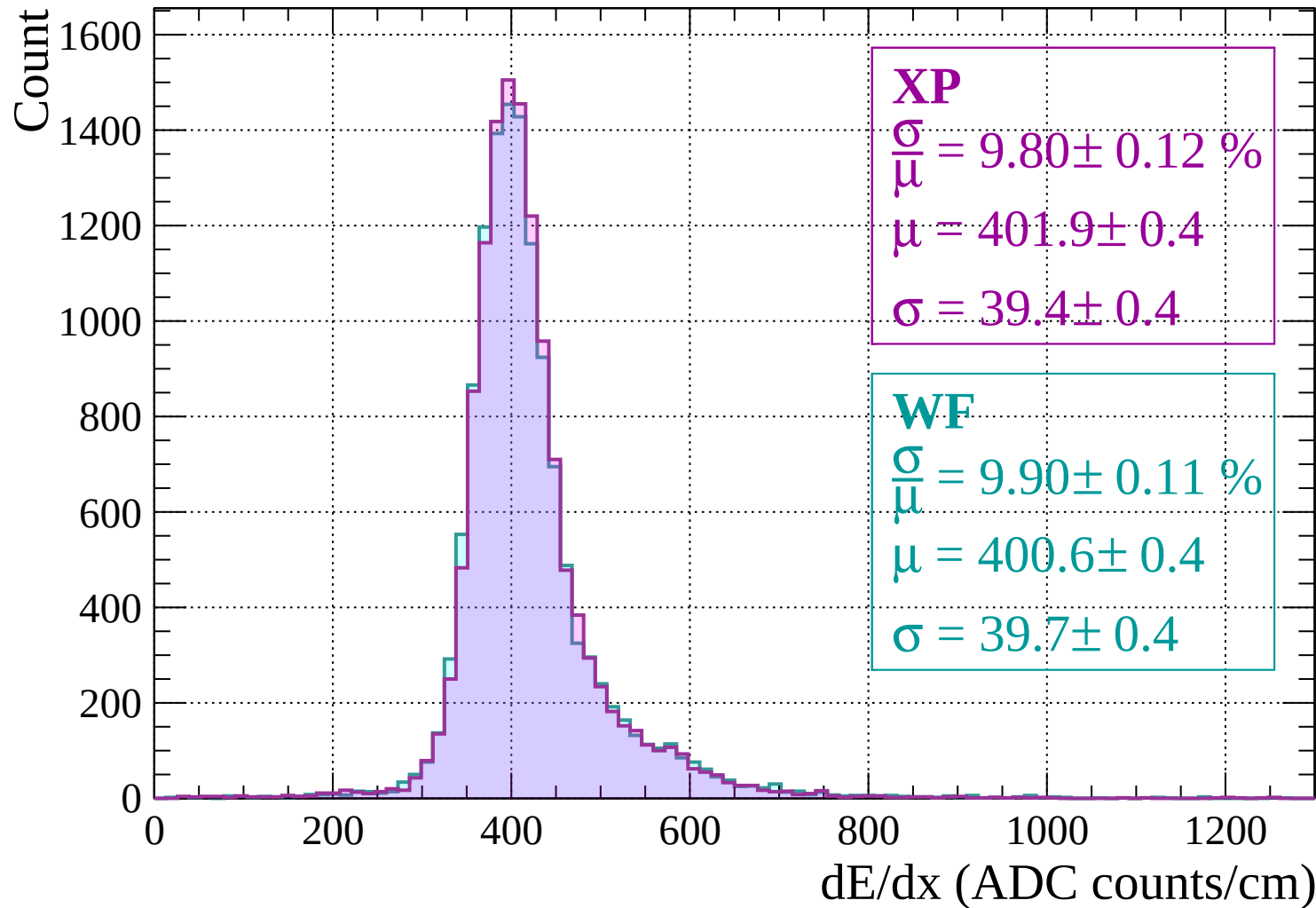
Energy loss | $320 < p < 360$



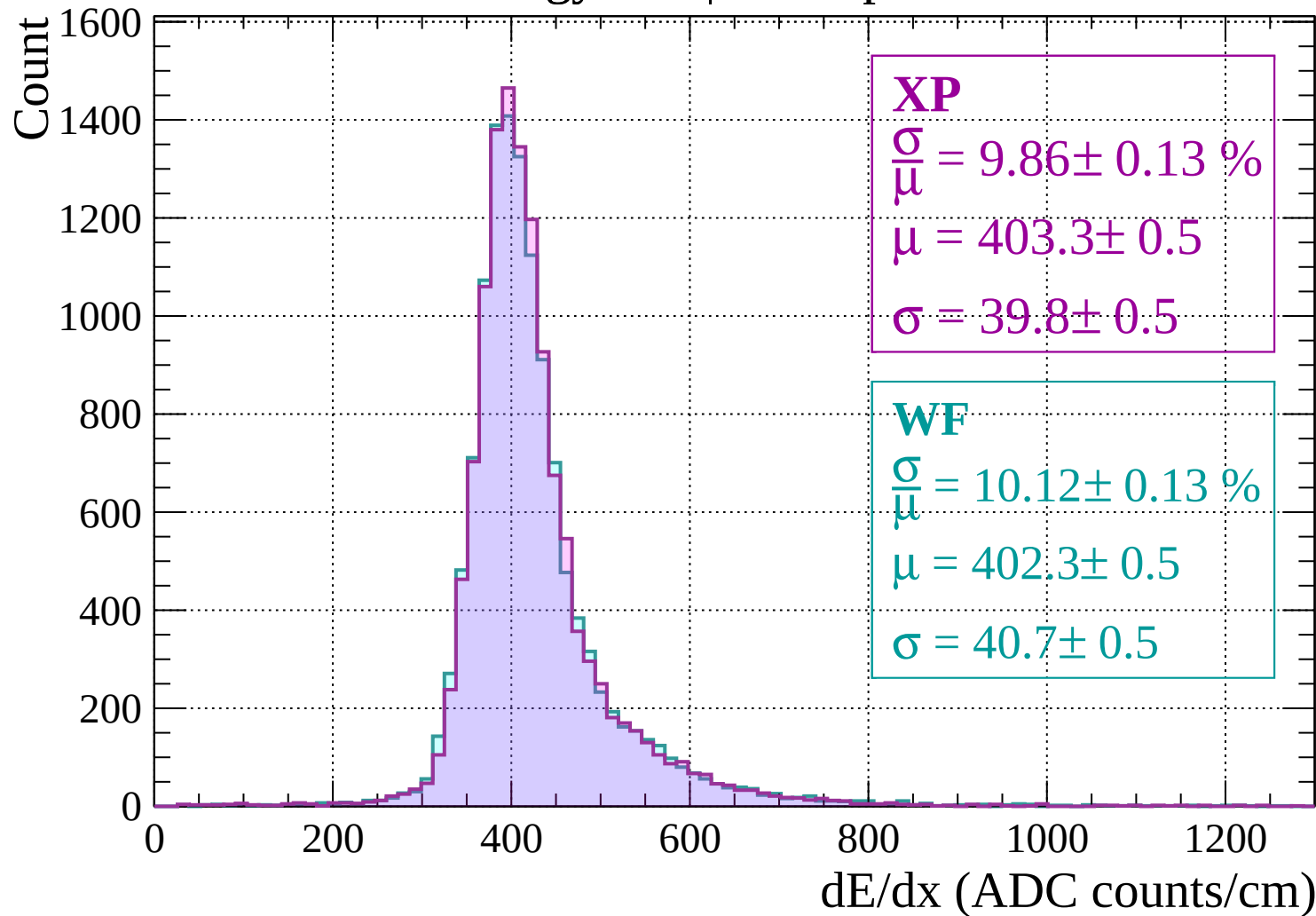
Energy loss | $360 < p < 400$



Energy loss | $400 < p < 440$



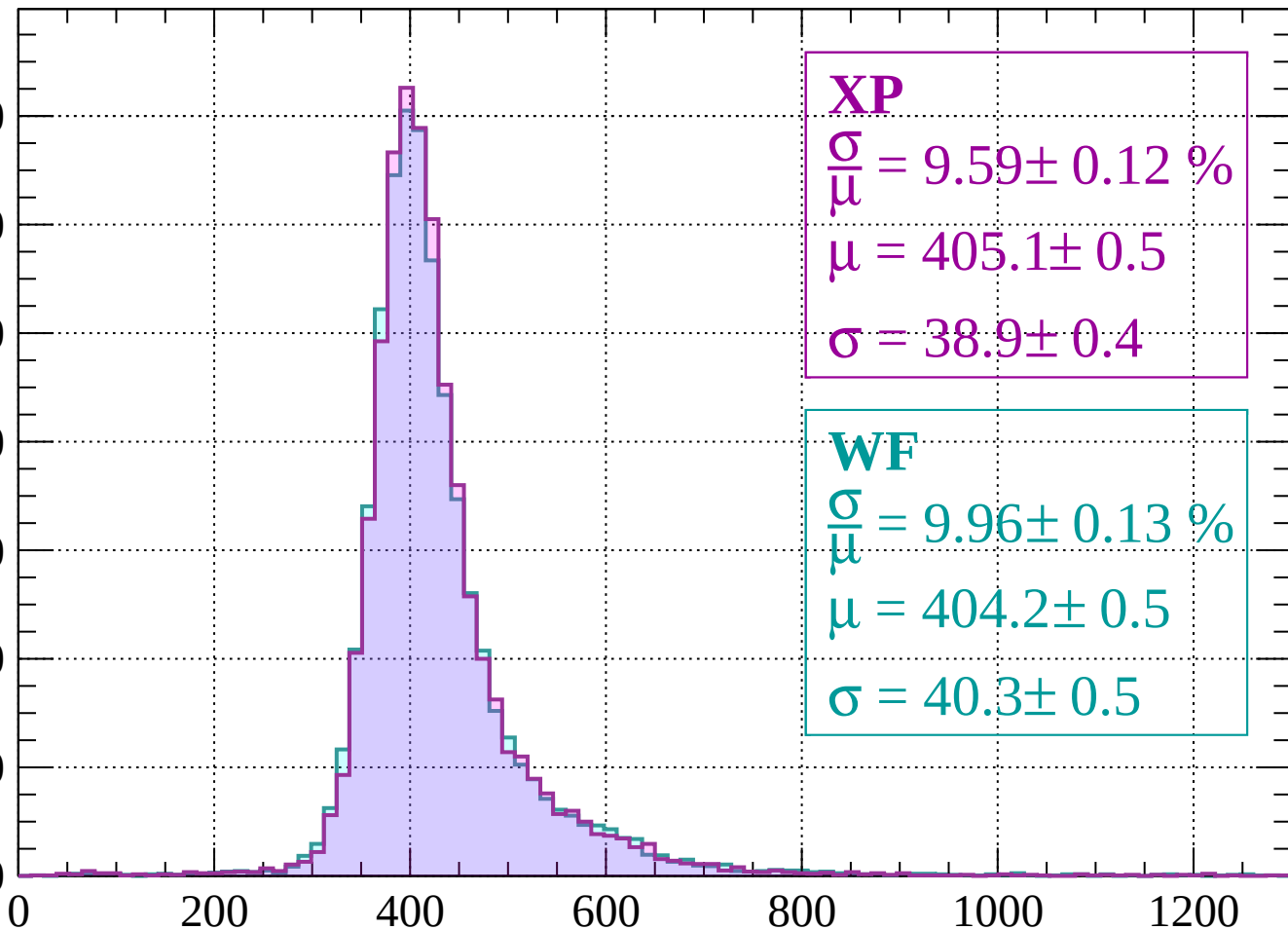
Energy loss | $440 < p < 480$



Energy loss | $480 < p < 520$

Count

1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.59 \pm 0.12 \%$$

$$\mu = 405.1 \pm 0.5$$

$$\sigma = 38.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 9.96 \pm 0.13 \%$$

$$\mu = 404.2 \pm 0.5$$

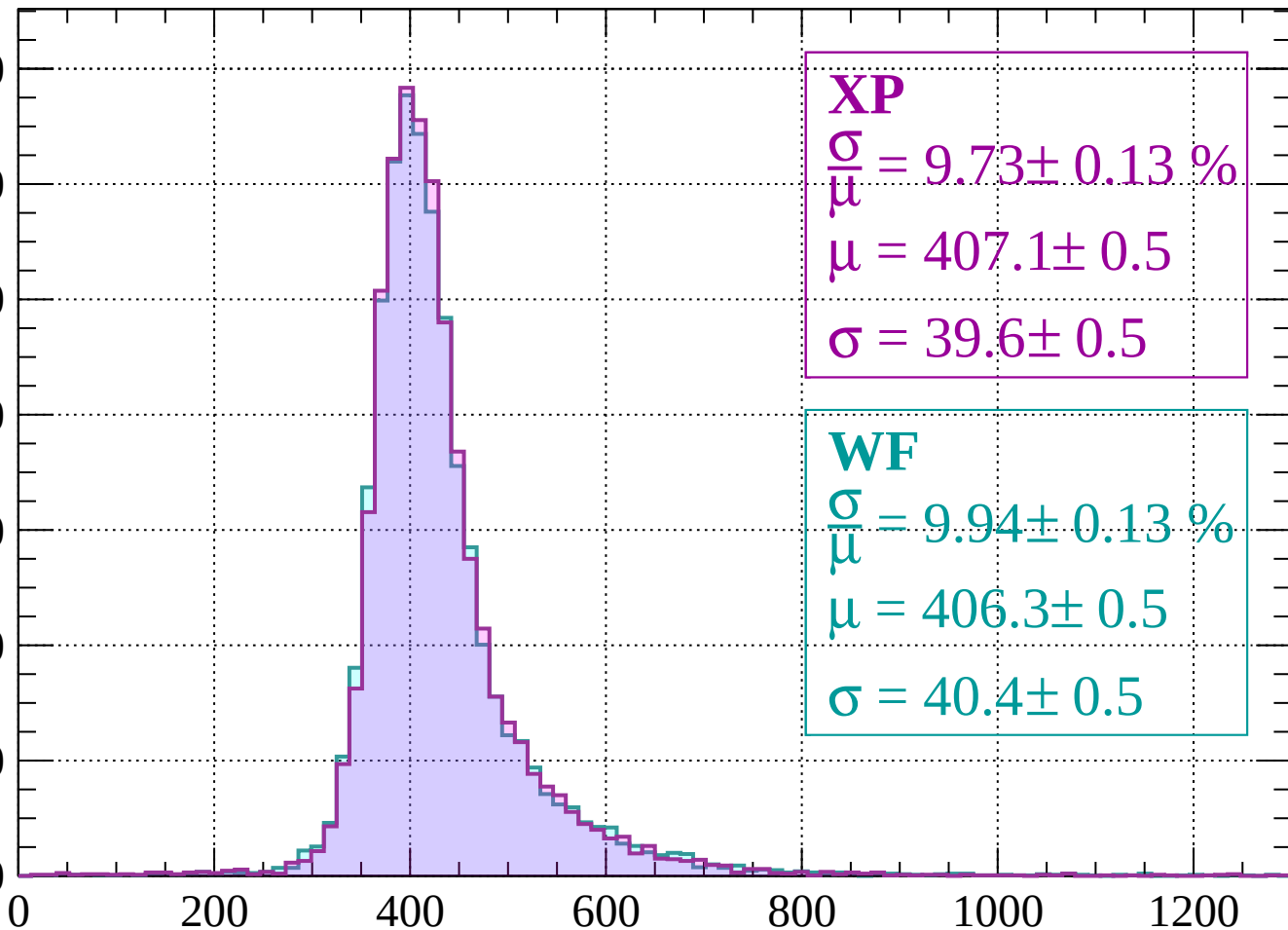
$$\sigma = 40.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$

Count

1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.73 \pm 0.13 \%$$

$$\mu = 407.1 \pm 0.5$$

$$\sigma = 39.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.13 \%$$

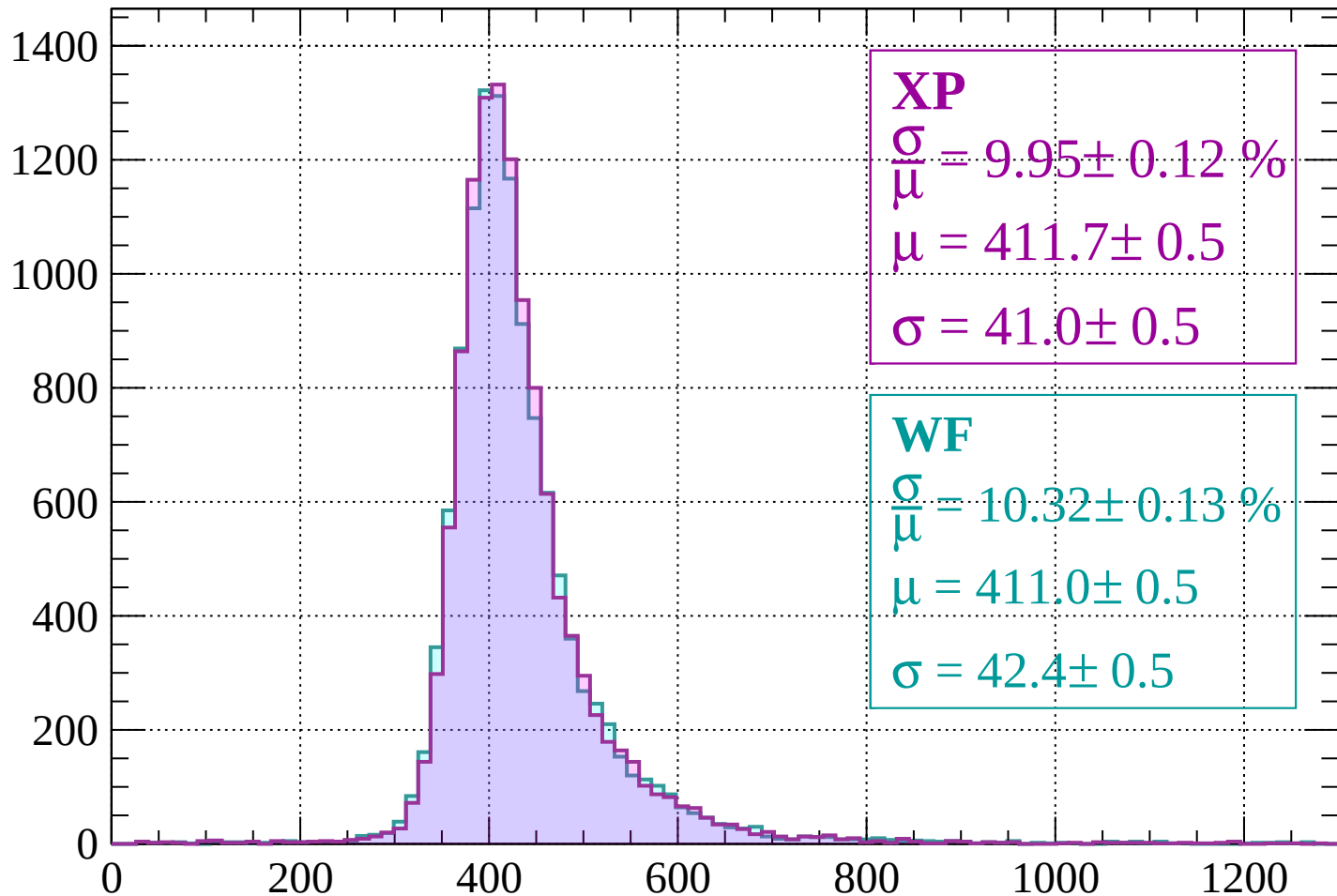
$$\mu = 406.3 \pm 0.5$$

$$\sigma = 40.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 9.95 \pm 0.12 \%$$

$$\mu = 411.7 \pm 0.5$$

$$\sigma = 41.0 \pm 0.5$$

WF

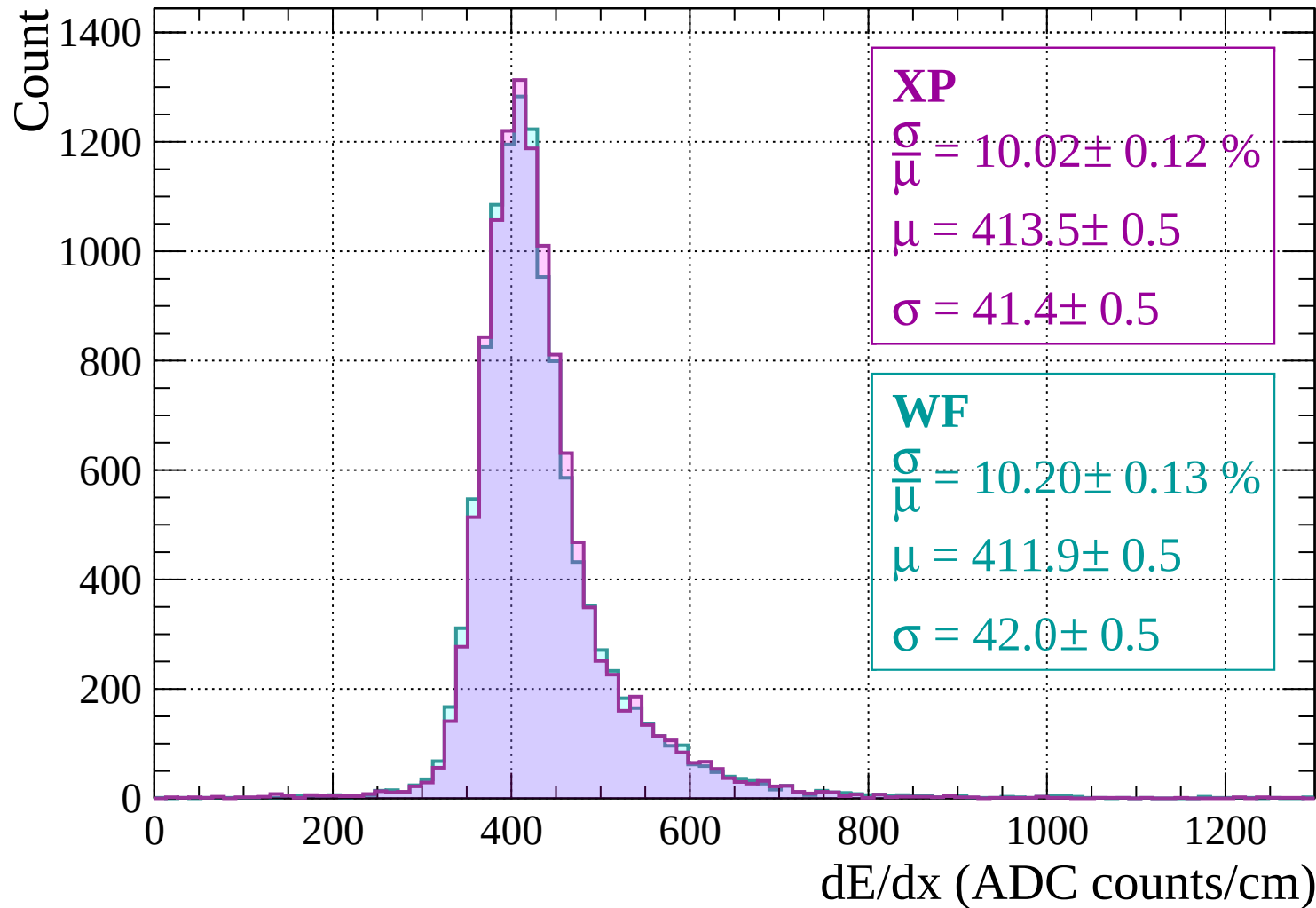
$$\frac{\sigma}{\mu} = 10.32 \pm 0.13 \%$$

$$\mu = 411.0 \pm 0.5$$

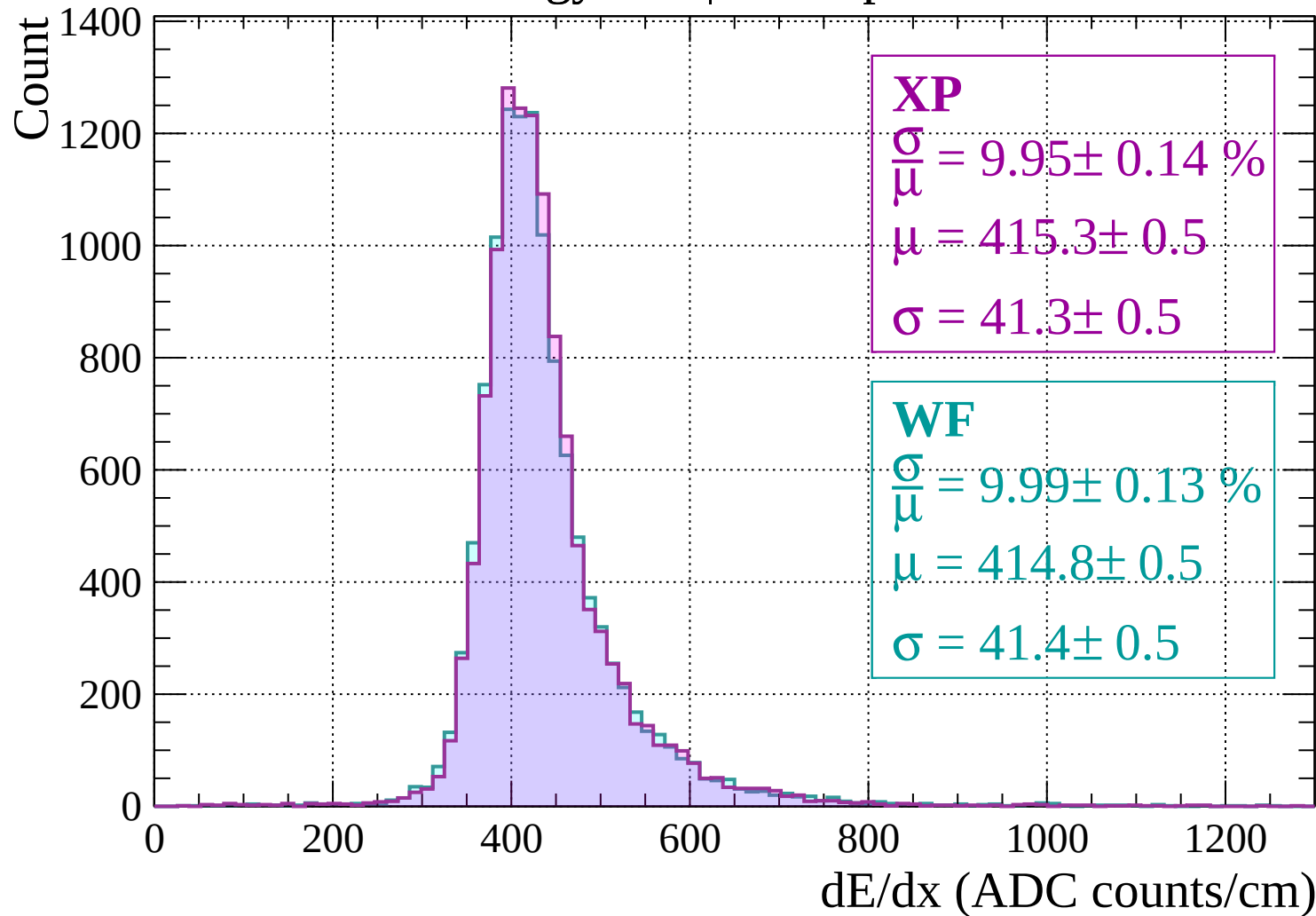
$$\sigma = 42.4 \pm 0.5$$

dE/dx (ADC counts/cm)

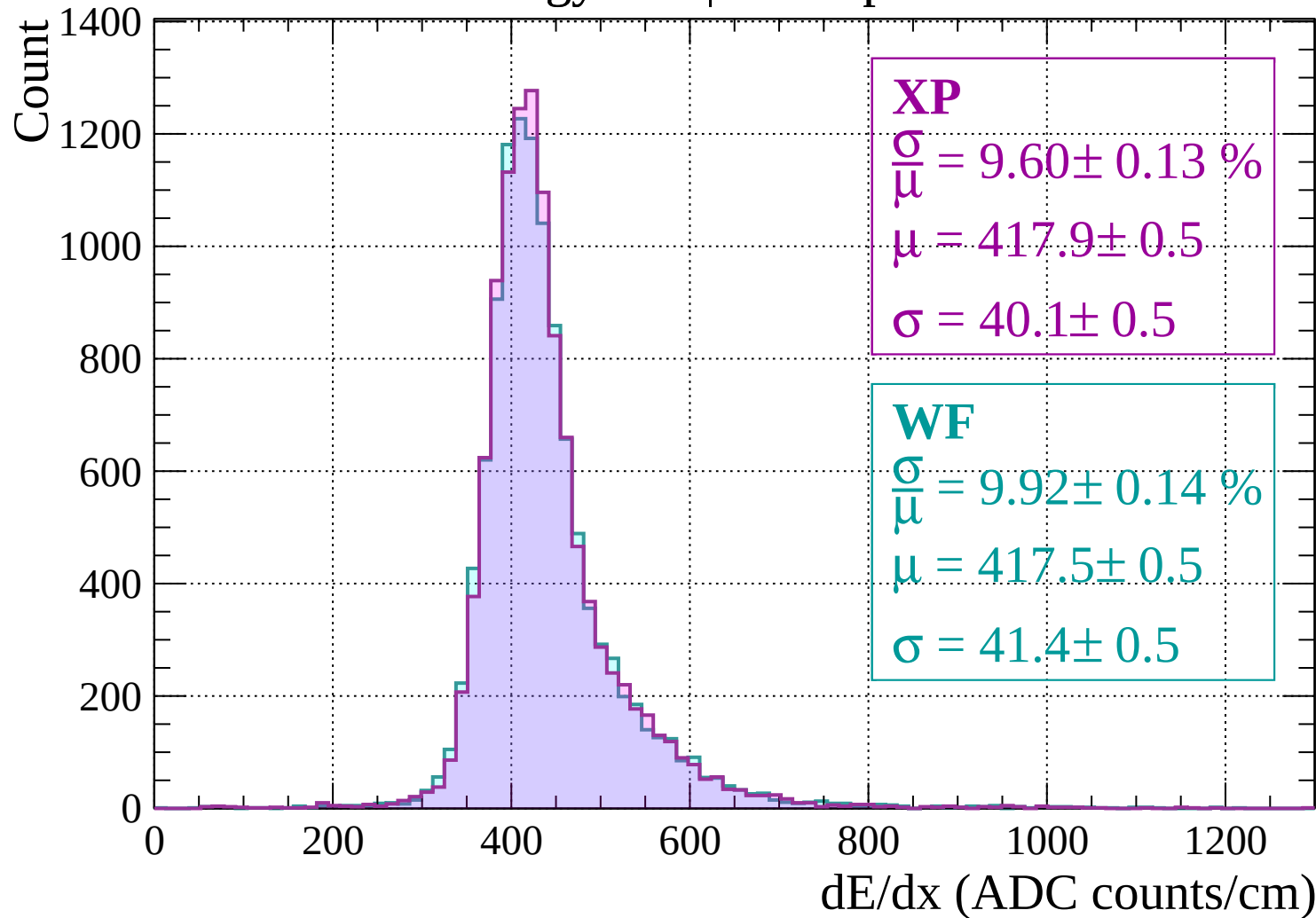
Energy loss | $600 < p < 640$



Energy loss | $640 < p < 680$

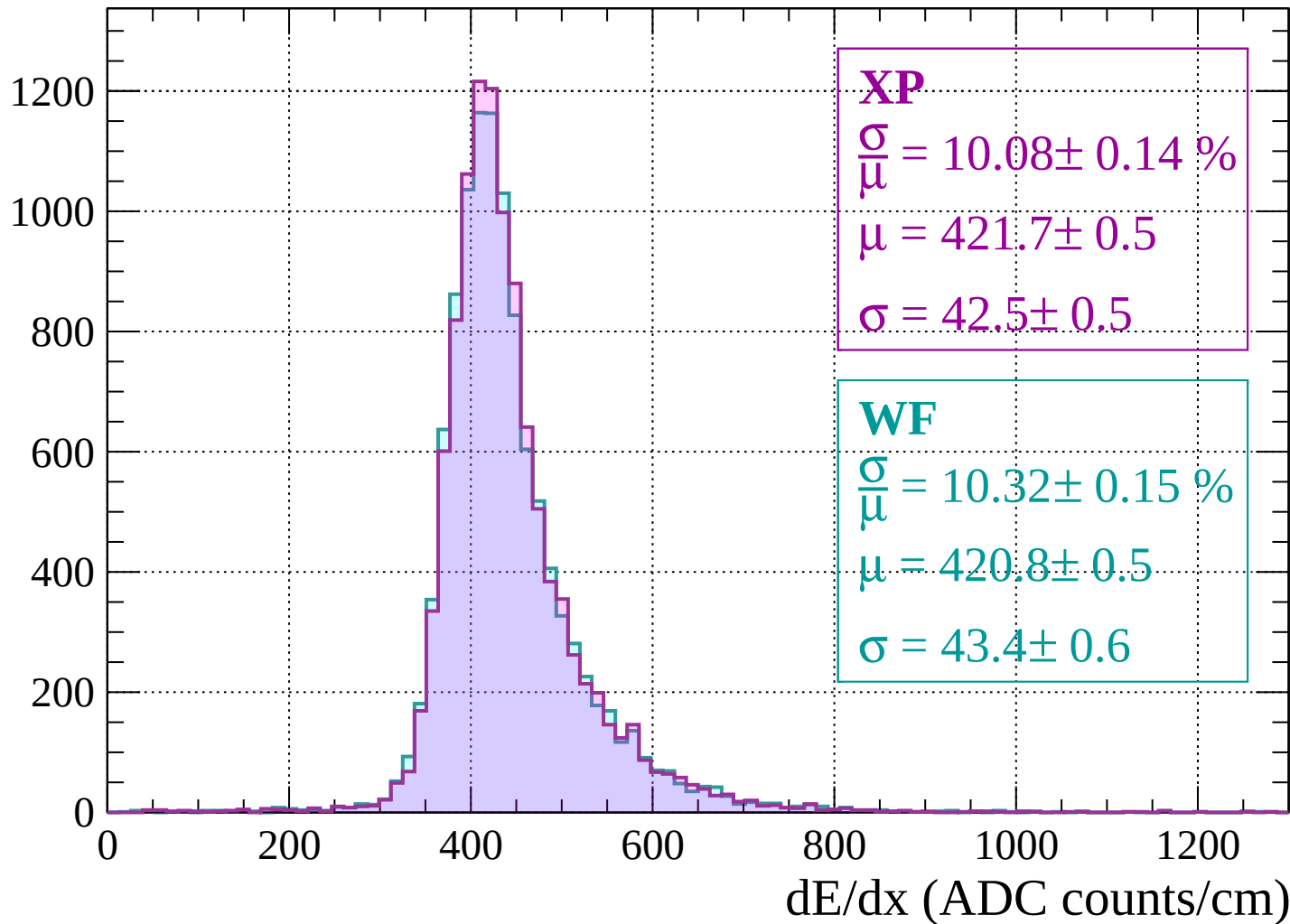


Energy loss | $680 < p < 720$



Energy loss | $720 < p < 760$

Count



Energy loss | $760 < p < 800$

Count

1200

1000

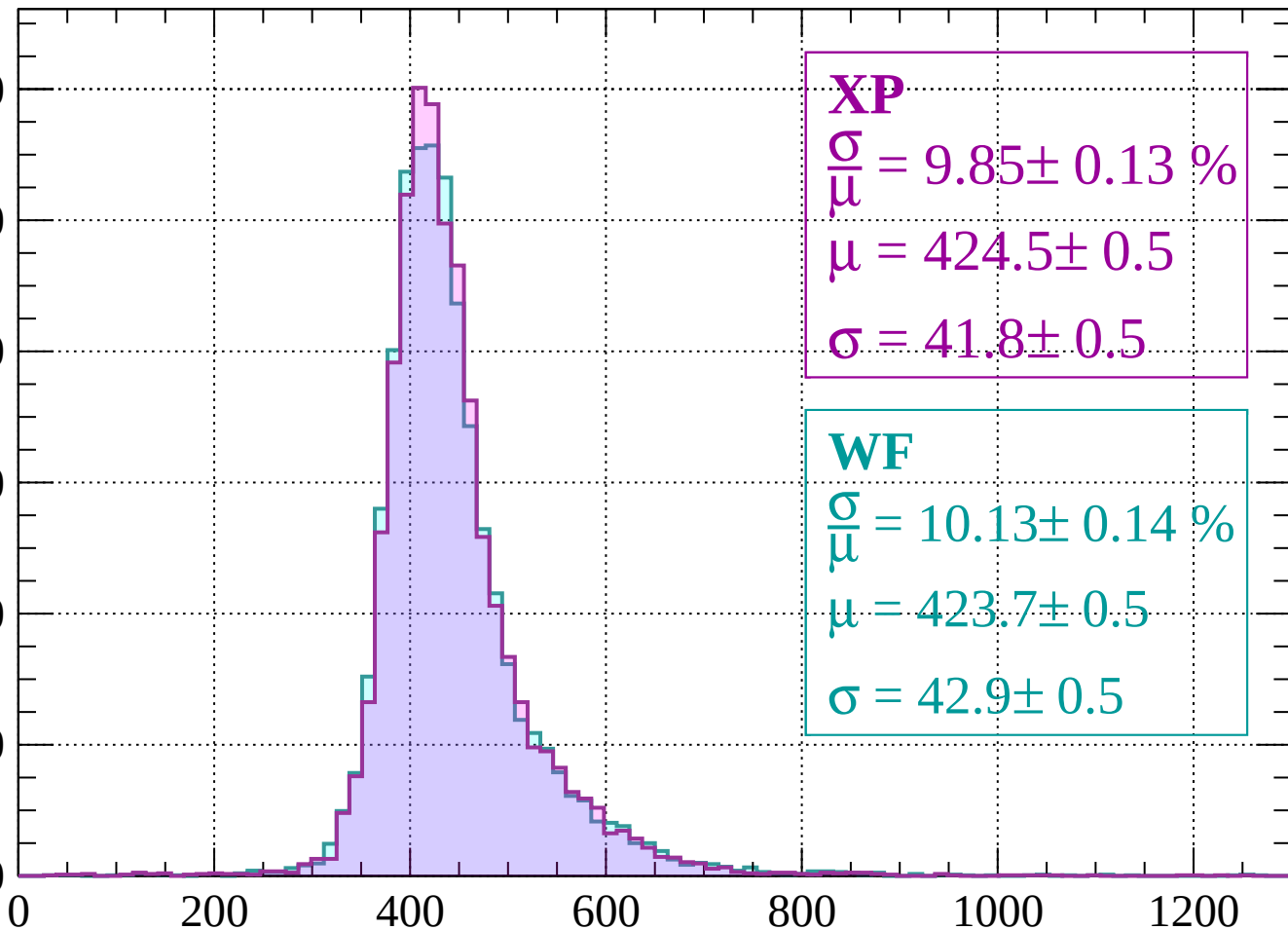
800

600

400

200

0



XP

$\frac{\sigma}{\bar{\mu}} = 9.85 \pm 0.13 \%$

$\mu = 424.5 \pm 0.5$

$\sigma = 41.8 \pm 0.5$

WF

$\frac{\sigma}{\bar{\mu}} = 10.13 \pm 0.14 \%$

$\mu = 423.7 \pm 0.5$

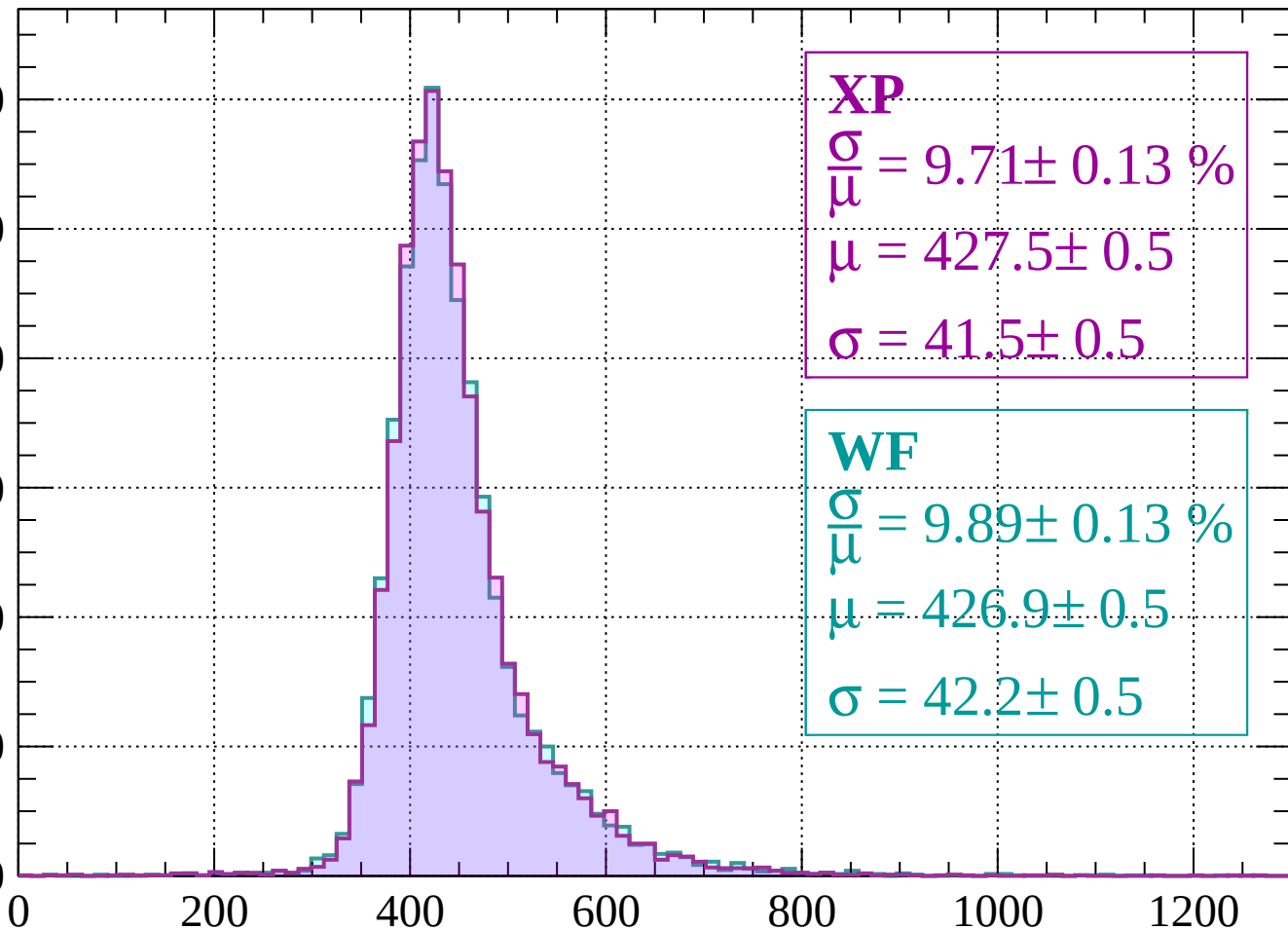
$\sigma = 42.9 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count

1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.13 \%$$

$$\mu = 427.5 \pm 0.5$$

$$\sigma = 41.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.89 \pm 0.13 \%$$

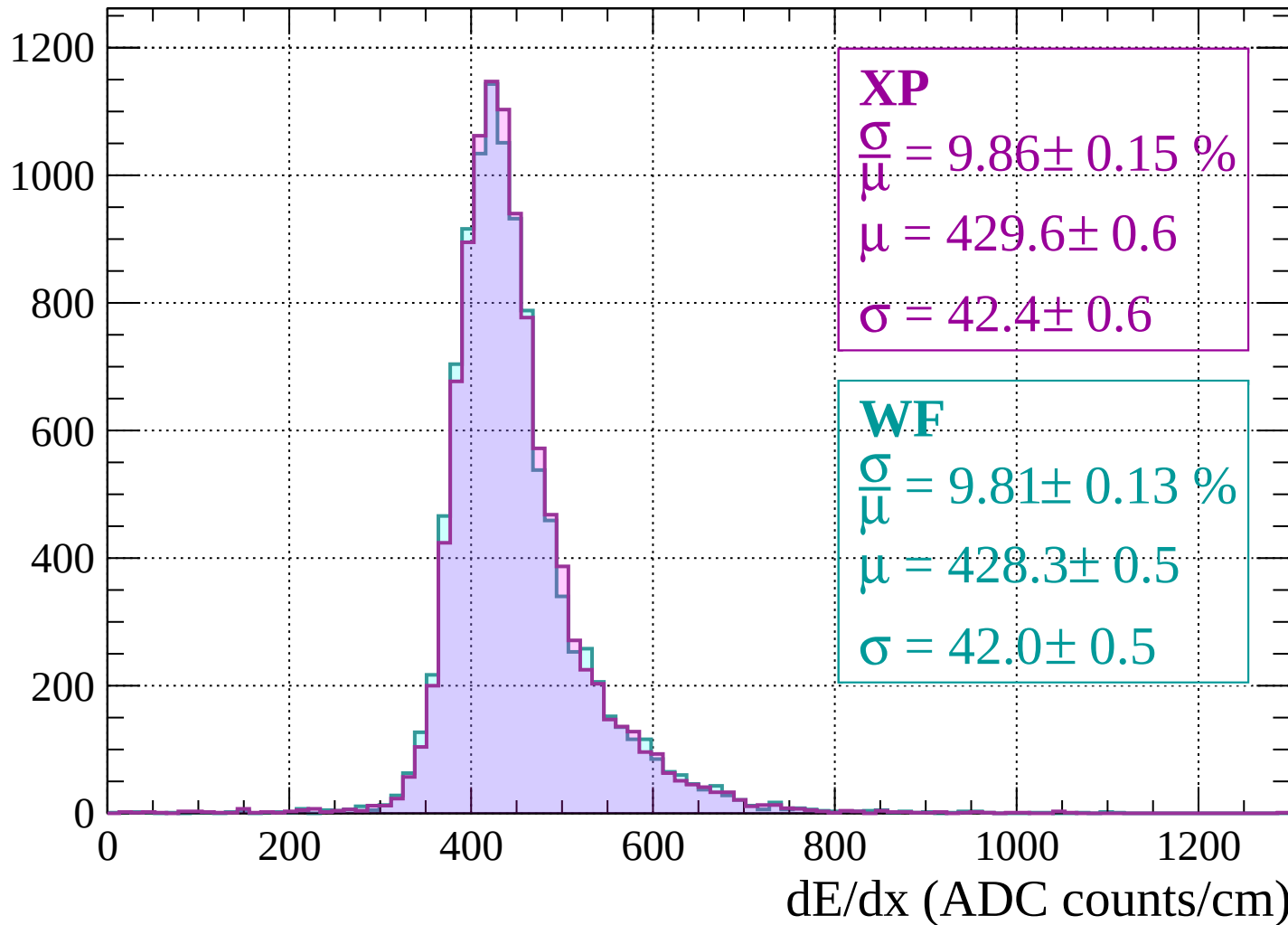
$$\mu = 426.9 \pm 0.5$$

$$\sigma = 42.2 \pm 0.5$$

dE/dx (ADC counts/cm)

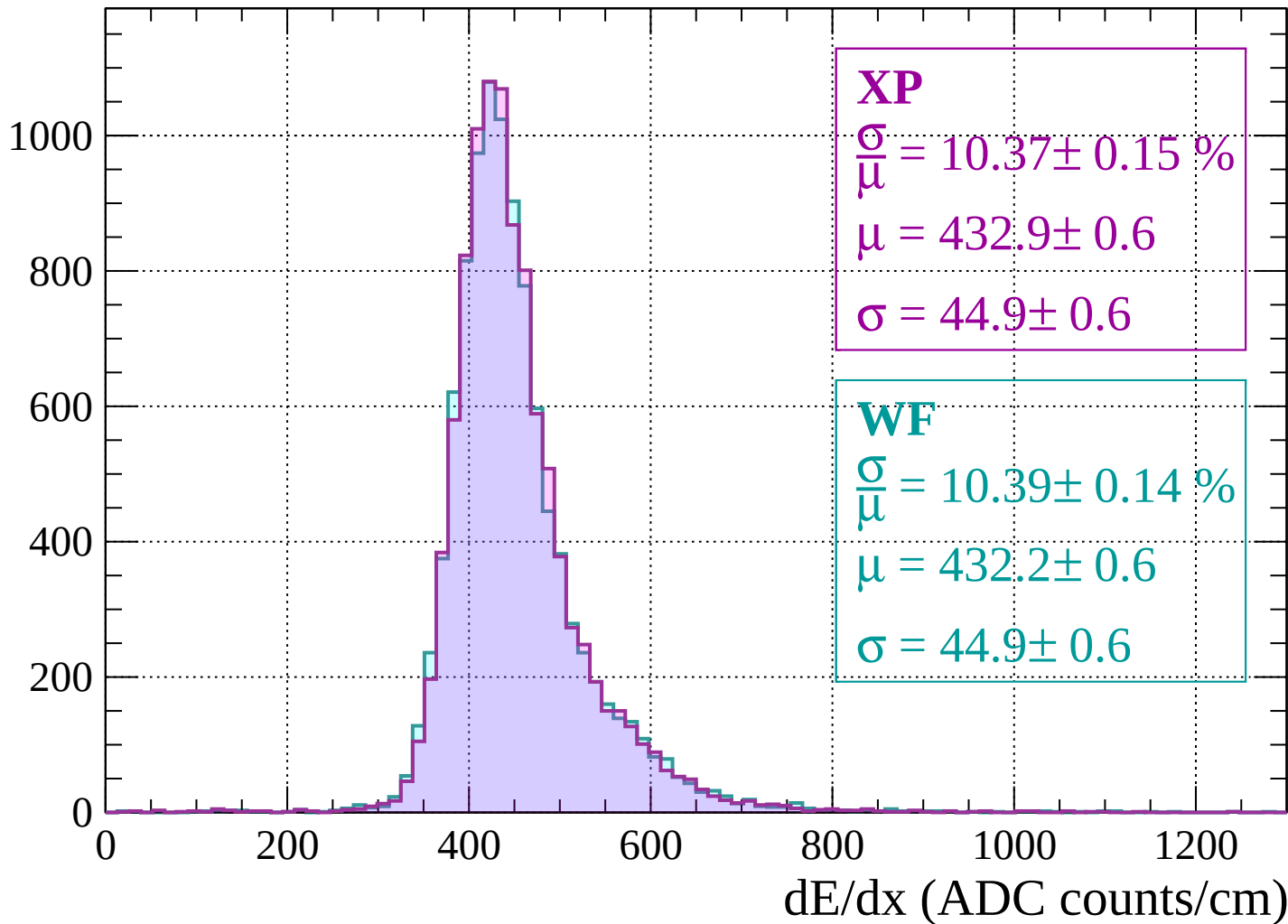
Energy loss | $840 < p < 880$

Count



Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count

1000

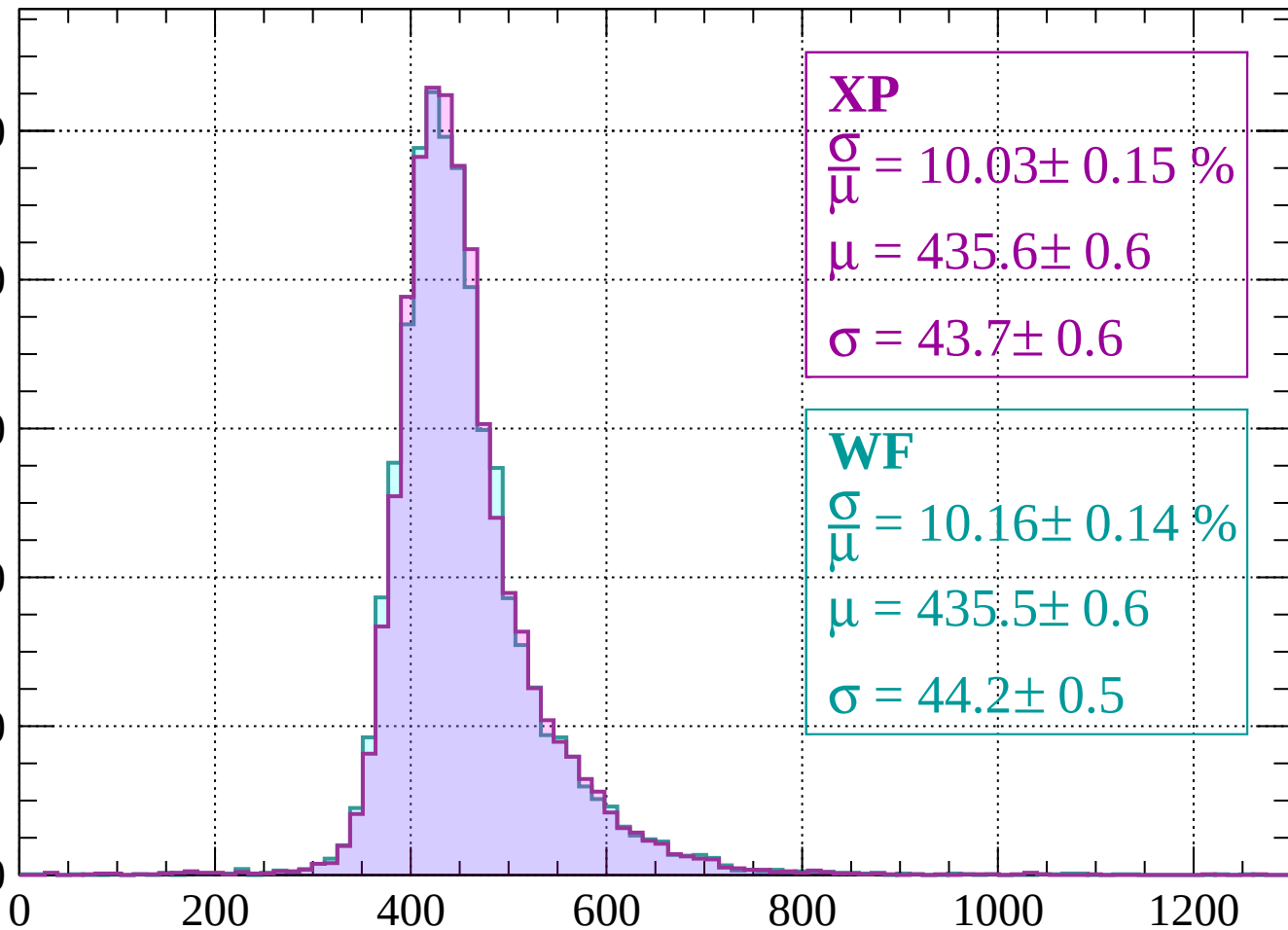
800

600

400

200

0



dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count

1000

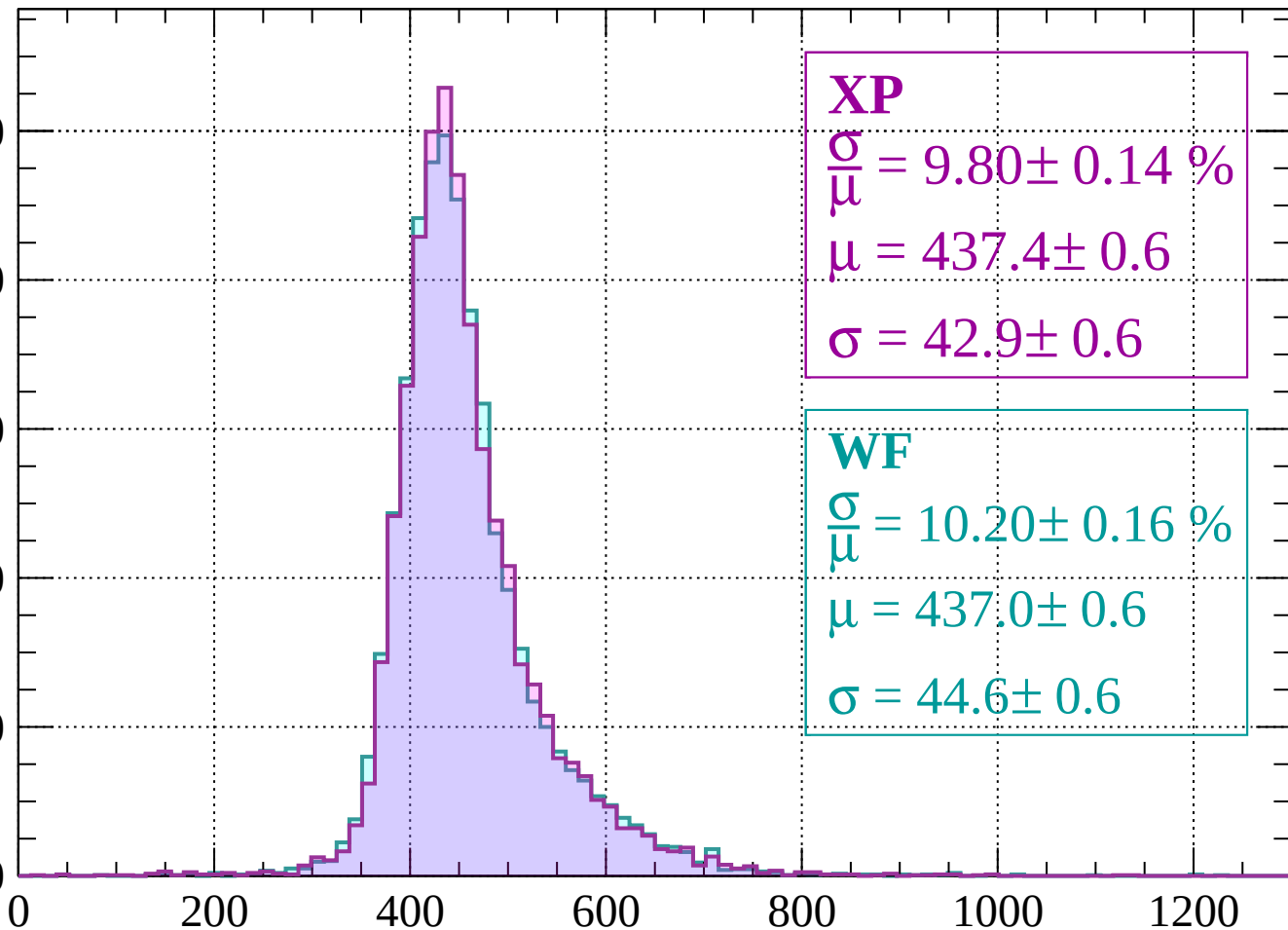
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 9.80 \pm 0.14 \%$$

$$\mu = 437.4 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.16 \%$$

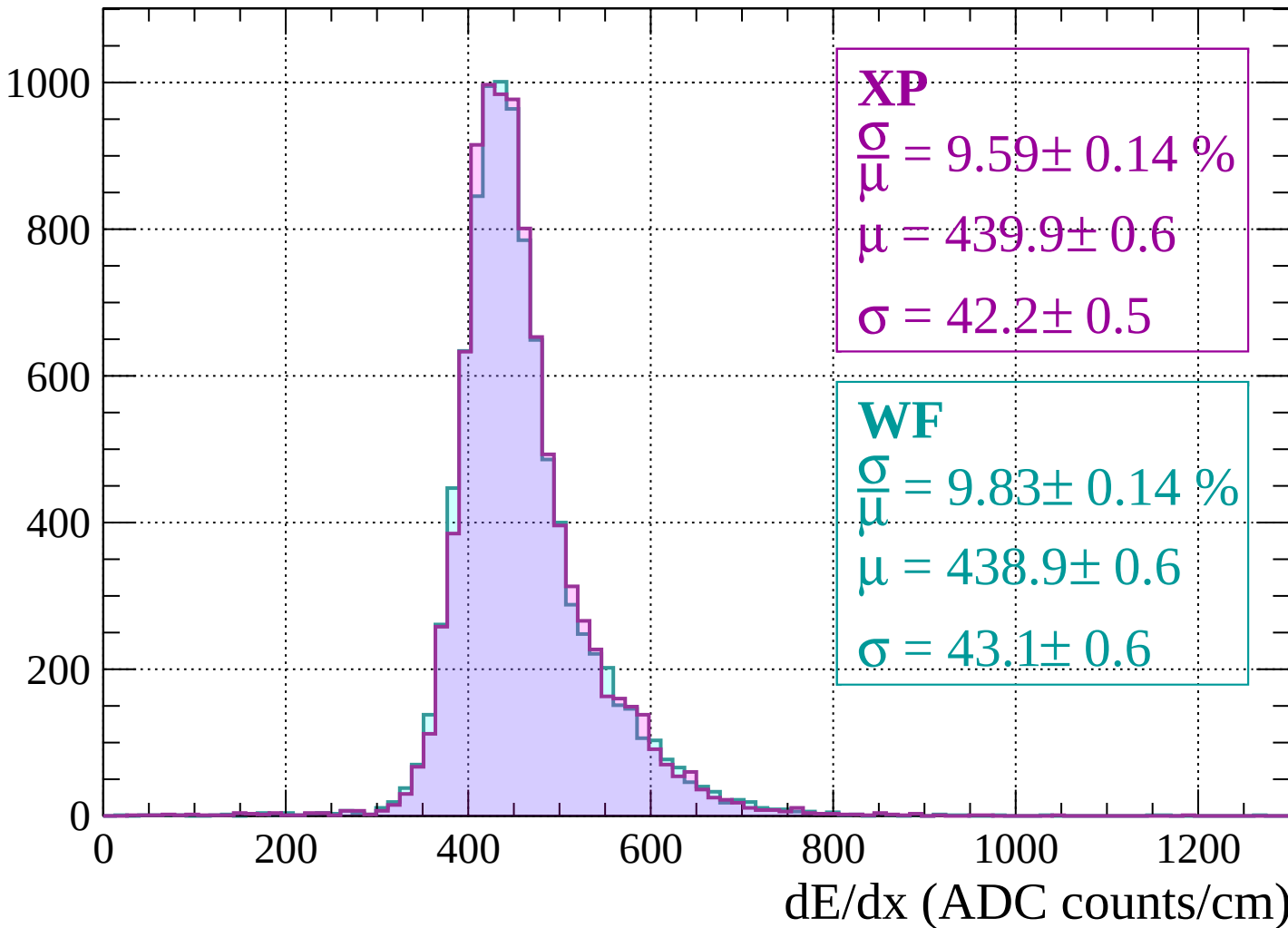
$$\mu = 437.0 \pm 0.6$$

$$\sigma = 44.6 \pm 0.6$$

dE/dx (ADC counts/cm)

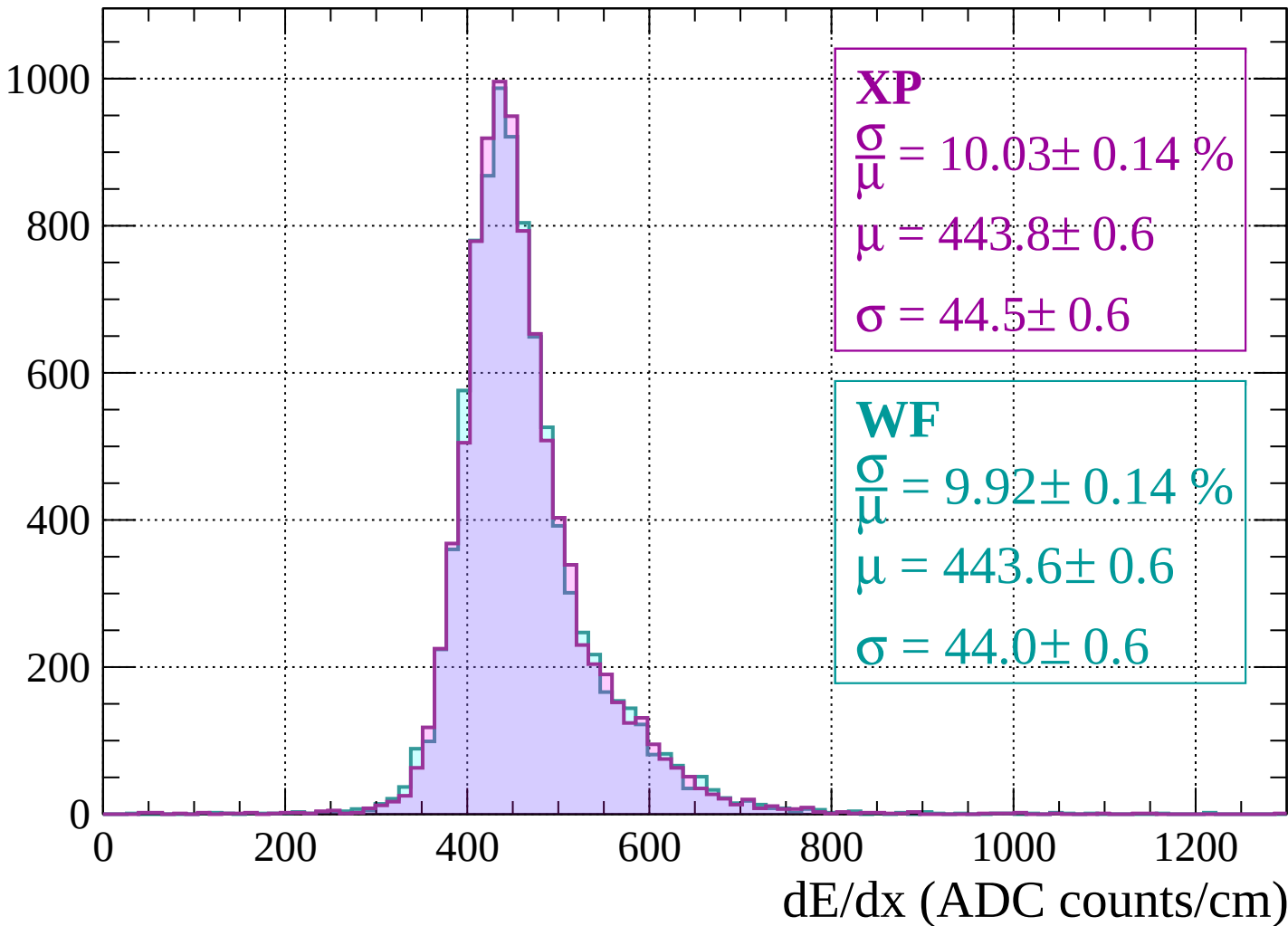
Energy loss | $1000 < p < 1040$

Count

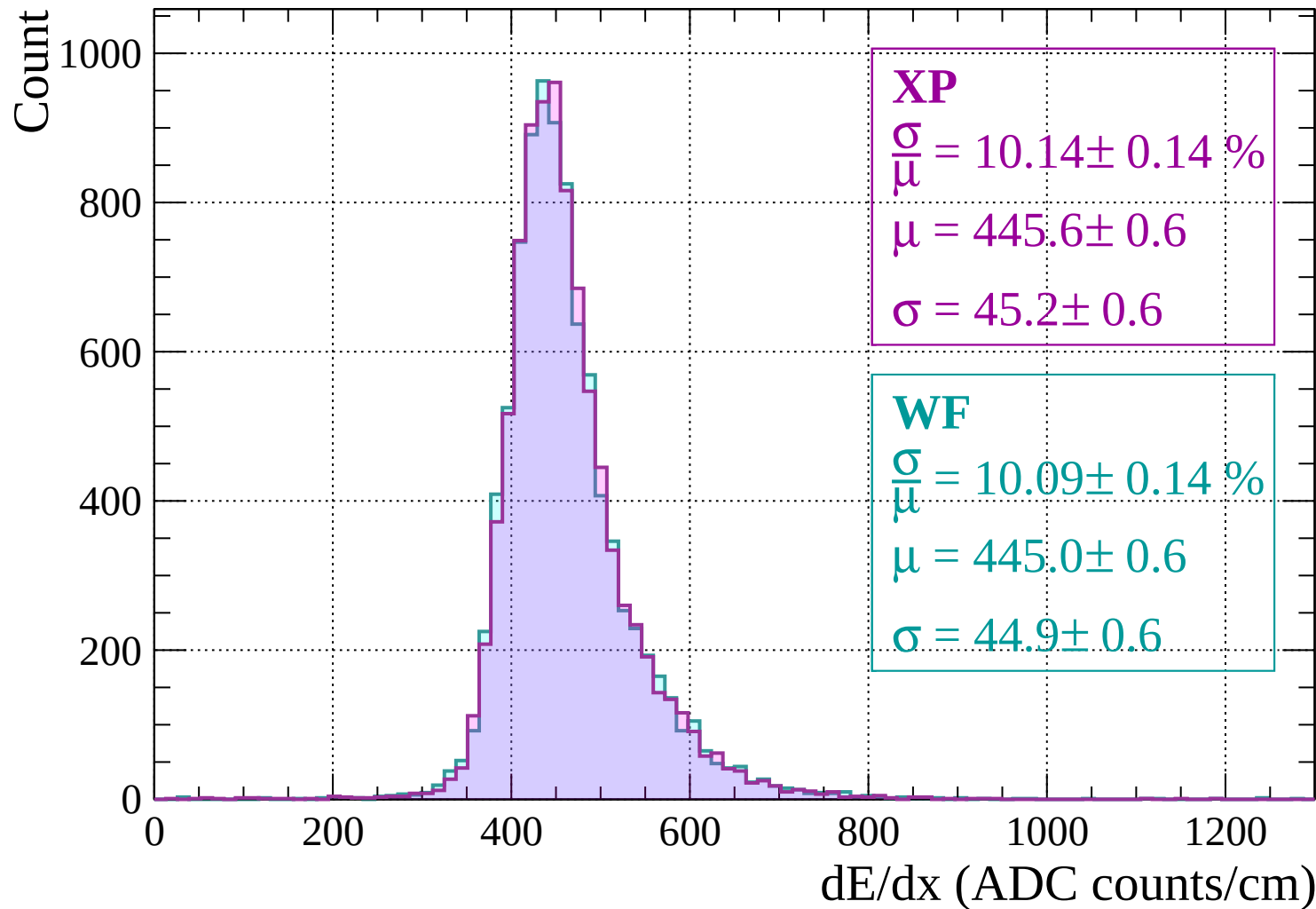


Energy loss | $1040 < p < 1080$

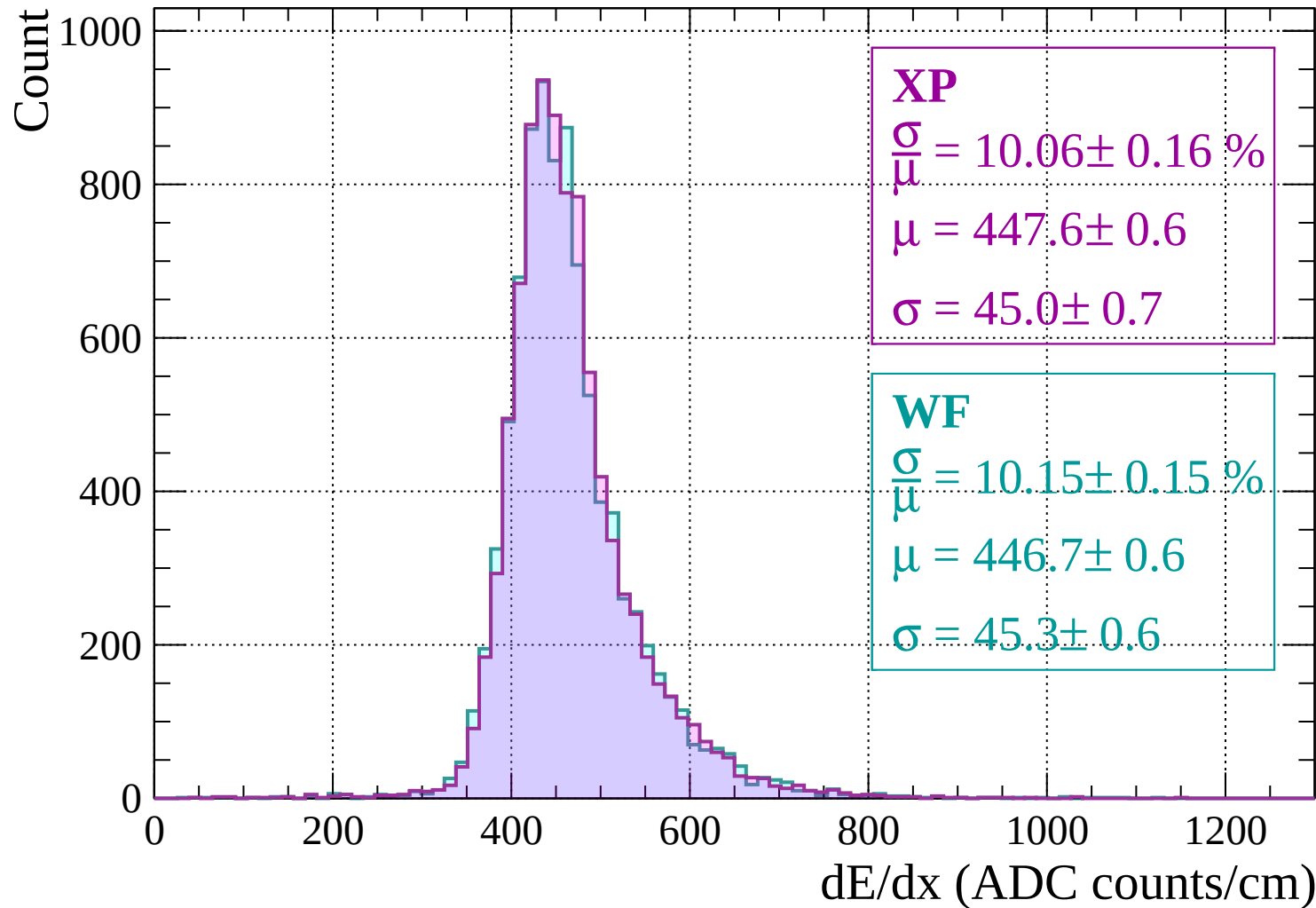
Count



Energy loss | $1080 < p < 1120$

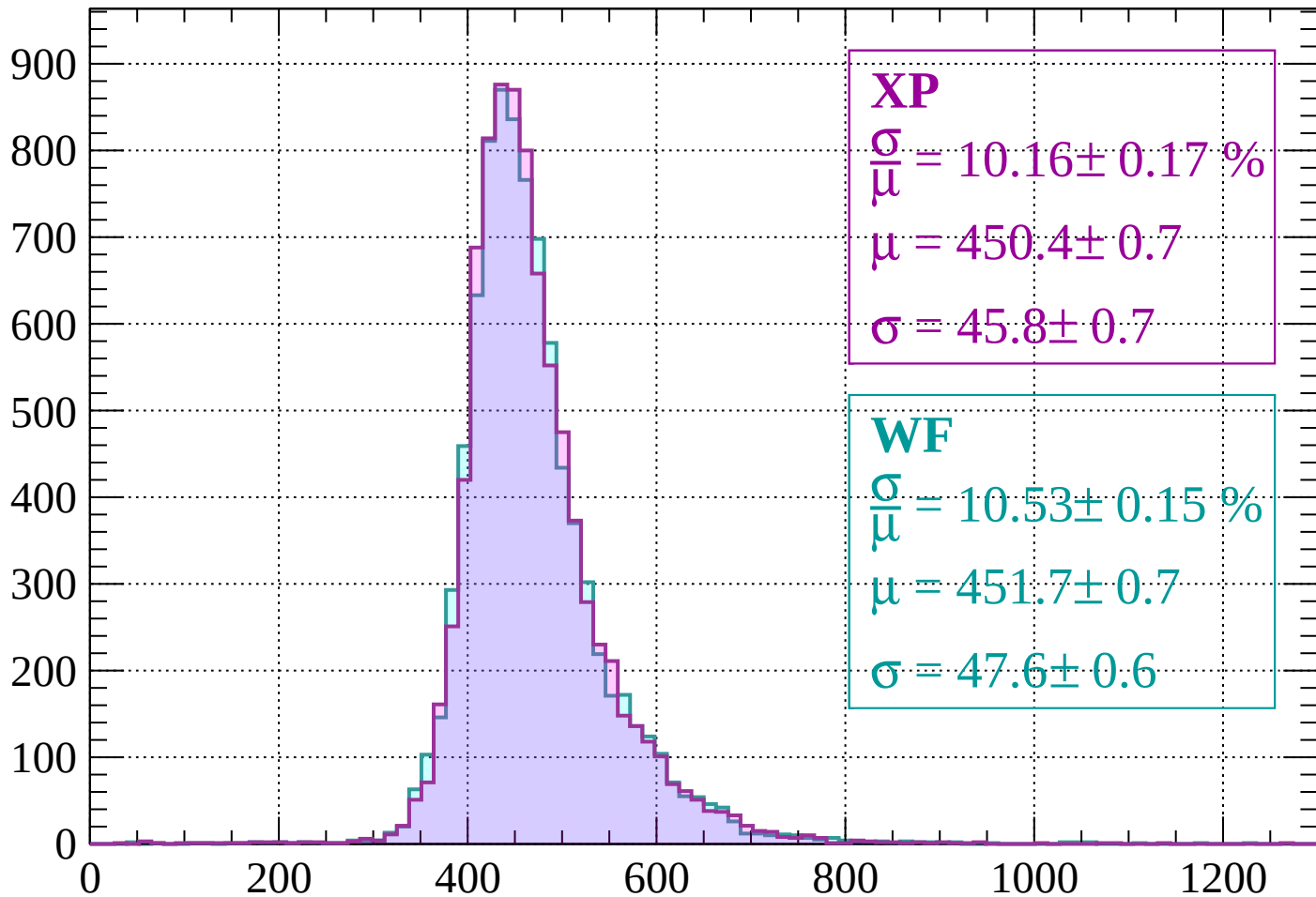


Energy loss | $1120 < p < 1160$



Energy loss | $1160 < p < 1200$

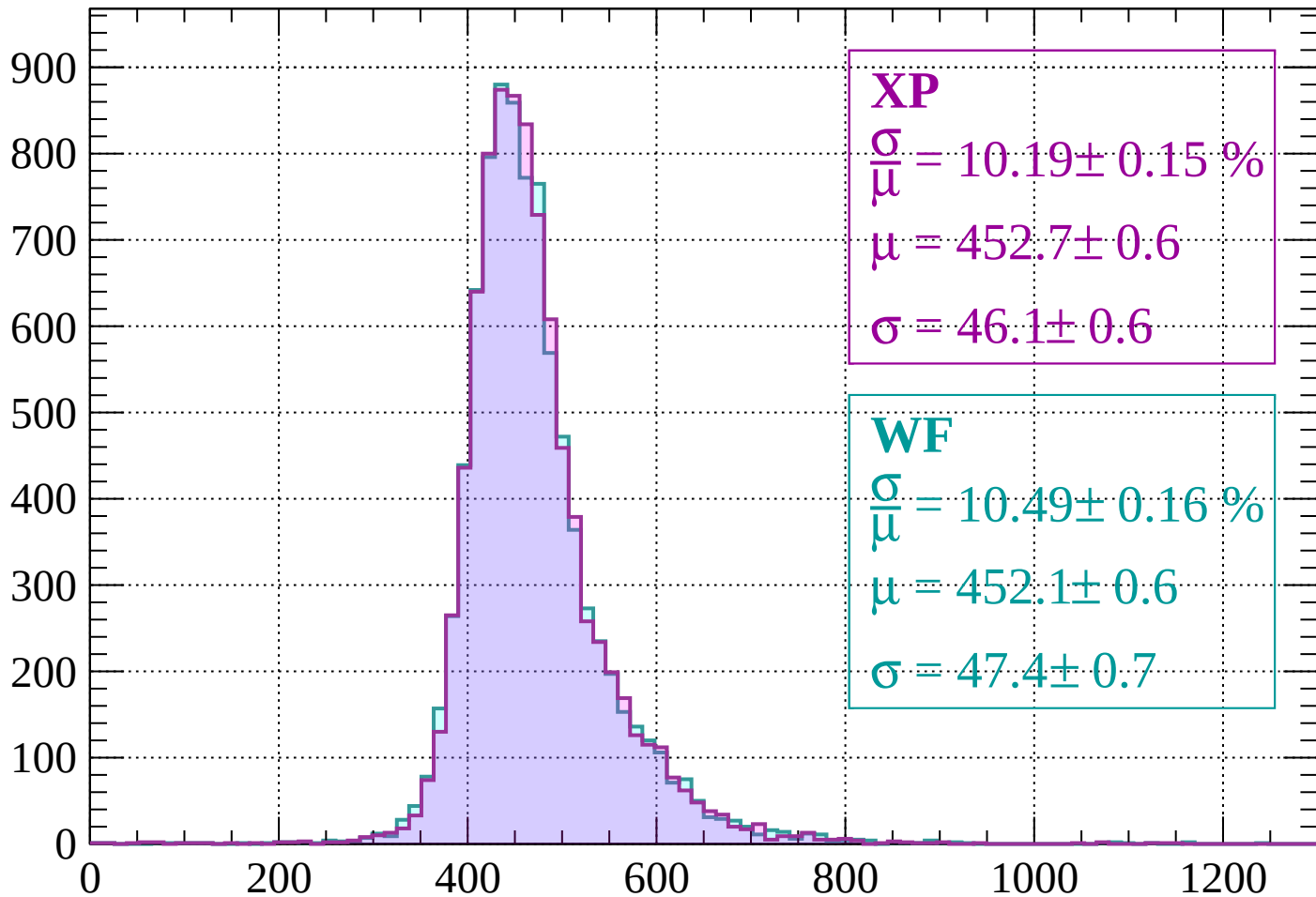
Count



dE/dx (ADC counts/cm)

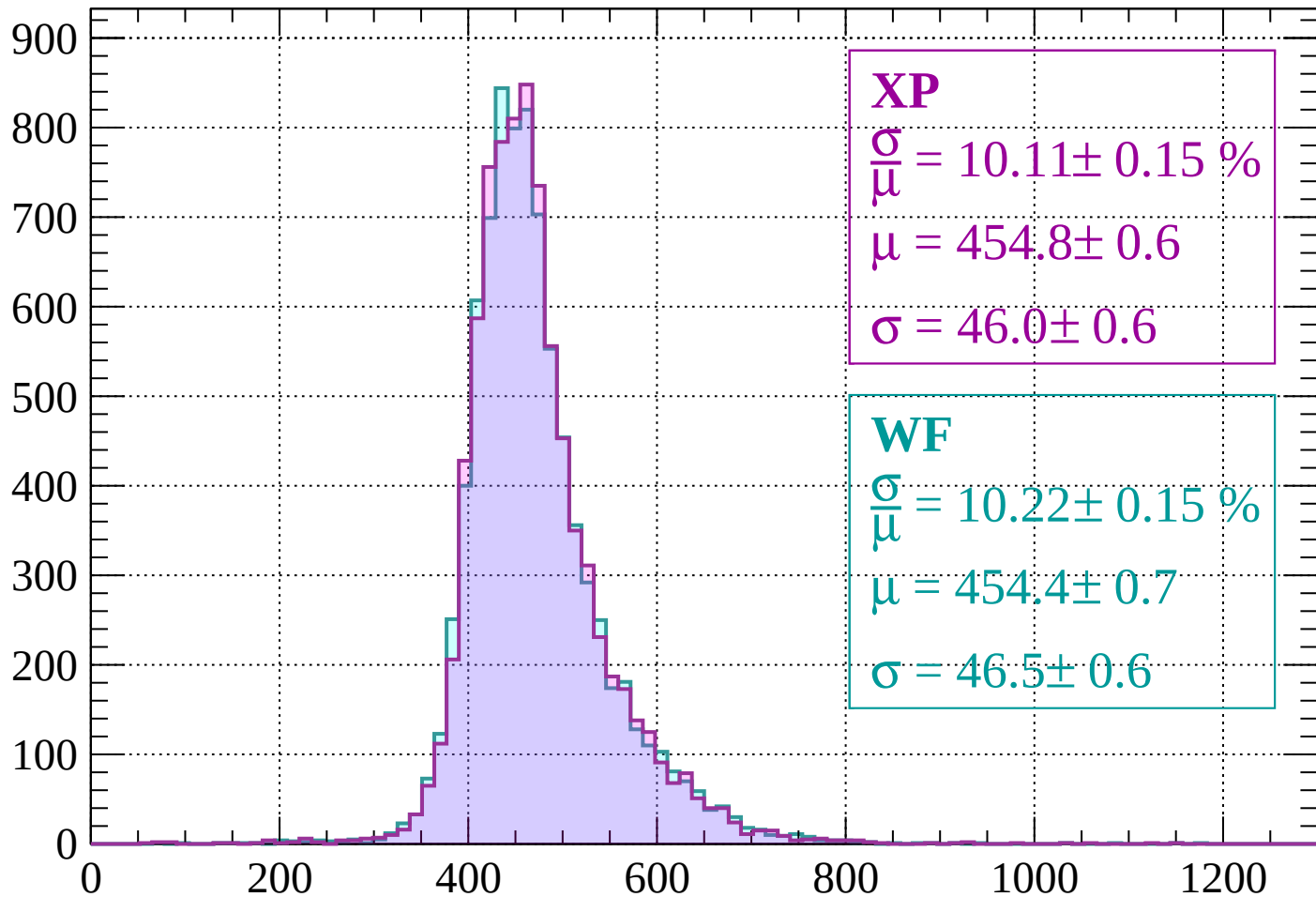
Energy loss | $1200 < p < 1240$

Count



Energy loss | $1240 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 10.11 \pm 0.15 \%$$

$$\mu = 454.8 \pm 0.6$$

$$\sigma = 46.0 \pm 0.6$$

WF

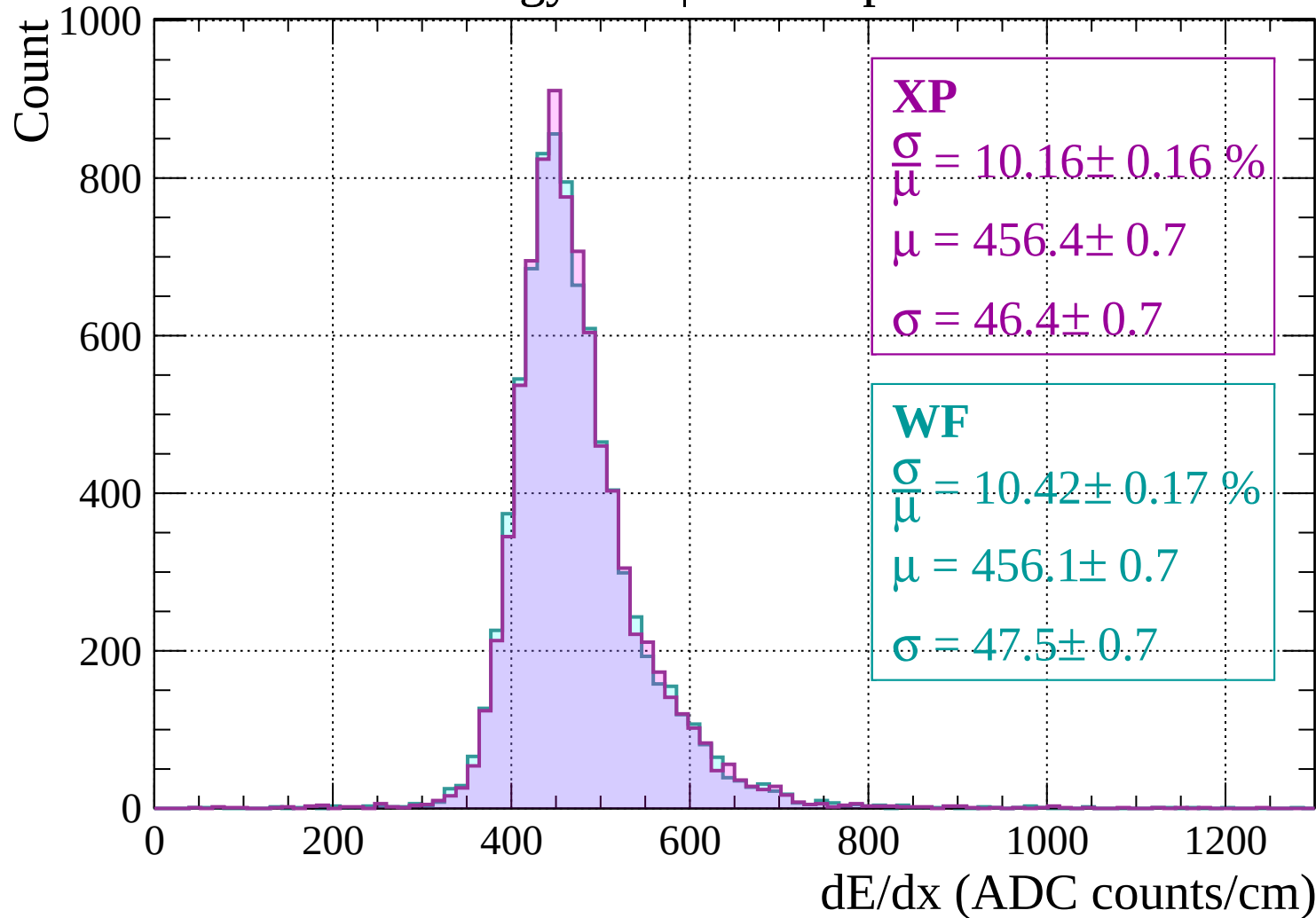
$$\frac{\sigma}{\mu} = 10.22 \pm 0.15 \%$$

$$\mu = 454.4 \pm 0.7$$

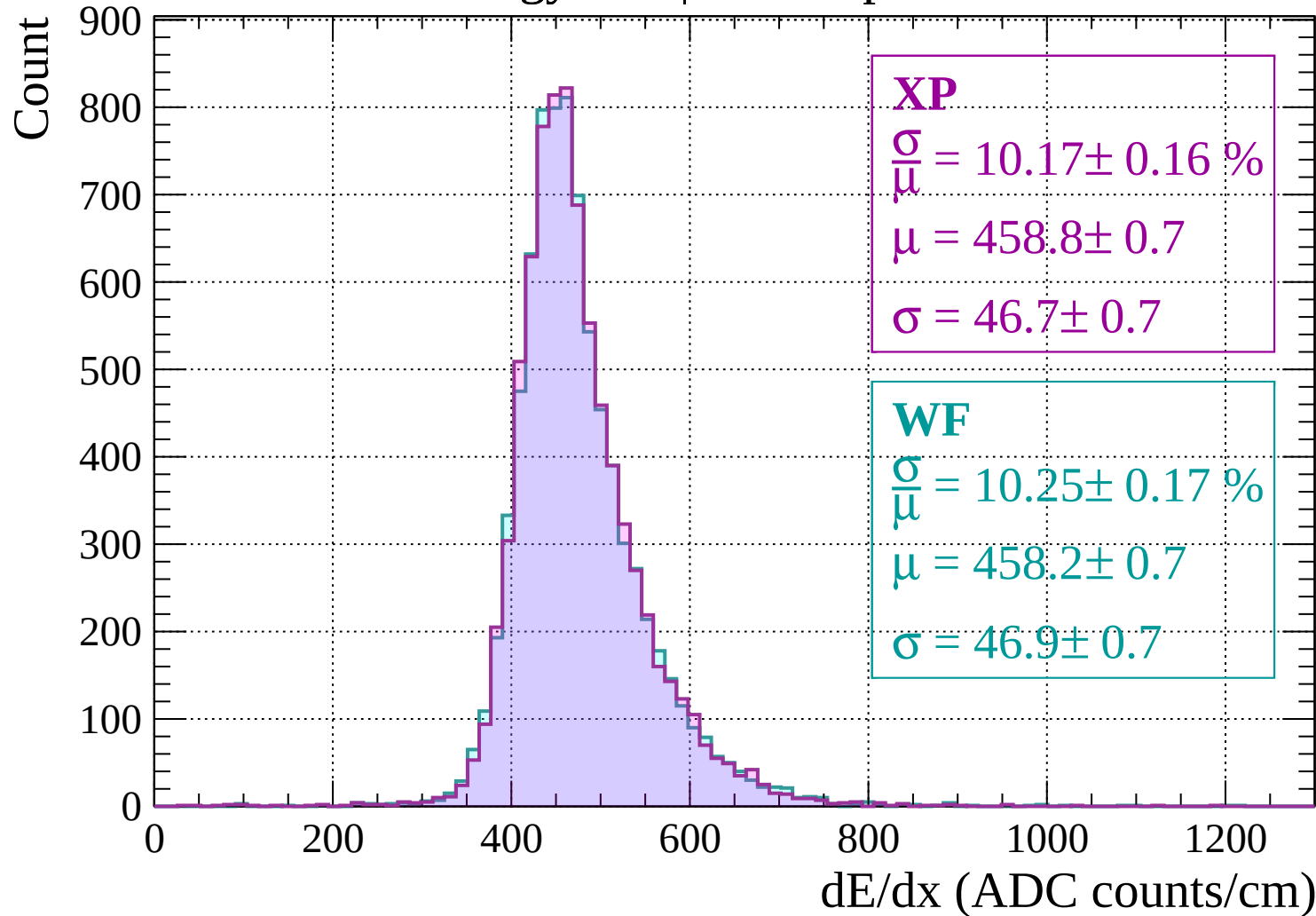
$$\sigma = 46.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

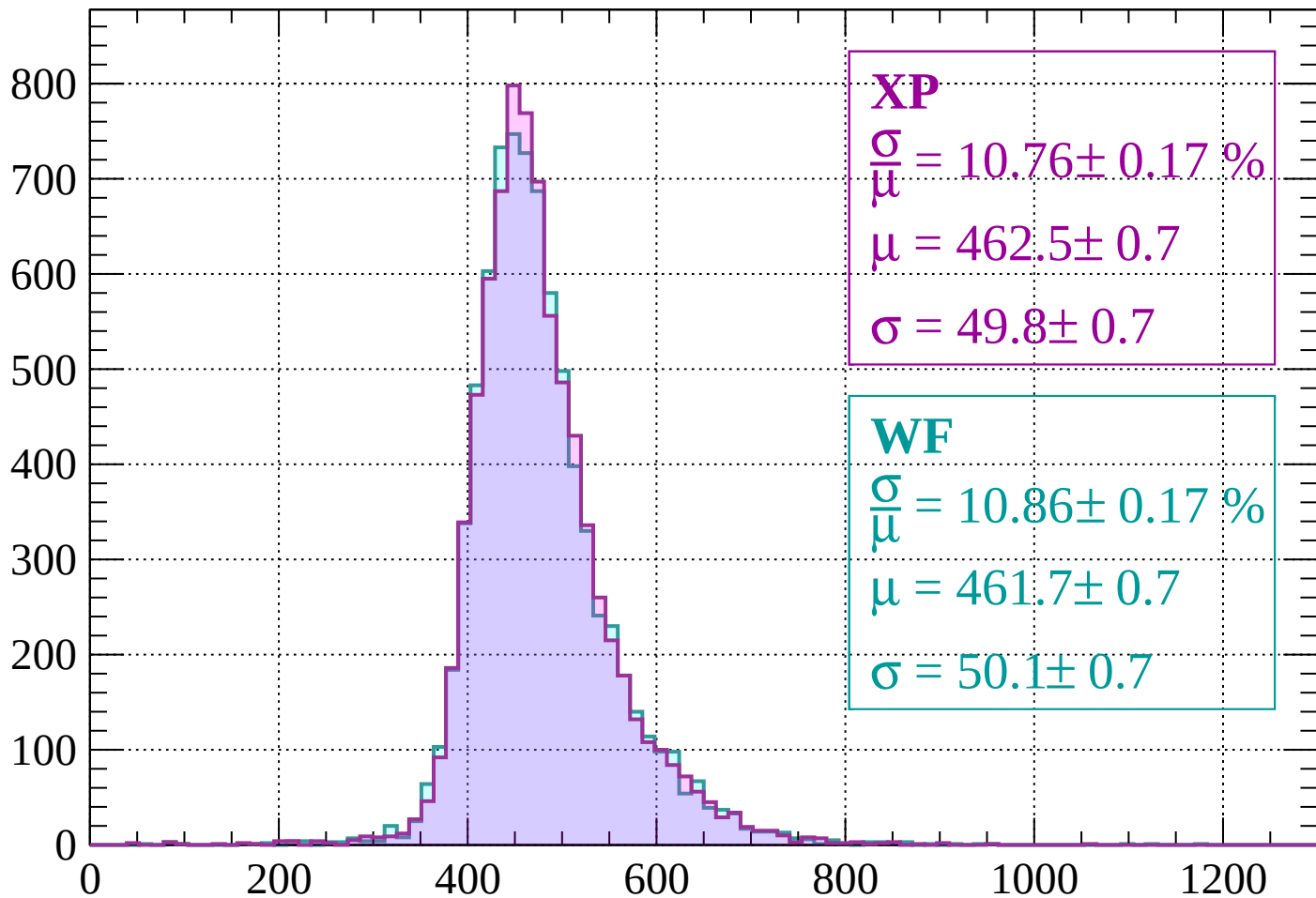


Energy loss | $1320 < p < 1360$



Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 10.76 \pm 0.17 \%$$

$$\mu = 462.5 \pm 0.7$$

$$\sigma = 49.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.17 \%$$

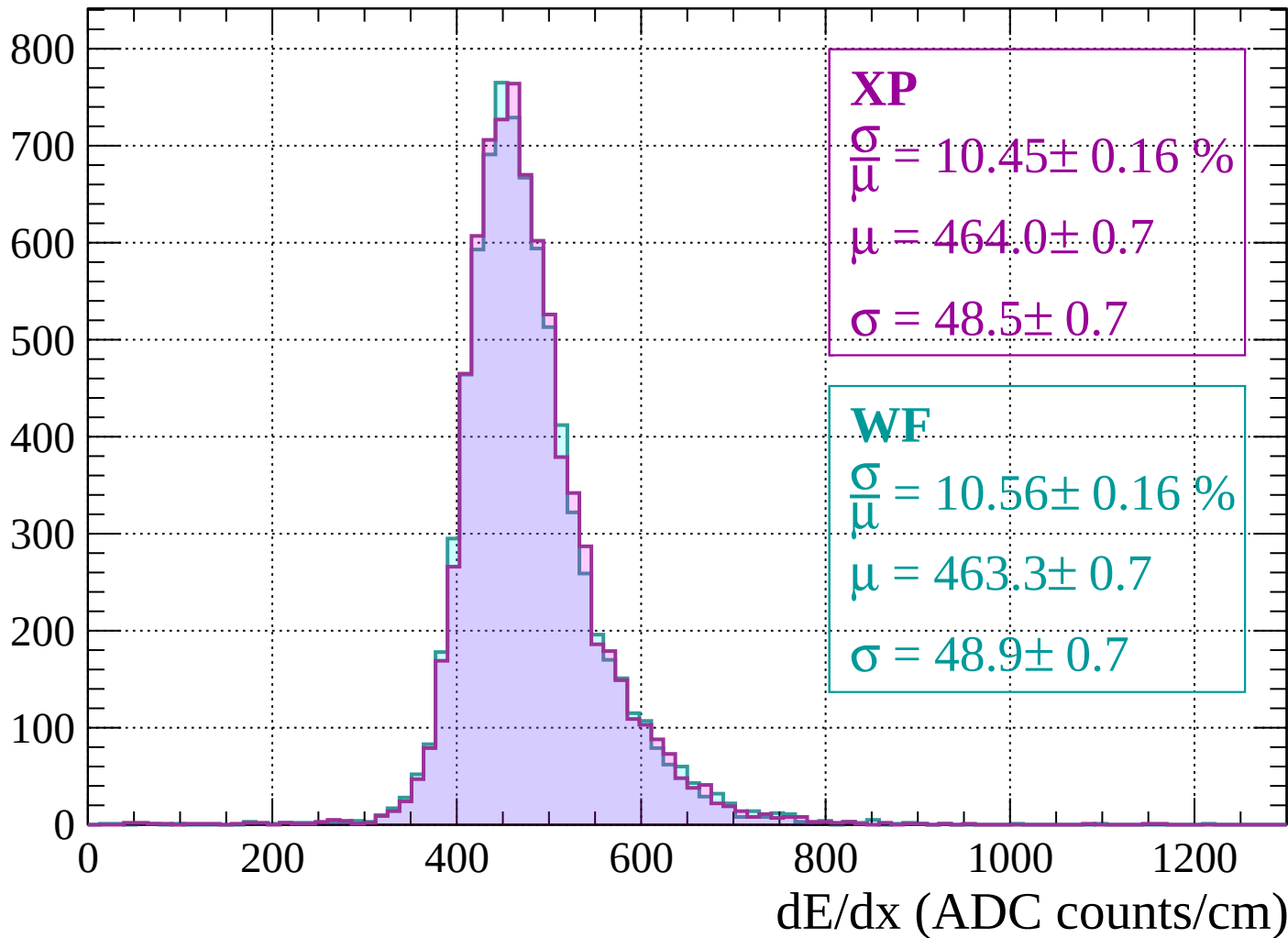
$$\mu = 461.7 \pm 0.7$$

$$\sigma = 50.1 \pm 0.7$$

dE/dx (ADC counts/cm)

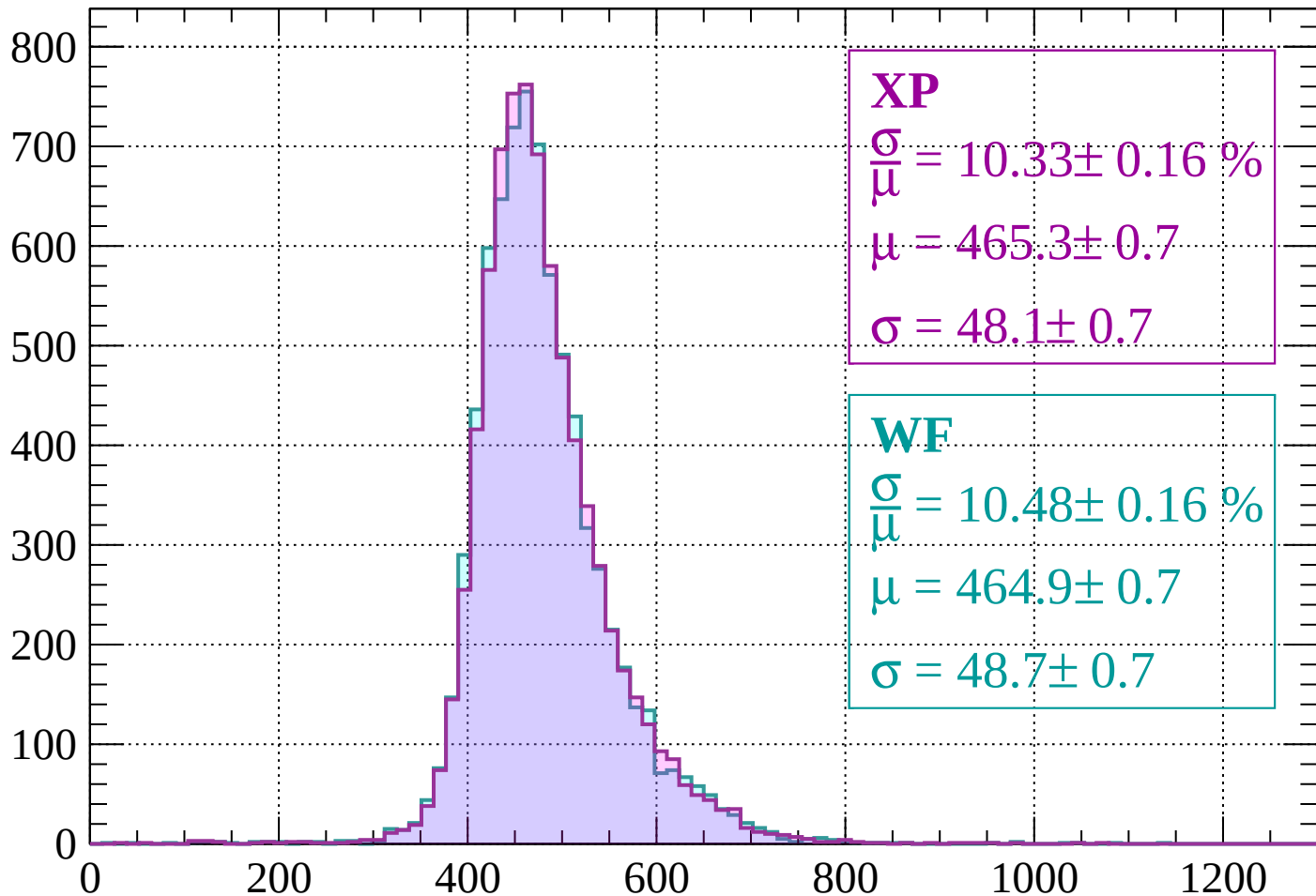
Energy loss | $1400 < p < 1440$

Count



Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.16 \%$$

$$\mu = 465.3 \pm 0.7$$

$$\sigma = 48.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.16 \%$$

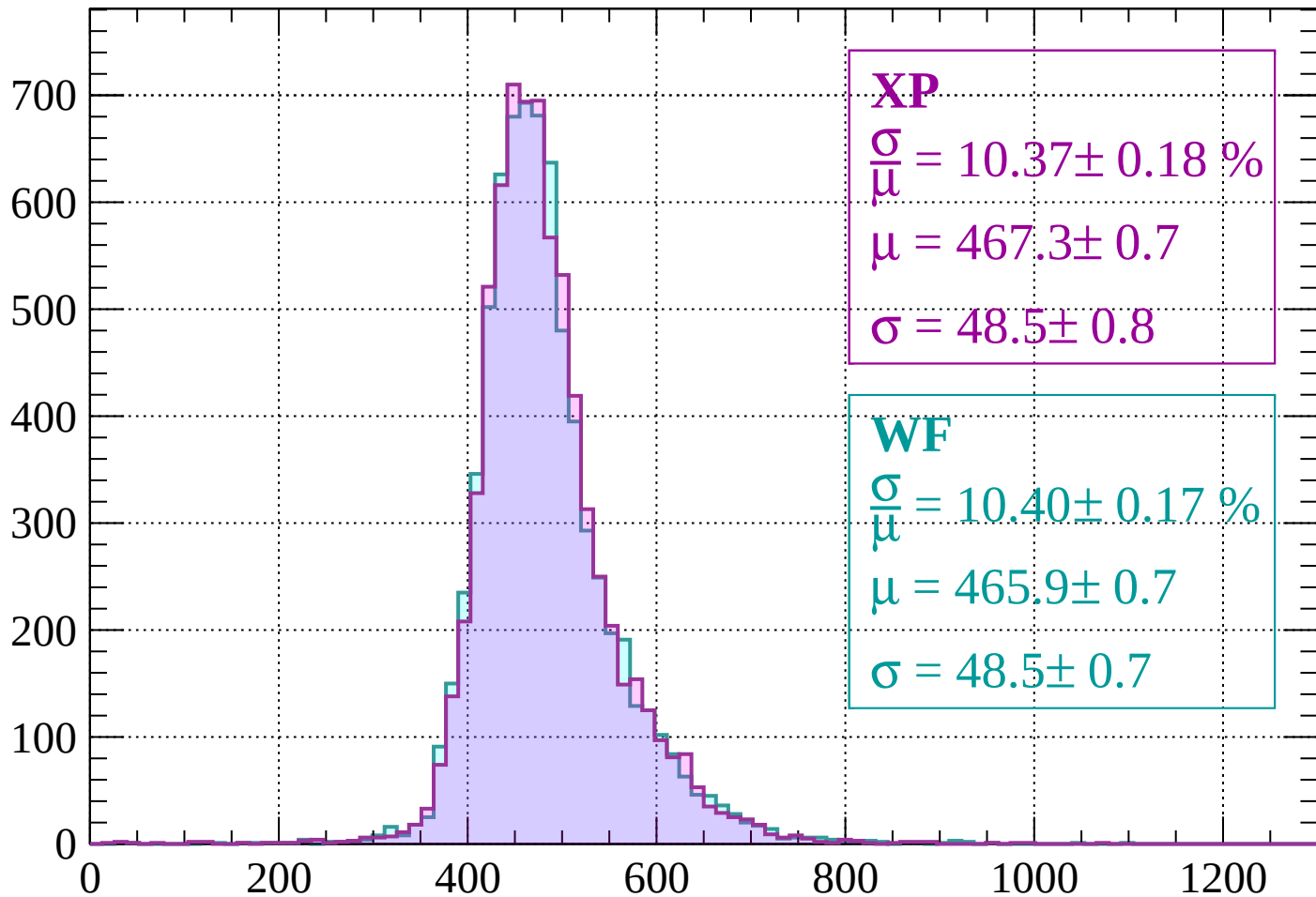
$$\mu = 464.9 \pm 0.7$$

$$\sigma = 48.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 10.37 \pm 0.18 \%$$

$$\mu = 467.3 \pm 0.7$$

$$\sigma = 48.5 \pm 0.8$$

WF

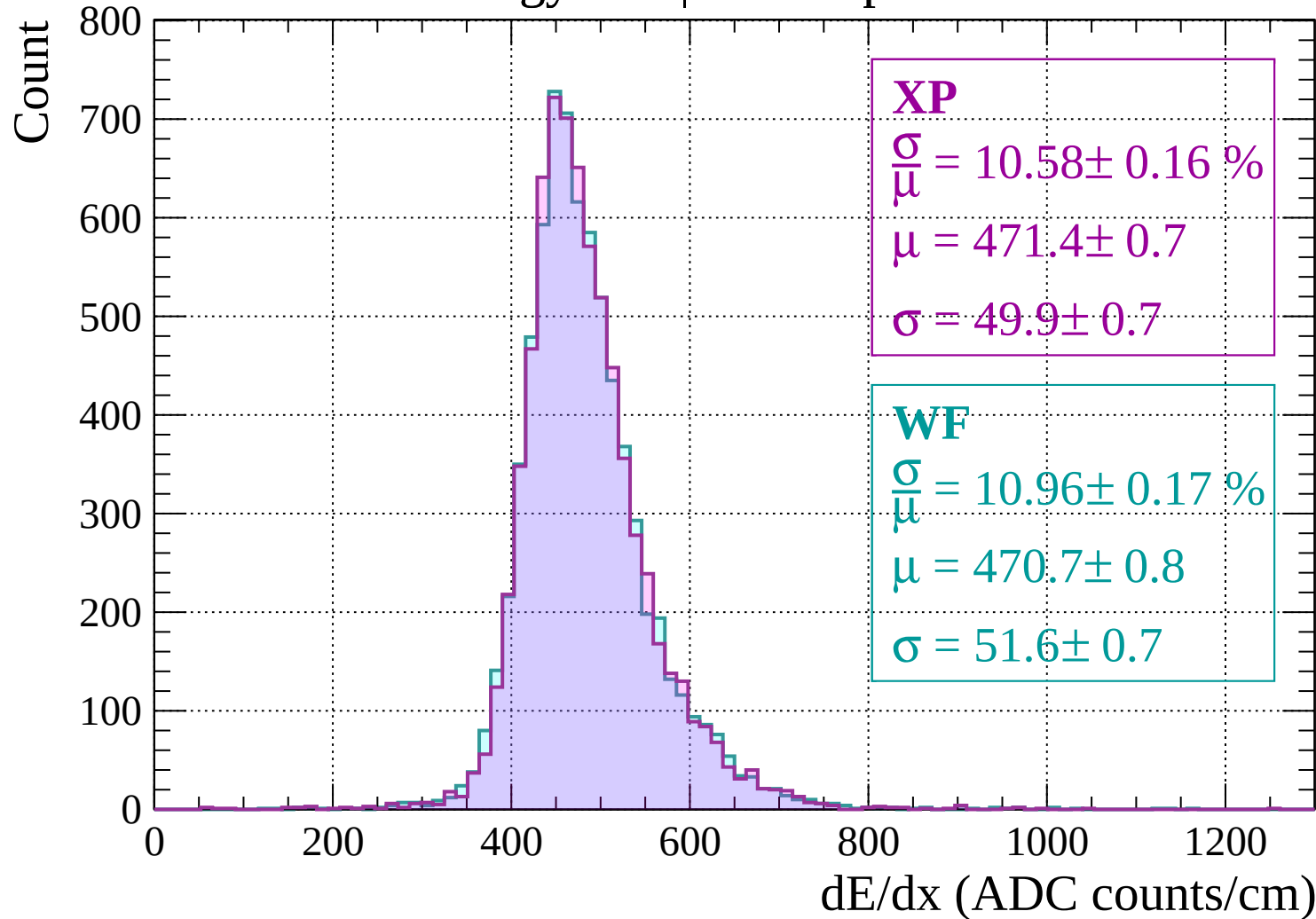
$$\frac{\sigma}{\mu} = 10.40 \pm 0.17 \%$$

$$\mu = 465.9 \pm 0.7$$

$$\sigma = 48.5 \pm 0.7$$

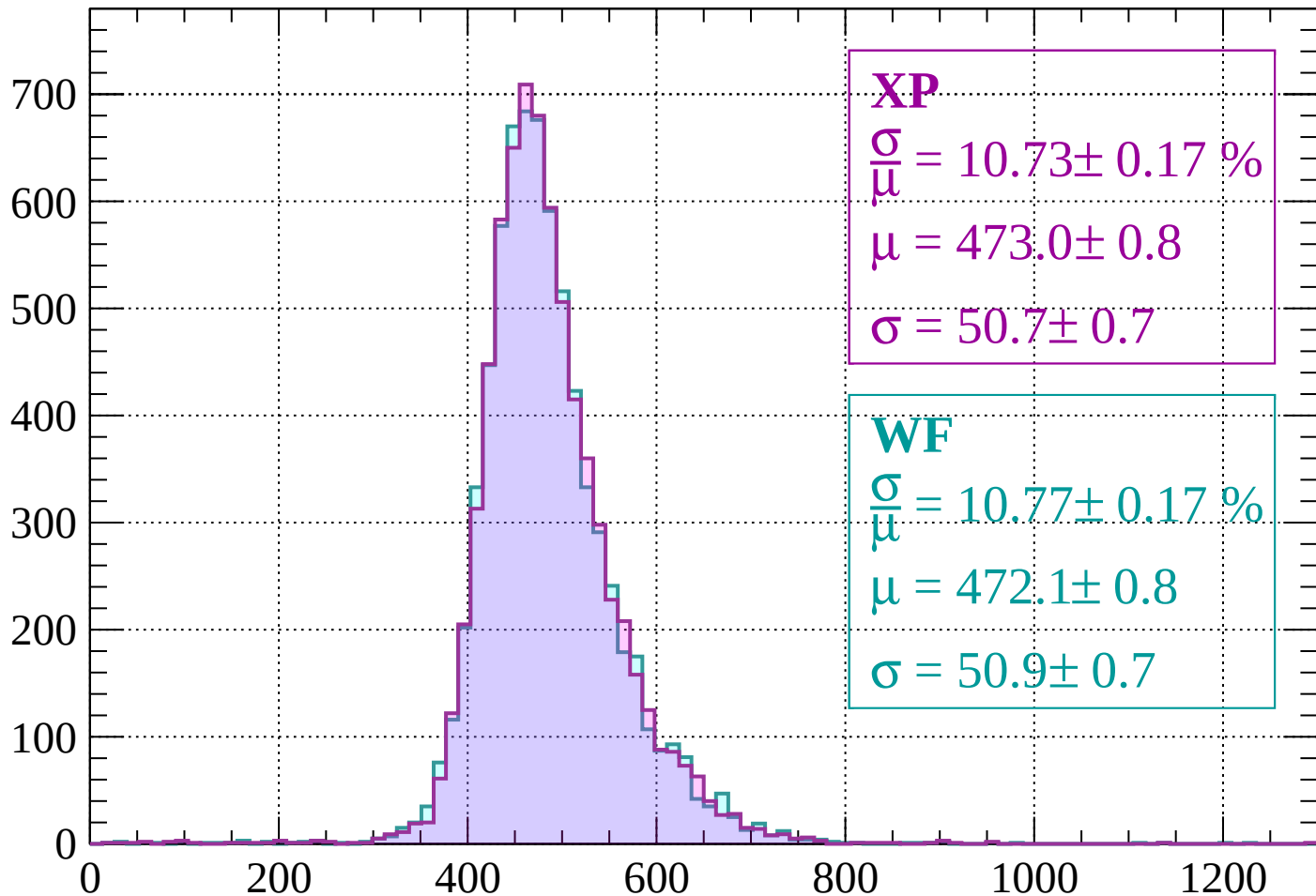
dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$



Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 10.73 \pm 0.17 \%$$

$$\mu = 473.0 \pm 0.8$$

$$\sigma = 50.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.17 \%$$

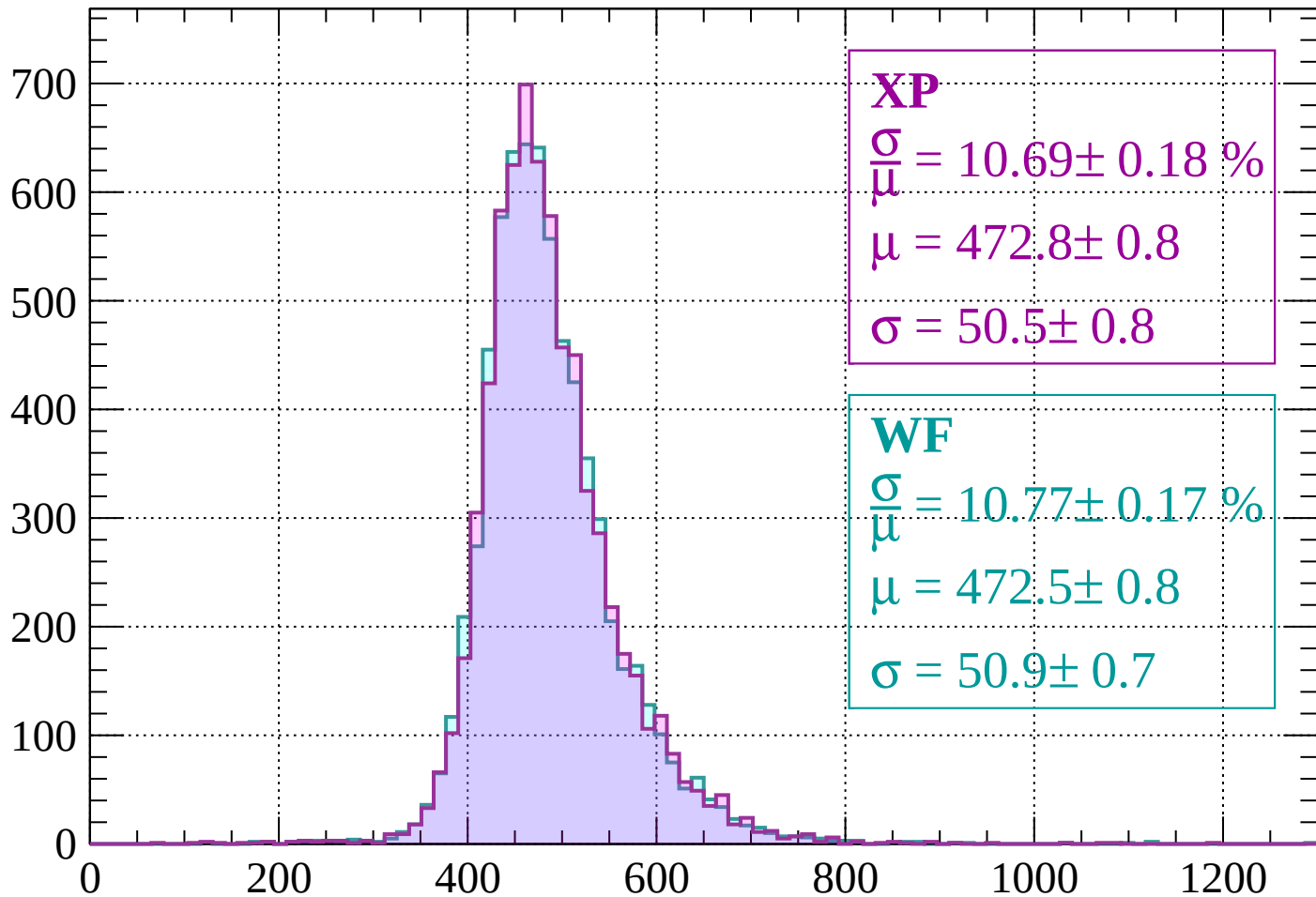
$$\mu = 472.1 \pm 0.8$$

$$\sigma = 50.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count



XP

$$\frac{\sigma}{\mu} = 10.69 \pm 0.18 \%$$

$$\mu = 472.8 \pm 0.8$$

$$\sigma = 50.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.77 \pm 0.17 \%$$

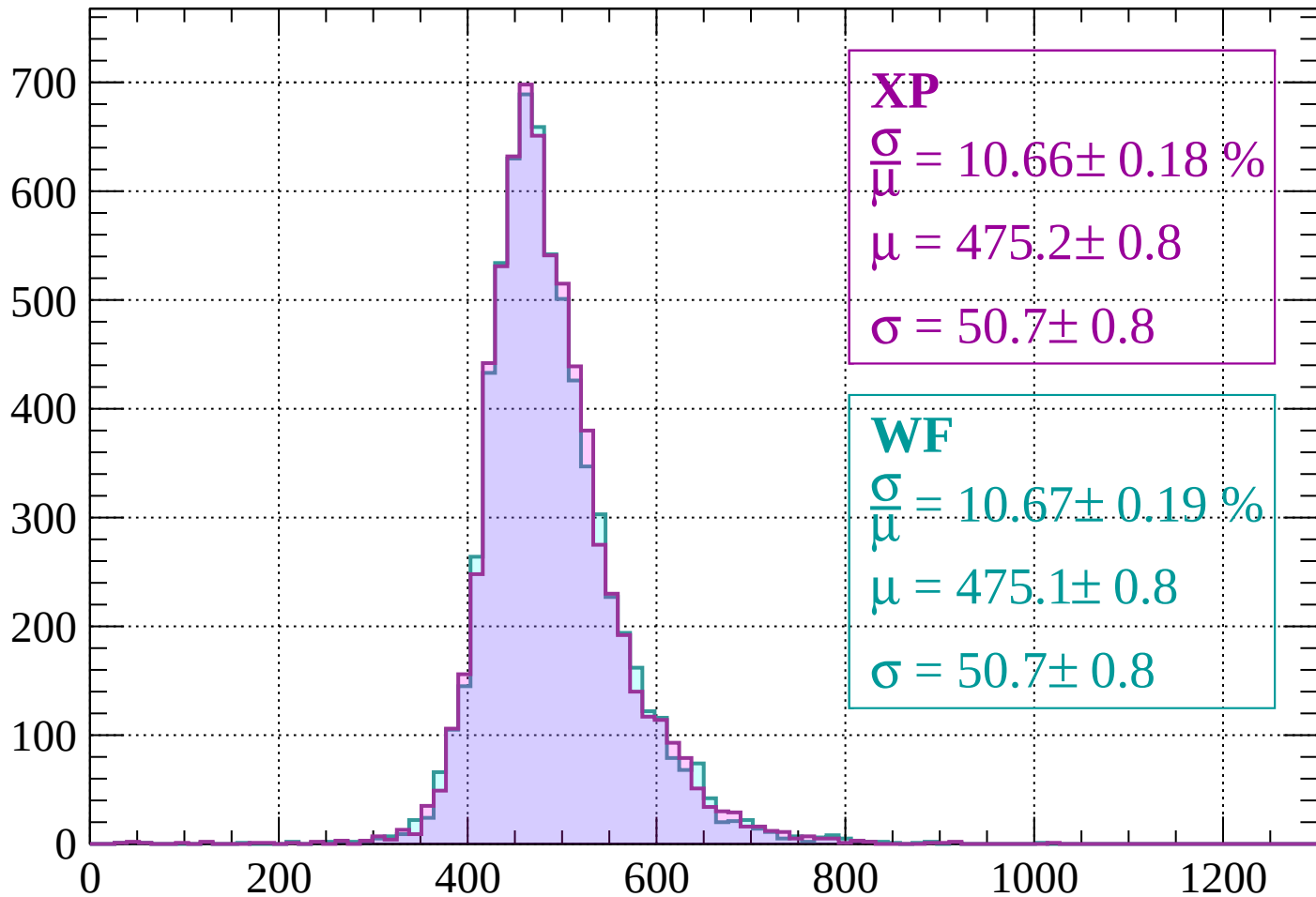
$$\mu = 472.5 \pm 0.8$$

$$\sigma = 50.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1640 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.66 \pm 0.18 \%$$

$$\mu = 475.2 \pm 0.8$$

$$\sigma = 50.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.19 \%$$

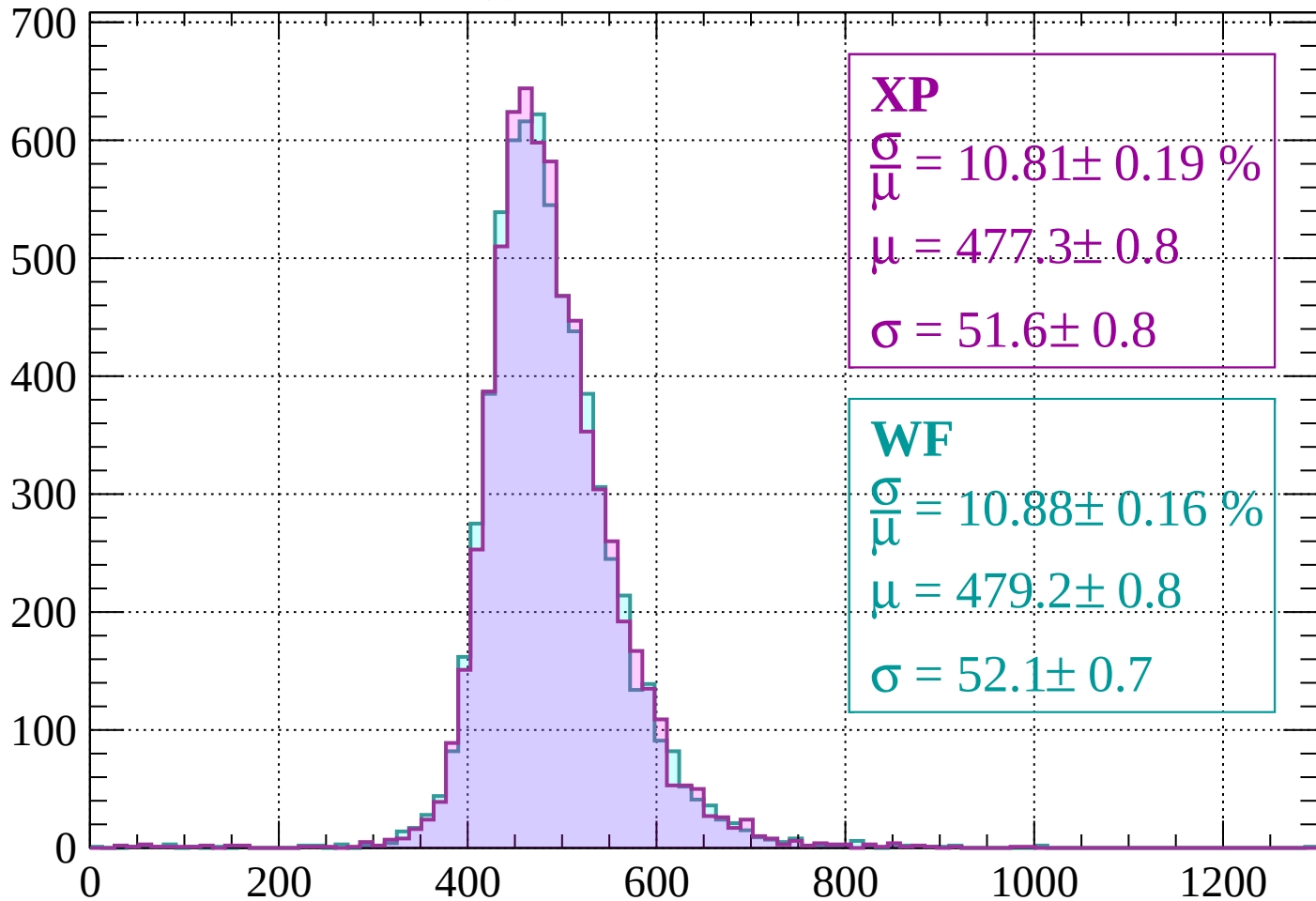
$$\mu = 475.1 \pm 0.8$$

$$\sigma = 50.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 10.81 \pm 0.19 \%$$

$$\mu = 477.3 \pm 0.8$$

$$\sigma = 51.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.88 \pm 0.16 \%$$

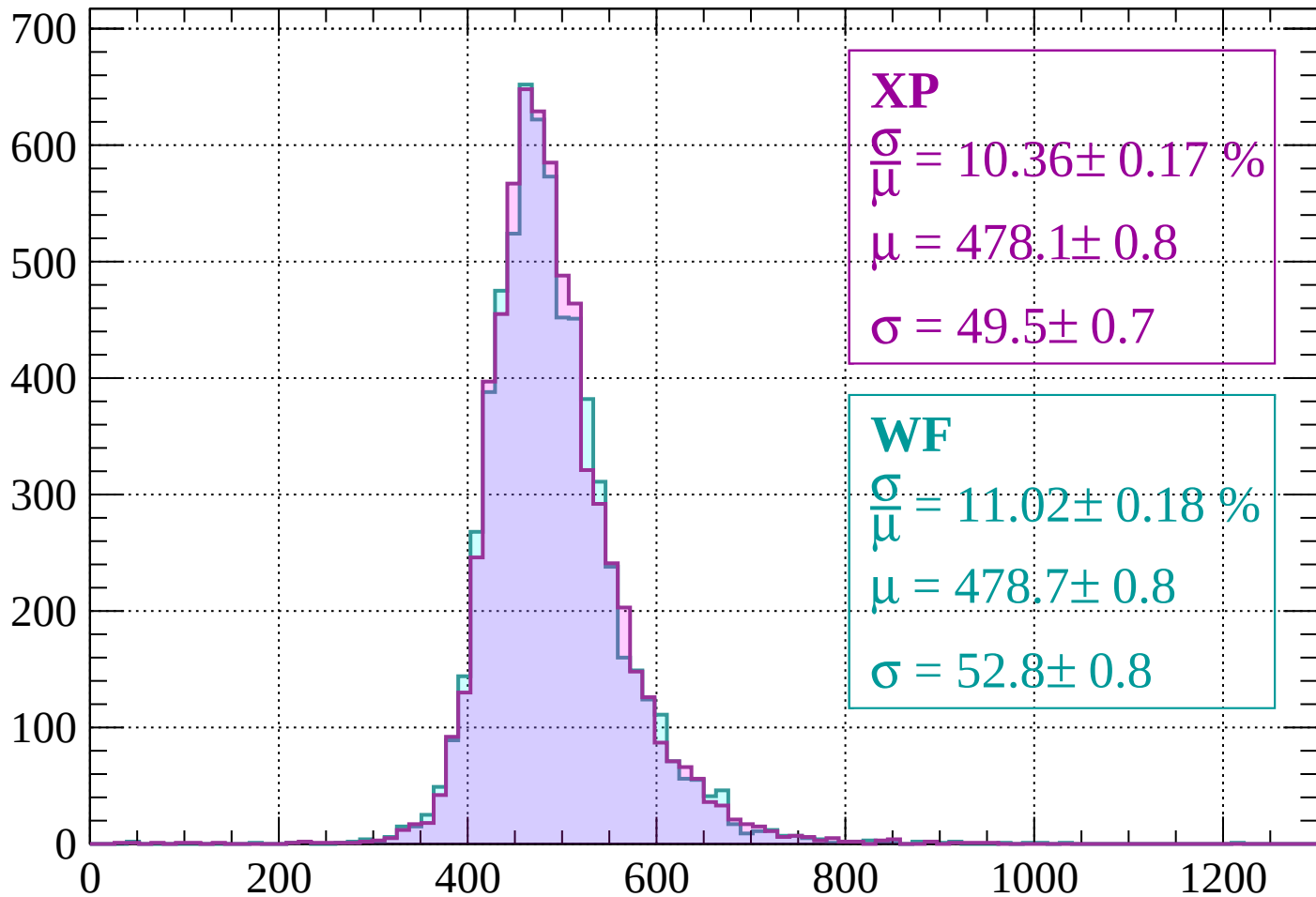
$$\mu = 479.2 \pm 0.8$$

$$\sigma = 52.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.17 \%$$

$$\mu = 478.1 \pm 0.8$$

$$\sigma = 49.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.02 \pm 0.18 \%$$

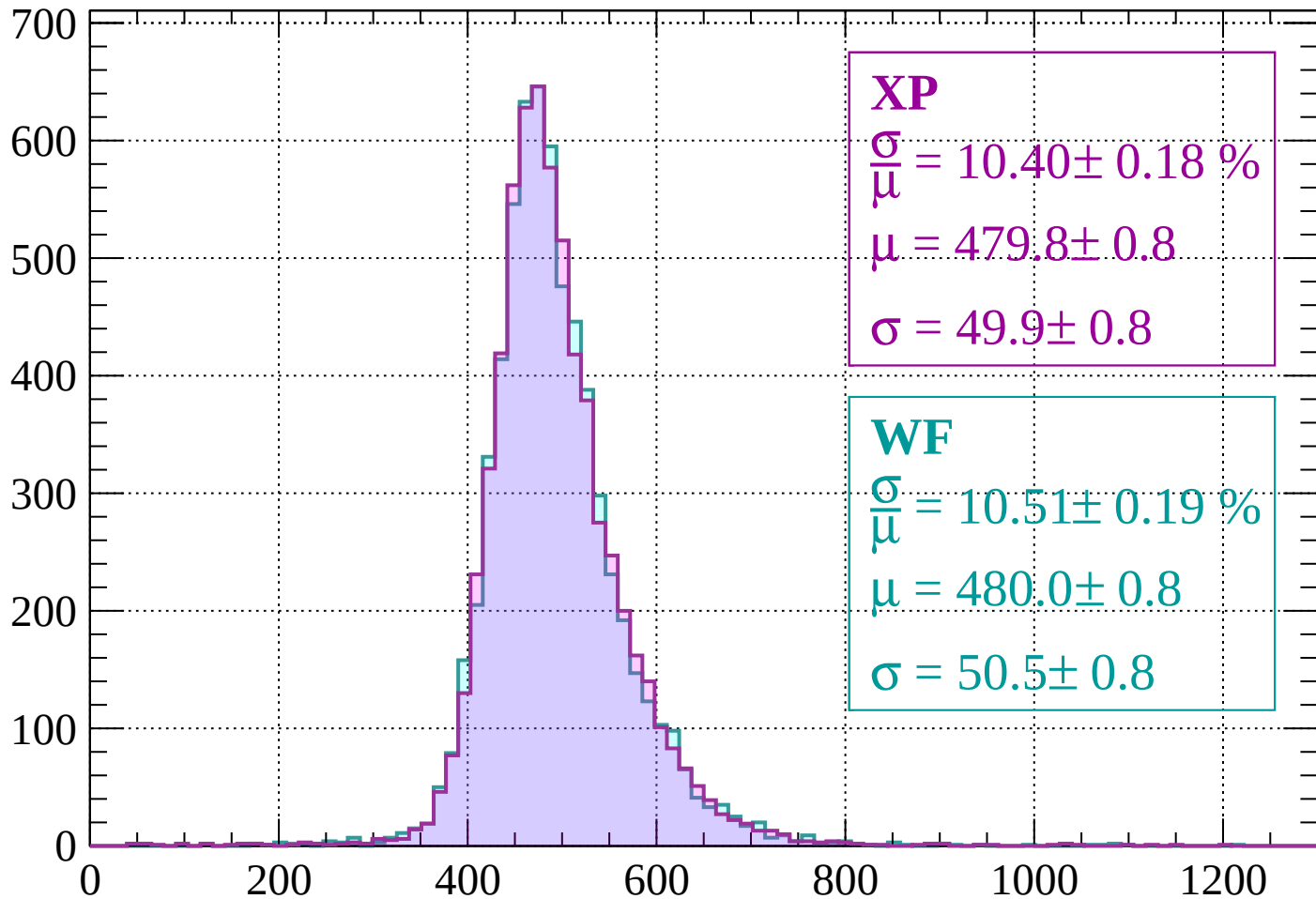
$$\mu = 478.7 \pm 0.8$$

$$\sigma = 52.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

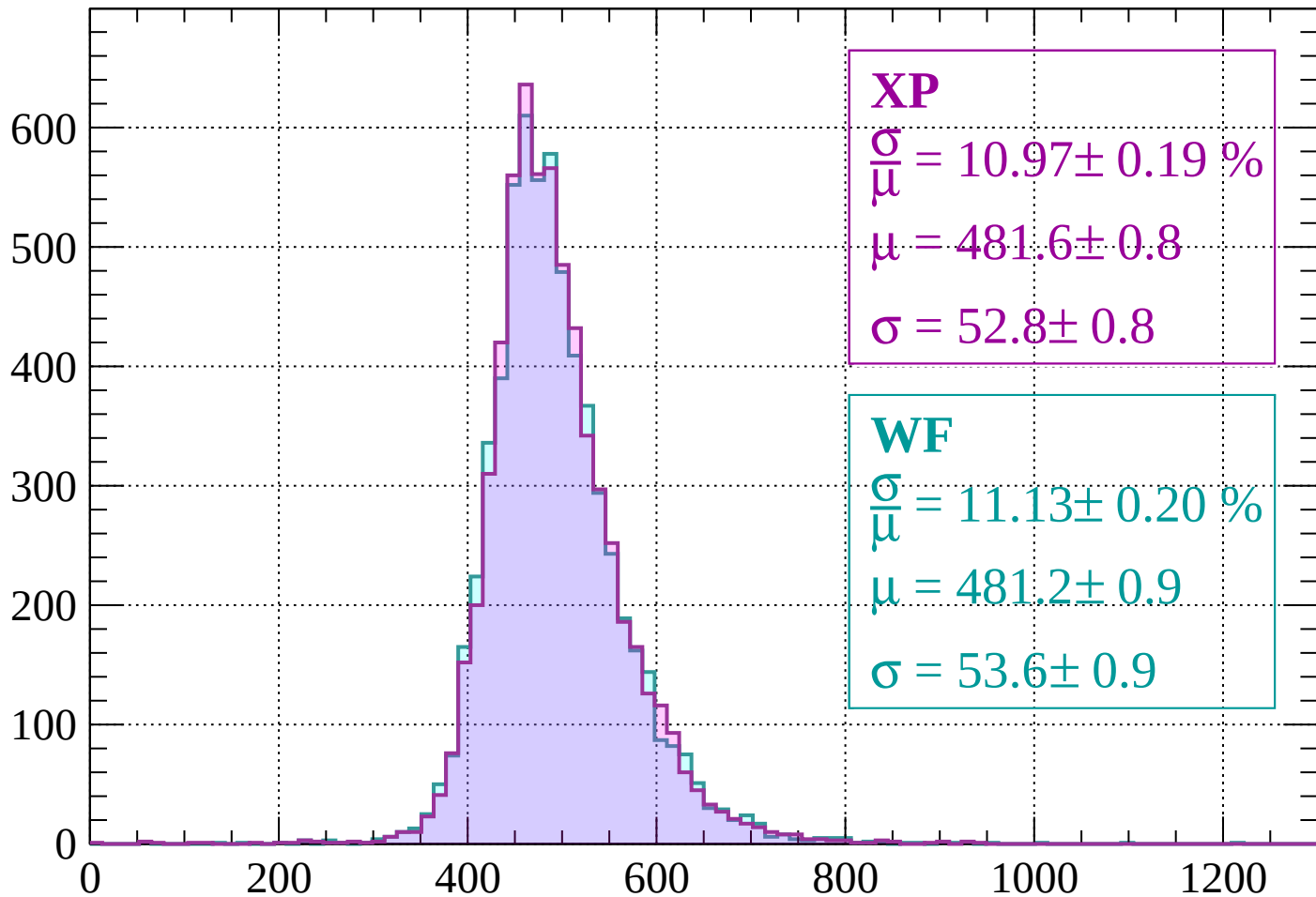
Count



dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 10.97 \pm 0.19 \%$$

$$\mu = 481.6 \pm 0.8$$

$$\sigma = 52.8 \pm 0.8$$

WF

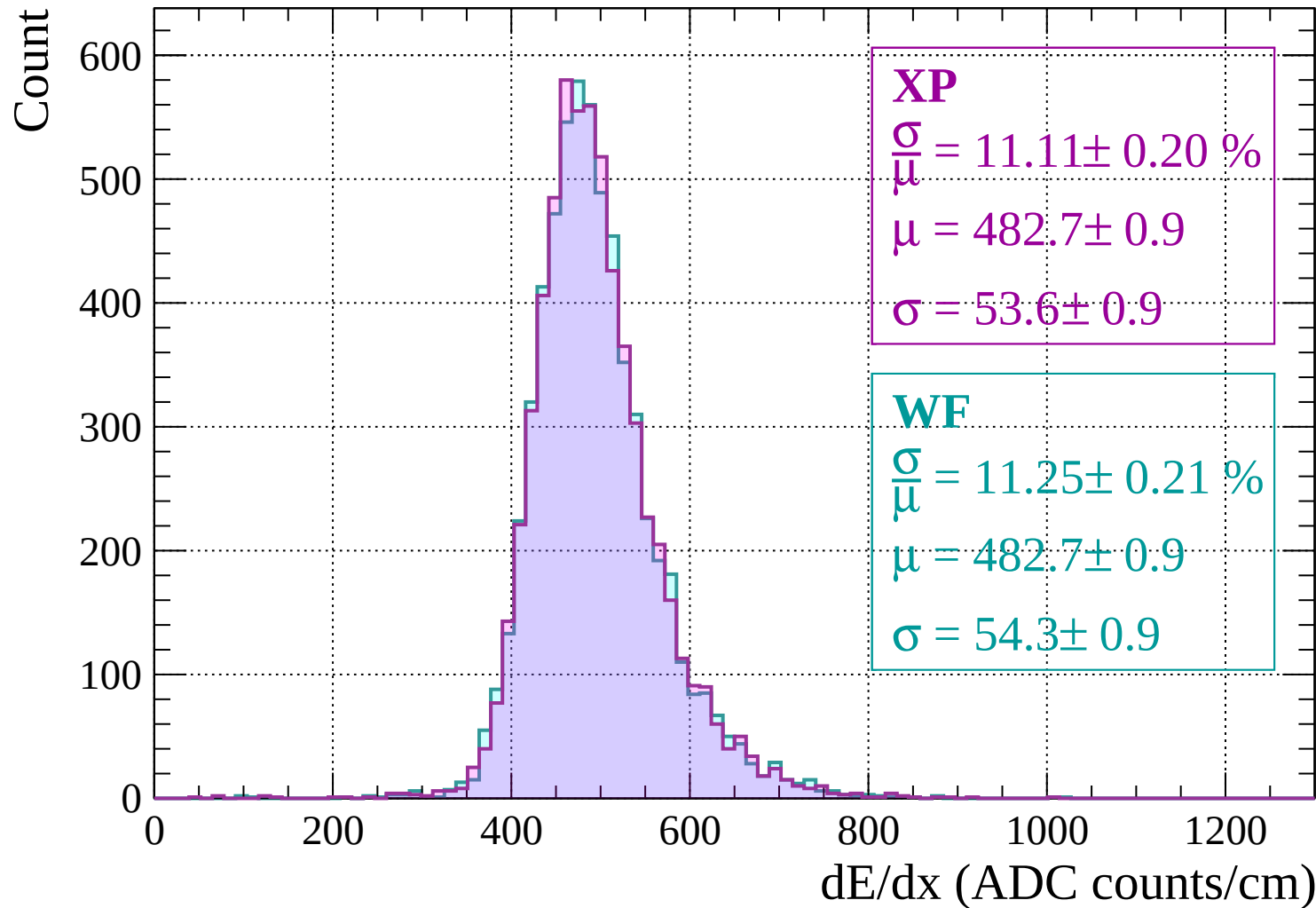
$$\frac{\sigma}{\mu} = 11.13 \pm 0.20 \%$$

$$\mu = 481.2 \pm 0.9$$

$$\sigma = 53.6 \pm 0.9$$

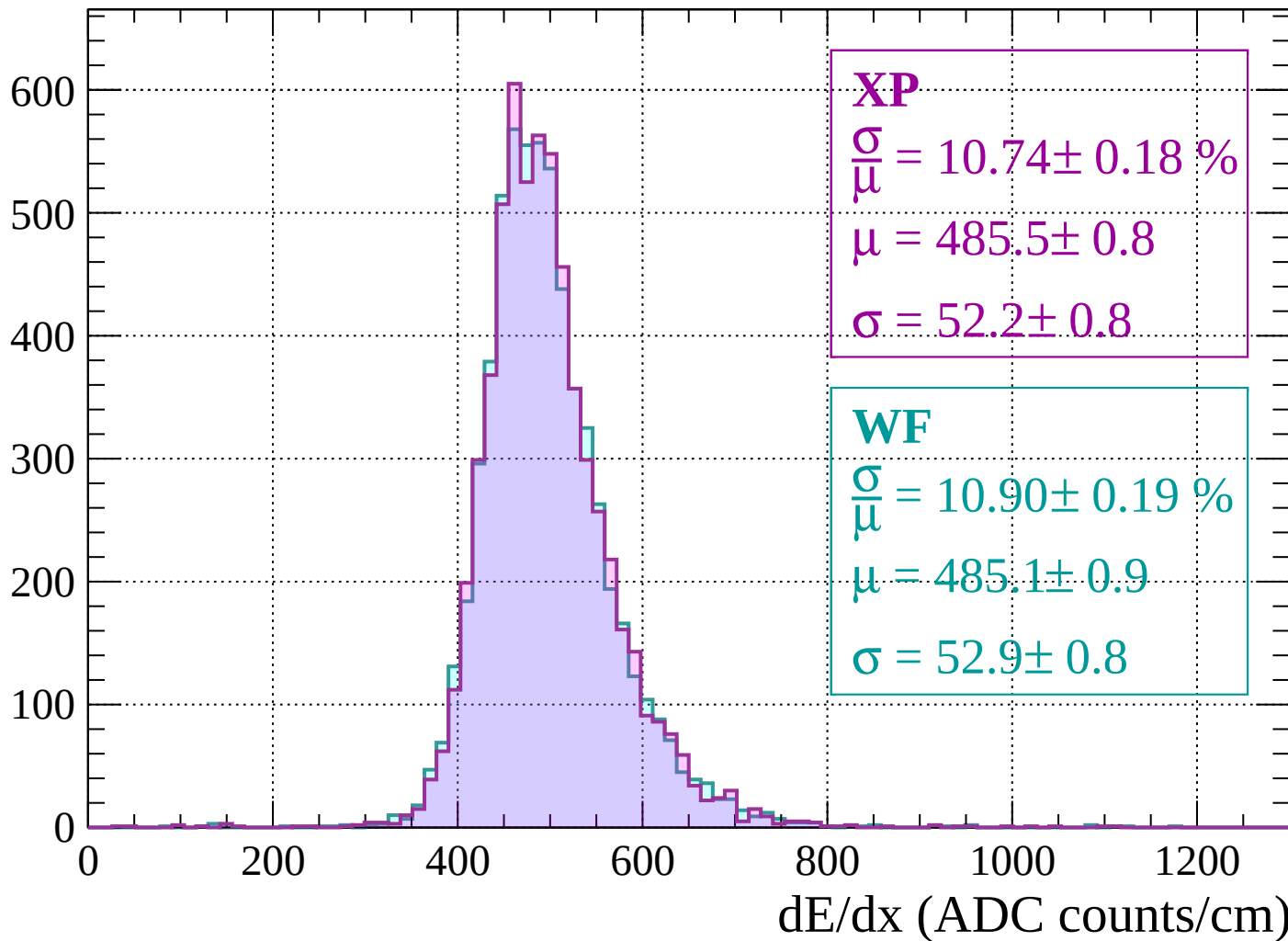
dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$



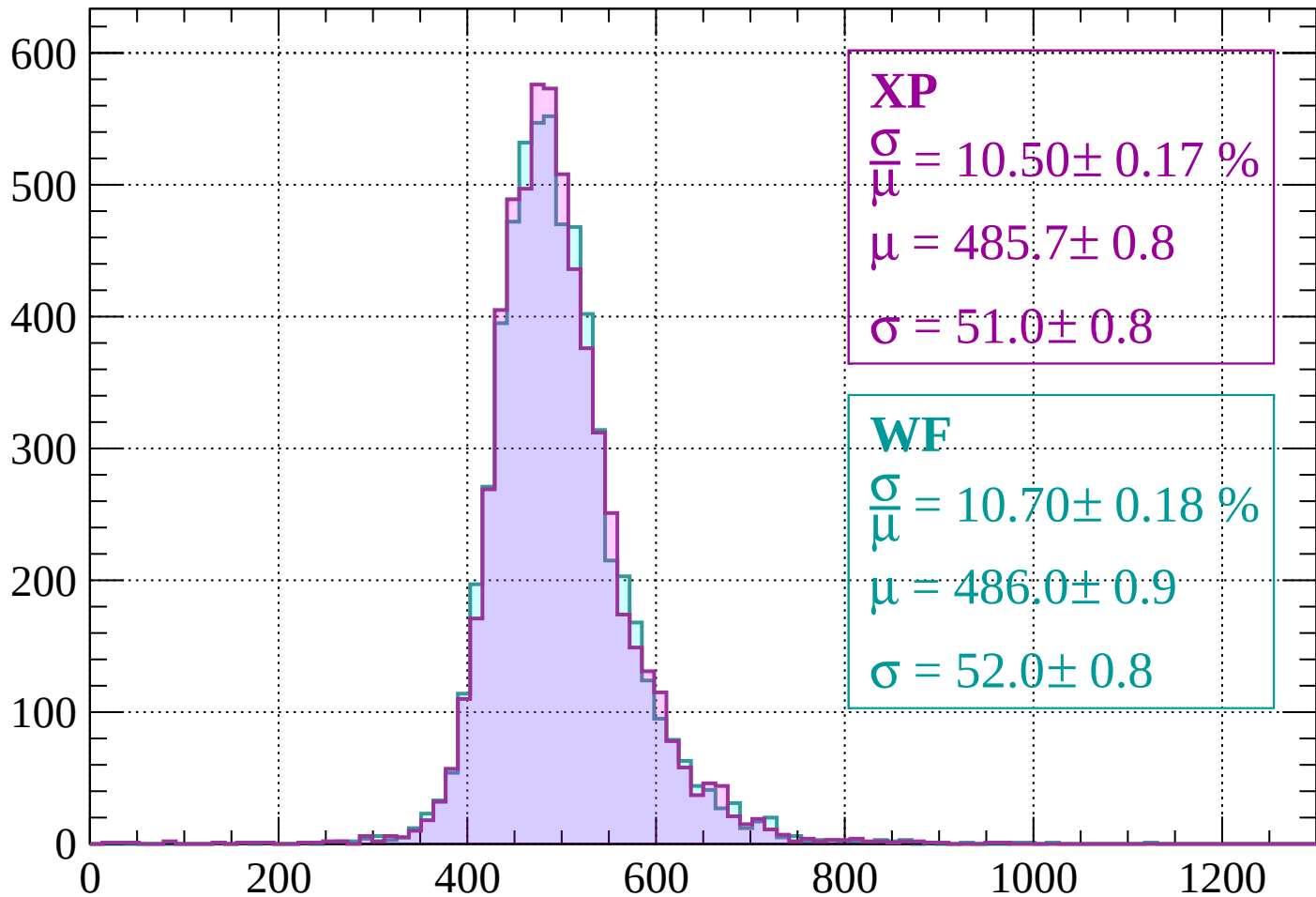
Energy loss | $1880 < p < 1920$

Count



Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 10.50 \pm 0.17 \%$$

$$\mu = 485.7 \pm 0.8$$

$$\sigma = 51.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.70 \pm 0.18 \%$$

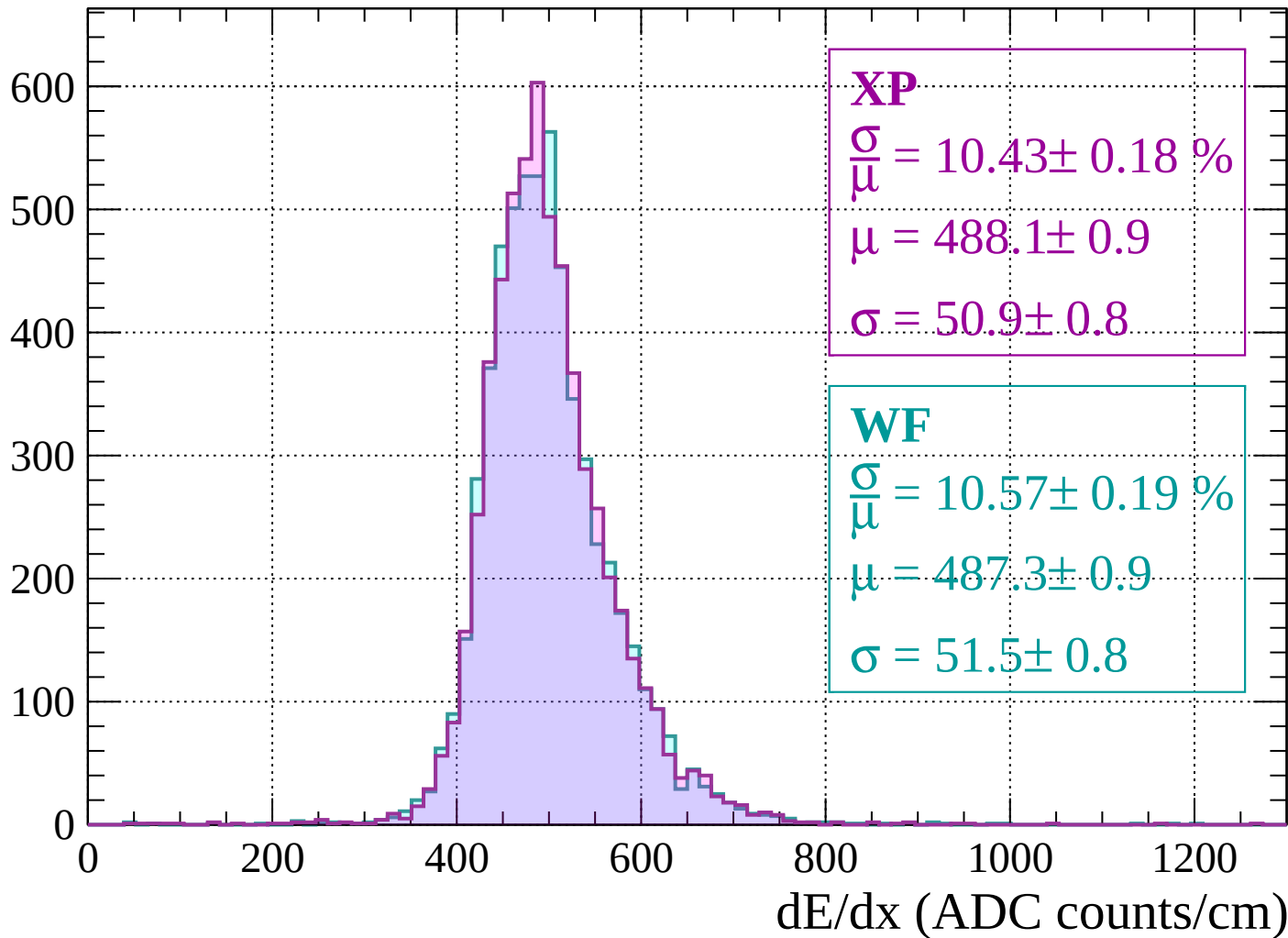
$$\mu = 486.0 \pm 0.9$$

$$\sigma = 52.0 \pm 0.8$$

dE/dx (ADC counts/cm)

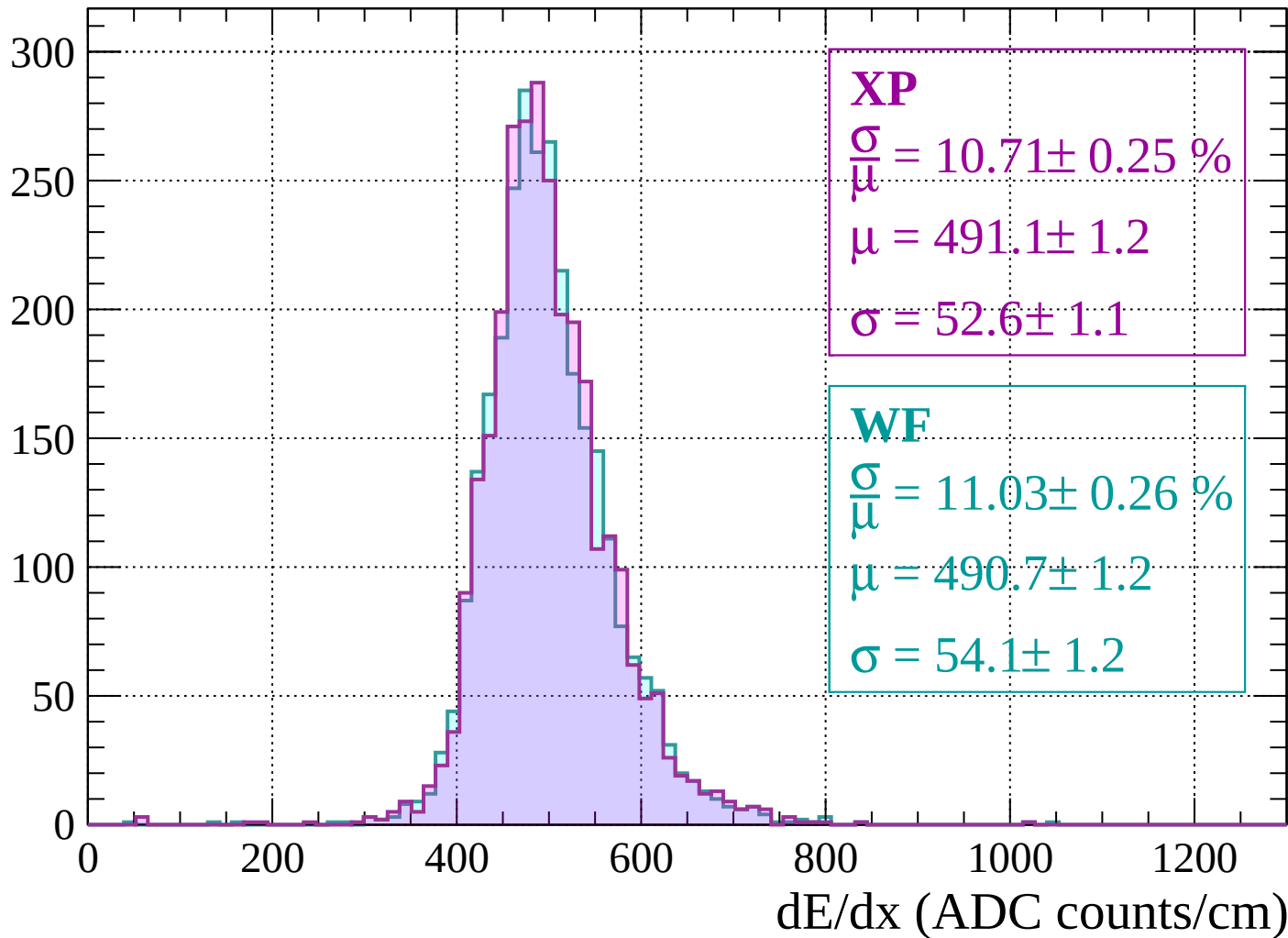
Energy loss | $1960 < p < 2000$

Count



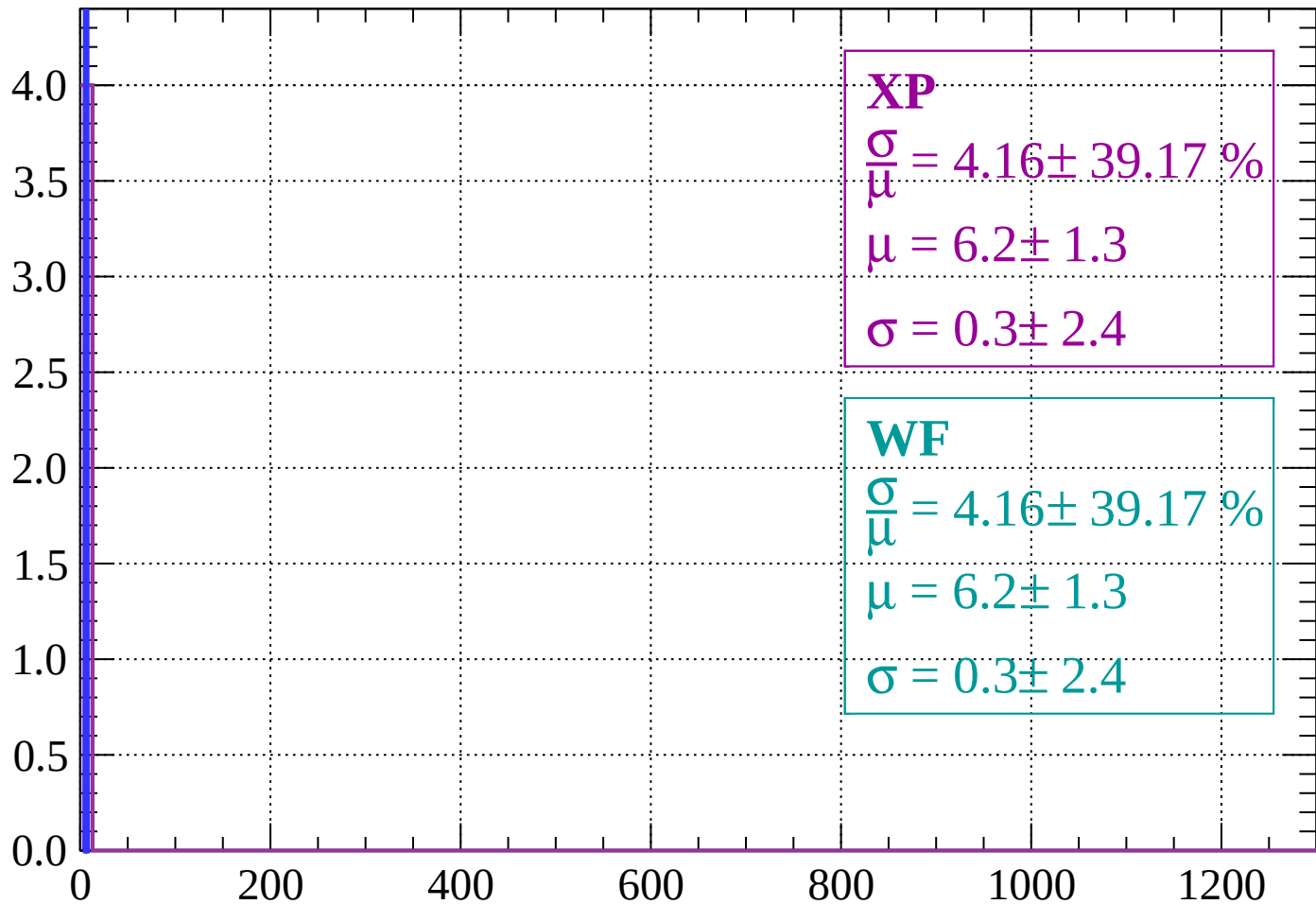
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

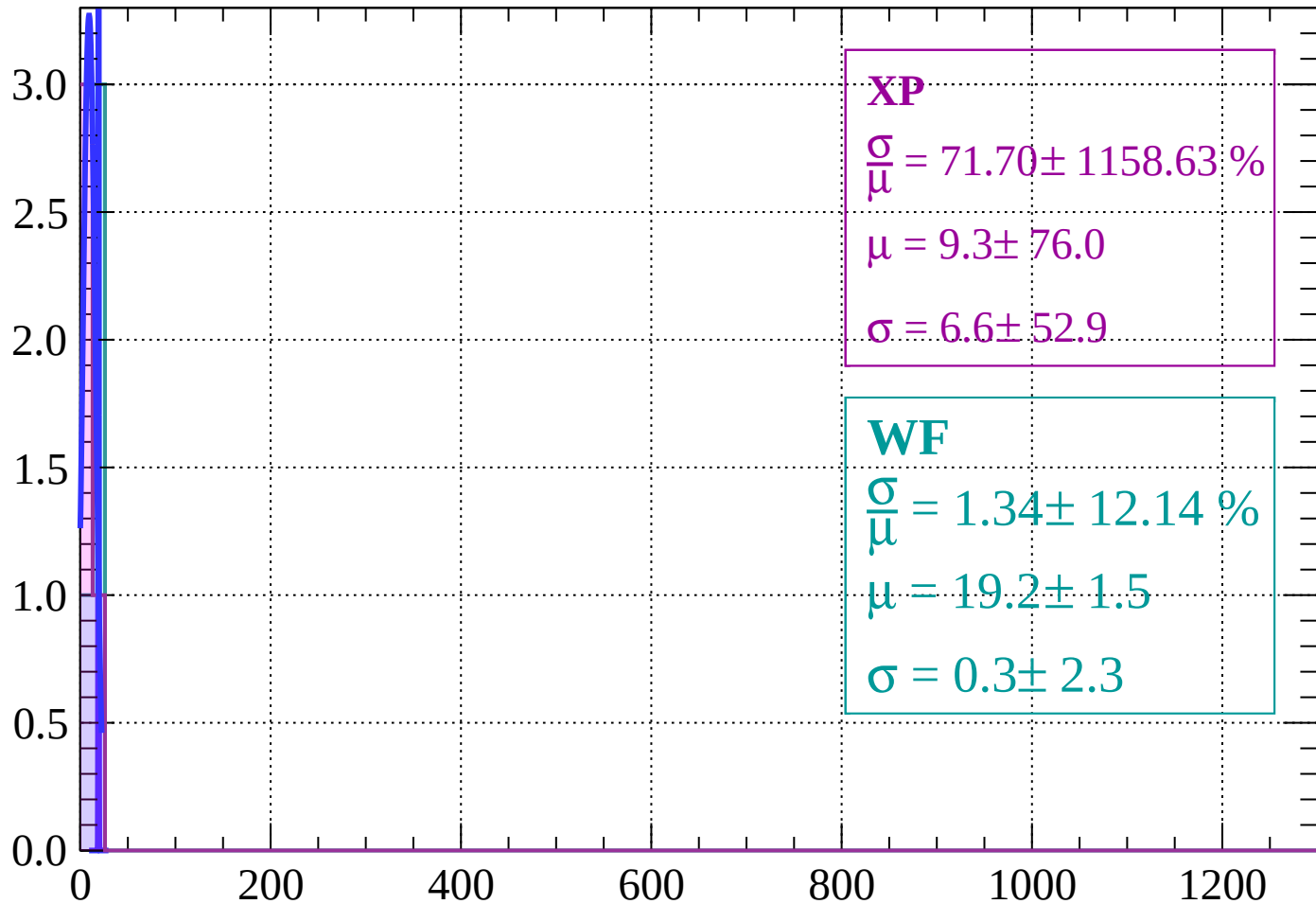
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

Count



XP

$$\frac{\sigma}{\mu} = 71.70 \pm 1158.63 \%$$

$$\mu = 9.3 \pm 76.0$$

$$\sigma = 6.6 \pm 52.9$$

WF

$$\frac{\sigma}{\mu} = 1.34 \pm 12.14 \%$$

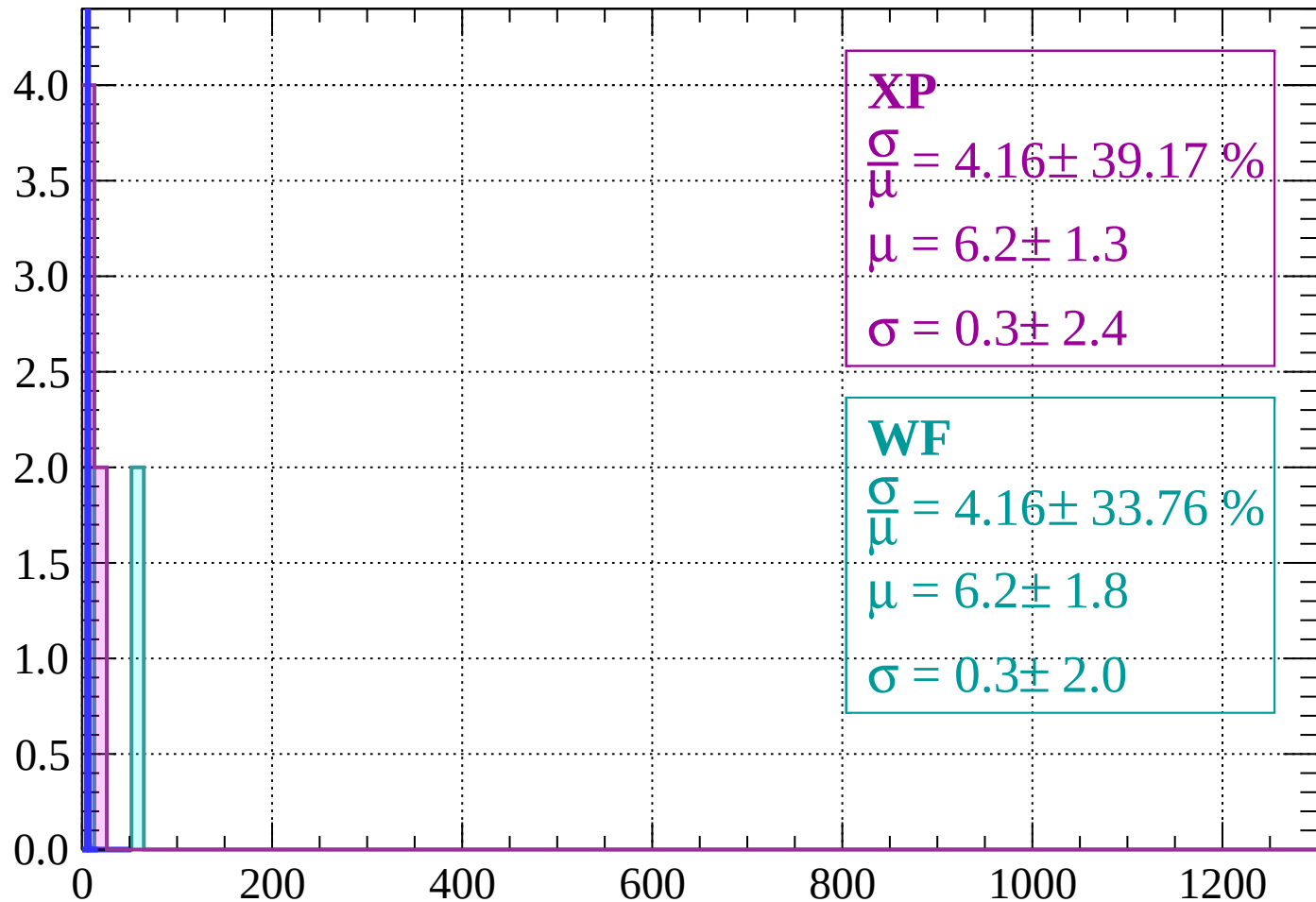
$$\mu = 19.2 \pm 1.5$$

$$\sigma = 0.3 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

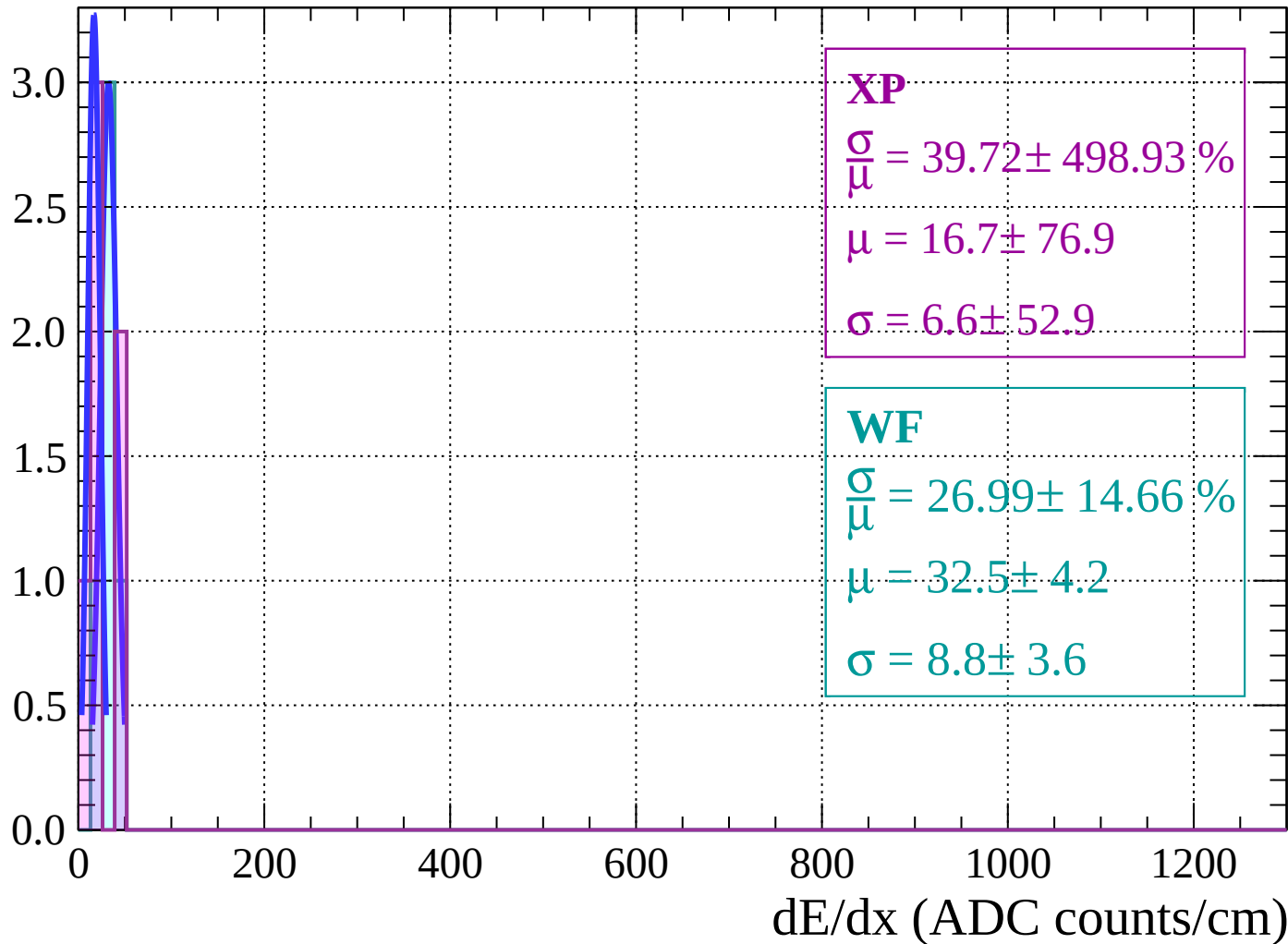
Count



dE/dx (ADC counts/cm)

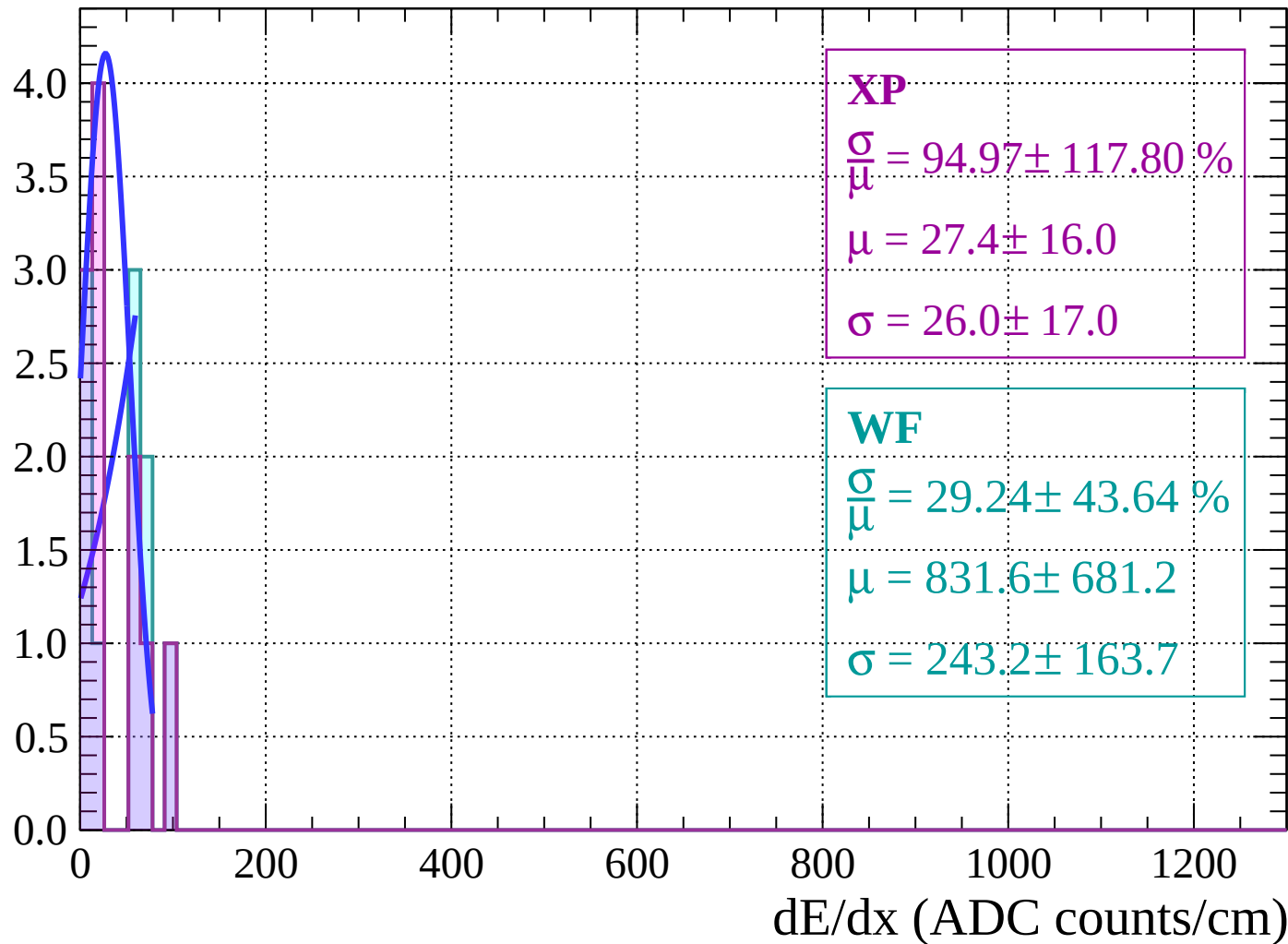
Energy loss | $-84 < \theta < -82$

Count



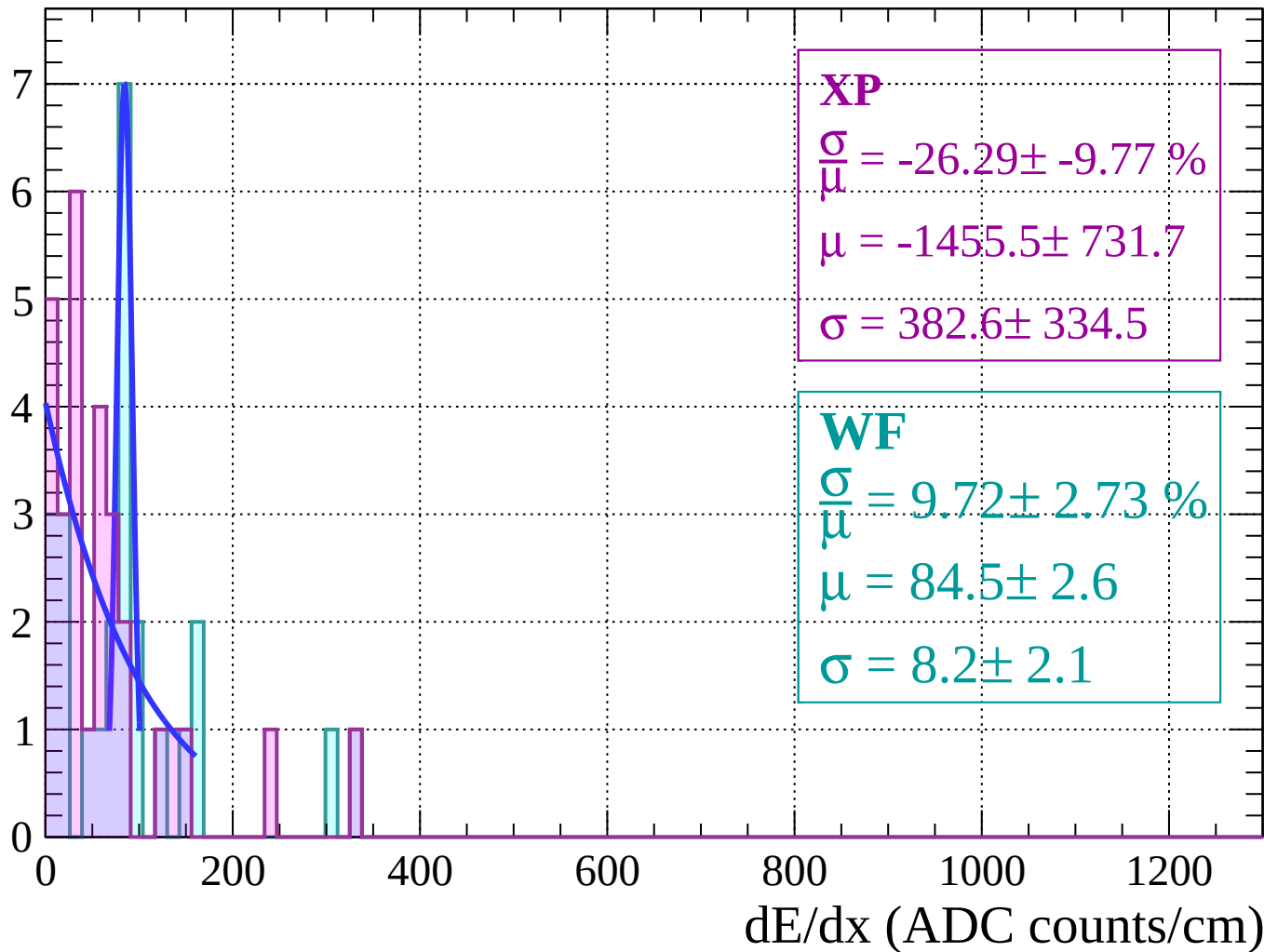
Energy loss | $-82 < \theta < -80$

Count



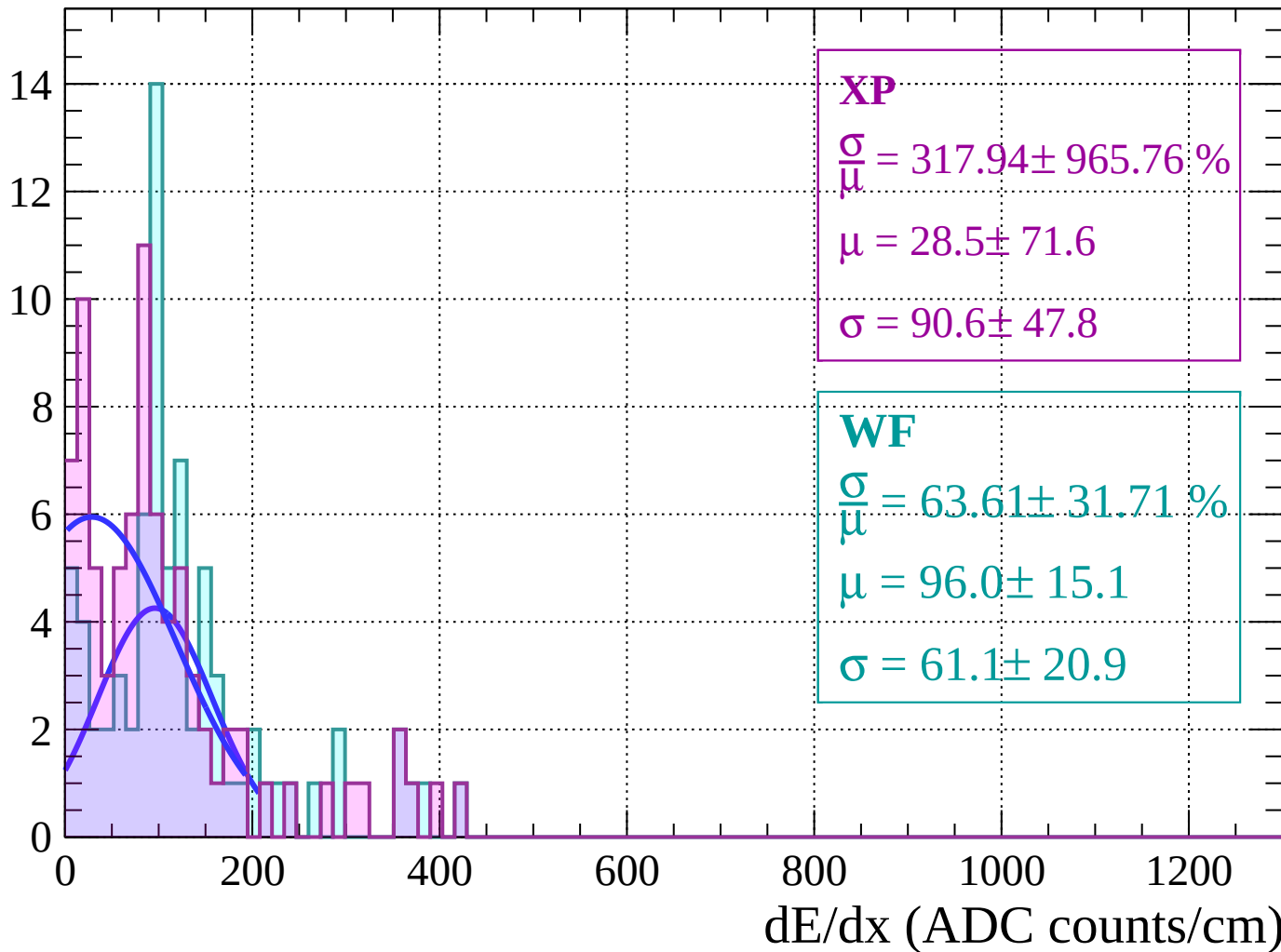
Energy loss | $-80 < \theta < -78$

Count



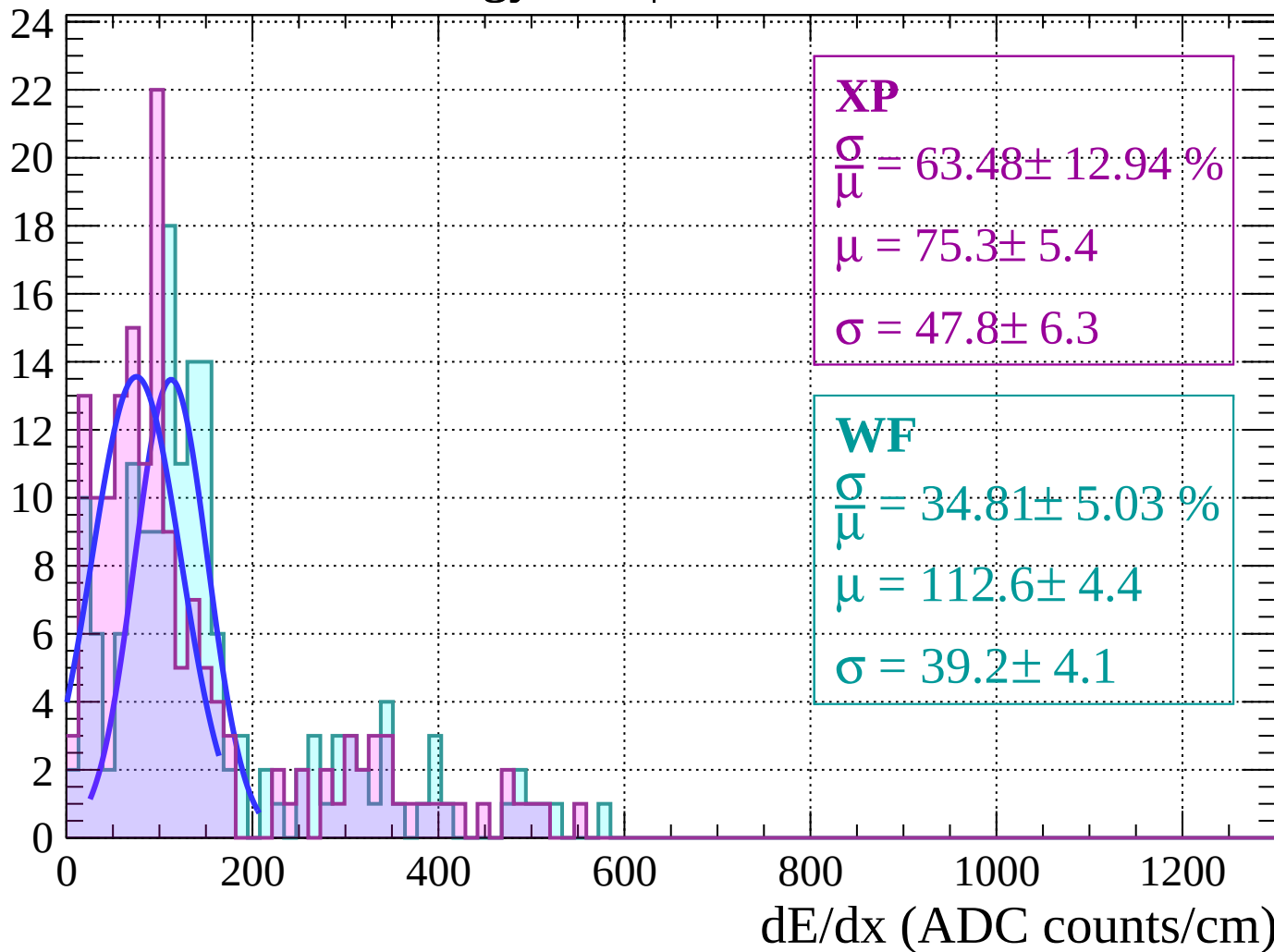
Energy loss | $-78 < \theta < -76$

Count



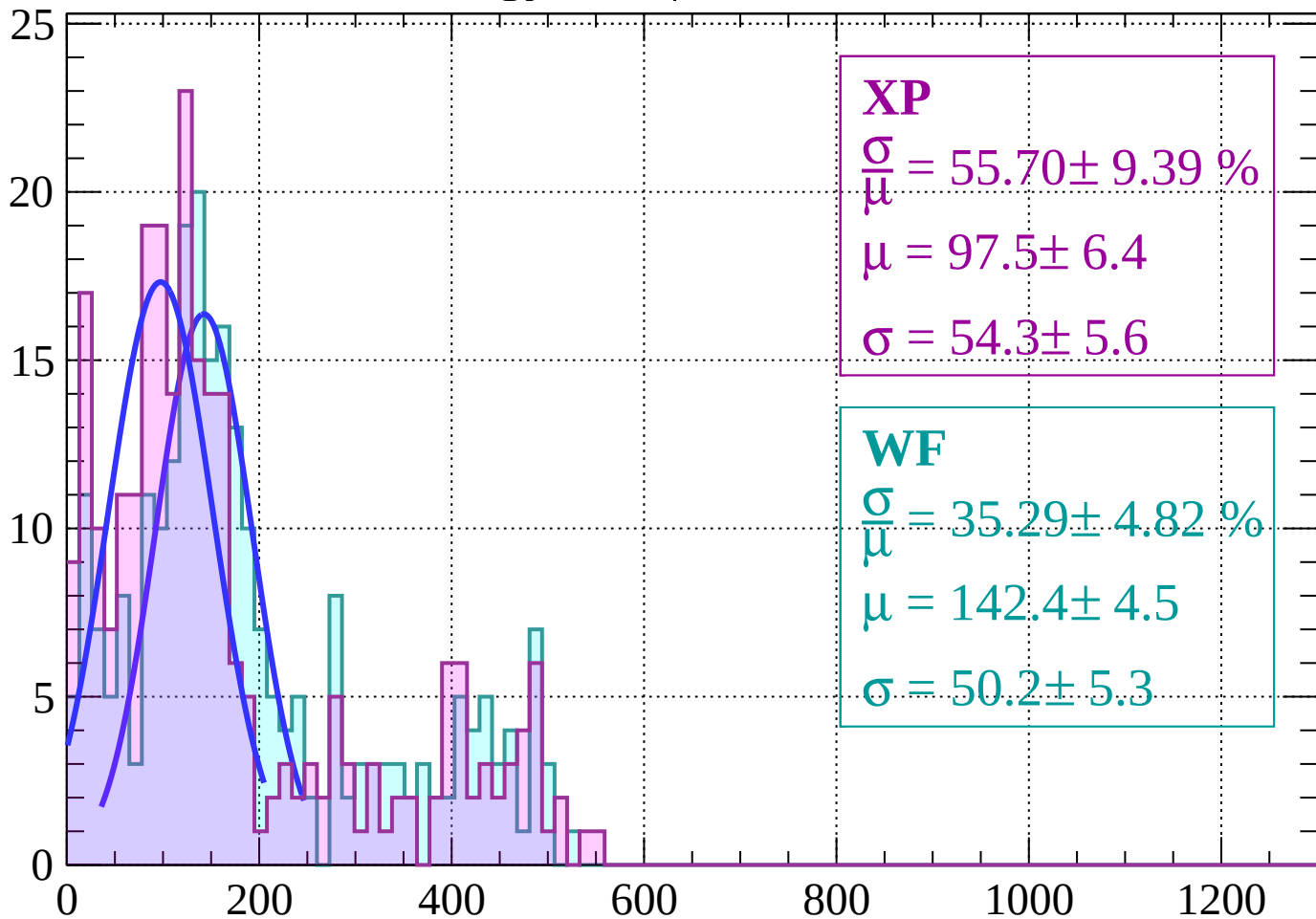
Energy loss | $-76 < \theta < -74$

Count



Energy loss $|-74 < \theta < -72$

Count



XP

$$\frac{\sigma}{\mu} = 55.70 \pm 9.39 \%$$

$$\mu = 97.5 \pm 6.4$$

$$\sigma = 54.3 \pm 5.6$$

WF

$$\frac{\sigma}{\mu} = 35.29 \pm 4.82 \%$$

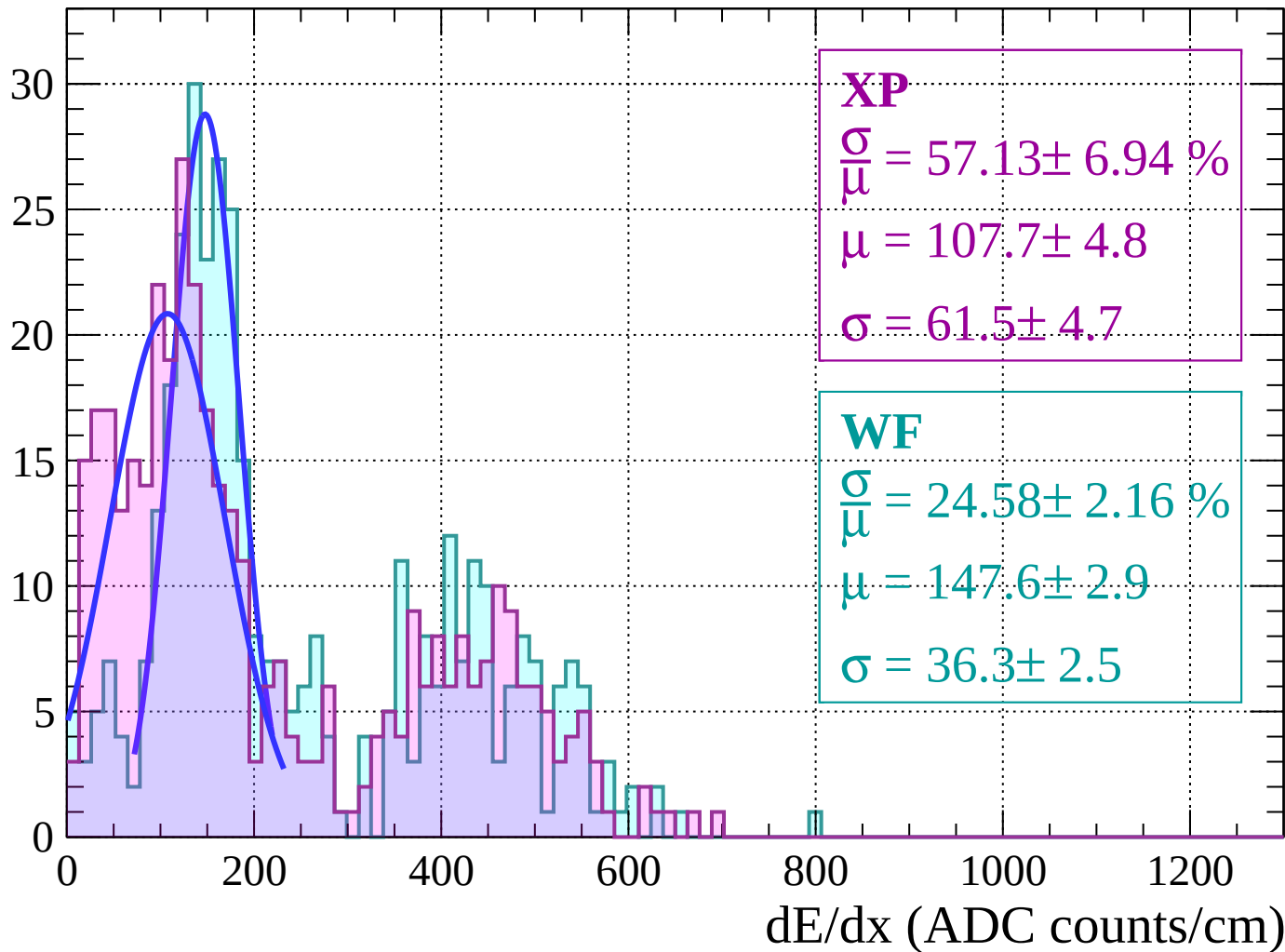
$$\mu = 142.4 \pm 4.5$$

$$\sigma = 50.2 \pm 5.3$$

dE/dx (ADC counts/cm)

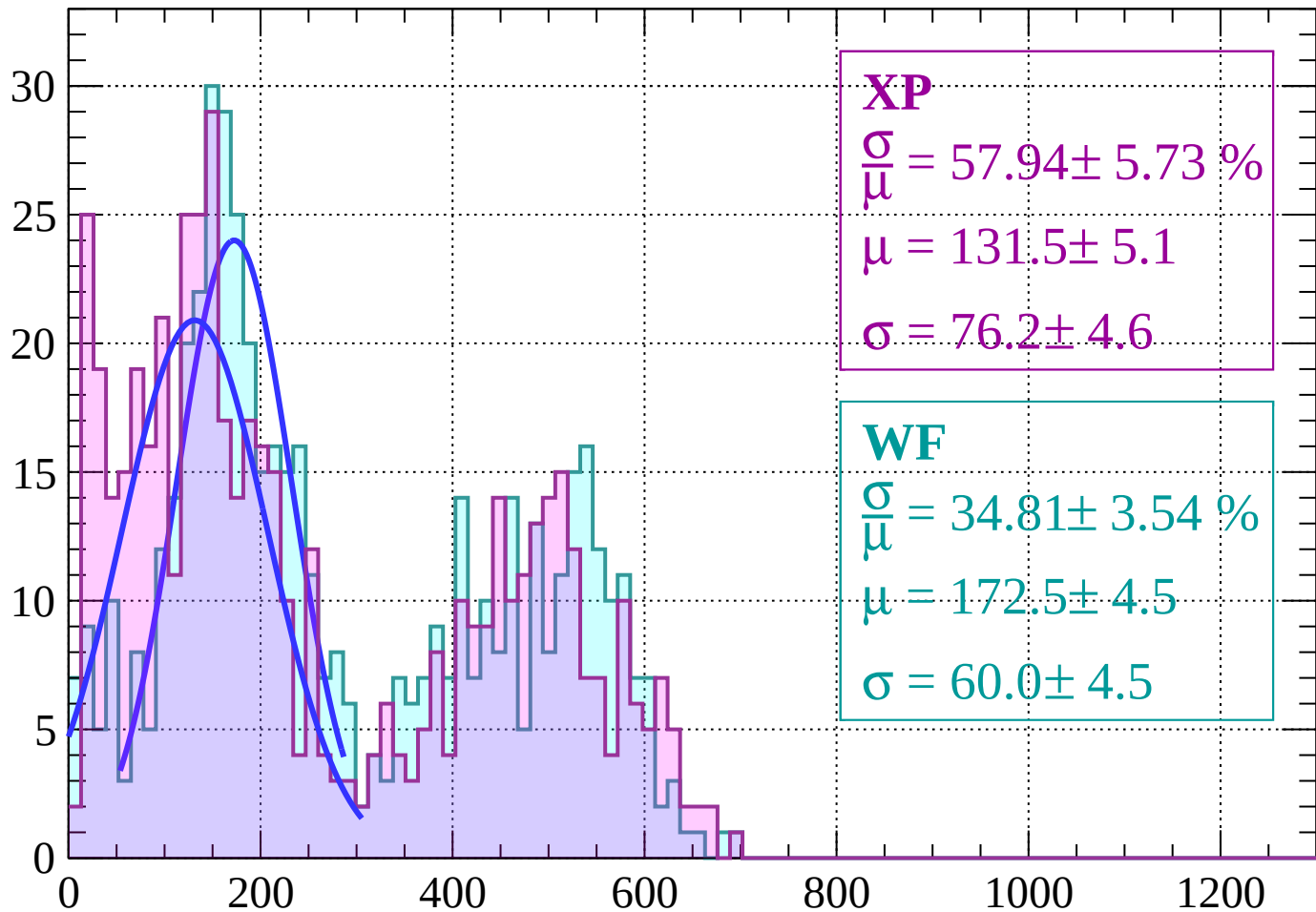
Energy loss $|-72 < \theta < -70$

Count



Energy loss | $-70 < \theta < -68$

Count



XP

$$\frac{\sigma}{\mu} = 57.94 \pm 5.73 \%$$

$$\mu = 131.5 \pm 5.1$$

$$\sigma = 76.2 \pm 4.6$$

WF

$$\frac{\sigma}{\mu} = 34.81 \pm 3.54 \%$$

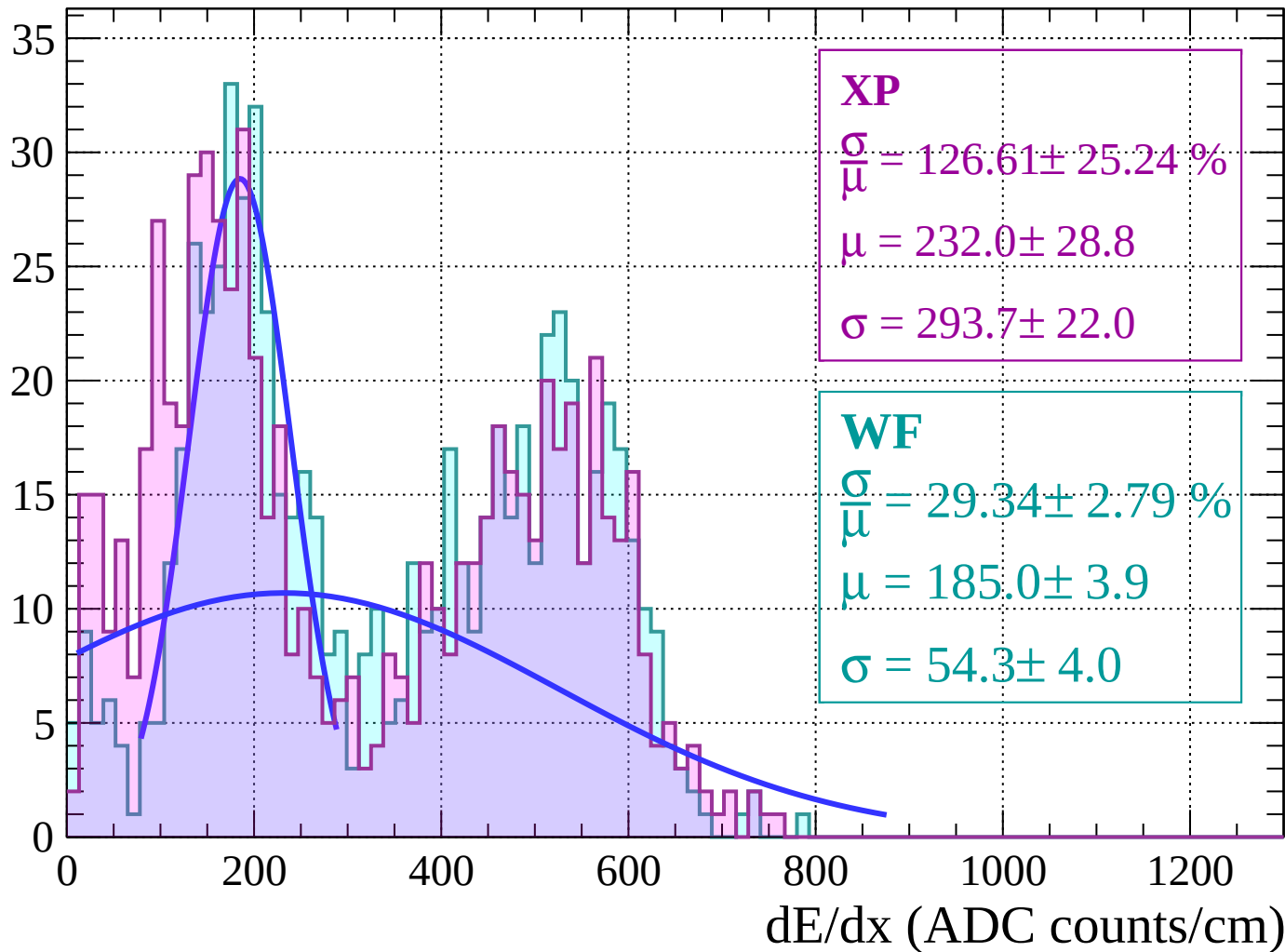
$$\mu = 172.5 \pm 4.5$$

$$\sigma = 60.0 \pm 4.5$$

dE/dx (ADC counts/cm)

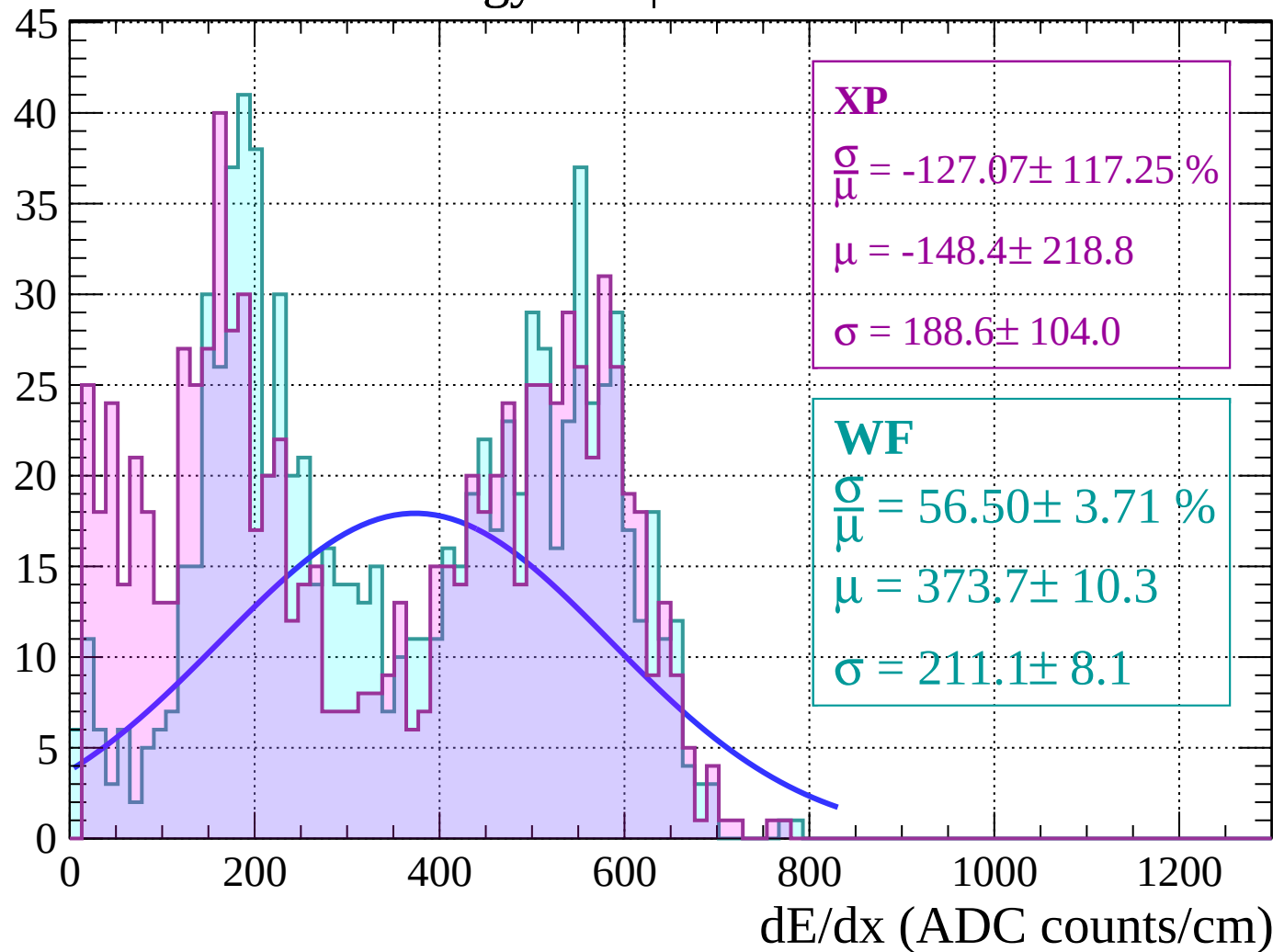
Energy loss | $-68 < \theta < -66$

Count



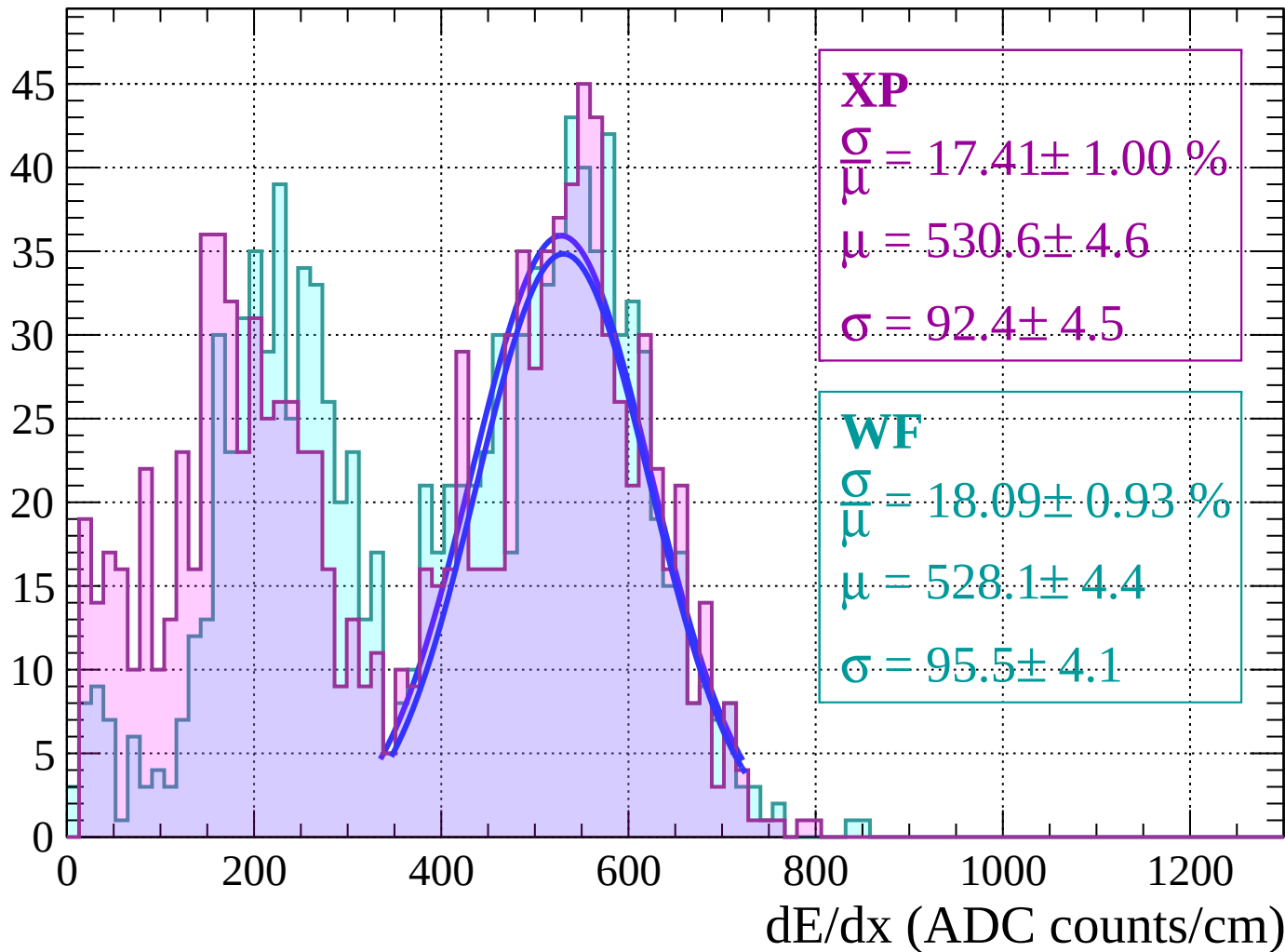
Energy loss | $-66 < \theta < -64$

Count



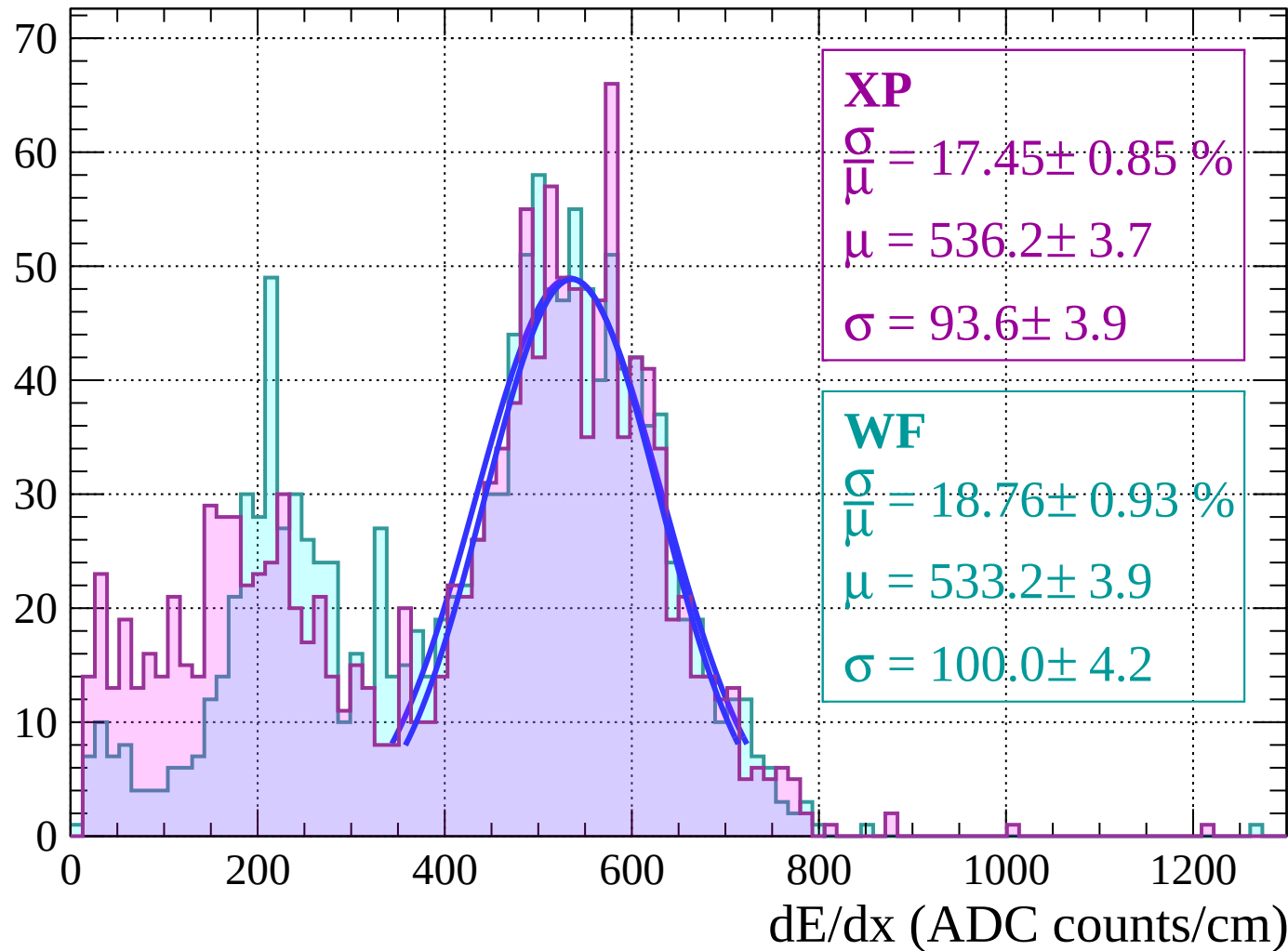
Energy loss | $-64 < \theta < -62$

Count



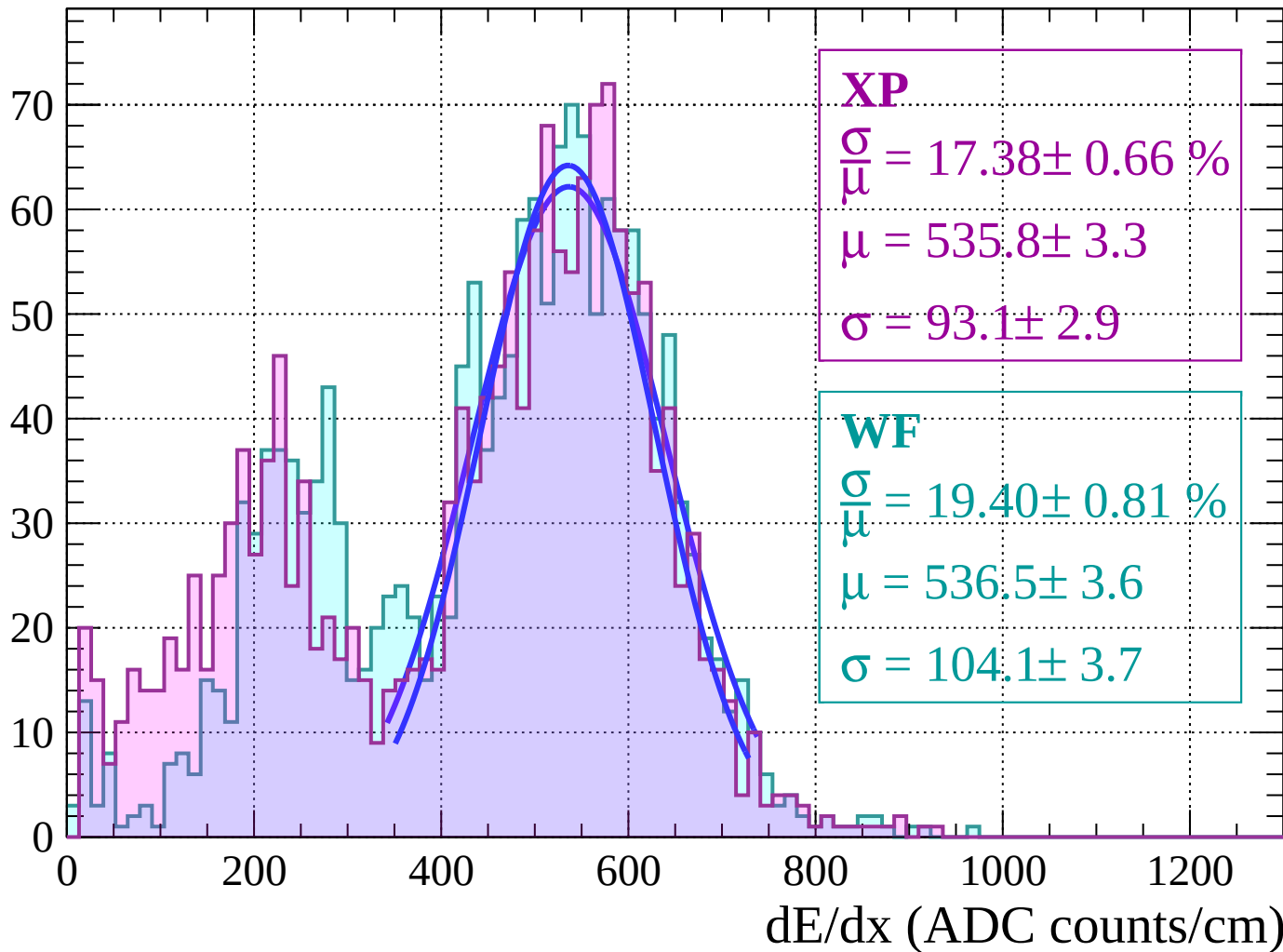
Energy loss | $-62 < \theta < -60$

Count



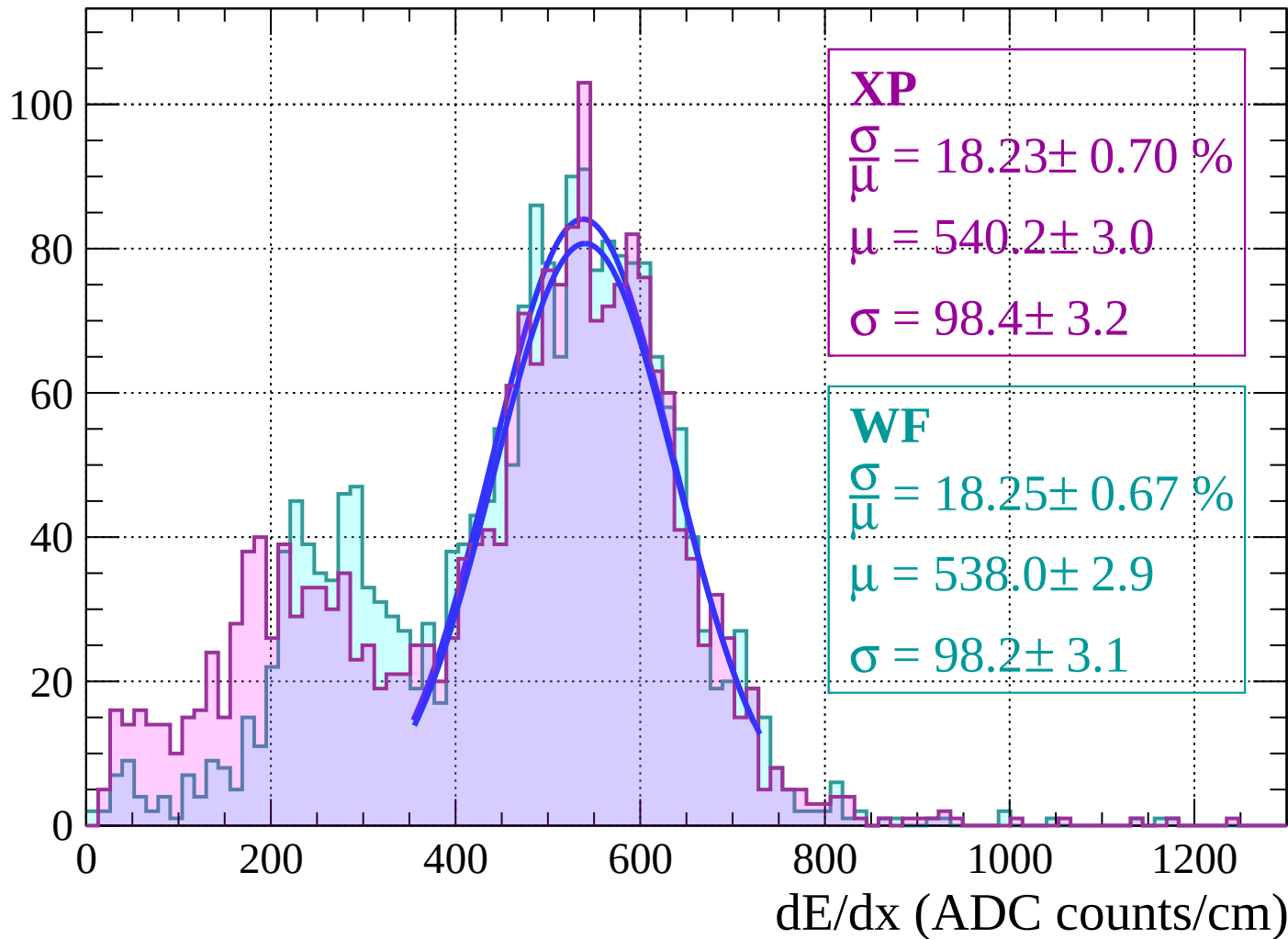
Energy loss $|-60 < \theta < -58$

Count



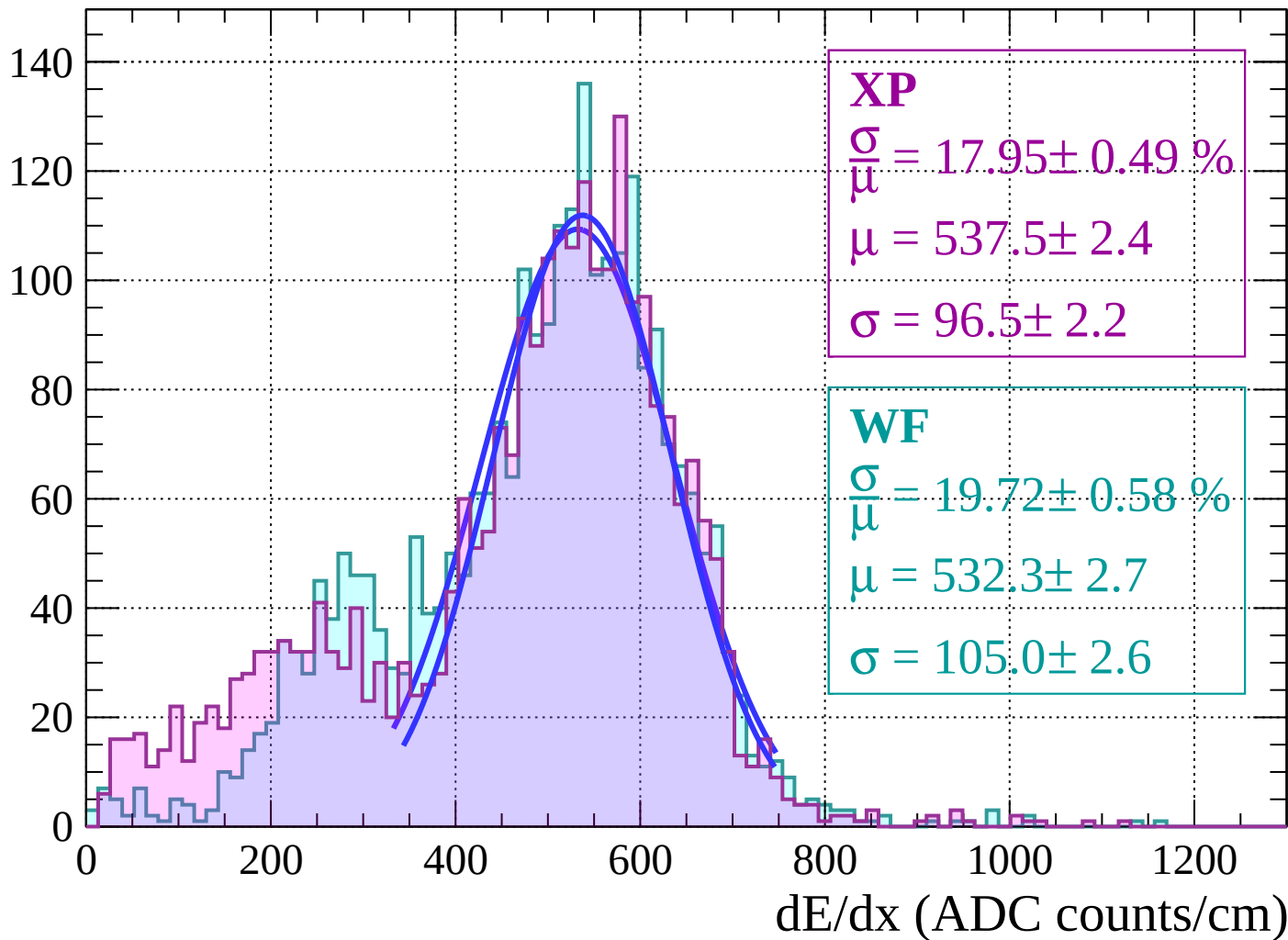
Energy loss $|-58 < \theta < -56$

Count



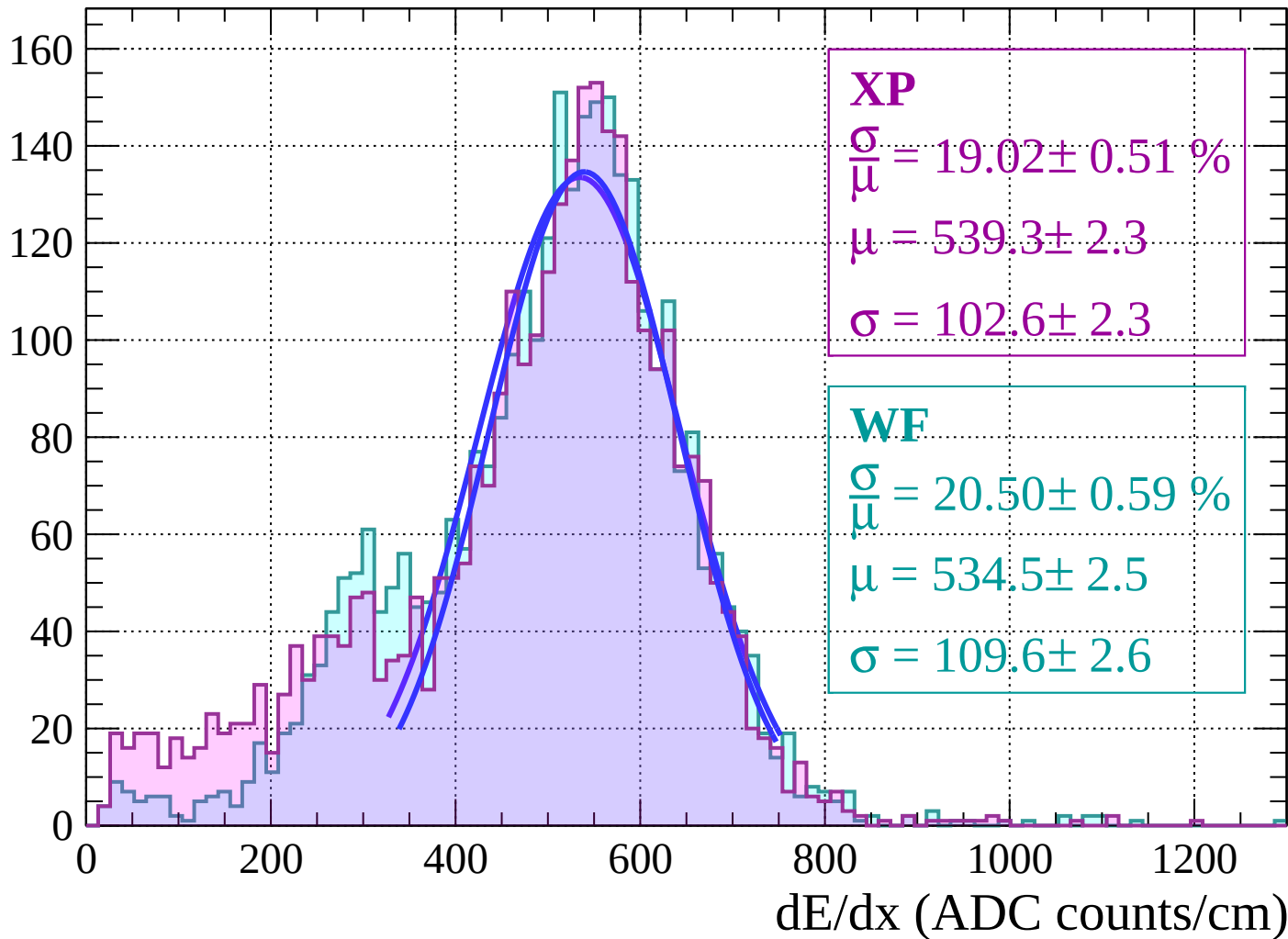
Energy loss $|-56 < \theta < -54$

Count



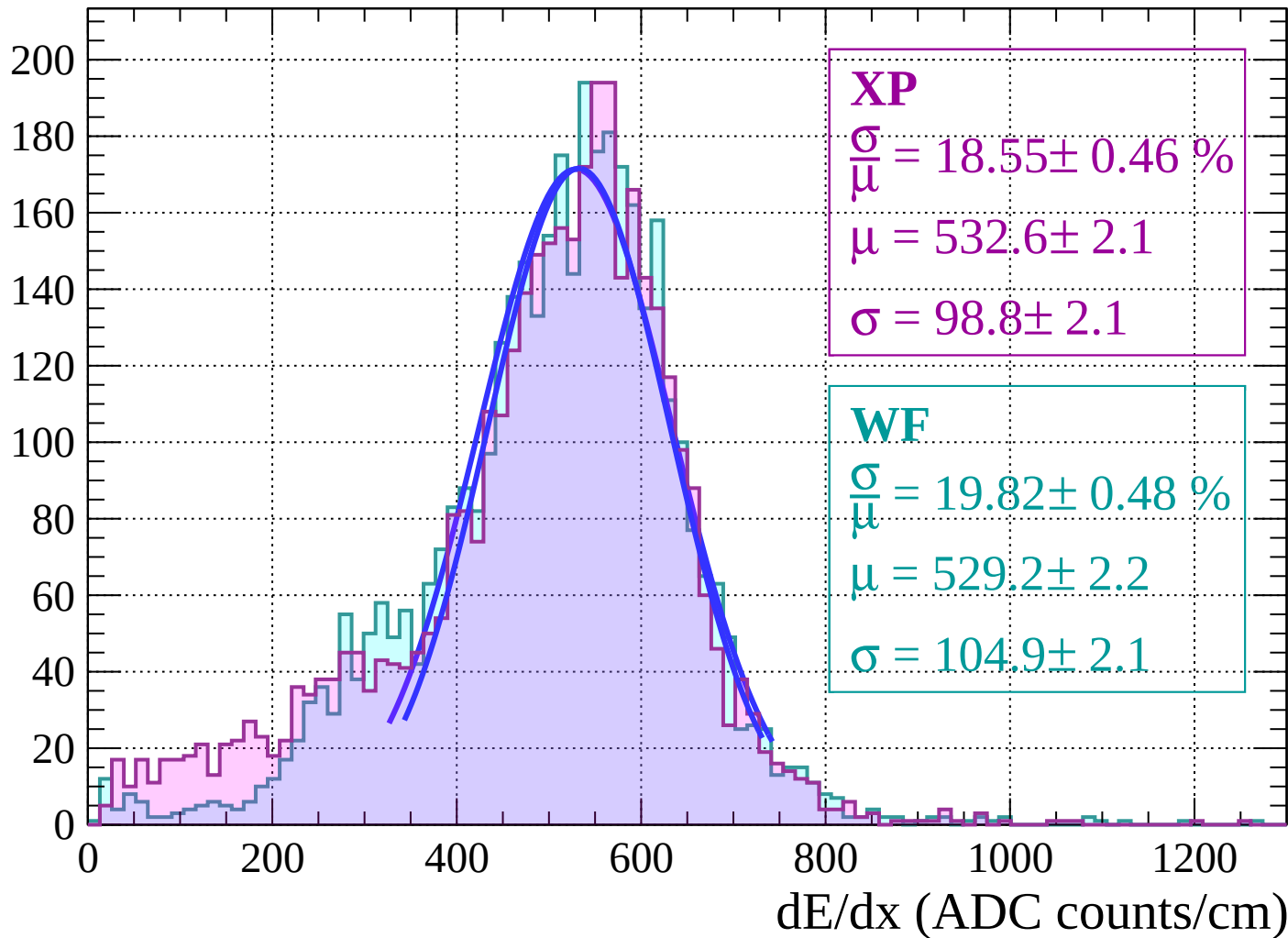
Energy loss $|-54 < \theta < -52$

Count



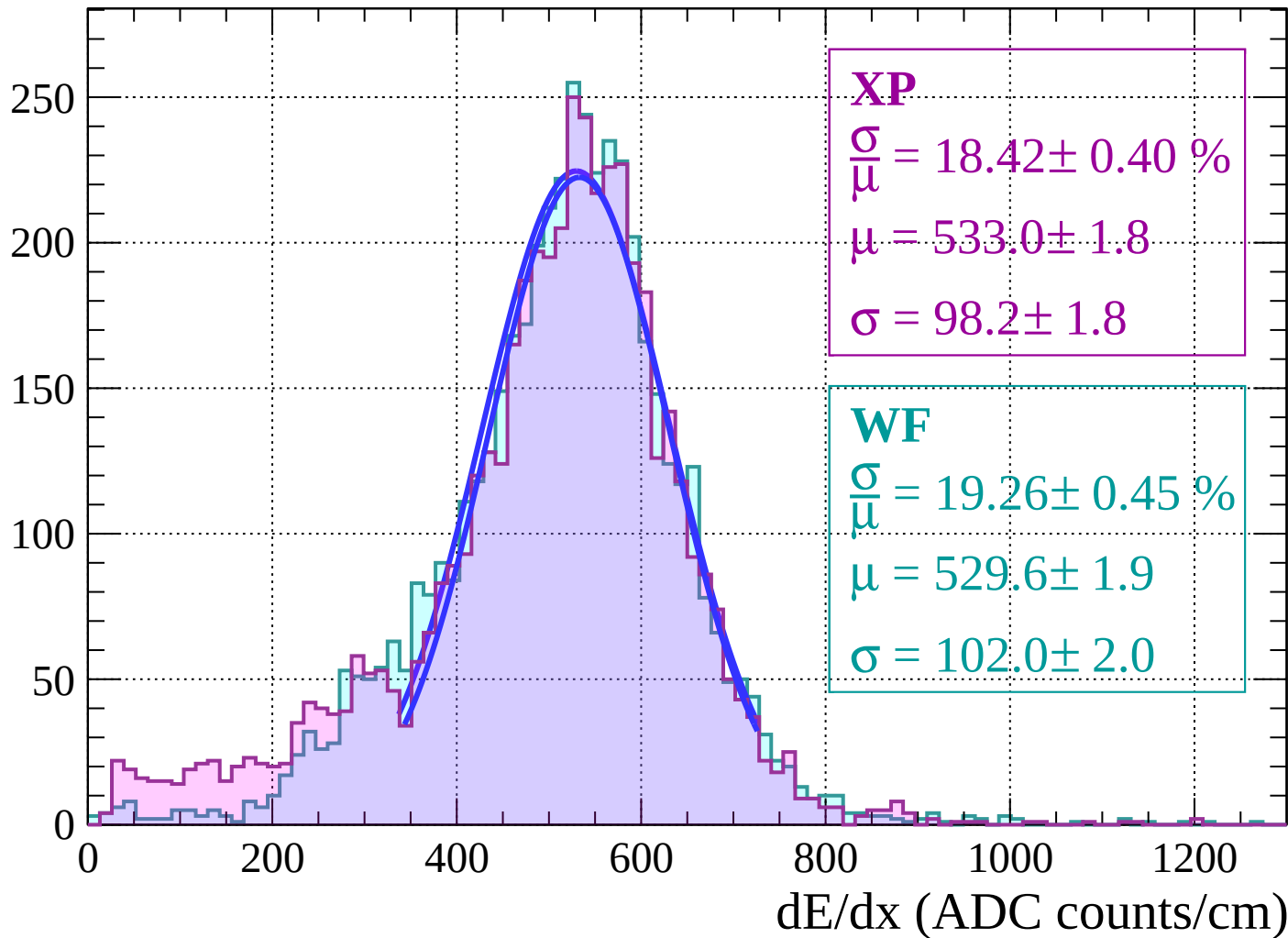
Energy loss $|-52 < \theta < -50$

Count



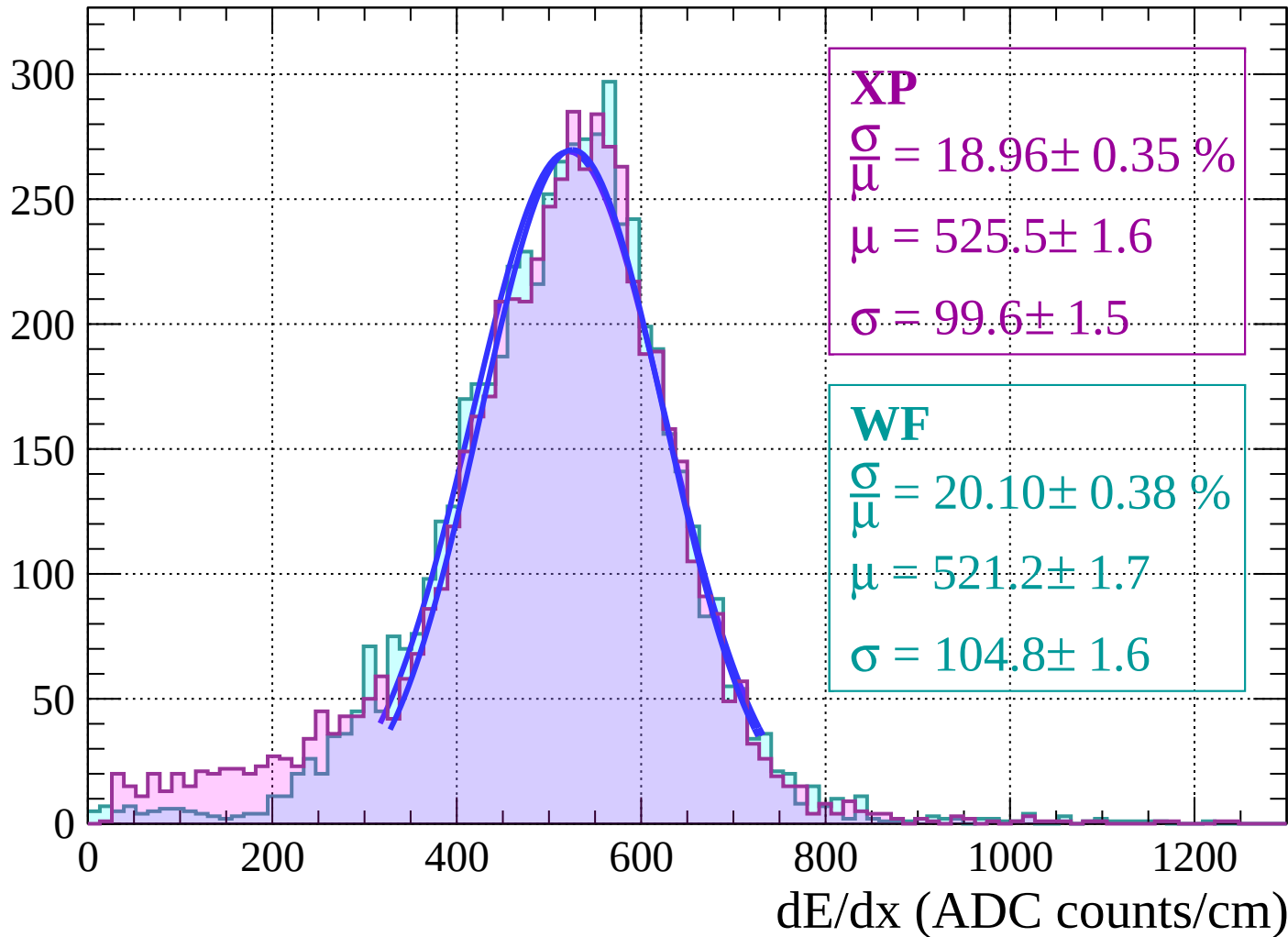
Energy loss | $-50 < \theta < -48$

Count

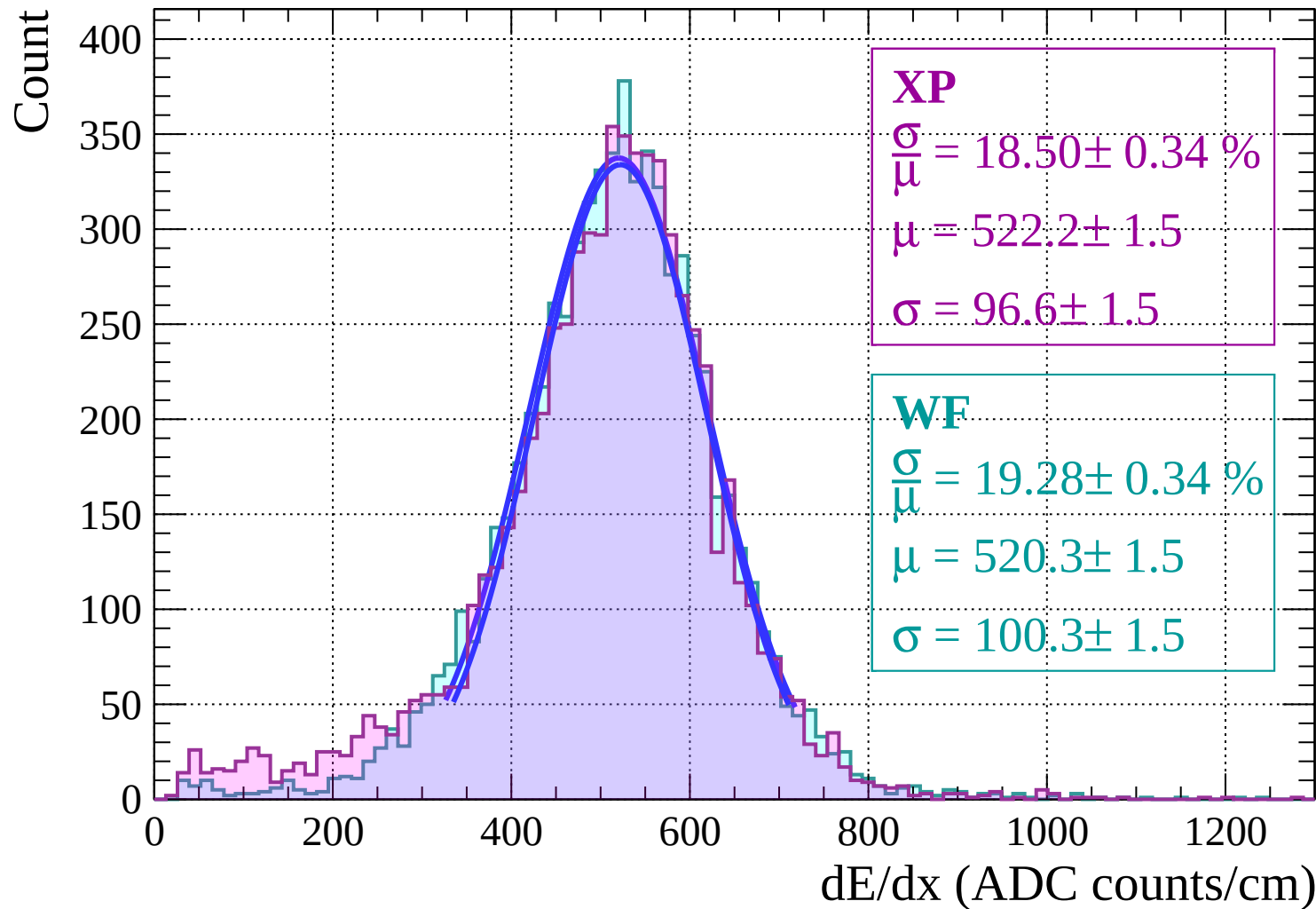


Energy loss | $-48 < \theta < -46$

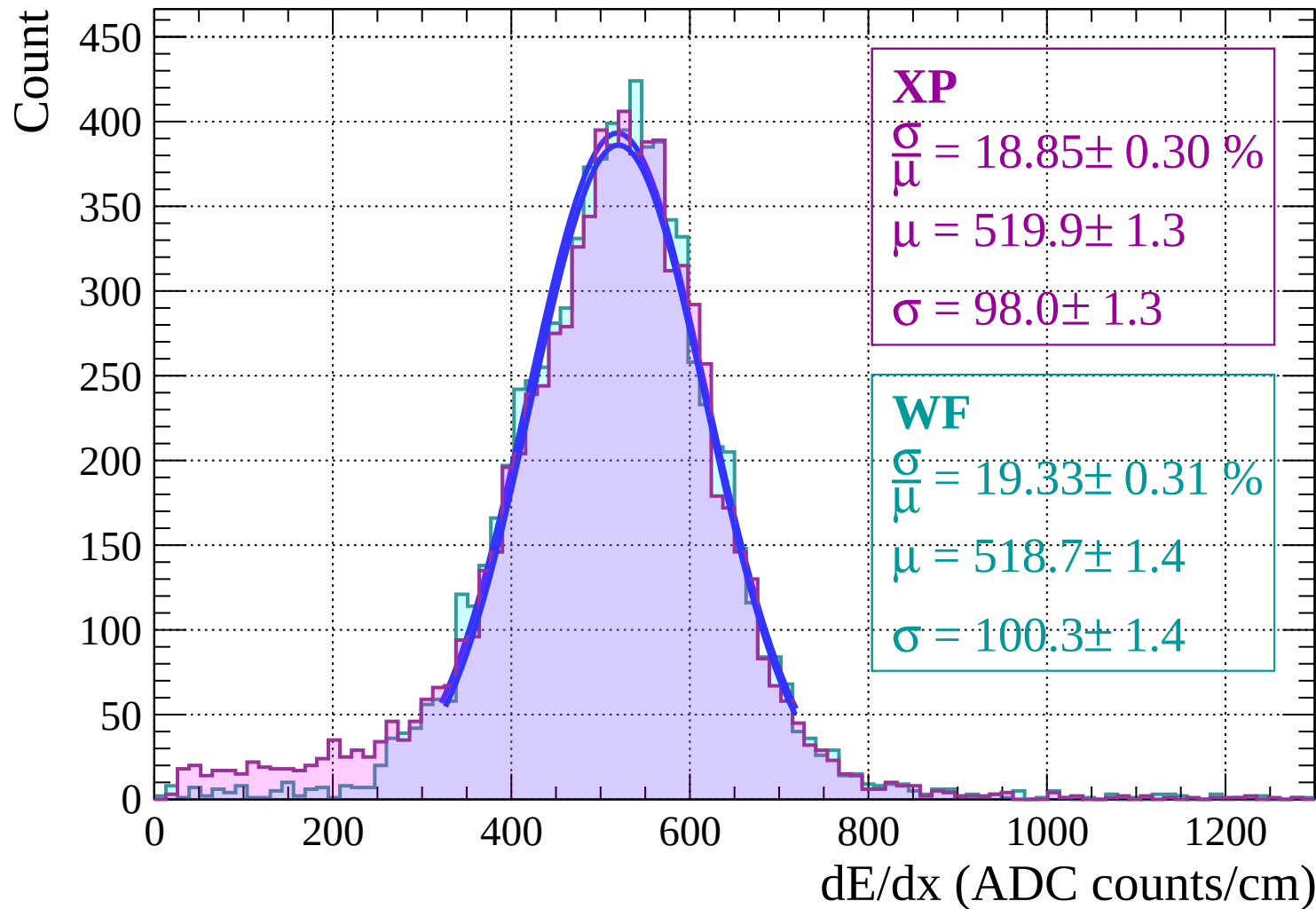
Count



Energy loss | $-46 < \theta < -44$

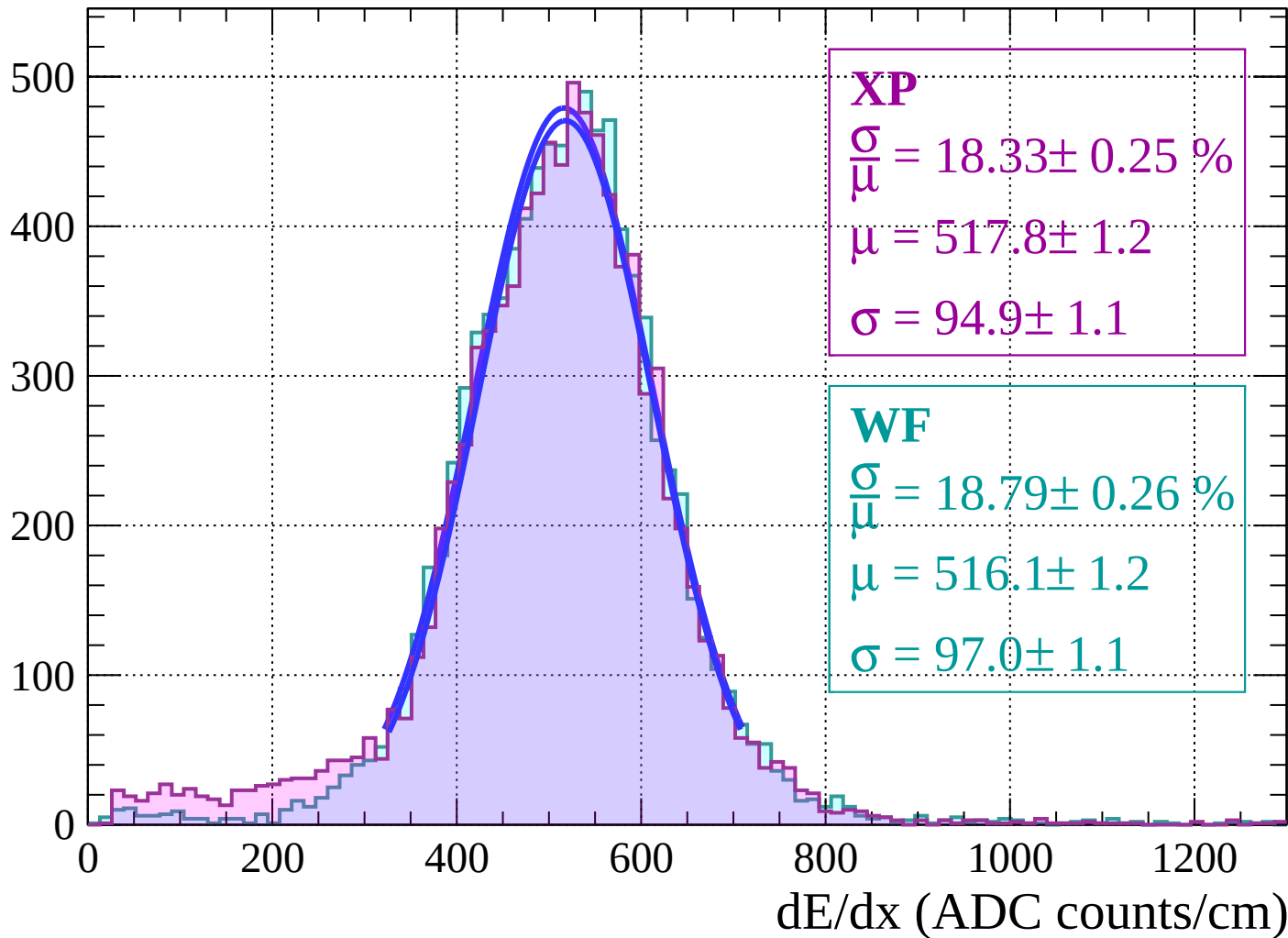


Energy loss $|-44 < \theta < -42$



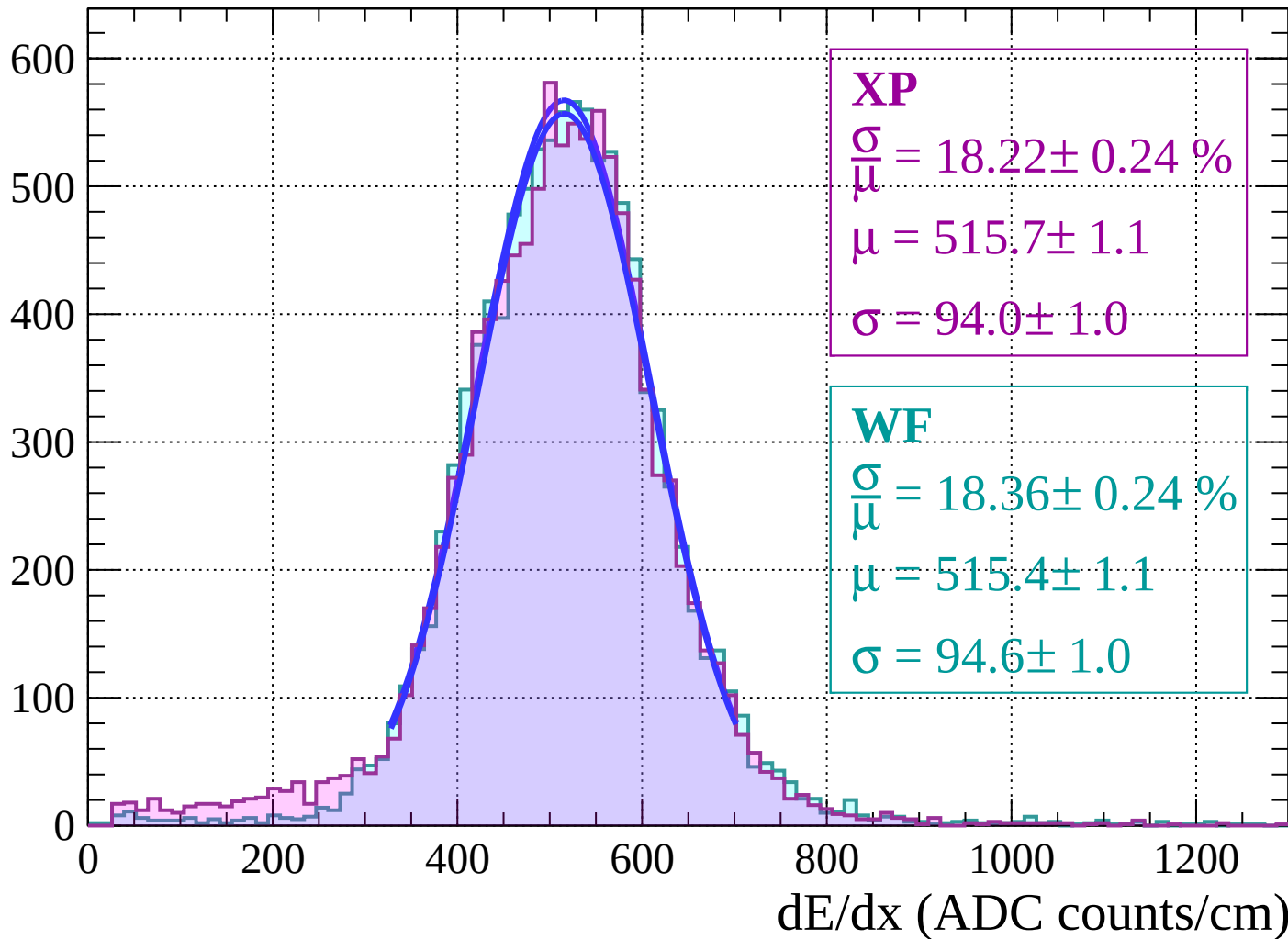
Energy loss $|-42 < \theta < -40$

Count



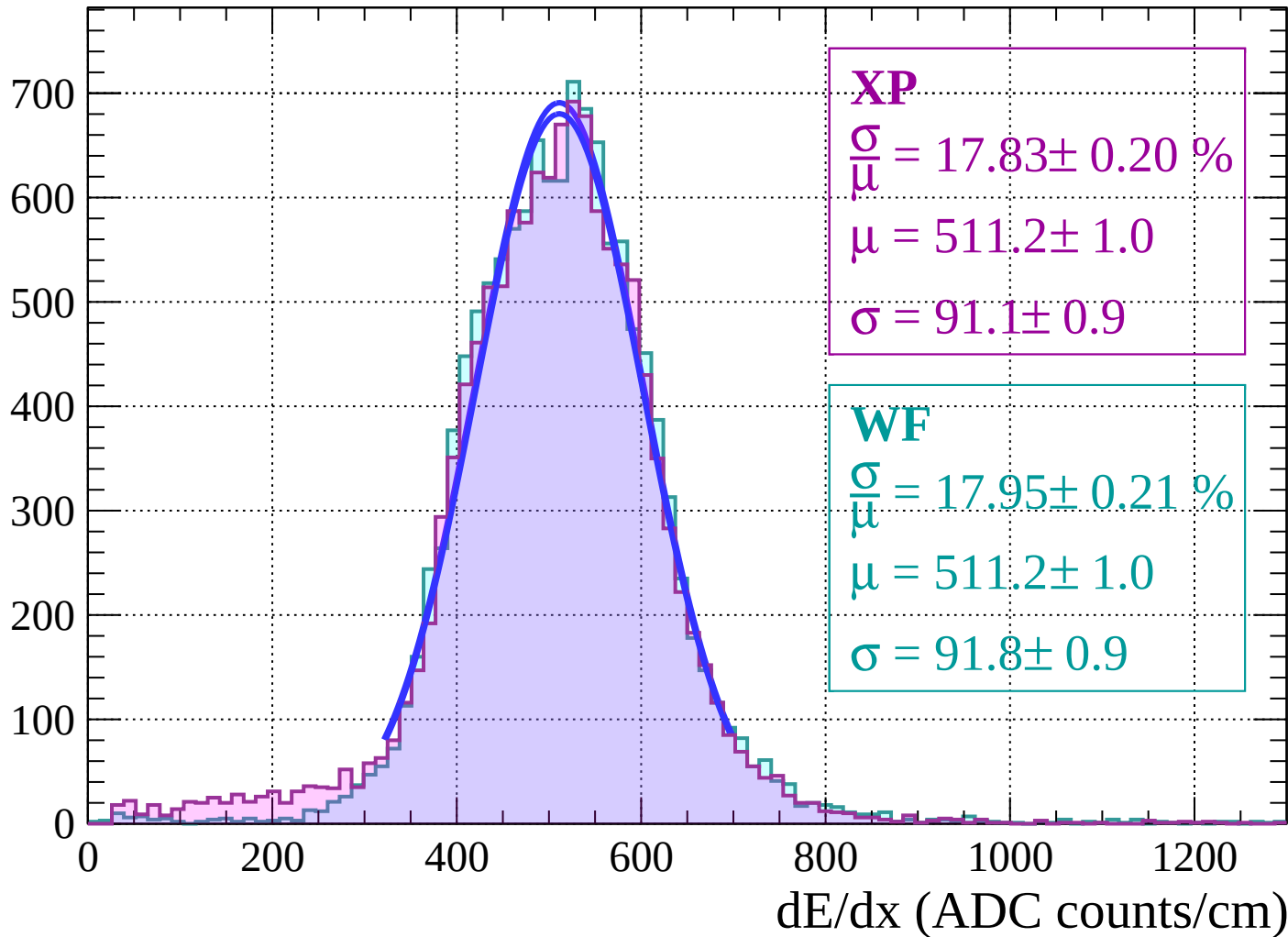
Energy loss | $-40 < \theta < -38$

Count



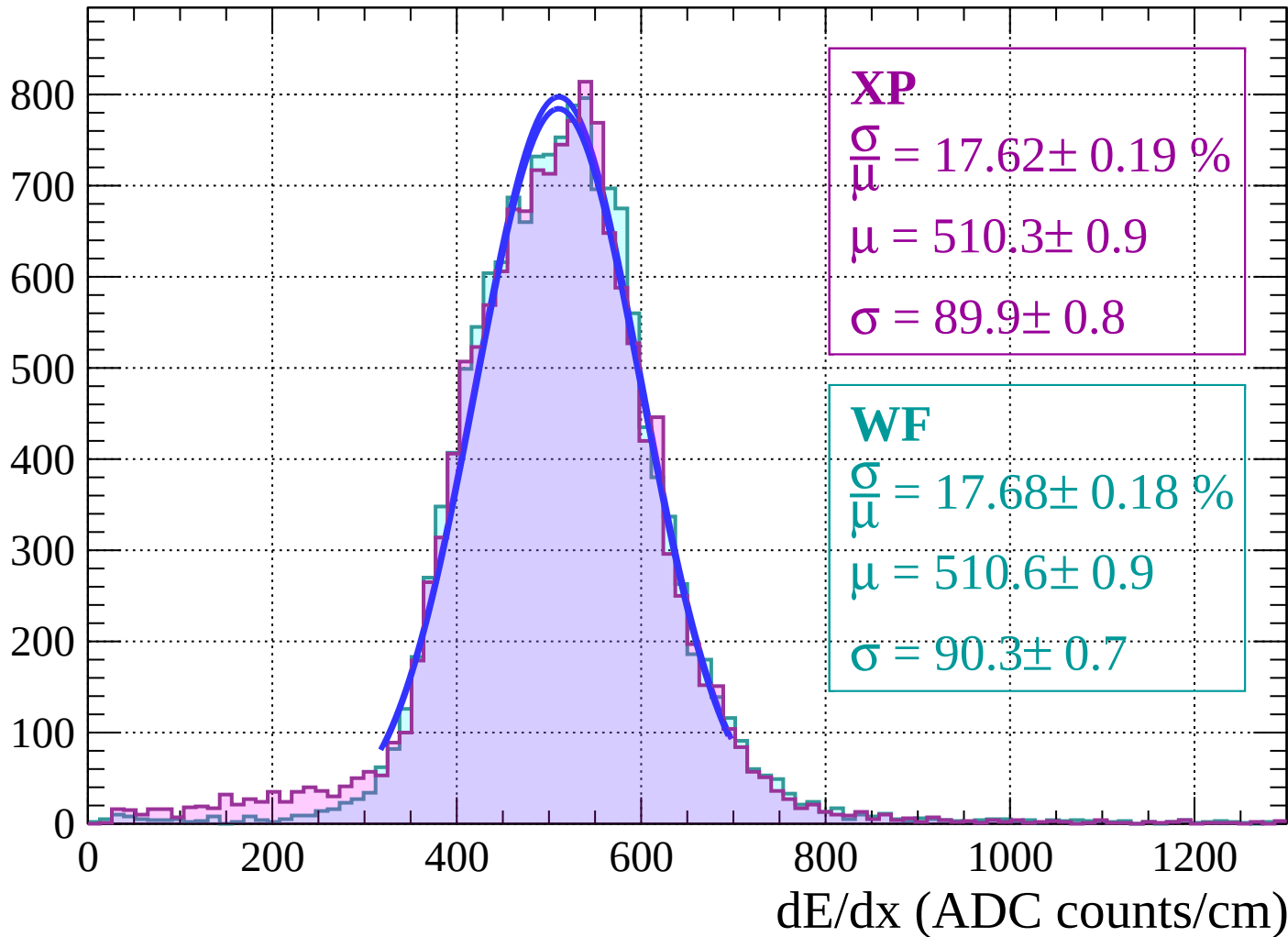
Energy loss $|-38 < \theta < -36$

Count

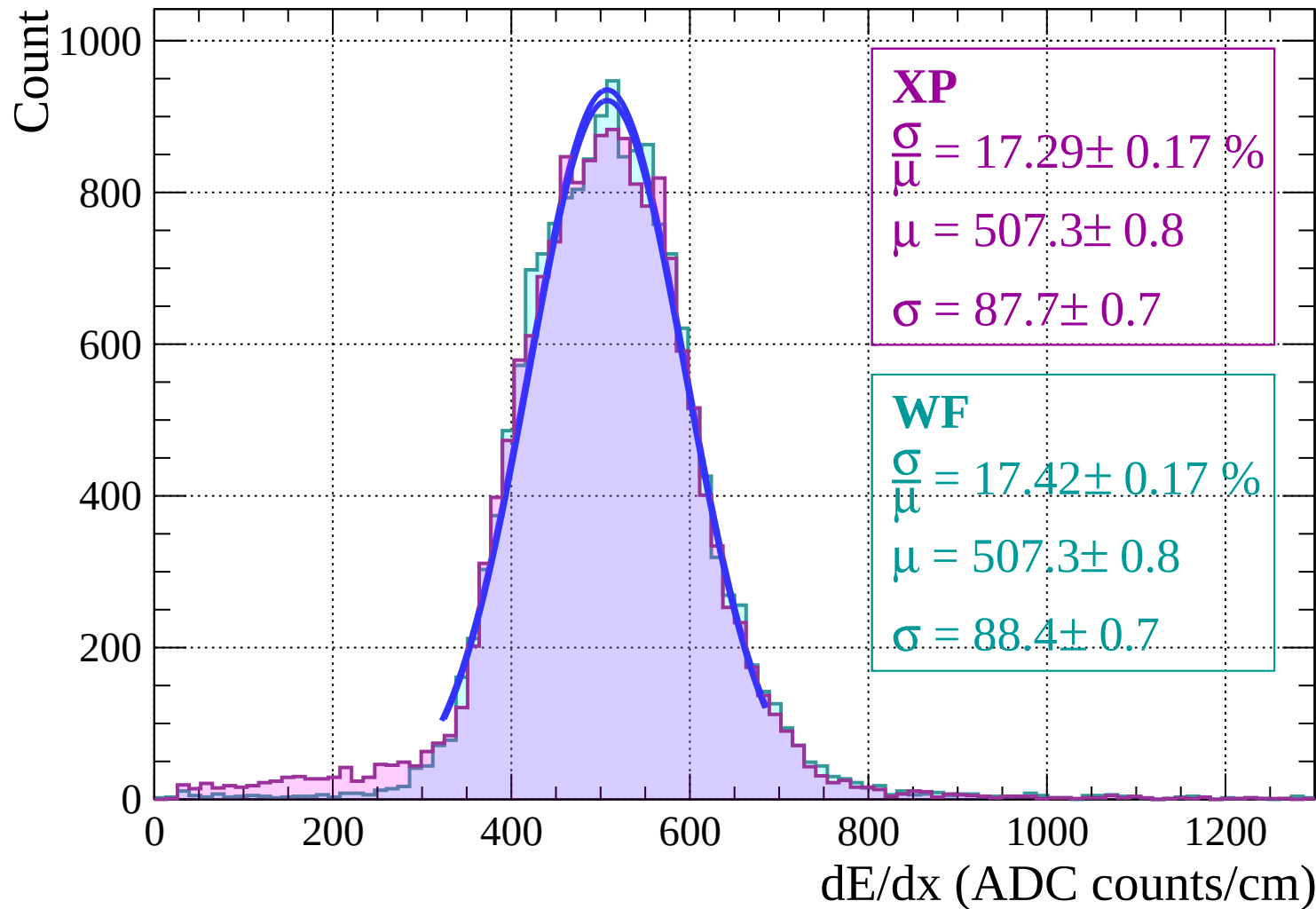


Energy loss $|-36 < \theta < -34$

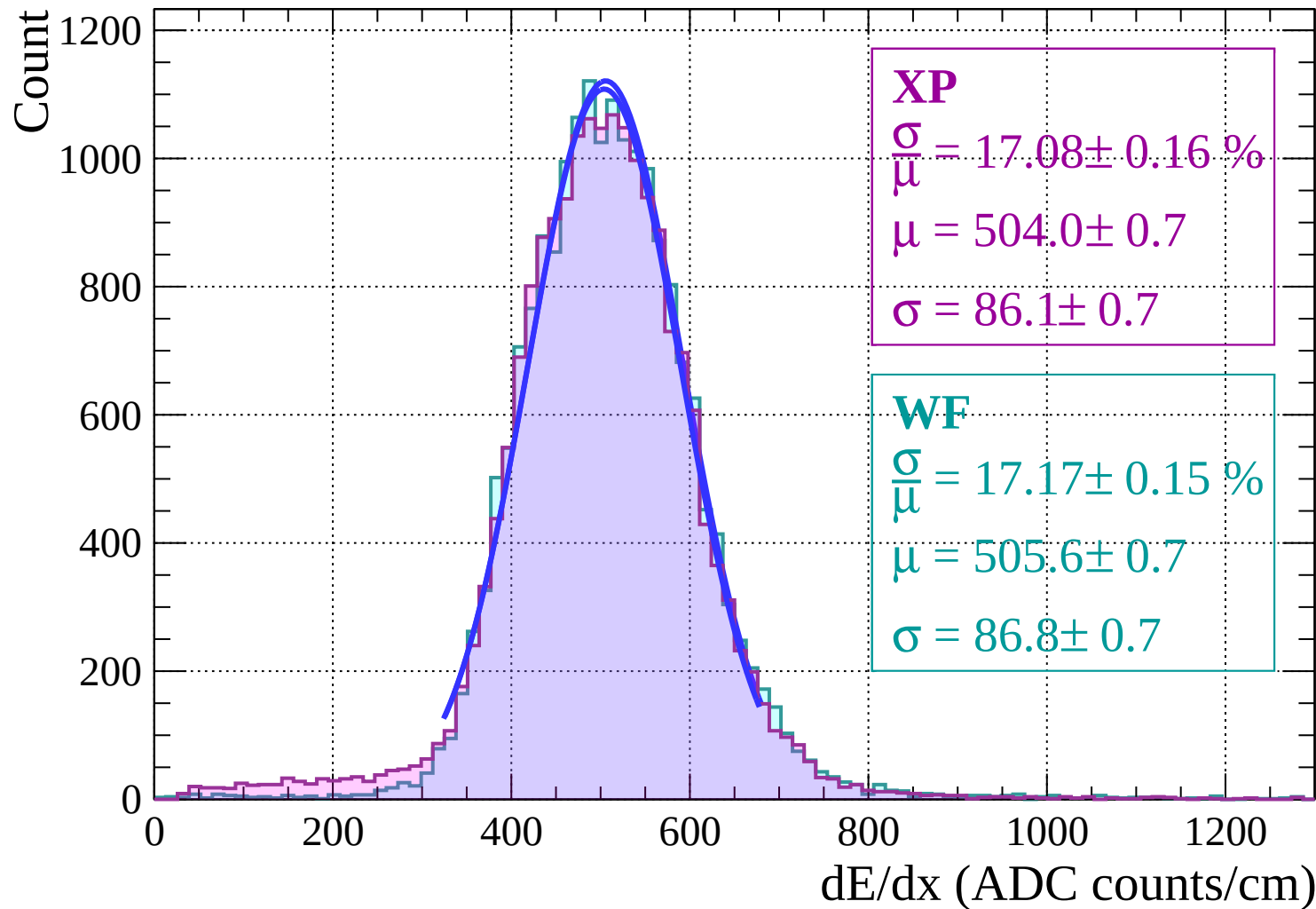
Count



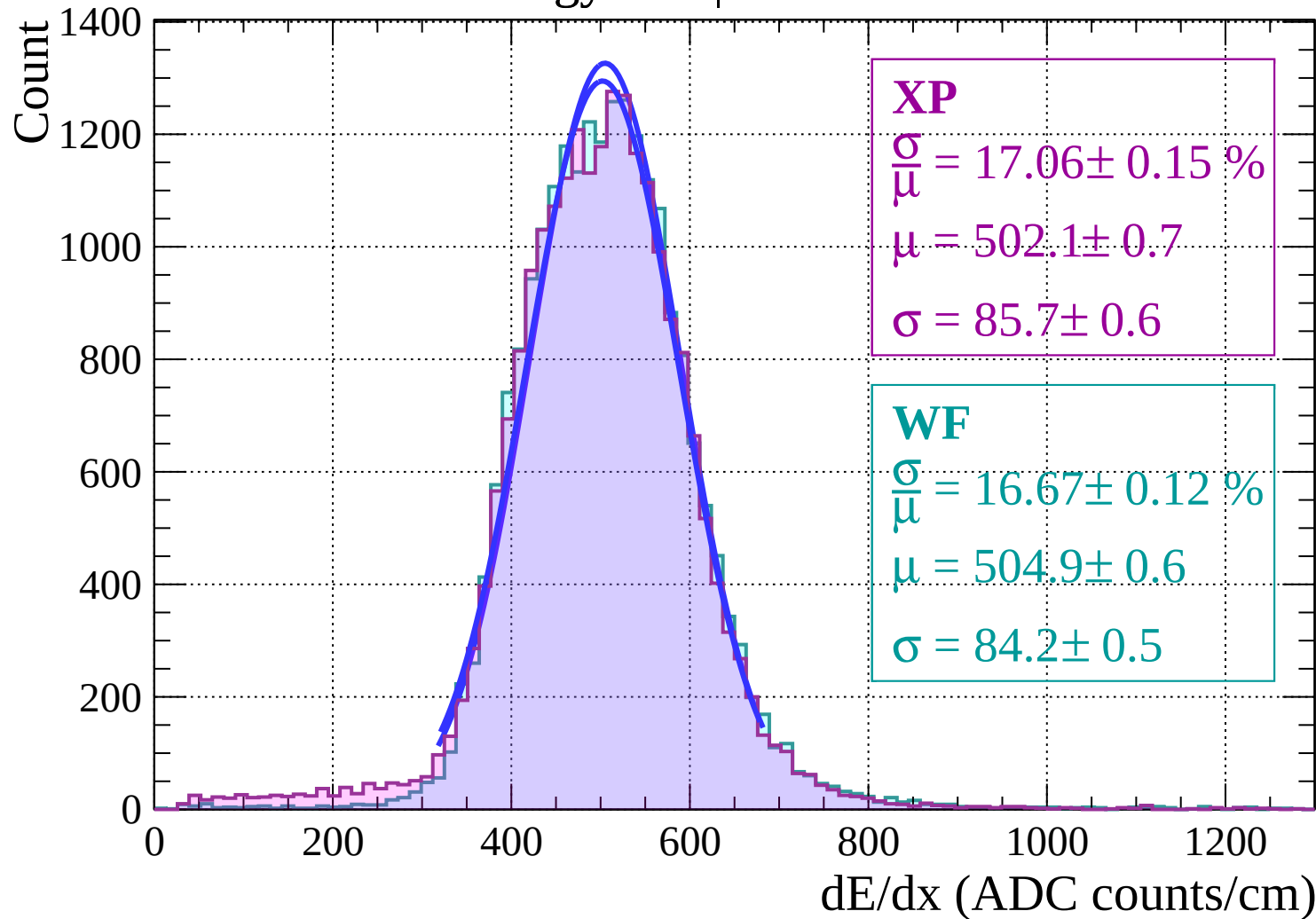
Energy loss | $-34 < \theta < -32$



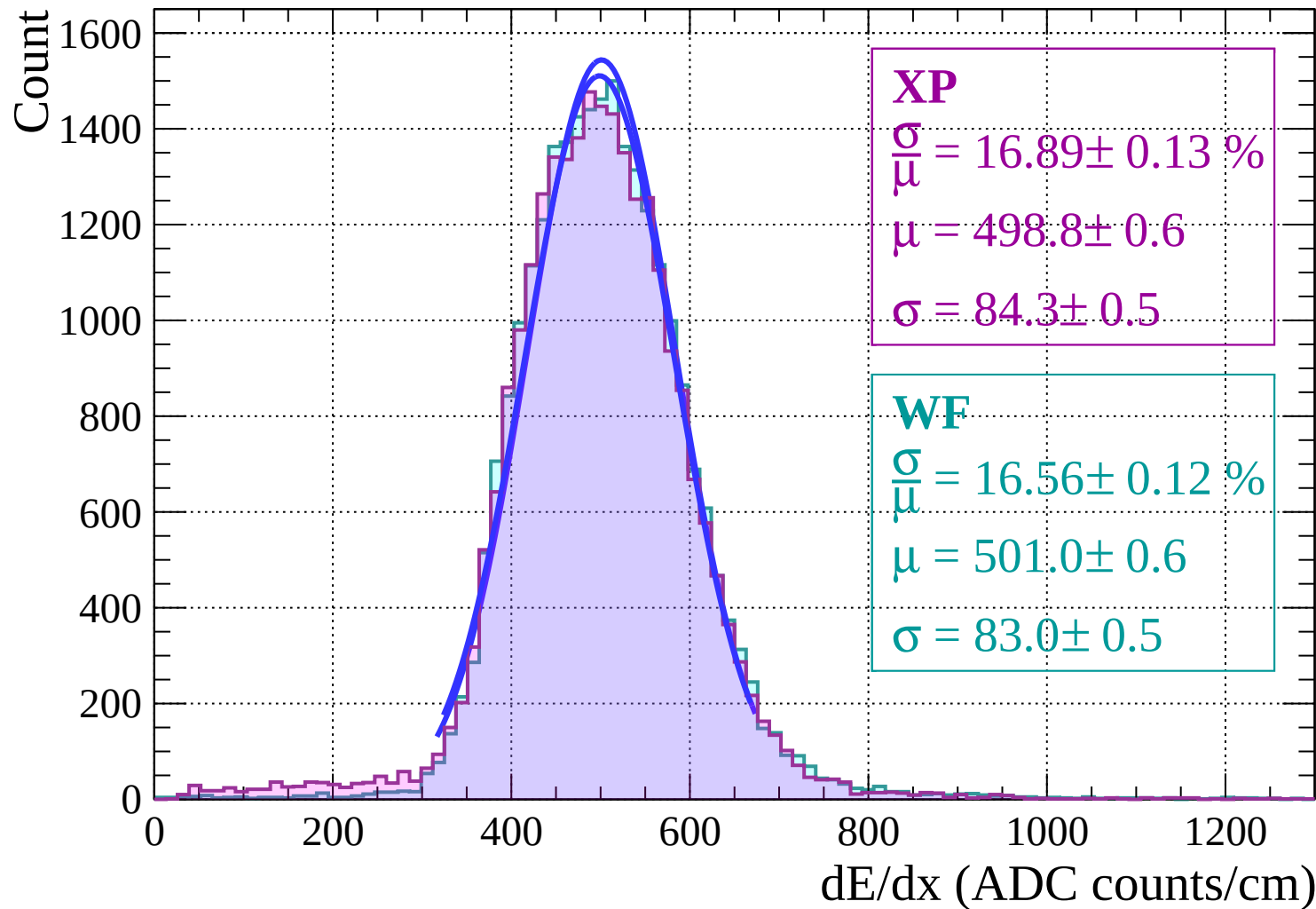
Energy loss | $-32 < \theta < -30$



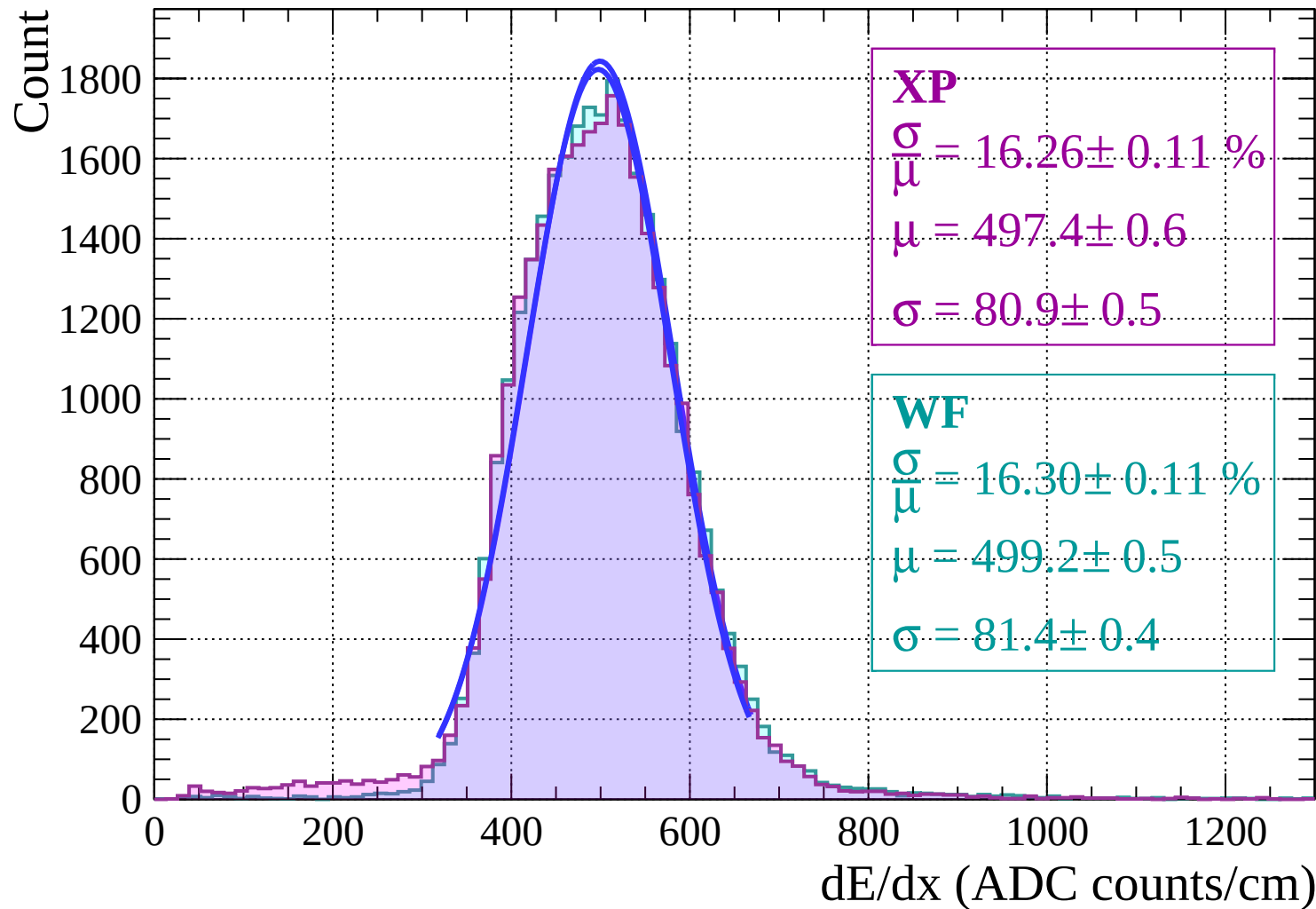
Energy loss $|-30 < \theta < -28$



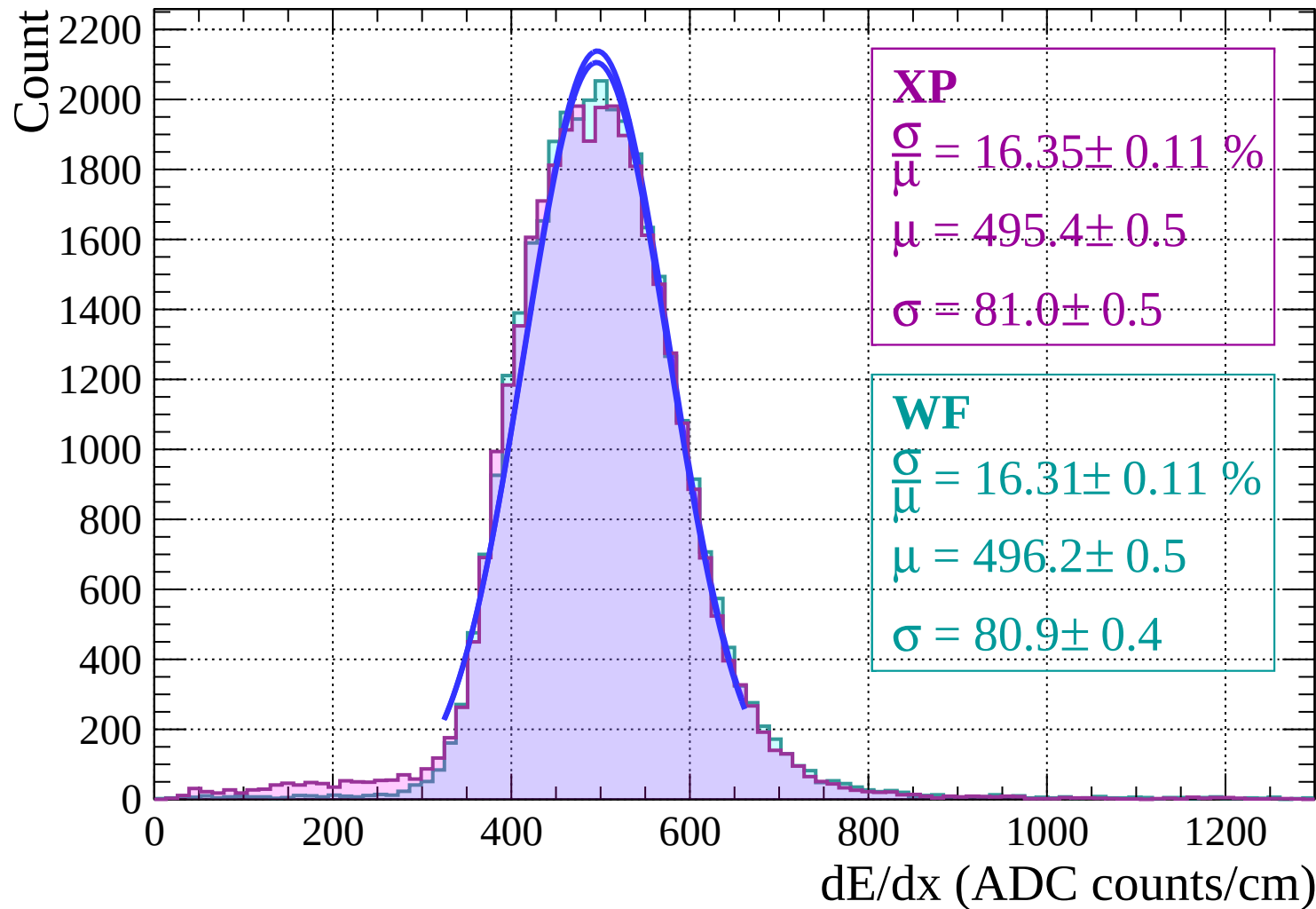
Energy loss | $-28 < \theta < -26$



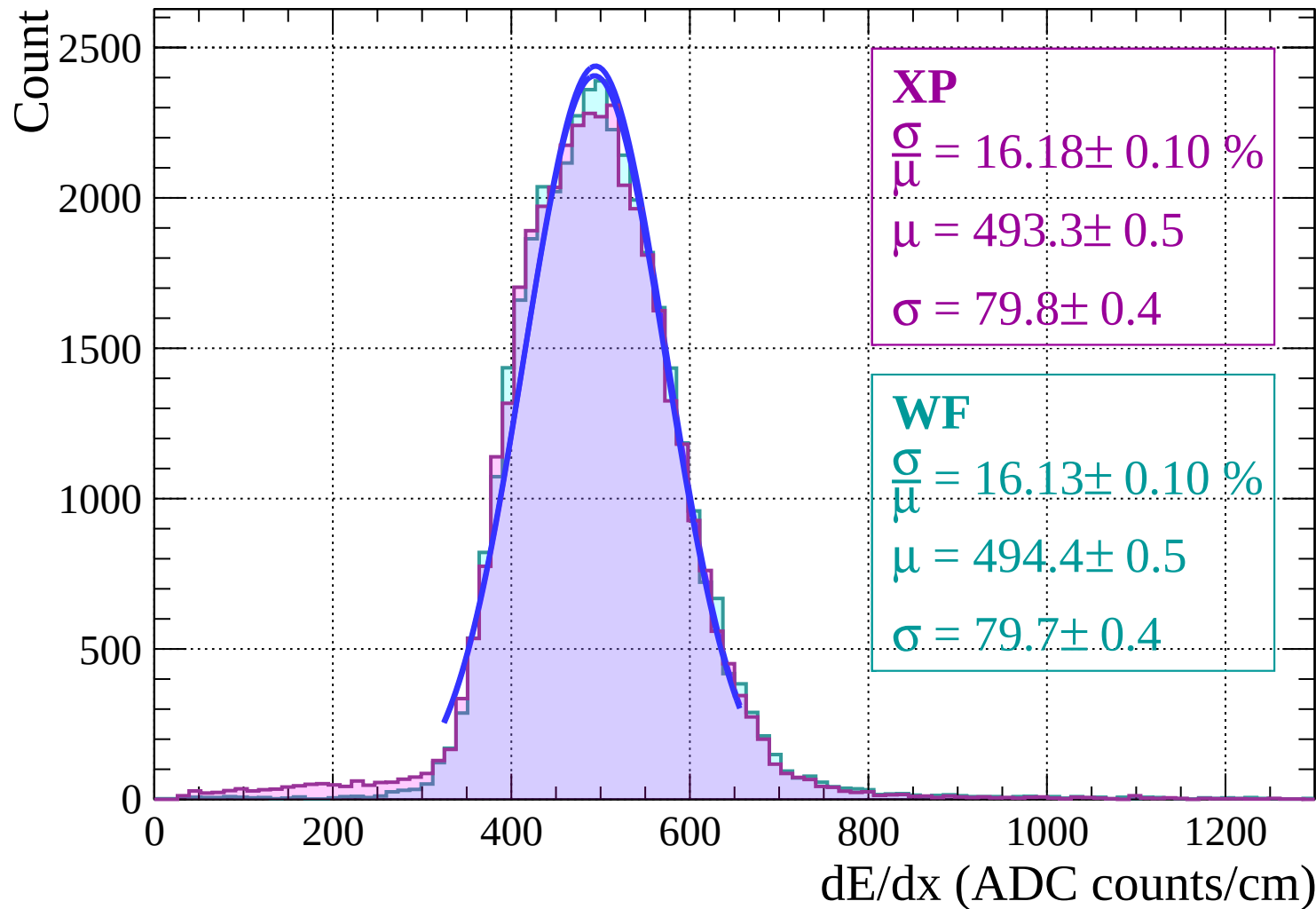
Energy loss $|-26 < \theta < -24$



Energy loss $|-24 < \theta < -22$

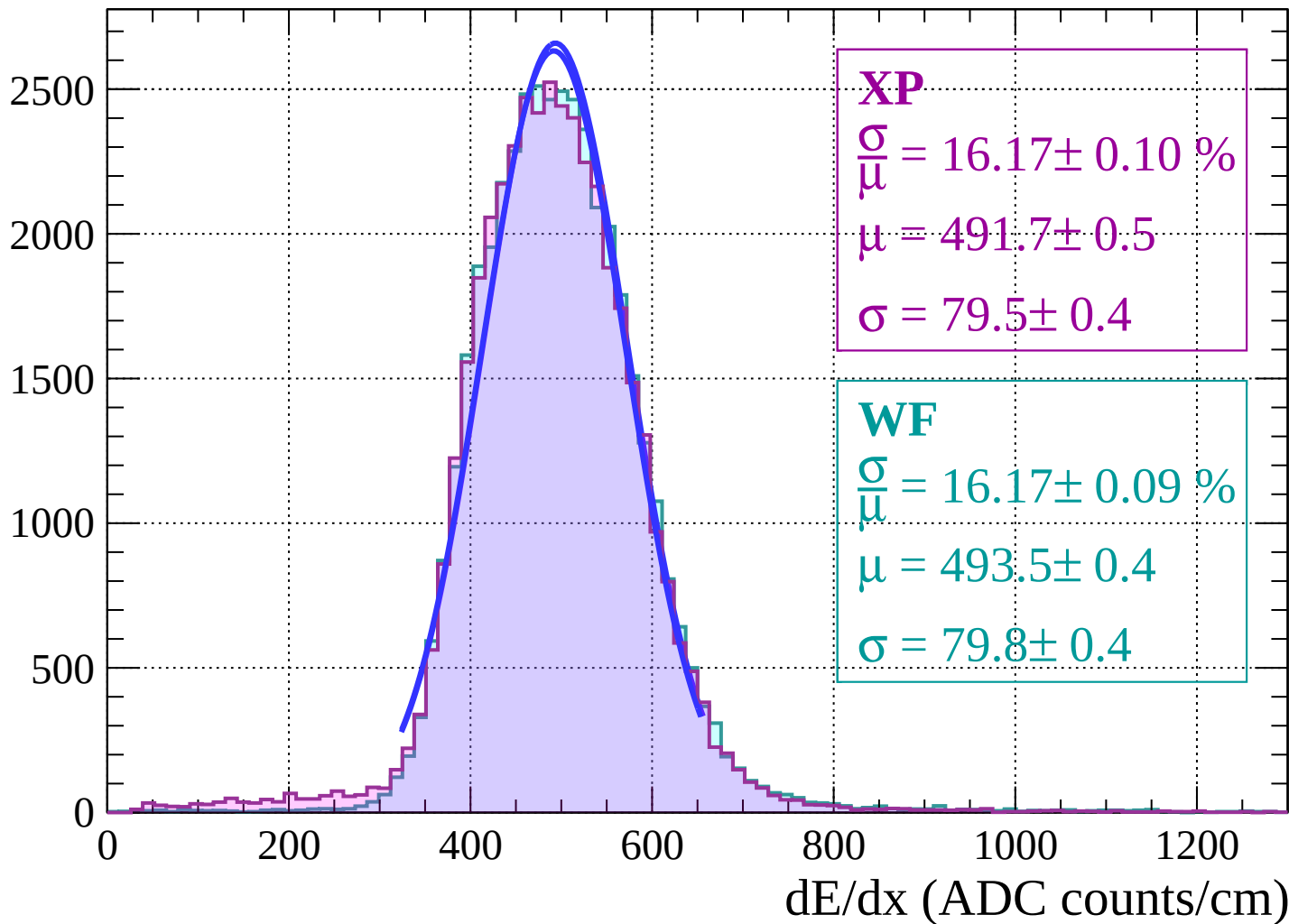


Energy loss $|-22 < \theta < -20$

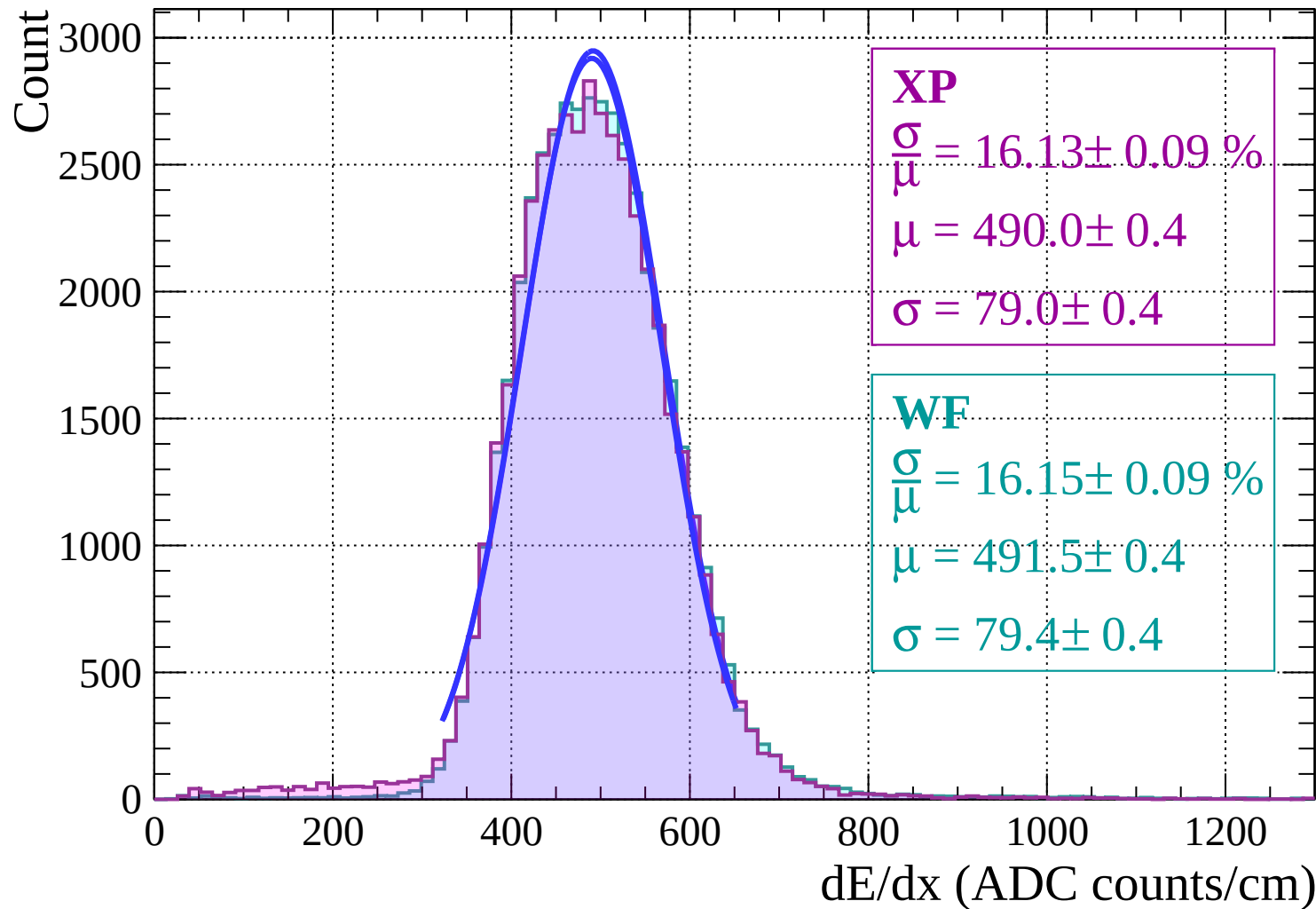


Energy loss | $-20 < \theta < -18$

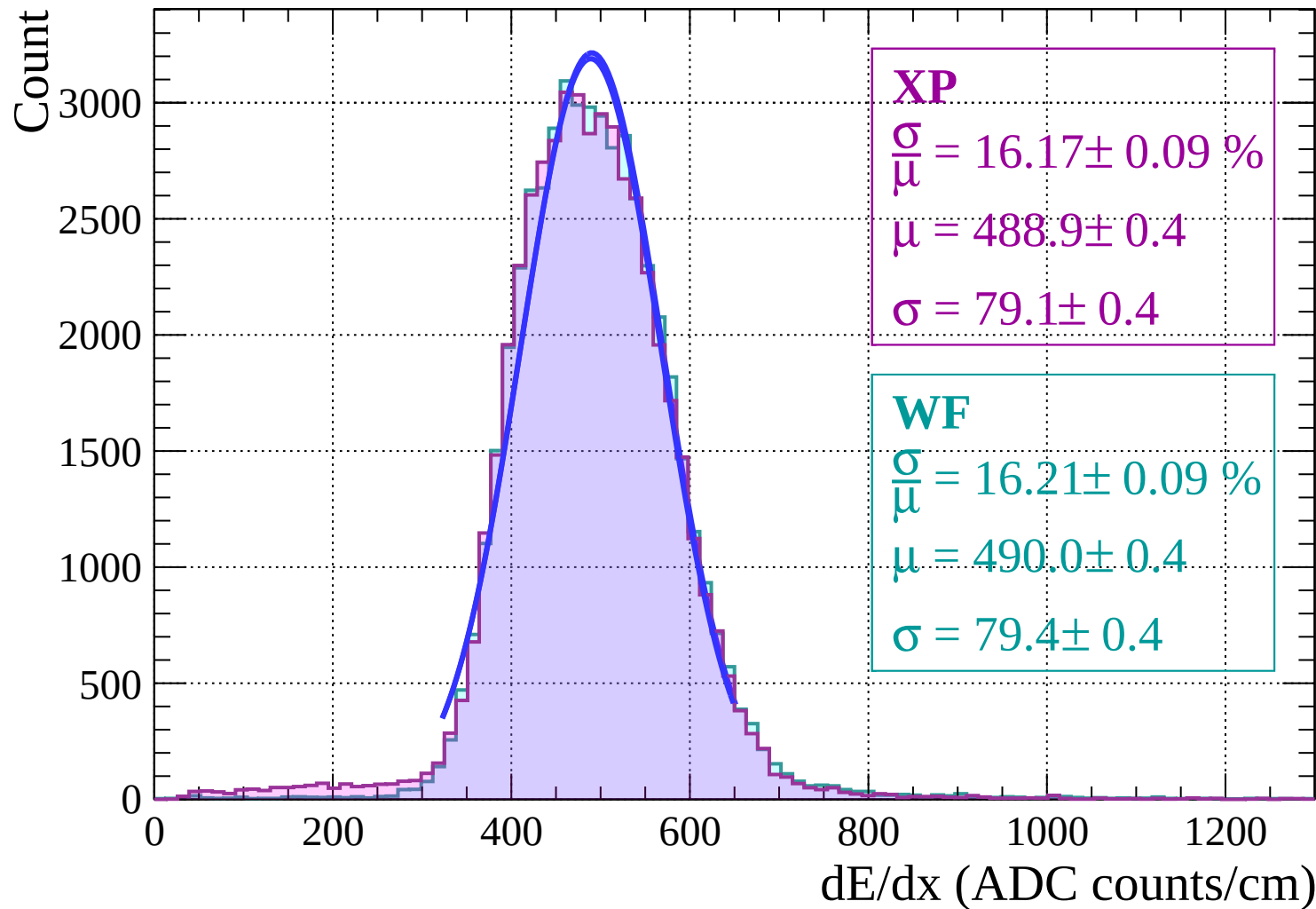
Count



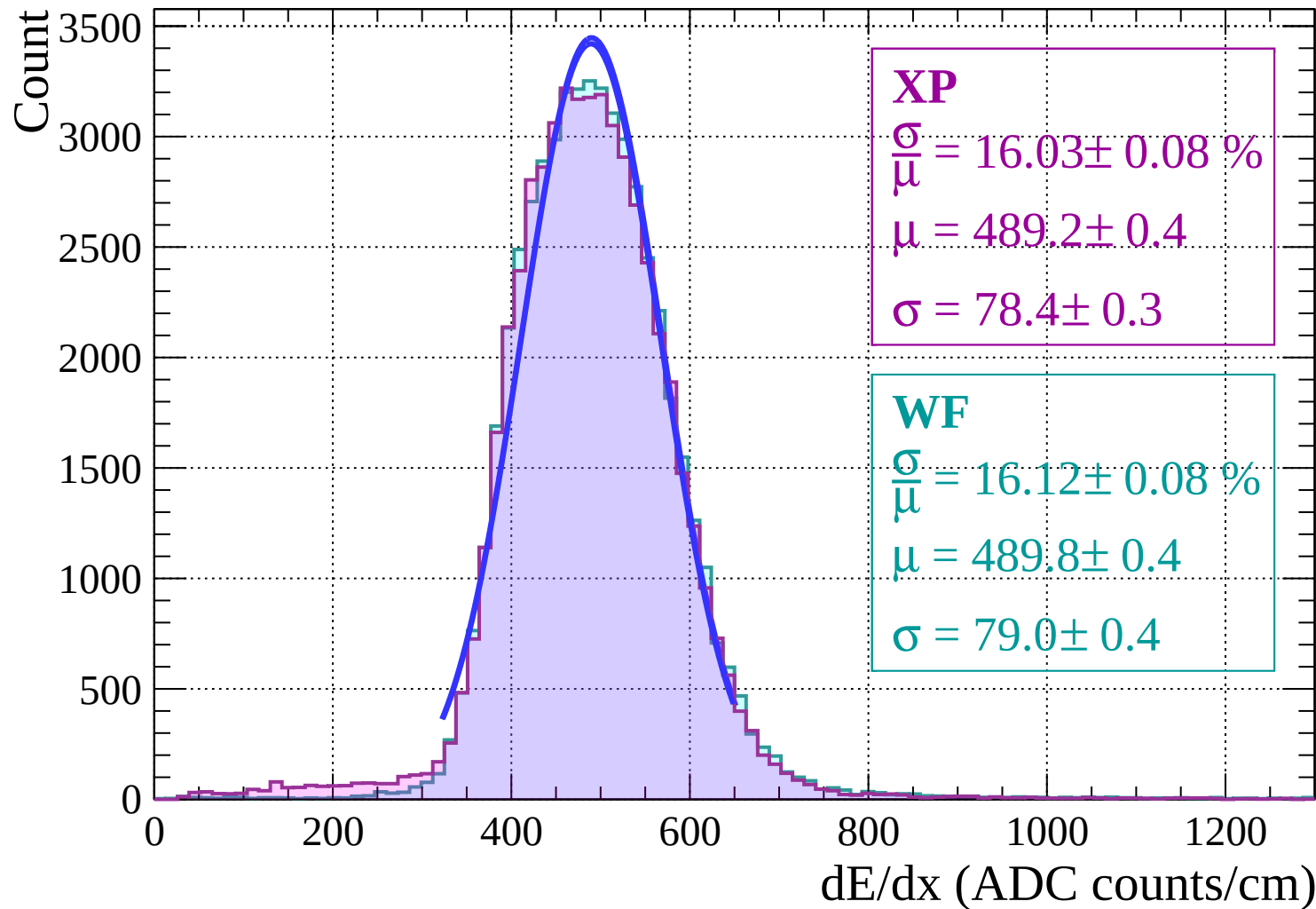
Energy loss | $-18 < \theta < -16$



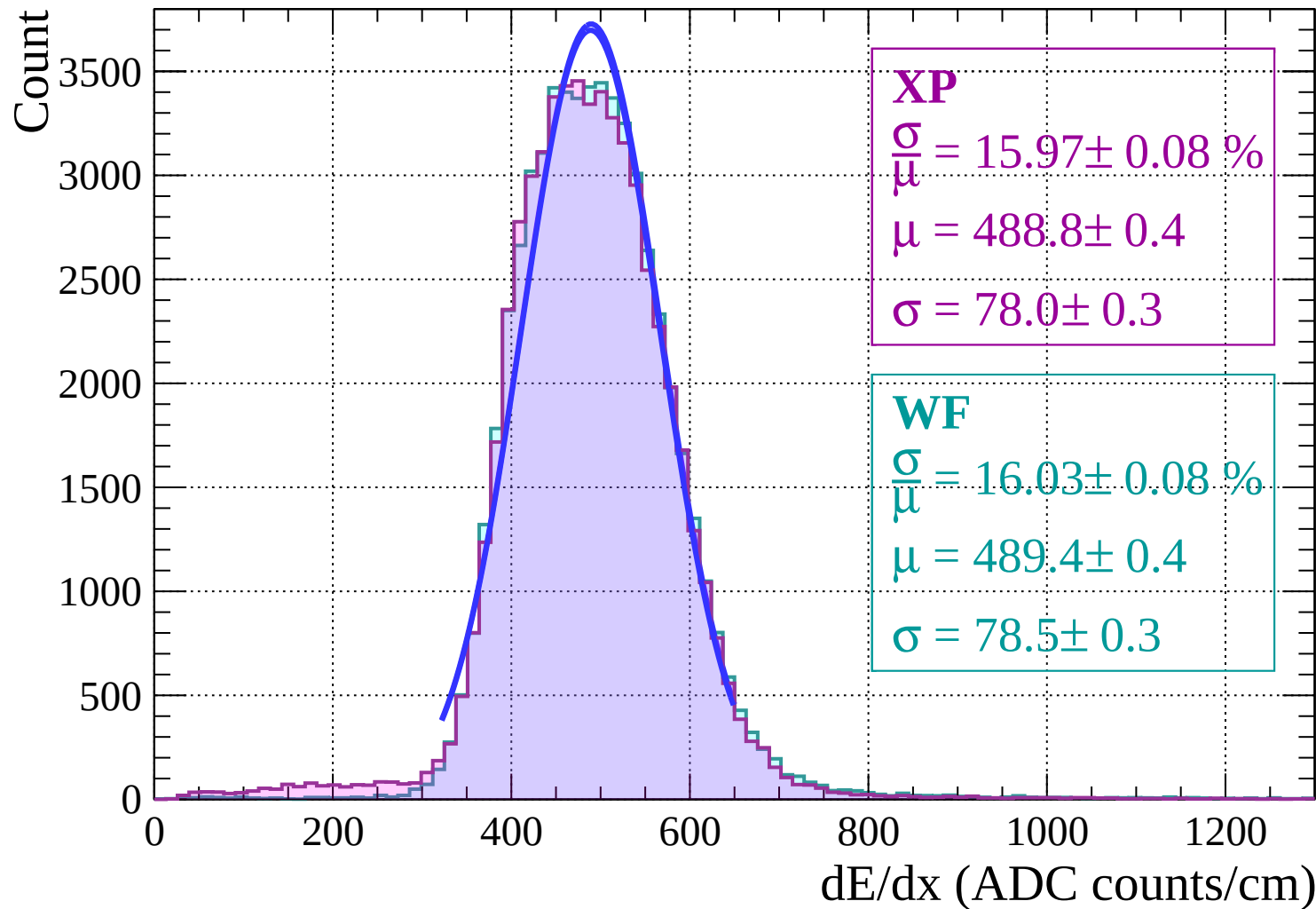
Energy loss | $-16 < \theta < -14$



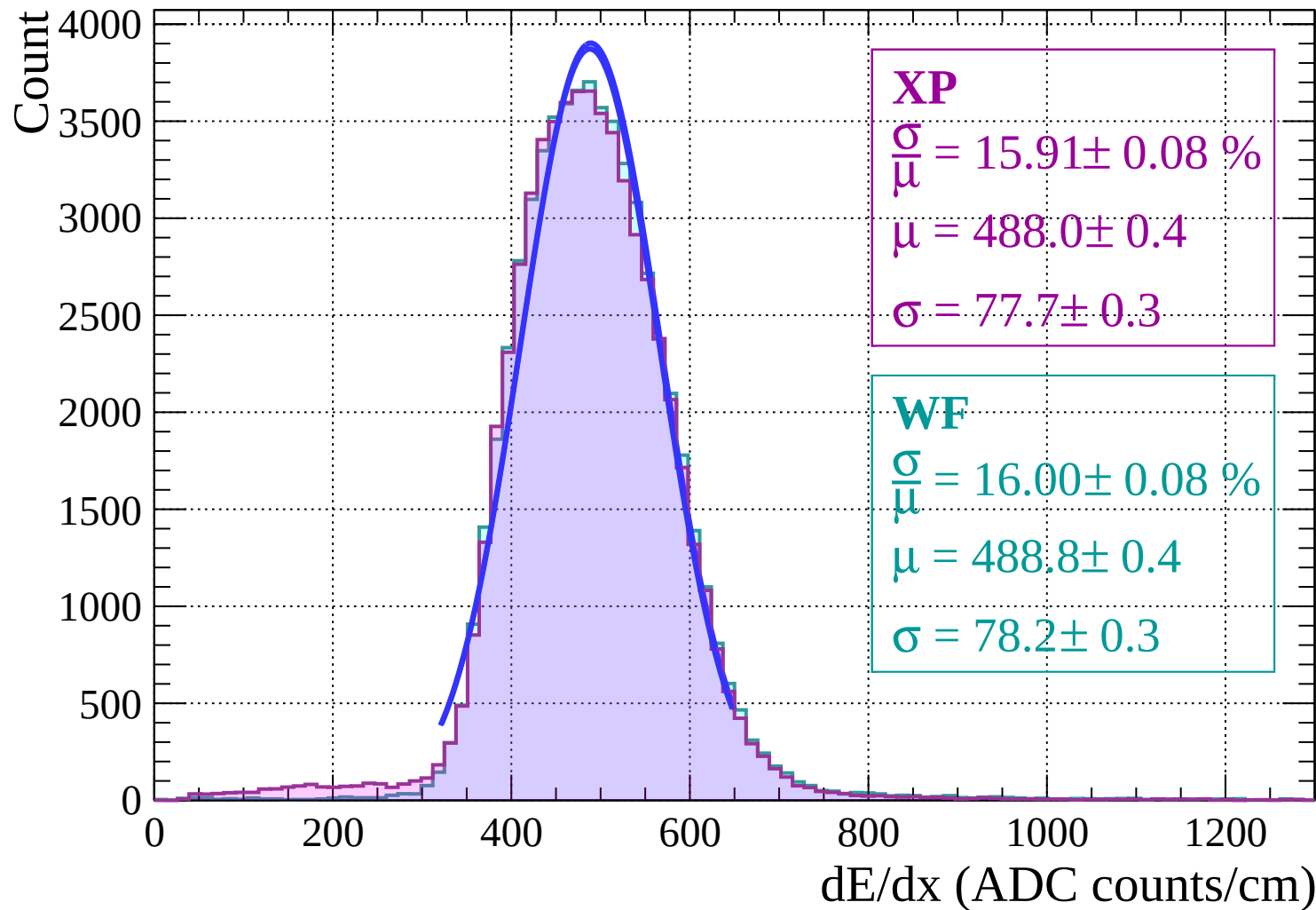
Energy loss | $-14 < \theta < -12$



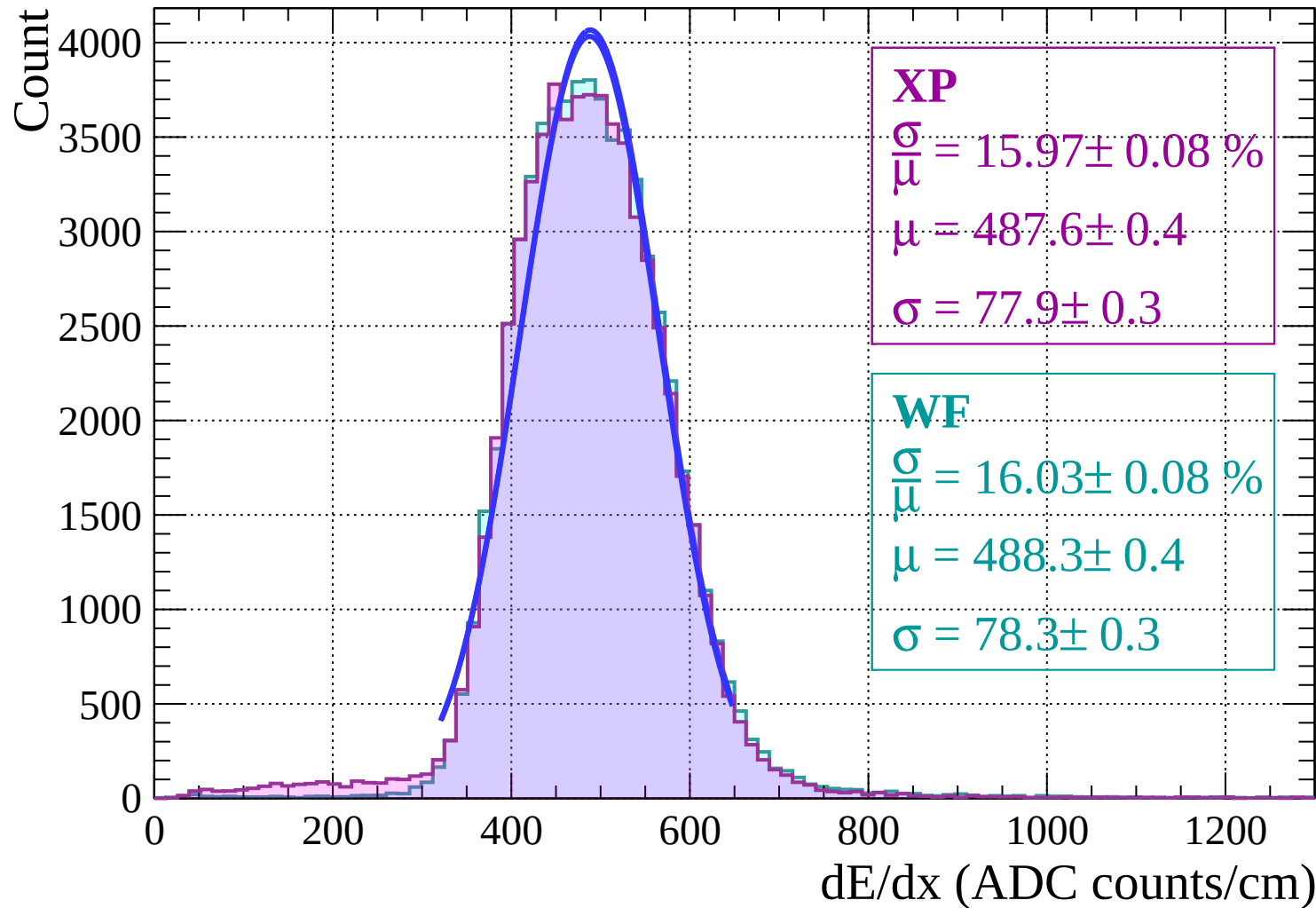
Energy loss | $-12 < \theta < -10$



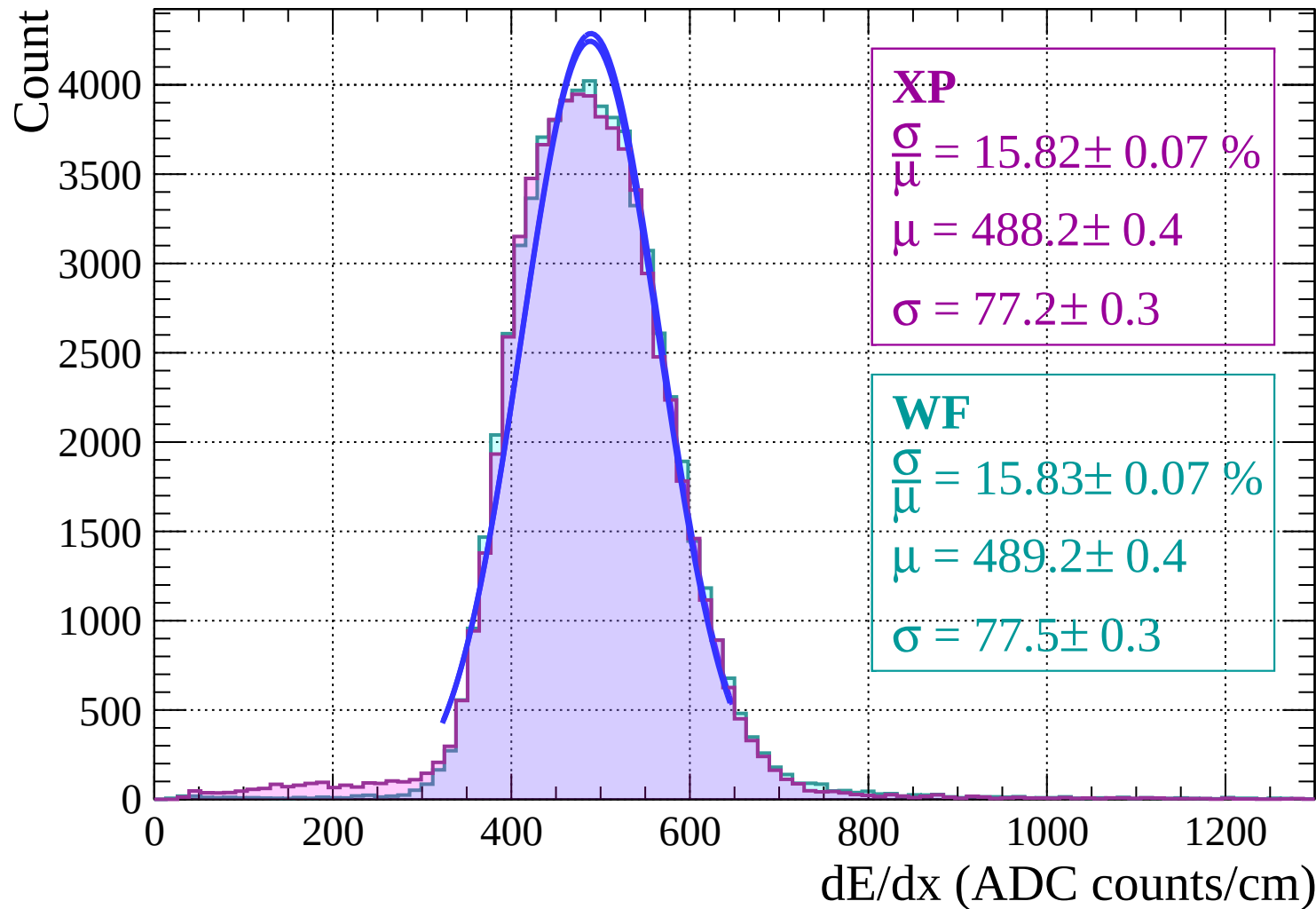
Energy loss | $-10 < \theta < -8$



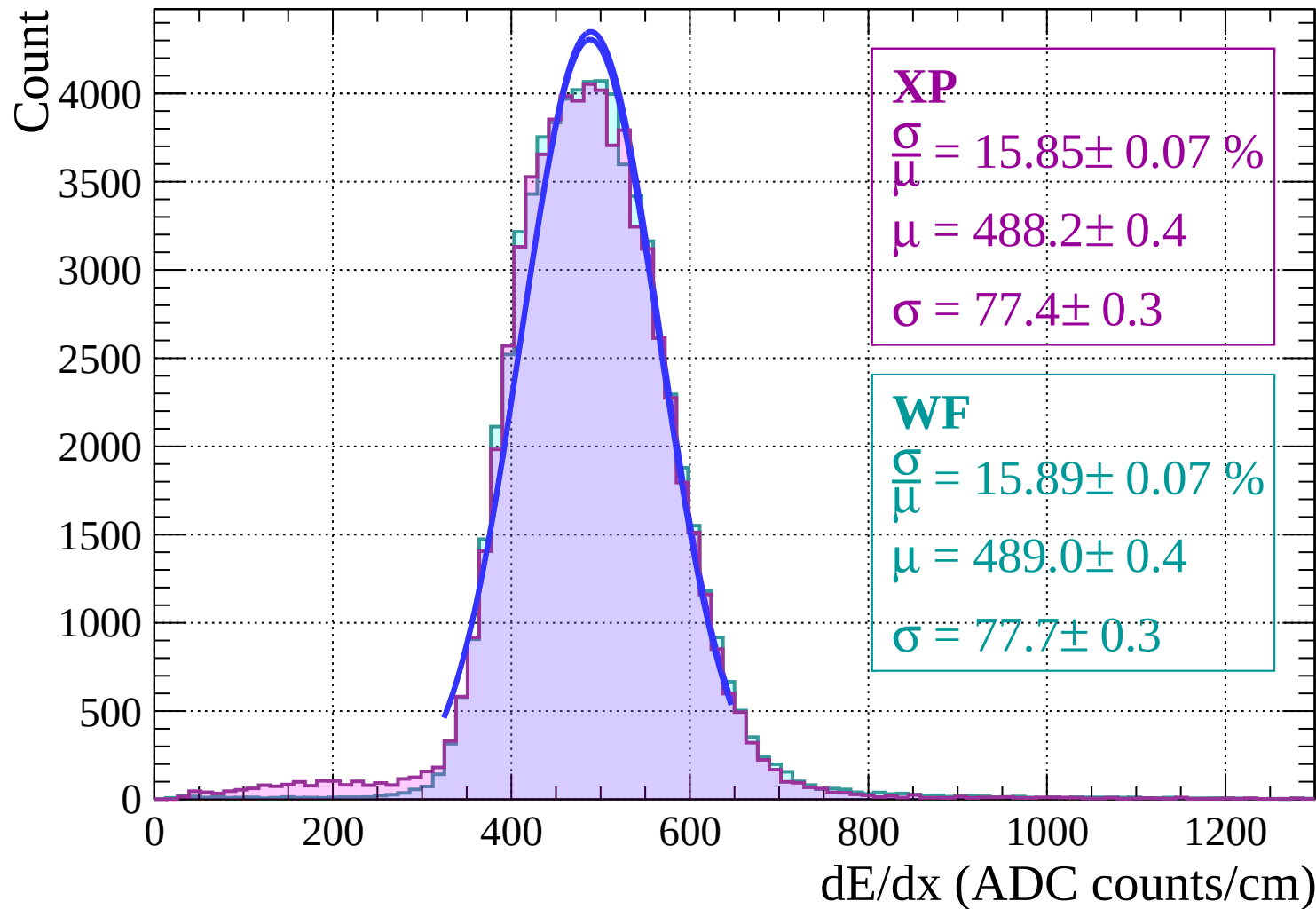
Energy loss | $-8 < \theta < -6$



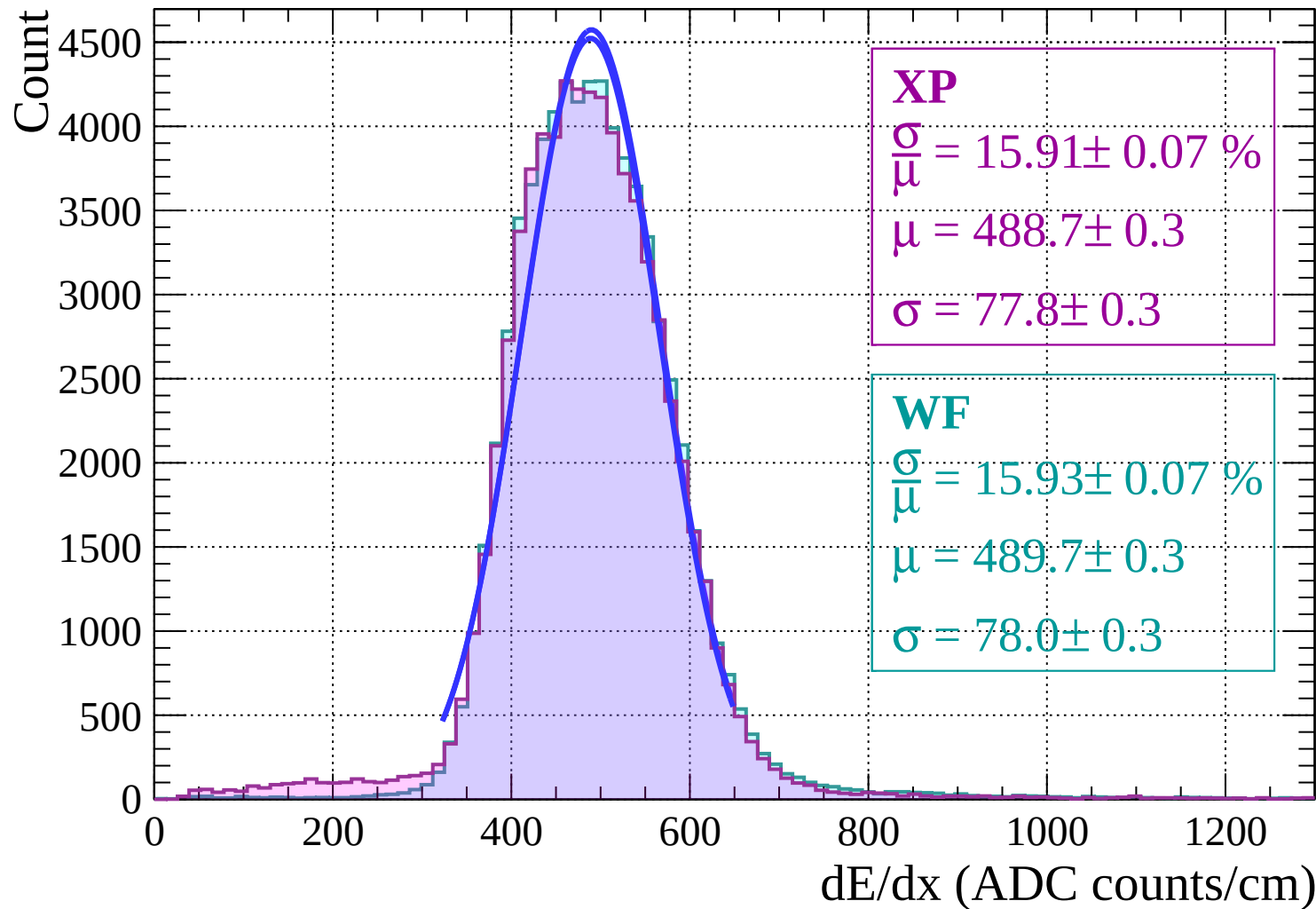
Energy loss | $-6 < \theta < -4$



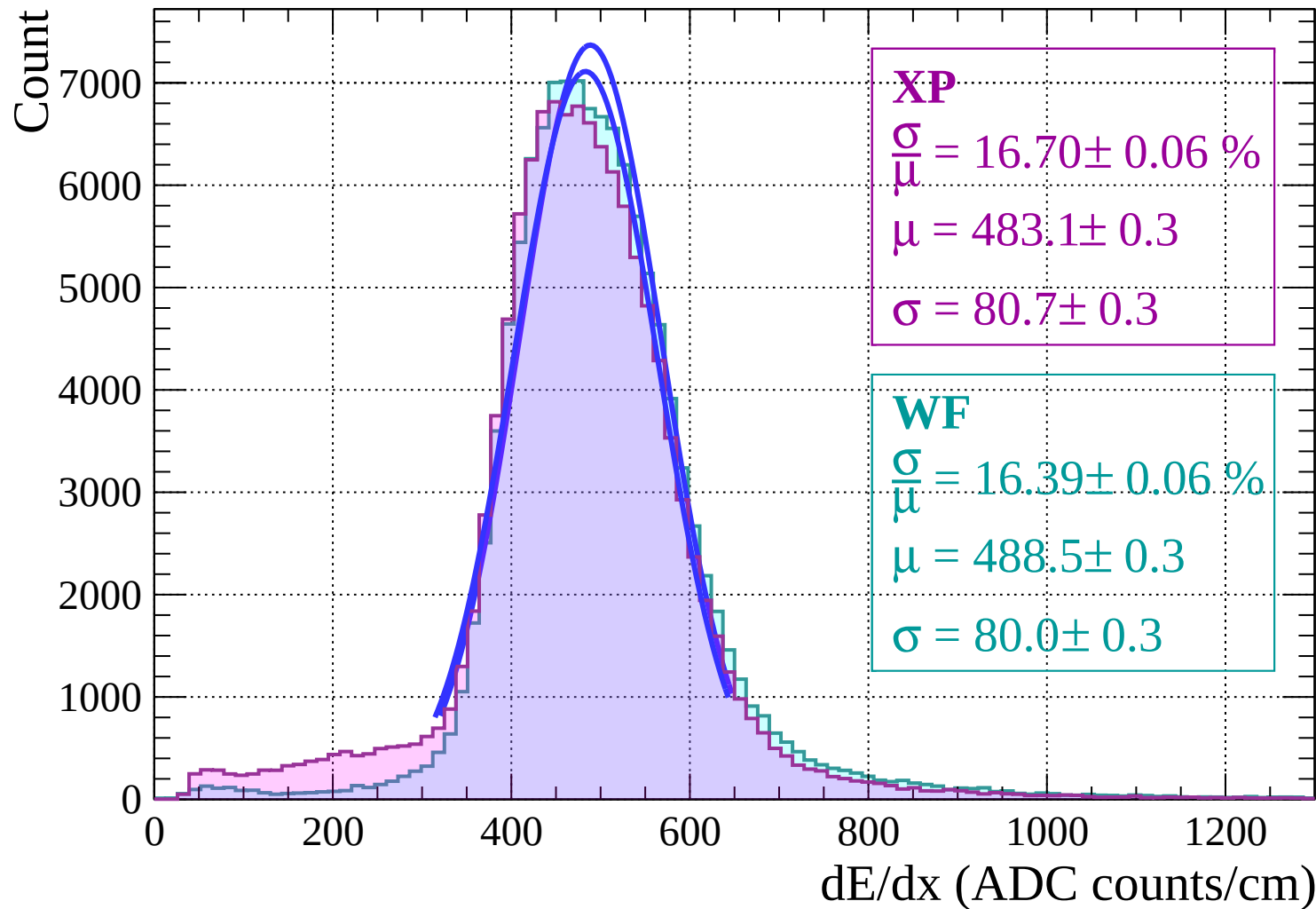
Energy loss | $-4 < \theta < -2$



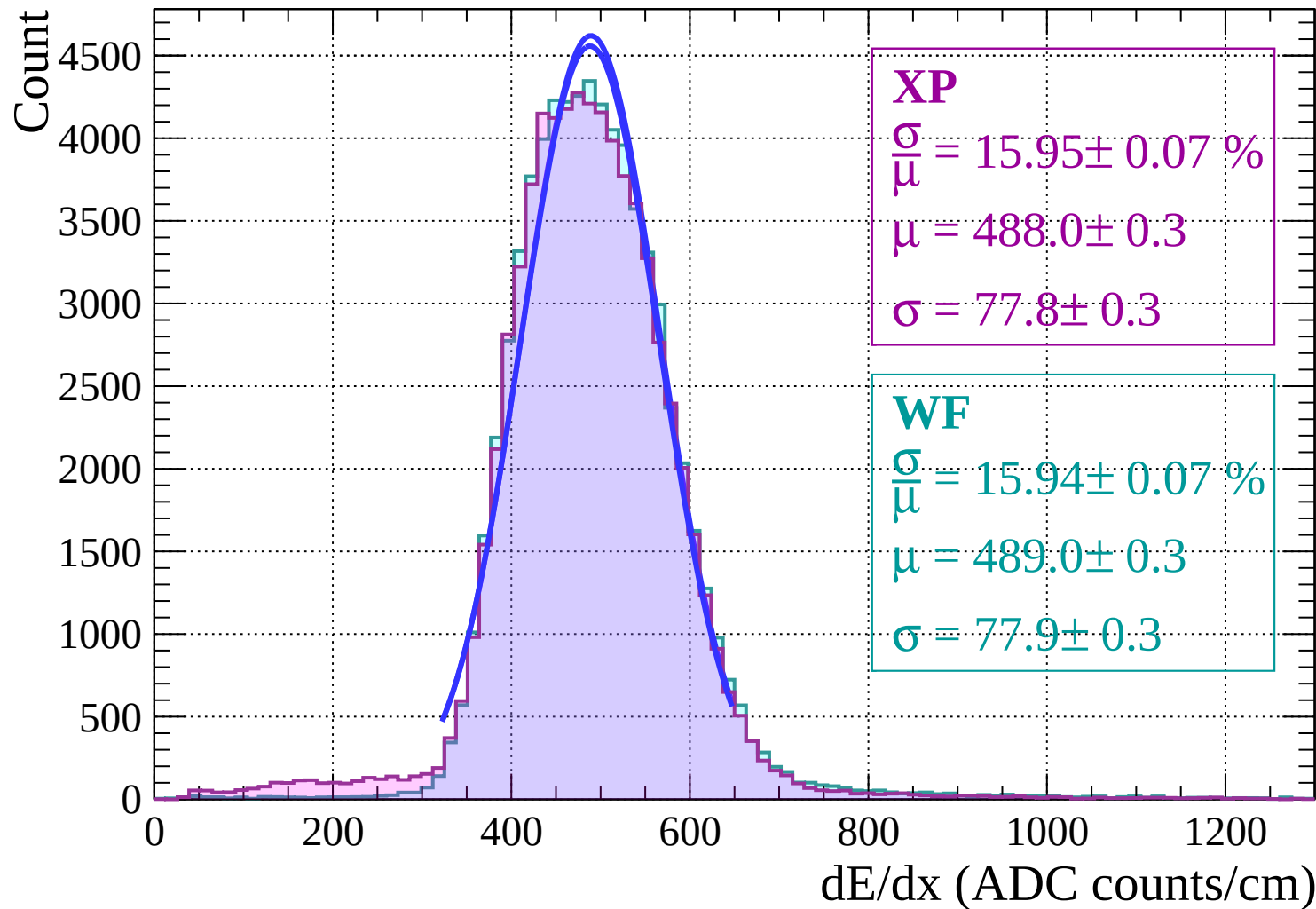
Energy loss | $-2 < \theta < 0$



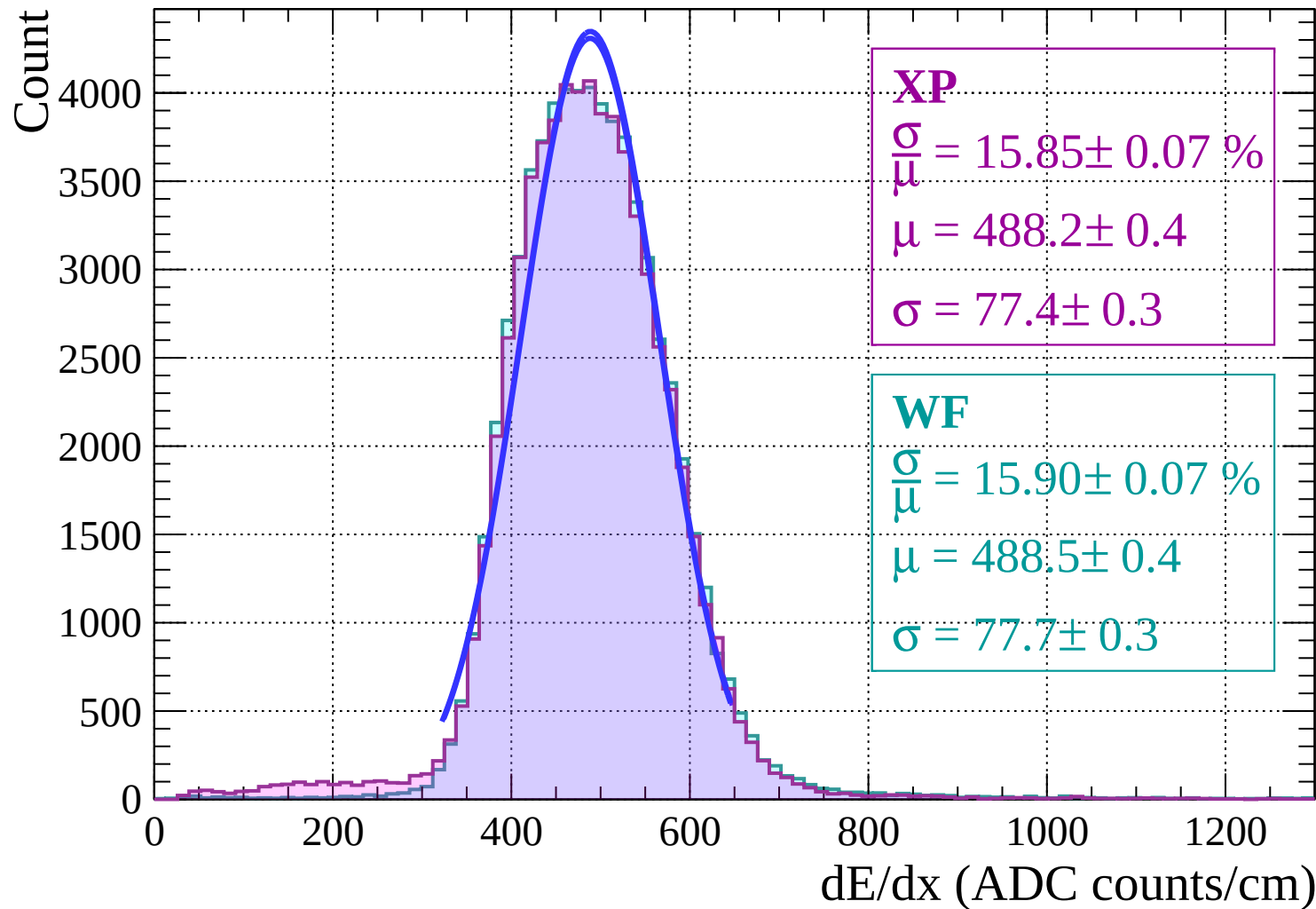
Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

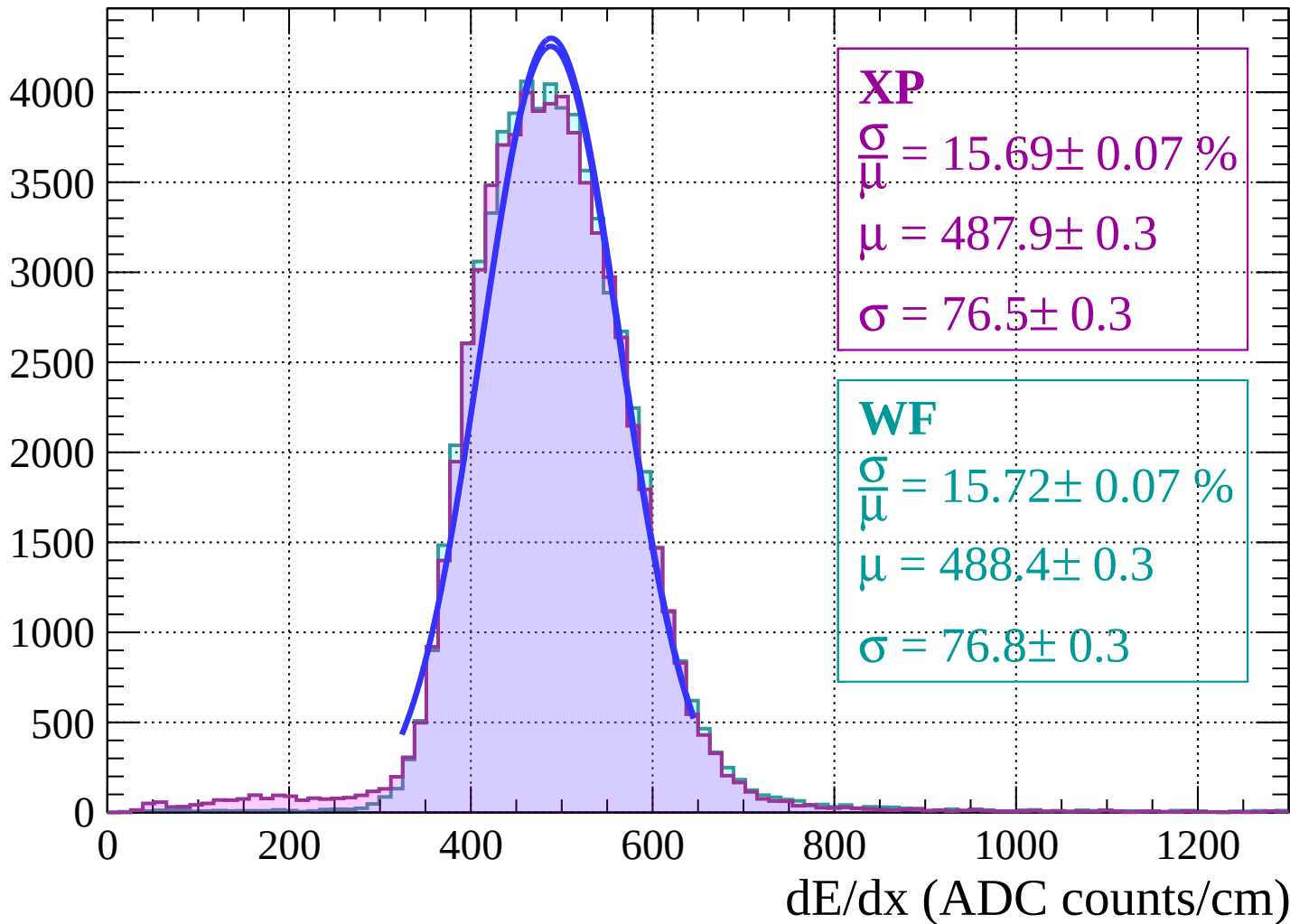


Energy loss | $4 < \theta < 6$

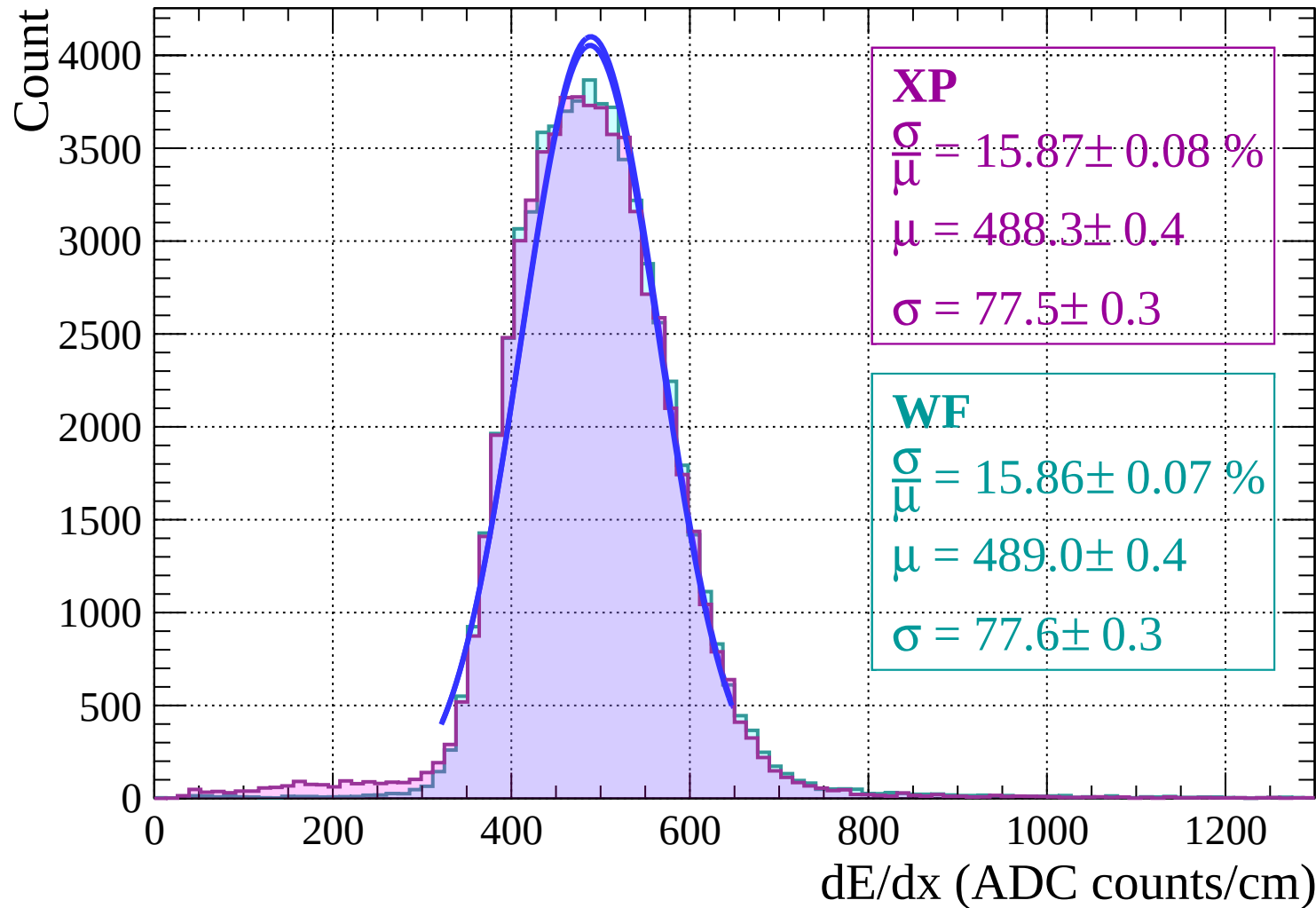


Energy loss | $6 < \theta < 8$

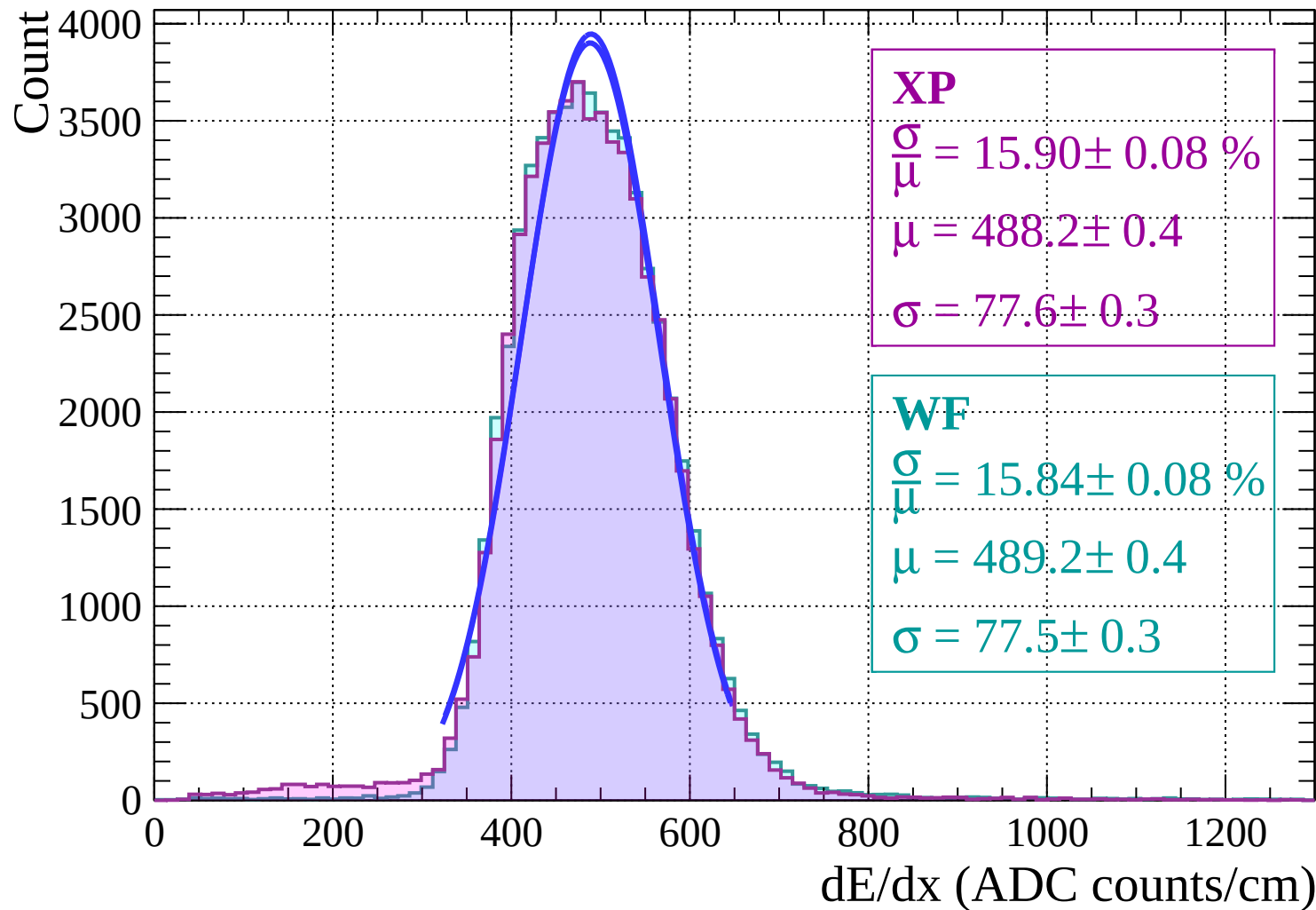
Count



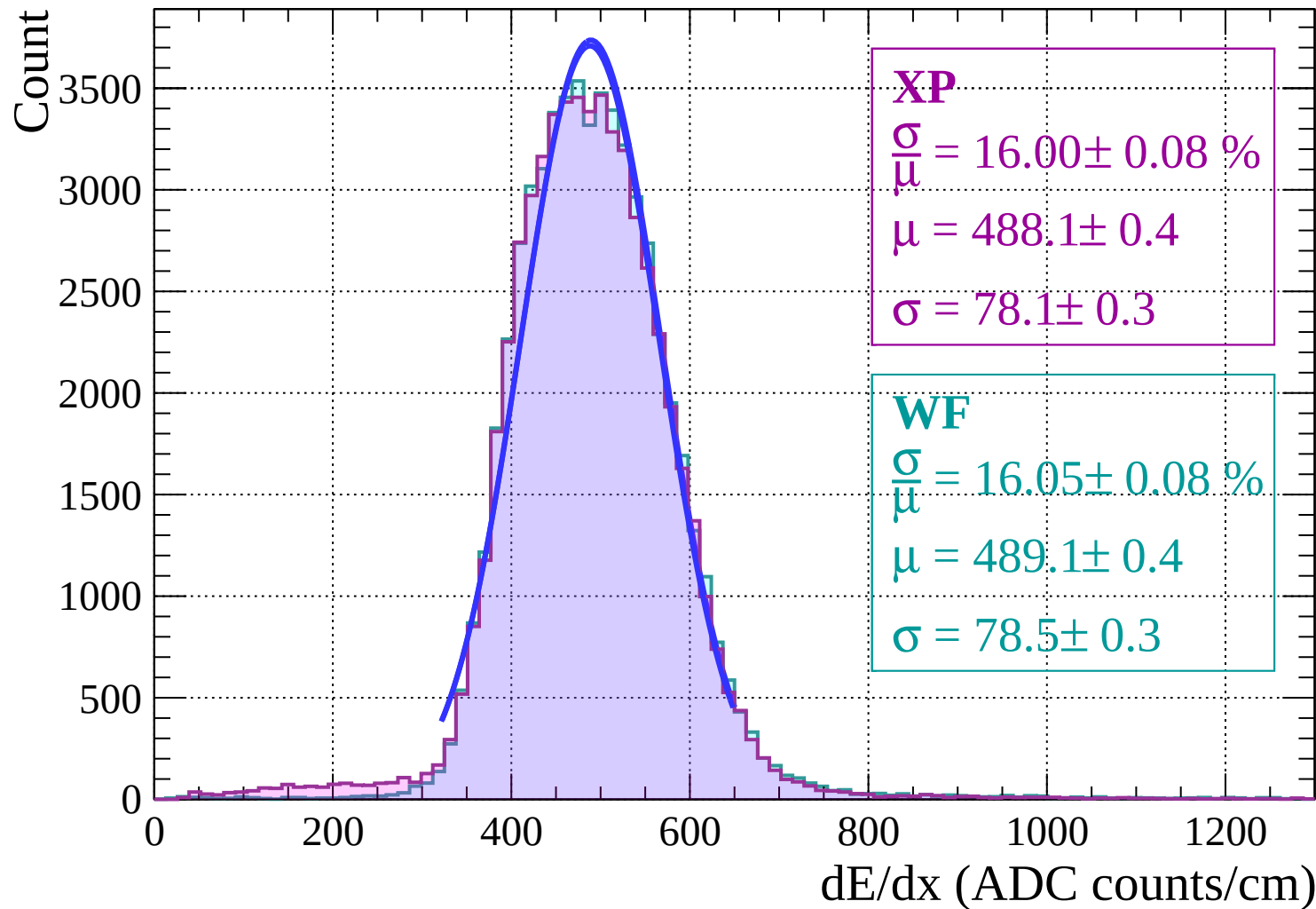
Energy loss | $8 < \theta < 10$



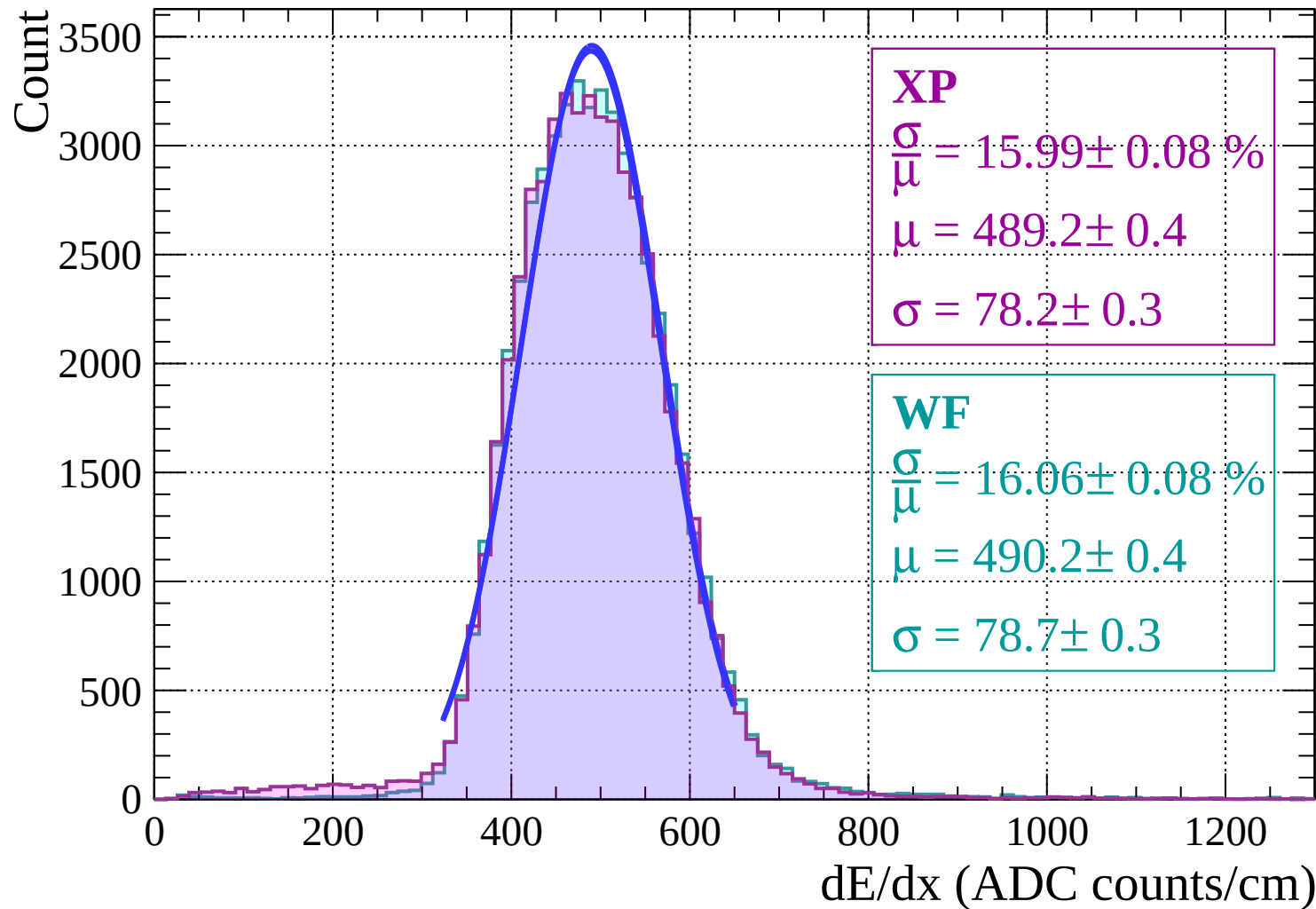
Energy loss | $10 < \theta < 12$



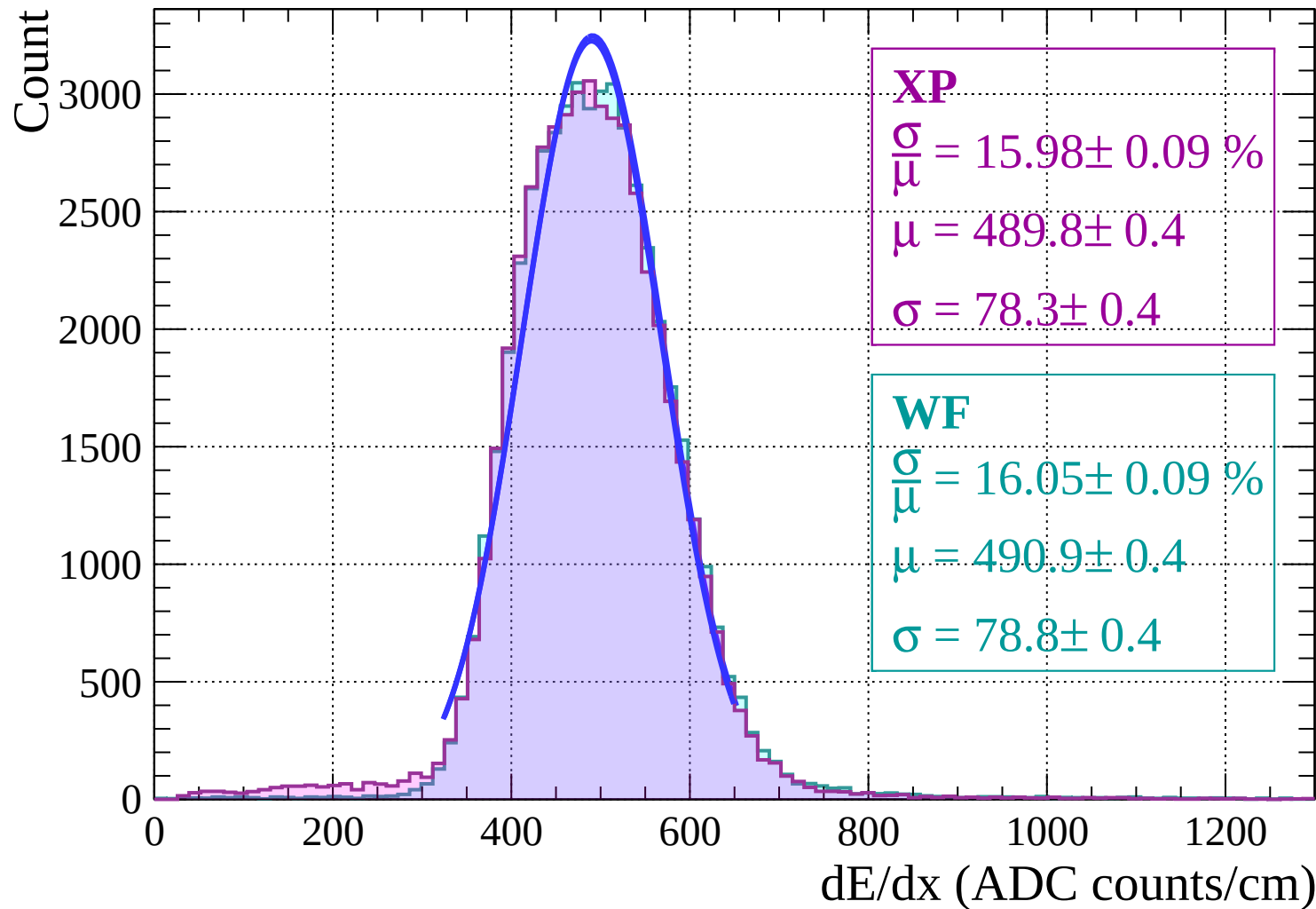
Energy loss | $12 < \theta < 14$



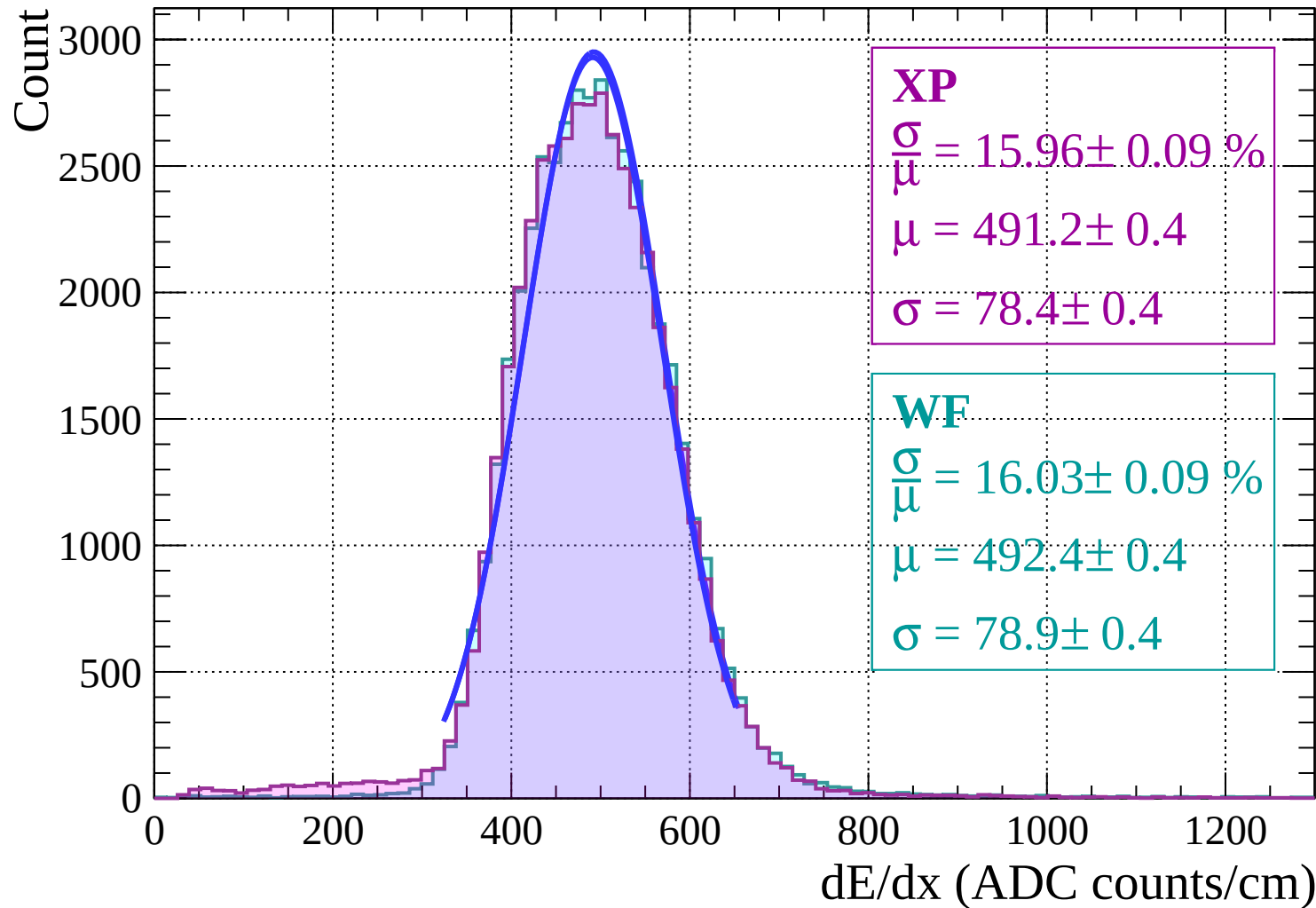
Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

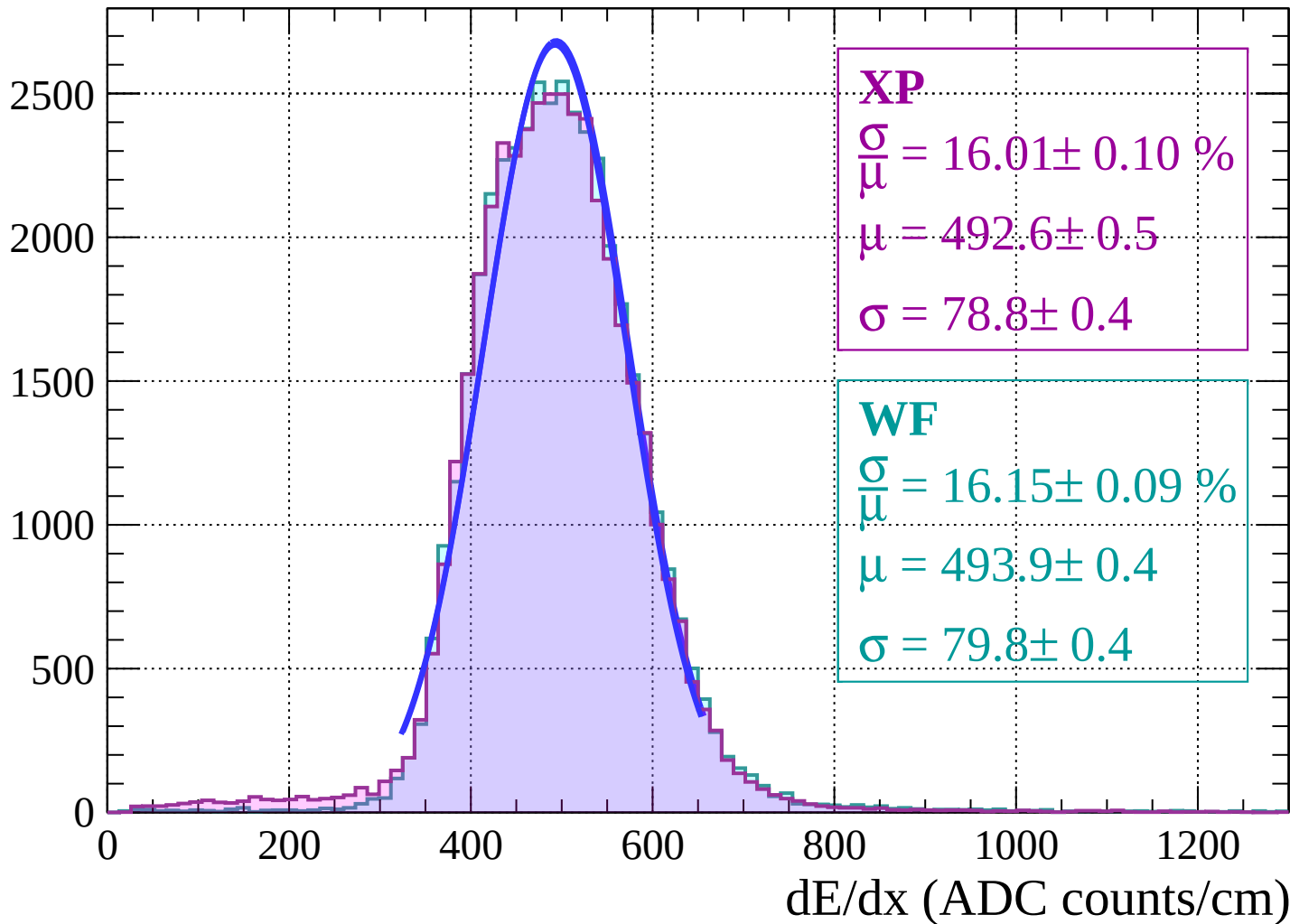


Energy loss | $18 < \theta < 20$

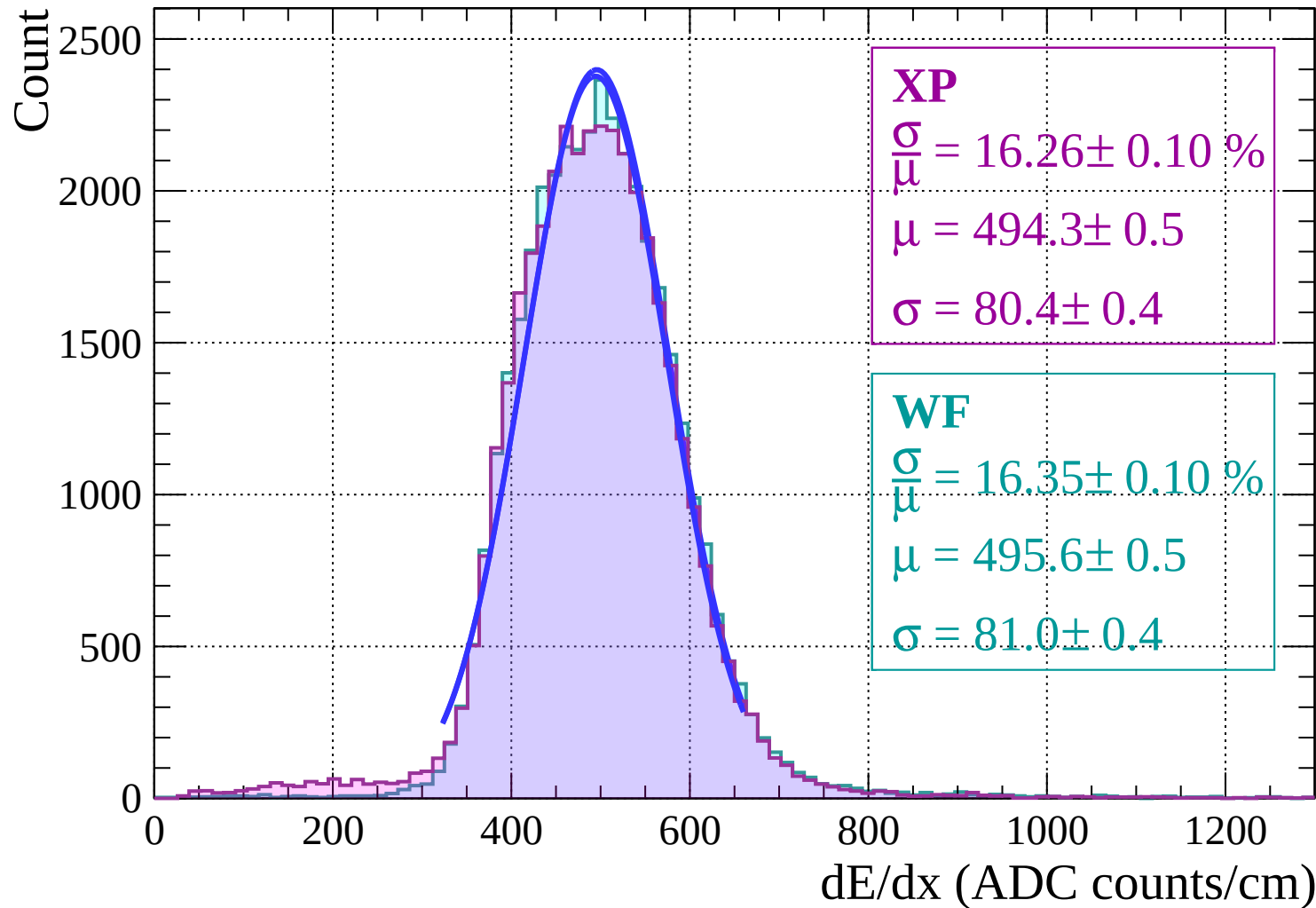


Energy loss | $20 < \theta < 22$

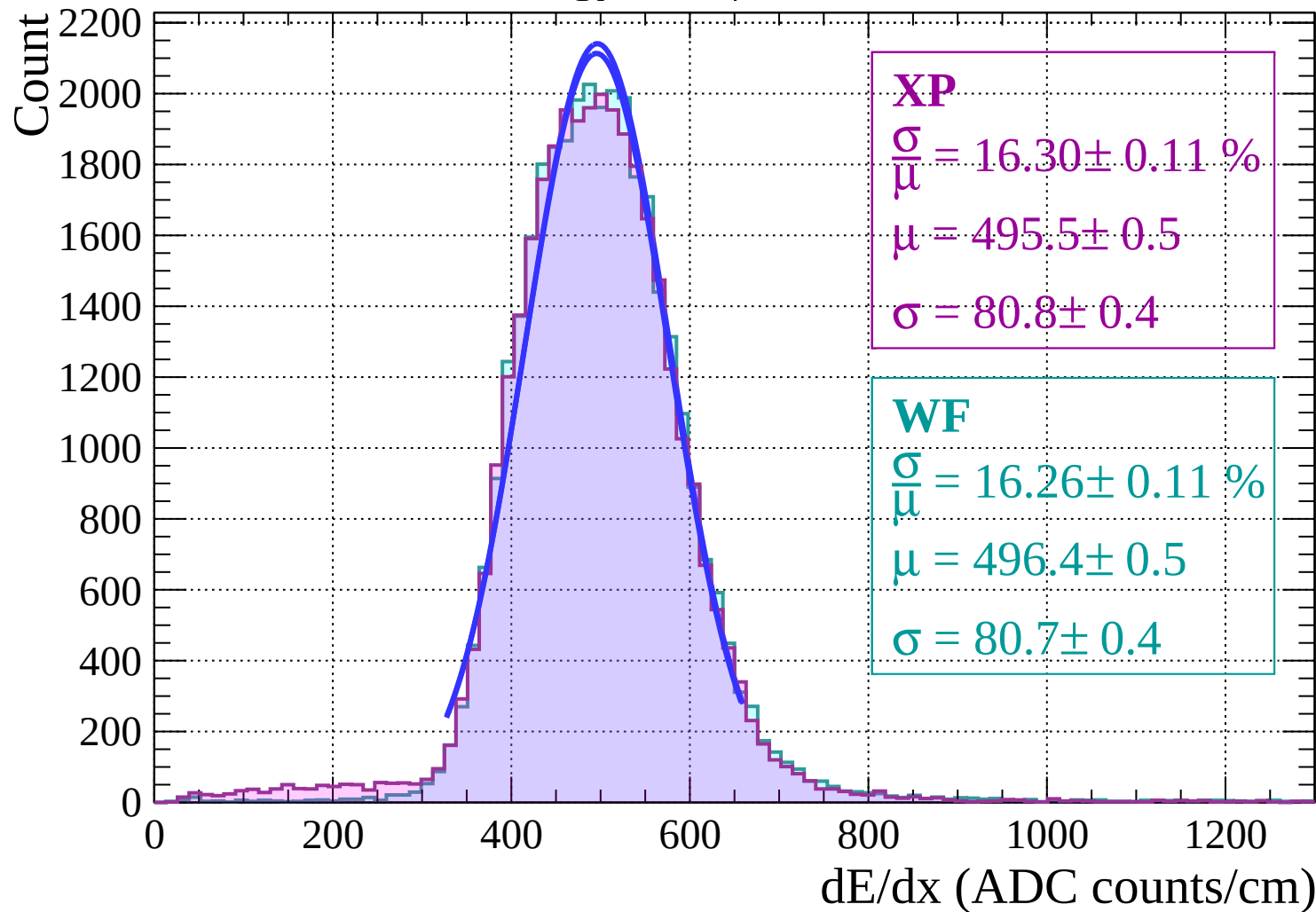
Count



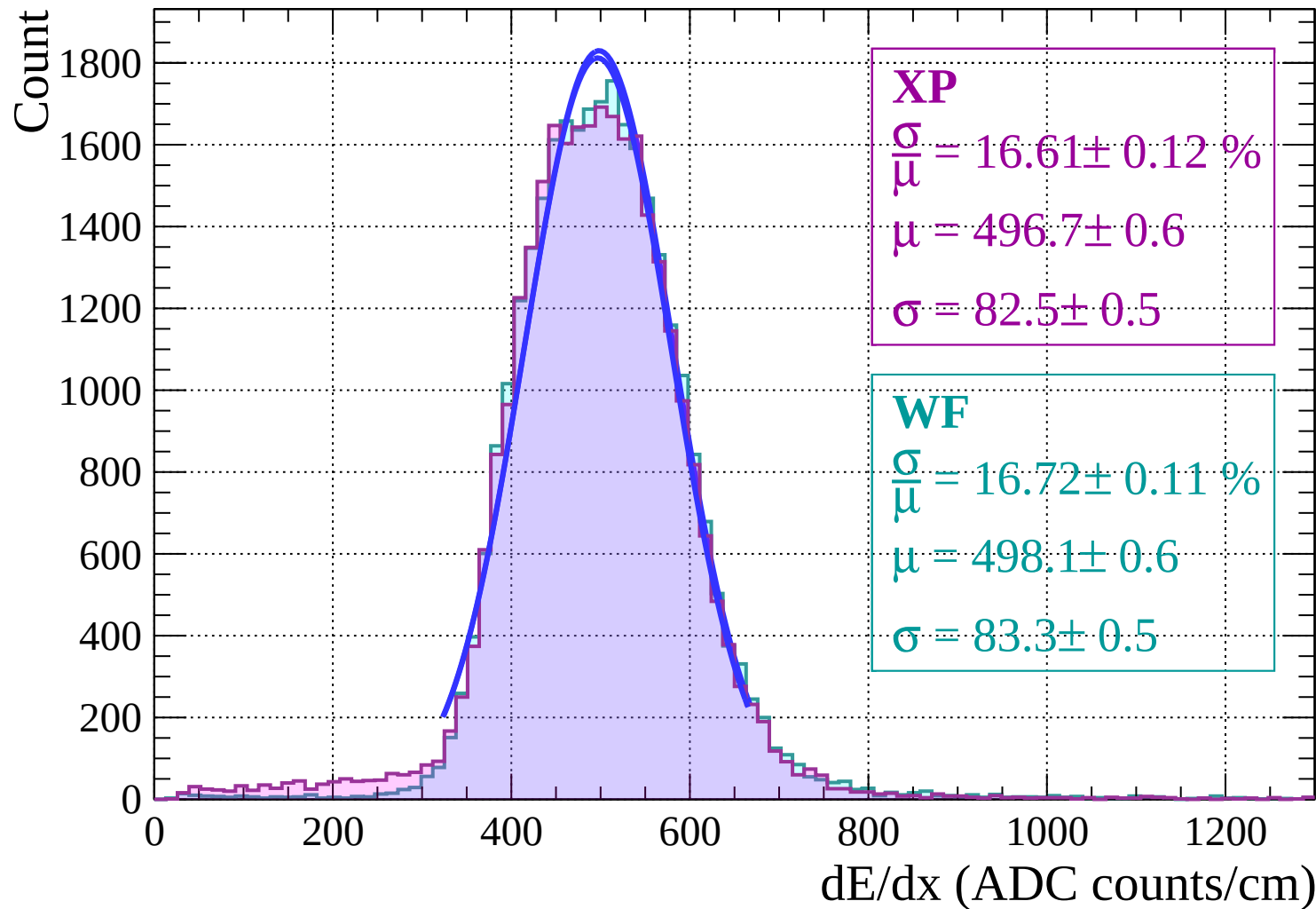
Energy loss | $22 < \theta < 24$



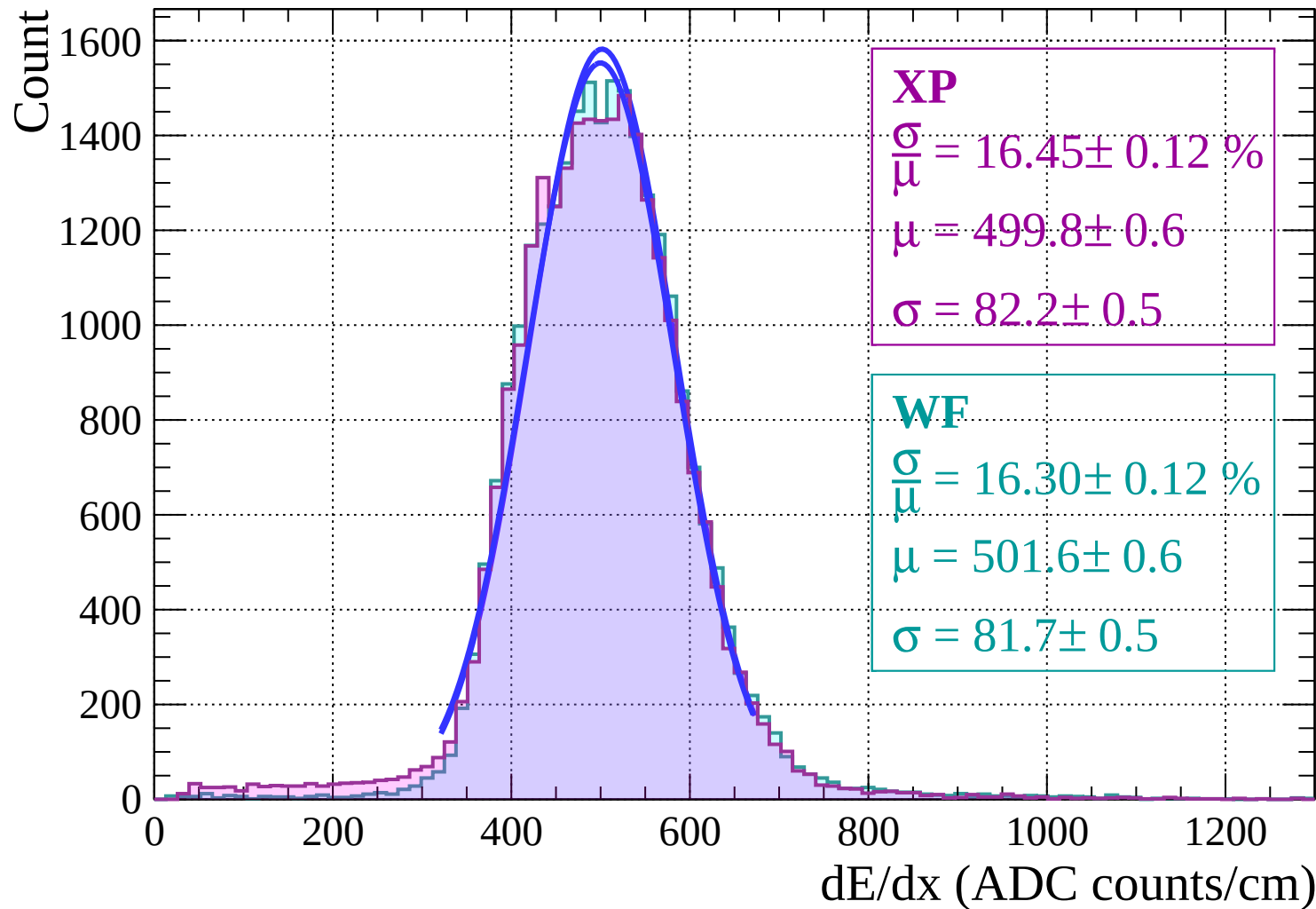
Energy loss | $24 < \theta < 26$



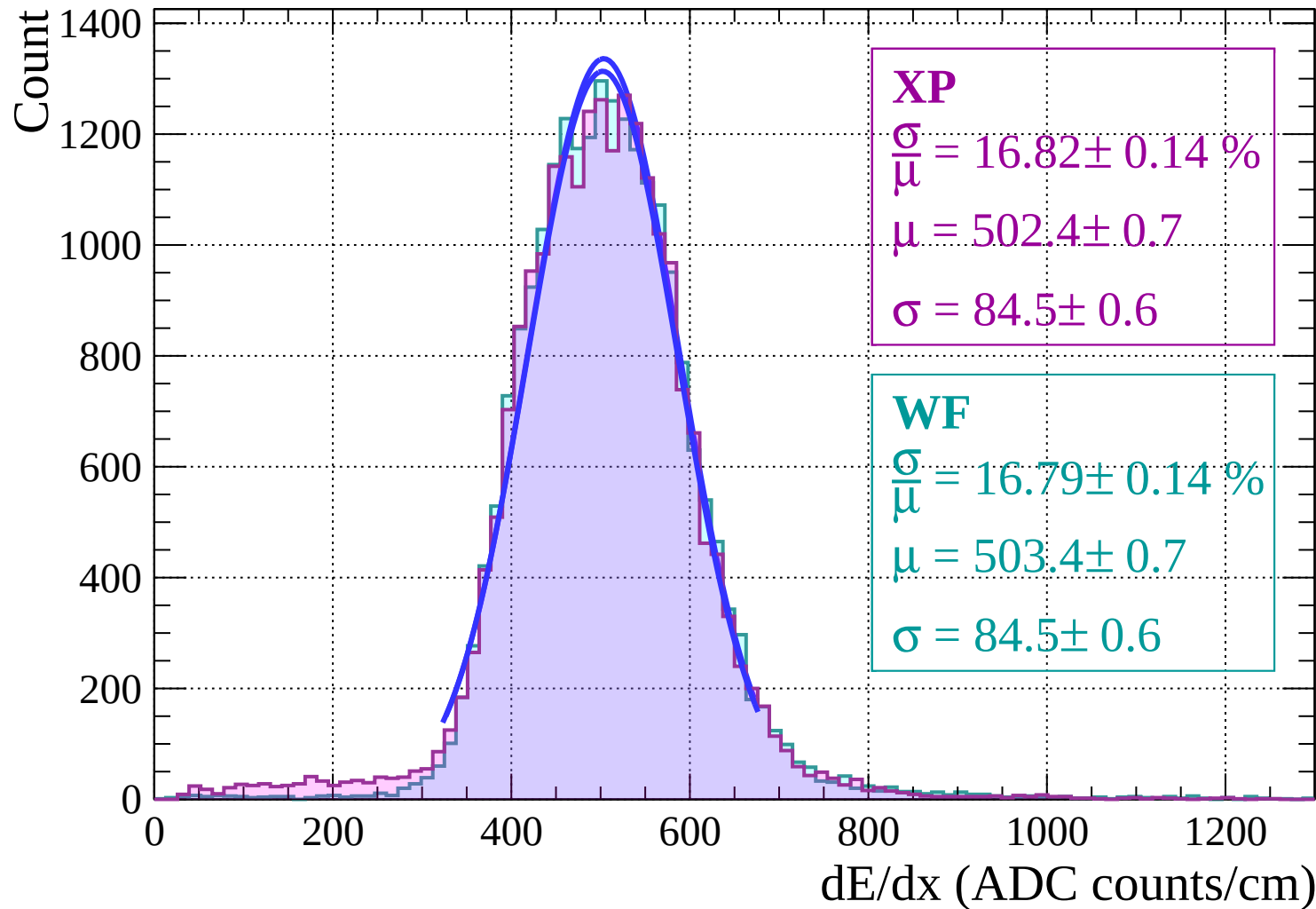
Energy loss | $26 < \theta < 28$



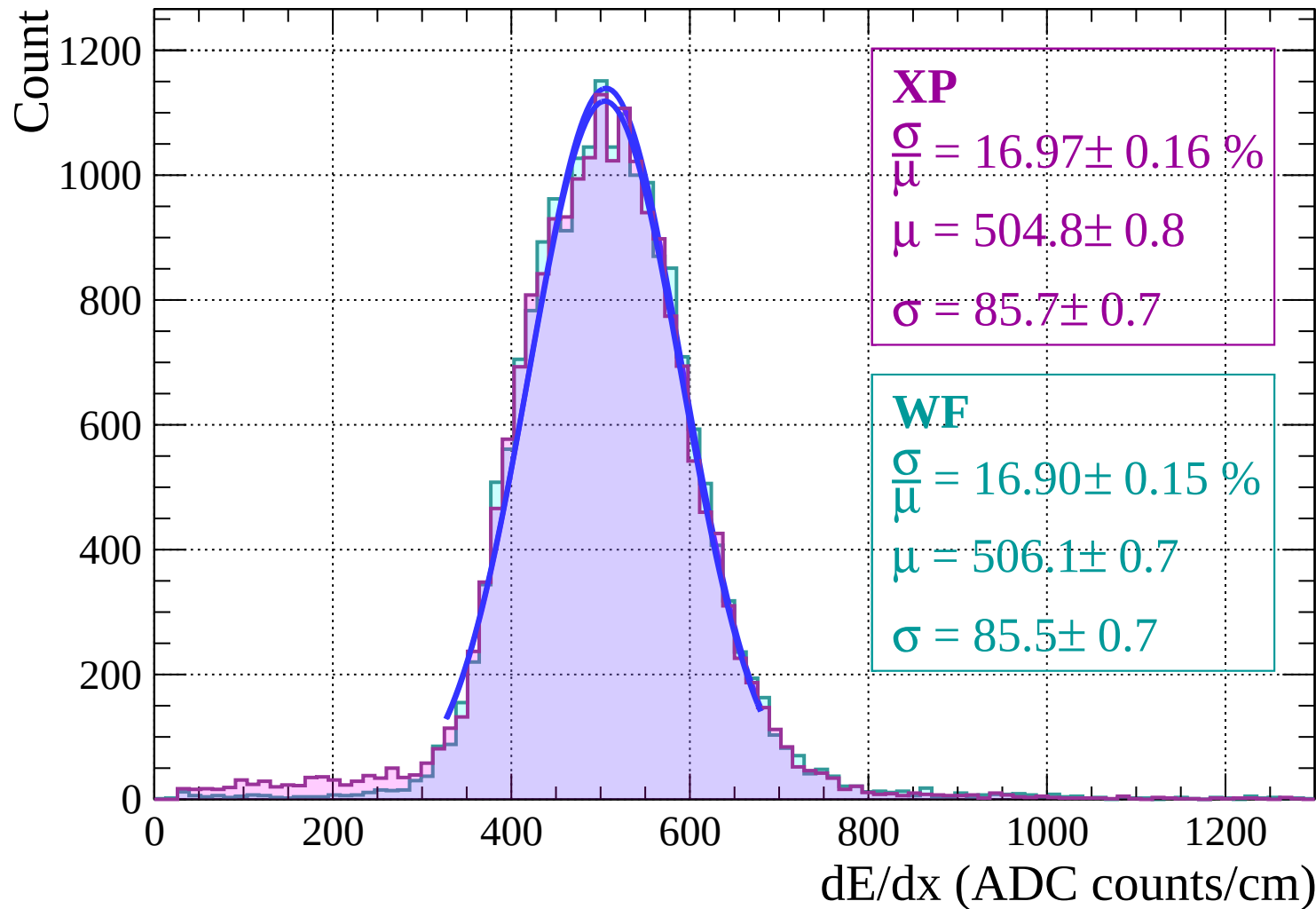
Energy loss | $28 < \theta < 30$



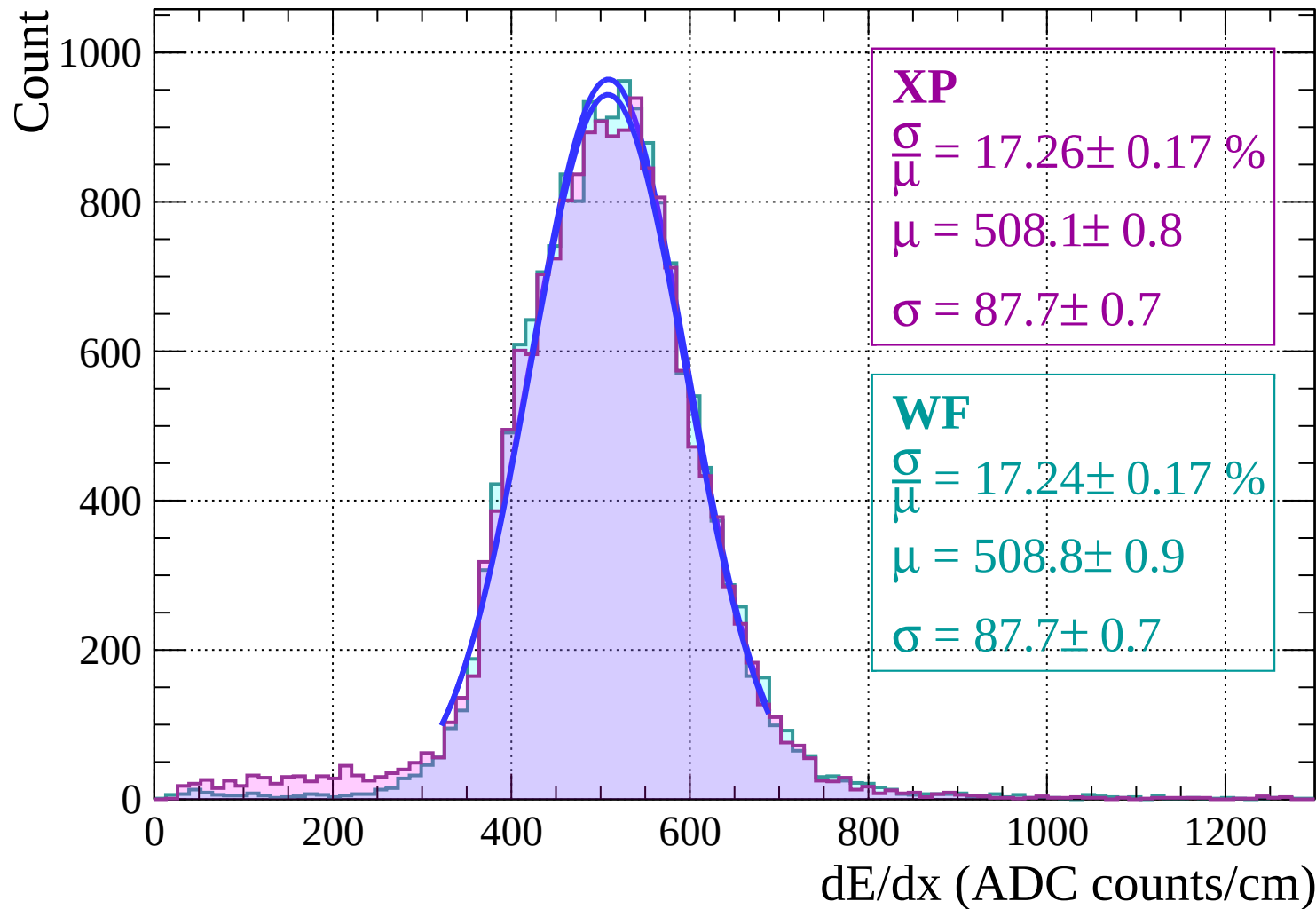
Energy loss | $30 < \theta < 32$



Energy loss | $32 < \theta < 34$

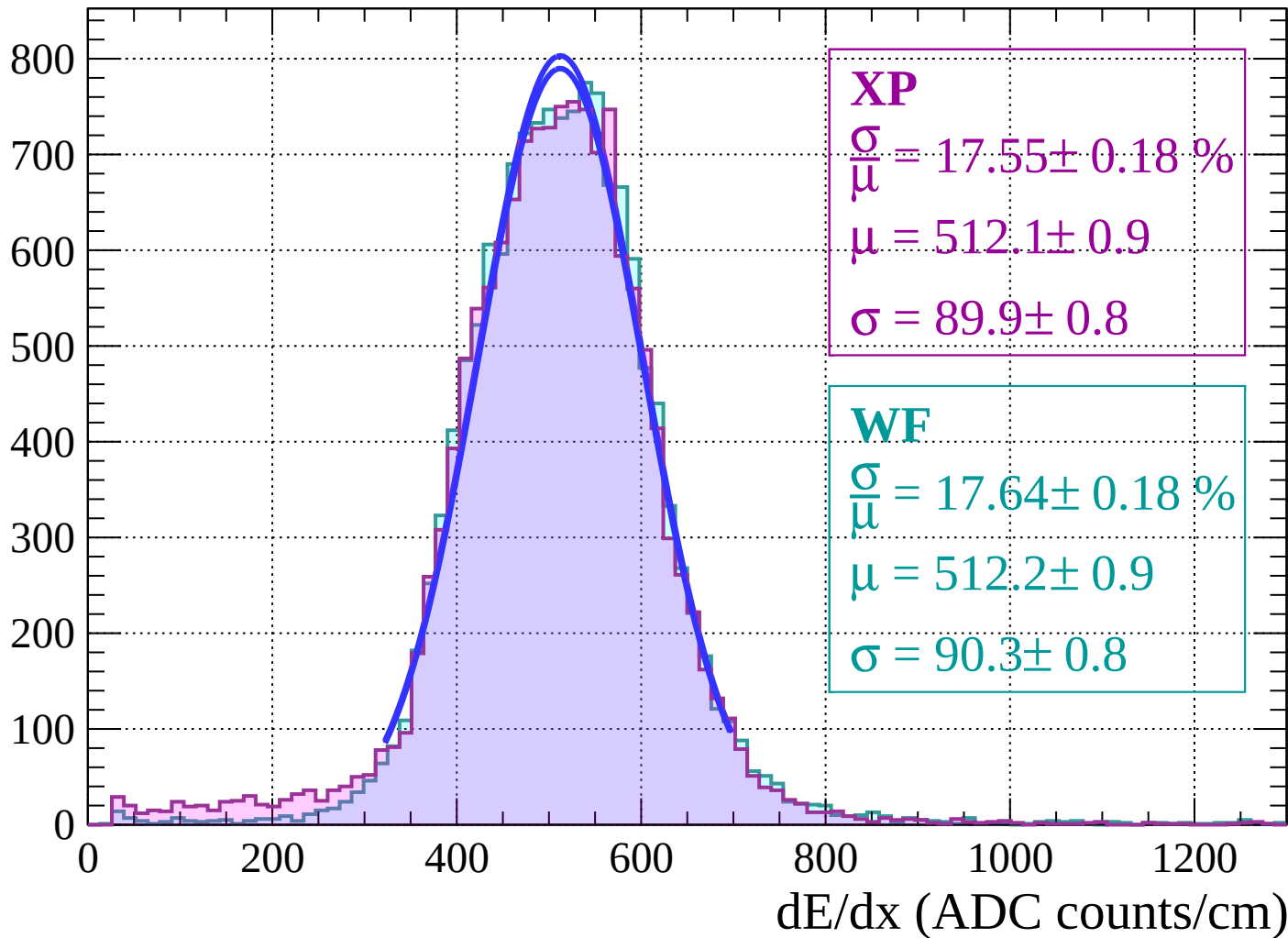


Energy loss | $34 < \theta < 36$



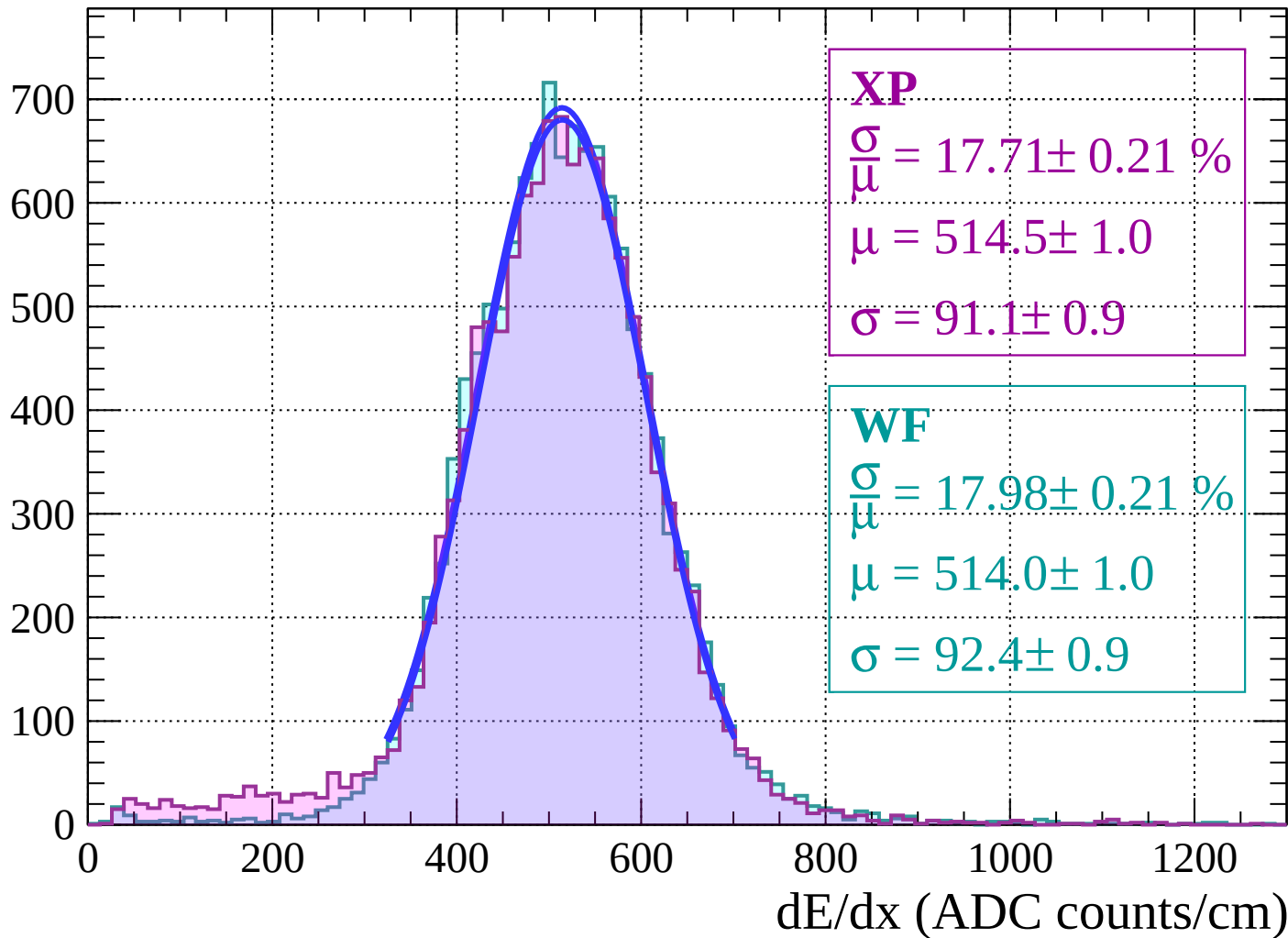
Energy loss | $36 < \theta < 38$

Count



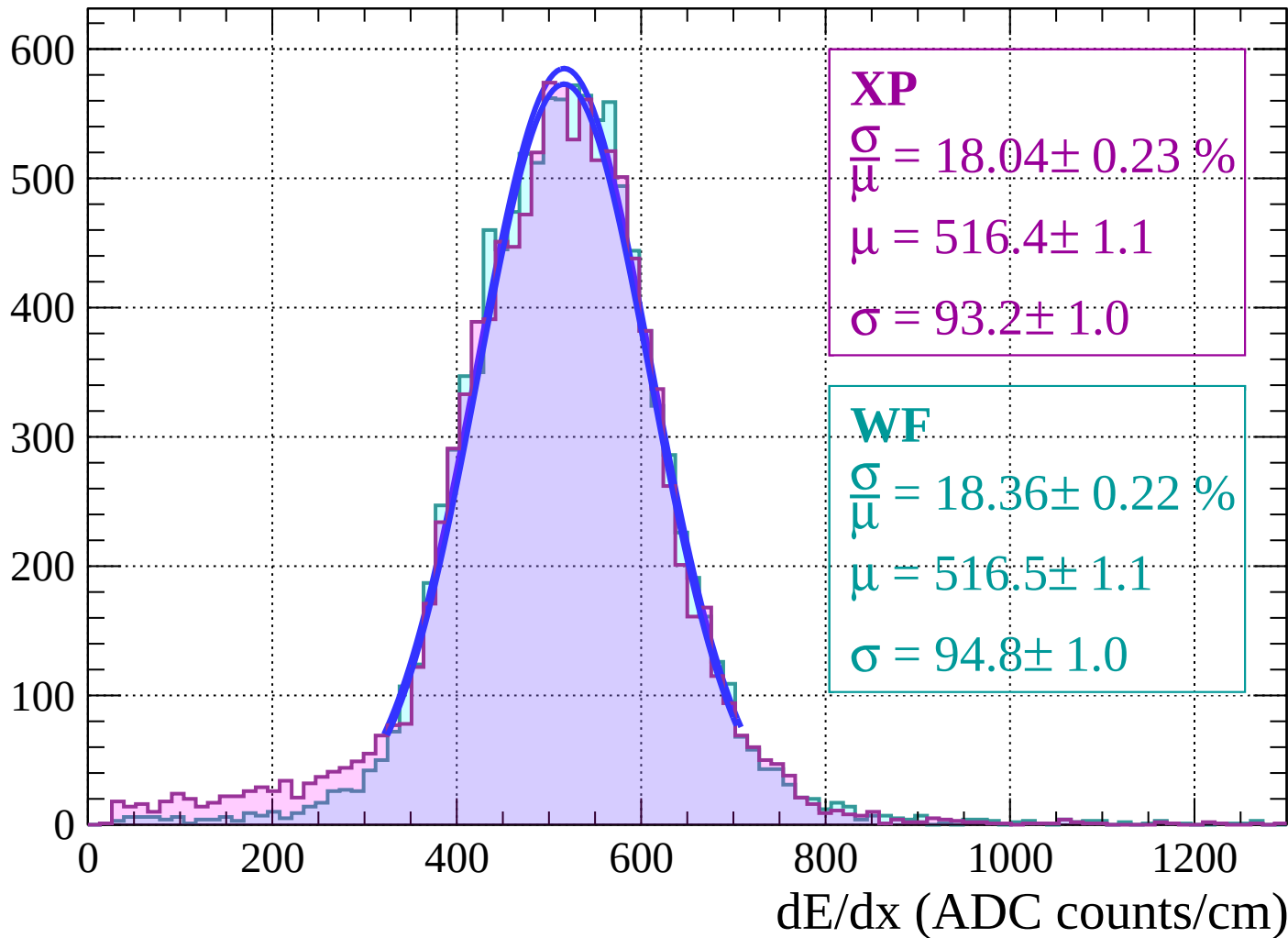
Energy loss | $38 < \theta < 40$

Count



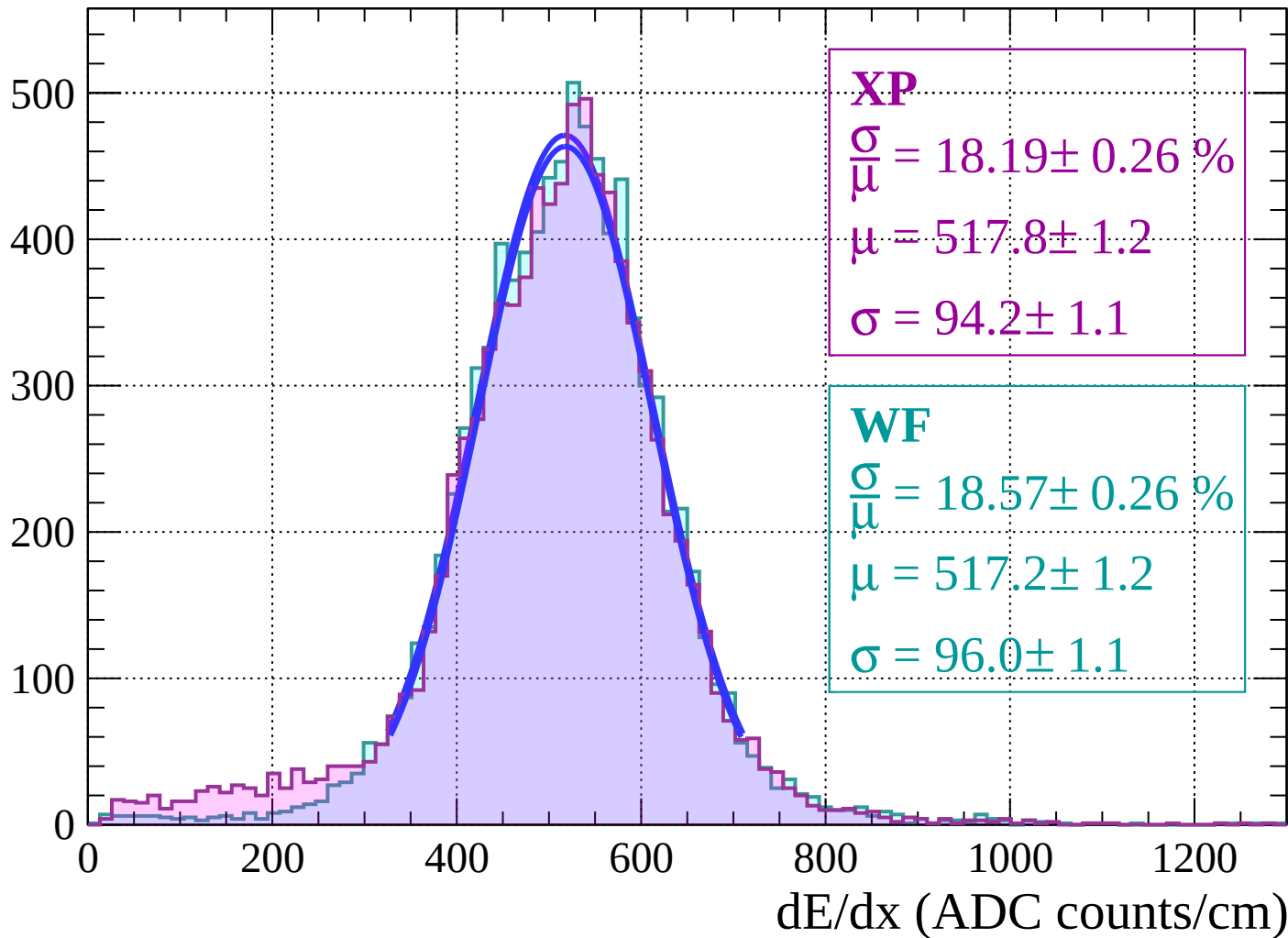
Energy loss | $40 < \theta < 42$

Count



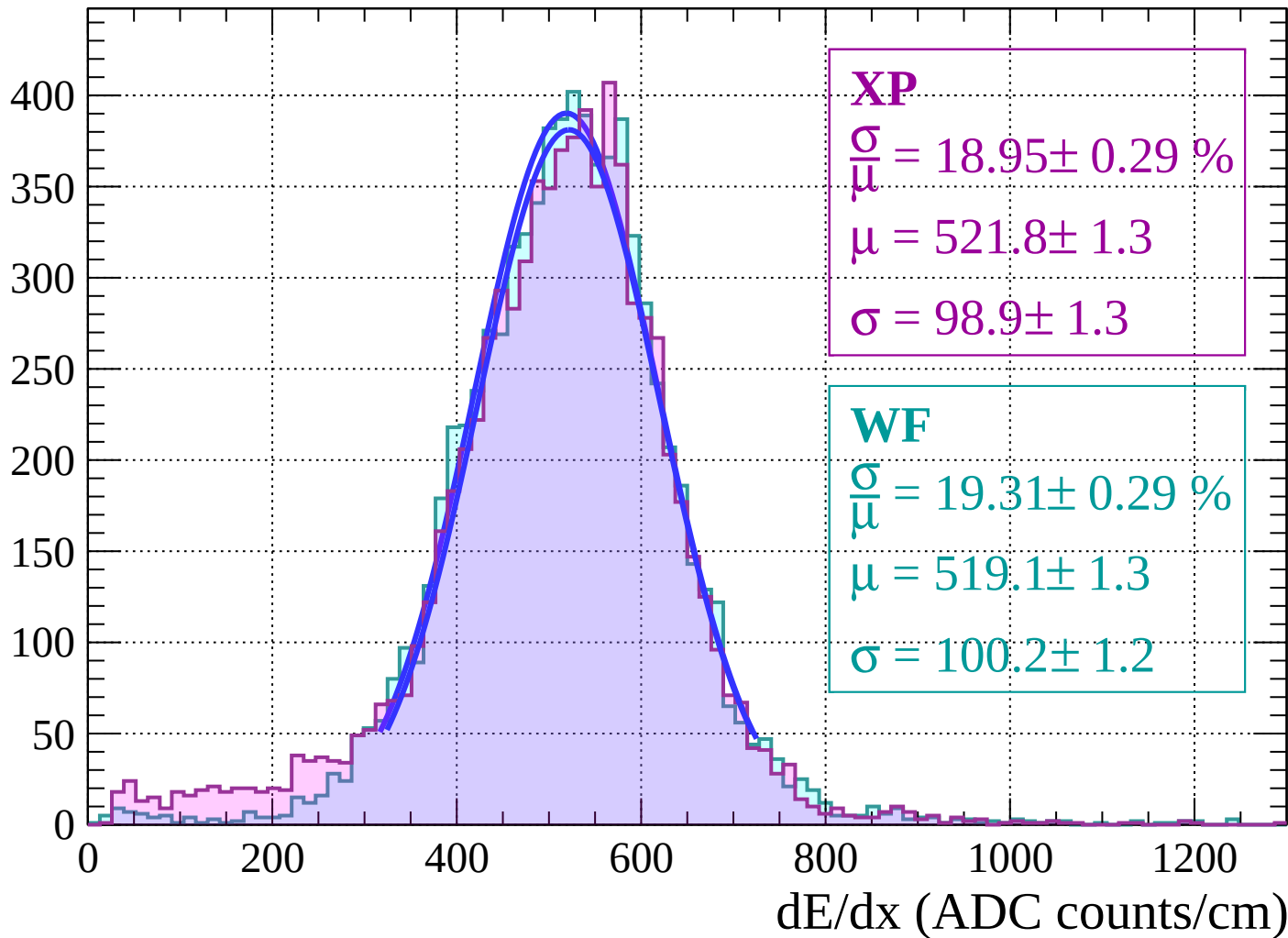
Energy loss | $42 < \theta < 44$

Count



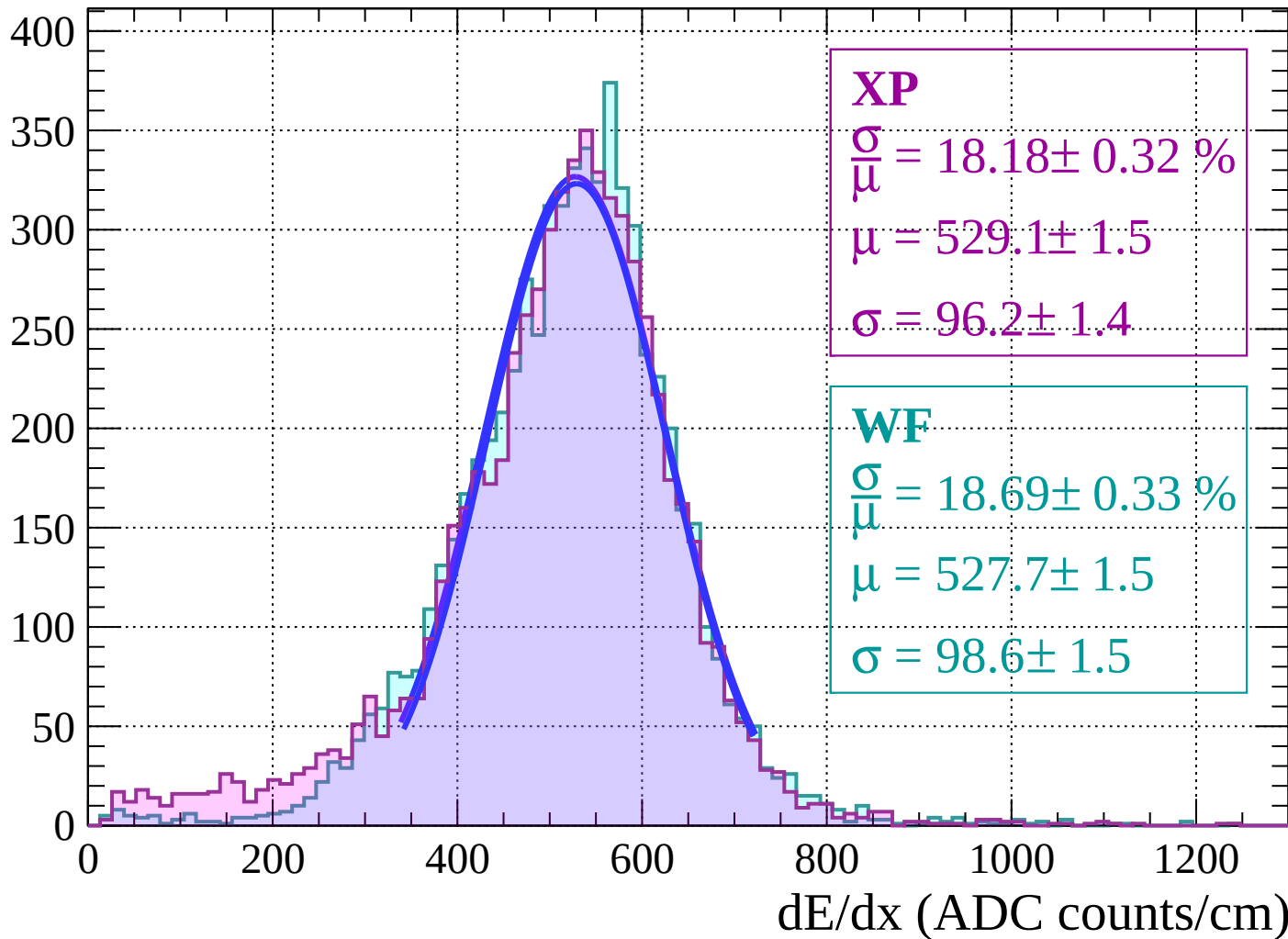
Energy loss | $44 < \theta < 46$

Count



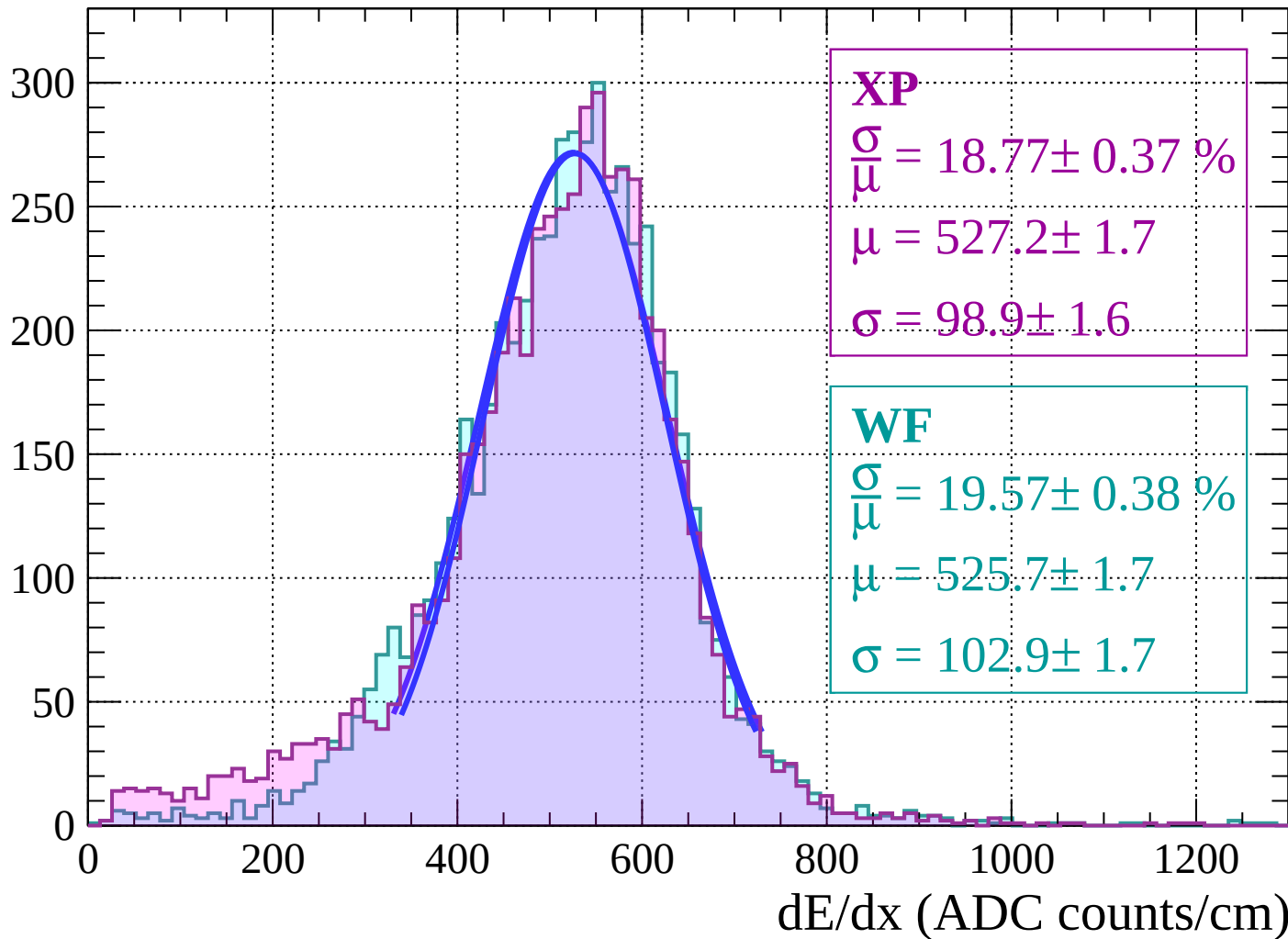
Energy loss | $46 < \theta < 48$

Count



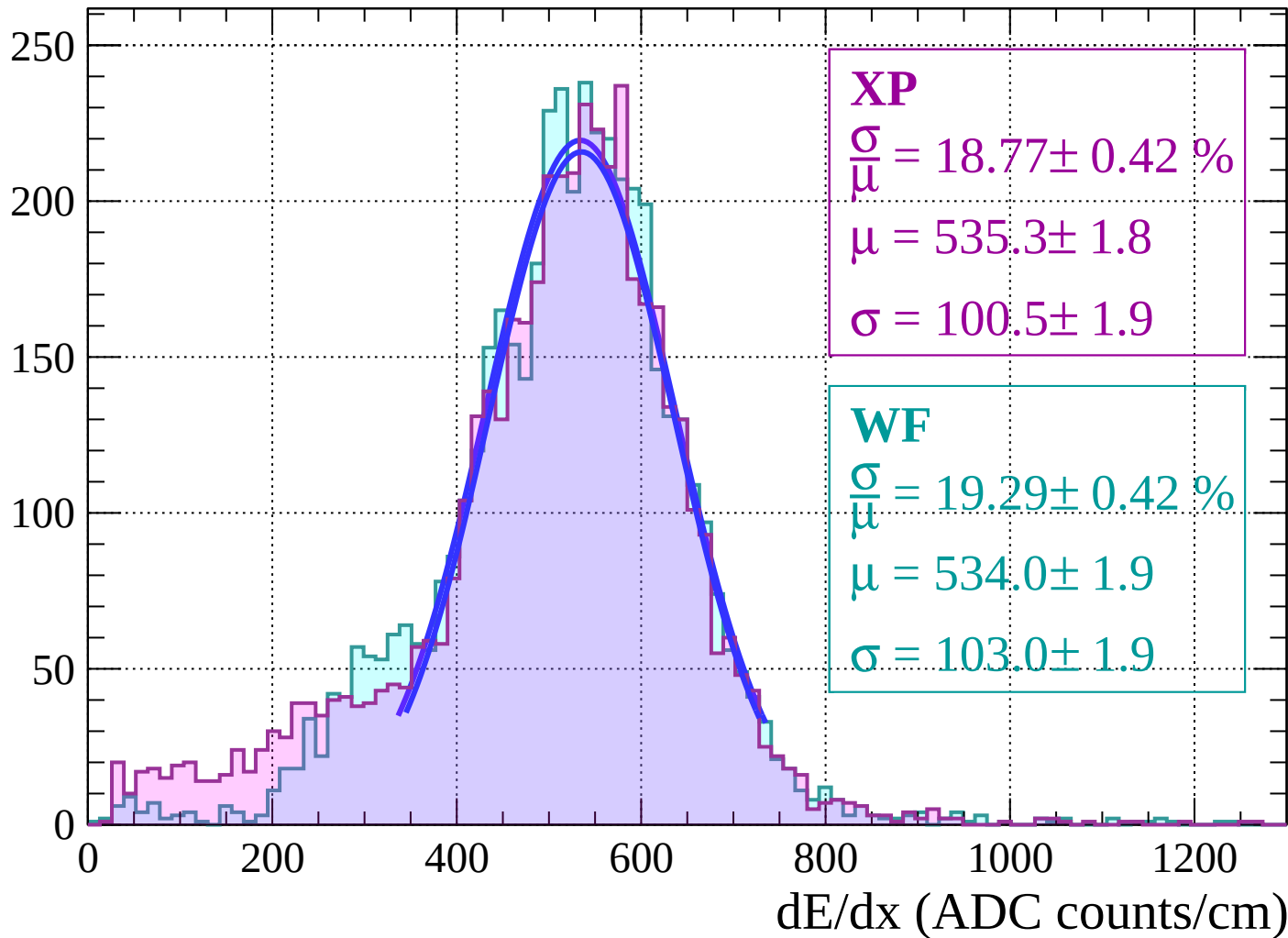
Energy loss | $48 < \theta < 50$

Count



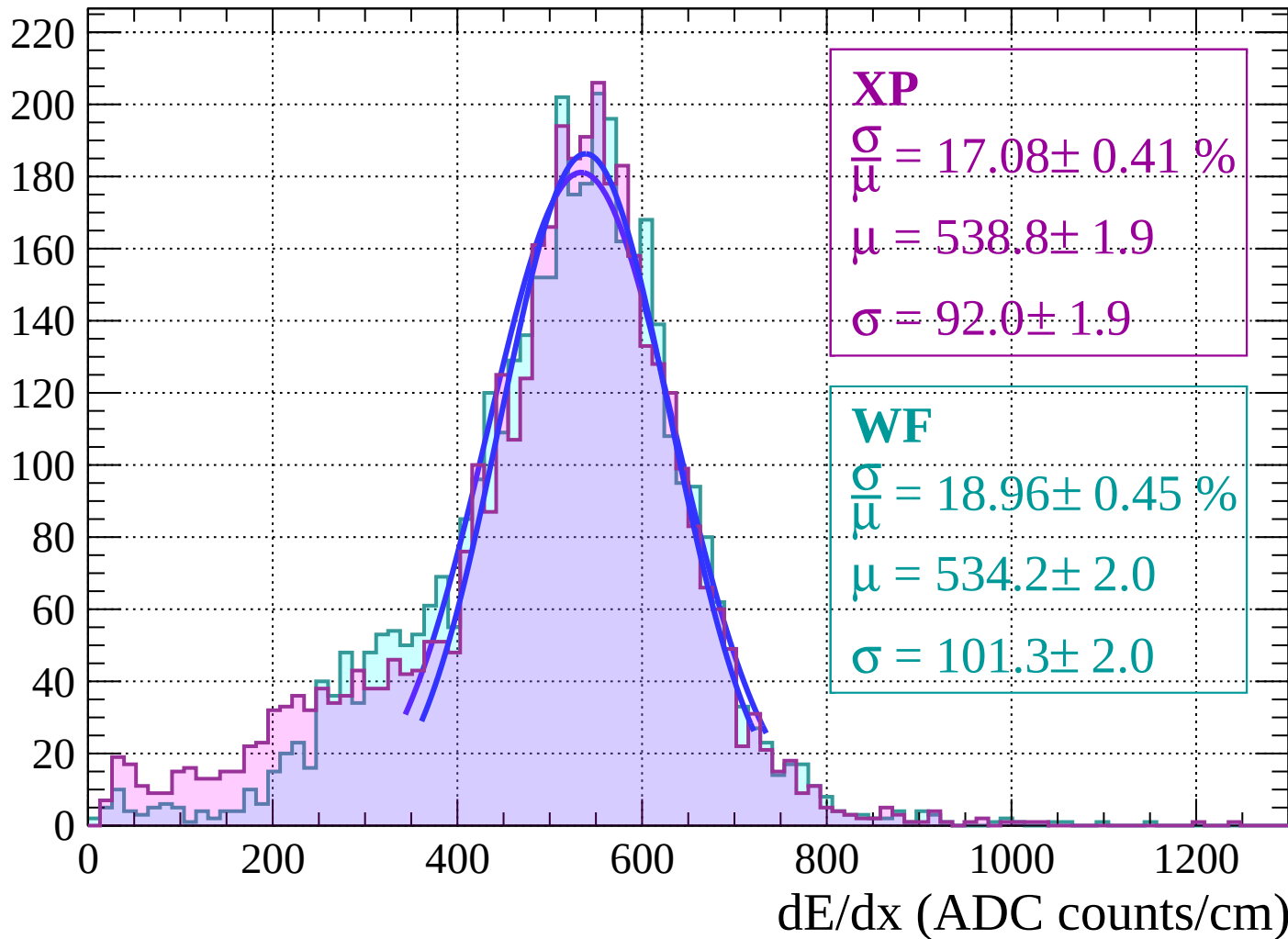
Energy loss | $50 < \theta < 52$

Count



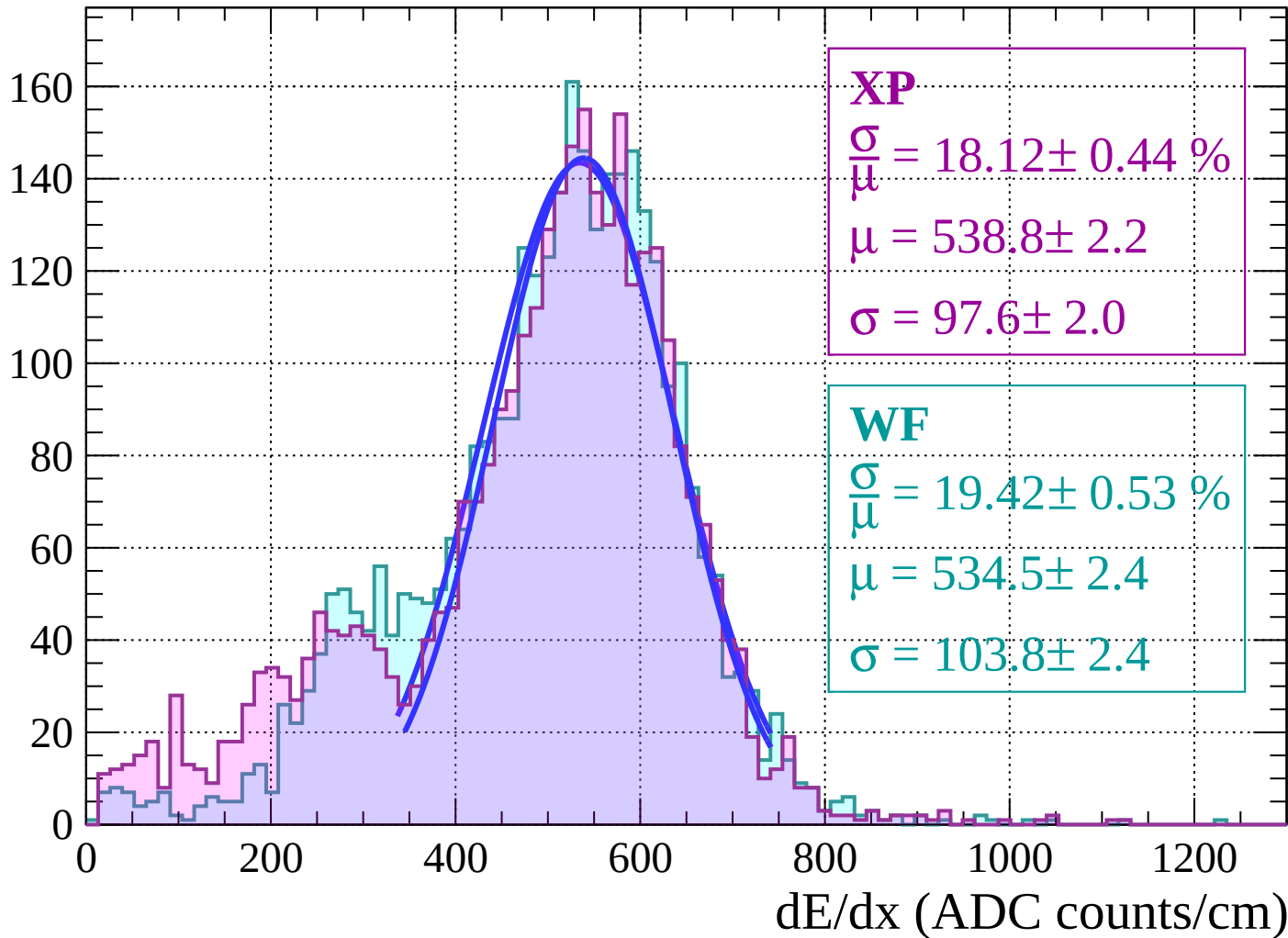
Energy loss | $52 < \theta < 54$

Count



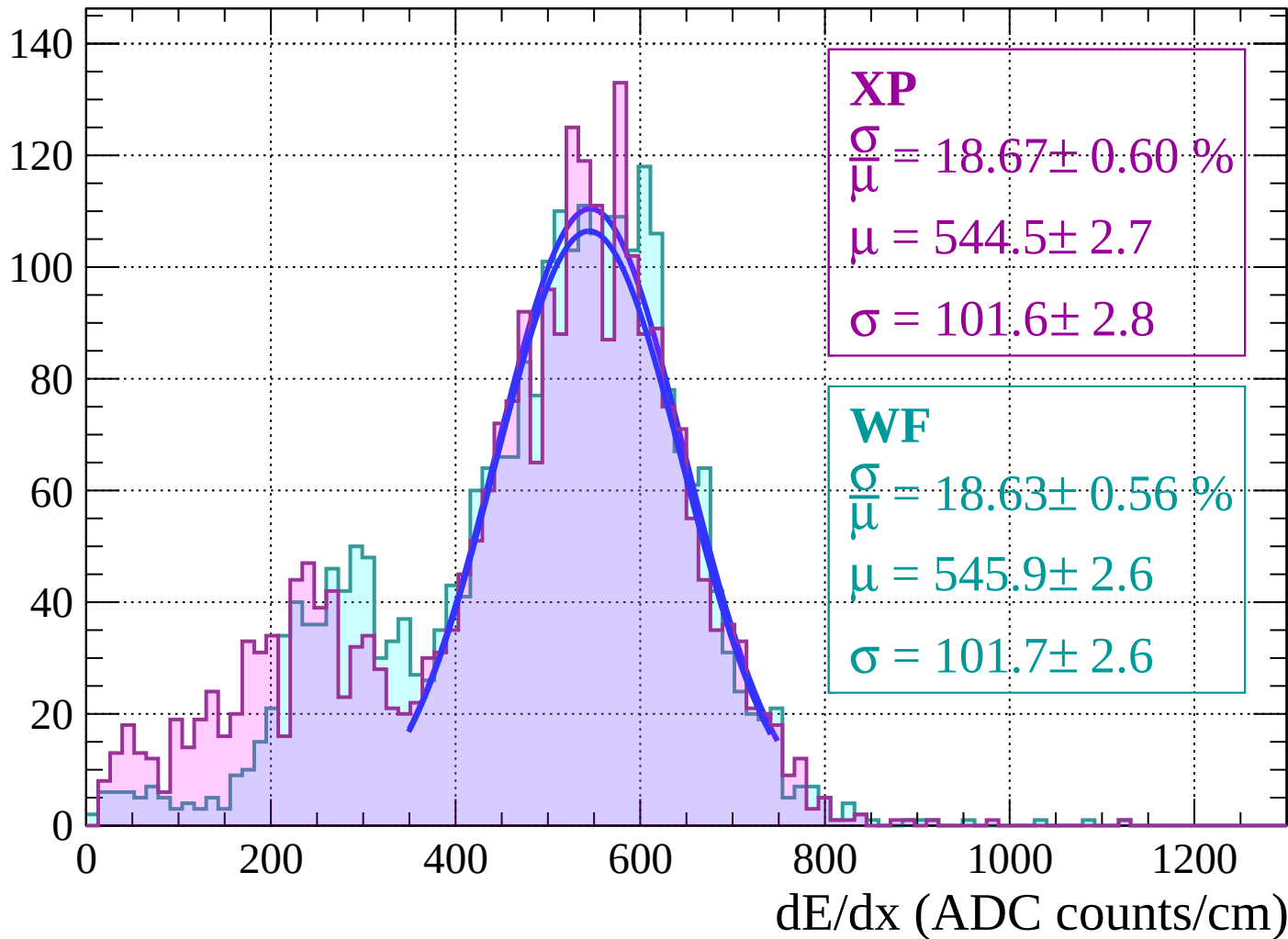
Energy loss | $54 < \theta < 56$

Count



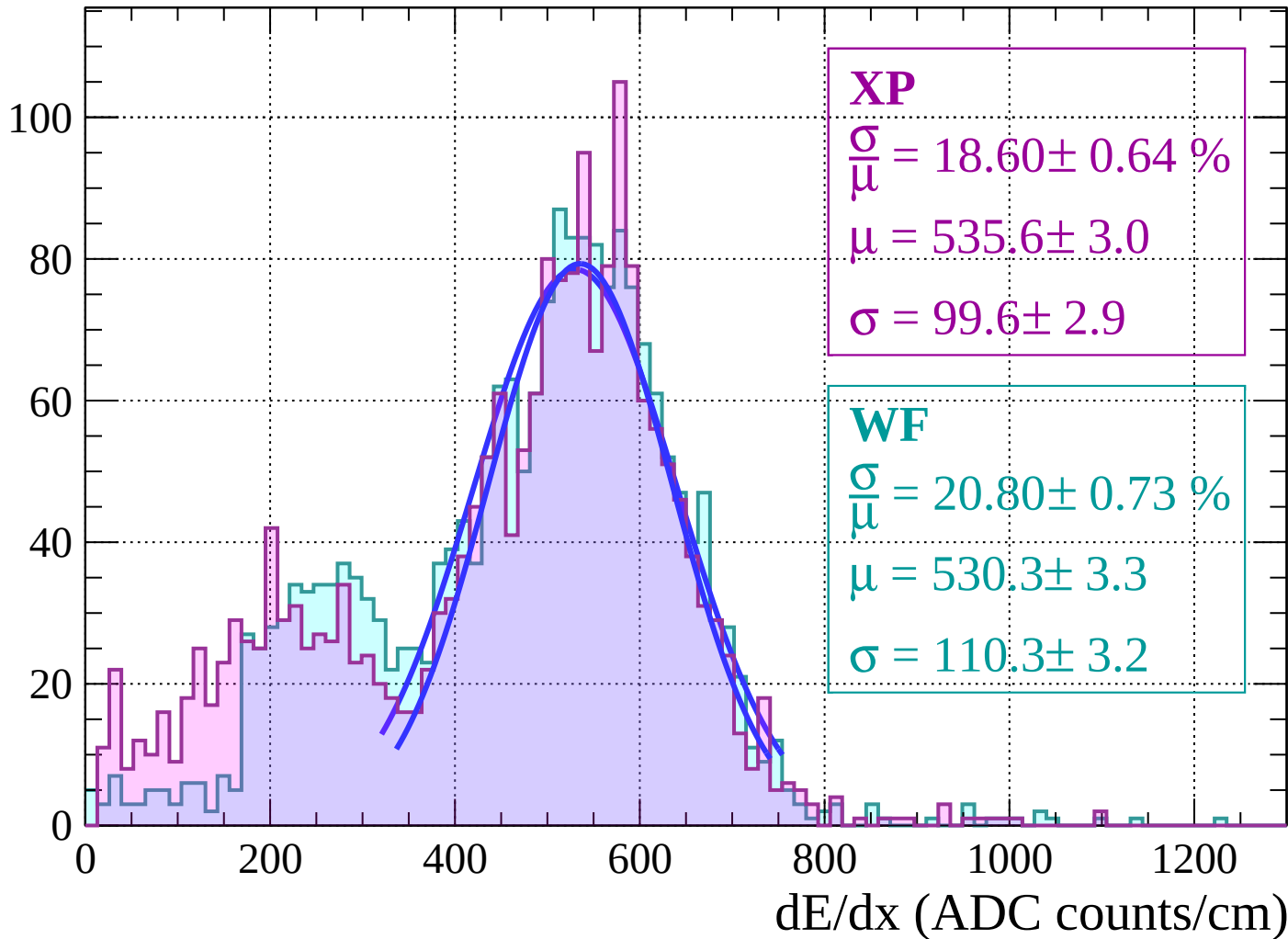
Energy loss | $56 < \theta < 58$

Count



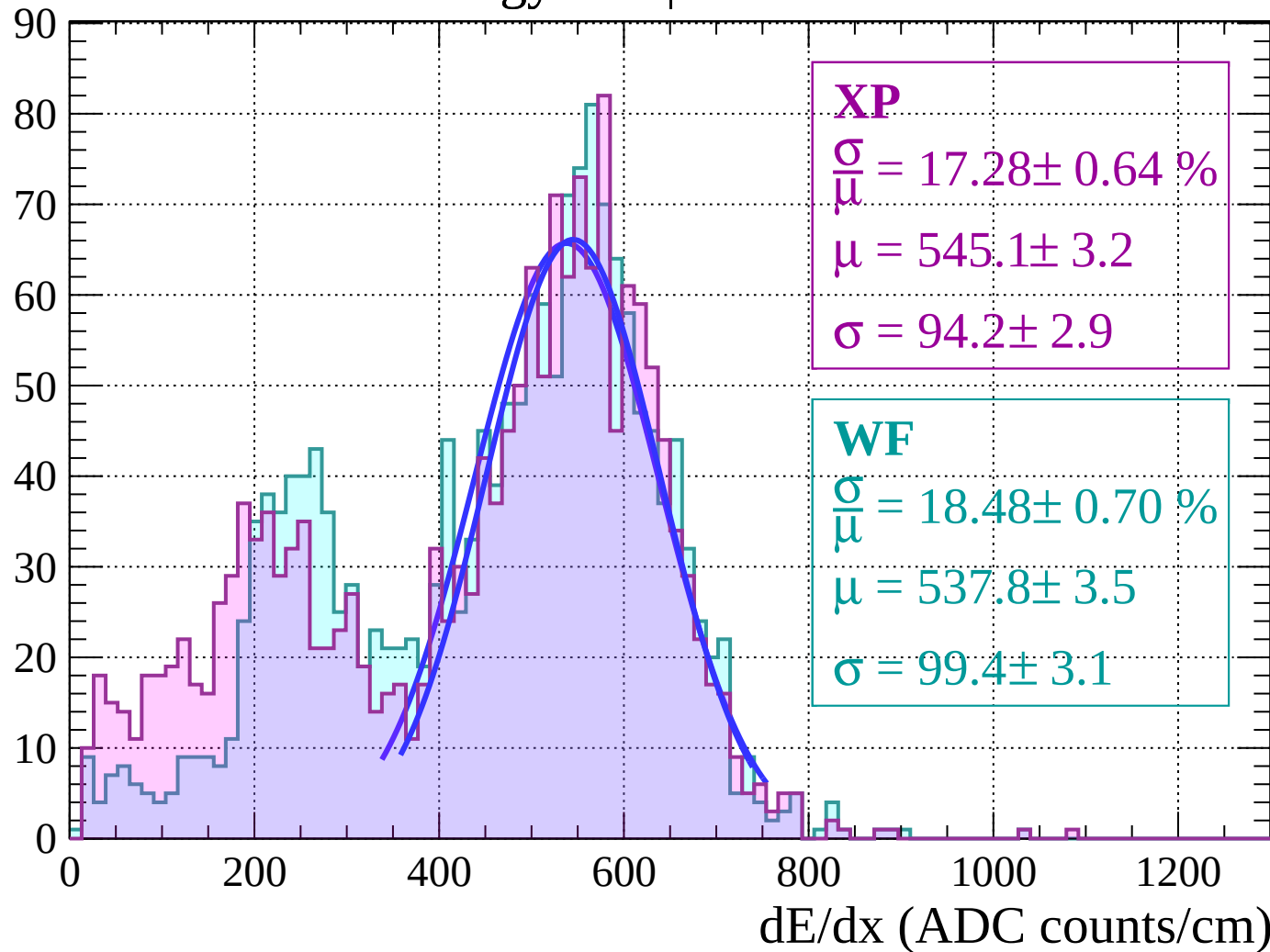
Energy loss | $58 < \theta < 60$

Count



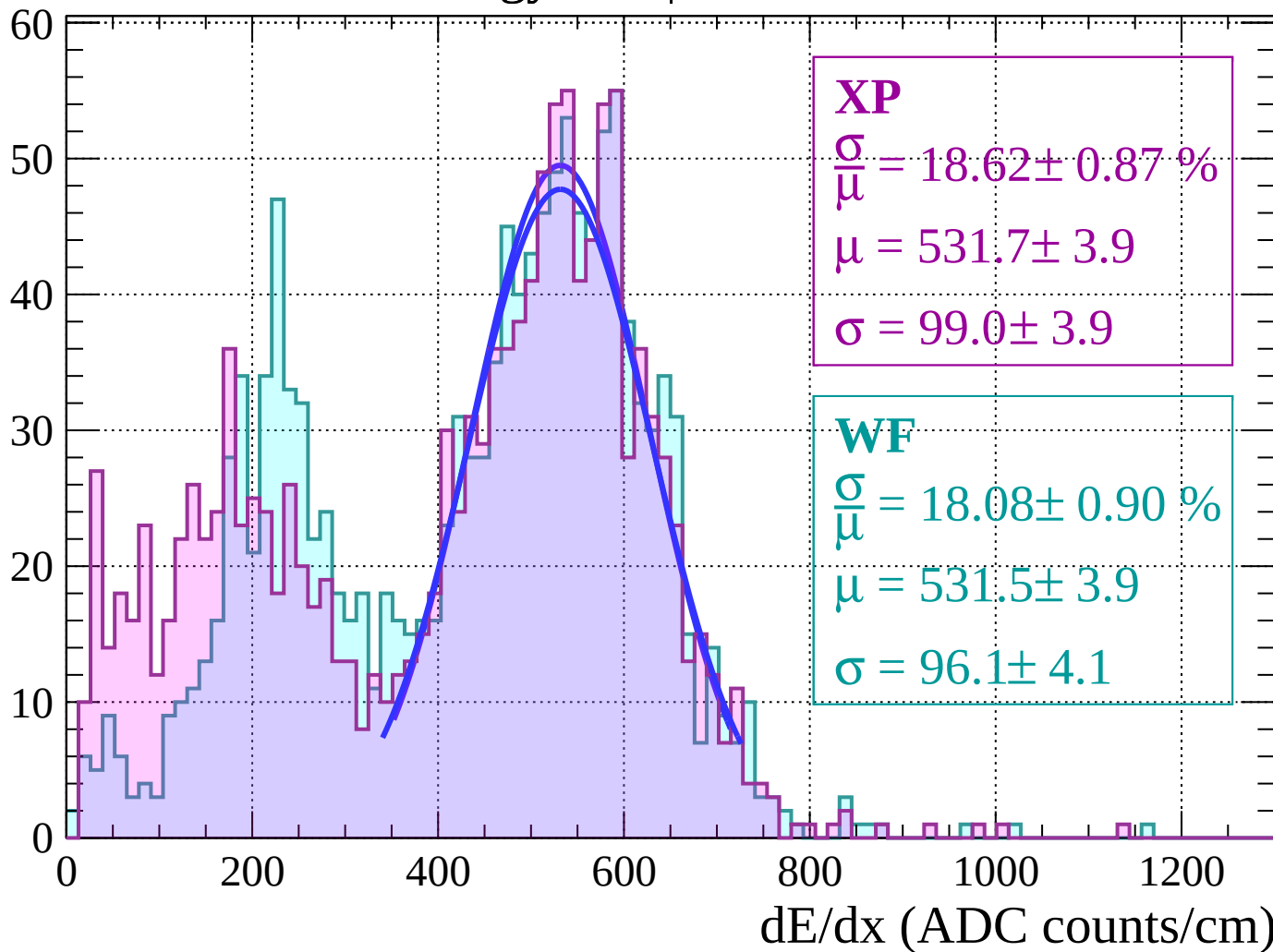
Energy loss | $60 < \theta < 62$

Count



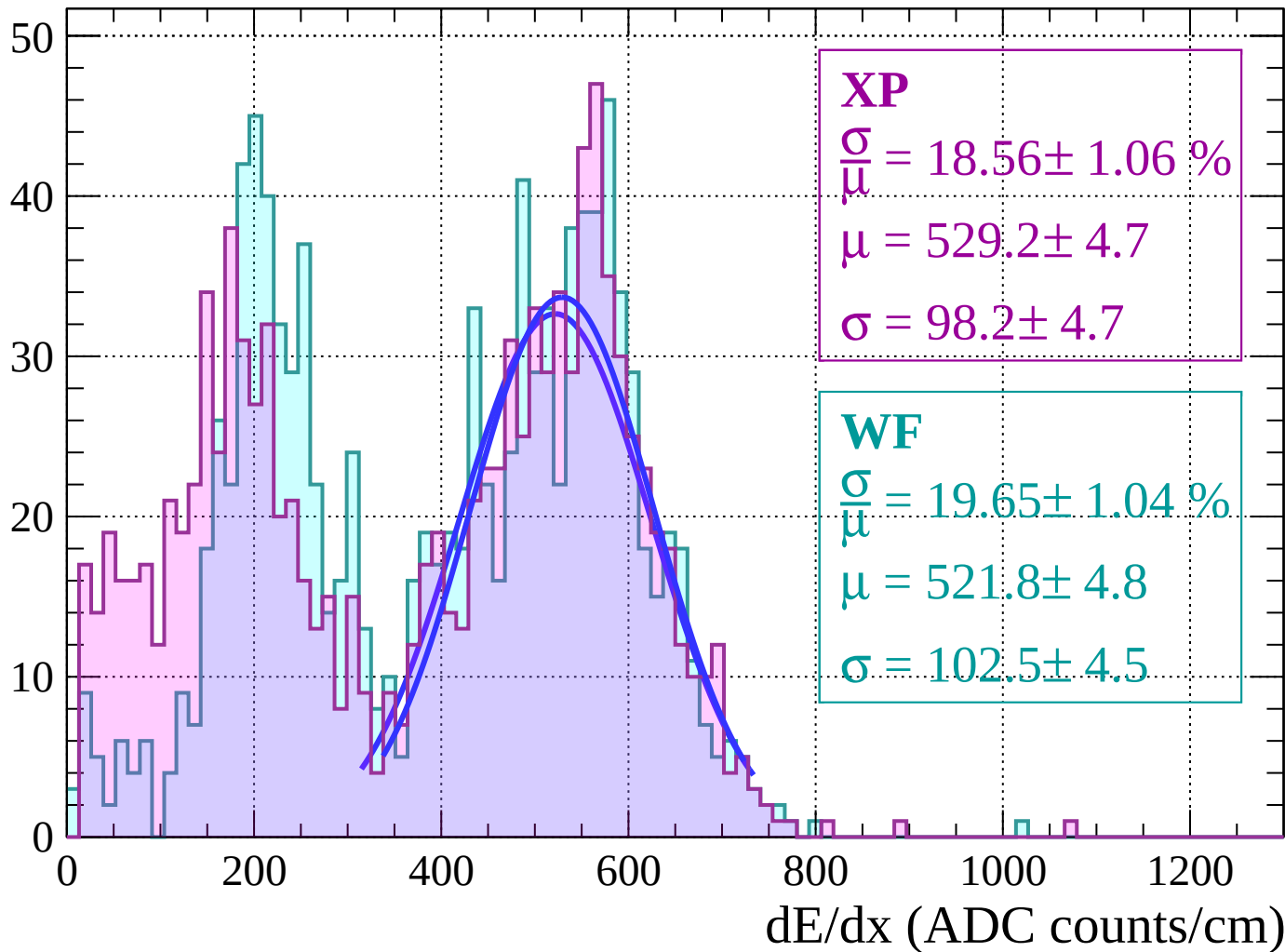
Energy loss | $62 < \theta < 64$

Count



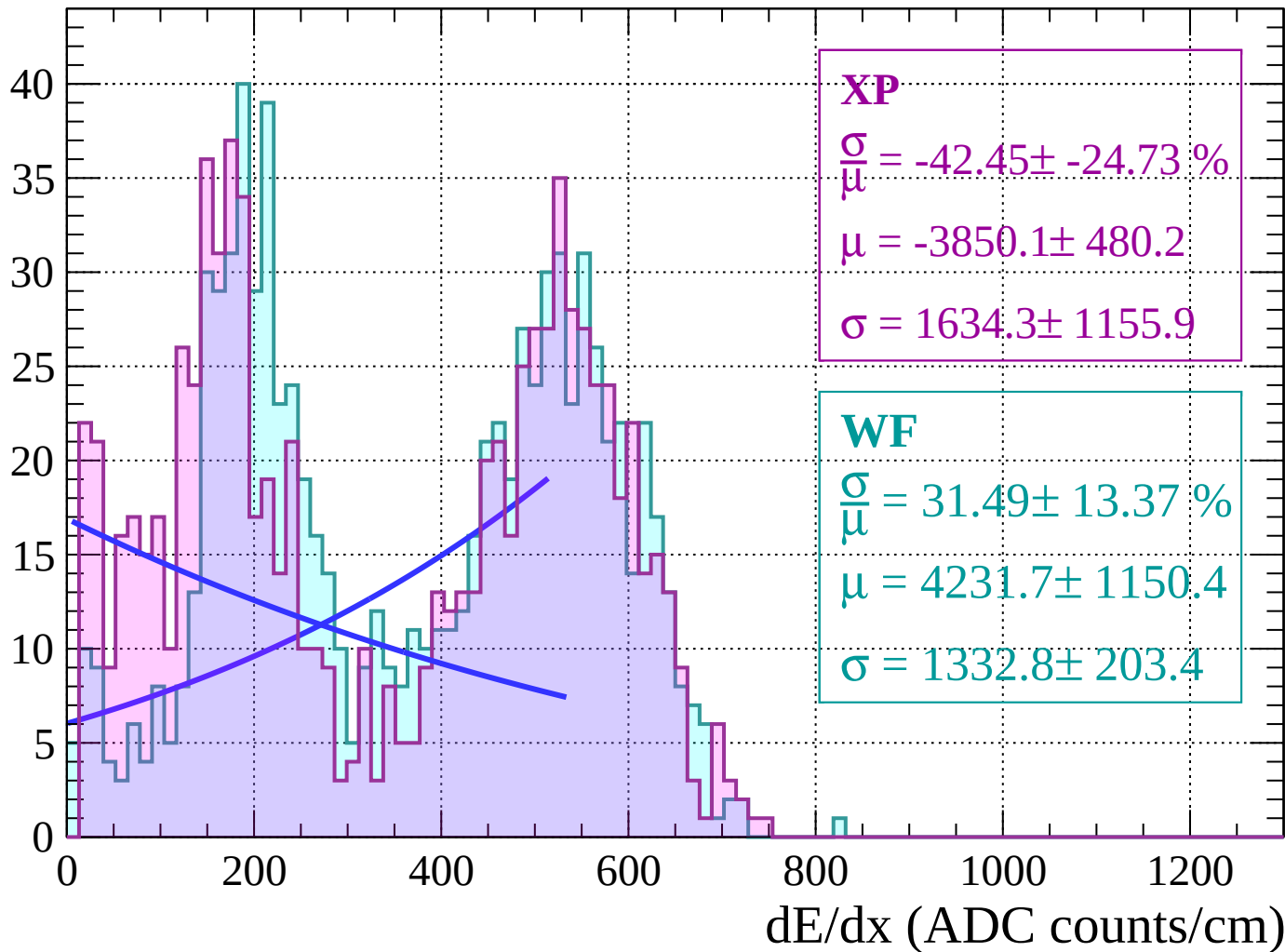
Energy loss | $64 < \theta < 66$

Count



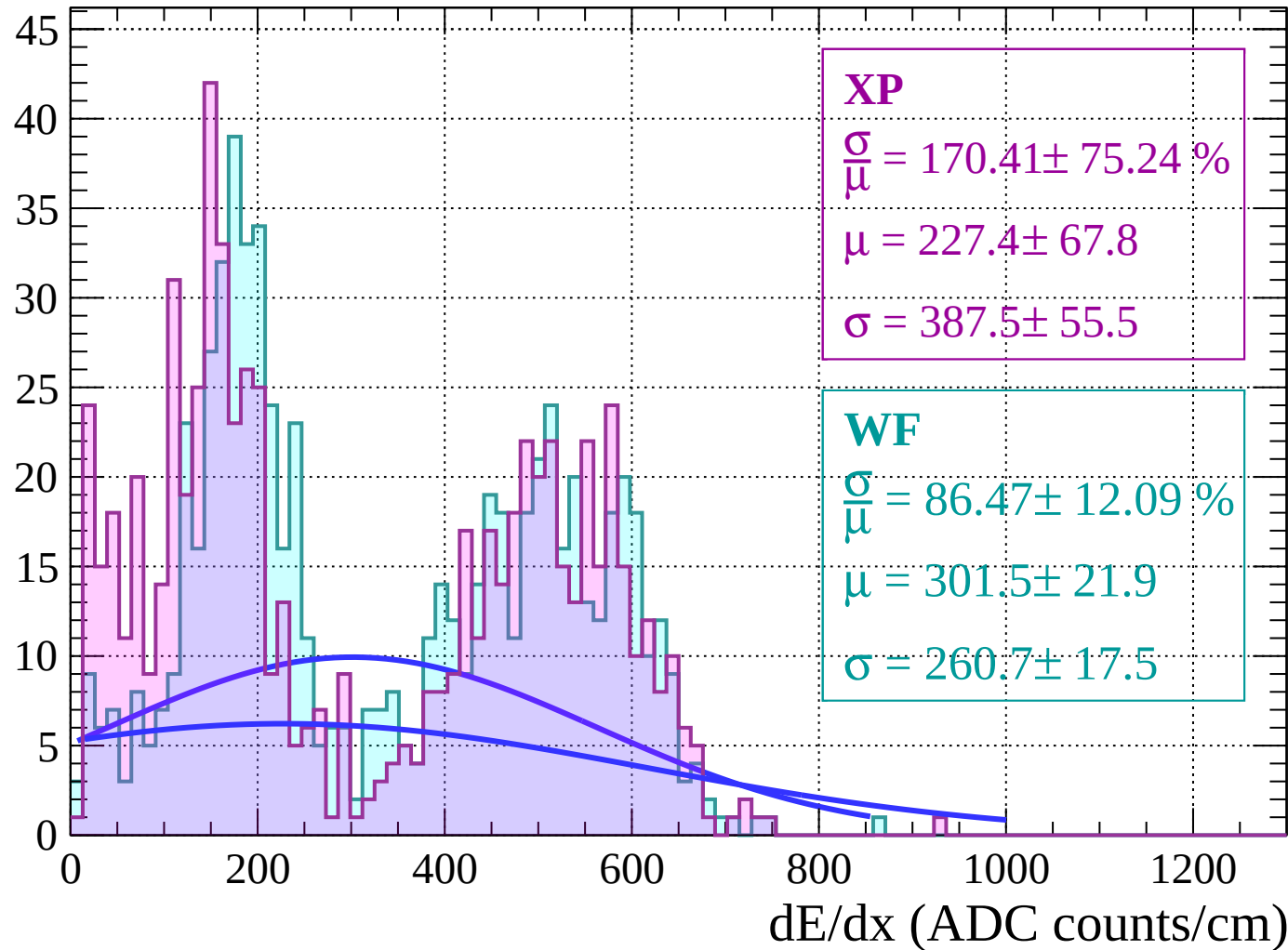
Energy loss | $66 < \theta < 68$

Count



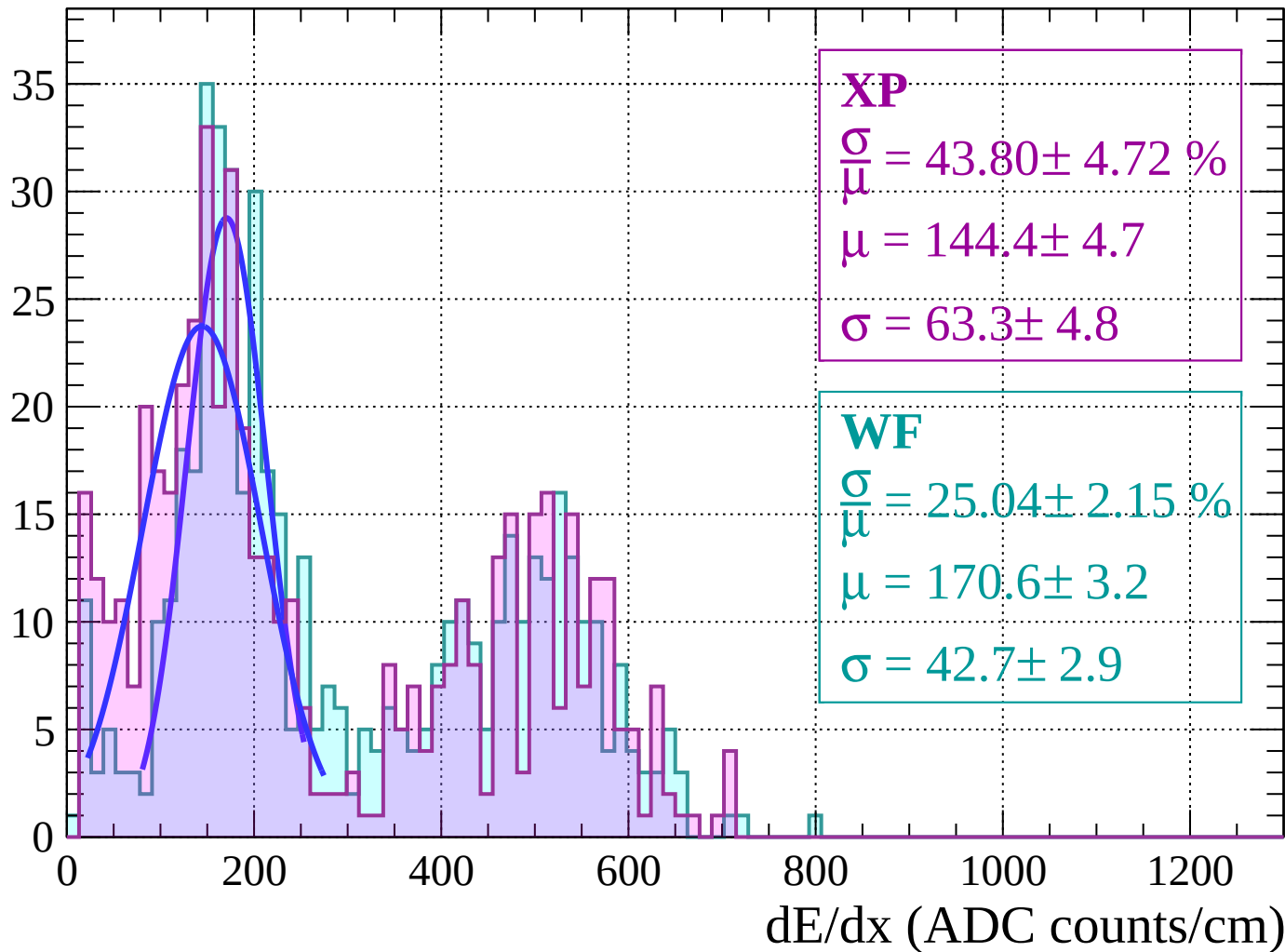
Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

Count



Energy loss | $72 < \theta < 74$

Count

XP

$$\frac{\sigma}{\mu} = 49.34 \pm 5.31 \%$$

$$\mu = 108.1 \pm 4.2$$

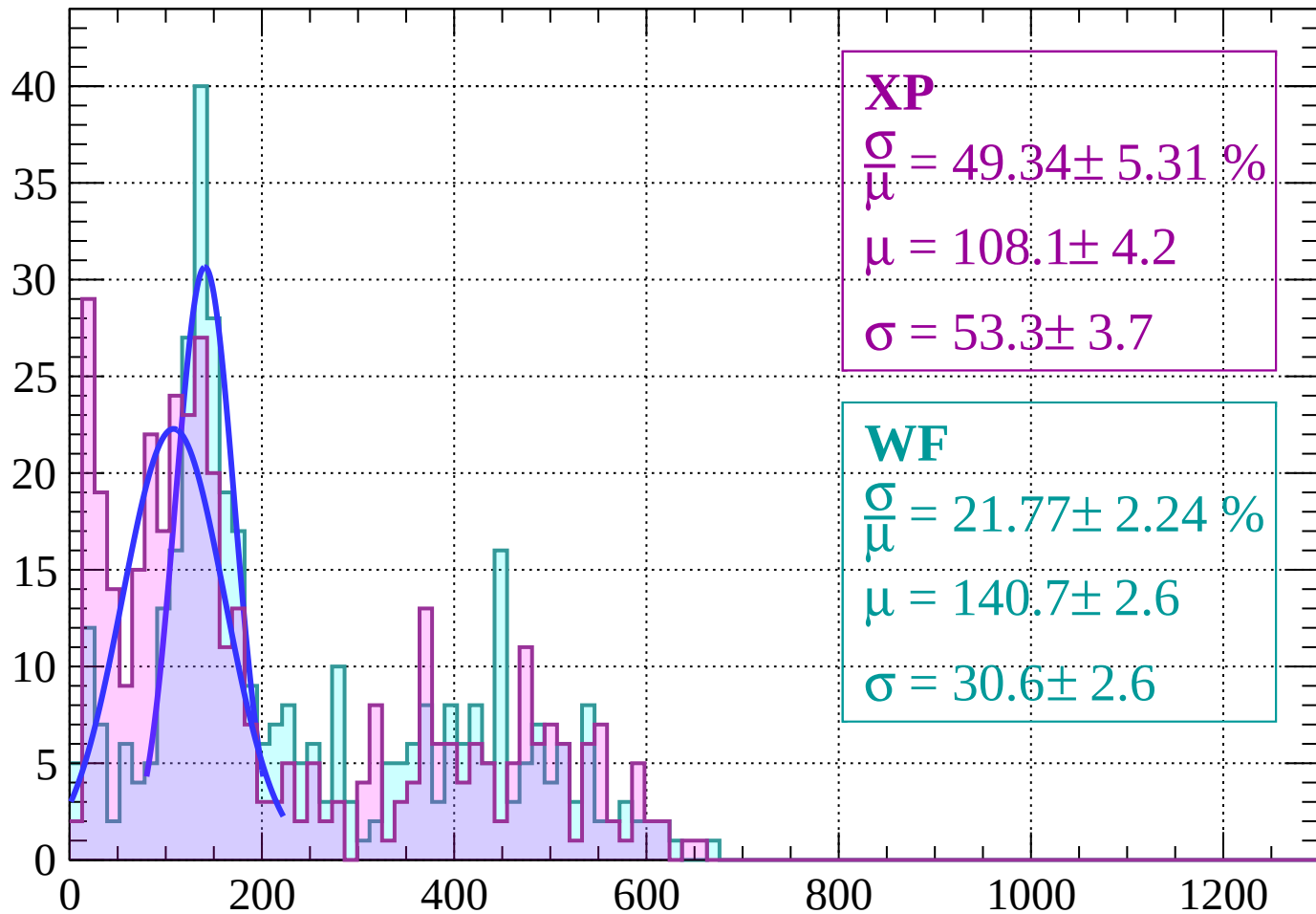
$$\sigma = 53.3 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 21.77 \pm 2.24 \%$$

$$\mu = 140.7 \pm 2.6$$

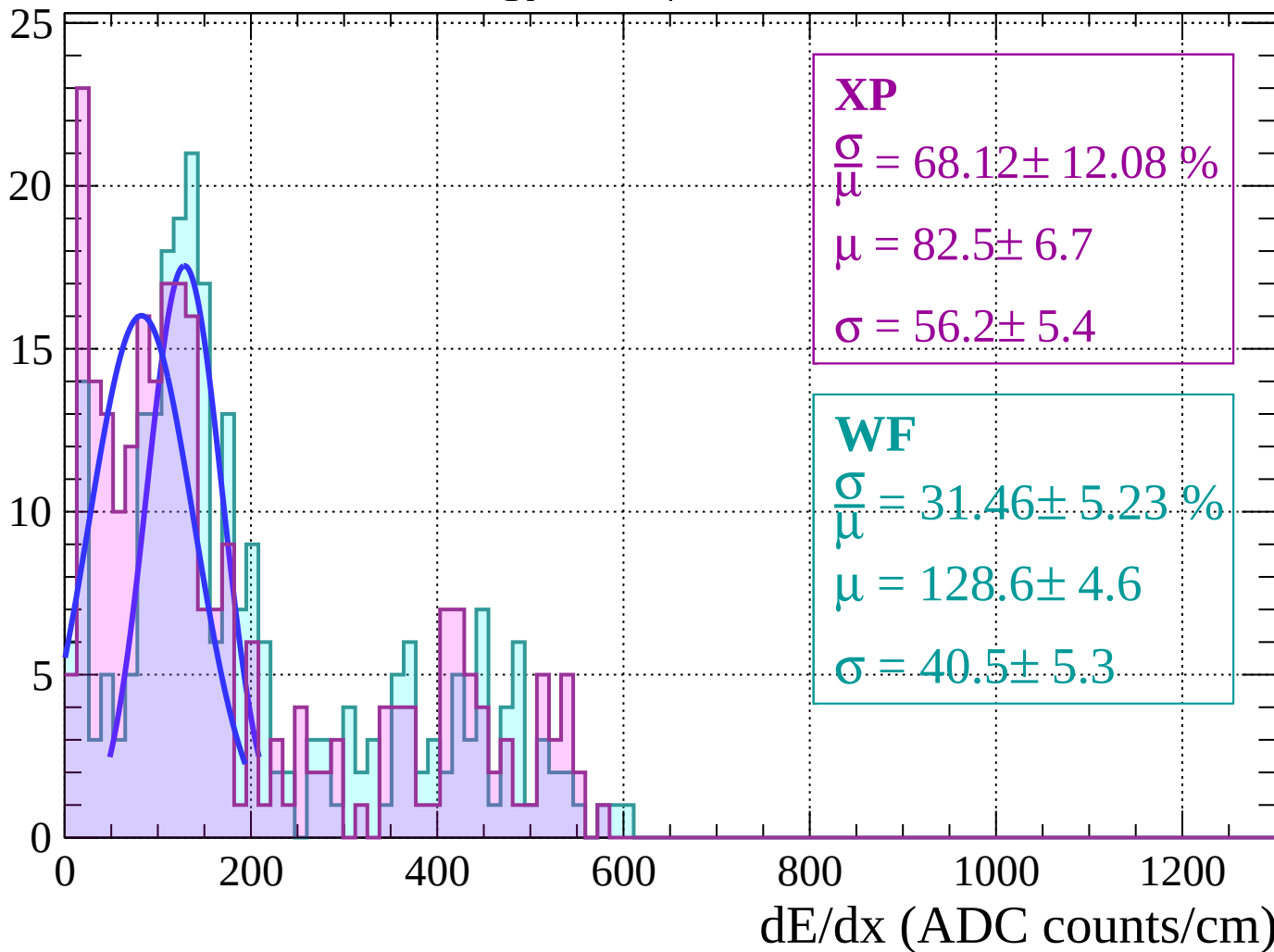
$$\sigma = 30.6 \pm 2.6$$



dE/dx (ADC counts/cm)

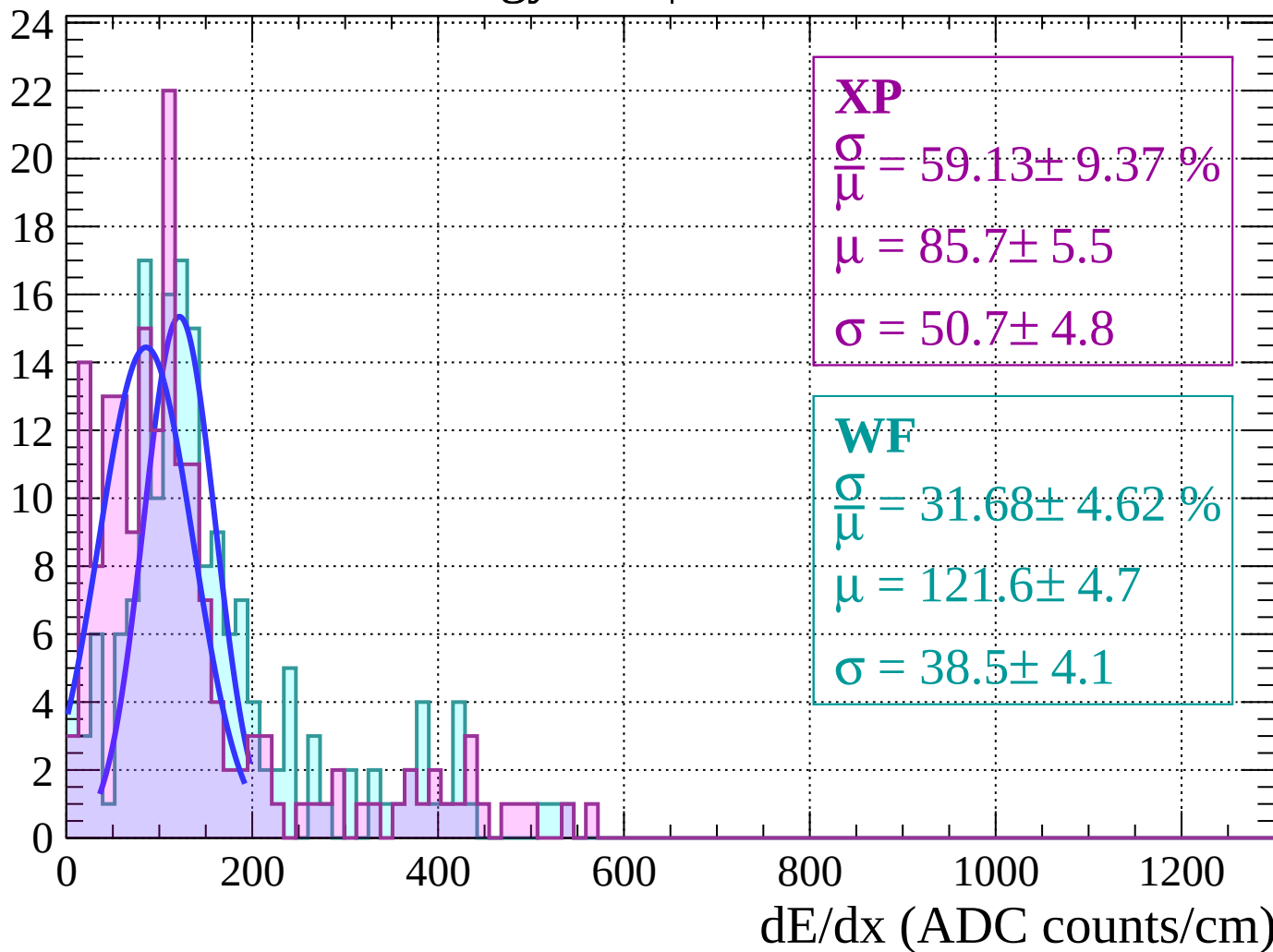
Energy loss | $74 < \theta < 76$

Count



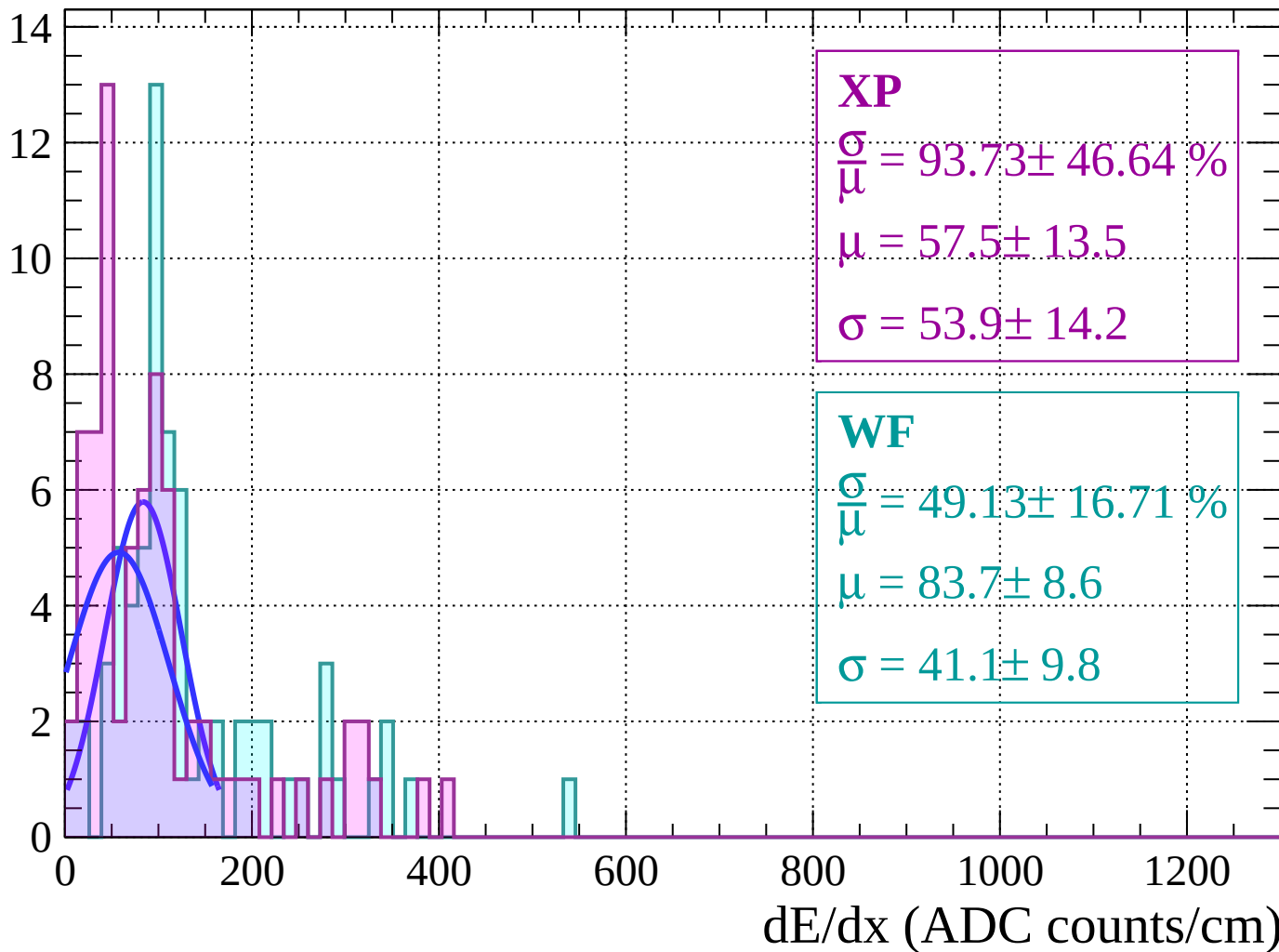
Energy loss | $76 < \theta < 78$

Count



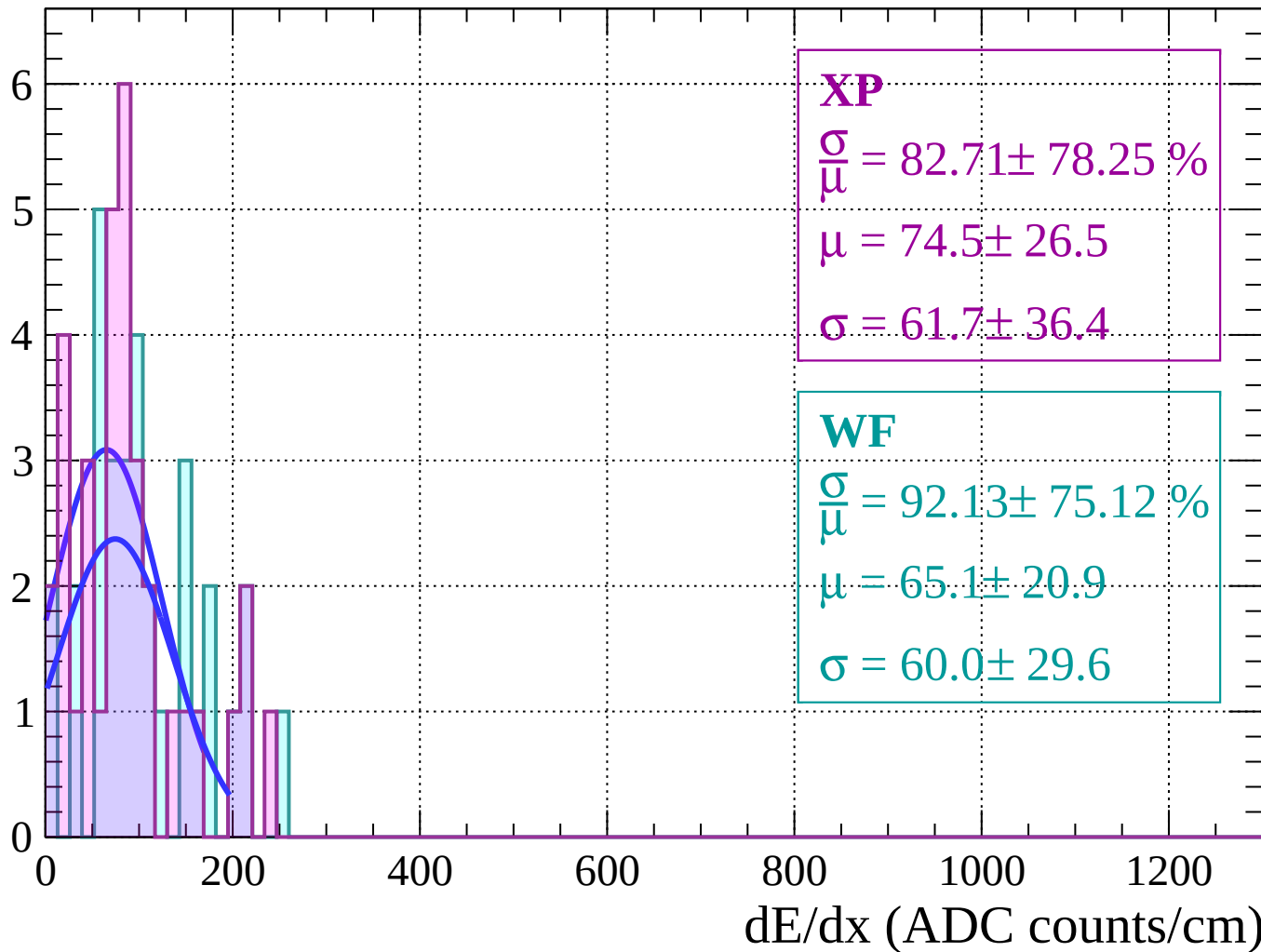
Energy loss | $78 < \theta < 80$

Count



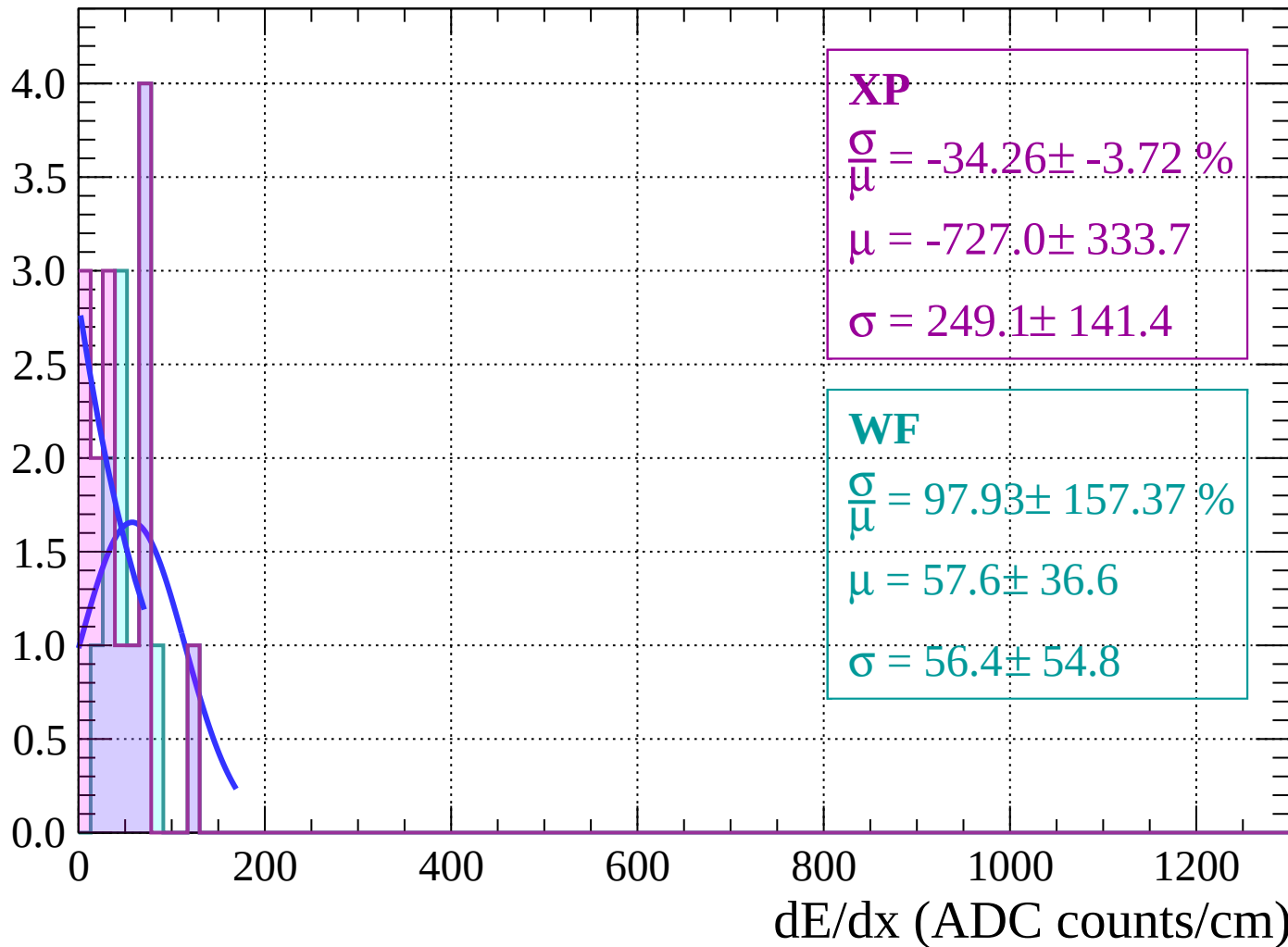
Energy loss | $80 < \theta < 82$

Count

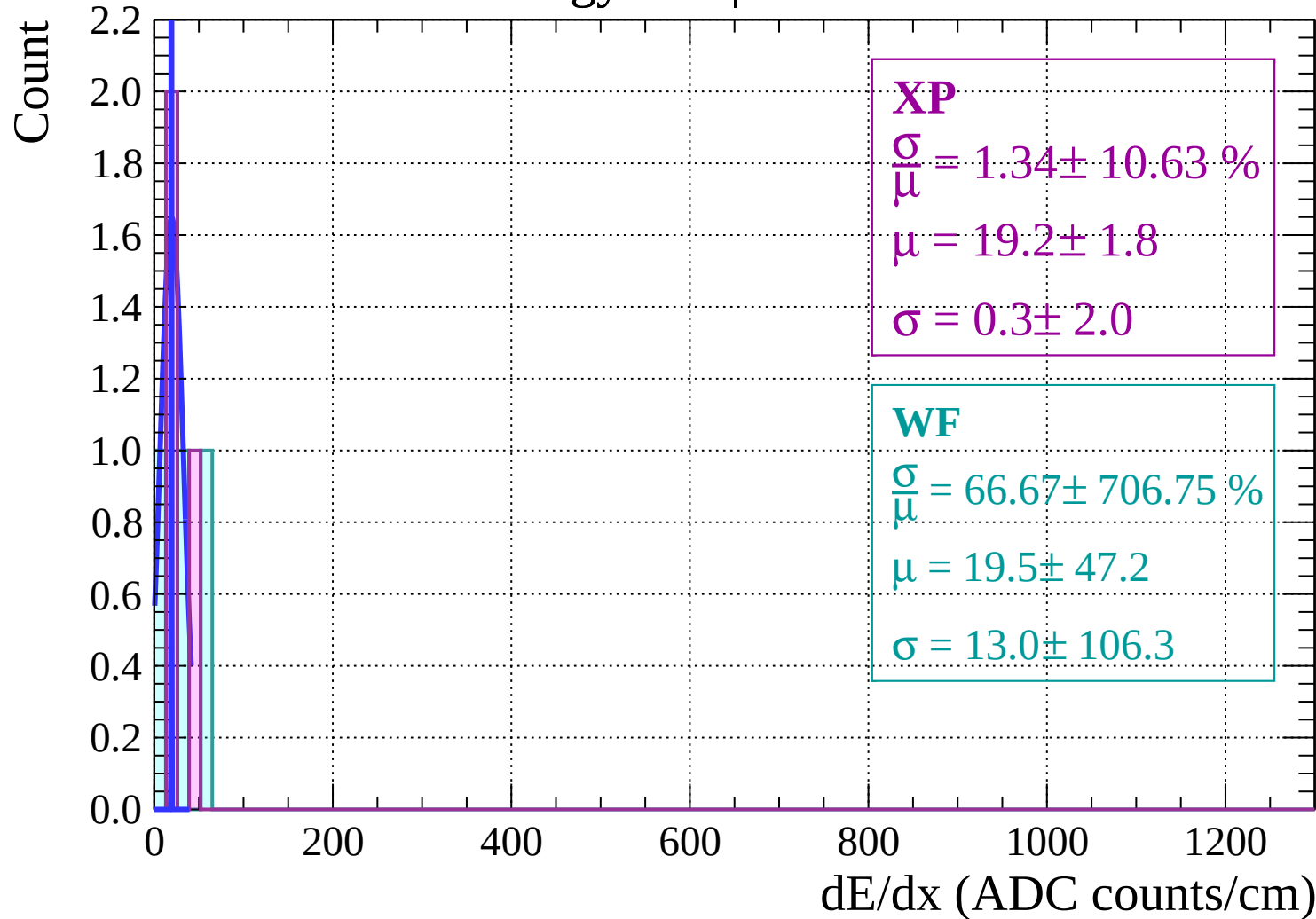


Energy loss | $82 < \theta < 84$

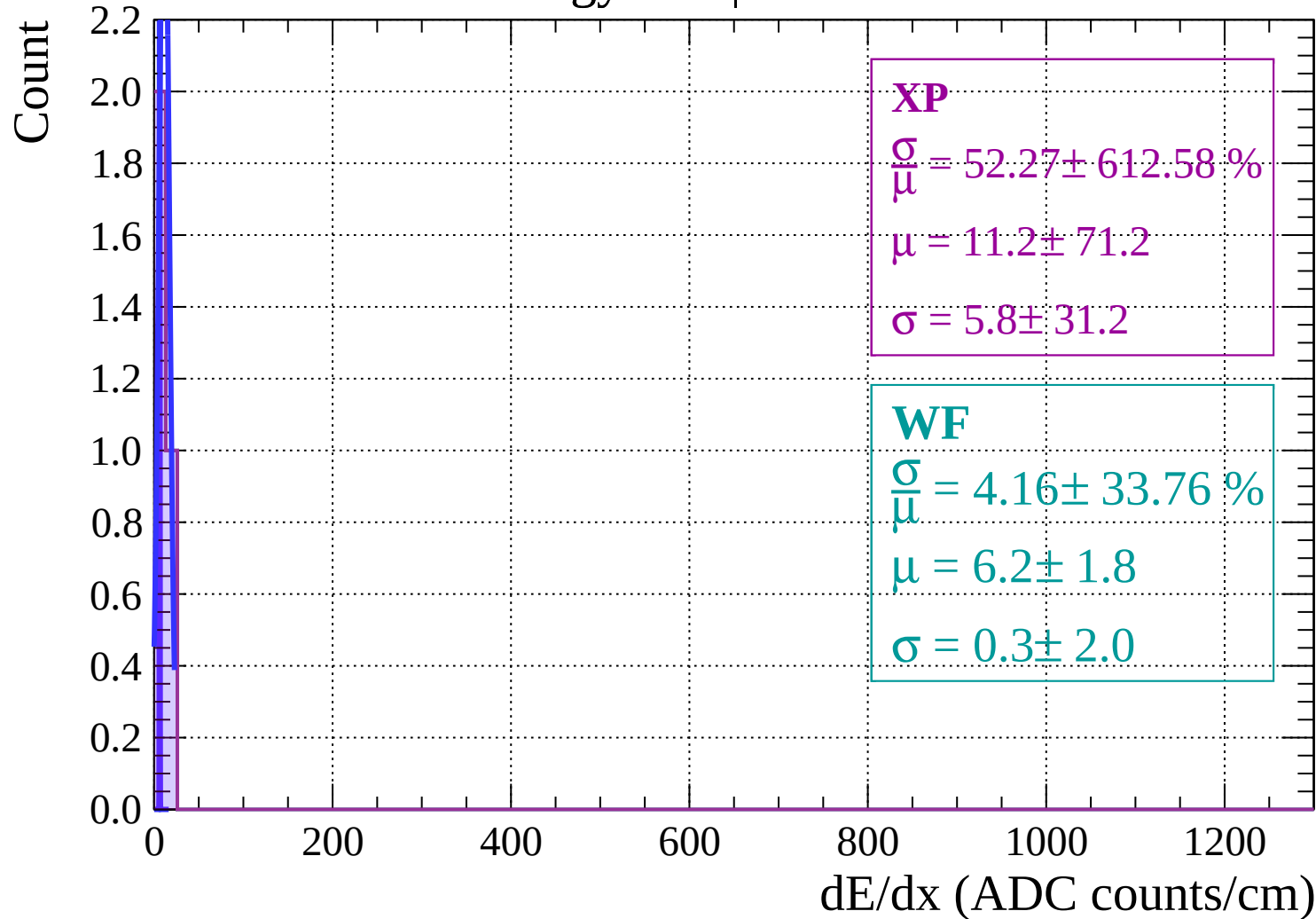
Count



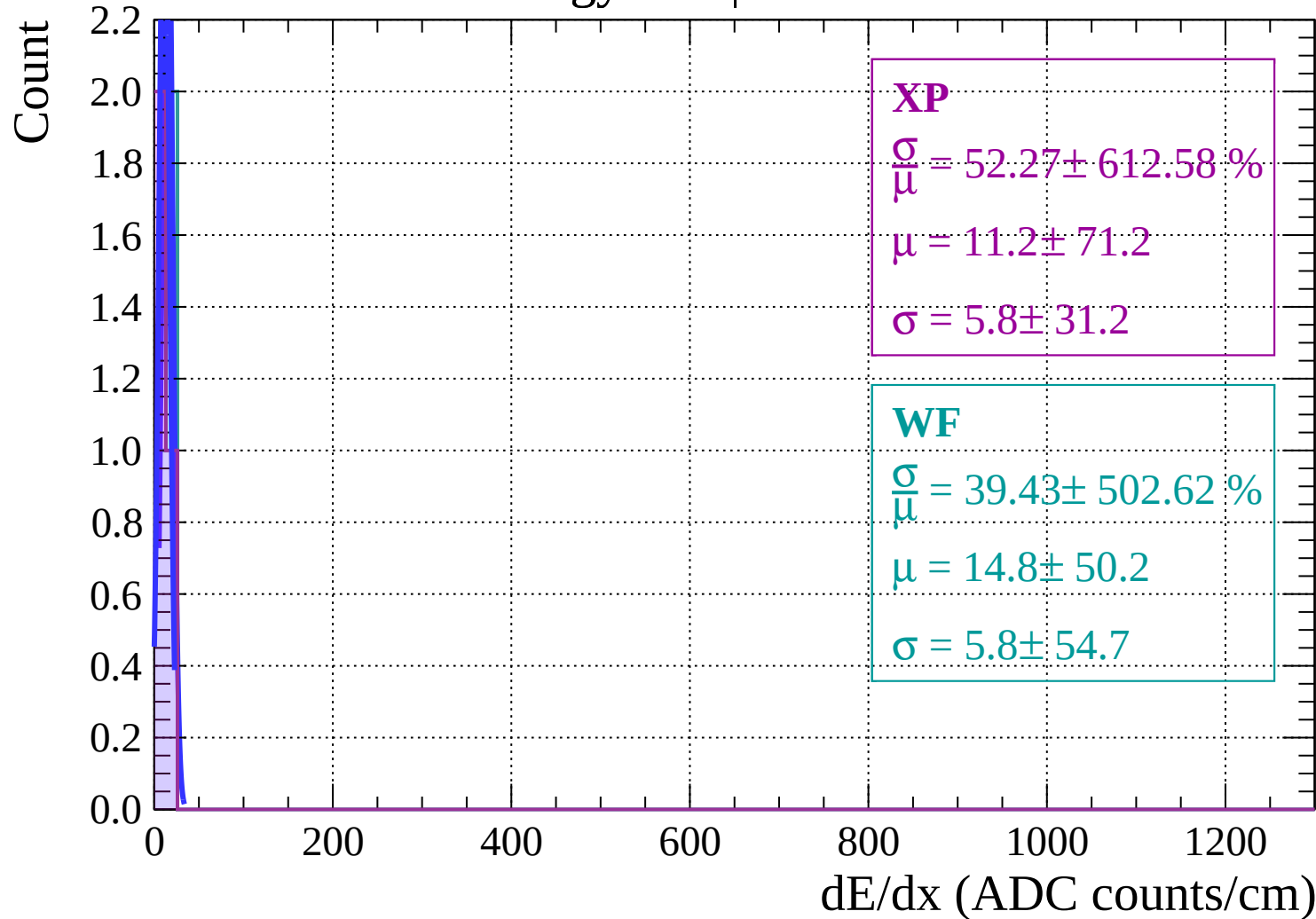
Energy loss | $84 < \theta < 86$



Energy loss | $86 < \theta < 88$

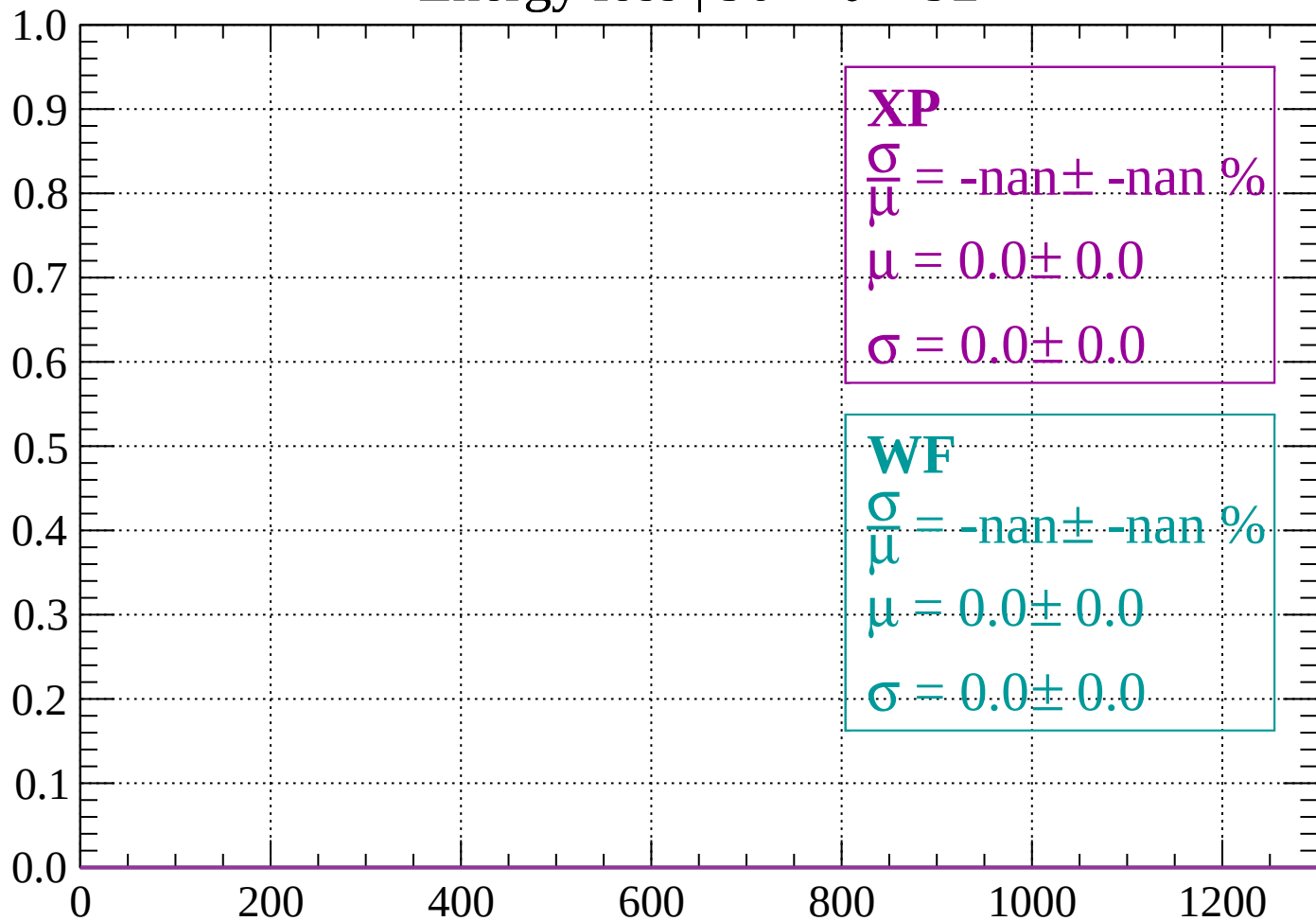


Energy loss | $88 < \theta < 90$



Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

